The Genesis Trilogy

World, interface and subject

Complete Collected Volume

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Introduction

How do a structured world, a physical interface and a subject capable of revising its own participation form together? HAFF, Q-RAIF and T-DOME retain these three responsibilities. The present volume follows their main sequence; Foundation, Consciousness, Imprint, Brain Anchor and the Q-RAIF companions develop its additional mathematical and explanatory interfaces.

The world line now puts physical selection before arbitrary redescription. Interactions select persistent blocks, coherence-decay rates give a coupling-dependent Euclidean relation, and conditions for a spatial interpretation become an explicit target. Transfer dynamics are not derived from dephasing; response-based distances require a separately identified reversible generator. Records can recover an unknown event order in a specified protocol, while time parametrization remains a different question. The gravitational line uses common-source and modular consistency conditions before turning to finite sourced-response constructions.

The interface line establishes conditional relationships among propagation, operator representation and the cost of control; it does not identify positive and Lorentzian metrics by name. The subject line gives task-information prices, a finite history-formation theorem and continued inquiry and repair from an acquired model. Those advances reduce specific supplied inputs without pretending that every device or every law has been generated.

Finite A has a precise meaning: one semiclassical mean-action scalar/JT history carries an acquired record into a physical source, geometric response and later X-basis reading, with a signal of order 10^{-9}. Sustained fixed-capacity thermal updating B and joint sharp heat export with effective return C retain stronger conjectural status. Neither establishes universal explanatory closure. The Imprint Chain's typed interface table and the postscript's cosmic formation-and-return conjecture keep the larger programme visible without borrowing those conjectures as proofs.

Reading map

HAFF-A Physical coarse-graining selects persistent distinctions, with a coupling-dependent coherence-decay relation. Transfer responses and a spatial dictionary require further inputs; finite geometry certificates test this generative route.

HAFF-B Interaction-selected blocks, protected content and finite lifetimes establish which relations survive. Stable sectors are not automatically tensor factors or spatial sites.

HAFF-C Acquired history changes the interventions available to a body. The relation between functional agency and experience remains a separate explanatory question.

HAFF-D One conserved physical source enters modular and geometric response constraints. Conditional field-law synthesis, finite physical return and stronger thermal conjectures have distinct assumptions.

HAFF-E Actual instruments connect permissions to stable and public records. Public agreement requires more than redundancy to certify correctness.

HAFF-F Endpoint records identify an unknown exchange-event order under specified gates. Ordinary copying can fail this test; acquisition, renewal and time parametrization remain distinct.

HAFF-G Internal programs alter permissions and sources. Protocol-relative indistinguishability constrains complete explanation without creating a universal final boundary.

HAFF-PS Cosmic origin and closure return as a formation-and-renewal conjecture, alongside conditional finite-body participation bounds.

QR-A Local propagation and the single quadratic shell constrain Clifford and Lorentzian read-out. A common cone still needs the stated speed-alignment dynamics.

QR-B A centred physical coupling gives a state-independent cost criterion and a conditional bridge to the propagation form. Alignment is an additional physical question.

QR-C Discarded control ports and closed propagation give different exact representation results. Finite source, clock and renewal witnesses test their physical implementation.

TD-I Task-relevant retained distinctions have usable work value. Compression prices a current feedback posterior, not every future learning task.

TD-II Finite paid training selects a two-bit reading frame that beats a history-independent direction at one test read, even with a free coefficient oracle. Total resources are not matched; the independent-replay control tests history relevance, not a separation from general learning.

TD-III Drift detection, inquiry and paid repair use different functions of the same acquired belief. Terminal compression and continuing self-revision need not retain the same information.

PHYSICAL COARSE-GRAINING AND THE PROBLEM OF EMERGENT GEOMETRY: THE HOLOGRAPHIC ALAYA–FIELD FRAMEWORK

A PREPRINT

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ABSTRACT

How can a physical process select the structures from which geometry is reconstructed? We study this question through implemented coarse-graining. For finite systems with commuting Hermitian noise, the dynamics determines a direct-sum decomposition through the centre of its fixed algebra. Cross-sector coherence decay determines a Euclidean rate geometry, and a finite-window certificate identifies persistent blocks from separated rates. Coherent probes can estimate these relations; sector-label measurements alone cannot. Identical system channels can also leave different readable environmental records when a purification is inaccessible. These results distinguish process-selected structure, physical permissions and geometric interpretation. Two quantitative tests then constrain reconstruction: an exact qubit criterion exposes reductions that entropy cannot distinguish, while a local interaction and recording protocol certifies non-isometric correlation metrics from a fixed GHZ source. The latter uses a declared site representation and distance rule. Neither its tensor sites nor spatial transport follow from the noise sectors. We formulate the missing bridge as a same-process locality conjecture: interaction histories must select persistent organization and produce intervention responses admitting a common spatial interpretation.

1 Introduction

A reconstruction of space already has parts to relate. Decoherence distinguishes a system, apparatus and environment; entanglement-based reconstructions assign observables to regions or tensor factors [1, 4, 3]. A Hilbert space alone selects none of these divisions. Accessible observables can induce subsystem structure [5, 6, 7]; conversely, tailoring the observables can give a fixed pure state any prescribed Schmidt decomposition [8]. The emergence problem therefore includes the physical selection of the parts, as well as the geometric reading of their relations.

We approach selection through the implemented process. A coupling determines a reduced channel and can protect a direct-sum organization of its carrier. In a commuting-noise class, this organization is the centre decomposition of the channel's fixed algebra; a supplied spatial factorization is unnecessary for that calculation. The rates at which its sectors lose coherence also determine an operational relation. Sections 3 and 4 make both statements precise and identify the measurements needed to test them. They use standard dephasing structure [28, 25], while separating it from the availability of instruments and from spatial transport.

The distinction changes what a geometric claim requires. The same observable algebra can arise from reductions with different resolution. Different channels can also have identical entropy functions. Section 5 characterizes this obstruction for qubits. Section 6 carries a fixed correlation reconstruction through noise interactions, environmental records and finite measurements. Its non-isometric metrics are physical consequences of that protocol, with the site representation and distance rule stated explicitly. Section 7 returns to the remaining question: when can process-selected relations acquire a spatial interpretation through actual propagation or intervention?

We use *Alaya-Field*, borrowed from Yogācāra's storehouse consciousness, as a technical name for an unfactorized source $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$, with Ω cyclic and separating. The name specifies no additional ontology. A Type III algebra has no intrinsic density operator or trace entropy [9]. All entropy results here instead concern finite-dimensional carriers; a faithful finite state has a cyclic and separating vector in its standard representation, whereas a pure carrier vector does not separate the full matrix algebra. No Type III entropy formula or Lorentzian metric is inferred from the finite calculations.

2 Coarse-graining structures, algebras and channels

2.1 Definition

An observable algebra does not by itself reinstate a tensor factorization: operators can act irreducibly on $\mathcal{H}_{\mathrm{total}}$, and in algebraic quantum field theory local algebras are assigned to regions without any global tensor product [9]. A coarse-graining here is meant to be realized by a physical interaction, not chosen by an agent.

Definition 2.1 (Coarse-graining structure)**.** *A coarse-graining structure is a pair* $\mathbf{c} = (\mathcal{A}_{\mathrm{sel},\mathbf{c}}, \Phi_{\mathbf{c}})$*, where*

1. $\mathcal{A}_{\mathrm{sel},\mathbf{c}} \subset \mathcal{B}(\mathcal{H}_U)$ *is a selected unital* $*$*-subalgebra used to define the effective representation. It is not identified with Paper E's probe-reachable effect system, pointer or record algebra, or environmental fixed set [26]; a physical model must state any map between them.*
2. *In the finite-dimensional results,* $\Phi_{\mathbf{c}}$ *is a completely positive trace-preserving (CPTP) map on states that implements an operational reduction; its Heisenberg dual* $\Phi_{\mathbf{c}}^*$ *is unital and completely positive. Partial trace over a specified factor and channels induced by a conditional expectation are examples. In a general von Neumann setting the corresponding object is a normal channel with a normal unital completely positive dual, and no entropy result below is claimed for that extension.*

No theorem guarantees a unique maximal observable algebra for a given physical system without additional structure. Several admissible algebras can coexist, and for a fixed algebra several channels can implement different losses of information, such as tracing out inaccessible degrees of freedom or effective decoherence. The two entries of $\mathbf{c}$ must therefore be related explicitly.

2.2 The algebra generated by a channel

For a channel $\Phi : \mathcal{B}(\mathcal{H})_* \to \mathcal{B}(\mathcal{K})_*$ put

$$\mathcal{S}_\Phi = \operatorname{ran} \Phi^*, \qquad \mathcal{A}_\Phi = C^*(\mathcal{S}_\Phi).$$

$\mathcal{S}_\Phi$ is a unital self-adjoint subspace, not necessarily closed under products. The effects that the channel actually reads out are $\Phi^*([0, I])$, which is in general smaller than the set of positive contractions in $\mathcal{S}_\Phi$. We call $\mathbf{c}$ weakly compatible if $\mathcal{S}_{\Phi_\mathbf{c}} \subseteq \mathcal{A}_{\mathrm{sel},\mathbf{c}}$ and minimally compatible if $\mathcal{A}_{\mathrm{sel},\mathbf{c}} = \mathcal{A}_{\Phi_\mathbf{c}}$.

Proposition 2.2 (Algebras and post-processing)**.**

1. *If two states agree on* $\mathcal{S}_\Phi$*, in particular if they agree on* $\mathcal{A}_\Phi$*, then their outputs under* Φ *coincide.*
2. *If* $\Phi_2 = \Lambda \circ \Phi_1$ *for a channel* Λ*, then* $\mathcal{S}_{\Phi_2} \subseteq \mathcal{S}_{\Phi_1}$ *and* $\mathcal{A}_{\Phi_2} \subseteq \mathcal{A}_{\Phi_1}$*.*
3. *The converse of (2) fails. For the depolarizing channels* $D_p(\rho) = p\rho + (1-p)I/2$ *on a qubit with* $0 < p < q \leq 1$*, both* $\mathcal{A}_{D_p}$ *and* $\mathcal{A}_{D_q}$ *equal the full matrix algebra,* $D_p = D_{p/q} \circ D_q$*, and* D_q *is not a post-processing of* D_p*.*

Proof. (1) For any output operator Y, $\mathrm{Tr}[\Phi(\rho)Y] - \mathrm{Tr}[\Phi(\sigma)Y] = \mathrm{Tr}[(\rho - \sigma)\Phi^*(Y)]$, and $\Phi^*(Y) \in \mathcal{S}_\Phi$. (2) $\Phi_2^* = \Phi_1^*\Lambda^*$, so the range of Φ_2^* lies in the range of Φ_1^*, and generated algebras inherit the inclusion. (3) Both maps are invertible on the operator space, so their dual ranges are everything. The identity $D_{p/q}(D_q(\rho)) = p\rho + (1-p)I/2$ is direct. Two orthogonal pure inputs have outputs at trace distance p under D_p and q under D_q; a channel cannot increase trace distance, so no Λ satisfies $\Lambda \circ D_p = D_q$. □

Two reductions with the same algebra can thus have different operational resolution. An algebra label is not a substitute for the channel and its allowed post-processing. If a conditional expectation is chosen as the implementation, that is an additional structure, not a synonym for every CPTP reduction.

3 Structures selected by a physical process

Consider a finite carrier $\mathcal{H}$ with commuting Hermitian couplings A_k at rates $\gamma_k > 0$. In a specified interaction picture the Heisenberg generator is

$$\mathcal{L}(X) = -\frac{1}{2}\sum_k \gamma_k [A_k, [A_k, X]], \qquad \Phi_t = e^{t\mathcal{L}}. \tag{1}$$

This generator is self-adjoint for the Hilbert–Schmidt inner product, so the same formula acts on states. A Hamiltonian contribution must either be included in a different analysis or removed by a justified interaction picture. The carrier, system–environment division, couplings and time scale are physical inputs.

Proposition 3.1 (Intrinsic sectors of a dephasing process)**.** *Let $\mathcal{B} = \mathrm{Fix}(\Phi) = \ker \mathcal{L}$, and let P_a, $a \in V$, be the minimal nonzero projections of $Z(\mathcal{B})$. Then*

$$\mathcal{H} = \bigoplus_{a\in V} P_a\mathcal{H}, \qquad \mathcal{B} = \bigoplus_{a\in V} \mathcal{B}(P_a\mathcal{H}).$$

The projections are determined by the semigroup, up to relabelling. For every $X \in P_a\mathcal{B}(\mathcal{H})P_b$,

$$\mathcal{L}(X) = -D_{ab}X, \qquad D_{ab} = \frac{1}{2}\sum_k \gamma_k(a_k(a) - a_k(b))^2, \quad A_kP_a = a_k(a)P_a. \tag{2}$$

These rates are intrinsic to the semigroup: $D_{aa} = 0$ and $D_{ab} > 0$ for distinct sectors. A common unitary conjugation rotates the P_a and preserves D.

Proof. The identity

$$-\langle X, \mathcal{L}(X)\rangle_{\mathrm{HS}} = \frac{1}{2}\sum_k \gamma_k \|[A_k, X]\|^2_{\mathrm{HS}}$$

gives $\mathcal{B} = \{A_k\}'$. Simultaneous diagonalization yields full joint eigenspaces $P_a\mathcal{H}$, on which every coupling is scalar. Their full matrix algebras form the commutant, and their projections are precisely its central atoms. No basis inside a degenerate eigenspace enters this description. Applying the double commutator to a cross-corner gives (2). Its scalar is fixed by $\mathcal{L}$, independently of the coupling representation; different atoms have different joint eigenvalue tuples. Conjugating the entire generator conjugates its corners and leaves their scalar eigenvalues unchanged. □

The fixed-algebra construction and its dephasing rates are known [28, 25]. Their role here is to remove a separate choice of effective sectors after the physical process has been supplied. A direct sum is not a tensor product. In particular, the P_a are not the sites of the correlation reconstruction below.

For finite t, the channel multiplies each corner by $e^{-tD_{ab}} > 0$, so it has full linear operator range and $\mathcal{A}_{\Phi_t} = \mathcal{B}(\mathcal{H})$. Its long-time limit is the pinching $\mathcal{E}_\infty(X) = \sum_a P_aXP_a$, whose range is $\mathcal{B}$. Thus $(\mathcal{B}, \mathcal{E}_\infty)$ is minimally compatible in Definition 2.1; replacing Φ_t by this projection at finite time requires an error estimate.

3.1 Selection on a finite window

Exact fixed algebras can jump under arbitrarily weak symmetry breaking. A finite capability instead asks how well an entire block survives over a stated time. Following Paper B, call a unital algebra ε-stable on $[0, T]$ if

$$\sup_{0\le t\le T} \sup_{\substack{X\in\mathcal{B}\\ \|X\|\le 1}} \|\Phi_t(X) - X\| \le \varepsilon.$$

Put $u_a = (\sqrt{\gamma_k/2}\, a_k(a))_k$. For a partition π of the intrinsic sectors into clusters C, define

$$Q_C = \sum_{a\in C} P_a, \qquad \mathcal{B}_\pi = \bigoplus_C \mathcal{B}(Q_C\mathcal{H}), \qquad r_C = \inf_c \max_{a\in C} \|u_a - c\|,$$

and $\kappa_{\mathrm{in}} = 4\max_C r_C^2$, $\Delta_{\mathrm{out}} = \min_{a,b \text{ in different } C} D_{ab}$. An empty minimum is $+\infty$.

Proposition 3.2 (A representation-independent window certificate)**.** *If $0 < \varepsilon < 1$, $T > 0$ and*

$$T\kappa_{\mathrm{in}} \le \varepsilon < 1 - e^{-T\Delta_{\mathrm{out}}}, \tag{3}$$

then $\mathcal{B}_\pi$ is ε-stable. No strictly containing unital algebra is ε-stable. At most one cluster partition satisfies this certificate; its algebra contains every stable block algebra in a common joint eigenbasis of the couplings.

The quantities in this certificate depend only on the rate matrix, since $D_{ab} = \|u_a - u_b\|^2$ determines its Euclidean configuration. The radius bound replaces Paper B's coordinate-box bound. Centring the noise coordinates on each cluster gives a generator norm at most $4r_C^2$; the separated cross-cluster matrix units exclude stable extensions. Appendix A gives the proof. The maximality statement does not compare all rotated, incomparable subalgebras.

Stability of individual coherences is insufficient. Three equally spaced points can have $TD_{01} = TD_{12} = .06$ but $TD_{02} = .24$: the first two matrix units lose less than .1 of their amplitude, while their product loses more than .1. Generating an algebra from all slow matrix units can therefore introduce unstable observables. The whole-block condition in (3) supplies a sufficient selection rule; absence of a gap does not determine a unique partition.

3.2 What a stable algebra makes available

A fixed observable need not have an available reader or gate. For example, a pointer-controlled collision can implement

$$\Gamma(\rho) = \sum_{z=0}^{1} \mathrm{Tr}(P_z\rho)|e_z\rangle\langle e_z|, \qquad \Gamma^*(F) = \sum_z \langle e_z|F|e_z\rangle P_z.$$

When $0 < |\langle e_0|e_1\rangle| < 1$, the dual range spans the pointer algebra, but its implementable effects exclude the sharp projections [26]. Source dephasing and the record channel arise from the same collision and have different outputs. Foundation I makes the same distinction between an ideal seed algebra and a finite instrument realizing a task [29].

For a faithful stationary state of (1), the density matrix is block diagonal and commutes with all P_a. Their modular orbit is therefore constant. Foundation I's stable hull of central seeds S is exactly $W^*(S)$; it is abelian and has no proper half-sided modular inclusion. The selected labels thus do not acquire a spacetime Cut by taking their hull. Noncentral seeds or an independently supplied infinite source require their own interface.

4 Which relations can the process identify?

4.1 A rate geometry and its measurements

Equation (2) gives a metric

$$d_D(a,b) = \sqrt{D_{ab}} = \|u_a - u_b\|. \tag{4}$$

If joint eigenvectors were labelled separately, this would be a pseudometric; equal-rate-zero labels must first be merged into the central sectors. If all couplings are scalar there is one sector. For $r = |V|$ and $J = I - \mathbf{1}\mathbf{1}^T/r$, the centred Gram matrix is $G = -JDJ/2$. Its positive eigenspaces reconstruct the minimal Euclidean configuration, up to isometry. Its dimension is a rank of the noise responses, not a spatial dimension. Moreover, d_D has units $\mathrm{time}^{-1/2}$; $\sqrt{t_0 D}$ is dimensionless for a supplied reference time t_0. Neither quantity is a physical length.

Proposition 4.1 (Visibility identifies the rate relation)**.** *Suppose a reader can choose unit vectors $x \in P_a\mathcal{H}$, $y \in P_b\mathcal{H}$, prepare $(x+y)/\sqrt{2}$, and measure $X_{ab} = |x\rangle\langle y| + |y\rangle\langle x|$ after known time $t > 0$. The mean is $v_{ab}(t) = e^{-tD_{ab}}$, hence $D_{ab} = -t^{-1}\log v_{ab}(t)$. If the true and estimated visibilities both belong to $[v_{\min}, 1]$ with $v_{\min} > 0$ and differ by at most ζ, then*

$$|\widehat{D}_{ab} - D_{ab}| \le \frac{\zeta}{t v_{\min}}, \qquad |\sqrt{\widehat{D}_{ab}} - \sqrt{D_{ab}}| \le \sqrt{\frac{\zeta}{t v_{\min}}}.$$

For M tested pairs with n independent preparations each, the visibility error bound holds simultaneously with failure probability at most $2Me^{-n\zeta^2/2}$. The sample mean may be clipped to $[v_{\min}, 1]$ when the true visibility is known to lie there.

Proof. The two off-diagonal entries of the prepared state are multiplied by $e^{-tD_{ab}}$, while its populations are constant. Their sum is the stated expectation. The logarithm has derivative bounded by $1/v_{\min}$ on the relevant interval, and $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x-y|}$ for $x, y \ge 0$. Each measurement has outcomes ± 1 on the prepared subspace; Hoeffding's inequality and a union bound give the final statement. Clipping to an interval containing the true mean cannot increase the error. □

The proposition assumes the stated coherent preparations and readers, rather than obtaining them from the fixed algebra. It identifies a relation of the physical dynamics. It cannot certify exact zero rates with finite data, and says nothing about the stability of the number of exact sectors.

Proposition 4.2 (Sector-only access cannot identify the rates)**.** *Fix projections P_a. If all allowed preparations, instruments and outputs act within the block-diagonal algebra $\bigoplus_a \mathcal{B}(P_a\mathcal{H})$, every adaptive record law is independent of the cross-sector rates of a dephasing semigroup with these projections.*

Proof. Every such semigroup restricts to the identity on this algebra. Each permitted step, conditional on the previous record, therefore has the same output in every model. Induction over the steps gives the same complete record law. □

Thus mathematical knowledge of a channel and its identification by a reader with finite permissions are different resources.

4.2 A conditional record interpretation

With a pure initial environment and access to its complete output, a dilation of this dephasing channel has

$$V_t\psi_a = \psi_a \otimes e_a(t), \qquad \langle e_b(t)|e_a(t)\rangle = e^{-tD_{ab}}.$$

Indeed, preserving every pure state in a sector forces its environment vector to be independent of the state within that sector; the cross-corner multiplier fixes the overlaps. Their Gram matrix is positive because it is the Schur kernel of the channel. The isometry has a unitary extension at each fixed time. Complete-environment binary discrimination then has trace distance

$$q_{ab}(t) = \sqrt{1 - e^{-2tD_{ab}}}. \tag{5}$$

This is another metric, supplied by an explicit discrimination task, and its inversion requires the appropriate environmental measurements. A restricted fragment can carry less information.

Example 4.3 (Same system channel, different readable records)**.** *Use the collision $U = P_0 \otimes I + P_1 \otimes e^{-i\theta Z_E}$ with $0 < \theta < \pi/2$. A memory initially in $|+\rangle$ has conditional states $|+\rangle$ and $e^{-i\theta Z}|+\rangle$, overlap $\cos\theta$ and trace distance $\sin\theta$. A memory initially in $I/2$ gives the identical system off-diagonal multiplier $\mathrm{Tr}[(I/2)e^{i\theta Z}] = \cos\theta$, but both its conditional states are $I/2$. No measurement on that memory distinguishes the labels. If its purifier is also accessible, the joint conditional pure states again have overlap $\cos\theta$ and trace distance $\sin\theta$.*

Taking $\cos\theta = e^{-tD}$ matches any positive two-sector rate at a fixed time. This example concerns a reduced channel, rather than claiming equivalence of complete multitime processes. It locates the missing information precisely: the specified environment port excludes a purification. Decoherence alone therefore does not fix readable records, in agreement with Paper B's leakage relation [25].

The metrics (4) and (5) describe response to noise and environmental discrimination. The dynamics has no sector population transport: $\mathrm{Tr}\, P_a\Phi_t(\rho) = \mathrm{Tr}\, P_a\rho$. Neither metric is consequently a propagation distance in this model. Establishing a spatial interpretation needs another observable relation, addressed in Section 7.

5 Entropy as a reconstruction test

A reconstruction may use entropies rather than coherent response data. Those entropies can distinguish channels only to a limited extent. The exact outputs of distinct channels agree on a measure-zero set of pure inputs, but entropy separation needs an additional nondegeneracy condition (Theorem B.4 in Appendix B). For qubits the condition can be decided exactly. The following results constrain any entropy-based dictionary, including those that retain the same generated algebra.

5.1 Qubit reductions: the criterion

A qubit channel acts on Bloch vectors as an affine map

$$\Phi_i:\ r \mapsto M_i r + t_i, \qquad i = 1, 2,$$

with M_i a real 3×3 matrix and $t_i \in \mathbb{R}^3$, subject to the complete-positivity conditions [15]. Put

$$A = M_1^T M_1 - M_2^T M_2, \qquad b = M_1^T t_1 - M_2^T t_2, \qquad c = \|t_1\|^2 - \|t_2\|^2.$$

The difference of the squared output Bloch lengths is

$$q(r) = \|M_1 r + t_1\|^2 - \|M_2 r + t_2\|^2 = r^T A r + 2b^T r + c. \tag{6}$$

The entropy of a qubit state is a strictly decreasing function of its Bloch length: $S = h\big((1 + \|s\|)/2\big)$ with h the binary entropy in nats.

Theorem 5.1 (Equal entropy on all pure inputs)**.** *Φ_1 and Φ_2 give equal output entropy on every pure input if and only if there is a real number α with*

$$A = \alpha I, \qquad b = 0, \qquad c = -\alpha. \tag{7}$$

Proof. Equal entropy on pure inputs means $q(r) = 0$ on the unit sphere. Comparing $q(r)$ and $q(-r)$ gives $b^T r = 0$ for all unit r, so $b = 0$. A real symmetric quadratic form that is constant on the unit sphere is a multiple of the identity, so $A = \alpha I$ and then $c = -\alpha$. Conversely, (7) gives $q(r) = \alpha(\|r\|^2 - 1)$, which vanishes on the sphere. No rank condition on the outputs is needed, and pure outputs on the boundary are included. □

Together with positivity of the Choi matrices, (7) decides membership for any given pair. It is a decision rule, not yet a classification of all solutions into physically distinct normal forms.

5.2 The two branches

Proposition 5.2 (Zero branch)**.** *Let $X_i = [M_i \mid t_i]$ be the 3×4 matrix with t_i appended. If $\alpha = 0$, then $X_2 = OX_1$ for an orthogonal $O \in O(3)$. If $\operatorname{rank} X_1 \leq 2$, O can be chosen with $\det O = +1$, so Φ_2 is Φ_1 followed by a unitary conjugation of the output. If $\operatorname{rank} X_1 = 3$, O is unique; $\det O = +1$ again gives a unitary conjugation, and $\det O = -1$ gives a unitary conjugation composed with transposition.*

Proof. For $\alpha = 0$, (7) says exactly $X_1^T X_1 = X_2^T X_2$. Equal Gram matrices define an isometry between the column spaces of X_1 and X_2, which extends to an orthogonal map of $\mathbb{R}^3$. If the column space has dimension at most two, the extension can be composed with a reflection in an unused direction to make the determinant positive. Rotations of the Bloch ball are unitary conjugations, and the reflection $\operatorname{diag}(1, -1, 1)$ is transposition in the standard basis. □

Transposition is not completely positive, but both endpoints of a transposed pair can still be channels.

Example 5.3 (Equal entropies everywhere, no simulation either way)**.** *Take $M_1 = pI$, $M_2 = p\operatorname{diag}(1, -1, 1)$ and $t_1 = t_2 = 0$. The first map is a depolarizing channel for $0 \leq p \leq 1$; by the conditions for unital qubit channels [16], the second is completely positive exactly when $0 \leq p \leq 1/3$. They give equal output entropy on every input, pure or mixed. For $0 < p \leq 1/3$ neither is a post-processing of the other. Indeed both are invertible on the Bloch ball, so a post-processing Λ with $\Lambda \circ \Phi_1 = \Phi_2$ would have to act as $r \mapsto \operatorname{diag}(1, -1, 1)r$ on an open set and hence everywhere; that map is transposition, whose normalized Choi matrix has smallest eigenvalue $-1/2$. The same argument excludes the reverse direction.*

Identical entropy functions on the whole input space therefore do not imply operational equivalence, and the two channels are incomparable in the order of Proposition B.2.

Solutions with $\alpha \neq 0$ are not rare. Every pair of complementary qubit channels satisfies (7): if $V : \mathcal{H} \to \mathcal{K} \otimes \mathcal{E}$ is an isometry with $\mathcal{K}$ and $\mathcal{E}$ two-dimensional, $\Phi(\rho) = \operatorname{Tr}_{\mathcal{E}} V\rho V^\dagger$ and $\Phi^c(\rho) = \operatorname{Tr}_{\mathcal{K}} V\rho V^\dagger$, then for a pure input the joint output is pure and its two marginals have the same entropy. For amplitude damping with decay probability γ and its complementary channel, $\alpha = 1 - 2\gamma$, which is nonzero unless $\gamma = 1/2$. A family with full Choi rank is

$$\Phi_{a,b} : r \mapsto ar + b\hat{z}, \qquad \Phi_{b,a} : r \mapsto br + a\hat{z}, \qquad a, b > 0,\ a + b < 1.$$

$\Phi_{a,b}$ is a convex combination of the identity and the replacer with Bloch vector $b\hat{z}/(1-a)$, so it is CPTP, and likewise $\Phi_{b,a}$. Both give squared pure-output length $a^2 + b^2 + 2abr_z$, and $\alpha = a^2 - b^2$. For $a = .3$, $b = .2$ both normalized Choi matrices have smallest eigenvalue .125.

5.3 Mixed inputs expose the nonzero branch

Proposition 5.4 (Entropy gap on mixed inputs)**.** *For any two qubit channels and any input,*

$$|S_1(r) - S_2(r)| \geq \tfrac{1}{2}|q(r)|. \tag{8}$$

If (7) holds, then $q(r) = \alpha(\|r\|^2 - 1)$ on the whole Bloch ball, so

$$|S_1(r) - S_2(r)| \geq \frac{|\alpha|}{2}\big(1 - \|r\|^2\big), \tag{9}$$

and $\operatorname{sgn}(S_1 - S_2) = \operatorname{sgn}\alpha$ for every strictly mixed input.

Proof. Write the entropy as a function of the squared Bloch length, $f(x) = h\big((1+\sqrt{x})/2\big)$. Then $-f'(x) = \operatorname{artanh}(\sqrt{x})/(2\sqrt{x}) \geq \frac{1}{2}$, with the continuous extension at $x = 0$ and the integral limit at $x = 1$. The mean value theorem gives (8), and substituting (7) into (6) gives (9). A larger Bloch length means smaller entropy, which fixes the sign. □

When all pure inputs agree, a single strictly mixed input with equal entropies therefore forces $\alpha = 0$ and the orthogonal relation of Proposition 5.2. The argument uses only the one-to-one relation between qubit entropy and Bloch length, not a polynomial form of the entropy.

Finite data can test a violation of this criterion through purity. Corollary B.5 and Proposition B.7 give a nine-probe finite-copy test with explicit confidence bounds; Appendix B supplies their proofs and sample assumptions. The test does not establish exact equality from finitely many outcomes.

6 A physical correlation geometry

Here the vertices are supplied tensor sites, not the central sectors of Section 3. The construction tests a correlation-to-distance rule within that representation.

6.1 Mutual information as connectivity

To compute correlations, declare a subsystem or region representation of the channel output and its relation to the selected source algebra. The channel and state alone do not determine that representation. For output factors $A, B, \ldots$ in this representation, set

$$I_{\mathbf{c}}(A:B) = S_{\mathbf{c}}(A) + S_{\mathbf{c}}(B) - S_{\mathbf{c}}(AB),$$

with entropies computed after coarse-graining.

Definition 6.1 (Connectivity)**.** *For disjoint output factors A, B in the declared representation, the connectivity is $\mu_{\mathbf{c}}(A,B) := I_{\mathbf{c}}(A:B)$. It defines a weighted graph but is not a distance: large mutual information denotes strong correlation rather than large separation, and the metric axioms generally fail.*

A state-parameter metric, such as the SLD Fisher metric in Appendix E, is a different construction. Neither it nor the graph weights alone supply a Lorentzian metric.

6.2 A finite reconstruction rule

For a declared family of finite-dimensional subsystems, a positive connected correlation graph supports a shortest-path metric once an edge-length rule has been chosen. We use $\ell_e = 1/I_e$ on edges with $I_e > 0$. An edge with $I_e = 0$ is absent; distinct connected components then have infinite separation under this rule. Mutual information itself is not a distance. The rule assigns longer distances to weaker correlations, but its physical interpretation depends on the model.

A positive lower bound on the used mutual informations makes this rule stable. No assumption of exponential decay is needed for this finite statement. Exponential decay relative to a pre-existing locality structure can help construct a spatial interpretation [24, 23]; it would supply an additional input rather than derive that locality from the graph.

Proposition 6.2 (Stability of a graph metric)**.** *On the complete graph with v vertices, give the edge e the length $1/I_e$ and use the shortest-path metric d. If every $I_e \geq m > 0$ and the perturbed weights satisfy $|I'_e - I_e| \leq \varepsilon < m$, then*

$$\max_{i,j} |d'_{ij} - d_{ij}| \leq (v-1)\frac{\varepsilon}{m(m-\varepsilon)}.$$

Proof. For each edge, $|1/I'_e - 1/I_e| = |I_e - I'_e|/(I_e I'_e) \leq \varepsilon/(m(m-\varepsilon))$. A shortest path with positive weights uses at most $v-1$ edges. Comparing the new shortest path with the old lengths, and the old shortest path with the new lengths, bounds the difference in both directions. □

6.3 One state, one operation family, one rule

Proposition 6.3 (Non-isometric finite geometries)**.** *Fix the three-qubit GHZ state. Allow only local bit-flip channels*

$$\Phi_{\mathbf{p}} = \bigotimes_{i=1}^{3} \big[(1-p_i)\,\mathrm{id} + p_i\,\mathrm{Ad}_{X_i}\big], \qquad 0 < p_i < \tfrac{1}{2},$$

which are generated by local bit-flip semigroups with $p_i = (1-e^{-2\gamma_i t})/2$. Fix the reconstruction rule: on the complete graph of the three qubits, the edge $\{i, j\}$ has length $1/I_{ij}$, and distances are shortest-path lengths. Then

$$I_{ij} = \ln 2 - h\Big(\frac{1+c_{ij}}{2}\Big) > 0, \qquad c_{ij} = (1-2p_i)(1-2p_j),$$

and different flip probabilities can give metric spaces that are not isometric under any relabelling of the vertices:

flip probabilities	*mutual informations of the three edges*	*diameter*
$(.1, .1, .1)$	$(.221753694,\ .221753694,\ .221753694)$	4.509507747
$(.05, .1, .2)$	$(.288183695,\ .153870839,\ .120090263)$	8.327069751
$(.05, .1, .15)$	$(.288183695,\ .214256042,\ .166239219)$	6.015427679

The first and third rows have the same expected number of applied flips, $\sum_i p_i = .3$, when each channel is implemented by applying X_i with probability p_i.

Proof. Every two-qubit marginal of the GHZ state is an equal mixture of $|00\rangle$ and $|11\rangle$. Local flips leave both bits uniform and make them agree with probability $(1+c_{ij})/2$, so $I_{ij} = 2\ln 2 - \big(\ln 2 + h((1+c_{ij})/2)\big)$. All edge lengths are finite and positive, so the rule defines a metric. The diameters differ, so no relabelling makes the spaces isometric. The numbers were also computed from the full eight-dimensional density matrix with the channels applied and partial traces taken; the analytic formula agrees to better than 3×10^{-16}. □

All three rows use the same source, operation family, sites and reconstruction rule. The finite-time flip channels have full operator range, so their algebra label is identical even when their metrics differ. Rows one and three locate the difference in the allocation of the same expected flip count. That count is a resource in this implementation, rather than a hardware-independent energy price. Proposition 6.2 bounds the distance error by $2\varepsilon/(m(m-\varepsilon))$ when the mutual informations vary by at most $\varepsilon < m$ and m is their smallest true value.

6.4 The reduction is a local interaction

For each source qubit prepare a noise ancilla N_i in $|0\rangle$ and apply

$$U_i(\theta_i) = e^{-i\theta_i X_i \otimes Y_{N_i}}, \qquad U_i(\theta_i)|\psi\rangle|0\rangle = \cos\theta_i|\psi\rangle|0\rangle + \sin\theta_i X_i|\psi\rangle|1\rangle.$$

Discarding N_i realizes exactly the local flip channel with $p_i = \sin^2\theta_i$. One possible interaction Hamiltonian is $\hbar g_i X_i \otimes Y_{N_i}$ applied for θ_i/g_i. The angles, tensor factors and timing are supplied by the apparatus. They do not encode a target metric.

Next attach, at each site, a fragment of k_i fresh memory qubits. Use Paper E's collisions $P_{0,i} \otimes I + P_{1,i} \otimes R_{i,a}$, with real conditional overlaps $c_{i,a}$, so the fragment overlap is $r_i = \prod_{a=1}^{k_i} c_{i,a}$. These collisions dephase the source in the Z basis. Every two-site marginal of the noisy GHZ state is already diagonal, so its probabilities and mutual information are unchanged by recording. The post-recording source therefore has the same graph metric as before, although its three-site coherence may have changed. This is preservation of the data used by this rule, not preservation of the entire source state.

The noise gate ends before recording begins. This sequencing and its switching resources matter: the flip couples through X, while the recorder uses Z. Their common fixed algebra is only the scalars. Indeed the composed single-site source channel sends a Bloch vector to $(r_i x, r_i(1-2p_i)y, (1-2p_i)z)$; for $0 < p_i < 1/2$ and $0 < r_i < 1$ it has no fixed nonidentity direction, although its linear range is still the full matrix algebra. The record channel nevertheless reads the final pointer probabilities with nonzero resolution. This supplies an actual finite-access example beyond an exact stable-algebra label.

Tracing out the source after the collisions gives the exact joint fragment state

$$\rho_{\mathrm{rec}} = \sum_{\mathbf{z}} P_{\mathbf{p}}(\mathbf{z}) \bigotimes_{i=1}^{3} |e_{z_i}^{(i)}\rangle\langle e_{z_i}^{(i)}|, \tag{10}$$

where $z_i = \pm 1$ labels the local pointer and $P_{\mathbf{p}}$ is the actual diagonal distribution of the noisy GHZ state. Thus conditional product structure follows from the unitary couplings and fresh product memories; it is not an extra assumption about independent answers. Each site has equal pointer priors.

For $r_i < 1$, the equal-prior Helstrom measurement on fragment i has a symmetric binary error probability $\epsilon_i = (1 - \sqrt{1 - r_i^2})/2$ [19, 26]. To see the symmetry directly, choose real vectors $e_0 = (1, 0)$ and $e_1 = (r, \sqrt{1 - r^2})$. The sign of $|e_0\rangle\langle e_0| - |e_1\rangle\langle e_1|$ has expectations $+\sqrt{1 - r^2}$ and $-\sqrt{1 - r^2}$ in the two states. Writing $Y_i = \pm 1$ for the actual outcome and $v_i = 1 - 2\epsilon_i = \sqrt{1 - r_i^2}$ therefore gives

$$\mathbb{E}(Y_i \mid \mathbf{z}) = v_i z_i, \qquad \mathbb{E}(Y_i Y_j) = v_i v_j c_{ij}. \tag{11}$$

Local measurement on (10) gives these identities even though the records are correlated before conditioning.

6.5 Finite records certify the geometric difference

Let $f(c) = \ln 2 - h((1 + c)/2)$, $-1 \le c \le 1$. Consider K noise configurations, with independently prepared source states and fresh ancillas in each run. For every configuration collect n triples (Y_1, Y_2, Y_3) from the local measurements above. Calibration supplies estimates $\widehat{v}_i \in [0, 1]$ with $\widehat{v}_i \ge \nu > 0$ and $|\widehat{v}_i - v_i| \le \xi$. The estimate

$$\widehat{c}_{ij} = \mathrm{clip}_{[-1,1]} \frac{n^{-1} \sum_{s=1}^{n} Y_i^{(s)} Y_j^{(s)}}{\widehat{v}_i \widehat{v}_j}$$

uses the measurement outcomes, not hidden pointer values.

Theorem 6.4 (Record-to-metric confidence certificate)**.** *Suppose the calibration event has failure probability at most δ_{cal}, and the record law is* (10) *with the stipulated local measurements. For every $\tau > 0$, with probability at least $1 - \delta_{\mathrm{cal}} - 6Ke^{-n\tau^2/2}$, all $3K$ correlations satisfy*

$$|\widehat{c}_{ij} - c_{ij}| \le e_c := \frac{\tau + 2\xi}{\nu^2}. \tag{12}$$

Intersect each interval $[\widehat{c}_{ij} - e_c, \widehat{c}_{ij} + e_c]$ with $[-1, 1]$ and minimize and maximize f on it, obtaining I_{ij}^-, I_{ij}^+. If every $I_{ij}^- > 0$, form shortest-path metrics d^- and d^+ from edge lengths $1/I_{ij}^+$ and $1/I_{ij}^-$. The true metric obeys $d^- \le d \le d^+$ entrywise, and its diameter lies in $[\operatorname{diam} d^-, \operatorname{diam} d^+]$. Disjoint diameter intervals for two configurations certify non-isometry on the same confidence event.

For a sample-size guarantee, if true and estimated correlations obey $|c_{ij}|, |\widehat{c}_{ij}| \le b < 1$, set $e_I = \operatorname{artanh}(b) e_c$. When $I_{ij} \ge \mu > e_I$,

$$\|\widehat{d} - d\|_\infty \le R := \frac{2e_I}{\mu(\mu - e_I)}. \tag{13}$$

For two configurations with true diameter gap $\Delta > 4R$, the intervals $[\operatorname{diam} \widehat{d} - R, \operatorname{diam} \widehat{d} + R]$ are disjoint on that event.

The proof and full calibration budgets are in Appendix C. For example, the equal-flip-count configurations $(.1, .1, .1)$ and $(.05, .1, .15)$ with fragment overlap .2 can be separated at confidence .95 using 1 217 920 runs per configuration when their readout visibilities are specified. Unknown visibilities require additional reference preparations, included in the appendix. Equal expected flip count does not imply equal total energy or a free reset.

6.6 Limits of record inversion

At $r_i = 1$ the record states coincide and $v_i = 0$: algebraic inversion of their span supplies no source information. Even for $v_i > 0$ the error grows as the inverse product visibility. A narrow effect set can span the pointer algebra while requiring too many trials for the available budget.

Calibration of individual error rates is also insufficient when the readers share an unmodelled error. Let a common random sign B multiply all otherwise ideal outcomes, with $\mathbb{E}B = v$. Each local reference test then reports visibility v, while the pair products are unchanged, since $B^2 = 1$. Dividing by v^2 overestimates the true correlations. A common inversion has $v = -1$ and is invisible to agreement. Joint reference tests, conditional error modelling or a verified local apparatus are needed to exclude such contamination. The confidence certificate is conditional on the joint record law, rather than an inference of that law from single-site calibration.

A spacetime reconstruction must solve a further problem: finite measurements determine a profile within an error region, not an exact profile. The relevant dictionary must be stable on that region and must quotient its coordinate freedom. Entropy equality from Section 5 supplies a genuine obstruction, including operationally different channels with identical entropy profiles. More accurate readout cannot repair a dictionary's failure to distinguish such profiles.

A coordinate change preserves the reconstructed geometry up to its coordinate equivalence. Changing a physical reduction can instead change its measured correlations. Algebraic difference alone does not prove this distinction: it needs a separating dictionary or an explicit invariant such as the diameter in Proposition 6.3.

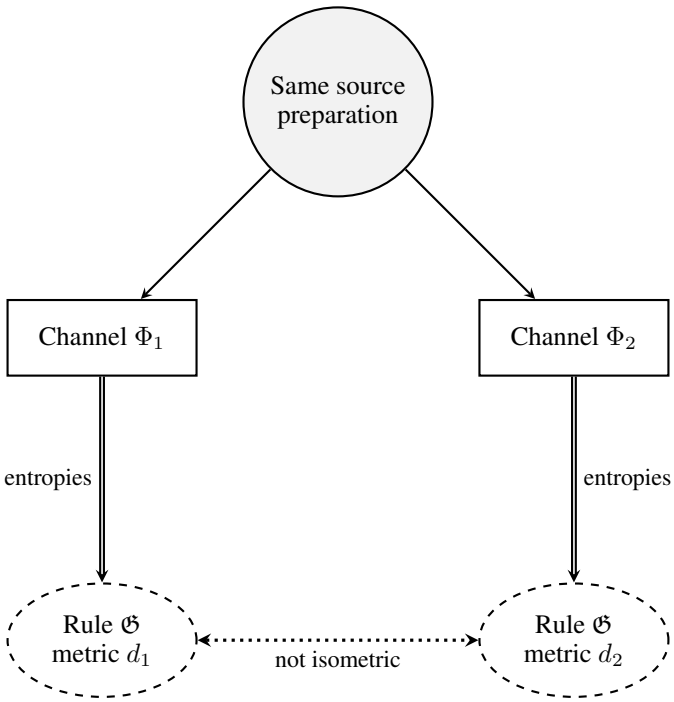


Figure 1: One global state is reduced by two channels from the same allowed family. The same reconstruction rule turns their entropy data into metrics. The tensor sites and the rule are fixed inputs. Proposition 6.3 exhibits different diameters; the process-selected sector geometry is a separate construction.

7 From relations to spatial structure

7.1 What the different geometries compare

The process-selected sectors and the GHZ sites answer different questions. The first are central atoms of a persistent algebra; the second are tensor factors supplied to a correlation protocol. There is no canonical identification between these two sets. In the rate geometry the channel determines a relation independently of the input state. In the correlation geometry the state, its reduction, the site representation and the reconstruction rule all matter.

A spatial interpretation needs evidence linking these relations to propagation or intervention. Pure dephasing supplies none of the required sector transport. A weak Hamiltonian connecting different sectors is a possible extension, but degenerate blocks generally retain quantum states that prevent a closed scalar equation for their populations. Paper B's protected Bell-pair interaction acts inside a joint noise sector and does not supply transitions between central labels [25]. Deriving a transfer law would require its own approximation, error bound and actual intervention ports.

If a reversible transfer process were identified through those ports, its stationary currents and responses could fix a conductance form and operational resistance distances. This would give a distance rule through a stated physical task, as (5) does for discrimination. An arbitrary support of the added Hamiltonian can still encode an arbitrary interaction graph. Inferring spatial locality, rather than inserting it into that support, remains the stronger problem.

For a precisely specified response candidate, suppose calibrated experiments identify a connected reversible jump generator with rates q_{ab} from a to b and stationary weights $\pi_a > 0$. The response fixes

$$\mathcal{E}(f,f) = \frac{1}{2}\sum_{a,b} \pi_a q_{ab}(f_b - f_a)^2, \qquad \Gamma(f)(a) = \frac{1}{2}\sum_{b} q_{ab}(f_b - f_a)^2.$$

A possible task convention fixes a reference rate $r_0 > 0$, with units time^{-1}, and bounds local quadratic variation:

$$d_{\Gamma,r_0}(a,b) = \sup\{f_a - f_b : \Gamma(f)(c) \le r_0 \text{ for all } c\}, \qquad f \in \mathbb{R}^V. \tag{14}$$

On a finite connected graph this is a finite separating metric: edgewise gradient bounds make every difference bounded, small multiples of separating functions are feasible, and the supremum gives the triangle inequality. For dimensionless

f, this distance is dimensionless; its conversion to length requires a separate calibration. The reference rate is part of the task convention. It is defined here by this task convention; it is not identified with the effective resistance or asserted to be the sole intrinsic metric of a jump process. For diffusion forms the energy-measure analogue defines a standard intrinsic pseudometric, whose nondegeneracy and agreement with the topology need extra conditions [30].

7.2 A same-process locality conjecture

The world-structure proposal is that the process selecting persistent organization can also constrain the relations that become spatial. Selection, record access and propagation should be tested within one history, rather than supplied by incompatible models.

Conjecture 7.1 (Locality from process-selected relations)**.** *There are families of finite carriers, specified by noise and Hamiltonian operators without assigning their matrix entries to spatially local gates, for which interaction histories produce refining persistent block descriptions and calibrated transfer responses on those blocks. Under refinement these responses admit a common continuum limit described by a nondegenerate strongly local Dirichlet form. After calibration, the distances* (14) *converge with these descriptions to the nondegenerate intrinsic distance of that limit. This distance is unchanged by alternative descriptions preserving the response, stationary weights and reference rate.*

The conjecture asks for an existence mechanism in one microscopic family, not a conclusion for every dephasing channel. Its substantial requirement is the joint formation of stable organization and a local response limit without a spatial dictionary supplied to the reconstruction. The finite selection result supplies a persistent-block ingredient; Paper B's interactions forming protected relations show that preservation and formation can be distinguished experimentally. Neither establishes the proposed limit.

A proposed realization must supply the refining blocks, allowed preparations and readers, measured transfer kernels, limiting form and its intrinsic distance. It would fail if block dynamics did not close at the claimed resolution, if persistent long-range shortcuts prevented a local limit, or if response-preserving descriptions produced inequivalent limiting distances after common calibration. Its strongest competing explanation is that any observed locality was already encoded in the supplied coupling architecture. A useful realization must show which locality constraints follow from the history and which remain in that architecture.

7.3 Reconstruction and the series

In holography, extremal-surface areas provide a specific entropy dictionary [2, 4]; tensor-network distances use another [22]. To infer distinct spacetimes from separated entropy profiles, a dictionary $\mathfrak{G}$ must identify its equivalence relation and satisfy, for example,

$$\mathfrak{G}[S_1] \simeq_{\mathrm{diff}} \mathfrak{G}[S_2] \implies S_1 \simeq_{\mathfrak{G}} S_2.$$

This separation property is an input about that dictionary. Finite readings also require its stability over a confidence region. The finite model in Section 6 proves separation under its declared rule rather than assuming a spacetime dictionary. Imprint's cross-scale analysis asks the related question of which readable relations survive a change of resolution and at what cost [27].

Conjecture 7.2 (Finite permissions approaching geometric access)**.** *In a source family with an independently justified local reconstruction dictionary, histories that enlarge the available record fragments and calibrate their readout can make the inferred geometry converge while the generated observable algebra remains constant. The required growth of visibility, preparation count and dictionary stability bounds must jointly dominate sampling and model error.*

The finite certificate supplies the mechanism and the costs of this candidate. For E edges with visibility at least ν, error in the correlations scales as $\nu^{-2}\sqrt{\ln(2E/\delta)/n}$ before calibration error. If reconstructed edge mutual informations have lower bound μ, distance error carries another factor of order μ^{-2} times the number of edges in a shortest path. Consequently, a growing graph or an approaching zero-information edge can defeat convergence even while all fixed-algebra labels are unchanged. The conjecture would fail in a proposed model if its dictionary is unstable, if common error survives calibration, or if its available budget cannot overcome these factors. This is a route from finite permissions to geometric access, not an assumption that a continuum or its dynamics is already generated.

The spatial/eigenmode example in Appendix E illustrates dependence of connectivity on factorization. Unlike Proposition 6.3, its empty eigenmode graph has no finite reconstructed paths.

8 Conclusion

A physical process can select an effective organization before spatial sites have been assigned. In the commuting-noise class, the organization is a direct sum fixed by the dynamics; its central sectors and cross-sector coherence rates are intrinsic to the channel. A finite stability window can select persistent blocks, and allowed coherent probes identify their rate relation.

What is selected must still be distinguished from what can be read. Equal system channels can leave different readable records at a restricted environmental port. Equal entropy functions can hide operationally incomparable reductions. The qubit criterion and finite-copy tests make those obstructions quantitative.

The GHZ protocol carries a separate correlation reconstruction through actual interactions and finite records. Changing the allocation of a fixed expected flip count changes its certified metric. Its tensor sites and distance rule remain specified inputs. Turning process-selected organization into spatial geometry requires a common account of persistence and actual transport. Conjecture 7.1 states that remaining mechanism as a testable target within the framework.

A Proof of the intrinsic window certificate

On a cluster C, choose a centre c of its smallest enclosing ball and put $B_k = \sum_{a\in C}(u_{a,k} - c_k)P_a$. Scalar shifts do not alter double commutators. The restricted generator is

$$\mathcal{L}_C(X) = 2\Psi(X) - \{R, X\}, \qquad \Psi(X) = \sum_k B_k X B_k, \qquad R = \sum_k B_k^2.$$

Here Ψ is completely positive and $\|\Psi\| = \|\Psi(I)\| = \|R\| \le r_C^2$. Thus $\|\mathcal{L}_C\| \le 4r_C^2$. The semigroup is a contraction on each invariant block, so Duhamel's formula gives $\|\Phi_t(X) - X\| \le t\kappa_{\mathrm{in}}\|X\|$ on $\mathcal{B}_\pi$.

A strictly containing unital algebra has a nonzero entry between two different Q_C blocks. Multiplication on both sides by matrix units already in $\mathcal{B}_\pi$ isolates a cross-cluster matrix unit E_{xy}. Its loss at time T is $1-e^{-TD_{ab}} \ge 1-e^{-T\Delta_{\mathrm{out}}} > \varepsilon$. Such a containing algebra cannot be stable.

Conversely, any stable block algebra in a common joint eigenbasis can put x and y together only if $1 - e^{-TD_{ab}} \le \varepsilon$. The separation in (3) forbids this across two clusters, so every stable block of that form lies in one Q_C block. Two certified partitions therefore refine each other and coincide.

The radii are determined by the Euclidean configuration of D. A computable sufficient upper bound uses the equal-weight centroid:

$$\|u_a - \bar{u}_C\|^2 = \frac{1}{|C|}\sum_{b\in C} D_{ab} - \frac{1}{2|C|^2}\sum_{b,c\in C} D_{bc}.$$

Replacing r_C^2 by its largest value over $a \in C$ retains the certificate. It need not be optimal.

B Channel order and entropy tests

B.1 Refinement

Definition B.1 (Refinement)**.** *For coarse-graining structures $\mathbf{c}_1, \mathbf{c}_2$ write $\mathbf{c}_1 \succeq \mathbf{c}_2$ if there is a CPTP map Λ from states on the output space of $\Phi_{\mathbf{c}_1}$ to states on the output space of $\Phi_{\mathbf{c}_2}$ with $\Phi_{\mathbf{c}_2} = \Lambda \circ \Phi_{\mathbf{c}_1}$. Then $\mathbf{c}_2$ is a further coarse-graining of $\mathbf{c}_1$.*

Proposition B.2 (Order on classes)**.** *The relation $\succeq$ is reflexive and transitive but not antisymmetric. Writing $\Phi \sim \Psi$ when each is a post-processing of the other, $\succeq$ induces a partial order on equivalence classes.*

Proof. Reflexivity uses $\Lambda = \mathrm{id}$, transitivity uses composition of channels. Two different replacer channels, which output fixed states $\omega_1 \neq \omega_2$, simulate each other by a further replacement, so antisymmetry fails. On classes the relation is well defined because composing with a simulating channel does not change membership, and antisymmetry holds by construction. □

If the allowed post-processings are restricted, for example to local, energy-bounded or specified control operations, the restricted class must be closed under composition before the same quotient can be formed. The comparison of channels by approximate post-processing has an operational characterization through success probabilities of discrimination tasks [14]; we use only exact post-processing.

B.2 Mutual information under refinement

Proposition B.3 (Local and global post-processing)**.** *For channels Λ_A, Λ_B acting on A and B separately,*

$$I(A' : B')_{(\Lambda_A \otimes \Lambda_B)(\rho)} \leq I(A : B)_\rho.$$

A global CPTP post-processing can increase mutual information: the channel that replaces every two-qubit input by a Bell state takes $|00\rangle$, with $I = 0$, to a state with $I = 2 \ln 2$.

Proof. $I(A : B)_\rho = D(\rho_{AB} \| \rho_A \otimes \rho_B)$, and $\Lambda_A \otimes \Lambda_B$ maps $\rho_A \otimes \rho_B$ to the product of the new marginals, so data processing for relative entropy gives the inequality. The Bell example is a direct computation. □

A requirement that mutual information decrease under refinement is therefore a theorem for local refinements and false for global ones. A geometric reading of mutual information must say which refinements it allows.

B.3 Generic separation

Theorem B.4 (Generic separation for a fixed pair of channels)**.** *Let $\mathcal{H}$ and $\mathcal{K} = \mathcal{K}_A \otimes \mathcal{K}_{A^c}$ be finite dimensional, and let $\Phi_1 \neq \Phi_2$ be CPTP maps from states on $\mathcal{H}$ to states on $\mathcal{K}$. The pure inputs on which Φ_1 and Φ_2 agree form a proper real-algebraic subset of Haar measure zero. If, in addition, the real-analytic function*

$$f_A(\Psi) := S(\mathrm{Tr}_{A^c}\, \Phi_1(|\Psi\rangle\langle\Psi|)) - S(\mathrm{Tr}_{A^c}\, \Phi_2(|\Psi\rangle\langle\Psi|))$$

is not identically zero on each connected component under consideration of the open set where both reduced outputs have full rank, then $f_A(\Psi) \neq 0$ for generic inputs in those components.

Proof. Pure-state projectors span the Hermitian operators, so $\Phi_1 - \Phi_2$ cannot vanish on all of them. The equality condition is polynomial in the real and imaginary parts of $|\Psi\rangle$, so its solution set is a proper real-algebraic subset and has Haar measure zero. On the full-rank domain the von Neumann entropy, and hence f_A, is real analytic, and a nonzero real-analytic function has a zero set of measure zero. □

The nondegeneracy hypothesis cannot be dropped: channels that differ by a unitary acting on A alone can give different outputs with identical entropies on every input. Channel inequivalence alone therefore does not yield an entropic or geometric difference. Section 5 gives the exact qubit criterion; the tests below turn its violation into a finite measurement problem.

B.4 Probes and random inputs

Corollary B.5 (Nine pure probes and one mixed probe)**.** *Let the probe set consist of the nine Bloch vectors*

$$\pm e_1,\ \pm e_2,\ \pm e_3,\ (e_1 + e_2)/\sqrt{2},\ (e_1 + e_3)/\sqrt{2},\ (e_2 + e_3)/\sqrt{2}.$$

If the two channels give equal output entropies on these nine inputs, they give equal output entropies on every pure input. If they also agree at $r = 0$, they agree on every input.

Proof. The six axial probes give $b_i = 0$ and $A_{ii} + c = 0$; the three diagonal probes then give $A_{ij} = 0$ for $i \neq j$. This is (7). At $r = 0$, $q = c = -\alpha$, so agreement there gives $\alpha = 0$. Then q vanishes on the entire Bloch ball, hence the output lengths and entropies agree everywhere. □

The corollary concerns exact entropy values on nine inputs, not nine measurements. On the unit sphere $q(r) = r^T A_0 r + 2b^T r + d$ with $\alpha = \frac{1}{3}\,\mathrm{tr}\, A$, $A_0 = A - \alpha I$ and $d = c + \alpha$, which is linear in nine real parameters.

Proposition B.6 (Random pure inputs)**.** *For Haar-random pure inputs,*

$$Q := \mathbb{E}\, q(r)^2 = \frac{2}{15}\|A_0\|_F^2 + \frac{4}{3}\|b\|^2 + d^2. \tag{15}$$

$Q = 0$ exactly when (7) holds. If $Q > 0$, then $\mathbb{E}(S_1 - S_2)^2 \geq Q/4$, and for $0 < \tau < 1$

$$\Pr\Big\{|S_1 - S_2| > \tfrac{1}{2}\sqrt{\tau Q}\Big\} \geq \Pr\{q^2 > \tau Q\} \geq \frac{(1-\tau)Q}{1-\tau Q}. \tag{16}$$

Proof. Use $\mathbb{E} r_i r_j = \delta_{ij}/3$ and $\mathbb{E} r_i r_j r_k r_l = (\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})/15$; odd moments and the cross terms with the traceless A_0 vanish. The second claim follows from (8). Both outputs are states, so $0 \le q^2 \le 1$; splitting $\mathbb{E} q^2 = Q$ at $q^2 = \tau Q$ gives $Q \le \tau Q(1-P) + P$ with $P = \Pr\{q^2 > \tau Q\}$, which is (16). □

With n independent Haar inputs, the probability of missing the difference at the level $\frac{1}{2}\sqrt{\tau Q}$ is at most $(1-p_\tau)^n$, where p_τ is the right side of (16). This assumes exact entropies; the next proposition removes that assumption. The average change of purity over pure inputs is also the basis of the unitarity measure of noise [17]; Eq. (15) concerns the difference between two channels rather than the coherence of one.

B.5 Finitely many copies

For a qubit, purity and entropy are both monotone functions of the Bloch length, so equal entropy is equivalent to equal purity, and purity can be measured directly. A SWAP test on two copies of a state σ returns the outcome 0 with probability $\frac{1}{2}\big(1 + \operatorname{Tr}\sigma^2\big)$ [18]. For a qubit with Bloch vector s this is $\frac{3}{4} + \frac{1}{4}\|s\|^2$. We assume that every channel use acts on a freshly prepared, known probe state, that the channels do not drift between uses, and that two output copies can be measured jointly.

Finite data cannot establish that the two entropy functions are exactly equal. They can separate equality from a violation of prescribed size, and they can bound how large a violation can be. Put

$$\kappa_9 = \frac{75 - 3\sqrt{465}}{8} = 1.28855\ldots.$$

Proposition B.7 (Finite-copy test and confidence bound)**.** *For each of the nine probes r_k of Corollary B.5, send $2m$ copies of the input through Φ_1, apply a SWAP test to each of the m output pairs, and do the same for Φ_2. Let D_k be the difference of the two observed frequencies of the outcome 0.*

1. *Given $\gamma > 0$, declare the channels different on pure inputs if $\max_k |D_k| > \gamma/8$. If (7) holds, a difference is declared with probability at most $18\, e^{-m\gamma^2/64}$. If $\max_k |q(r_k)| \ge \gamma$, the difference is missed with probability at most $2\, e^{-m\gamma^2/64}$.*
2. *$\sum_k q(r_k)^2 \ge \kappa_9\, Q$, and κ_9 is the largest constant with this property. Hence $Q \ge Q_0 > 0$ forces $\max_k |q(r_k)| \ge (\kappa_9 Q_0/9)^{1/2}$, and*
$$m \ge \frac{576\, \ln(18/\delta)}{\kappa_9\, Q_0}$$
output pairs per probe and channel keep both error probabilities below δ for the hypotheses $Q = 0$ and $Q \ge Q_0$.
3. *For every $t > 0$, with probability at least $1 - 18\, e^{-mt^2}$,*
$$Q \le \frac{16}{\kappa_9} \sum_{k=1}^{9} \big(|D_k| + t\big)^2.$$

Proof. The expected value of D_k is $q(r_k)/4$. D_k is a sum of $2m$ independent Bernoulli variables with coefficients $\pm 1/m$, so Hoeffding's inequality [20] gives $\Pr\{|D_k - \mathbb{E} D_k| \ge t\} \le 2e^{-mt^2}$. Under (7) every $\mathbb{E} D_k$ vanishes, and $t = \gamma/8$ with a union bound over the nine probes gives the first error bound. If $|q(r_k)| \ge \gamma$ for some k, then $|\mathbb{E} D_k| \ge \gamma/4$, and missing it requires $|D_k - \mathbb{E} D_k| \ge \gamma/8$, which gives the second.

The smallest generalized eigenvalue of the two quadratic forms $\sum_k q(r_k)^2$ and Q is κ_9, as computed in Appendix B.6. The largest of nine terms is at least their mean; setting $\gamma^2 = \kappa_9 Q_0/9$ and requiring $18e^{-m\gamma^2/64} \le \delta$ gives item 2. For item 3, on the event that all nine deviations are at most t, which fails with probability at most $18e^{-mt^2}$, we have $|q(r_k)| = 4|\mathbb{E} D_k| \le 4(|D_k| + t)$, and $Q \le \kappa_9^{-1} \sum_k q(r_k)^2$. □

If the pure-input criterion is known to hold, the same argument at the mixed probe $r = 0$ tests the size of α: the expected difference there is $-\alpha/4$, and $m \ge 64 \ln(2/\delta)/\alpha_0^2$ pairs per channel separate $\alpha = 0$ from $|\alpha| \ge \alpha_0$ with error at most δ in each direction. The pure test uses $18m$ channel uses per channel, $36m$ in all. The constants are conservative. Beyond qubits, purity and entropy are different functionals and this reduction no longer applies.

B.6 The nine-probe constant

On the unit sphere, $\sum_k q(r_k)^2$ and Q are quadratic forms in the nine parameters (A_0, b, d), and both are positive definite; the first because the nine probe values determine the parameters, as in the proof of Corollary B.5. Their generalized eigenvalues do not depend on the basis chosen for the parameters, and a direct computation gives the characteristic polynomial, up to a constant factor,

$$(\lambda - 6)\,(4\lambda^2 - 75\lambda + 90)\,(8\lambda^3 - 225\lambda^2 + 1620\lambda - 2700)^2.$$

Its roots are 1.2886, 2.3962, 6, 7.9001, 17.4614 and 17.8287, the three roots of the cubic each counted twice. The smallest is the smaller root of the quadratic factor, κ_9, and it is attained.

C Record certificate: proof and calibration budgets

C.1 Proof of Theorem 6.4

Proof. Each $Y_iY_j \in [-1, 1]$ has expectation $v_iv_jc_{ij}$. Hoeffding's inequality gives a two-sided tail $2e^{-n\tau^2/2}$; a union bound covers the $3K$ edge estimates. The three edges within a run need not be independent. On the calibration event, $|v_iv_j - \widehat{v}_i\widehat{v}_j| \le 2\xi$, by telescoping and the bounds $v_i, \widehat{v}_i \le 1$. Dividing by $\widehat{v}_i\widehat{v}_j \ge \nu^2$, using $|c_{ij}| \le 1$, and clipping cannot increase the error, proving (12). The interval construction contains the true mutual informations. Shortest-path lengths and their diameter are monotone in positive edge lengths, proving the metric enclosure. Finally $f'(c) = \operatorname{artanh} c$, so $|\widehat{I} - I| \le e_I$. Proposition 6.2, with three vertices, gives (13). Each estimated diameter differs from its true value by at most R, so the estimated gap is at least $\Delta - 2R$; two radius-R intervals are disjoint when $\Delta > 4R$. □

C.2 Specified and empirically calibrated readers

For the equal-flip-count configurations $(.1, .1, .1)$ and $(.05, .1, .15)$, take one recording qubit per site with overlap $r_i = .2$. Then $v_i = \sqrt{.96}$ and the correlations lie between $.56$ and $.72$. With exactly specified v_i, use $\nu = \sqrt{.96}$, $\xi = 0$, $\tau = .003$, $b = .8$, $\mu = .16$ and $\delta = .05$. The choice

$$n = \left\lceil \frac{2\ln(12/\delta)}{\tau^2} \right\rceil = 1\,217\,920$$

runs per configuration gives $e_c = .003125$, $e_I = .0034332$ and $R < .275$. The true diameter gap is $1.505919931 > 4R$. Thus a finite record-only experiment separates the two geometries with confidence at least $.95$, under the stated source and apparatus model. The sample bound is conservative and does not claim optimal tomography.

A known unitary specification is not free empirical calibration. If the v_i are unknown, prepare reference sources in $Z = +1$ and apply the same record couplings and readers. Each reference outcome has mean v_i. For N independent reference runs per site, clipping the empirical mean to $[0, 1]$ yields

$$\Pr\{\max_i |\widehat{v}_i - v_i| > \xi\} \le 6e^{-N\xi^2/2}$$

for three shared readers; separately calibrated readers for K configurations replace 6 by $6K$. Thus $N \ge \lceil 2\ln(6/\delta_{\mathrm{cal}})/\xi^2 \rceil$ per site suffices. Reject calibrations below ν. For the same two geometries, take $\xi = .001$, $\delta_{\mathrm{cal}} = .025$, $\nu = .978$ and $\tau = .002$. Then $N = 10\,961\,278$ reference runs per site and $n = 3\,086\,894$ main runs per configuration give total failure at most $.05$, $e_c < .004182$, $R < .36955$ and $4R < 1.4782 < \Delta$. On the calibration event the true $v_i = \sqrt{.96}$ ensures $\widehat{v}_i > \nu$; the correlation bound also stays inside $b = .8$. The reference budget is therefore $3N$ additional preparations and readouts, not part of the $2n$ main runs. Equation (12) includes their residual error. Prepared reference pointers, their basis labels and the stability of the reader between calibration and use are additional physical resources.

The comparison uses $2n$ preparations of the same GHZ source, $6n$ fresh noise ancillas, $6n$ fresh recording qubits, $6n$ local binary readouts and communication of the outcomes to a comparator, plus any calibration runs. The expected noise-flip count is $.3$ per source in both configurations; their recording and readout schedule is identical. Preparation, switching work, storage, communication and renewal do not vanish because the counts agree. No heat or total-energy equality follows from the flip count, and this finite protocol supplies no reservoir reset.

D No canonical tensor factorization

For a Hilbert space $\mathcal{H}_{\text{total}}$ of dimension $d_A d_B$ with $d_A, d_B > 1$, there is no unique way to write $\mathcal{H}_{\text{total}} = \mathcal{H}_A \otimes \mathcal{H}_B$ without further physical input. Given any pure state $|\Psi_U\rangle$, choose a unitary U with $U(|0\rangle_A \otimes |0\rangle_B) = |\Psi_U\rangle$ and pull the reference factors through U; in the resulting factorization $|\Psi_U\rangle$ is a product. More generally, observables can be chosen so that a given pure state has any prescribed Schmidt decomposition [8].

What survives is typicality for a family of factorizations fixed in advance.

Theorem D.1 (Entanglement typicality for a specified family)**.** *Let $\mathcal{H}_{\text{total}}$ be finite dimensional and let $\mathfrak{F} = \{\mathbf{f}_1, \ldots, \mathbf{f}_m\}$ be a finite family of nontrivial tensor factorizations selected by a physical rule that does not depend on the sampled state. Write $\mathcal{H}_{\text{total}} = \mathcal{H}_{A_j} \otimes \mathcal{H}_{B_j}$ with $d_{A_j} \le d_{B_j}$. For $\delta > 0$ let*

$$p_j(\delta) := \Pr_{\Psi \sim \text{Haar}} \left[S(\rho_{A_j}) < \log d_{A_j} - \delta \right].$$

Then

$$\Pr\left[S(\rho_{A_j}) \ge \log d_{A_j} - \delta \text{ for every } j \right] \ge 1 - \sum_{j=1}^{m} p_j(\delta),$$

and standard concentration estimates make each $p_j(\delta)$ small at large dimension under the usual conditions on the dimension ratio.

Proof. For each fixed $\mathbf{f}_j$, Page's formula gives [10]

$$\mathbb{E}[S(\rho_{A_j})] = \sum_{k=d_{B_j}+1}^{d_{A_j} d_{B_j}} \frac{1}{k} - \frac{d_{A_j} - 1}{2 d_{B_j}}, \tag{17}$$

and concentration of measure bounds the lower tail [11]; this tail is $p_j(\delta)$. The union bound over the finite, preselected family proves the inequality. □

The size of the statement matters. For a balanced split $d_A = d_B$, Eq. (17) gives a mean entropy of $\log d_A - \frac{1}{2} + o(1)$, so typical states are close to maximally entangled but keep a gap of about half a nat. Drawing the factorization at random with the state fixed is equivalent to drawing the state at random with the factorization fixed. Hence for a fixed state, product factorizations form a set of measure zero, while factorizations with small but nonzero entanglement have small positive measure.

Theorem D.1 says nothing about all factorizations and does not pick out any member of $\mathfrak{F}$. Which family is admissible, and why, is a question of quantum mereology [12, 13]. Earlier work showed that subsystem structure is induced by the accessible observables rather than given by the Hilbert space [5, 6]. The channel-level results in the main text retain the accessible-observable view while distinguishing its algebra from its resolution.

E Other connectivity constructions

When a metric on a parameter space of coarse-grained states is wanted, the finite-dimensional construction is the quantum Fisher metric. For a smooth faithful family $\lambda \mapsto \rho_{\mathbf{c}}(\lambda)$, define the symmetric logarithmic derivatives by $\partial_a \rho_{\mathbf{c}} = \frac{1}{2}(L_a \rho_{\mathbf{c}} + \rho_{\mathbf{c}} L_a)$ and

$$g^{\text{QF}}_{ab}(\lambda) := \frac{1}{2} \operatorname{Tr}[\rho_{\mathbf{c}}(\lambda)\{L_a, L_b\}]. \tag{18}$$

This is a positive-semidefinite Riemannian metric on identifiable parameter directions [21]. It is not a Lorentzian spacetime metric, and the pairwise numbers $\mu_{\mathbf{c}}(A, B)$ do not determine it without an additional map from graph or state parameters to geometry.

E.1 Spatial and eigenmode descriptions of a free-fermion chain

Example E.1 (A dimerized free-fermion chain)**.** *Take a periodic chain of $2L$ sites with alternating hopping amplitudes $t_1 = 1$ and $t_2 = 1/2$,*

$$H = -\sum_{c=0}^{L-1} \left(t_1 \, c^\dagger_{2c} c_{2c+1} + t_2 \, c^\dagger_{2c+1} c_{2c+2} + \text{h.c.} \right),$$

at half filling. The single-particle spectrum has two bands separated by the gap $2|t_1 - t_2| = 1$*, and the ground state fills the lower band,* $|\Psi_0\rangle = \prod_k a^\dagger_{k,-}|0\rangle$*, where* $a_{k,-}$ *are the lower-band Bloch modes. In the description by occupation numbers of the band-resolved Bloch modes the ground state is a product of filled and empty modes: every set of modes has a pure reduced state, disjoint sets have zero mutual information, and the connectivity graph is empty. In the spatial description blocks of sites are correlated. For* $L = 60$ *and blocks of four sites, the block entropy saturates at* $.3554$ *as the block grows, as the area law for gapped one-dimensional systems requires [24, 23]. The mutual information between two blocks falls from* $.333$ *for adjacent blocks to* 9.0×10^{-5} *at a separation of ten sites and* 5.2×10^{-8} *at twenty, close to* $e^{-r/1.32}$*. The connectivity graph is dominated by edges between neighbouring blocks.*

The second description has to be chosen with care. Plain Fourier modes are not enough when the Hamiltonian is not translation invariant: for two disconnected dimers on four sites, $H = -(c_0^\dagger c_1 + c_2^\dagger c_3 + \text{h.c.})$ at half filling, the gap at fixed particle number is 2, yet the Fourier modes $k = \pi/2$ and $3\pi/2$ share one particle in superposition and have mutual information $2\ln 2$. A translation-invariant single-band chain with particle conservation is gapless unless its band is full or empty, so a gapped example whose ground state is a product of mode occupations needs band-resolved modes, as above, or in general the single-particle eigenmodes of the Hamiltonian.

The example shows that connectivity depends on the factorization: the spatial description has short-range edges and the eigenmode description has none. It does not exhibit two reconstructed geometries, since its zero-connectivity graph supplies no finite paths under the chosen rule; Proposition 6.3 is the example in which both descriptions produce geometries.

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COUPLING, RECORDS, AND STABLE RELATIONS: FORMATION, PROTECTED CONTENT AND THE LIFETIME OF A GEOMETRY

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ABSTRACT

A coupling can preserve an inherited relation without having formed it. We ask how interactions select stable structure and how records make that structure usable. Commuting Hermitian noise selects a direct-sum algebra on a given carrier; its central sectors do not require an independently chosen spatial basis. A separated range of decay rates further selects a unique block partition over a finite window within the stated block class. We distinguish this selection from the formation of correlations and from learning the operations that use them. A collective recorder creates a flagged protected Bell branch from a product input, while a physical link creates a deterministic protected Bell pair across noise-disconnected components. Both retain the resources that produce the relation. Block-constant records preserve internal quantum content, whereas independent leakage imposes a sharp $R^{-1/2}$ symmetry threshold for R records. An exact entropy identity separates sector and within-sector contributions to a supplied correlation geometry. The GHZ example then exhibits a selected geometry, its finite plateau and its loss under symmetry breaking. Finite experimental capabilities remain continuous under perturbations even when the exact fixed algebra changes discontinuously. These results support a process-selected subsystem programme: an effective component must acquire reproducible operations as well as stable correlations. The carrier, available interactions and spatial interpretation remain distinct inputs to be tested rather than inferred from a noise graph alone.

1 Introduction

A stable distinction is a physical achievement. It must survive the interactions that would erase it, be readable through an available instrument, and last long enough to matter. An algebra of invariant observables describes one part of that achievement; it does not specify a reader, a preparation history or a set of available gates.

Paper A [1] shows why the distinction matters for geometry. Channels generating the same observable algebra can resolve different states and produce non-isometric finite correlation geometries. Its local bit-flip family has a Stinespring realization and a finite fragment-readout certificate. Here we ask a complementary question: what structure a coupling protects, how actual records and interactions form usable relations within that structure, and when those relations lose their support.

Commuting Hermitian noise gives an exact starting point. The fixed observables are the commutant of the coupling operators, familiar from einselection and noiseless codes [9, 10, 12, 13]. The infinite-time channel pinches onto this algebra. At finite times its operator range is full, so a protected block algebra is additional stability data rather than the range of the actual channel. A window and a tolerance give a unique block partition when intra-block and inter-block rates are separated (Theorem 4.2). The maximality statement concerns block algebras in the specified joint eigenbasis, not every rotated algebra or every useful finite task.

Recording adds a different restriction. A coupling constant on each block can record its label without learning its quantum content. Independent fragments that resolve two states within a block destroy their coherence according to a multiplicative visibility budget (Theorem 5.2). The centre and the blocks thus perform different jobs within one stable structure. They are not disjoint sources of geometry: classical sector weights can sustain mutual information even when every block is one-dimensional. Section 6 gives the exact decomposition needed to distinguish the contributions.

Selection alone also says too little about formation. Two disconnected collective-noise components can preserve a logical Bell pair shared at the start. Conversely, local independent processes cannot create cross-component correlations from independent inputs. Section 7 supplies the missing process distinction and two explicit formation mechanisms. One retains a collective measurement flag and all its branches; the other uses an actual cross-component Hamiltonian link. Both generate relations from product inputs, and both expose the interaction resources responsible for them. Foundation III [6] supplies a further mechanism in which acquired records select an encoder and decoder for the same pre-existing quantum content. Storage, formation and learned use then have separate, testable meanings.

The GHZ calculation in Section 8 shows how a coupling changes a finite correlation geometry and how weak symmetry breaking ends its plateau. For noncommuting perturbations, Section 4.1 bounds complete adaptive record laws and task risks without requiring a stable algebra. A transverse qubit exhibits a diverging record lifetime despite a scalar exact fixed algebra.

The Hilbert space and coupling permissions are supplied. The formation and GHZ experiments additionally specify site factors; the exact sector-selection result does not need a prior spatial factorization. Consciousness I [4] selects a Virasoro fixed net inside a given current-net representation; Foundation's hulls concern closure under declared operations. Neither result derives the ambient carrier. Our formation results address how relations and usable protected sectors arise on that carrier. Section 2 states the effective-subsystem question before the constructions. Section 10 examines its extension beyond commuting noise.

2 What has to form

The generation question has three physically different outputs. A coupling can select a direct-sum sector without supplying a tensor factorization. An interaction can create a relation between supplied components without teaching a reader how to use it. Acquiring a component also requires repeatable operations whose errors stay controlled over the intended task. These distinctions set the target of the constructions below.

Conjecture 2.1 (Process-selected effective subsystems)**.** *On an initially specified carrier, robust effective subsystems form through a history of interactions that both builds relations and makes a noncommuting set of internal operations reproducible across a finite task window. The relevant connectivity is that of the complete physical process, including recording ports and Hamiltonian controls, not the collective-noise hypergraph alone. Within a separated-rate regime, a learned encoding can turn the process's indistinguishable sectors into stable operational components; their boundaries change when the reproducible operations or their lifetime change.*

The proposal is tested by changing the preparation history, recording port or coupling while keeping the task fixed. A stable Bell pair present initially demonstrates preservation. A product preparation followed by the actual collective port or Hamiltonian of Section 7 demonstrates formation of a relation. A diagnostic record that selects a working encoder demonstrates learned use. None of these tests alone establishes a complete subsystem gate algebra. This is why the complete interaction history, rather than just the noise graph or the final state, is the candidate source of effective organization.

The mathematical starting point needs only a carrier and its specified couplings. Site factors are introduced later for correlation and formation experiments. The general tensor-factorization question is therefore kept separate from the exact selection of stable sectors.

3 Reductions selected by a coupling

3.1 Setting

Let $\mathcal{H}$ be a finite-dimensional system Hilbert space. The environment couples to the system through Hermitian operators $A_1, \ldots, A_K$ with $[A_k, A_l] = 0$, at rates $\gamma_k > 0$. In the Markovian limit the reduced dynamics is generated by

$$\mathcal{L}(\rho) = \sum_{k=1}^{K} \gamma_k \Big(A_k \rho A_k - \tfrac{1}{2}\{A_k^2, \rho\} \Big), \qquad \Phi_t = e^{t\mathcal{L}}. \tag{1}$$

A system Hamiltonian that commutes with every A_k only rotates states inside the blocks defined below; we work in the corresponding interaction picture and return to Hamiltonians that do not commute in Remark 3.3. Section 5 treats the same couplings in an exact unitary model with environmental fragments.

Choose an orthonormal basis $\{|x\rangle : x \in \mathcal{X}\}$ of joint eigenvectors, $A_k|x\rangle = a_k(x)|x\rangle$, and put

$$D(x,y) = \tfrac{1}{2}\sum_k \gamma_k\big(a_k(x) - a_k(y)\big)^2 = \|u(x) - u(y)\|^2, \qquad u(x) = \big(\sqrt{\gamma_k/2}\, a_k(x)\big)_{k=1}^{K} \in \mathbb{R}^K. \tag{2}$$

We call $u(\mathcal{X}) \subset \mathbb{R}^K$ the rate image of the basis; decoherence rates are squared Euclidean distances in it. Write $x \approx y$ when $a_k(x) = a_k(y)$ for every k, let π_∞ be the partition of $\mathcal{X}$ into $\approx$-classes a, and let P_a be the projection onto $\mathcal{H}_a = \operatorname{span}\{|x\rangle : x \in a\}$. For any partition π of $\mathcal{X}$ into classes C, put

$$\mathfrak{B}_\pi = \bigoplus_{C\in\pi} \mathcal{B}(\mathcal{H}_C), \qquad \mathbb{E}_\pi(\rho) = \sum_{C\in\pi} P_C \rho P_C.$$

$\mathfrak{B}_\pi$ is a unital $*$-algebra, $\mathbb{E}_\pi$ is a channel (the pinching), and $\mathbb{E}_\pi$ is its own dual, the conditional expectation onto $\mathfrak{B}_\pi$.

Proposition 3.1 (The reduction selected by a commuting coupling)**.** *1. Φ_t acts as a Schur multiplier in the eigenbasis: $\Phi_t(\rho)_{xy} = e^{-tD(x,y)}\rho_{xy}$, and $\Phi_t^* = \Phi_t$.*

2. *The observables fixed by every Φ_t^* form the algebra $\mathfrak{B}_{\pi_\infty} = \{A_1, \dots, A_K\}'$.*
3. *$\Phi_t \to \mathbb{E}_{\pi_\infty}$ as $t \to \infty$, and the algebra generated by the dual range of $\mathbb{E}_{\pi_\infty}$ is $\mathfrak{B}_{\pi_\infty}$. In the terminology of Paper A, $(\mathfrak{B}_{\pi_\infty}, \mathbb{E}_{\pi_\infty})$ is a minimally compatible coarse-graining structure.*
4. *For every finite t, Φ_t is invertible on the operator space, so the algebra generated by its dual range is $\mathcal{B}(\mathcal{H})$.*

Proof. (1) In the eigenbasis, $(A\rho A - \frac{1}{2}\{A^2, \rho\})_{xy} = (a(x)a(y) - \frac{1}{2}a(x)^2 - \frac{1}{2}a(y)^2)\rho_{xy} = -\frac{1}{2}(a(x) - a(y))^2\rho_{xy}$, so $\mathcal{L}$ multiplies the (x, y) entry by $-D(x, y)$. The multiplier is real and symmetric, hence $\Phi_t^* = \Phi_t$. (2) X is fixed for all t exactly when $X_{xy} = 0$ whenever $D(x, y) > 0$, that is, whenever $x \not\approx y$. An operator commutes with every diagonal A_k exactly when it has no entries between basis vectors with different eigenvalue tuples. (3) Entries with $D(x, y) > 0$ tend to zero and the others are constant. The dual range of $\mathbb{E}_{\pi_\infty}$ is $\mathfrak{B}_{\pi_\infty}$, which is already an algebra. (4) All multipliers $e^{-tD(x,y)}$ are nonzero. □

Part (2) is the elementary commuting case of the known structure of fixed points of quantum dynamical semigroups [14, 15]. For this series the consequence matters more than the statement: the given coupling determines both its channel and its invariant algebra. Part (4) is Paper A's result that an algebra label does not record resolution, now in dynamical form. Before the long-time limit the channel generates everything, and the structure that a coupling makes accessible has to be identified by how well observables survive over time.

Corollary 3.2 (The selected sectors are intrinsic)**.** *In the interaction picture of* (1)*, the minimal central projections of* $\operatorname{Fix}(\Phi)$ *are precisely the complete joint-eigenvalue projections P_a. They are determined by the semigroup up to relabelling. A common unitary conjugation of the carrier and all couplings conjugates these projections and preserves the rates between them.*

Proof. Proposition 3.1 gives $\operatorname{Fix}(\Phi) = \bigoplus_a \mathcal{B}(P_a\mathcal{H})$. The centre of this algebra is $\operatorname{span}\{P_a\}$; its minimal nonzero projections are unique. On $P_a\mathcal{B}(\mathcal{H})P_b$, the generator is multiplication by $-\frac{1}{2}\sum_k \gamma_k(a_k(a) - a_k(b))^2$. Conjugating the entire model preserves these scalars. □

The eigenbasis used in the proof is not an extra choice of spatial sites. What has been selected is a direct sum, including quantum content within its blocks. It neither supplies the instruments for that content nor identifies the blocks with spatial tensor factors.

The objects of Paper E [3] can be located in this class. Its environmental fixed set $\operatorname{Fix}(\mathcal{E})$ is $\mathfrak{B}_{\pi_\infty}$. In Paper E's collision model, with a qubit pointer, the pointer algebra equals the fixed algebra of the induced dephasing and contains every effect a fragment can report. The centre of $\mathfrak{B}_{\pi_\infty}$ is generated by the block projections P_a; it is commutative, and Section 5 shows that it is what the environment records, so it serves as Paper E's pointer or record algebra. The inclusion of the record algebra in the fixed set is thus a conclusion of the model, as Paper E requires of any such overlap. The fixed set is strictly larger than its centre as soon as some block has dimension greater than one; a nontrivial two-sector qubit pointer has only one-dimensional blocks. Such blocks are decoherence-free subspaces [12, 13]. They retain quantum content; sector weights can retain classical correlations as well.

Remark 3.3 (Hamiltonians that do not commute with the couplings)**.** *If the system Hamiltonian H has a part H_{off} connecting different blocks, $\mathfrak{B}_{\pi_\infty}$ need not remain exactly invariant. Strong dephasing can then confine the dynamics to the blocks, with leakage on a time scale of order $\Gamma_{\mathrm{out}}/\|H_{\mathrm{off}}\|^2$, where Γ_{out} is the smallest between-block rate; this is the mechanism of quantum Zeno subspaces [16]. Section 4.1 gives an exact two-level lifetime and a finite-time stability bound without a commutativity assumption. Paper D's chain with a transverse field requires its own many-body estimate.*

4 Selection over a finite window

Definition 4.1 (Stable algebra on a window)**.** *A unital $*$-subalgebra $\mathfrak{B} \subseteq \mathcal{B}(\mathcal{H})$ is ε-stable on $[0,T]$ if*

$$\sup_{0\le t\le T}\ \sup_{X\in\mathfrak{B},\ \|X\|\le 1} \|\Phi_t^*(X) - X\| \le \varepsilon.$$

The diagnostic tests the whole unit ball of an algebra, rather than selected states or a list of slow observables. For a partition π define the width of a class C in the direction k, $w_k(C) = \max_{x\in C} a_k(x) - \min_{x\in C} a_k(x)$, and the two rates

$$\Gamma_{\mathrm{in}}(\pi) = \max_{C\in\pi}\ \tfrac{1}{2}\sum_k \gamma_k w_k(C)^2, \qquad \Gamma_{\mathrm{out}}(\pi) = \min\{D(x,y) :\ x,y \text{ in different classes of } \pi\}. \tag{3}$$

The minimum over an empty set is $+\infty$, so a single-block partition has $\Gamma_{\mathrm{out}} = +\infty$. If every coupling is scalar, the full algebra is fixed and selected for every window. In the rate image, $\Gamma_{\mathrm{in}}(\pi)$ is the largest squared diagonal of the axis-parallel boxes enclosing the classes, and $\Gamma_{\mathrm{out}}(\pi)$ is the smallest squared distance between points of different classes.

Theorem 4.2 (Selection by a window)**.** *Let $0 < \varepsilon < 1$ and $T > 0$.*

1. *For every $X \in \mathfrak{B}_\pi$ and $t \ge 0$, $\|\Phi_t^*(X) - X\| \le t\,\Gamma_{\mathrm{in}}(\pi)\|X\|$. For every state $\rho \in \mathfrak{B}_\pi$, $\|\Phi_t(\rho) - \rho\|_1 \le t\,\Gamma_{\mathrm{in}}(\pi)$. In particular $\mathfrak{B}_\pi$ is ε-stable on $[0,T]$ when $T\Gamma_{\mathrm{in}}(\pi) \le \varepsilon$.*
2. *If $1 - e^{-T\Gamma_{\mathrm{out}}(\pi)} > \varepsilon$, no unital $*$-algebra that contains $\mathfrak{B}_\pi$ properly is ε-stable on $[0,T]$.*
3. *Call π selected by (T,ε) if*
$$T\,\Gamma_{\mathrm{in}}(\pi) \le \varepsilon < 1 - e^{-T\Gamma_{\mathrm{out}}(\pi)}. \tag{4}$$
At most one partition is selected by (T,ε). If π is selected by (T,ε) and π' by (T',ε) with $T \le T'$, then π' refines π, so $\mathfrak{B}_{\pi'} \subseteq \mathfrak{B}_\pi$.
4. *For $T > -\ln(1-\varepsilon)/\Gamma_{\mathrm{out}}(\pi_\infty)$ the selected partition exists and is π_∞, so the selected algebra is the fixed algebra of Proposition 3.1.*

Proof. (1) $\mathfrak{B}_\pi$ is invariant under $\mathcal{L}$, and on it $\mathcal{L}(X) = -\frac{1}{2}\sum_k \gamma_k [A_k,[A_k,X]]$. On a class C, replacing A_k by $A_k - m_k$ with m_k the midpoint of its eigenvalues on C does not change the commutator and gives an operator of norm $w_k(C)/2$; hence $\|[A_k,X_C]\| \le w_k(C)\|X_C\|$ and $\|[A_k,[A_k,X_C]]\| \le w_k(C)^2\|X_C\|$. Since $\|X\| = \max_C \|X_C\|$, $\|\mathcal{L}(X)\| \le \Gamma_{\mathrm{in}}(\pi)\|X\|$. Then $\Phi_t^*(X) - X = \int_0^t \Phi_s^*(\mathcal{L}(X))\,ds$ and Φ_s^* is a contraction in operator norm. For a state $\rho \in \mathfrak{B}_\pi$ and $\|Y\| \le 1$, $\mathrm{Tr}[(\Phi_t(\rho) - \rho)Y] = \mathrm{Tr}[\rho(\Phi_t^* - \mathrm{id})(\mathbb{E}_\pi Y)]$, because $\rho = \mathbb{E}_\pi\rho$, $\mathbb{E}_\pi$ is self-dual and commutes with the Schur multiplier Φ_t^*; since $\mathbb{E}_\pi Y \in \mathfrak{B}_\pi$ and $\|\mathbb{E}_\pi Y\| \le 1$, the first statement bounds this by $t\Gamma_{\mathrm{in}}(\pi)$.

(2) Let $\mathfrak{B} \supsetneq \mathfrak{B}_\pi$ and $X \in \mathfrak{B} \setminus \mathfrak{B}_\pi$. Some entry $X_{x'y'}$ with $x' \in C$, $y' \in C'$, $C \ne C'$, is nonzero. For any $x \in C$, $y \in C'$ the matrix units $E_{xx'}$ and $E_{y'y}$ lie in $\mathfrak{B}_\pi \subseteq \mathfrak{B}$, so $E_{xx'}XE_{y'y} = X_{x'y'}E_{xy} \in \mathfrak{B}$. Then $\|\Phi_T^*(E_{xy}) - E_{xy}\| = 1 - e^{-TD(x,y)} \ge 1 - e^{-T\Gamma_{\mathrm{out}}(\pi)} > \varepsilon$.

(3) Let π and π' be selected by (T,ε) and (T',ε), $T \le T'$. If x,y lie in one class of π', then $D(x,y) \le \Gamma_{\mathrm{in}}(\pi') \le \varepsilon/T' \le \varepsilon/T$. If x,y lay in different classes of π, then $D(x,y) \ge \Gamma_{\mathrm{out}}(\pi) > -\ln(1-\varepsilon)/T \ge \varepsilon/T$. Hence every class of π' lies in a class of π. For $T = T'$ the same argument with the roles exchanged gives $\pi = \pi'$.

(4) $\Gamma_{\mathrm{in}}(\pi_\infty) = 0$ and $\Gamma_{\mathrm{out}}(\pi_\infty) > 0$, so π_∞ satisfies (4) for such T; uniqueness is part (3). □

A window and a tolerance therefore pick out at most one algebra of the form $\mathfrak{B}_\pi$, and they pick one out when the rate image of the basis splits into clusters that are small compared with ε/T and far apart compared with $-\ln(1-\varepsilon)/T$. This is a separation of time scales, the mechanism behind metastability in open quantum systems [17]; for commuting couplings it can be stated exactly. The selected algebra is moreover the largest ε-stable algebra of block form: if $\mathfrak{B}_\sigma$

is ε-stable on $[0,T]$, every matrix unit E_{xy} with x,y in one class of σ satisfies $1-e^{-TD(x,y)} \le \varepsilon$, so x and y lie in one class of the selected partition π, and $\mathfrak{B}_\sigma \subseteq \mathfrak{B}_\pi$. As the window lengthens, the selected algebra shrinks in discrete steps located at the gaps of the rate spectrum and ends at the fixed algebra. In the order of Paper A this is a chain of further coarse-grainings, since $\mathbb{E}_{\pi'} = \mathbb{E}_{\pi'} \circ \mathbb{E}_\pi$ when π' refines π.

Without a gap there is no selected algebra at that window. Selection uses the sufficient bound of part (1), so between that bound and the necessary bound of part (2) a unique maximal stable block algebra can still exist; but the window may also fail to determine one, and two incomparable maximal stable block algebras can occur. The set of ε-stable matrix units can then be an operator system whose generated algebra contains unstable elements: if $D(x,y)$ and $D(y,z)$ are small, $D(x,z)$ can still be as large as $(\sqrt{D(x,y)}+\sqrt{D(y,z)})^2$, and a chain of small steps can connect distant points of the rate image. Theorem 4.2 is a sufficient selection certificate. Failure of its gap condition does not by itself decide existence or uniqueness of a maximal stable algebra. Nor does stability of a block algebra supply the instruments needed to use every element as a gate.

4.1 Continuous capabilities when the fixed algebra jumps

A weak coupling can change the exact commutant discontinuously. It cannot change every finite experiment discontinuously. The distinction matters when a selected algebra is used to describe a finite observer.

Proposition 4.3 (Finite-time operational stability)**.** *Let $\mathcal{L}_0, \mathcal{L}_1$ be finite-dimensional Lindblad generators, and put $v = \|\mathcal{L}_1 - \mathcal{L}_0\|_\diamond$. Then*

$$\|e^{t\mathcal{L}_1} - e^{t\mathcal{L}_0}\|_\diamond \le \min\{2, tv\}. \tag{5}$$

Consider any experiment with the same initial states, retained ancillas and permitted instruments in both models. Between its instruments the system evolves for prescribed intervals $t_1,\dots,t_m$, with total $T=\sum_i t_i$. Instruments may depend on earlier outcomes. The total variation distance between its complete record laws is at most $\min\{1, Tv/2\}$. Consequently, for a common decision problem with loss in $[0,L]$, the two optimal risks differ by at most $L\min\{1,Tv/2\}$. In particular, a Hamiltonian perturbation $-i\epsilon[K,\cdot]$ gives $v \le 2|\epsilon|\|K\|$.

Proof. Duhamel's formula writes the channel difference as $\int_0^t e^{(t-s)\mathcal{L}_1}(\mathcal{L}_1-\mathcal{L}_0)e^{s\mathcal{L}_0}\,ds$. CPTP maps have diamond norm one, giving tv; two channels differ by at most two. A hybrid argument replaces the evolution intervals one at a time, keeping outcome-controlled instruments as CPTP maps on the system and classical record. Contractivity and the triangle inequality give a trace-norm difference at most Tv. The bounded-loss estimate holds uniformly over each common policy, so it also holds for the infima. Finally, $\|[K,A]\|_1 \le 2\|K\|\|A\|_1$ holds with every retained ancilla. □

The estimate needs neither a gap nor a selected algebra. It bounds distinguishing, storing and acting through the permitted experiment, not just the state of an isolated system. The available instruments and the total time remain the same on both sides. A discontinuous change of permissions, or an unbounded observation time, is a different comparison.

For an exact noncommuting example, take

$$\mathcal{L}_\epsilon(\rho) = -i\epsilon[X,\rho] + \kappa(Z\rho Z - \rho), \qquad \kappa > 0. \tag{6}$$

The fixed algebra is $\operatorname{span}\{I,Z\}$ at $\epsilon = 0$ and only $\mathbb{C}I$ at every $\epsilon \neq 0$. To see the latter, the Hilbert–Schmidt dissipative form of a fixed observable forces $[Z,O]=0$; the remaining Hamiltonian equation then forces $[X,O]=0$. Nevertheless, the two initially perfect records $(I \pm Z)/2$ remain readable for a long time. If $0 < 2|\epsilon| < \kappa$, put $\omega = \sqrt{\kappa^2 - 4\epsilon^2}$. Their Bloch vectors at time t are $\pm(0, y(t), z(t))$, where

$$z(t) = e^{-\kappa t}\left(\cosh\omega t + \frac{\kappa}{\omega}\sinh\omega t\right), \qquad y(t) = -\frac{2\epsilon}{\omega}e^{-\kappa t}\sinh\omega t. \tag{7}$$

These expressions solve $\dot y = -2\kappa y - 2\epsilon z$, $\dot z = 2\epsilon y$, with $(y,z) = (0,1)$ initially. Reading only Z has error $(1-z(t))/2$; allowing the optimal system measurement gives error $(1-\sqrt{y(t)^2+z(t)^2})/2$. The slow decay rate is exactly

$$r = \kappa - \sqrt{\kappa^2 - 4\epsilon^2} = \frac{2\epsilon^2}{\kappa} + O(\epsilon^4/\kappa^3). \tag{8}$$

Thus arbitrarily weak transverse coupling erases the infinite-time record, while its finite-time lifetime diverges as $\kappa/(2\epsilon^2)$. Exact hulls and fixed algebras describe the ideal endpoint; finite-time risks describe what remains available on the way to it. This is also the distinction needed by Foundation's hulls and the maintenance tasks of T-DOME and Brain Anchor. Neither the general stability estimate nor this example asserts that every slow operator space is an algebra or a physical projection.

5 Records live on the centre

5.1 A recording model

To see what the environment learns we replace the Markovian generator by an exact unitary model. Environmental fragments $E_1, \dots, E_K$ start in states ω_k, and the interaction is

$$U = \sum_{x \in \mathcal{X}} |x\rangle\langle x| \otimes V_1(x) \otimes \cdots \otimes V_K(x), \tag{9}$$

with unitaries $V_k(x)$ that depend on the basis state through the coupling, for example $V_k(x) = \exp[-i\theta_k a_k(x) G_k]$ for a fragment generator G_k. The joint state after the interaction is

$$\sigma = \sum_{x,y} \rho_{xy} |x\rangle\langle y| \otimes \bigotimes_k V_k(x)\, \omega_k\, V_k(y)^\dagger,$$

so the system undergoes the Schur multiplier

$$\rho_{xy} \mapsto c(x,y)\, \rho_{xy}, \qquad c(x,y) = \prod_k c_k(x,y), \qquad c_k(x,y) = \mathrm{Tr}\big[\omega_k V_k(y)^\dagger V_k(x)\big]. \tag{10}$$

The generator (1) arises from such models in weak-coupling and repeated-collision limits. Here the collision statements are exact for the specified unitary and an initial product $\rho \otimes \bigotimes_k \omega_k$. Fresh independent fragments are an input: recollisions or shared initial fragment correlations do not obey the product formula automatically.

Proposition 5.1 (Records carry only the block label)**.** *Let π be a partition of $\mathcal{X}$ such that every $V_k(x)$ depends on x only through its class $a(x) \in \pi$. Then:*

1. *$c(x,y) = 1$ whenever x and y lie in one class, so $\mathbb{E}_\pi(\rho)$, and with it the state on every block, is unchanged.*
2. *For every $O \in \mathfrak{B}_\pi$ and every operator F on any collection $\mathcal{F}$ of fragments,*

$$\mathrm{Tr}[\sigma(O \otimes F)] = \sum_{a \in \pi} \mathrm{Tr}[P_a \rho P_a\, O]\ \mathrm{Tr}[\omega_{\mathcal{F}}(a) F], \qquad \omega_{\mathcal{F}}(a) = \bigotimes_{k \in \mathcal{F}} V_k(a)\, \omega_k\, V_k(a)^\dagger.$$

The state of the fragments is $\sum_a p(a)\, \omega_{\mathcal{F}}(a)$ with $p(a) = \mathrm{Tr}\, P_a \rho$; it depends on ρ only through the distribution of the label, and given the label, block observables and fragments are uncorrelated.

Proof. (1) $V_k(x) = V_k(y)$ when x, y lie in one class, so $c_k(x,y) = 1$. (2) $O_{yx} \neq 0$ only when x and y lie in one class a, and then the fragment factor is $\omega_{\mathcal{F}}(a)$ for both. Summing $\rho_{xy} O_{yx}$ over $x, y \in a$ gives $\mathrm{Tr}[P_a \rho P_a O]$. □

Whatever the environment learns under such couplings is the block label, which generates the centre of $\mathfrak{B}_\pi$. Many fragments can hold readable copies of that label while leaving the blocks alone. Block dependence alone does not guarantee distinguishability: the conditional states $\omega_k(a)$ must also be different, and a permitted measurement must resolve them. Records and the correlations inside the blocks are thus carried by one selected algebra, the first by its centre and the second by its blocks. That the redundantly available information is classical, and is information about one measurement, holds far more generally [18]; a two-qubit system with a decoherence-free subspace that proliferates part of its information while the remaining subspace keeps non-classical features was studied in [19]. What the commuting class adds is the exact location of both kinds of information in one algebra and, next, the price of imperfect symmetry.

5.2 Leakage costs coherence, and the costs multiply

Theorem 5.2 (Leakage and coherence)**.** *For basis states x, y and a fragment k, let $D_k(x,y) = \frac{1}{2}\big\| V_k(x)\omega_k V_k(x)^\dagger - V_k(y)\omega_k V_k(y)^\dagger \big\|_1$ be the fragment's distinguishability of x and y, so that an optimal measurement on the fragment identifies one of the two equally likely states with probability $\frac{1}{2}(1 + D_k)$ [20]. Then*

$$|c_k(x,y)|^2 + D_k(x,y)^2 \leq 1, \tag{11}$$

with equality when ω_k is pure. Consequently

$$|c(x,y)| \leq \prod_k \big(1 - D_k(x,y)^2\big)^{1/2} \leq \exp\Big(-\tfrac{1}{2} \sum_k D_k(x,y)^2\Big), \tag{12}$$

and if the coherence between x and y must stay above $1 - \varepsilon$, then

$$\sum_k D_k(x,y)^2 \leq 2 \ln \frac{1}{1-\varepsilon}. \tag{13}$$

Proof. Purify $\omega_k = \mathrm{Tr}_R\,|\Omega\rangle\langle\Omega|$. Then $c_k(x,y) = \langle\Omega|(V_k(y)^\dagger V_k(x)\otimes I)|\Omega\rangle$ is the overlap of the pure states $(V_k(x)\otimes I)|\Omega\rangle$ and $(V_k(y)\otimes I)|\Omega\rangle$, whose trace distance is $(1-|c_k|^2)^{1/2}$. Partial trace over R does not increase trace distance, which gives (11), with equality when there is nothing to trace out. Inequality (12) follows from (10) and $\ln(1-u)\le -u$, and (13) by taking logarithms. □

Inequality (11) is Englert's relation between visibility and which-way distinguishability [21], applied to one fragment. It bounds what a fragment learns by the coherence it destroys, not the reverse: a fragment in a mixed state can destroy coherence while learning nothing, as in Paper E's maximally mixed memory example. Purity is one useful preparation resource; mixed fragments can also record when their conditional states differ. For many fragments it becomes the budget (13): the sum of their squared pairwise distinguishabilities is at most $2\ln(1/(1-\varepsilon))$ for a protected pair of basis states. This is not a bound on their total Shannon information about an arbitrary block.

Corollary 5.3 (Redundancy against symmetry)**.** *Suppose that R fragments each distinguish two states x, y of one block with $D_k(x,y)\ge\beta$. Then $|c(x,y)|\le(1-\beta^2)^{R/2}$, and keeping the coherence above $1-\varepsilon$ requires*

$$\beta \le \Big(\frac{2\ln(1/(1-\varepsilon))}{R}\Big)^{1/2}. \tag{14}$$

In a sequence of models with $R\to\infty$ and $D_k(x,y)=\beta_R=\lambda R^{-\alpha}$, $\alpha>0$, for the R fragments that couple to the block, the coherence tends to zero if $\alpha<\frac12$. If the fragment states are pure and no other fragment distinguishes x from y, it tends to 1 for $\alpha>\frac12$ and to $e^{-\lambda^2/2}$ for $\alpha=\frac12$.

Proof. The bound and (14) are (12) and (13). For pure fragments $|c(x,y)| = (1-\beta_R^2)^{R/2}$ by the equality case, and $\frac{R}{2}\ln(1-\lambda^2R^{-2\alpha})\to 0,\ -\lambda^2/2,\ -\infty$ for $\alpha>\frac12, =\frac12, <\frac12$. □

A coupling of the form (9) is block-constant only by symmetry: the fragments must respond identically to all states of a block. Real couplings break such a symmetry slightly, and each recording fragment then leaks a little. Corollary 5.3 says that objectivity by redundancy and protection of the blocks can coexist, but only if the symmetry error per fragment falls as $R^{-1/2}$. In the limit of many records this is a sharp threshold in the exponent α. It is the closest analogue of a transition in this class. It is a statement about how a symmetry error scales with the number of records, and it is not a phase transition in the thermodynamic sense.

6 Sector records and protected correlations

A mutual-information geometry does not require entanglement. Even local dephasing preserves $\rho = \frac12|00\rangle\langle 00| + \frac12|11\rangle\langle 11|$, for which $I(1:2)=\ln 2$. With local Z couplings every selected block is one-dimensional and every sector state is a product. Its entire edge comes from sector weights.

For a selected state $\bar\rho = \mathbb{E}_\pi(\rho) = \sum_k p_k\rho_k$, with $\rho_k = P_k\rho P_k/p_k$, retain the label in a classical register:

$$\Omega_{SK} = \sum_k p_k\rho_k\otimes|k\rangle\langle k|_K. \tag{15}$$

This extension is physically realized by a perfectly resolving block-constant recorder followed by measurement or dephasing of its label register. It is not the coherent source–memory state before that readout. If the actual fragment states overlap, K is still a useful mathematical label, but it is not a freely readable record. Zero-weight sectors are omitted.

Proposition 6.1 (The two contributions to correlation)**.** *For any supplied disjoint subsystems A, B of S,*

$$I(A:B)_{\bar\rho} = \sum_k p_k I(A:B)_{\rho_k} + I(A:K)_\Omega + I(B:K)_\Omega - I(AB:K)_\Omega. \tag{16}$$

The last three terms need not have a nonnegative sum.

Proof. For $X = A, B, AB$, the classical–quantum entropy formula gives $I(X:K) = S(\bar\rho_X) - \sum_k p_k S((\rho_k)_X)$. Substitution in $S(A)+S(B)-S(AB)$ proves the identity. □

The first term locates correlations protected within sectors; it can include classical as well as quantum correlations. The correction describes what mixing sector-conditioned states does to those correlations. In the locally dephased example above it equals $\ln 2$. In the next recording example it is negative. Thus neither a noncommutative block nor a classical centre alone determines the geometry. The sector states, their weights, the site marginals and the dictionary all enter.

7 How protected relations are formed

A noise commutant specifies what can be stored. It does not determine whether that content was inherited or created. The distinction already invalidates a noise-only mereology rule.

7.1 Disconnected noise can preserve a shared Bell pair

Take four qubits with only $A_{12} = Z_1 + Z_2$ and $A_{34} = Z_3 + Z_4$. Each pair has a zero-magnetization sector with logical basis $|0_L\rangle = |01\rangle$, $|1_L\rangle = |10\rangle$. The state

$$|\Psi_L\rangle = \frac{|0_L\rangle_{12}|0_L\rangle_{34} + |1_L\rangle_{12}|1_L\rangle_{34}}{\sqrt{2}} \tag{17}$$

is annihilated by both noise operators and is fixed exactly by (1). Yet $I(12:34) = 2\ln 2$. The two noise-hypergraph components are disconnected. Applying a Hadamard to each qubit gives the same counterexample for the X couplings of Section 8. Classical and quantum correlations can both cross disconnected noise components when they are present initially.

7.2 A collective recorder forms a flagged protected relation

Start two qubits in the physical product $|+_X\rangle|+_X\rangle$ and a fresh three-level memory in $|0\rangle$. Let $M = Z_1 + Z_2$ and let P_+, P_0, P_- project onto its eigenvalues $2, 0, -2$. For the shift $S_3|k\rangle = |k+1 \bmod 3\rangle$, use

$$U_{\rm rec} = P_+ \otimes I + P_0 \otimes S_3 + P_- \otimes S_3^2. \tag{18}$$

This is a block-constant recording interaction, not a product of independent local readers. It has a finite Hamiltonian implementation: choose a Hermitian logarithm G with $e^{-iG} = S_3$ and $\|G\| = 2\pi/3$, put $a(M) = (2I - M)/2$, and apply $H_{\rm rec} = (\hbar/t_g)a(M) \otimes G$ for a time $t_g > 0$. The same memory couples to both sites. Its preparation, shared physical port, interaction time, control and subsequent readout are resources.

The complete output is

$$|\Omega\rangle = \tfrac{1}{2}|00\rangle|0\rangle + \tfrac{1}{\sqrt{2}}|\Psi^+\rangle|1\rangle + \tfrac{1}{2}|11\rangle|2\rangle, \qquad |\Psi^+\rangle = (|01\rangle + |10\rangle)/\sqrt{2}. \tag{19}$$

A computational-basis measurement of the memory produces the classical–quantum state with probabilities $(1/4, 1/2, 1/4)$ and conditional source states $(|00\rangle\langle 00|, |\Psi^+\rangle\langle\Psi^+|, |11\rangle\langle 11|)$. All branches remain in the protocol. Flag 1, of probability $1/2$, certifies a Bell pair in the $M = 0$ protected sector; flags $0, 2$ certify product states.

If the flag is discarded, the source is instead

$$\rho_* = \tfrac{1}{4}|00\rangle\langle 00| + \tfrac{1}{2}|\Psi^+\rangle\langle\Psi^+| + \tfrac{1}{4}|11\rangle\langle 11|, \qquad I(1:2)_{\rho_*} = \tfrac{1}{2}\ln 2. \tag{20}$$

It is separable. Indeed it is the uniform collective Z-rotation twirl of $|+_X, +_X\rangle\langle +_X, +_X|$, a mixture of product rotations. Its partial-transpose eigenvalues are $0, 1/4, 1/4, 1/2$. The unconditional source therefore gains a mutual-information edge without entanglement; separability does not imply that every correlation measure is classical. The retained flagged ensemble permits a conditional protected Bell use, with herald probability and every unsuccessful branch charged. The pure source–memory state (19), the measured flagged state and the unconditional source are three different outputs.

For this flagged ensemble, $\sum_k p_k I(1:2)_{\rho_k} = \ln 2$, $I(1:K) = I(2:K) = \frac{1}{2}\ln 2$ and $I(12:K) = H(K) = \frac{3}{2}\ln 2$. Equation (16) recovers the unconditional $\frac{1}{2}\ln 2$ with correction $-\frac{1}{2}\ln 2$. Collective dephasing of the product input alone reaches the same ρ_* but supplies no accessible flag. A flag requires the specified recording interaction and actual readout.

7.3 A real link forms a deterministic protected Bell pair

The distinction is sharper in the four-qubit example. Define

$$L^x_{ab} = \tfrac{1}{2}(X_aX_b + Y_aY_b), \qquad H_{\rm link} = \hbar J L^x_{12}L^x_{34}, \quad J > 0.$$

L^x_{ab} exchanges $|01\rangle$ and $|10\rangle$ and vanishes on $|00\rangle, |11\rangle$. Consequently $[H_{\rm link}, A_{12}] = [H_{\rm link}, A_{34}] = 0$. Starting from the physical product $|01\rangle_{12}|01\rangle_{34}$, the full noisy evolution remains in the joint zero sector and is exactly

$$|\psi(t)\rangle = \cos(Jt)|0_L0_L\rangle - i\sin(Jt)|1_L1_L\rangle. \tag{21}$$

At $t = \pi/(4J)$ it has $I(12:34) = 2\ln 2$ and one ebit across the pair partition. The noise term vanishes throughout the trajectory.

This is new protected correlation from an initially uncorrelated state. It uses an actual cross-pair Hamiltonian link, here a four-body control term. The noise graph remains disconnected but the complete interaction graph does not. The factorization, prepared input, gate permissions, coupling strength and duration are supplied; there is no inferred free implementation or heat price for the link.

7.4 The process constraint and learned use

Proposition 7.1 (Correlation provenance across independent components)**.** *Let a supplied partition of a carrier have initially independent states and auxiliary resources. Suppose its full channel, including record registers, factors across these components. Then it creates no cross-component mutual information. If the input is correlated, a product channel cannot increase the mutual information across any bipartition of the components, but it can preserve initially shared classical or quantum correlations.*

Proof. A product channel sends a product input to a product output, including its local records. For a correlated input, $I(A:B) = D(\rho_{AB}\|\rho_A \otimes \rho_B)$, so the second statement is data processing under the product channel. □

An implementation satisfies the factor premise when its Hamiltonian and noise generators act separately on the components, its fresh ports are independent, and every control and instrument is component-local. Shared random controls, communication or common memories must be added to the process description: they can correlate unconditional outputs. Conditioning on a joint event can also change correlations and is not a product unconditional channel. The proposition is a diagnostic, not a claim that every process with a disconnected noise graph factors.

The constructions above do more than add initial-state qualifications to a conjecture. They exhibit how the missing relations can form: a collective port records an indistinguishability sector, while an interaction inside protected sectors builds usable quantum content. The other indispensable step is learning which encoding an unknown physical plant protects. Foundation III [6] implements that step for $A_s = n_1 + sn_2$ with unknown $s = \pm 1$. A diagnostic sharing the plant's actual sign writes a record; that record, not s, selects odd or even encoding and the matching decoder. The same logical content and its old reference survive from before diagnosis. The record-controlled frame changes the later contact, rather than merely changing a description of the fixed commutant.

That result assumes a physical relation between diagnostic and storage couplings. A diagnostic carrying an unrelated sign would not learn the protected frame. Its retained-history scalar/JT implementation also contains real body forces, a source and a second acquisition; it is not identified with the full quantum material model. The complementary contribution here is the separation of formation, protection and use: each can fail while the others hold. A noiseless block can be unreadable, a readable label can outlive its quantum content, and a learned encoder can protect content without supplying every logical gate.

8 A geometry selected by the coupling

We now apply these results to the state and the reconstruction rule of Paper A. The rule acts on the complete graph of the sites: the edge $\{i,j\}$ has length $1/I_{ij}$, where I_{ij} is the mutual information of sites i and j, and distances are shortest-path lengths. We extend it to $I_{ij} = 0$ by giving that edge infinite length, so that a site whose edges all have zero mutual information is disconnected. The state is the three-qubit GHZ state $(|000\rangle + |111\rangle)/\sqrt{2}$. In the eigenbasis $|\pm\rangle$ of the Pauli operator X it reads

$$|\mathrm{GHZ}\rangle = \tfrac{1}{2}\big(|{+}{+}{+}\rangle + |{+}{-}{-}\rangle + |{-}{+}{-}\rangle + |{-}{-}{+}\rangle\big).$$

All couplings below are sums of X_i, hence diagonal in this basis, and Sections 3–5 apply. A local coupling $A = X_i$ with rate γ_i is the bit-flip noise of Paper A, with flip probability $p_i = (1 - e^{-2\gamma_i t})/2$. A collective coupling $A = X_1 + X_2 + X_3$ dephases between different values of the X-magnetization $M = \sum_i x_i$ and leaves states of equal M coherent.

Proposition 8.1 (Three couplings, three geometries)**.** *Let* $\Psi^+ = (|{+}{-}\rangle + |{-}{+}\rangle)/\sqrt{2}$ *and* $\tilde{W} = (|{+}{-}{-}\rangle + |{-}{+}{-}\rangle + |{-}{-}{+}\rangle)/\sqrt{3}$*. The long-time selected states* $\mathbb{E}_{\pi_\infty}(|\mathrm{GHZ}\rangle\langle\mathrm{GHZ}|)$ *and their geometries are:*

1. *Collective coupling* $X_1 + X_2 + X_3$*:* $\frac{1}{4}|{+}{+}{+}\rangle\langle{+}{+}{+}| + \frac{3}{4}|\tilde{W}\rangle\langle\tilde{W}|$*. Each pair has* $I_{ij} = \frac{1}{2}\ln 2$*; the geometry is an equilateral triangle with side* $2/\ln 2 = 2.8854$*.*

2. *Collective coupling $X_1 + X_2$ and local coupling X_3:* $\frac{1}{4}|++\rangle\langle+++| + \frac{1}{2}|\Psi^+\rangle\langle\Psi^+|_{12} \otimes |-\rangle\langle-|_3 + \frac{1}{4}|--+\rangle\langle--+|$. *Here $I_{12} = \frac{1}{2}\ln 2$ and $I_{13} = I_{23} = 0$; the geometry is a single edge, and site 3 is disconnected.*

3. *Local couplings X_1, X_2, X_3: the uniform mixture of the four even strings $|+++\rangle, |+--\rangle, |-+-\rangle, |--+\rangle$. Every $I_{ij} = 0$; no edge survives.*

Proof. The pinching keeps the amplitudes inside each class of equal eigenvalue tuples. (1) The classes met by the GHZ state are $M = 3$ and $M = -1$. The marginal of sites $1, 2$ is $\frac{1}{4}|++\rangle\langle++| + \frac{1}{2}|\Psi^+\rangle\langle\Psi^+| + \frac{1}{4}|--\rangle\langle--|$, with eigenvalues $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$ and entropy $\frac{3}{2}\ln 2$; each single-site marginal is $I/2$; so $I_{12} = 2\ln 2 - \frac{3}{2}\ln 2$, and the other pairs follow by symmetry. (2) The classes are labelled by $(x_1 + x_2, x_3)$; the strings $|+--\rangle$ and $|-+-\rangle$ share the label $(0, -)$. The marginal of sites $1, 2$ is as in (1). The marginal of sites $1, 3$ is uniform on the four strings $|++\rangle, |+-\rangle, |--\rangle, |-+\rangle$, and so is that of sites $2, 3$. (3) Every pair marginal of the uniform mixture of even strings is uniform. Direct evaluation of the full eight-dimensional density matrices gives the same values. □

The state and the rule are the same in all three cases; only the coupling differs. At zero time each pair has $I_{ij} = \ln 2$, Paper A's GHZ geometry at zero noise. Under local noise Paper A's triangles relax toward the empty graph; a collective coupling stops the relaxation at an equilateral triangle, and a coupling that is collective on one pair keeps exactly that edge. The surviving correlations come from blocks of dimension greater than one: the three-dimensional block $M = -1$, which contains the entangled state $\tilde{W}$, in case (1), and the two-dimensional block spanned by $|+--\rangle$ and $|-+-\rangle$ in case (2). In case (3) the selected algebra is commutative with product minimal projections, every selected state is classically correlated, and for this state the classical pair correlations vanish as well. By Proposition 3.1(4) the channels at any finite time generate the full matrix algebra in all three cases, so an algebra label cannot tell them apart.

8.1 The lifetime of the selected geometry

Real couplings are not exactly collective. Add weak local couplings X_i with rates γ_i to the collective coupling with rate γ_c. The exact fixed algebra is then commutative and the long-time geometry is empty, but on a window the magnetization blocks are still selected. With π the partition by M, the rates (3) are $\Gamma_{\text{in}}(\pi) = 2(\gamma_1 + \gamma_2 + \gamma_3)$ and $\Gamma_{\text{out}}(\pi) = 2\gamma_c + 2\min_i \gamma_i$, while the coherences of the GHZ state between its two blocks decay at the rates $8\gamma_c + 2(\gamma_i + \gamma_j)$.

Lemma 8.2 (Plateau bound)**.** *For a state ρ and a partition π, let $\Gamma_{\text{out}}(\rho, \pi)$ be the smallest $D(x, y)$ over pairs in different classes with $\rho_{xy} \neq 0$, and let r be the number of basis vectors with $\rho_{xx} \neq 0$. Then for all $t \geq 0$*

$$\|\Phi_t(\rho) - \mathbb{E}_\pi(\rho)\|_1 \leq \sqrt{r}\, e^{-t\Gamma_{\text{out}}(\rho,\pi)}\, \|\rho - \mathbb{E}_\pi(\rho)\|_2 + t\,\Gamma_{\text{in}}(\pi). \tag{22}$$

Proof. Split $\Phi_t(\rho) - \mathbb{E}_\pi\rho = \Phi_t(\rho - \mathbb{E}_\pi\rho) + (\Phi_t(\mathbb{E}_\pi\rho) - \mathbb{E}_\pi\rho)$. The second term is bounded by Theorem 4.2(1). The first is the Schur product of $\rho - \mathbb{E}_\pi\rho$ with multipliers of modulus at most $e^{-t\Gamma_{\text{out}}(\rho,\pi)}$; its rank is at most r, because a row of a positive matrix with zero diagonal entry vanishes, and $\|Z\|_1 \leq \sqrt{\operatorname{rank} Z}\, \|Z\|_2$. □

The bound turns into a certified geometry through continuity of entropy. If two states of a qubit pair are at trace distance $\delta = \frac{1}{2}\|\rho - \sigma\|_1 \leq \frac{1}{2}$, then by the Fannes–Audenaert inequality [22] and Winter's bound for conditional entropy [23] their mutual informations differ by at most

$$f(\delta) = h(\delta) + 2\delta\ln 2 + (1 + \delta)\, h\Big(\frac{\delta}{1 + \delta}\Big), \tag{23}$$

with h the binary entropy in nats. Partial traces do not increase trace distance, and Paper A's stability bound for graph metrics converts a uniform change $f < m$ of mutual informations bounded below by m into a change of at most $2f/(m(m - f))$ of every distance among three sites.

Proposition 8.3 (A certified plateau)**.** *For $\gamma_c = 1$ and $(\gamma_1, \gamma_2, \gamma_3) = 10^{-4}(1, 2, 4)$, the bounds* (22) *and* (23) *certify that every distance of the GHZ geometry lies within 25% of the plateau value $2/\ln 2$ for $0.716 \leq t \leq 3.58$. Numerical root finding of the exact evolution gives the approximate 25% window $0.053 \lesssim t \lesssim 53.6$ and 5% window $0.163 \lesssim t \lesssim 8.86$. If the local rates are scaled to $s(1, 2, 4)$ with $s \leq 10^{-3}$, the late edges scale to leading order as $1/s$: approximately $5.36 \times 10^{-3}/s$ for the 25% window, $8.86 \times 10^{-4}/s$ for the 5% window and, when $s \leq 10^{-4}$ so that it exists, $3.58 \times 10^{-4}/s$ for the certified 25% window. The windows begin at times of order $1/\gamma_c$.*

Proof. For this state $r = 4$, $\|\rho - \mathbb{E}_\pi\rho\|_2 = \sqrt{6}/4$, $\Gamma_{\text{out}}(\rho, \pi) = 8.0006$ and $\Gamma_{\text{in}}(\pi) = 1.4 \times 10^{-3}$. The right side of (22) is the convex function $b(t) = (\sqrt{6}/2)e^{-8.0006t} + 0.0014t$. On $[0.716, 3.58]$ its maximum is at an endpoint and is

t	$\|\Phi_t(\rho) - \mathbb{E}_\pi\rho\|_1$	bound (22)	distances (d_{12}, d_{13}, d_{23})
0	0.8660	1.2247	(1.4427, 1.4427, 1.4427)
1	0.0010	0.0018	(2.8968, 2.9034, 2.9066)
10	0.0094	0.0140	(2.9728, 3.0221, 3.0458)
100	0.0897	0.1400	(3.5620, 3.9855, 4.2034)
300	0.2442	0.4200	(4.8943, 6.5032, 7.4508)
1000	0.5969	1.4000	(12.559, 28.867, 41.426)

Table 1: GHZ state under a collective coupling with $\gamma_c = 1$ and local couplings with $(\gamma_1, \gamma_2, \gamma_3) = 10^{-4}(1, 2, 4)$. The plateau value of every distance is $2/\ln 2 = 2.8854$.

less than 0.0050121. The trace distance used in (23) is $\delta(t) = \frac{1}{2}\|\Phi_t(\rho) - \mathbb{E}_\pi\rho\|_1 \le b(t)/2$, not $b(t)$. The increasing entropy and graph bounds therefore give a distance error less than $0.72130 < \frac{1}{4} \cdot 2/\ln 2$ throughout that interval. The exact-evolution window edges were located by root finding of the Schur multipliers. Once the coherences between blocks have decayed, the in-block multipliers depend on t only through st, which gives the late-edge scaling. The root values are numerical diagnostics, distinct from the conservative analytic interval. □

Table 1 shows the three stages. The GHZ geometry, with side $1/\ln 2$, relaxes on the time scale $1/\gamma_c$ to the selected triangle with side $2/\ln 2$; the triangle persists while $t\Gamma_{\text{in}}(\pi)$ is small; then the unequal local rates deform it into scalene triangles, which are not isometric to the plateau, before every edge diverges. This is Paper A's non-isometry produced by one time-ordered dynamics rather than by a comparison of separately chosen channels. Proposition 8.3 gives a state-specific stability certificate for Paper A's declared graph dictionary on an explicit window. The plateau exists when its end lies after its beginning, that is, when the ratio of in-block to between-block rates is small, as Theorem 4.2 requires.

The constants in Proposition 8.3 are small. At the end of the exact 25% window the three in-block coherences have fallen by only 3.2, 5.2 and 6.2 per cent. The selected pair marginal has a zero eigenvalue, and in-block decoherence populates it, so pair entropies respond to a coherence loss δ like $\delta\ln(1/\delta)$. Geometries read from entropies are therefore far more fragile than the coherences that carry them, and the gap between the certified and the exact window comes mostly from this logarithmic response together with worst-case continuity bounds.

8.2 Imperfect redundant records detach a vertex

Finally we let fragments record the magnetization with a small asymmetry. Each of K fragment qubits starts in $|0\rangle$ and interacts through $V_k(x) = \exp[-i\theta(M(x) + \eta x_1)Y]$ with $\theta = \pi/8$, where Y is the Pauli operator of the fragment. For $\eta = 0$ the coupling is block-constant. The two magnetizations present in the GHZ state, $M = 3$ and $M = -1$, then produce orthogonal fragment states, so every fragment holds a perfect copy of the label, and by Proposition 5.1 the equilateral triangle survives for any number of fragments. For $\eta > 0$ each fragment distinguishes strings of the block $M = -1$ that differ at site 1, with $D_k = |\sin 2\theta\eta|$.

η	K	coherence of $\|+--\rangle, \|-+-\rangle$	$I_{12} = I_{13}$	I_{23}	label error
0.02	1	0.99988	0.34631	0.34657	6.2×10^{-5}
0.02	100	0.98774	0.32790	0.34657	6.2×10^{-5}
0.02	1000	0.88393	0.23581	0.34657	6.2×10^{-5}
0.05	1	0.99923	0.34525	0.34657	3.9×10^{-4}
0.05	100	0.92577	0.26724	0.34657	3.9×10^{-4}
0.05	1000	0.46243	0.05555	0.34657	3.9×10^{-4}

Table 2: Recording the magnetization with a site-1 asymmetry η. The last column is the minimal error probability with which a measurement on one fragment identifies the label M, whose prior is $(\frac{1}{4}, \frac{3}{4})$. Coherences are relative to their initial values; the coherence between $|-+-\rangle$ and $|--+\rangle$ stays equal to 1.

Table 2 shows the effect. Every fragment is a nearly perfect and readable record of the label for both asymmetries, so the loss of geometry is not a price paid for better records. It is the small leakage about site 1, multiplied over fragments as in Theorem 5.2. The coherences between strings that differ at site 1 decay as $(1 - D_k^2)^{K/2}$, the edges to site 1 weaken, and as $K \to \infty$ the state approaches $\frac{1}{4}|+++\rangle\langle+++| + \frac{1}{4}|+--\rangle\langle+--| + \frac{1}{2}|-\rangle\langle-|_1 \otimes |\Psi^+\rangle\langle\Psi^+|_{23}$. Then

$I_{12} = I_{13} = 0$ and $I_{23} = \frac{1}{2}\ln 2$: site 1 is detached and the geometry becomes the single edge of Proposition 8.1(2), now on sites $2, 3$. With $\eta_K = 0.5K^{-\alpha}$ the coherence after $K = 10^6$ fragments is 0 for $\alpha = \frac{1}{4}$, $0.9258 = e^{-(\pi/8)^2/2}$ for $\alpha = \frac{1}{2}$, and 0.9999 for $\alpha = \frac{3}{4}$, as Corollary 5.3 predicts with $\lambda = 2\theta \cdot 0.5$ to leading order.

9 What selection settles

The selected partition is fixed by the supplied dynamics, window and tolerance, and is robust to perturbations preserving the rate gap. This differs in emphasis from QBism's state assignments and relational quantum mechanics' reference-system dependence [26, 25], without deciding their wider interpretations. Public agreement further needs actual fragments and calibrated readers; Paper A provides one finite realization. A stable algebra alone does not grant those permissions.

The GHZ comparison changes a state under one graph dictionary, not the coordinates of an unchanged metric. Sector mixing can sustain edges after quantum content is lost; the leaking GHZ block instead supports the specific edges destroyed in Table 2. Formation requires the collective port or Hamiltonian link of Section 7, not noise selection alone. Removing either resource supplies its corresponding opposing control.

AQFT and Consciousness I start with their local or current-net carriers [24, 4]. Observable-induced factorization and quantum mereology address when retained relations support an effective subsystem description [28, 29, 30]. Decoherence and Everettian accounts share the underlying dynamics [10, 27]; the issue here is whether its resulting structure is readable, operable and persistent for a finite task.

10 Beyond the commuting class

Conjecture 2.1 separates the existence of a stable block from its acquisition as an operational component. We now examine the further conditions required outside the exactly commuting class.

The conjecture concerns effective factorization, not the existence of elementary sites. It predicts more than correlation alone: successful formation should yield an encoder, decoder and repeatable operations on the formed component, with calibration records and finite error bounds. The exact formation examples give protected relational content; Foundation III adds record-controlled encoding of unknown content. Together they support a route from coupling through history to changed capability. They do not yet supply a general autonomous acquisition of a complete internal gate set. A proposed component that has only a state-dependent correlation but no reproducible internal operations would fail this criterion. The disconnected Bell counterexample rules out noise-only connectivity as a necessary condition, while Proposition 7.1 identifies the real no-creation obstruction for product processes.

For general Lindblad dynamics, a separated spectrum defines slow states and an effective metastable evolution [17]. A finite-window extension should select the operational structure of this manifold rather than assume that slow observables close under multiplication. Even complete positivity and idempotence do not ensure closure in the original multiplication. For example,

$$P(X) = \operatorname{diag}\big(x_{11}, x_{22}, (x_{11} + x_{22})/2\big) \tag{24}$$

is a unital completely positive idempotent on M_3. It is completely positive because it is a sum of two positive evaluation maps into positive diagonal matrices. Its range contains $E = \operatorname{diag}(1, 0, 1/2)$ but not $E^2 = \operatorname{diag}(1, 0, 1/4)$. A completely positive projection gives its range a Choi–Effros product $x \circ y = P(xy)$ [7, 8]. This abstract product does not make xy a physically permitted observable or control in the original representation. An ordinary conditional expectation onto a represented subalgebra needs the additional algebraic closure. A metastable spectral projection may not even be completely positive. Thus a general selection theory must provide actual instruments or recovery maps, their errors and their operational closure, rather than infer them from a slow spectral space.

The physical price of that capability is the next link to Q-RAIF and the gravitational line. A passive GHZ geometry has lifetime of order $1/s$ under symmetry breaking, while independent redundant fragments require an $R^{-1/2}$ protecting symmetry. Q-RAIF B realizes correction as field coupling with work and stress [5]; HAFF D distinguishes entropy removal, usable side information and supplied purity, and gives specified source–response constructions [2]. These results suggest a maintained-structure mechanism: actual records learn a protected frame, operations install it, maintenance changes the physical source, and the response changes subsequent records. Foundation III realizes a finite retained-history scalar/JT return, whereas the full quantum material source requires its own support and preparation account. Connecting those implementations and renewing their resources is a substantive requirement, not an irreducible heat flux inferred from algebra inclusion.

Many-site and continuum selection require further work when rate gaps close. Consciousness I's ideal fixed-net closure and finite-time noise estimates are not already a continuum uniqueness theorem for window-selected blocks. Likewise, the graph dictionary is supplied here. A general emergence of spacetime needs a physical family of readers and a reconstruction law that remains controlled as the operational structure grows.

11 Conclusion

A commuting coupling determines its fixed block algebra and its long-time pinching. On a separated finite window it selects a unique block partition in the supplied joint eigenbasis; the selected algebra contains every stable block algebra of that form. The actual finite-time channel has full operator range. Block-constant record couplings report sector weights without resolving protected content. Independent leakage multiplies the coherence loss and imposes a sharp redundancy–symmetry threshold.

The relation to geometry has two parts. Sector mixing and within-sector content contribute through the exact identity (16); neither alone determines the mutual information graph. For the GHZ source, collective, pair-collective and local couplings yield a triangle, one edge and no edge under one rule. Weak asymmetry gives a finite plateau and then destroys its supporting edges. Noncommuting perturbations can erase the exact fixed algebra without abruptly erasing a finite record or task capability.

Formation requires a history. An initially shared logical Bell pair can survive between noise-disconnected protected components, so a noise graph cannot delimit all surviving quantum relations. A collective recorder forms a flagged protected Bell branch from a product input, with all branches retained; an actual Hamiltonian link forms a deterministic Bell pair protected throughout its generation. These constructions reduce the missing formation step to identifiable physical operations without hiding their costs. Record-controlled encoding supplies the further route from learned relations to changed capability. The remaining general problem is to derive reproducible subsystem operations and their sustained maintenance from such histories, rather than merely rename correlated states as subsystems.

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Causation, Agency, and Existence: Historical Control and Maintained Intervention

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Abstract

A cause changes what an allowed intervention can bring about; a record alone need not identify that change. We define interventional influence relative to a process, an intervention set and a readout. It contracts under post-processing and along channel chains. A direct cause and a common cause can have identical declared record laws and influences 1 and 0.

Agency requires more than a persistent register. In a binary tracking task, the memoryless failure probability is bounded by $\frac{1}{2}[1-(1-2q)\kappa_{\mathrm{s}}\kappa_{\mathrm{c}}]$. Three retained readings beat every present-reading controller in the specified sensor menu, despite a smaller influence of the present world on the action. We then give the same agent a damageable body: an actual mixed-port diagnosis controls repair, and the repaired body determines the next sensing channel. The finite reversible protocol maintains tracking error at most 0.15820 from a healthy start, against a long-run lower bound 0.22172 for damage-independent maintenance calendars at the same repair rate. All spent ports and old records remain; fresh blank supply is not a thermal reset. This is a mechanism for maintaining intervention capability, not a derivation of the initial body or its couplings.

These results support relational existence: structures can be objective within their coupling, history and observation window without being intrinsic to a state alone. The essay develops implications for will, self-maintenance and structural realism, then compares traces and dependence with Yogācāra and Mādhyamaka. These comparisons have their own arguments and do not establish phenomenal consciousness.

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1 Introduction

A register can remain unchanged while doing nothing. An agent can change almost every internal value while continuing to sense, act and preserve the conditions for doing so again. The distinction matters when a quantum state does not by itself select a division into things, causes and agents.

The earlier papers provide different parts of the physical account. Paper A [1] separates process-selected sectors, identification through actual responses and reconstruction of spatial geometry. Its finite correlation geometries keep their source and dictionary explicit; its continuum target requires more than relabelling those examples. Paper B [2] derives stable block content in a commuting coupling class and distinguishes pre-existing relations from relations formed by actual recording or Hamiltonian links. Block-constant couplings record central labels; arbitrary environmental records need not have that form. Paper E [3] derives measurement permissions, partly mixed records and calibrated public readout. Paper F [4] distinguishes event-order identification from record retention and updating; Paper G [5] asks how much of the selection can be performed from inside the process.

The present question is how these structures acquire causal and agentive roles. Section 2 separates observation from intervention. Section 3 bounds viable control with and without history, then closes a missing step: records act on the same body whose condition controls later sensing. Section 4 develops relational objectivity rather than identifying reality with a reader's answer. Section 5 compares the argument with Buddhist accounts of traces and dependence. The initial partition and gate permissions are stated inputs to the finite mechanism; their primitive formation remains a broader question.

2 Part I: Causation, records and interventions

2.1 Observational asymmetry and its limits

Standard quantum mechanics assigns no joint density operator to non-commuting observables at two times without a process formalism, so a cross-time mutual information is defined only relative to declared instruments.

Definition 2.1 (Instrument-relative directional asymmetry)**.** *Fix a two-time process $\mathcal{N}_{t_1\to t_2}$, an initial preparation and instruments $\mathcal{I}_A, \mathcal{I}_B$ with classical outcomes $X_A^{t_i}, X_B^{t_i}$, measured in separate forward and reverse arrangements. Put*

$$\Delta^{\mathcal{I}}_{A\to B} = I(X_A^{t_1} : X_B^{t_2}) - I(X_B^{t_1} : X_A^{t_2}),$$

with Shannon mutual informations of the outcome distributions.

$\Delta^{\mathcal{I}}$ is a well-defined observational diagnostic. Its sign depends on the instruments, the preparation and the process, and it can arise from a common cause or from asymmetric noise. Other conventions for quantum two-time objects exist [9, 10]; none turns a correlation into a cause without an account of intervention, which we now give.

2.2 Interventional influence

Definition 2.2 (Interventional influence)**.** *Fix a process in which a declared set $\mathcal{J}_A$ of interventions can be applied at A, each replacing the mechanism that ordinarily determines A while everything else is held fixed, and fix a readout at B. Let ρ_B^J be the state, or the outcome distribution, at B under the intervention J. The interventional influence of A on B is*

$$\kappa_{\mathcal{J}}(A \to B) = \sup_{J,J' \in \mathcal{J}_A} D\big(\rho_B^J, \rho_B^{J'}\big),$$

where D is the trace distance, or the total-variation distance for classical outcomes.

For several steps, with memory in the environment and controls at several times, the natural setting is a quantum comb or process tensor [11, 12]; the definition applies to any slot of such an object.

Proposition 2.3 (Properties of interventional influence)**.** *1. $0 \le \kappa \le 1$, and $\kappa = 0$ exactly when no allowed intervention changes the state at B.*

2. Post-processing the readout at B by any channel, or restricting $\mathcal{J}_A$, cannot increase κ.

3. If the interventions at A affect B only through M, in the sense that $\rho_B^J = \Lambda(\rho_M^J)$ for a channel Λ that does not depend on J, then

$$\kappa(A \to B) \le \eta(\Lambda)\, \kappa(A \to M), \qquad \eta(\Lambda) = \sup_{\rho \neq \sigma} \frac{D(\Lambda\rho, \Lambda\sigma)}{D(\rho, \sigma)}.$$

For a classical channel with kernel $\Lambda(\cdot \mid m)$, $\eta(\Lambda) = \max_{m,m'} D(\Lambda(\cdot \mid m), \Lambda(\cdot \mid m'))$, Dobrushin's coefficient [13], which is the influence of M on B under all point interventions on M.

Proof. (1) is immediate. (2) Trace and total-variation distances do not increase under channels, and a supremum over a smaller set is smaller. (3) $D(\Lambda\rho_M^J, \Lambda\rho_M^{J'}) \le \eta(\Lambda) D(\rho_M^J, \rho_M^{J'})$ by the definition of η. For classical kernels the supremum is attained on pairs of point masses, which is Dobrushin's formula. □

Part (3) is the causal counterpart of the data-processing inequality: influence passed along a chain is at most the product of the influences of its links. Section 3 uses it to show that an agent needs both sensing and control.

2.3 Records do not reveal causes

Proposition 2.4 (Same observations, different causes)**.** *Consider two processes on bits A, B observed in the computational basis at times $t_1 < t_2$.*

1. Direct cause: A is a uniform bit at t_1, $B = 0$ at t_1, and a controlled NOT from A to B acts between t_1 and t_2.

2. Common cause: a hidden uniform bit C is copied into A before t_1 and into B between t_1 and t_2; $B = 0$ at t_1.

The joint distributions of $(X_A^{t_1}, X_B^{t_1}, X_A^{t_2}, X_B^{t_2})$ *coincide, both equal to* $\frac{1}{2}$ *on* $(0,0,0,0)$ *and on* $(1,0,1,1)$*, and both give* $\Delta^{\mathcal{I}}_{A\to B} = \ln 2$*. For the interventions "set* A *to* 0*" and "set* A *to* 1*" at* t_1*, the influence on* B *at* t_2 *is* 1 *in the first process and* 0 *in the second.*

Proof. In both processes A keeps its value, B starts at 0 and ends equal to the uniform bit carried by A. Under the intervention, B at t_2 equals the set value in the first process and the hidden bit in the second. □

Every statistic of the declared observed variables is the same in the two processes, so accumulating more of those records cannot settle which structure produced them. For quantum variables the situation is partly different: some causal structures can be inferred from observations alone when entanglement or coherence is available [14]. That is an advantage of particular quantum protocols, not a general replacement for interventions.

2.4 Records and interventions in one process

A record carries a recoverable difference from an earlier process. An intervention changes a declared mechanism or operation. They can occur within the same interaction without being the same role. Paper E shows that a memory need not be pure to record: the relevant budget is its purity deficit $\ln d - S(\eta)$, and partly mixed pointer ports have nonzero readable information. Records of a commuting pointer label are a specific acquisition model, not a classification of every record.

For an already detached port, a later trace-preserving operation on the source leaves its marginal unchanged, by no-signalling. A renewed interaction with the port, or a direct operation on it, can change its state. Reversible relabelling, repair, transfer into another memory and reset have different resource balances. Paper F treats them through the joint state; they cannot all be priced as erasure of a marginal entropy. The Imprint Chain [6] makes a complementary distinction: a passive record and a feedback can carry the same information about an interface while only the feedback changes the world state.

In Proposition 2.4, the recorded correlation between A and B does not make A a cause of B. Conversely, an influence can change a state without a record being retained. Memory consists here of physical records that remain available to later decisions. Their availability may itself depend on repair, storage and permission.

The influence comparisons here already have a time order. Paper F also asks whether an unknown contact order can be recovered from endpoint relations. Ordinary pointer-copying gates fail that test even with complete endpoint access; its retained-label record–swap protocol passes within a known one-pass architecture. That theorem does not make the observational process in Proposition 2.4 causally identifiable. The permitted contacts, interventions and preparation are different. Recontact can restore coherence and reduce local entropy; neither recovery of a permutation nor record growth produces an external time parameter.

3 Part II: Historical control and maintained capability

3.1 What a boundary criterion must allow

A tempting boundary criterion asks every internal observable to stay nearly unchanged: $\sup_t \sup_{\|O\|\le 1} \|\mathcal{E}_t^*(O) - O\| \le \epsilon$. It admits an isolated frozen register and excludes a changing tracker. In the example below, the internal $Z_r = (-1)^r$ has deviation $1+(1-2\nu)(1-2q) = 1.768$, although the tracker succeeds. What must persist is a capability and the physical organization that supports it, not every stored value.

A stable algebra of the kind derived in Paper B can hold part of that organization. It does not by itself provide the sensor, actuator or maintenance procedure. We first specify the task and then construct the maintenance mechanism.

Definition 3.1 (Viable agent on a window)**.** *Fix a declared partition into a world variable W, an agent register R and an actuated variable b, declared interventions on W and on R, a viability condition $v(W, b)$ and a window. The register is a viable agent on the window, with margin $\eta < \frac{1}{2}$, if*

1. *the viability condition fails with probability at most η at every step of the window;*
2. *the sensing influence $\kappa_{\mathrm{s}} = \kappa(W \to R)$ is positive;*
3. *the control influence $\kappa_{\mathrm{c}} = \kappa(R \to b)$ is positive.*

The agent is memoryless if its action at a step depends on the world only through its reading at that step.

3.2 Sensing and control multiply

Consider a world bit W_t that flips with probability $q < \frac{1}{2}$ at each step. The agent reads it into R, acts on b, and is viable when $b_{t+1} = W_{t+1}$.

Proposition 3.2 (Viability needs sensing and control)**.** *Let a uniform world bit be read through a fixed sensor channel with independent fresh randomness. The actuator receives the world only through that reading, uses independent randomness, and the next world flip is independent of both. Thus the sensor's ordinary conditional output law equals its output under interventions setting W_t, with no initial world–controller common cause or retained world-correlated side input. For this memoryless menu, with κ_{s} computed for interventions setting W_t to 0 or 1 and κ_{c} for all point interventions on R,*

$$\Pr(b_{t+1} \neq W_{t+1}) \;\geq\; \tfrac{1}{2}\big(1 - (1 - 2q)\,\kappa_{\mathrm{s}}\kappa_{\mathrm{c}}\big). \tag{1}$$

For a sensor that errs with probability ν and an actuator that errs with probability μ, both symmetric, equality holds and the misalignment is $\frac{1}{2}(1 - (1 - 2\nu)(1 - 2\mu)(1 - 2q))$.

Proof. Let K be the channel from W_t to b_{t+1}. With $W_{t+1} = W_t \oplus f$, $\Pr(f = 1) = q$, the success probability is $q + \frac{1}{2}(1 - 2q)(K(0|0) + K(1|1))$, and $K(0|0) + K(1|1) \leq 1 + D(K(\cdot|0), K(\cdot|1)) = 1 + \kappa(W_t \to b_{t+1})$. By Proposition 2.3(3), $\kappa(W_t \to b_{t+1}) \leq \kappa_{\mathrm{s}}\kappa_{\mathrm{c}}$. For symmetric channels $\kappa_{\mathrm{s}} = 1 - 2\nu$, $\kappa_{\mathrm{c}} = 1 - 2\mu$, and the misalignment is the probability that an odd number of the three independent events sensor error, actuator error and flip occurs. □

Corollary 3.3 (Records without control, control without records)**.** *In this fresh-sensor task, a passive recorder with $\kappa_{\mathrm{c}} = 0$, an open-loop actuator with $\kappa_{\mathrm{s}} = 0$, and an initially uncorrelated frozen register fail with probability $\frac{1}{2}$.*

The channel condition is necessary. If a hidden fair bit is copied into W_t and a frozen R before the slot, then $b_{t+1} = R$ has ordinary error q, although intervening on current W_t does not change R: $\kappa_{\mathrm{s}} = 0$. The common-cause record is useful without a new sensing influence. It is not a fresh sensor channel from the intervened world. Treating observed $\Pr(R \mid W_t)$ as that channel would erase the distinction proved in Proposition 2.4. Retained world-correlated inputs must instead be included as memory, as in the next subsection.

The magnitudes $\kappa_{\mathrm{s}}, \kappa_{\mathrm{c}}$ alone do not certify the labels needed for the directional task: a reversed sensor has the same influence as a correctly oriented one. T-DOME III shows that acting cannot reveal parameters moved by a symmetry of the complete record law. Certifying this directional task therefore needs a symmetry-breaking labelled operation or retained standard [31]. It may be physically implemented inside the agent; the requirement concerns the information it supplies, not its location outside the loop.

With $\nu = 0.1$, $\mu = 0.05$ and $q = 0.02$ the memoryless agent fails with probability 0.1544, while its register changes value in one step with probability 0.193. It is viable in Definition 3.1 and fails the pointwise-invariance criterion; an initially uncorrelated frozen register passes that criterion and has no sensing route.

3.3 Memory changes what an agent can do

Let the agent instead act on the majority of its last three readings. Exact enumeration gives a misalignment of 0.1082. The influence of the present world state on the action falls at the same time: an intervention on W_t changes only the newest reading, which decides the majority only when the two older readings disagree, so the influence is $(1-2\nu)(1-2\mu)\Pr(\text{older readings disagree}) = 0.139$, against 0.72 for the memoryless agent.

Corollary 3.4 (Memory is necessary for this level of viability)**.** *In the specified drifting-bit menu with one fresh sensor reading per slot, every memoryless agent fails with probability at least* $\frac{1}{2}(1-(1-2q)(1-2\nu))$*, which is* 0.116 *for* $\nu = 0.1$*,* $q = 0.02$*. The majority-of-three agent fails with probability* 0.1082 *even with an imperfect actuator,* $\mu = 0.05$*. Within this menu, a failure probability below* 0.116 *therefore requires dependence on past readings.*

Proof. A memoryless agent's register depends on W_t only through the reading, so $\kappa_{\mathrm{s}} \le 1-2\nu$ by Proposition 2.3(2); Proposition 3.2 with $\kappa_{\mathrm{c}} \le 1$ gives the bound; the majority value follows from enumerating the 2^8 values of the initial world bit, three flips, three reading errors and actuator error. The exact value is 0.10822784; Appendix A gives the verification method. □

The memory agent tracks the world better while the present world influences it less. Its success rests on records of past world states, which are correlated with the present one because the world persists; an intervention on the present world cannot reach those records. Agency therefore involves two different relations to the world: causal influence through current sensing and control, and correlation carried by records. Proposition 3.2 bounds what the first can achieve alone. This is the task-level form of the memory resource studied in T-DOME I [8], and it is the step at which the self-reading of Imprint W [7] and the internal selection rules of Paper G become relevant: an agent that also keeps a model of its own sensor and actuator, and revises it from its records, can choose which readings to take.

3.4 Records maintain the body that reads next

The tracker so far has a supplied sensor. Give it a body bit D_t: 0 is functional and 1 damaged. The same physical body remains throughout the run; its condition modulates the world-sensing coupling, and it can be repaired before sensing. Damage is independent of the world, arrives after each task slot with probability $0 < d < 1$, and is absorbing until repair. This is a specified failure mechanism, not a biological model.

Each slot begins with diagnosis. A fresh port in $\eta_{\mathrm{d}} = [I + (1-2e)Z]/2$, $0 < e < 1/2$, interacts by controlled X with D_t. Its Z record is $Y_t = D_t \oplus N_t$, $\Pr(N_t = 1) = e$, from the mixed-port instrument of Paper E. Exactly when $Y_t = 1$, the controller swaps a fresh $|0\rangle$ into D_t; the old body state goes into that port and remains in the spent bank. The post-repair body is $D'_t = D_t(1-Y_t)$. The decision uses Y_t, not hidden D_t.

The world sensor then gives

$$R_t = W_t \oplus N_t^{\mathrm{s}} \oplus D'_t Q_t, \qquad \Pr(N_t^{\mathrm{s}} = 1) = \nu, \quad \Pr(Q_t = 1) = \tfrac{1}{2}. \tag{2}$$

For a healthy body it is the original sensor; for a damaged body its reading is fair. Two gates realize this law: controlled X from W_t to a port in $[I+(1-2\nu)Z]/2$, followed by a flip controlled by D'_t and a fresh fair bit Q_t. The spent Q_t is retained but outside the controller's readout menu. Joint access to it would change the sensing permission.

The reading drives the same actuated variable b. A fresh blank is swapped into b, retaining its old value; copying R_t and adding independent actuator error gives $b_{t+1} = R_t \oplus N_t^{\mathrm{a}}$, $\Pr(N_t^{\mathrm{a}} = 1) = \mu$. The world then flips with probability q, and finally the body suffers damage: $D_{t+1} = D'_t \vee J_t$, $\Pr(J_t = 1) = d$. Fresh ports are independent of the retained body, world and history. Diagnostic orientation and error are known or separately calibrated; a common unknown sign is not eliminated by a self-certifying reference.

Proposition 3.5 (Maintenance of a sensing route)**.** *Let* $x_t = \Pr(D_t = 1)$*,* $z_t = \Pr(D'_t = 1)$*, and* $r_t = \Pr(Y_t = 1)$*. Then*

$$z_t = ex_t, \qquad r_t = e + (1-2e)x_t, \qquad x_{t+1} = d + (1-d)ex_t, \tag{3}$$

$$x_* = \frac{d}{1-(1-d)e}, \qquad z_* = ex_*, \qquad r_* = e + (1-2e)x_*. \tag{4}$$

With $\rho = (1-d)e$*,* $x_t = x_* + \rho^t(x_0 - x_*)$*. For a uniform world independent of the initial body, the raw sensing influence and tracking failure are*

$$\kappa_{\mathrm{s},t} = (1-z_t)(1-2\nu), \tag{5}$$

$$\mathcal{R}_t = \mathcal{R}_0 + (\tfrac{1}{2} - \mathcal{R}_0)z_t, \qquad \mathcal{R}_0 = \tfrac{1}{2}[1-(1-2q)(1-2\nu)(1-2\mu)]. \tag{6}$$

A healthy start gives $z_t \le z_*$ *and* $\mathcal{R}_t \le \mathcal{R}_0 + (\frac{1}{2} - \mathcal{R}_0)z_*$ *at every slot. Without repair, sensing influence tends to zero and tracking failure tends to* $1/2$*.*

Proof. Only a damaged body missed by diagnosis remains damaged, with probability ex_t. The diagnostic is positive on a healthy body with probability e and on a damaged body with probability $1-e$. Independent damage gives $x_{t+1} = z_t + d(1-z_t)$. Solving the recurrence gives the fixed point and convergence. Repair and damage do not depend on the world, so the body remains independent of W_t. The sensor mixes the original channel, with weight $1-z_t$, and a fair channel, with weight z_t, giving its influence and risk. Without repair, $x_t = 1-(1-x_0)(1-d)^t$. □

This is intervention rather than relabelling. Replacing diagnosis by *never repair* or *always repair*, with the pre-slot distribution fixed, changes post-repair damage from x_t to zero and risk by $(\frac{1}{2} - \mathcal{R}_0)x_t$. Setting the body to healthy or damaged after repair changes the available sensing channel. The record maintains a physical condition of later intervention, not merely confidence in a previous answer.

Proposition 3.6 (A matched calendar comparison)**.** *Consider repair calendars independent of damage and of all information about it, including diagnostic or world-sensor histories. They may use private randomness and have positive long-run repair frequency* r*, with empirical mean gap* $1/r$ *and vanishing end effects in the occupancy average. In the same present-reading control menu, their long-run mean post-repair damage is at least*

$$z_{\mathrm{cal}}(r) = r[(1-\omega)G(L) + \omega G(L+1)], \quad L = \lfloor r^{-1} \rfloor, \quad \omega = r^{-1} - L, \tag{7}$$

where

$$G(\ell) = \sum_{j=0}^{\ell-1}[1-(1-d)^j] = \ell - \frac{1-(1-d)^\ell}{d}.$$

Gaps $L, L+1$ *in these proportions attain the bound. Frequency at most* r *gives the same lower bound.*

Proof. A gap of length ℓ, starting with repair, has expected damaged occupancy $1-(1-d)^j$ at age j, with sum $G(\ell)$. Conditioning on an independent calendar preserves the damage law. The differences $G(\ell+1) - G(\ell) = 1-(1-d)^\ell$ increase. The convex envelope on the integers minimizes average gap occupancy at fixed mean gap $1/r$ by balancing adjacent lengths. Multiplication by gaps per slot gives (7). Adding repairs cannot increase occupancy, so the allowance is optimally used in full. □

For $d = e = 0.1$ and $q = 0.02, \nu = 0.1, \mu = 0.05$,

$$x_* = \frac{10}{91}, \quad z_* = \frac{1}{91}, \quad r_* = 0.18791209, \quad \mathcal{R}_* = 0.15819780.$$

The optimal independent calendar at this repair frequency has $z_{\mathrm{cal}} = 0.19479225$ and mean error 0.22172020. Both receive the same diagnostic and world-reading ports and the same repair and actuator menu. One unit is charged per actual repair, with other per-slot inputs equal. A controller inferring damage from sensor history is another record-guided strategy, outside the calendar class. Equal long-run rate does not mean equal repair counts on every finite trajectory.

3.5 Finite stock and a maintained boundary

For n slots, the realization has three continuing bits D, W, b and nine fresh ports per slot: diagnostic, repair blank, world sensor, fair sensor-noise, actuator-noise, world-flip, damage coin, damage-dilation blank and actuator blank. Maximum repair count is n; a healthy start gives expected count at most nr_*. Supplying the bank, storing spent ports and routing records belong to the finite-window budget. Mean activation count is not stock capacity.

Each of the three blank ports per slot supplies initial purity deficit $\ln 2$. Diagnostic and sensor ports supply $\ln 2 - h(e)$ and $\ln 2 - h(\nu)$; actuator, world-flip and damage coins supply $\ln 2 - h(\mu), \ln 2 - h(q), \ln 2 - h(d)$; fair noise supplies none. Here h is binary entropy in nats. Unused repair blanks may remain pure; supplied purity is not necessarily exhausted on every slot. Old states and correlations remain; the reversible circuit in Appendix A conserves joint entropy. No bath is reset and no work or heat price follows from counting swaps. For the numerical parameters, the nine-port input purity is $8\ln 2 - 3h(0.1) - h(0.05) - h(0.02) = 4.27337417$ nats per slot. For degenerate port Hamiltonians, $k_B T$ times this quantity is the free-energy difference from a maximally mixed preparation at temperature T, not heat dissipated or net work consumed by the run. The initialized healthy body and blank actuator in the explicit check add two initial pure bits; the world starts fair. Gate strengths, preparation, duration, clock, storage and later bank renewal remain physical inputs.

The body's condition regulates whether a reading is informative, and a record of that condition regulates the operation preserving the route. Organization survives while its values change. Continued intervention capability is produced by record-conditioned action, rather than merely imposed as a viability condition. Repair uses the current diagnosis only; earlier diagnoses remain stored but are not read by this policy. The majority tracker establishes the value of longer history, whereas this circuit establishes the maintenance of later permissions. Neither result alone constructs a long-term self-model.

Remark 3.7 (Maintained organization)**.** *A broader agency hypothesis is that an effective agent boundary is the organization of sensing and action routes whose continued availability is causally regulated by retained history. Disrupting use of a bodily diagnostic should impair later intervention capability even when current discrimination is spared. At a matched budget, an independent calendar should fail where a condition-dependent policy succeeds.*

The binary body establishes one instance. Formation must also explain how routes and the repair menu arise from earlier interactions and how histories coordinate several demands. The hypothesis explains no additional boundary if a fixed external controller accounts for every apparent self-maintenance effect, or if manipulating history never changes later permissions. BRAIN supplies learned protection and routing for a different body model [32]; Imprint W and Imprint SH price reliability-based action [7, 33]. Their dynamics and budgets must match before they form a larger construction; they are not assumed to be this circuit.

3.6 Which aspects of a body are being formed?

The models connected to this account change different physical relations. In the present circuit, D'_t gates the next sensor channel and Y_t controls actual replacement of the damaged body state. The repair menu and its known diagnostic orientation are supplied. T-DOME III instead asks what a retained model must know to choose a priced repair: histories with the same sign

confidence can give different values of future access, because gain, physical body state and their correlations differ [31]. BRAIN learns relations among several ports and uses their connected components to route and maintain access [32]. These are distinct mechanisms, not repeated observations of one neural system.

There is a common intervention question. Hold current world discrimination and available repair operations fixed, then change whether the policy can use its own diagnostic history. A difference in later sensing capability tests the path from history through maintenance to future access. In this paper the matched calendar comparison isolates that path within a declared repair budget. A loss of performance alone does not identify it: destroying a port, changing the world or reducing the available repair stock can produce the same loss. The comparisons therefore match those inputs and use the intermediate body state and sensor law as separate readouts.

A formation claim requires a further change: history must select some relation used by later sensing or control, rather than merely operate a preselected repair schedule. The learned route in BRAIN and the retained foreground of T-DOME address that step in their own task classes. Combining them with this circuit would require a common transition law, diagnostic permission and resource ledger. Their conceptual fit is a reason to seek that construction, not evidence that it has already been made.

3.7 Will, degrees of agency and free will

On this account, "will" names the control route from an agent's records to the world: the influence κ_c, together with the dependence of the action on the agent's records. A thermostat has a sensing route, a control route and a viability condition, and it is a viable agent in the weak sense of Definition 3.1. Organisms differ from it in degree along several axes: the length of the window over which viability is maintained, the margin, the strength of sensing and control, the depth of memory, and whether the agent can revise its own selection rules. Agency is graded along these axes and is not an all-or-nothing property.

The traditional free-will question presupposes well-defined agents and causal histories. The present account supplies both relative to a declared partition and intervention set, and it turns part of the question into questions that can be answered: how strongly the agent's records, rather than the present state of the world, determine what it does, and whether its selection rules are themselves revised from its records. It does not settle the metaphysical question and is not offered as doing so.

The account says nothing about what it is like to act. The explanatory gap between functional organization and experience [15, 16] remains. One empirical route connects these notions to neuroscience, where habitual and flexible control have been associated with distinct cortico-basal ganglia loops [17, 18] and the sense of agency with motor prediction [19]; the relevant times there are classical control times, not quantum coherence times [20]. Whether the quantities defined here are useful for that research is open.

4 Part III: Existence without substance

4.1 Relational existence

Parts I and II treat causes and agents as real within declared contexts and dependent on them. This raises the question of their ontological status. Two extremes are available and both are mistaken: treating context-dependent structures as fundamental, and denying them objective reality because they depend on context. We propose structural realism about relational patterns [21, 22].

Definition 4.1 (Relational existence)**.** *An entity X exists relative to a context $\mathcal{S}$ if X is well defined and stable within $\mathcal{S}$, participates in objective relational patterns there, and may be absent or differently constituted in another context $\mathcal{S}'$.*

Definition 4.2 (Emptiness, technical sense)**.** *An entity is empty in the technical sense if it lacks being independent of every context while exhibiting stable patterns within specified contexts. Emptiness in this sense entails neither non-existence nor illusion.*

Paper B supplies physical instances of a context. In its commuting-coupling class, the stated stability and observability criteria select a unique largest admissible block algebra on a window. Smaller admissible algebras also exist. Selected content depends on coupling and window, yet the same inputs define the same criterion for all readers. Readable redundant labels additionally require actual recording couplings, fragment preparation and calibrated readout, as in Paper E; stability alone does not make them public.

The maintained body adds history and budget to the context. The sensor's operational distinction survives under record-guided repair and disappears under unmaintained damage. That difference is real in the task outcomes even for a reader unaware of the controller's policy. Relational objectivity concerns these reproducible consequences, not agreement on a chosen label.

4.2 Examples

Temperature depends on which degrees of freedom are treated as macroscopic, yet given that choice it is measurable and predictive. Quasiparticles such as phonons and magnons are collective excitations with no counterpart among the microscopic constituents; operationally they are stable peaks in spectral functions, dispersion relations, lifetimes and scattering amplitudes, all of which depend on temperature, doping and resolution. Subsystems are the case closest to this framework. A state alone does not privilege one of its many factorizations, and Paper A shows that entropies, mutual informations and reconstructed geometries depend on the channel through which the state is read; Paper B shows how couplings select one structure over a window.

4.3 What is invariant

Objectivity does not require independence from every context. It requires that patterns be stable and predictive within a context and that the transformations between contexts be stated. What stays invariant depends on that class of transformations. Under a change of factorization, implemented by a unitary relabelling of the Hilbert space, the spectrum of the global state is invariant, while subsystem entropies, mutual informations and the geometries built from them are not. For readers applying the same selection criterion with couplings and window fixed, the selected algebra is unchanged. Access to its labels requires the declared record protocol; readers with different measurement permissions need not obtain the same information. A mere unitary change of coordinates is also different from physically changing a coupling.

5 Part IV: Interpretive bridges to Buddhist philosophy

5.1 Method

The comparison below is between formal patterns. It does not claim historical influence in either direction, endorsement of Buddhist doctrine, or that physics confirms any religious view; "Buddhism" is not treated as one doctrine, and the discussion is limited to some Yogācāra and Mādhyamaka texts. One limitation must be stated plainly. The framework was named after the ālaya-vijñāna, so a resemblance between the two was partly built in by the naming and cannot count as independent evidence for either. The mathematical results of Parts I–III do not depend on the comparison; the comparison depends on them.

Kind	Characterization	Example
Intrinsic existence	independent of every context	classical particles, if fundamental
Relational existence	dependent on context, objective within it	temperature, selected subsystems, records
Conventional existence	dependent on context and on agreement among agents	money, legal rights
Non-existence	absent from every relevant context	phlogiston, luminiferous ether

Table 1: Kinds of existence claim. The causal influences, agents and geometric features of this series belong to the second kind.

5.2 Yogācāra: traces that condition later appearance

Yogācāra texts describe the ālaya-vijñāna, the store consciousness, as the stream in which the impressions (*vāsanā*) of past actions are kept as seeds (*bīja*) that condition later experience [23, 24, 26, 27]. Vasubandhu's *Triṃśikā* says that it proceeds like a torrent; it is a changing stream, not a fixed container. The following correspondences are formal analogies.

Yogācāra notion	Counterpart in this series (analogy only)
Ālaya-vijñāna, a stream that keeps traces	the global state together with the records its history has formed
Bīja and vāsanā, seeds and impressions	records that change what later readouts can distinguish (Papers E, F; the Imprint papers)
Pravṛtti-vijñāna, the active consciousnesses	selected readouts and the effective subsystems they define (Papers A, B)
Āśraya-parāvṛtti, transformation of the basis	a change of couplings that changes which structure is selected, as opposed to a change of description (Paper B)

Table 2: Formal analogies. Yogācāra concerns consciousness and liberation; HAFF concerns physical structure, and its global state carries no phenomenal properties.

The analogy concerns the role of traces. In both pictures what can appear later depends on what earlier processes have left behind, and the traces are not inert stores but conditions on later access. The differences are as important. The Yogācāra substrate is described as consciousness; the HAFF state is physical and the framework is neutral on mind and matter. The analogy does not establish panpsychism, idealism or the claim that quantum information is conscious.

5.3 Mādhyamaka: dependence without a foundation

For Nāgārjuna, whatever arises dependently is empty of intrinsic nature (*svabhāva*), and that emptiness is itself a dependent designation, the middle way (*Mūlamadhyamakakārikā* 24:18) [25, 28, 29]. The technical emptiness of Definition 4.2 has the same form: denial of intrinsic being together with affirmation of functional reality.

Dependence also applies to the apparatus of selection. Selected content depends on couplings and observation windows (Paper B). Paper G constructs choices inside a fixed executor and separates that execution closure from an explanation of the executor's own organization. It does not prove that every possible model has an unrevisable last selector. The supplied dynamics of a finite construction is an explanatory boundary of that construction, not an independently existing ground. A larger model may explain some of it. Reading this revisability as dependence without an ultimate base is a philosophical proposal, not a consequence of a finite-register count.

The comparison with Mādhyamaka concerns this form of dependence; the tradition's use of it to loosen attachment to views has a different practical purpose.

5.4 Karma: actions, traces and later access

Buddhist accounts of karma hold that intentional actions leave traces that condition future experience [30, 34]. In the stated fresh-port process, an intervention on the source changes later records while detached earlier ports retain their marginals (Paper F). Records then condition later reading and action. In Section 3.4, they condition whether later sensing remains informative. The analogy is formal and stops at normativity. Karma classifies actions as wholesome or unwholesome; interventional influence and records carry no ethical valuation.

5.5 Where the traditions and the framework differ

The differences are substantive. Buddhist philosophy starts from first-person experience and aims at liberation from suffering; the physical results used here begin with third-person structure. Rebirth, karmic continuity across lives, cosmological realms and meditative attainment have no counterpart here, and the comparison neither supports nor undermines them. Yogācāra readings that give consciousness priority differ from the framework's neutrality. And the methods differ: contemplative inquiry, textual interpretation and dialectic on one side, mathematics and experiment on the other. The comparison is worth making because both traditions and the framework deny intrinsic subsystem structure while affirming stable relational patterns, and because precise statements on one side can sharpen questions on the other.

6 Conclusion

Causal influence is a property of a declared intervention process, not of correlation alone. It contracts through links and distinguishes processes whose observed records coincide. In the tracking task, sensing and control jointly limit memoryless viability; history can surpass that limit without raising the influence of the current world on the action.

The same finite task now includes a damageable body. An actual diagnostic controls repair of that body, whose condition determines the next sensor channel. The recurrence and matched calendar comparison show how a record maintains later intervention capability. Fresh blanks and spent history are explicit resources; this is a finite-stock mechanism, not a perpetual thermal cycle or the primitive origin of the initial body.

Agency so understood is a maintained relational organization. Partition, loss and permissions specify the task; success and failure are physical consequences, not matters of agreement. A fuller formation account must generate and coordinate the routes rather than supply all of them. The Buddhist comparison clarifies a related philosophical possibility: traces and dependent structures can have functional reality without an independently existing substrate. Its experiential and ethical questions remain independent of the physical mechanism.

A Reversible realization and checks

The body protocol is an explicit basis-permutation unitary on twelve bits. Besides continuing D, W, b, use fresh $(C, B, R, Q, A, F, J, G, V)$ with initial one-probabilities

$$(e, 0, \nu, \tfrac{1}{2}, \mu, q, d, 0, 0).$$

Here C, B are the diagnostic and repair ports; R, Q the sensor and fair-noise ports; A, F, J actuator, world and damage randomness; G, V blank damage and actuator ports. On diagonal states perform, in order:

1. $C \leftarrow C \oplus D$; swap D, B conditional on $C = 1$.

2. $R \leftarrow R \oplus W$; $R \leftarrow R \oplus DQ$.

3. Swap b, V; $b \leftarrow b \oplus R \oplus A$.

4. $W \leftarrow W \oplus F$.

5. $G \leftarrow G \oplus D$; $D \leftarrow D \oplus [(1 - G)J]$.

Each gate is reversible on all basis inputs, not just the initialized subspace. Since G starts at zero, the last step stores the post-repair body in G and gives $D_{\text{next}} = D' \vee J$. Repair uses $C = Y$ alone. The physical control of noise by the body does not grant a hidden diagnostic read. All ports remain in the spent bank, and the next slot uses fresh ones. Z reading of C, R is compatible with the diagonal protocol and does not change its joint law. Realizing these gates and their known orientations is a charged hardware permission.

The check enumerates all 4096 input strings and verifies that the composite is a permutation. For diagonal product inputs with $x = \Pr(D = 1)$ equal to $0, 1/5, 10/91, 1$, exact rational probabilities reproduce every term of (3) and (6). Old b does not affect the output: it remains in V. Joint entropy is unchanged. This is a circuit distribution check for the stated classical task, not unknown quantum-source or bath certification.

The calendar check verifies discrete convexity of G and the optimal $5, 6$ gap mixture at the stated rate. Independently, enumeration of the original three-reading task gives 0.10822784. The probability that the two older readings disagree is $\frac{1}{2}[1 - (1 - 2q)(1 - 2\nu)^2] = 0.1928$, giving present-world influence 0.138816. Analytic contraction and calendar proofs, rather than numerical sampling, establish the general statements.

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Selection, Maintenance and Gravitational Response: Modular Constraints and a Physical Return

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Abstract

Can the physical selection of observable algebras account for gravitational dynamics, rather than merely change what an observer can reconstruct? We formulate this question as an adiabatic evolution of selected localization structures and a common-action consistency requirement. Modular stability alone does not select the algebra. A supplied half-sided modular family and holographic dictionary give a conditional linearized gravity route. In the same-spacetime conformal scalar model of Consciousness II, the improved stress and its Wald surface coefficient connect accessible preparations to an all-ball test of the Einstein response. The test still requires modular consistency; changing a stress-containing algebra at fixed matter history does not change its first-order sourced Einstein tensor.

The source side is constructive. Calibrated correction, selected projections and learned protection change stress through actual interactions. In a specified Jackiw–Teitelboim theory, an acquired quantum record drives a material force and radiation, whose response changes later operations. The finite mean-action return retains old references, clock energy, support work and recoil. Separate spatial and full-mode bulk constructions extend the source account without identifying their distinct models.

Maintenance supplies a possible selection pressure. For complete entropy removal at rate $\dot{s}$ through a thermal chiral exhaust of fixed capacity c_{ex}, the heat rate obeys $\dot{Q} \geq k_B T \dot{s} + 3\hbar \dot{s}^2/(\pi c_{\mathrm{ex}})$. Pointer alignment lowers the complete renewal debit, and matched horizon transport reads its price. Sustained renewal and joint sharp exhaust with persistent causal return remain open. The proposed gravitational mechanism must connect formed access, a conserved total source and geometric consistency in one dynamics; neither a visibility filter nor a finite response in an assigned gravity theory supplies that connection.

Technical appendices and their original numbering are retained in HAFF-D technical supplement (`HAFF_D_supplement.pdf`). The certificate guide at the end of this paper identifies their claims.

1 The problem: what makes selection gravitational?

An apparatus changes which distinctions it can preserve and which operations it can execute. This can reorganize a reconstructed geometry. It can also change matter through

the selecting interaction. A theory of gravity based on observable-algebra selection must connect these two effects: the localization structure selected by physical histories must constrain a geometric law, and the same histories must supply the source and respond to it.

The central proposal is that gravitational dynamics describes a slow evolution of the selected geometric algebra, not just motion of matter within a fixed localization structure. This is a mechanism conjecture, developed in Section 4, rather than an identification of every algebra change with curvature. Its strongest obstruction is already visible in a conformal model: two selected algebras that retain the same stress readings can give different relative entropies while fixing the same matter-sourced Einstein tensor. A physical change of localization must therefore be distinguished from a change of description.

HAFF-A, B and E relate channels, stable content and actual records [19, 20]. Foundation I constructs a stable hull relative to a readable seed [54]. These results make parts of algebra selection precise; they do not determine a spacetime metric or a universal gravitational dynamics. Here we ask what further relation would make that selection gravitational.

Sections 2–5 state the algebraic question, its central conjecture and the modular constraints. The holographic route needs an explicit region family and dictionary. Section 6 gives a different conditional bridge in the same spacetime: one conformal scalar source, its surface charge, and its all-ball geometric test. That bridge identifies which law is compatible with the supplied consistency condition, while leaving the physical origin of that condition to the common-action conjecture.

Sections 8 and 9 then test the source and return edges through actual interactions in a specified two-dimensional JT theory. The decisive finite mechanism is an acquired record that changes a material source, not a command label attached to a separate emitter. Section 10 prices the full maintained state. Section 11.2 asks whether the source, renewal and informative return can persist together. Long controller, clock, support and source proofs are in the appendices.

Table 1 separates these constructions. Its rows are not pieces of one already realized apparatus. In particular, a supplied JT response cannot establish the higher-dimensional consistency law, and a modular constraint cannot manufacture a controller.

2 The Factorization Problem

2.1 No canonical tensor factorization

A Hilbert space of dimension $d = d_1 d_2$ admits a tensor factorization, but its Hilbert structure alone does not select one. Commuting observable algebras can induce the subsystem structure [17, 18]. Which of those algebras an apparatus can use is a physical question about its couplings and resources.

2.2 Coarse-graining and effective descriptions

Paper A [19] gives the following conditional separation.

Proposition 2.1 (Conditional coarse-graining separation)**.** *Let $\Phi_1 \neq \Phi_2$ be finite-dimensional coarse-graining channels with a common output space. Their outputs differ for generic pure inputs. Entropy profiles differ only under the additional nondegeneracy*

Object and supplied inputs	Result	Remaining interface
Modular family, dictionary and ball laws	Conditional translations and linearized geometric consistency (Sections 2–5)	Formation of the candidate family and its geometric dictionary.
Conformal scalar, source reference, collar prescription and MC	One prepared source, matched Wald coefficient and all-ball Einstein test (Section 6)	Physical origin of MC; apparatus source and finite-resource readings.
Calibrated field operations and a JT action	Derived outgoing source, retarded response and finite echo/contact readout (Section 8; Appendix G)	Total apparatus stress and physical references; the action is an input.
Quantum core with classical mechanics and mean-action source	Finite retained-history return and strong-domain perturbation bounds (Section 9; Foundation III)	Fixed-window mean-action result; protected content, old references and initial work remain charged. No quantum stress tensor for the scalar fields or support is supplied.
Spatial quadratic fields and real-quartet bulk	Work-corrected full-mode stress matching, a finite spatial return, and a separately constrained quartet interior	Joining their receivers, moving endpoints, preparation and thermal ports in one material model.
Complete retained entropy and paid finite carrier	Long-window heat floor and neutral-sector sorting of a resettable source (Section 10; Proposition 11.2)	Actual adaptive source concentration, protected-content compatibility and local delayed readout.
One reused body over growing runs	Conjecture C.1 jointly requires complete-output sharpness, normalized window contrast and causal transition gain	Sustained inlet, support, sensing and resource account in that same dynamics; none is supplied by sorting alone.

Table 1: Objects, results and interfaces. Finite constructions retain their stated inputs; the joint growing-window target is a conjecture.

hypothesis stated in Paper A, and metric inequivalence follows only when a supplied reconstruction dictionary separates those profiles modulo diffeomorphisms.

Channel inequivalence therefore establishes a difference of effective descriptions, not by itself a difference of spacetime metrics. Paper A also decides, for qubit reductions, exactly when two channels give the same output entropy on every pure input, and it constructs one global state, one family of local noise channels and one fixed reconstruction rule that yield non-isometric finite metrics. Metric inequivalence of spacetime geometries remains a conditional model claim throughout this paper.

2.3 Geometry from entanglement

In a holographic model the Ryu–Takayanagi formula [27] relates an extremal area to boundary entropy,

$$S_A = \frac{\mathrm{Area}(\gamma_A)}{4G_N\hbar}, \tag{1}$$

and motivates an entanglement account of connectivity [28]. Such a reconstruction requires its dictionary. Outside a specified model, the accessible data determine operator relations rather than a spacetime metric.

3 A Structural Distinction: Gauge Forces and Gravity

Gauge interactions evolve matter and connection fields within a supplied localization structure. The algebraic proposal asks whether gravity instead describes how physical correlations, accessible operations and their stability select that structure. With a geometric dictionary, changing those relations can change effective distances and causal structure. The source constructions below test a necessary interface: changed access reaches geometry through a specified matter process.

If the proposal holds, quantum gravity must describe selection of localization as well as fields within it. This is not an explanation of why quantization fails. Background independence [29, 30] asks for covariance under changes of selected geometric structure, not equality of the metrics assigned to operationally different algebras. Selection is constrained by interactions and stability, rather than subjective choice [20]; changed knowledge affects stress only through an operation.

A common reconstruction dictionary gives every matter representation in one selected geometric sector the same metric. This conditional universality is not a derivation of free fall or equality of inertial and gravitational mass. Degrees outside the sector cannot be declared geometrically irrelevant while changing its metric: a dark-sector application would need an enlarged source algebra and explicit backreaction map. No dark-matter or dark-energy conclusion follows here.

4 Algebra selection as a gravitational mechanism

4.1 Setup

We separate the operator-algebraic and finite-dimensional regimes result by result. Modular statements use a Type III_1 ambient factor with a faithful normal state and explicit cyclic and separating hypotheses. Density matrices, traces, the quantum Fisher information metric and the qubit chain are finite-dimensional statements. The Rindler example is an algebraic quantum field theory check. Theorem 5.6 uses the holographic regime; Section 6 uses a distinct same-spacetime conformal model. The JT model of Section 8 is a supplied two-dimensional gravity theory.

Definition 4.1 (Selected geometric algebra)**.** *A* ***selected geometric algebra*** $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ *is a von Neumann subalgebra chosen as input to a specified algebra-to-geometry reconstruction model. It is not Paper E's probe-reachable effect system, pointer or record algebra, or environmental fixed set. A coarse-graining model must declare the channel or inclusion relating any of those objects to* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$*.*

No canonical tensor factorization is assumed. State restrictions and relative-entropy data on $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ can be associated, as in holography and tensor networks, with an effective distance structure. That structure depends on the choice of $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$, on the stability of correlations under coarse-grained dynamics and on the redundancy of the encoding; it is an effective description valid in the context the algebra defines.

4.2 Central conjecture

Conjecture 4.2 (Gravity as adiabatic algebra evolution)**.** *Gravitational dynamics and the* ***adiabatic flow*** *of selected geometric algebras* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(t)$ *arise from one physical selection process:*

1. *the global state evolves unitarily,* $|\Psi_U(t)\rangle = U(t)|\Psi_U(0)\rangle$*;*
2. *a model-dependent rule selects a geometric candidate* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(t) \subset \mathcal{M}_U$ *at each time;*
3. *as the state evolves, the selected candidate may shift,* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(t) \to \mathcal{A}_{\mathrm{geo},\mathbf{c}}(t+dt)$*;*
4. *the interaction that changes the selected algebra also determines the complete physical source and the compatible localization data. A change of algebra alone, with the matter history and source readings held fixed, supplies no additional first-order sourced Einstein tensor in the common-source regime below. A changed sourced response must be carried by the changed total source under that regime's field-law conditions.*

The proposed distinction from ordinary matter feedback is that the same dynamics selects the localization structure in which the source and response are represented, rather than merely evolving matter on a permanently supplied stage. Evolution within a fixed selected localization can describe gauge dynamics, but neither unitarity nor a moving algebra by itself identifies an interaction as gauge or gravitational. The selection claim must be established through the common-action conditions below; finite sourced return alone does not establish it.

The proposed flow needs a physical selector. A readable history changes which controls and correlations remain stable; those controls determine a selected hull and the matter interactions that sustain it. The changing hull must then admit compatible geometric region data, rather than merely receive an arbitrary distance assignment. In an adiabatic sector, the relaxation time of the selected structure must be short compared with its change time. This separation is a condition to establish in a model, not a consequence of writing $\mathcal{A}(t)$.

Two explanations compete here. In the reconstruction explanation, the ambient geometry and complete source stay fixed, while smaller algebras resolve less of them. In the dynamical explanation, history changes an actual coupling or preparation, changes the complete source, and changes later physical readings. The latter has a finite realization below; the stronger claim that the localization law itself is selected requires Section 7. Algebra variation alone distinguishes neither explanation.

A useful test compares histories under one action and one source calibration. Erasing command access before execution should remove the record-mediated part of the source; erasing a log after execution should not. If two selected structures contain the same total stress and surface data, a first-order difference of their sourced Einstein tensors would contradict the conformal bridge in Section 6. A difference only in relative entropy supports a reconstruction claim unless a corresponding physical intervention is demonstrated. Conversely, a history-dependent source that fails the same all-ball response test does not realize the proposed gravitational flow, even if it produces a detectable JT signal. Rapid algebra transitions lie outside the adiabatic claim; whether they admit a different geometric regime is open.

4.3 Comparison with quantum information geometry

For comparison consider the quantum Fisher information metric (QFIM) on a smooth finite-dimensional family of faithful effective states $\rho(\lambda)$ [36, 37],

$$g_{\mu\nu}(\lambda) = \frac{1}{2}\mathrm{Tr}\left(\rho(\lambda)\{L_\mu, L_\nu\}\right), \qquad \partial_\mu\rho = \frac{1}{2}(\rho L_\mu + L_\mu\rho). \tag{2}$$

It is coordinate independent, reduces to the classical Fisher metric in commuting limits, and measures how hard nearby states are to distinguish. It is also positive semidefinite, whereas a Lorentzian spacetime metric is indefinite and carries causal structure. Nothing in Eq. (2) supplies the negative direction, so the QFIM is a metric on model space and an analogy, not the spacetime metric. A change $\mathcal{A}_{\mathrm{geo},\mathbf{c}} \to \mathcal{A}_{\mathrm{geo},\mathbf{c}} + \delta\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ may change this state-space metric; calling the change a spacetime perturbation requires an additional Lorentzian reconstruction map. In AdS/CFT, relative entropy and the entanglement first law constrain bulk perturbations [38, 39]; those results use the holographic dictionary and do not make an arbitrary boundary QFIM the bulk metric.

5 Algebraic Foundation: Modular-Stable Candidates

This section places one candidate class in Tomita–Takesaki modular theory. It does not translate measurement accessibility or environmental redundancy into modular conditions, and it does not select a unique algebra.

Standard modular definitions and their two checks are in Appendix E.

5.1 Modular definition of candidate algebras

Let $\mathcal{M}_U$ be a Type III_1 factor in standard form on $\mathcal{H}_U$, let $|\Psi_U\rangle$ be cyclic and separating, and write $\omega(\cdot) = \langle\Psi_U| \cdot |\Psi_U\rangle$. Then ω is faithful and normal and its ambient modular group $\sigma_t^{\omega,U}$ exists. Let $\{\mathcal{E}_t\}_{t\geq 0}$ be a normal unital completely positive (UCP) semigroup acting in the Heisenberg picture on $\mathcal{M}_U$.

Definition 5.1 (Modular-stable geometric candidate)**.** *A von Neumann subalgebra $\mathcal{A}_{\mathrm{geo},\mathbf{c}} \subset \mathcal{M}_U$ is a* ***modular-stable geometric candidate*** *if:*

(U1) ***Faithful restriction.*** *$\omega|_{\mathcal{A}_{\mathrm{geo},\mathbf{c}}}$ is faithful and normal. In the faithful ambient setting this holds for every unital subalgebra, so it is a regularity condition, not a selector.*

(U2) ***Ambient modular stability.*** *$\sigma_t^{\omega,U}(\mathcal{A}_{\mathrm{geo},\mathbf{c}}) = \mathcal{A}_{\mathrm{geo},\mathbf{c}}$ for all real t.*

(U3) ***Semigroup stability.*** *$\mathcal{E}_t(\mathcal{A}_{\mathrm{geo},\mathbf{c}}) \subseteq \mathcal{A}_{\mathrm{geo},\mathbf{c}}$ for all $t \geq 0$.*

Remark 5.2 (Typed relation to the measurement objects)**.** *Conditions (U1)–(U3) do not derive Paper E's implementable effects* Eff*, their operator system* $\mathrm{Eff}_{\mathrm{span}}$ *or the pointer algebra $\mathcal{P}_{\mathrm{env}}$, and setwise semigroup stability is not the exact fixed set* $\mathrm{Fix}(\mathcal{E})$*. A downstream model may specify a channel or inclusion from those objects into $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$; without it the relation is motivational. Environmental redundancy does not imply faithfulness of a state restriction. A causal restriction map and a Lindblad evolution are different types, and each model must supply its own $\mathcal{E}_t$ and verify invariance.*

5.2 Selection is not settled by these conditions

Remark 5.3 (No uniqueness from (U1)–(U3))**.** *Conditions (U1)–(U3), even with only scalars fixed by both dynamics, do not determine a unique candidate. There are four independent obstructions.*

1. *Uniqueness of the modular flow for a fixed pair $(\mathcal{A}, \omega)$ does not select the algebra $\mathcal{A}$.*

2. *A normal completely positive map may preserve two subalgebras separately without preserving their join, because it is not multiplicative. Join preservation holds for modular automorphisms but not for a general $\mathcal{E}_t$; it does hold for subalgebras of the fixed algebra of a state-preserving dynamics, on which $\mathcal{E}_t$ acts as the identity [54].*

3. *Pointwise ergodicity constrains fixed elements, while (U2)–(U3) are setwise conditions; no implication to a unique maximal element of the lattice was proved.*

4. *With a faithful ambient state, $\mathcal{M}_U$ itself satisfies (U1)–(U3), so a bare maximality rule selects the ambient algebra unless an independent seed, properness, read-out or minimal-hull condition is supplied.*

Replacing $\mathcal{B}(\mathcal{H}_U)$ by a faithful Type III factor repairs the existence of the ambient modular flow but none of these selection failures. The class of candidates may contain several members that are not unitarily equivalent, and classifying it requires additional structure. A nontrivial selection principle could start from a foreground seed and construct a minimal invariant hull, or impose an explicit read-out constraint. Foundation I carries out the first

route for a supplied seed: under state-preserving dynamics the hull of a seed in the fixed algebra lies between that seed and the fixed algebra. For a minimal compact symmetry, it equals the seed-stabilizer fixed algebra when that algebra is preserved by the hull generators and the hull contains the group-fixed algebra [54]. These hypotheses establish uniqueness relative to the seed; the seed's physical origin is a further question.

5.3 Connection to linearized gravity

Theorem 5.4 (Entanglement first law in a density-matrix setting)**.** *Let ρ_0 be a faithful finite-dimensional density matrix and $\rho_\varepsilon = \rho_0 + \varepsilon\,\delta\rho + O(\varepsilon^2)$ a normalized perturbation, with $K_0 = -\log\rho_0$. Then $\delta S = \delta\langle K_0\rangle$. An infinite-dimensional extension requires entropy differentiability and control of the modular-energy variation; faithfulness alone is insufficient. For a Type III local algebra the statement must instead use Araki relative entropy.*

Proof sketch. $S(\rho_\varepsilon\|\rho_0) = -S(\rho_\varepsilon) + \langle K_0\rangle_{\rho_\varepsilon}$ plus a constant is nonnegative and stationary at $\varepsilon = 0$. □

Conjecture 4.2 posits a family $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(\lambda)$. Modular theory treats two kinds of perturbation differently. For a state perturbation on a fixed algebra the Connes cocycle $u_t = (D\omega' : D\omega)_t$ relates the automorphism groups by $\sigma_t^{\omega'}(A) = u_t\sigma_t^{\omega}(A)u_t^*$ [50, 40]. This is not a left-multiplication formula for the two full modular operators. If the algebra family is generated by the positive translation of an HSMI, $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(a) = e^{iaP}\mathcal{A}_{\mathrm{geo},\mathbf{c}}(0)e^{-iaP}$, and the vacuum is fixed by this translation, its full modular generator obeys $K(a) = e^{iaP}K(0)e^{-iaP}$. On a common smooth domain,

$$\frac{dK(a)}{da}\Big|_{a=0} = i[P, K(0)]. \tag{3}$$

Equation (3) concerns an HSMI-generated family and a derivative of matrix elements on that domain, not arbitrary subalgebra changes or an operator-norm Taylor expansion. The one-sided CHM charge used for a ball below is distinct from the full modular generator; their relevant local-unitary expectation differences agree, but the operators themselves do not.

Theorem 5.5 (Linearized gravity from entanglement, [39])**.** *In a holographic CFT dual to Einstein gravity in asymptotically AdS spacetime, the entanglement first law $\delta S = \delta\langle K_{\mathrm{mod}}\rangle$ for all boundary balls is equivalent to the linearized bulk Einstein equations $G^{(1)}_{\mu\nu} + \Lambda g^{(1)}_{\mu\nu} = 8\pi G_N\, T^{(1)}_{\mu\nu}$.*

The derivation uses the ball modular Hamiltonian [47]

$$K^B_{\mathrm{mod}} = 2\pi \int_B d^{d-1}x\, \frac{R^2 - |\mathbf{x} - \mathbf{x}_0|^2}{2R}\, T_{00}(x), \tag{4}$$

together with the Ryu–Takayanagi formula and the relation between bulk and boundary modular Hamiltonians [48]; complementary arguments are due to Jacobson [49].

Theorem 5.6 (Conditional holographic matching for an HSMI-generated family)**.** *Let $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ be a modular-stable candidate in a holographic CFT vacuum representation, and assume:*

(H1) the AdS/CFT correspondence holds;

(H2) the Ryu–Takayanagi formula and its quantum corrections hold;

(H3) the JLMS relation between bulk and boundary modular Hamiltonians holds;

(H4) the compared algebra family is an explicitly supplied HSMI/ball-region family whose generator P implements the boundary-region variation used in the holographic first-law argument;

(H5) the regulated entropy variation, or its relative-entropy replacement, satisfies the hypotheses of the Faulkner et al. theorem for all required balls.

Then Eq. (3) *describes the modular-generator variation within that family, and the holographic first law matches the linearized Einstein equations by the imported result:*

$$\begin{aligned} \textit{HSMI/ball family (H4)} &\Longrightarrow \delta K_{\mathrm{mod}} \xrightarrow{(H5)} \delta S = \delta\langle K_{\mathrm{mod}}\rangle, \\ \textit{(H1)–(H3)} &\Longrightarrow \delta(G_{\mu\nu} + \Lambda g_{\mu\nu}) = 8\pi G_N T^{(1)}_{\mu\nu}. \end{aligned} \tag{5}$$

The theorem does not show that a generic stability-selected algebra variation is an HSMI translation, nor that HAFF algebra flow is gravitational dynamics.

5.4 The circularity issue

Jacobson's thermodynamic derivation [34] presupposes a background geometry, while HAFF wants geometry to emerge from algebra. Three strategies are discussed in the literature. (A) In the algebraic programme, wedges are defined through modular inclusions without a background metric, and Theorem E.5 recovers a positive translation from an HSMI; constructing a sufficient net and showing that the candidates of this paper form such inclusions remain open tasks. (B) A bootstrap starts from a seed geometry, derives linearized equations from the first law, updates the geometry and iterates. (C) In holography the boundary theory supplies the algebra and the bulk geometry is derived from boundary data. Theorem 5.6 uses strategy (C) and assumes the HSMI/ball interface. A geometry-free reconstruction from the present candidate class remains open.

6 One source and a same-spacetime geometric test

The holographic matching above is not the only conditional route. Consciousness II [24] uses a free massless conformally coupled complex scalar in $D = d + 1 \geq 3$ dimensions. Its ball algebras, physical translation generator and vacuum are supplied in Minkowski spacetime. This matters: they are not the finite pointer algebra of a JT reader. What can be combined in this model is the prepared source, its modular charge and a geometric consistency test in that same spacetime.

In units $\hbar = c = 1$, the scalar curvature coupling is $-\xi\mathcal{R}\Phi^2/2$, with $\Phi^2 = \phi_1^2 + \phi_2^2$ and $\xi = (d-1)/(4d)$. Metric variation gives the conformally improved stress. For a ball $B = B(x_0, R, u)$, write

$$\begin{aligned} K_B &= 2\pi \int_B w_R T_{uu}\, d^d x, \qquad w_R = \frac{R^2 - |x - x_0|^2}{2R}, \\ H_B &= K_B + Q_B, \qquad Q_B = 2\pi\xi \int_{\partial B} {:}\Phi^2{:}\; dA. \end{aligned} \tag{6}$$

These are vacuum-subtracted expectation data with a fixed scalar boundary prescription. Surface readings mean controlled collar limits. H_B is the physical one-sided energy, K_B its bulk charge; neither is the full modular generator $-\log\Delta_B$. Keeping the same curvature coupling with Einstein coefficient $1/(16\pi G)$ gives Wald density $1/(4G) - 2\pi\xi\Phi^2$. The surface coefficient is therefore fixed by the same matter action, not chosen separately for every reader.

Proposition 6.1 (A common-source conditional bridge)**.** *Use the conformal scalar representation and surface prescription above. Let δT and h be smooth source and metric perturbations on an open domain. For the intrinsic geodesic ball of fixed proper radius set*

$$\delta F_B[h] = \delta\left(A(\partial B) - \frac{d-1}{R}V(B)\right), \qquad \eta_A > 0.$$

Then the following statements hold.

1. *The bulk condition*
$$\eta_A \delta F_B[h] + \delta\langle H_B\rangle - \delta\langle Q_B\rangle = 0 \tag{7}$$
for all centres, radii and rest frames is equivalent to $G^{(1)}[h] = (2\pi/\eta_A)\delta T$. It suffices to test the leading small-ball term at every centre and an open set of rest frames.

2. *At $\eta_A = 1/(4G)$ the same condition is $\mathcal{B}_B + \delta\langle H_B\rangle = 0$, where*
$$\mathcal{B}_B = \delta S_{\mathrm{W}}[\partial B] - \eta_A\frac{d-1}{R}\delta V(B) = \eta_A\delta F_B - \delta\langle Q_B\rangle$$
at linear order about the vacuum reference. The source normalization is $8\pi G$ in both forms.

3. *If selected ball subalgebras have vacuum-preserving normal conditional expectations and contain the bounded stress readings and the required surface readings, restricting one fixed physical state to them leaves Eq.* (7) *and its sourced Einstein tensor unchanged. A metric additionally requires common gravitational initial or boundary data and gauge identification.*

4. *Suppose a preparation has a smooth source with integrable energy density of strictly positive total energy, and that the source and response Fourier transforms belong to the tempered tensor-valued Radon measure class of Consciousness II. If consistency is tested at all centres and an open frame set, its nonzero-frequency data identify the canonical intrinsic response up to one normalization at each tested radius, within that paper's intrinsic, gauge-invariant, isotropic, compactly supported linear response class.*

Proof. The exact flat ball identity is $\delta F_B = -\int_B w_R G^{(1)}_{uu}$ [24, Theorem on the exact identity]. Subtracting Q_B from H_B in Eq. (6) gives the other integrand $2\pi\delta T_{uu}$. The leading coefficient of their sum is a nonzero multiple of $R^{d+1}(-\eta_A G^{(1)} + 2\pi\delta T)_{uu}$. An open frame set recovers the symmetric tensor; the field equation conversely makes every ball residual zero. The curvature derivative of $(1/(16\pi G) - \xi\Phi^2/2)\mathcal{R}$ gives the displayed Wald density, so its scalar part is exactly $-Q_B$ at this order. Conditional expectations preserve the vacuum modular restriction and the common stress readings, proving the third statement. For the fourth, a Radon Fourier measure supported only at zero would

give a constant source. Integrability forces its energy density to vanish, contrary to the positive energy. The nonzero-frequency identification theorem of Consciousness II now applies with its stated response hypotheses. These are consequences of that model's results, not an independent derivation of modular consistency. □

The preparation hypotheses are nonempty. In the same scalar theory, a local Weyl pulse gives a smooth improved stress with positive finite energy and a rapidly decreasing mass-shell convolution measure. A neutral pulse $U_\theta = \exp[i\theta(\phi_1(f)^2 + \phi_2(f)^2)/2]$ belongs to a proper symmetry-fixed stable hull when its bounded seed is permitted. It too has a smooth positive-energy nonzero-frequency source [24, Sections on prepared sources and selected algebras]. The permission to execute that pulse is an input; containing stress observables alone need not contain its generator.

There is a direct first-order comparison using this same preparation. Fix $\theta \neq 0$ and take the one-sided family $\psi_\epsilon = (1-\epsilon)\omega + \epsilon\omega_{U_\theta}$, $0 \leq \epsilon \leq 1$. Its source and collar tangents are exactly the neutral pulse's finite-amplitude differences. Choosing the constraint-compatible linear metric response to that conserved source makes Eq. (7) hold for every ball, with the same Wald coefficient. The mixture can be prepared by a recorded random choice of the permitted pulse or identity; that choice and the apparatus are charged. The prepared matter source is not the whole apparatus source when controls occupy the tested domain. This common-model comparison uses the Einstein coupling and gravitational constraint data as inputs, not outputs of the pulse.

A bounded non-scalar accessible observable with a permitted finite translation interval can instead be time-smoothed to give a bounded pulse of strictly positive finite energy. Local stress regularity and the Radon condition still require separate proof in that general route. Exact positive-energy translation permission on a nontrivial separated finite-dimensional algebra is impossible in the unique-vacuum setting; a finite clock is not that translation generator.

The fixed-source statement also survives finite adaptive reading within closed permissions. If $E : \mathcal{A} \to \mathcal{B}$ is a normal conditional expectation and every history-conditioned Heisenberg instrument preserves $\mathcal{B}$, initial states ρ and $\rho \circ E$ give the same record probabilities and, on positive-probability histories, conditional bounded source readings in $\mathcal{B}$ [24]. For each history, the pulled-back record effect and source observable stay in $\mathcal{B}$, where E is identity. Unbounded readings need absolute expectation convergence of spectral truncations. This permits real feedback that changes a common source; it forbids that same restricted feedback from recovering a discarded initial distinction. Enlarged controls or side information change the premise.

The bridge constrains the law rather than merely measuring one response: within its intrinsic response class, a nonzero spectral witness removes the higher-spatial-derivative ambiguity left by a homogeneous source. It does not select an entire metric from matter data; homogeneous gravitational solutions and common boundary data remain. Raw scalar H_B cannot replace the subtracted charge on all small balls: its surface term is $O(R^{d-1})$, while smooth bulk stress and geometry begin at $O(R^{d+1})$. Nor is the Wald coefficient match a modular first law for an arbitrary curved diamond. The area coefficient, region family, source reference, preparation permissions and Eq. (7) are still supplied.

6.1 Conservation before geometric interpretation

A source eligibility rule cannot remove unread stress at will. Q-RAIF NV [26, Section on physical selection and reaction] proves a limited obstruction: on a connected flat domain

of $D \geq 2$, a filter $T \mapsto a(x)T + b(x)\,\mathrm{tr}(T)g$ that conserves every smooth symmetric conserved T has constant a and $b = 0$. If it also kills every constant vacuum source, it is zero. This is a statement about that local algebraic filter family, not a no-go theorem for additional-field or nonlocal gravity models.

The same paper exhibits the appropriate alternative at the classical level. For an actual coupling field χ and canonical matter $\mathcal{L}_m = -\frac{1}{2}(\nabla\psi)^2 - U(\psi)$, let

$$S = \int \sqrt{-g}\left[-\tfrac{1}{2}(\nabla\chi)^2 - V(\chi) + f(\chi)\mathcal{L}_m\right], \qquad f > 0.$$

On both field equations,

$$\nabla^\mu(fT^m_{\mu\nu}) = \mathcal{L}_m\partial_\nu f = -\nabla^\mu T^\chi_{\mu\nu}.$$

The changed material coupling carries a compensating physical stress. This standard action-and-reaction identity is not a quantum total-source construction or a history-derived selector. It also does not cancel a vacuum term: shifting $\mathcal{L}_m$ by $-\rho$ changes both the total stress and the χ equation.

There is a concrete incompatibility to resolve before combining this mechanism with Proposition 6.1. A varying $f(\chi)$ changes the matter theory, so its ball modular Hamiltonian is not automatically the free conformal one. One may compare the prepared source in a causal domain where that interaction has ended and the specified conformal dynamics applies, with selector stress absent there or explicitly accounted for; or one must derive the interacting modular response anew. A separately prepared switching field plus the free CHM formula is not a common-model proof.

6.2 The second-order test

At the next order the source must follow the changed geometry. Consciousness II writes $g_\epsilon = \eta + \epsilon h + \epsilon^2 j + \cdots$ and $T_\epsilon = \epsilon t_1 + \epsilon^2 t_2 + \cdots$. Its corrected small-ball test yields

$$G^{(1)}[j] + G^{(2)}[h,h] = (2\pi/\eta_A)t_2$$

only after adding the slice term $-\frac{V_d}{2(d+2)}R^{d+1}[(\mathrm{tr}\,K)^2 - K_{ij}K^{ij}]$ to the area–volume functional, or choosing a geodesic slice with $K(p) = 0$. The source coefficient obeys the expanded covariant Ward identity, including its metric variation and changed propagation. A flat coherent pulse kept unchanged in the perturbed metric generally fails this identity. The same nonminimal action supplies the scalar propagation correction, the $\xi G_{\mu\nu}\Phi^2$ stress and the Wald coefficient.

This is a conditional leading two-jet stress law, not an exact finite-radius curved modular identity. An arbitrary bent slice already gives a false source without its extrinsic correction. The full quantum reference stress and preparation apparatus must also enter the same renormalized source. Thus a common action supplies necessary source and coefficient relations, while the actual curved modular response remains an additional problem.

7 The mechanism to be realized

A physical history must do two jobs at once: form and maintain the accessible structure, and supply the total source to which the geometric law responds. The finite return

of Section 9 establishes a source-to-operation edge inside an assigned gravity theory. Proposition 6.1 establishes what an all-ball consistency law requires in a different, higher-dimensional matter theory. Their relation is the following strong target.

Conjecture 7.1 (Common-action selection and geometric consistency)**.** *There is a local matter–controller theory in which readable histories form and maintain a nontrivial selected geometric structure, and whose causal effective action uses the same prepared fields, support and clock in its source and response. In a low-energy sector with a common cone, its selected ball data satisfy the geometric consistency condition of Consciousness II. Its first-order response is the corresponding Einstein response with one source normalization; its higher-order response includes the interaction, contact and clock work that produced the selection. Selection changes this response through physical couplings and sources, while a change of description at fixed matter history leaves the source unchanged. The sustainable family of Conjecture C.1 is a finite operating sector of this theory, with a response-to-operation gain that survives the growing-window limit.*

The conjecture has tests beyond assigning gravity to a massive apparatus. Removing the history's access to a command must remove the record-mediated component of the source, while holding the commanded interaction fixed and merely discarding a log must preserve it. Source identification must survive switched physical references; logical noise records alone fail this test in Proposition F.1. The same calibrated source must then pass the all-ball consistency test with fixed initial or boundary data. A law that reproduces only the JT dilaton or one binary return has not passed the higher-dimensional test. At second order the source, geometry and changed preparation must be varied together; entropy Hessians do not become SLD or spacetime metrics by name.

A model satisfying the conjecture must output a nontrivial selected sector from a family of retained histories; a prescribed sequence of couplings is not enough. With a supplied noise and readable seed, Foundation I gives a relative stable-hull construction. The next demand is that the same history dynamics prepares those inputs and that the resulting region data satisfy Eq. (7), with the total source and one area coefficient. The scalar surface match removes an independent matter boundary coefficient, but does not remove MC. This is the precise gap between common-action source formation and a selected gravitational law.

The hardest competing explanation is ordinary matter feedback on an unchanged gravitational stage. It explains every finite JT return here. To go beyond it, the selected localization family must predict a common geometric response without refitting the source reference, area coefficient or region prescription for each history. A permanent mismatch of that test would refute this common-action realization, even if its controller remained useful.

8 From a Selected Interface to a Physical Source

A calibrated interface acts through a field coupling whose outgoing stress sources the supplied JT theory. This gives a physical selection–source map; the origin of the gravitational law in Conjecture 4.2 is a separate question.

8.1 What records can and cannot do to a source

Let a classical record $R \sim \mathrm{Bernoulli}(p)$ control whether a physical operation prepares the matter state ψ or leaves the reference ω. Without reading R, the matter state is $\rho_p = (1-p)\omega + p\psi$ and its stress is the mixture of the two stresses. Deleting or ignoring R after the preparation does not act on the matter, so ρ_p and its source are unchanged. Changing the preparation probability p changes the source. The information side has a matching statement. For the joint state $\rho_{RM} = (1-p)|0\rangle\langle 0| \otimes \omega + p|1\rangle\langle 1| \otimes \psi$ and the reference $\pi_R \otimes \omega$, with π_R the marginal of R, additivity of relative entropy on direct sums and data processing give

$$D(\rho_{RM} \| \pi_R \otimes \omega) = p\, D(\psi\|\omega), \qquad D(\rho_p\|\omega) \le p\, D(\psi\|\omega) \tag{8}$$

[51]. Losing the record lowers the distinguishability available to a reader; it does not remove the matter source. A gravitational consequence of selection must therefore pass through a preparation or interaction, and the conditional geometry of a branch, the average of branch geometries and the geometry of the averaged source are three different objects.

8.2 The source of a calibrated feedback actuator

The companion paper [21] realizes a corrective flip of a degenerate bit as a field coupling. A right-moving free massless scalar with null coordinate u, current $j = \partial_u \phi$ and stress $T_{uu} =: j^2 :$ is coupled through a real smooth switch g. We require a finite nonzero one-particle norm $\lambda^2 = \int_0^\infty \omega |\hat{g}(\omega)|^2 d\omega/(4\pi)$ and $\int (1+u^2)|g'(u)|^2 du < \infty$, where $\hat{g}(\omega) = \int g(u) e^{-i\omega u} du$ in the L^2 sense when needed. For example, $g \in H^1(\mathbb{R})$ together with the weighted derivative condition is sufficient. Smooth compact switches and the full-line packet below both belong to this class. The smeared current defines a Weyl unitary,

$$U = \exp\Big(-iX \int g(u) j(u)\, du\Big) = \exp\big[-i\lambda X (b + b^\dagger)\big], \tag{9}$$

where b is a normalized field mode with mean frequency $\bar{\omega}$ fixed by g, and $\lambda^2 \bar{\omega} = \frac{1}{4}\int g'^2 du$. With the drive mode in the coherent state $|\sqrt{N}\rangle$ and $2\lambda\sqrt{N} = \pi/2$, the coupling flips the bit with success probability $s_N = (1 + e^{-\pi^2/(8N)})/2$. From the current commutator $[j(u), j(v)] = \frac{i}{2}\delta'(u-v)$ one has $U^\dagger j(u) U = j(u) + \frac{1}{2} X g'(u)$ and

$$U^\dagger T_{uu}(u) U = T_{uu}(u) + g'(u) X j(u) + \tfrac{1}{4} g'(u)^2. \tag{10}$$

The cross term has zero expectation if the bit and the fresh field input are initially independent and $\mathrm{tr}(\rho X) = 0$, or if their joint state has no coherence in the bit's Z basis; a Z-diagonal bit marginal alone does not suffice [21]. Writing $j = \psi(u) b + \psi^*(u) b^\dagger + j_\perp$, the incoming coherent beam has stress $T_{\mathrm{in}} = 4N[\mathrm{Re}\,\psi(u)]^2$ and each activation adds

$$F_g(u) = \tfrac{1}{4} g'(u)^2 = 4\lambda^2 [\mathrm{Im}\,\psi(u)]^2. \tag{11}$$

This added stress is a property of the coupling. Its integral is the switching work $\lambda^2 \bar{\omega}$ supplied by the external switching agent.

The feedback task is that of the Imprint Chain [22] with the flip realized by this coupling. A hidden instrument label is estimated from calibration records with mismatch

probability q; each task step reads the bit with reliability ν, accepts feedback with probability h, switches g on when accepted and the reading is negative, and lets a bath with contraction c act. Each opportunity receives its own scheduled incoming beam, independent of the bit and earlier records. In the companion paper the stationary loss $L_\infty(q)$ and the stationary coupling rate satisfy [21]

$$p(q) = h\nu L_\infty(q) + \tfrac{h(1-\nu)}{2}. \tag{12}$$

Proposition 8.1 (Source–loss relation)**.** *At fixed drive, switch and task parameters, the stationary mean source per opportunity outside the apparatus is* $\langle T_{uu}\rangle_q = T_{\rm in} + p(q)F_g$. *For two calibrations,*

$$\langle T_{uu}(u)\rangle_q - \langle T_{uu}(u)\rangle_{q'} = h\nu\,[L_\infty(q) - L_\infty(q')]\,F_g(u). \tag{13}$$

Proof. The incoming beam is present at every opportunity. A coupling occurs with stationary probability $p(q)$ and, under the joint initial condition, adds F_g in expectation by Eq. (10). Equation (12) gives the difference. □

Better calibration lowers the task loss and, through it, the source. The coefficient comes from the same interaction that performs the correction; no separate rule maps a calibration error to an emitted pulse. The relation does not assign stress to knowledge. Deleting the calibration record after the commands have been executed leaves the outgoing source unchanged, in line with Section 8.1. The relation is a stationary average per opportunity; a single trajectory, with its fixed hidden match label, does not follow the population mean step by step.

8.3 A source set by learned quantum protection

The activation-rate mechanism is not the only way that retained records can change a source. Brain Anchor [57] constructs two degenerate qubits with a programmable coupling

$$A_a = Z_1 + \alpha a Z_2, \qquad U_a = e^{-iA_aQ_f}, \quad Q_f = \int f(u)j(u)du = \sqrt{k}(b + b^\dagger). \tag{14}$$

The real smooth compact switch f and positive gain interval are supplied. The gain α is unknown, and each encounter receives a fresh independent vacuum field. On $\mathrm{span}\{|01\rangle, |10\rangle\}$ the coupling is $(1 - \alpha a)Z_L$. Calibration records from the commands $\pm h$ have means $r_\pm = e^{-2k(1\mp\alpha h)^2}$, so $\log(r_+/r_-) = 8k\alpha h$. The finite-record certificate in that paper bounds $|1 - \alpha a|$, including command rounding and known drift. It does not promise an exact fixed algebra after finite training.

Proposition 8.2 (One residual controls protection and source)**.** *Fix the acquired command* a, *write* $m = 1 - \alpha a$ *and* $F_f = f'^2/4$, *and keep the initial field independent of the logical body and retained ancillas. For every logical input, one encounter has added mean stress* m^2F_f *and switching work* m^2E_f, *with* $E_f = \int F_f du$. *After* M *independent identical encounters, with the output ports discarded and no interleaved gates or active recovery, its passive protection error is*

$$\epsilon_M = \tfrac{1}{2}(1 - e^{-2Mkm^2}). \tag{15}$$

For a single pulse with stress moments $E_j = \int u^j F_f du$ and the initial data of Proposition 8.4, the late invariant is

$$\mathcal{C}(m) = H_0^2 + 2\kappa H_0 m^2 E_1 + \kappa^2 m^4 (E_1^2 - E_0 E_2). \tag{16}$$

Thus retained history can change the physical protection and the gravitational response at fixed pulse shape and activation schedule.

Proof. The current shift is $j \mapsto j + A_a f'/2$. The centered independent field input makes the stress cross term vanish for every logical state, and $A_a^2 = m^2 I_L$ on the code. Its characteristic function multiplies the logical off-diagonal by e^{-2km^2} per encounter; a dephasing channel's half-diamond error is its phase-flip probability. Time ordering may give a factor $e^{i\vartheta_f A_a^2}$, which is scalar on the code but not on the whole two-qubit register. Finally substitute $m^2 F_f$ in the fixed-initial-data source solution below. □

This link does not require a rule that emits a pulse only after a wrong answer. The same interaction produces both the residual error and its source, even when the logical body is entangled with retained memory. At $m = 0$ the body decouples in the code; a noiseless stored qubit need not continuously export entropy. The finite-rate entropy-export bound of Section 10 applies to actual renewal, not to every protected state. Calibration itself adds field energy $2N(1 + \alpha^2 h^2)E_f$ for N independent tests at each sign. Preparations, measurement, the controller and switching mechanism add further charges. The source suppression during operation alone is not a total-energy saving.

Brain Anchor also treats an unknown centered Gaussian variance K_ω. Logical records at $0, \pm h$ give $K_\omega = -\log r(0)/2$ and $\alpha = [\log r(h) - \log r(-h)]/(8hK_\omega)$; two settings suffice when the signs of $1 \mp \alpha h$ are known. Its finite-record mismatch certificate needs no supplied temperature label. The coherence is now $e^{-2K_\omega m^2}$, but the coupling-added source remains $m^2 F_f$ on centered independent input.

Incoming stress must still be added. A u-clock chiral KMS input has background $\pi/(12\beta^2)$ in natural units. Within a supplied bounded one-temperature family, strictly decreasing K_β permits a finite-error background estimate; a general Gaussian variance does not determine incoming energy. At common preparations and initial data, the linear dilaton equation compares coupling-source differences. Equation (16) cannot be used for the full source after dropping its thermal background, nor is u thermality the rapidity-clock reference of Section 10.

The same command can also operate on one reused vacuum field. For nonnegative compact switches f_j on disjoint intervals, with no noncommuting logical gates, the smeared currents commute but have negative correlations. Brain Anchor gives exact coherence $e^{-2m^2 k_M}$ with $k_M = \langle (\sum_j Q_{f_j})^2 \rangle \le Mk$, and added source $m^2 \sum_j f_j'^2/4$ of energy $Mm^2 E_f$. Its JT response uses the actual pulse-train moments and initial coefficients in a valid operating domain. No independent-port approximation, thermal input or infinite-lifetime worldline is inferred.

There is a finite causal pulse in this construction. For example, set $f(u) = A \exp[-1/(1 - (u-3)^2)]$ on $2 < u < 4$, zero outside, and normalize its one-particle norm to $k = .2$. Numerical quadrature gives $(E_0, E_1, E_2) = (.4568522, 1.3705565, 4.3215400)$. For $H_0 = 1$, $\kappa = .05$ and $\alpha = 1.1$, the supplied command $a = 48/55$ has residual $m = .04$ and gives $\mathcal{C} \simeq 1.00021929$; the comparison $a = 1$ has $m = -.1$ and $\mathcal{C} \simeq 1.00137053$. These are deterministic evaluations of the displayed pulse, with code and outputs in `supplemental/pulse_debit`, not statistics of a calibration pilot. For a random learned

command, the invariant of the mean source uses $(\mathbb{E}m^2)^2$, whereas the mean branch invariant uses $\mathbb{E}m^4$ in the quadratic term. The compact source ends before a later boundary echo is scheduled, but the calibrated wall and timing instrument remain the inputs of Section G.1.

Stress readback also has an exact blind direction. At fixed $a_0 > 0$, the gains $\alpha_\pm = (1 \pm d)/a_0$ give residuals $\mp d$ and therefore the same protection channel and source. Field parity conjugates their full code interactions while leaving the vacuum and stress invariant. No parity-covariant stress instrument with invariant outcome labels can distinguish them, nor can their mean JT responses at common initial data. Their correcting commands $1/\alpha_\pm$ differ. Probes at $a_0 \pm h$ break this ambiguity through the opposite signs of $8k\alpha h(1-\alpha a_0)$ in their log contrast. A gravitational response is therefore not a complete calibration reference for the interface that produced it. Off-frame coupling probes, or a signed field quadrature, are additional access, with their own sources and costs. Full output-field parity and a conditional logical pulse can instead restore the stored state without identifying the gain, as shown in Ref. [57]. This recovery leaves the emitted source unchanged. Active state correction and learning a match that suppresses the actual source are therefore distinct operations.

The gain is a physical actuator parameter, not a freely changed description. The selected finite code is not the geometric algebra of Section 5, and the supplied JT action still supplies the gravitational response. A link to continuous modular selection requires that further interface; it is not obtained by identifying a protected M_2 sector with a half-sided modular inclusion.

8.4 A physical seed changes the source

The rate-controlled model changes how often a fixed interaction occurs. The coupled observable can itself change, giving a direct relation between an exactly represented algebra and a source. Keep the incoming field, switch and activation schedule fixed. On a qubit put $P_\theta = (I + \cos\theta\, Z + \sin\theta\, X)/2$ and use $U_\theta = \exp\big[-i\lambda P_\theta(b + b^\dagger)\big]$. The projection selects a pointer algebra and its source population under this fixed schedule.

Proposition 8.3 (Selected projection and outgoing stress)**.** *For $\lambda \neq 0$, an initially independent qubit state ρ and normal field state σ, the exactly represented algebra of the induced qubit channel is $\{P_\theta\}'$. Let $r_\theta = \mathrm{tr}(\rho P_\theta)$. The stress formula below requires its displayed current moments to exist. The outgoing field is*

$$(1 - r_\theta)\sigma + r_\theta e^{-i\lambda Q}\sigma e^{i\lambda Q}, \qquad Q = b + b^\dagger,$$

and its stress is

$$T_\theta(u) = T_{\mathrm{in}}(u) + r_\theta \left[g'(u)\langle j(u)\rangle_\sigma + \tfrac{1}{4}g'(u)^2\right]. \tag{17}$$

Thus two selected projections can change the source at fixed global input state and fixed activation schedule.

Proof. The quadrature has non-atomic spectral distribution in a normal state, so the multiplicative-domain theorem of Q-RAIF C gives $\{P_\theta\}'$ [14]. Expanding U_θ in its two projection blocks and tracing the qubit gives the field mixture; off-diagonal blocks have zero trace even when ρ does not commute with P_θ. In the current identity replace the Pauli X by P_θ and use $P_\theta^2 = P_\theta$. Initial independence gives (17). □

For $\rho = (I + 0.8Z)/2$, the seeds P_0 and $P_{\pi/2}$ select different pointer algebras, with $r_0 = 0.9$ and $r_{\pi/2} = 0.5$. In the worked coherent packet, the cross profile is $8\lambda\sqrt{N}\,\mathrm{Re}\,\psi\,\mathrm{Im}\,\psi$. Its first three moments vanish, so the invariant increment is (70) with p replaced by r_θ. With the table's $N = 16$ and background it is 2.0807581×10^{-5} versus 1.1559770×10^{-5}. The full source also contains the cross profile, which enters the local dilaton even though its first three moments do not enter the final invariant.

Here the source weight is derived from the state's population in the selected projection; it is not assigned to an external mismatch label. Changing the projection means changing a physical interaction, not renaming an unchanged algebra. The two algebras are abstractly isomorphic but are differently embedded relative to the fixed state and coupling. This supplies a seed–interface–source link. It does not identify that finite-dimensional pointer algebra with a Type III geometric candidate or explain why the coupling selects this seed rather than another. Those are the remaining selection and geometry bridges.

8.5 Response of a given Jackiw–Teitelboim model

Take the JT action [52, 53]

$$I = \frac{1}{16\pi G}\int \sqrt{-g}\,\Phi(R + 2/L^2) + I_{\mathrm{matter}}, \qquad ds^2 = -\frac{4L^2\,du\,dv}{(1+uv)^2}, \tag{18}$$

with the usual boundary terms, $\kappa = 8\pi G$, $D = 1 + uv$ and the vacuum dilaton $\bar\Phi = H_0(1 - uv)/D$, $H_0 > 0$. The field equations are $\nabla_\mu\nabla_\nu\Phi - g_{\mu\nu}\Box\Phi + g_{\mu\nu}\Phi/L^2 = -\kappa T_{\mu\nu}$, and in these coordinates

$$\Phi_{uu} + \tfrac{2v}{D}\Phi_u = -\kappa T_{uu}, \qquad \Phi_{vv} + \tfrac{2u}{D}\Phi_v = -\kappa T_{vv}, \qquad \Phi_{uv} + \tfrac{2}{D^2}\Phi = \kappa T_{uv}. \tag{19}$$

The matter is a conformal theory with matched left and right central charges, and the state-independent anomaly is removed by subtracting the same background. The left-moving input is unchanged, and the state-dependent source is right-moving, $T_{uu} = F(u)$ with $T_{uv} = T_{vv} = 0$.

Proposition 8.4 (Response with fixed initial data)**.** *Let F be integrable with $\int |u|^2|F|du < \infty$. On a connected coordinate domain with $D \neq 0$, write the perturbation as $\Phi - \bar\Phi = (A(u) + vB(u))/D$ and impose the initial coefficient data $A(u), B(u) \to 0$ as $u \to -\infty$. For a source bounded below in support this is equivalent to zero perturbation before the source. The unique solution with these data is*

$$\Phi = \bar\Phi + \frac{A(u) + vB(u)}{D}, \qquad A = -\kappa\int_{-\infty}^{u}(u-z)F(z)dz, \qquad B = -\kappa\int_{-\infty}^{u} z(u-z)F(z)dz. \tag{20}$$

The perturbation is linear in F. With $E_j = \int u^j F\,du$, after the source (or in the asymptotic limit for sources with tails) the invariant $\mathcal{C} = \Phi^2 - L^2(\nabla\Phi)^2$ takes the vacuum value

$$\mathcal{C} = H_0^2 + 2\kappa H_0 E_1 + \kappa^2(E_1^2 - E_0E_2), \tag{21}$$

and inside the support $\partial_u\mathcal{C} = \kappa F\,(2H_0u + uA - B)$.

Proof. Write $\psi = D\,(\Phi - \bar\Phi)$. The vv equation of (19) reads $\psi_{vv} = 0$, so $\psi = A(u) + vB(u)$, and the other two equations reduce to $A'' = -\kappa F$, $B'' = uA''$ and $B' = uA' - A$, which

Eq. (20) satisfies. The absolute first and second moments make both retarded coefficients tend to zero at $-\infty$. Two solutions differ by $\alpha + \beta u + \gamma v - \alpha uv$, whose coefficients are $A_h = \alpha + \beta u$ and $B_h = \gamma - \alpha u$. Their two zero limits force $\alpha = \beta = \gamma = 0$. Every vacuum solution has numerator $\alpha + \beta u + \gamma v - \alpha uv$ and invariant $\alpha^2 + \beta\gamma$; after the source $\alpha = H_0 + \kappa E_1$, $\beta = -\kappa E_0$, $\gamma = \kappa E_2$. The balance law follows by differentiation. □

The coefficient condition cannot be replaced by $\Phi - \bar{\Phi} \to 0$ at each fixed v. The homogeneous solution $\eta v/(1 + uv)$ has that limit for every fixed v along nonsingular points. It even tends to zero uniformly on the subdomain $u < 0$, $v \leq 0$, since its absolute value is at most $|\eta|/|u|$; no uniform limit near $D = 0$ is implied. Adding it leaves the source unchanged but changes the late invariant by $-\kappa E_0 \eta$. It is excluded by $B(-\infty) = 0$. Source comparisons below hold H_0, the coordinate matching and both coefficient boundary data fixed.

The JT curvature stays $-2/L^2$; the dilaton and the invariant respond. A dilaton difference is not by itself a higher-dimensional curvature or a mass, and comparing boundary proper times needs a boundary normalization.

The matching condition matters. Set $A_F, B_F \to 0$ at $+\infty$, or exactly to zero after a compact source, instead of the initial data of Proposition 8.4. The final-state-matched solution differs from the retarded solution by

$$A_F = A_R + \kappa(uE_0 - E_1), \qquad B_F = B_R + \kappa(uE_1 - E_2).$$

For the chiral diffeomorphism states of Consciousness I [25], with a Cut at a to the left of the support, the relative entropy is $D_a = 2\pi \int (u - a)F(u)\, du$. The final-state-matched solution gives $\delta\Phi(a, 0)/(4G) = -D_a$, while the initial-state-matched solution gives $\delta\Phi(a, 0) = 0$ at a Cut before the pulse. The first identity compares an earlier section with a matched final state; it does not say that a pulse removes area before it arrives. The dilaton–modular-energy relation itself is known [58]; what is used here is its type-correct pairing with a specified source and matching condition.

The same equations show that a null-only record of the source is incomplete. A genuine vacuum-density shift $T_{\mu\nu} = -\delta\rho\, g_{\mu\nu}$ has zero null components but gives the particular solution $\delta\Phi = \kappa L^2 \delta\rho$. A probe that reads only null energy cannot certify that a source is absent, which is the source-completeness condition relevant to the vacuum curvature test of Ref. [26].

9 Response that changes the next operation

A source-to-geometry calculation becomes a return only when the geometry changes an actual record and that record changes a later interaction. The readout permission matters. In the minimally coupled JT sector, instruments on the same matter metric with the same inputs have identical adaptive record laws even when the dilaton differs (Proposition G.1). A boundary tied to a fixed dilaton value evades that obstruction. Its echoes reconstruct the three vacuum coefficients, and a source-compensated inverse includes finite probe backreaction (Propositions G.2 and G.3). An exact current contact instead obtains an invariant-dependent bit, including the contact's own source (Propositions G.5 and G.6). The bit selects the next physical projection. Its same-field conditional covariance remains part of the history; detector replacement does not reset the field.

These are finite physical records, not permission to read an exact metric. The worked current contact has total variation only 2.3650×10^{-4} between its two initial geometries. Its fixed-cutoff energy budget also forbids indefinitely many contacts of unchanged positive source charge. Appendix G contains the timing, stress and accumulated-source bounds.

9.1 An acquired record drives a new material source

A record can act without first being converted into a supplied mechanical initial condition. Keep the two classical scalar fields and the material action of Section J.4, and append $i\langle\Psi,\dot\Psi\rangle - \langle\Psi, H_c\Psi\rangle$ to its proper-time Lagrangian. The normalized state Ψ contains a finite history clock, probe P, record R, replacement blank B and an old purification Q. Set

$$H_c(\Phi,y) = H_{\rm hist} + E_{\rm off}\mathbf{1} + \lambda(\Phi-\phi_0)Z_P + \frac{\chi_R}{2}P_R(y-d)^2, \qquad P_R = |1\rangle\langle 1|_R, \quad \chi_R \geq 0. \tag{22}$$

$H_{\rm hist}$ is the positive gate-conjugated clock of Q-RAIF C. Its first nonidentity edge performs the Y_P acquisition C_Y, the next performs a true SWAP of P and B, and the remaining edges are identities. There is no proper-time switch. Initially $R = 0$, so the spring has zero expectation. The actual acquisition opens its force channel. $H_c \geq 0$ while $E_{\rm off} > |\lambda(\Phi - \phi_0)|$. The target d, couplings and initial clock are prepared hardware. The scalar fields, mechanical coordinates and geometry use an Ehrenfest mean-action approximation [61, Section 2]; the finite quantum core and its old reference are retained in full, without sampling or collapsing R.

Here θ is a material angle, y its storage coordinate, $I, J, k, K > 0$ their inertia and elastic parameters, and $U(\Phi)$ the material support potential. Proper-time derivatives obey $I\ddot\theta + k(\theta - y) = -\gamma\dot\theta$, $\gamma = g_+^2 + g_-^2$, with no incoming classical waves. Their emitted powers are $P_\pm = g_\pm^2\dot\theta^2$. The mechanical stored energy is $e = I\dot\theta^2/2 + J\dot y^2/2 + k(\theta-y)^2/2 + Ky^2/2$, and $w = \dot\Phi$ along the wall. The positive wall junction and its two invariant charges are specified by Eqs. (128) and (137); all of these mechanical and gravitational data are supplied.

Let $p = \langle P_R\rangle$, $z = \langle Z_P\rangle$ and $E_c = \langle H_c\rangle$. Variation of this one action gives

$$\begin{aligned} i\dot\Psi = H_c\Psi, \qquad F_y = \chi_R(y-d)p, \qquad F_\Phi = \lambda z, \\ J\ddot y + k(y-\theta) + Ky = -F_y, \qquad \dot e = -P_+ - P_- - F_y\dot y, \qquad \dot E_c = F_y\dot y + F_\Phi w. \end{aligned} \tag{23}$$

The angle equation and $P_\pm = g_\pm^2\dot\theta^2$ are unchanged. Here p is an expectation obtained by variation, not a decoded number fed to a separate control policy. The mass and its tangent balance are

$$M = m_{\rm rest} + U + e + E_c, \qquad \dot M = (U' + F_\Phi)w - P_+ - P_-. \tag{24}$$

Consequently Eq. (137) holds with U' replaced by $U' + F_\Phi$ in its acceleration equation; both invariant and null-clock equations retain their directions. Differentiating the mass junction with these equations gives zero. The separate work balances are

$$\begin{aligned} e(\tau) + Q_{\rm out}(\tau) + \int_0^\tau F_y\dot y\, ds = e(0), \\ E_c(\tau) - E_c(0) = \int_0^\tau (F_y\dot y + F_\Phi w)\, ds. \end{aligned} \tag{25}$$

The clock interaction and conditional spring are included in E_c. The spring breaks the symmetry of Proposition J.6; F_Φ cannot be set to zero in this coupled calculation.

Proposition 9.1 (A nonzero record-mediated source return)**.** *Take two initial geometries with the same $\Phi = \phi_0$, $w = 0$, mass, mechanical data and quantum input, but different $\mathcal{C}_-$; set $\mathcal{C}_+$ by the mass junction. In the positive exterior domain, either $\lambda = 0$ or $\chi_R = 0$ makes their proper emitted powers identical. With both interactions present, there are positive finite-clock data for which the emitted powers differ on a nonempty finite interval. Their record, mechanical response, source and geometry obey the same action and work account.*

Proof. For $\chi_R = 0$, the mechanical subsystem is independent of the reader and geometry. For $\lambda = 0$, the joint mechanical and quantum equations are independent of geometry. Either condition fixes identical mechanical sources for the two initial geometries.

For a nonzero example choose $\theta(0) = \dot\theta(0) = 0$, $y_0(y_0 - d) \neq 0$ and probe state $(I + \upsilon X_P)/2$, $\upsilon \neq 0$, with a blank R and its old purification held. Let $r \geq 1$ be the C_Y edge, $a_j > 0$ the clock edge strengths and $G_r = \prod_{j=1}^{r} a_j$. The initial acceleration difference δa follows from the junction equations, so that $\delta\Phi(t) = \delta a\, t^2/2 + O(t^3)$. The shortest-path Dyson expansion gives

$$\delta p(t) = K_r t^{2r+3} + O(t^{2r+4}), \qquad K_r = \frac{\upsilon G_r^2 \lambda \delta a}{6r!(r-1)!(r+3)}. \tag{26}$$

An insertion of $\lambda\delta a\, t^2 Z_P/2$ after $j < r$ forward edges contributes the time integral $t^{r+3}(j+1)(j+2)/(r+3)!$. Their sum is $t^{r+3}r/[3(r+3)r!]$. Using $C_Y^* P_R C_Y = (I + Y_P)/2$ in the interference with the unperturbed shortest path gives the coefficient above. The insertion after all r edges has Hermitian $P_R Z_P$ in that interference and zero real part. Further graph paths and spring insertions raise the order; the spring annihilates the blank record before acquisition.

Integrating the mechanical equations then yields

$$\delta(P_+ + P_-)(t) = -\frac{2\gamma k^2 y_0 \chi_R (y_0 - d) K_r}{I^2 J (2r+4)(2r+5)(2r+6)} t^{2r+7} + O(t^{2r+8}). \tag{27}$$

The baseline angle velocity is $k y_0 t/I + O(t^2)$. Different initial $\mathcal{C}_-$ give a nonzero δa for generic supported data, since initially $\dot w = -U'\beta_+\beta_-/M - \phi_0/L^2$. The displayed coefficient is nonzero for the stated nonzero parameters. The coupled finite-dimensional equations are analytic on their strict positive exterior domain. Local existence and continuity therefore give a nonempty admissible family, not just a numerically resolved difference. □

The proof isolates the causal mechanism analytically, rather than attributing a small numerical difference to geometry. Blocking either sensing or record actuation removes the source contrast. Keeping the record but blocking actuation preserves a readout difference while removing the later source difference. Its numerical realization and complete mass/work checks are in Appendix J. The two histories here differ in initial gravitational data; they do not purport to generate those data from algebra selection.

For histories with common initial geometric constraints, the distinct spatial construction of Proposition J.7 has a nonzero source–dilaton–sensor path and a later material response. Its finite-width regulator, support and synchronization are physically varied.

The complete material response also contains ordinary clock and force effects; only the separately isolated path is gravitational. A third construction, Proposition J.8, couples the actual full-mode quartet state to paid clock/director fields and constrained JT bulk evolution. It supplies a local Ward identity and a causal prepared bulk domain, not moving-wall gate matching or the full thermal apparatus.

A finite record error alone does not control an unbounded stress. The response comparison uses weighted source bounds (Proposition A.1), and the coupled trajectory–reader bound requires domain-preserving strong solutions and the stated bounded-coupling derivative assumptions (Proposition A.2). Appendix A gives these estimates and the separate full-mode clock-work matching result. None converts fixed-window existence into sustained operation.

10 Maintenance and its gravitational price

10.1 Entropy removal and the source it can constrain

A fixed unitary correction can return its control port coherently. Removing entropy injected by unobserved noise is a different task. To price it, the state whose entropy is removed must include the controller's usable side information. The heat account must also distinguish a thermal reservoir from a supply of pure memories.

HAFF G supplies a record-driven program and executor in which selecting a physical coupling changes its source under one fixed executor [55]. The dephased quartet below adds a specified full renewal debit. It does not identify the two material realizations or inherit a thermal controller from that program loop.

Proposition 10.1 (Maintenance entropy account)**.** *Let $V = SC$ include the maintained system and every retained controller and side-information register. A correction acts unitarily on finite-dimensional VAB, initially in $\rho_V \otimes \alpha_A \otimes \tau_B$, where $\tau_B = e^{-\beta H_B}/Z$, $\beta = (k_B T)^{-1} > 0$. Put*

$$s = S(V) - S(V'), \qquad p = S(A') - S(A), \qquad Q = \operatorname{tr} H_B(\rho'_B - \tau_B).$$

Then, with $I_{V:A:B} = S(V) + S(A) + S(B) - S(VAB)$,

$$\beta Q + p = s + I_{V':A':B'} + D(\rho'_B \| \tau_B) \ \geq s. \tag{28}$$

If the marginal entropy of C is returned, s equals the decrease of $S(S|C)$. In a sequence of such corrections, a bounded total purity consumption and a positive asymptotic entropy-removal rate $\dot{s}$ imply $\liminf Q_{[0,t]}/t \geq k_B T \dot{s}$.

Proof. Unitary invariance of total entropy and the initial product give $\Delta S(V) + \Delta S(A) + \Delta S(B) = I_{V':A':B'}$. Also $\beta Q = \Delta S(B) + D(\rho'_B \| \tau_B)$. Combine the identities and sum over corrections. The rate statement follows when the accumulated p divided by elapsed time tends to zero. □

This is the Landauer entropy balance, with the memory-purity term displayed explicitly; the equality and the role of accessible side information are standard [13, Theorem 3 and Section 5.1]. For initially correlated or nonthermal inputs the same calculation instead gives

$$\beta Q + p = s + (I_{\text{out}} - I_{\text{in}}) + D(\rho'_B \| \tau_B) - D(\rho_B \| \tau_B).$$

These consumed correlations and bath free energy can pay for correction. A classical copy of the error permits a reversible conditional flip with $s = 0$, although the system's marginal entropy decreases. Negative quantum conditional entropy gives another reason to include the side information. A pure-memory swap has $p = s$ and can have $Q = 0$. A retained oscillator's closed unitary loop also has no entropy removal by itself. Thus a count of corrections or a marginal noise entropy is not automatically $\dot{s}$. In a cyclic device $\dot{s}$ must be established from its actual noise and controller dynamics. For nondegenerate systems, heat and supplied work are further separated by the change in internal energy.

Proposition 10.2 (A finite-rate chiral exhaust floor)**.** *Consider a unitary, zero-chemical-potential chiral conformal exhaust with central charge $c_{\mathrm{ex}} > 0$, thermal input at $T \geq 0$, and an extensive entropy-current account. Energy and temperature are defined by the same translation generator and clock. Assume that the outgoing modes export net dimensionless entropy current $x \geq \dot{s} > 0$, where $\dot{s}$ is the actual maintenance entropy-removal rate after all retained side information and sublinear resource consumption are accounted for. The entropy-current maximum at fixed energy is the thermal one. Then the excess heat current over the incoming thermal stream obeys*

$$\dot{Q} \geq k_B T \dot{s} + \frac{3\hbar}{\pi c_{\mathrm{ex}}} \dot{s}^2. \tag{29}$$

Proof. Write $a = \pi c_{\mathrm{ex}} k_B^2/(12\hbar)$. The thermal energy and dimensionless entropy currents are $J_E(T) = aT^2$ and $J_S(T) = 2aT/k_B$. The first is the standard chiral CFT current [11, Eq. (1)]; the second follows from its extensive Gibbs entropy. Entropy maximization at fixed energy is the Gibbs variational principle, underlying the single-channel information-flow bound [10]. Choose $T_* = T + k_B x/(2a)$ so that the thermal output has the required entropy. Any output with at least that entropy has energy current at least aT_*^2. Subtracting aT^2 gives

$$\dot{Q} \geq k_B T x + \frac{k_B^2 x^2}{4a} = k_B T x + \frac{3\hbar x^2}{\pi c_{\mathrm{ex}}}.$$

The expression increases for $x \geq 0$. □

The current account is a physical condition on the exported modes, not the entropy of a Type III local interval. Equation (44) establishes the needed joint entropy rate in a fresh-port account; matching those ports to a conformal current must preserve its capacity and entropy budget. At fixed capacity and positive rate the quadratic term survives even at $T = 0$. Thermal output saturates the channel inequality, but this alone does not construct a correcting apparatus. The currents are long-window, extensive Gibbs limits. For an observation window τ with net exported entropy X_τ, bounded rates and sublinear boundary corrections give

$$Q_\tau \geq k_B T X_\tau + \frac{3\hbar X_\tau^2}{\pi c_{\mathrm{ex}} \tau} + o(\tau).$$

The current formula is not an exact inequality for an arbitrarily short finite packet. Applying the Gibbs bound to the entire output also includes correlations between packets; it does not require each output to be a separate thermal state.

10.2 A conditional gravitational reading

The price becomes a geometric reading only with a physical transport map. The following JT coefficient reads the total routed heat in one matched clock and reference. It does not assign curvature to entropy that remains in a memory or exits through another channel.

Proposition 10.3 (A heat account read through the dilaton)**.** *Compare two JT solutions with the same H_0, boundary coefficient data and coordinate and clock normalization. Suppose their source difference F is the exhaust of the correction in Proposition 10.1, with finite absolute second moment and $\int F du = Q$. This last equality requires that the specified outgoing channel carry all the heat in the same energy normalization. Then its retarded coefficient difference obeys*

$$-\frac{A'(+\infty)}{\kappa} = Q \;\geq\; k_B T(s-p). \tag{30}$$

For a compact exhaust and any $u_2 > u_1$ after it, the left side is exactly $-[A(u_2) - A(u_1)]/[\kappa(u_2-u_1)]$. No pointwise sign of F is required.

Proof. Differentiate Eq. (20): $A'(u) = -\kappa \int_{-\infty}^{u} F(z)dz$. Take the late limit and apply Eq. (28). □

The reference for Q is the incoming thermal reservoir transported through the same exhaust, before its heat uptake; H_B is its u-translation generator. A thermal background has a nonintegrable stationary flux, so the JT solution here is the difference from that same transported thermal background. The identification $E_0 = Q$ is an energy-transport hypothesis: other exhaust channels, energy stored in the device, work beams crossing the observation surface and redshift must be accounted for separately. With a fixed transport factor $E_0 = \zeta Q$, Eq. (30) acquires the same factor ζ. The coefficient slope is read from the dilaton in the declared matched coordinate frame, not an invariant under arbitrary changes of boundary clock. The relation links an actual entropy-removal process to a specified dilaton reading. The JT dynamics and heat-routing map are its geometric inputs.

For a matched running single-chiral source, the marginal-horizon balance is $dS_h/d\eta = 2\pi F\xi^2$. If the charged heat is exactly $F\xi^2$ and the physical input is thermal at $T_H = 1/(2\pi)$ in the same rapidity clock, the long-window bound becomes $\overline{\dot{S}_h} \geq \dot{s} + 6\dot{s}^2/c_{\rm ex}$. The actual thermal-source and reference-clock terms, positive-domain conditions and nonthermal contact excess are retained in Appendices B and H. A reparametrized vacuum is not a prepared heat reservoir.

11 Selection and response over growing windows

11.1 A conjecture on a sustainable selection–response loop

The unresolved question is whether one reused local body can combine selection, renewal, sharp exhaust and informative geometric feedback. Fix two physical preparations on a logical receiver A and its current purifier R:

$$|\Phi_Z\rangle = \sqrt{1-\epsilon}|00\rangle + \sqrt{\epsilon}|11\rangle, \qquad |\Phi_X\rangle = (H_A \otimes I_R)|\Phi_Z\rangle, \quad 0 < \epsilon < 1/2.$$

The logical labels denote Q-RAIF C's neutral pair codes, with their spectator sea retained. Here R is the paid, resettable current purifier in the third and fourth body species, available to joint neutral gates. Old references outside this source quartet are denoted R_{old} and are not acted upon. The quartet is a resettable working source alongside the series' protected unknown content H, not a replacement for it; access to the current R comes from those paid body species and joint neutral gates, not from an old-purifier permission. Each cycle's inaccessible environment applies the same pointer-aligned channel

$$\mathcal{D}_q(\rho) = (1-q)\rho + qZ_A\rho Z_A, \qquad 0 < q < 1/2, \qquad h(x) = -x\ln x - (1-x)\ln(1-x).$$

No noise error label is available to the apparatus. Renewal returns the complete quartet, including the current purifier.

Proposition 11.1 (Pointer alignment and the complete renewal debit)**.** *For $\rho_p = \mathcal{D}_q(|\Phi_p\rangle\langle\Phi_p|)$, return to its specified pure quartet removes entropy $h_p = S(\rho_p)$, where*

$$h_Z = h\left(\frac{1-\sqrt{1-16q(1-q)\epsilon(1-\epsilon)}}{2}\right) > 0, \qquad h_X = h(q) > h_Z. \tag{31}$$

The receiver-only debits are 0 for Z and $h(\epsilon + q(1-2\epsilon)) - h(\epsilon)$ for X. Approximate quartet return subtracts its residual joint entropy from h_p, with finite-dimensional entropy continuity controlling the error. Program updates, sensors and usable controller registers enter the full maintenance debit in Proposition 10.1 as well.

Proof. The mixture of $|\Phi_p\rangle$ and $Z_A|\Phi_p\rangle$ has nonzero eigenvalues $(1 \pm d_p)/2$, with $d_p^2 = 1 - 4q(1-q)(1 - |\langle\Phi_p|Z_A|\Phi_p\rangle|^2)$. The overlaps are $1-2\epsilon$ and zero. Strict monotonicity of h on $(0, 1/2)$ proves the contrast. Tracing R gives the receiver debits. The reference marginal is unchanged by the noise, so the difference between each complete and marginal debit equals the receiver–purifier mutual information lost to the noise. □

Pointer alignment lowers the complete renewal price without making it zero. For $\epsilon = .1, q = .05$, $h_Z = .0877516288$ and $h_X = .1985152433$ nats per cycle. The marginal Z state is unchanged, but its purifier correlations are not. The loop must establish that its additional charged operations preserve an actual complete-debit contrast $\bar{s}_X > \bar{s}_Z$.

11.2 Three distinct targets

The finite return is target A: an actual acquisition changes a source, its response changes another acquisition, and the retained history remains in the same finite mean-action evolution. Target B is sustained thermal renewal of the body, controls and ports over growing runs while the independent unknown content stays held. Target C combines that renewal with a sharp complete exhaust and nonvanishing causal response gain. A does not prove B, and B alone does not prove C.

Conjecture C.1, stated in full in Appendix C, makes C quantitative. At fixed cycle rate and exhaust capacity, it requires complete-output sharpness, normalized whole-window response contrast, and a positive average gain at *all* declared decision opportunities on the actual-run history measure. Every sensor, program update, delay, buffer, work bank and current purifier enters the account; old inaccessible references are not reset. Resource size may grow with the run, while uncharged purity and correlation consumption have no positive asymptotic rate.

This target has real obstructions. Accurate direct purifier dumping retains cross-species redundancy and cannot make the complete Gibbs deficit vanish. Individually sharp program phases still pay a positive Jensen variance cost when their entropy rates differ at nonzero duty fractions. An exploration scheme whose only response updates occur on a vanishing fraction of opportunities has vanishing average gain, even with infinitely many such updates. These facts do not exclude every joint protocol; they exclude tempting combinations of successful components. The complete statements and resource clocks remain in Appendix C.

The following fixed-source converter supports one part of the target. It does not supply the adaptive source's product structure.

Proposition 11.2 (Finite-carrier sorting of the dephased source)**.** *Q-RAIF C's charge-sector sorter [14] applies to L independently realized copies of either ρ_p under its finite-carrier hypotheses: the source marginal is $\rho_p^{\otimes L}$, initially a product with the complete thermal carrier; their external purifications are retained; joint source operations act on both the receiver and its current R pair, while identity on old references refers to systems outside that quartet; the source is neutral in the original zero-mode quartet sector; the original neutral gates and paid loading, unloading and pipeline stages are available; and weighted energy tails control the implementation limit. It returns the source near its blank with entropy slack $o(L)$ and complete output Gibbs deficit $o(V)$, $V = L/\nu$, at*

$$T_{*,p} = T_H + \frac{6\nu h_p}{\pi c_{\mathrm{ex}}}, \qquad c_{\mathrm{ex}} = 4.$$

The same spectral argument applies to a supplied deterministic word with product marginal $\bigotimes_{j=1}^{L} \rho_{p_j}$ and actual entropy $s_L = \sum_j h_{p_j}$. The finite-bank target uses s_L plus the same subextensive margin. If $s_L/L \to s$, its temperature tends to $T_H + 6\nu s/(\pi c_{\mathrm{ex}})$, with no convergence rate required; otherwise there need not be one limiting temperature. This is a block converter, not an adaptive moving apparatus or a per-cycle heat-reading law.

The spectral proof is in Appendix H.1. It uses a supplied product marginal and controlled energy tails, not a local reset certificate for an adaptive history.

12 Relation to Existing Approaches

AdS/CFT [31] and its area–entropy relation [27] supply the dictionary used in Theorem 5.6; it is not derived here or granted to arbitrary selected algebras. MERA [32, 33] implements coarse-graining through disentanglers and isometries, with network depth resembling a radial AdS direction. Different networks can describe the same boundary data. Relating either construction to physical selection requires the explicit algebra-to-geometry map.

Jacobson's horizon Clausius argument [34] shares the entropy-based approach to gravitational dynamics. Section 8 addresses its source-side input: which selecting processes produce the stress whose response is computed. Loop quantum gravity [29, 35] instead quantizes the metric with background independence. Its relation to a path of selected algebras would require a representation map from spin-network states. These connections motivate the proposal without supplying its missing selection law.

13 Conclusion

The gravitational proposal is that physical histories form a localization structure whose slow evolution admits a common geometric law. Modular stability does not select that structure uniquely. A specified half-sided modular family supports conditional holographic matching; the conformal scalar bridge instead relates one prepared source and its Wald surface coefficient to an all-ball test in the same spacetime. Nonzero-frequency preparations can identify the intrinsic response within its stated class. The consistency condition itself still needs a physical explanation.

The source mechanism is explicit. Changed correction, protection and projection interactions change stress. A finite JT mean-action model retains the acquired quantum record, old reference, mechanics, clock, support and recoil, and proves a nonzero record-mediated source return. The spatial return and full-mode quartet bulk strengthen different interfaces. They are evidence for a common theory, not one already joined higher-dimensional apparatus.

Maintenance can favor one selected structure over another. Pointer alignment lowers the complete renewal debit, and a thermal chiral channel of fixed capacity places a quadratic rate price on its export. Matched horizon transport can read that price, provided the actual source, clock reference and heat routing are supplied. Sharp conversion of a fixed product source does not establish sustained renewal or sharp informative adaptive return.

The remaining mechanism is therefore specific: histories must form the accessible sector, the selecting interactions must carry their reaction stress, and one calibrated source must satisfy the same geometric consistency test across those histories. At second order the geometry, propagation, source and region prescription must vary together. A visibility filter cannot replace this dynamics, and changing an algebra at fixed stress cannot create a second first-order sourced Einstein tensor. Establishing this common selection–source–geometry law would turn the adiabatic algebra proposal from an interpretation of supplied gravity into a mechanism for its effective dynamics.

Certificate guide: HAFF-D technical supplement

The main argument and its core results remain in this paper. The complete technical appendices are in `HAFF_D_supplement.pdf`, *Sources, Return and Apparatus Certificates: Technical Supplement to HAFF-D*. This is supplementary documentation for the present paper, not a separate paper. Appendix letters and all existing result and equation numbers are retained. Cross-document references below link to that file; the supplement links back to the main argument.

Main claim or question	Technical certificate	Scope of support
Conditional modular route	Appendix E; Theorem E.5	Definitions and candidate checks, not a physical selection law.
Source changed by actual interaction	Appendix F; Propositions F.1–F.3	Source identification requires the stated reference permissions.
Finite source–response return	Appendices A, G; Propositions A.2, G.7	A fixed-window mean-action return; every source and clock debit remains.
Complete maintenance price	Appendix B; Proposition B.1	Correlations and the complete output deficit enter the export bill.
Horizon transport and clock	Appendix H; Propositions H.1, H.5	A matched source and running-clock reference are required.
Physical preparation and source	Appendix I; Proposition I.2	Finite preparation and the stated material/source matching.
Support, recoil and later reading	Appendix J; Proposition J.6	Retained support and old reference; not a fresh-body replacement.
Spatial and full-mode extensions	Sections J.7, J.8, J.9	Distinct layouts and domains; not one joined apparatus.
Sustained renewal and sharp return	Appendices C, D	Conjecture C.1 remains open; component certificates do not close targets B and C.

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Measurement as Accessibility: Measurement Permissions, Stable Records and Public Readout

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Abstract

A measurement context supplies probes, environmental dynamics and physical record ports. These determine different objects: implementable effects, persistent content and readable records. We compute their relations in a fresh-memory collision model. For two pure conditional fragment states of overlap r, the exact pointer-effect permission set is $\sqrt{ab} + \sqrt{(1-a)(1-b)} \geq r$; the closest permitted sharp pointer test has error $(1 - \sqrt{1-r^2})/2$. Its operator span does not determine this resolution. Holevo and single-measurement information count different numbers of readable records: one environment has 12 Holevo records but only 8 certified readable ones. Conditional product fragments give explicit public-agreement laws, and pooled one-sided local tests attain the joint pure-state discrimination exponent. Recording does not require a pure memory. We derive a purity-deficit budget and exhibit partially mixed qubits whose exact permissions form a parallelogram, whose local measurement saturates Holevo information, and whose reduced source channel is the same complete dephasing for every memory polarization. At zero polarization this channel produces no record. Reusing a spent port can erase its record and restore coherence, so freshness cannot be inferred from a channel label. For a common unknown reader orientation, agreement alone cannot certify correctness; prepared-pointer references give a finite calibration bound and an explicit residual common-error term. The same actual fragment readout supports Paper A's finite correlation-geometry certificate. The construction accounts for preparation, readout and shared standards without selecting a unique global outcome or deriving the Born rule.

1 Introduction

1.1 Three questions about measurement

Standard quantum mechanics gives rules for outcome probabilities but does not say how definite records arise from states that assign amplitudes to several possibilities. Collapse models change the dynamics, many-worlds accounts deny that one outcome is selected, and decoherence explains the suppression of interference without selecting a result. This

paper does not add to that list. It asks a set of narrower questions that every account has to answer in some form, and it answers them inside a declared physical context.

The first question is one of *permission*. A probe is a physical channel. Which effects on the system can the probe actually implement, given its coupling and initial state? The second concerns *stability*: which effects, or which record values, survive the environmental dynamics over the relevant time window? The third concerns *public readout*: when can several readers, each with access to a different part of the environment, obtain the same value, and how often is a shared value wrong?

The three answers are different kinds of object. Each probe's implementable effects form a convex set inside an operator system; a family of probes gives their union, which is convex only if the permitted family also includes the required randomized choices. Exact fixed points of a channel form an operator system, and under further hypotheses a von Neumann algebra. Records concern a classical label and the information that environmental fragments carry about it. Treating the three as one "accessible algebra" obtained by intersection loses the information that matters, and the paper keeps them apart.

The central claim of the paper is that measurement, in the operational sense used here, is a property of an interaction context: what a declared probe family can implement, what the declared dynamics leaves fixed, and what declared fragments record. We do not explain why one global outcome occurs, and we assume the Born rule.

The formation claim is nevertheless physical. Initially independent memories acquire correlations with a pointer through actual unitary couplings; their conditional distinguishability determines what a later reader can report. Record formation is therefore more than attaching a label to a supplied channel. Its limits are also physical: the same reduced dephasing may be produced with a port that records nothing, and a second contact can undo a record. The apparatus and elementary system–port split remain inputs. Their selection is the separate problem of Papers A and B, not an unexplained premise hidden inside the word *measurement*.

1.2 Results

Section 2 fixes the typed data of a context. Section 3 states the three tests and separates two notions of redundancy, one based on Holevo information and one based on what a measurement can read. Section 4 introduces a record model in which fresh memories collide once with a pointer, and computes:

- the exact set of pointer effects that a fragment can implement (Theorem 4.2) and the distance from a sharp pointer projection to that set (Corollary 4.3);
- a nesting of the three tests: fragment effects lie in the pointer algebra, which equals the fixed algebra of the induced dephasing (Proposition 4.4);
- the gap between Holevo and accessible information, with closed-form fragment lengths for both notions of redundancy (Propositions 4.5 and 4.6);
- a purity-deficit budget and exact nonzero readout from partially mixed memories (Propositions 4.7 and 4.8);

Section 5 turns to several readers: conditional independence, the error exponents of joint, pooled local and majority readout, and a finite calibrated public-readout bound when

the readers share an unknown orientation. Section 6 collects the model's conditions and interfaces. Section 6.2 compares the approach with decoherence theory, quantum Darwinism and other frameworks, and Section 6.3 identifies the actual record interface consumed elsewhere.

Several ingredients are standard: the exact Holevo quantity of a branching state, the Helstrom bound, and the role of redundancy in quantum Darwinism [10, 11, 12]. What the paper adds is the typed separation, the exact permission set of a fragment, the nesting of the tests in one dynamics, the counting difference between Holevo and readable records, the mixed-memory realization and a calibration-sensitive public-readout certificate.

2 Typed accessibility data

2.1 Algebraic background

In the algebraic formulation of quantum theory a system is described by an algebra of observables, and states are positive normalized functionals on it [8, 9]. In algebraic quantum field theory local algebras are attached to regions without a global tensor product, and tensor product structures can be induced by the accessible observables [20, 22]. HAFF builds on this view and asks which operator structures a physical context selects. Paper A treats the reduction of a global state by physical channels and shows that two channels can generate the same algebra while resolving different things [1]. The present paper applies the same caution to measurement.

2.2 The objects of a context

Let $\mathcal{M}_U$ be the ambient von Neumann algebra. A context $\mathbf{c}$ specifies allowed probe channels, an environmental dynamics, a record algebra, time scales and resolution thresholds.

Definition 2.1 (Probe channel and implementable effects)**.** *For each allowed probe α, let*

$$\mathcal{R}^*_{\alpha,\mathbf{c}} : \mathcal{B}(\mathcal{K}_\alpha) \longrightarrow \mathcal{M}_U$$

be its normal unital completely positive Heisenberg map. The set of ***implementable effects*** *of the context is*

$$\mathrm{Eff}(\mathbf{c}) := \{\mathcal{R}^*_{\alpha,\mathbf{c}}(F) : 0 \le F \le \mathbf{1},\ \alpha \text{ allowed}\},$$

and the ***probe-reachable operator system*** *is its ultraweakly closed linear span,*

$$\mathrm{Eff}_{\mathrm{span}}(\mathbf{c}) := \overline{\mathrm{span}}^{\,uw}\,\mathrm{Eff}(\mathbf{c}).$$

The family $\mathfrak{R}_{\mathrm{probe}}(\mathbf{c}) := \{\mathcal{R}_{\alpha,\mathbf{c}}\}_\alpha$ is part of the context, not an observable algebra.

For a single probe Eff is convex, contains 0 and $\mathbf{1}$, and is closed under $E \mapsto \mathbf{1} - E$. The span is the natural object for linear reconstruction of expectation values from many runs. It is not a list of effects that one run of an allowed probe implements: Section 4 gives probes whose span is the whole pointer algebra while the sharp pointer projections are not implementable. The same distinction appears for reductions in Paper A, where the effects a channel reads out, $\Phi^*([0, \mathbf{1}])$, are in general fewer than the positive contractions in the range of Φ^* [1].

Definition 2.2 (Environmental fixed set)**.** *For a normal UCP semigroup $\{\mathcal{E}_t\}_{t\geq 0}$ on $\mathcal{M}_U$, set*

$$\mathrm{Fix}(\mathcal{E}) := \{O \in \mathcal{M}_U : \mathcal{E}_t(O) = O \text{ for every } t \geq 0\}.$$

This is always an ultraweakly closed operator system. It is a von Neumann subalgebra, for example, when the semigroup admits a faithful normal invariant state: equality in the Schwarz inequality then places every fixed element in the multiplicative domain. A discrete-time channel is treated in the same way with $t \in \mathbb{N}$.

Definition 2.3 (Pointer and record algebra)**.** *A **pointer/record algebra** $\mathcal{P}_{\mathrm{env}} \subset \mathcal{M}_U$ is a designated commutative von Neumann algebra generated by a classical record PVM $Z(B)$, or by a finite-resolution coarse graining. For a finite label set, a fragment record channel acts on label states by*

$$\Gamma_{k,*}\big((p_x)_x\big) = \sum_x p_x \rho^x_{E_k}.$$

Its normal UCP Heisenberg dual is

$$\Gamma_k^* : \mathcal{B}(\mathcal{H}_{E_k}) \longrightarrow \mathcal{P}_{\mathrm{env}}, \qquad \Gamma_k^*(F) = \sum_x \mathrm{Tr}(\rho^x_{E_k} F) Z_x.$$

More generally the state map acts on the predual of the commutative record algebra. Approximate pointer observables are treated by their distance to $\mathcal{P}_{\mathrm{env}}$.

There is no default equality among Eff, $\mathrm{Fix}(\mathcal{E})$ and $\mathcal{P}_{\mathrm{env}}$. The declared maps are the probe maps, the dynamics, the inclusions of chosen subalgebras into $\mathcal{M}_U$, and the record channels. A model may establish relations among the objects, for example that every implementable effect lies in $\mathcal{P}_{\mathrm{env}} \cap \mathrm{Fix}(\mathcal{E})$; such relations are results or assumptions of that model.

Definition 2.4 (Measurement-accessibility profile)**.** *The **measurement-accessibility profile** of a context is the typed datum*

$$\mathbf{Acc}(\mathbf{c}) := \big(\mathrm{Eff}(\mathbf{c}),\ \mathrm{Eff}_{\mathrm{span}}(\mathbf{c}),\ \mathfrak{R}_{\mathrm{probe}}(\mathbf{c}),\ \mathcal{P}_{\mathrm{env}}(\mathbf{c}),\ \mathrm{Fix}(\mathcal{E}_{\mathbf{c}});\ \Gamma_{k,*}, T_{\mathbf{c}}, \epsilon, \delta\big). \tag{1}$$

The operator-system and fixed-set statements above hold for general von Neumann algebras under the hypotheses shown. Entropic statements are made for finite alphabets or finite-resolution labels and specified states.

3 Three tests

3.1 Probe reachability

Constraint 3.1 (Probe reachability)**.** *An effect $M(B) \in \mathcal{M}_U$ is implementable in context $\mathbf{c}$ if $M(B) = \mathcal{R}^*_{\alpha,\mathbf{c}}(F)$ for some allowed probe α and some probe effect $0 \leq F \leq \mathbf{1}$. A POVM M is implementable if one probe and one probe POVM F give $M(B) = \mathcal{R}^*_{\alpha,\mathbf{c}}(F(B))$ for every outcome set B.*

Implementability is a property of a channel and an effect. A probe Hamiltonian may realize the channel, but the condition $[O, H_{\mathrm{probe}}] \neq 0$ is neither necessary nor closed under sums and products; ideal nondemolition couplings commute with the measured pointer observable. Effects outside $\mathrm{Eff}(\mathbf{c})$ are not ruled out as physical quantities. The context simply supplies no implementation of them, and enlarging the context, for example by allowing more fragments or repeated runs, is a change of context.

3.2 Dynamical stability

Constraint 3.2 (Stability)**.** *An effect O passes the* ***finite-window stability test*** *if*

$$\sup_{0\le t\le T_{\mathbf{c}}} \|\mathcal{E}_t(O) - O\| < \epsilon. \tag{2}$$

Exact stability is the separate statement $O \in \mathrm{Fix}(\mathcal{E})$.

In decoherence theory the pointer observables are those left stable by the system–environment interaction [10]. The threshold ϵ and the window $T_{\mathbf{c}}$ are set by the recording mechanisms and time scales of the context. An implementable effect need not be stable, and an exact fixed point need not be implementable by any allowed probe.

3.3 Environmental redundancy: Holevo and readable records

Let X be the classical label of a pointer PVM, or of a finite coarse graining of a POVM, with distribution (p_x). For each environmental fragment E_k form the classical–quantum state $\rho_{XE_k} = \sum_x p_x |x\rangle\langle x| \otimes \rho^x_{E_k}$. Two quantities measure what E_k carries about X. The Holevo information

$$\chi(X : E_k) = S\Big(\sum_x p_x \rho^x_{E_k}\Big) - \sum_x p_x S(\rho^x_{E_k})$$

bounds what any measurement on E_k can extract. The accessible information

$$I_{\mathrm{acc}}(X : E_k) = \sup_{\text{POVM } N \text{ on } E_k} I(X : Y_N)$$

is what a single measurement on the fragment does extract at best, where Y_N is the outcome. Always $I_{\mathrm{acc}} \le \chi$ [14].

Constraint 3.3 (Redundancy)**.** *Fix $0 < \delta < 1$ and a family of disjoint fragments. The label X has R* ***Holevo records*** *if R disjoint fragments satisfy $\chi(X : E_k) \ge (1-\delta)H(X)$, and R* ***readable records*** *if R disjoint fragments satisfy $I_{\mathrm{acc}}(X : E_k) \ge (1-\delta)H(X)$. For a declared measurement N_k on each fragment one may use $I(X : Y_{N_k})$ in place of I_{acc}.*

Readable records are Holevo records, but not conversely, and the two counts can differ (Proposition 4.6). For continuous spectra X denotes a declared finite-resolution partition. No von Neumann entropy is assigned to an operator, and no expression such as $I(\hat{O} : E_k)$ is used.

Redundancy is what allows several observers to read the same label from different fragments without disturbing one another. It does not by itself say how often their common value is wrong, nor which global outcome is realized. Section 5 computes the former in the model.

4 A record model in which the three tests can be computed

4.1 Fresh memories colliding once with a pointer

The system S is a qubit with pointer projections $P_x = |x\rangle\langle x|$, $x \in \{0, 1\}$. The environment consists of n memory qubits $E_1, \ldots, E_n$, each prepared in $|0\rangle$. The k-th memory collides

once with S through

$$U_k = P_0 \otimes \mathbf{1} + P_1 \otimes R_k, \qquad R_k = \begin{pmatrix} c_k & -\sqrt{1-c_k^2} \\ \sqrt{1-c_k^2} & c_k \end{pmatrix}, \qquad 0 \le c_k \le 1, \tag{3}$$

and does not interact again. Starting from $\sqrt{p}\,|0\rangle + \sqrt{1-p}\,|1\rangle$ with $0 < p < 1$, the state after n collisions is

$$|\Psi_n\rangle = \sqrt{p}\,|0\rangle|e_0\rangle + \sqrt{1-p}\,|1\rangle|e_1\rangle, \qquad |e_x\rangle = \bigotimes_{k=1}^{n} |e_{x,k}\rangle, \tag{4}$$

with $|e_{0,k}\rangle = |0\rangle$ and $|e_{1,k}\rangle = R_k|0\rangle$, so that $\langle e_{0,k}|e_{1,k}\rangle = c_k$. A fragment F is a declared set of memories; write

$$r_F = \prod_{k\in F} c_k, \qquad H_p(r) = h\left(\frac{1+\sqrt{1-4p(1-p)(1-r^2)}}{2}\right),$$

where $h(x) = -x\ln x - (1-x)\ln(1-x)$. All entropies are in nats. The label X is the value a pointer measurement on S would return.

The inputs are explicit: the pointer basis, the split into system and memories, the order of collisions, the fragment partition, and the purity of the memories. None of them is derived here. The model is a version of the standard branching model of quantum Darwinism [11, 12]; its role in this paper is to make all components of the profile of Definition 2.4 computable at once.

Proposition 4.1 (Record quantities)**.** *For every fragment F,*

$$\chi(X : E_F) = H_p(r_F), \qquad (\rho_S)_{01} = \sqrt{p(1-p)}\, r_{[n]},$$

$$S(\rho_S) = H_p(r_{[n]}), \qquad I(S : E_{[n]}) = 2H_p(r_{[n]}).$$

Proof. Given $X = x$ the fragment is in the pure state $|e_x^F\rangle = \bigotimes_{k\in F} |e_{x,k}\rangle$, and $\langle e_0^F|e_1^F\rangle = r_F$. The average state $p|e_0^F\rangle\langle e_0^F| + (1-p)|e_1^F\rangle\langle e_1^F|$ has two nonzero eigenvalues with sum 1 and product $p(1-p)(1-r_F^2)$, so its entropy is $H_p(r_F)$, and the conditional entropies vanish. Tracing the memories out of (4) multiplies the off-diagonal element by $\langle e_1|e_0\rangle = r_{[n]}$, and the global state is pure, so $S(\rho_S) = S(\rho_{E_{[n]}}) = H_p(r_{[n]})$ and $I(S : E_{[n]}) = 2S(\rho_S)$. □

The reduced dynamics of S is the dephasing channel $\mathcal{D}_k(\rho) = P_0\rho P_0 + P_1\rho P_1 + c_k(P_0\rho P_1 + P_1\rho P_0)$ at each step. Its fixed algebra, for any $c_k < 1$, is the diagonal algebra $\mathcal{P} = \mathrm{span}\{P_0, P_1\}$, which is also the pointer algebra.

4.2 Measurement permissions of a fragment

A reader with access to fragment F implements effects on S through the Heisenberg map

$$\Phi_F^*(A) = \langle e_0^F|A|e_0^F\rangle\, P_0 + \langle e_1^F|A|e_1^F\rangle\, P_1, \qquad A \in \mathcal{B}(\mathcal{H}_F), \tag{5}$$

obtained by coupling S to F through the collisions and tracing out the rest. The span of the image is all of $\mathcal{P}$ as soon as $0 < r_F < 1$, since it contains $\mathbf{1}$ and the operator $\Phi_F^*(|e_0^F\rangle\langle e_0^F|) = P_0 + r_F^2 P_1$. The implementable effects are fewer.

Theorem 4.2 (Permission set of a fragment)**.** *Let* $|e_0\rangle, |e_1\rangle$ *be unit vectors with* $|\langle e_0|e_1\rangle| = r$*. Then*

$$\begin{aligned}&\big\{(\langle e_0|A|e_0\rangle, \langle e_1|A|e_1\rangle) : 0 \le A \le \mathbf{1}\big\}\\ &\qquad = L_r := \big\{(a,b) \in [0,1]^2 :\ \sqrt{ab} + \sqrt{(1-a)(1-b)} \ge r\big\}.\end{aligned}$$

Hence $\mathrm{Eff}_F := \Phi_F^*([0,\mathbf{1}]) = \{aP_0 + bP_1 : (a,b) \in L_{r_F}\}$.

Proof. Inclusion. For an effect A put $a = \langle e_0|A|e_0\rangle$ and $b = \langle e_1|A|e_1\rangle$. By the Cauchy–Schwarz inequality applied to $A^{1/2}e_0$ and $A^{1/2}e_1$, $|\langle e_0|A|e_1\rangle| \le \sqrt{ab}$, and likewise $|\langle e_0|(\mathbf{1}-A)|e_1\rangle| \le \sqrt{(1-a)(1-b)}$. Adding and using the triangle inequality gives $r \le \sqrt{ab} + \sqrt{(1-a)(1-b)}$.

Equality. The function $f(a,b) = \sqrt{ab} + \sqrt{(1-a)(1-b)}$ is a sum of geometric means, hence concave, so L_r is a compact convex set. The set of attained pairs is convex as the image of the convex set of effects under an affine map. At $r = 0$ the states are orthogonal and all pairs in the square are attained by diagonal effects; at $r = 1$ the states are identical up to phase and the attained set is the diagonal. For $0 < r < 1$, a compact convex subset of the plane is the convex hull of its boundary, so it suffices to attain every boundary point of L_r. Choose phases and an orthonormal basis of $\mathrm{span}\{e_0, e_1\}$ with $e_0 = (1,0)$ and $e_1 = (\cos\theta, \sin\theta)$, $\cos\theta = r$, $0 \le \theta \le \pi/2$; effects supported on this span suffice. For $u = (\cos\varphi, \sin\varphi)$ with $\theta - \pi/2 \le \varphi \le 0$, the projection $|u\rangle\langle u|$ gives $a = \cos^2\varphi$ and $b = \cos^2(\theta - \varphi)$, and since $\cos\varphi \ge 0$, $\sin\varphi \le 0$, $\cos(\theta-\varphi) \ge 0$ and $\sin(\theta-\varphi) \ge 0$,

$$f(a,b) = \cos\varphi\cos(\theta-\varphi) - \sin\varphi\sin(\theta-\varphi) = \cos\theta = r.$$

This traces the arc of $\{f = r\}$ from $(1-r^2, 0)$ to $(1, r^2)$. The map $A \mapsto \mathbf{1} - A$ sends it to the arc from $(r^2, 1)$ to $(0, 1-r^2)$. The remaining boundary consists of segments of the square: $b = 0$ with $0 \le a \le 1 - r^2$, attained by $A = t|e_1^\perp\rangle\langle e_1^\perp|$; $a = 1$ with $r^2 \le b \le 1$, attained by $A = |e_0\rangle\langle e_0| + t|e_0^\perp\rangle\langle e_0^\perp|$, $0 \le t \le 1$; and their images under $A \mapsto \mathbf{1} - A$. On each edge of the square f reduces to $\sqrt{1-a}$, $\sqrt{b}$, $\sqrt{1-b}$ or $\sqrt{a}$, so these segments are exactly the parts of the edges in L_r. For $r < 1$, $f(a,a) = 1 > r$, so the level set $\{f = r\}$ inside the square consists of the two arcs. Every boundary point is therefore attained. □

For $0 < r < 1$ the permission set is bounded by two curved arcs and four square-edge segments. The corners $(0,0)$ and $(1,1)$ lie on those segments, not on the arcs $f = r$. It fills the square when $r = 0$ and collapses to the diagonal $\{a = b\}$, the trivial effects $a\mathbf{1}$, when $r = 1$. For $0 < r < 1$ it contains no sharp pointer projection: $(1,0) \notin L_r$ because $f(1,0) = 0$.

Corollary 4.3 (Closest implementable pointer test)**.** *For* $0 \le r \le 1$*,*

$$\min_{E \in \mathrm{Eff}_F} \|E - P_0\| = \varepsilon(r) := \frac{1 - \sqrt{1-r^2}}{2},$$

with $r = r_F$*, attained at* $E = (1-\varepsilon)P_0 + \varepsilon P_1$*. The value* $\varepsilon(r)$ *is the minimal error probability for distinguishing* $|e_0^F\rangle$ *from* $|e_1^F\rangle$ *with equal priors.*

Proof. For $E = aP_0 + bP_1$ one has $\|E - P_0\| = \max(1-a, b)$. If $\max(1-a, b) \le t < 1/2$, then $a \ge 1 - t > 1/2 > t \ge b$. In that region $\partial f/\partial a = \frac{1}{2}\big(\sqrt{b/a} - \sqrt{(1-b)/(1-a)}\big) < 0$

and $\partial f/\partial b = \frac{1}{2}\big(\sqrt{a/b} - \sqrt{(1-a)/(1-b)}\big) > 0$, so $f(a,b) \le f(1-t,t) = 2\sqrt{t(1-t)}$. Since $2\sqrt{\varepsilon(1-\varepsilon)} = r$ and $t \mapsto 2\sqrt{t(1-t)}$ is increasing on $[0,1/2]$, the region contains points of L_r only if $t \ge \varepsilon(r)$, and $(1-\varepsilon,\varepsilon) \in L_r$. The identification with the Helstrom error is the formula $(1-\sqrt{1-4p(1-p)r^2})/2$ at $p = 1/2$ [15]. □

For a fragment of m memories with equal overlaps c, $r_F = c^m$ and $\varepsilon(c^m) = c^{2m}/4 + O(c^{4m})$. Measurement permission for the pointer grows with the number of memories a reader controls, and an η-approximate pointer test becomes available once $c^{2m} \le 4\eta(1-\eta)$. Enlarging F is a change of context, in the sense of Constraint 3.1.

4.3 The three tests in one dynamics

Proposition 4.4 (Nesting)**.** *Let a probe act through a pointer-controlled unitary $U = \sum_x P_x \otimes V_x$ on $\mathcal{H}_S \otimes \mathcal{K}$ with probe state $|\phi\rangle$. Then for every $A \in \mathcal{B}(\mathcal{K})$*

$$\mathcal{R}^*(A) = \sum_x \langle\phi|V_x^* A V_x|\phi\rangle\, P_x \in \mathcal{P}_{\mathrm{env}}.$$

In the record model, for every fragment F with $0 < r_F < 1$ and every $c_k < 1$,

$$\mathrm{Eff}_F \subsetneq \mathrm{Eff}_{\mathrm{span},F} \cap [0,\mathbf{1}] = \mathcal{P} \cap [0,\mathbf{1}] = \mathrm{Fix}(\mathcal{D}_k) \cap [0,\mathbf{1}].$$

Proof. $U^*(\mathbf{1}\otimes A)U = \sum_{x,x'} P_x P_{x'} \otimes V_x^* A V_{x'} = \sum_x P_x \otimes V_x^* A V_x$, and the partial expectation in $|\phi\rangle$ gives the formula. In the model $\mathrm{Eff}_{\mathrm{span},F} = \mathcal{P}$ as noted before Theorem 4.2, the fixed algebra of $\mathcal{D}_k$ is $\mathcal{P}$, and Eff_F is the lens of Theorem 4.2, which omits P_0. □

In this model every effect that the environment can report is a stable pointer effect, and the stable pointer effects are exactly the positive contractions of the span. The tests still differ. The lens measures resolution, the fixed algebra measures persistence, and redundancy, below, counts how many disjoint readers have access. A probe that is not pointer-controlled, for example one that rotates the system before recording, can reach effects outside $\mathrm{Fix}(\mathcal{D}_k)$; those effects are then not stable under the environment that produced the records.

4.4 Holevo records and readable records

Proposition 4.5 (Accessible information of a fragment)**.** *Let*

$$P_e(r) = \frac{1 - \sqrt{1 - 4p(1-p)r^2}}{2}$$

be the Helstrom error for two pure states with overlap r and prior $(p, 1-p)$. For every fragment,

$$I_{\mathrm{acc}}(X : E_F) = h(p) - h(P_e(r_F)) \le \chi(X : E_F) = H_p(r_F).$$

For $p = 1/2$,

$$I_{\mathrm{acc}}(X : E_F) = \ln 2 - h\big(\varepsilon(r_F)\big), \qquad \chi(X : E_F) = h\Big(\frac{1+r_F}{2}\Big),$$

and the two differ whenever $0 < r_F < 1$.

Proof. This binary pure-state optimum is standard [16, Sec. 3.5.4]; the following argument proves it for all POVMs. Write their conditional outcome laws as a_y, b_y, put $q_y = pa_y + (1-p)b_y$, and omit zero-weight outcomes. Their posterior probabilities are $\pi_y = pa_y/q_y$; set $C_y = 2\sqrt{\pi_y(1-\pi_y)}$ and

$$E(C) = h\Big(\frac{1-\sqrt{1-C^2}}{2}\Big).$$

Then $H(X \mid Y) = \sum_y q_y E(C_y)$. For $t = \sqrt{1-C^2}$, $E'(C) = C\,\mathrm{artanh}(t)/t \geq 0$ and $E''(C) = (\mathrm{artanh}(t) - t)/t^3 \geq 0$ for $0 < C < 1$; continuity extends convexity to the endpoints. The outcome-wise Cauchy–Schwarz argument of Theorem 4.2 gives

$$\sum_y q_y C_y = 2\sqrt{p(1-p)} \sum_y \sqrt{a_y b_y} \geq 2\sqrt{p(1-p)}\, r_F.$$

Jensen and monotonicity imply $H(X \mid Y) \geq E(2\sqrt{p(1-p)}r_F) = h(P_e(r_F))$. Diagonalizing $p|e_0^F\rangle\langle e_0^F| - (1-p)|e_1^F\rangle\langle e_1^F|$ shows that its Helstrom measurement has posterior error $P_e(r_F)$ on every nonzero outcome, so it attains the bound. The cases $r_F = 0, 1$ follow directly. The Holevo bound and value are those of Proposition 4.1. At equal priors the displayed forms agree only at $r_F = 0, 1$. □

At $p = 1/2$ and $r_F = 0.8$, $\chi = 0.325083$ while $I_{\mathrm{acc}} = 0.192745$. With $\delta = 0.6$ the threshold $(1-\delta)\ln 2 = 0.277259$ lies between them: the fragment is a Holevo record and not a readable one. The threshold for a Holevo record and the threshold for a measurement-based record are therefore not equivalent, even when the measurement is optimal; the difference concerns one measurement on one fragment, and joint coding over many independent preparations is a different task.

Proposition 4.6 (Fragment lengths)**.** *Let all $c_k = c \in (0,1)$ and $0 < \delta < 1$, and let h^{-1} denote the inverse of h on $[0, 1/2]$. Put*

$$q_\delta = h^{-1}\big((1-\delta)h(p)\big), \qquad a_\chi = \frac{(1-2q_\delta)^2 - (2p-1)^2}{4p(1-p)},$$

$$e_\delta = h^{-1}\big(\delta h(p)\big), \qquad a_{\mathrm{op}} = \frac{e_\delta(1-e_\delta)}{p(1-p)}.$$

A fragment of m memories is a Holevo record if and only if $c^{2m} \leq a_\chi$, and it is a readable record if and only if $c^{2m} \leq a_{\mathrm{op}}$. The smallest such lengths are

$$m_\chi = \Big\lceil \frac{\ln a_\chi}{2\ln c} \Big\rceil, \qquad m_{\mathrm{op}} = \Big\lceil \frac{\ln a_{\mathrm{op}}}{2\ln c} \Big\rceil,$$

so m_{op} is the smallest readable length for every $0 < p < 1$.

Proof. $H_p(r)$ is a decreasing function of r^2, and $H_p(r) = (1-\delta)h(p)$ exactly when the larger eigenvalue $\lambda = (1 + \sqrt{1 - 4p(1-p)(1-r^2)})/2$ equals $1 - q_\delta$, that is when $r^2 = a_\chi$. For the readable case, Proposition 4.5 gives $I_{\mathrm{acc}} \geq (1-\delta)h(p)$ exactly when $h(P_e(r)) \leq \delta h(p)$, that is when $P_e(r) \leq e_\delta$, which rearranges to $4p(1-p)r^2 \leq 4e_\delta(1-e_\delta)$. Since $e_\delta \leq \min(p, 1-p)$, $a_{\mathrm{op}} \leq 1$. □

When fragments are unequal, the counts depend on the chosen partition. With equal overlaps, m_χ gives the largest Holevo count. m_{op} gives the largest readable count for the same pure binary ensemble at every prior.

p	c	n	δ	m_χ	m_{op}	Holevo records	certified readable
0.5	0.80	60	0.10	5	7	12	8
0.2	0.80	60	0.10	5	7	12	8
0.5	0.95	100	0.05	27	38	3	2

Table 1: Fragment lengths and the number of disjoint equal-length records in n memories. The binary pure-state optimum in Proposition 4.5 makes every listed m_{op} minimal over all POVMs. At $p = 0.2$, I_{acc} is .439321 for $m = 6$ and .458263 for $m = 7$, across the threshold .450362.

4.5 The preparation resource for a record

Proposition 4.7 (Purity-deficit budget)**.** *Replace the memories by systems of dimension d_k in product states $\bigotimes_k \sigma_k$, and let the collisions be pointer-controlled unitaries $P_0 \otimes \mathbf{1} + P_1 \otimes W_k$. Then for every fragment*

$$\chi(X : E_F) = S\Big(p \bigotimes_{k\in F} \sigma_k + (1-p) \bigotimes_{k\in F} W_k \sigma_k W_k^*\Big) - \sum_{k\in F} S(\sigma_k) \le \sum_{k\in F} \big[\ln d_k - S(\sigma_k)\big],$$

while each collision multiplies the off-diagonal element $(\rho_S)_{01}$ by $\mathrm{tr}(\sigma_k W_k^*)$. If $\sigma_k = \mathbf{1}/2$ and $W_k = R_k$, no fragment carries any information about X, and the coherence of S is still multiplied by c_k.

Proof. The two conditional states of the fragment are unitarily related products, so both have entropy $\sum_{k\in F} S(\sigma_k)$; the average state has entropy at most $\sum_{k\in F} \ln d_k$. The block $P_0\rho_S P_1 \otimes \sigma_k$ is mapped to $P_0\rho_S P_1 \otimes \sigma_k W_k^*$, so tracing out the memory multiplies $(\rho_S)_{01}$ by $\mathrm{tr}(\sigma_k W_k^*)$. For $\sigma_k = \mathbf{1}/2$ both conditional states equal $\mathbf{1}/2$, and $\mathrm{tr}(R_k^*)/2 = c_k$. □

The initial purity deficits bound the information that these unitary ports can record. They do not require every memory to be pure. Dephasing is not evidence of a readable record: a maximally mixed port can generate the same source channel as a pure port and carry no label information. Paper F uses this resource budget as a quantitative low-record boundary condition [5]. Preparation, correlation and renewal must then be accounted for separately; this entropy bound alone is not a universal work price.

4.6 Partially mixed memories: exact permission and readout

Let each fresh qubit memory start in $\eta_\alpha = (I + \alpha Z)/2$, $0 \le \alpha \le 1$, and use

$$U = P_0 \otimes I + P_1 \otimes X.$$

The conditional states are $\eta_0 = \eta_\alpha$ and $\eta_1 = X\eta_\alpha X = (I - \alpha Z)/2$. Put $\varepsilon_\alpha = (1-\alpha)/2$.

Proposition 4.8 (A mixed-memory record)**.** *For one fresh memory, its exact effect permission set is*

$$\mathrm{conv}\{(0,0), (1,1), (1-\varepsilon_\alpha, \varepsilon_\alpha), (\varepsilon_\alpha, 1-\varepsilon_\alpha)\}. \tag{6}$$

For $\alpha > 0$ the effect span is the full pointer algebra, but the closest permitted P_0 test has distance ε_α. At equal label priors,

$$\chi(X : E) = I_{\mathrm{acc}}(X : E) = \ln 2 - h((1+\alpha)/2) = \ln 2 - S(\eta_\alpha). \tag{7}$$

Measurement of Z_E attains this value and has symmetric error ε_α. For every α, tracing the memory completely dephases the source.

Proof. Only the diagonal entries $u, v \in [0,1]$ of a memory effect matter. The reported pair is $((1-\varepsilon_\alpha)u + \varepsilon_\alpha v, \varepsilon_\alpha u + (1-\varepsilon_\alpha)v)$. Diagonal effects attain every such pair, proving the parallelogram. For any reported pair (a,b), $a - b = \alpha(u-v) \le \alpha$; hence $\max(1-a, b) \ge (1-\alpha)/2$, with equality at $u = 1, v = 0$. The conditional states commute, have entropy $h((1+\alpha)/2)$ and average to $I/2$. Their common eigenbasis measurement attains the Holevo quantity. Finally the source coherence multiplier is $\mathrm{Tr}(\eta_\alpha X) = 0$. □

Records in partially mixed spin environments are already part of quantum Darwinism [12]. For $0 < \alpha < 1$ this construction gives their exact effect permissions and an explicit readout saturating the one-port purity-deficit budget. At $\alpha = 0$ its effects collapse to the trivial diagonal (a, a) and it carries no record, although its source channel is unchanged. At $\alpha = 1$ it records perfectly. Thus neither an effect span nor the complete source channel fixes record quality.

After m fresh collisions, the conditional memory states are $\eta_x^{\otimes m}$. Measuring every Z_E reads the entire commuting ensemble and still saturates Holevo information. The reports are binary independent errors conditional on x. Majority vote has exact error

$$e_m = \Pr\{\mathrm{Bin}(m, \varepsilon_\alpha) > m/2\} + \tfrac{1}{2}\Pr\{\mathrm{Bin}(m, \varepsilon_\alpha) = m/2\} \le e^{-m\alpha^2/2}. \tag{8}$$

The bound is Hoeffding's for the sample mean of signs in $[-1, 1]$. Fano's inequality then gives a sufficient readable-record length: choose m so that the bound is at most $1/2$ and $h(e^{-m\alpha^2/2}) \le \delta \ln 2$. This constructive mixed-state extension needs no classification of all mixed conditional-state permission sets.

4.7 A spent port is not a fresh channel input

The same controlled-X unitary also makes the port distinction exact. Since $U^2 = I$, applying it a second time to the same memory, without intervening measurement, export or dynamics, returns the complete state to its pre-collision value. The first collision dephases the source and writes the α-dependent record; the second removes that record and restores the original source coherence. Two independent fresh collisions instead preserve complete source dephasing and leave two conditional memory copies. This reversal does not erase an external measurement record or a copy already transferred to another port.

The one-step reduced channel cannot be applied again to the source while silently replacing the spent memory by η_α. After the first collision the source and memory are correlated. A second actual interaction acts on that joint state, including its old purification. Paper F studies repeated contact and joint memory updating [5]; the distinction also applies to the retained-port acquisition used by Q-RAIF C [4]. Storage, overwrite, reset and acquisition are different operations even when their isolated channel labels are similar.

5 Public readout

5.1 Agreement, correctness and common error

Proposition 5.1 (Conditional independence of readers)**.** *Let readers $j = 1, \ldots, J$ apply POVMs $N^{(j)}$ to disjoint fragments F_j. Then the joint distribution of the pointer value*

and the outcomes is

$$\Pr(x, y_1, \ldots, y_J) = p_x \prod_{j=1}^{J} \langle e_x^{F_j} | N_{y_j}^{(j)} | e_x^{F_j} \rangle.$$

For two readers who each report a value $\hat{x}_j(y_j) \in \{0, 1\}$ *with error probabilities* $\varepsilon_j(x) = \Pr(\hat{x}_j \neq x \mid x)$*, the agreement rate* A*, the correctness rates* C_j *and the common error rate* $E = \Pr(\hat{x}_1 = \hat{x}_2 \neq x)$ *are*

$$E = \sum_x p_x \varepsilon_1(x) \varepsilon_2(x), \qquad A = \sum_x p_x \big[(1 - \varepsilon_1(x))(1 - \varepsilon_2(x)) + \varepsilon_1(x)\varepsilon_2(x)\big],$$

and $C_1 + C_2 = 1 + A - 2E$*, so that* $\Pr(\textit{correct} \mid \textit{agree}) = 1 - E/A$*.*

Proof. Tracing S out of (4) gives $\rho_E = \sum_x p_x |e_x\rangle\langle e_x|$ with $|e_x\rangle$ a product over memories. A pointer measurement on S commutes with the fragment measurements and returns x with probability p_x, leaving the memories in $|e_x\rangle$. The product form of $|e_x\rangle$ over disjoint fragments gives the factorization. The formulas for E and A follow by summing over the four combinations of right and wrong reports, and the identity $C_1 + C_2 = 1 + A - 2E$ holds event by event: an agreeing correct pair contributes two correct reports, an agreeing wrong pair none, and a disagreeing pair one. □

With equal priors, $c = 0.8$ and two five-memory fragments read with their Helstrom measurements, each reader errs with probability $\varepsilon(0.8^5) = 0.02761$, the readers agree with probability $A = 0.94631$, and an agreeing pair is wrong with probability $E/A = 8.05 \times 10^{-4}$. In this model agreement certifies correctness up to the product of private errors. The identity $C_1 + C_2 = 1 + A - 2E$ and the separation of agreement from correctness are the subject of Foundation II, where a contamination shared by two instruments raises the common error far above the product of private errors [7]. Independent private errors already produce a nonzero common error. An excess beyond the conditional-product formula requires an additional shared influence or a failure of that record law, such as shared instrument contamination or calibration error. T-DOME III makes this precise: under a symmetry acting on all readers at once, their relative sign is detectable from agreement and a common inversion is not, so the error that agreement can never expose is the direction shared by the readers' standards [21].

5.2 Calibrated public readout with a common orientation

Conditional independence is a property of the prepared ports and their actual instruments. It is not implied by agreement or commuting effects. The mixed-memory family gives an explicit calibration problem. Encode the pointer value as $z = (-1)^x$. Suppose all readers share an unknown fixed orientation $C \in \{-1, 1\}$, so each reported elementary sign has conditional mean $C\alpha z$. The physical response $\alpha > 0$ is supplied and stable across reference and target runs. A common inversion leaves every agreement rate unchanged. With equal target priors and no independently anchored reference, the transformations $(C, z) \mapsto (-C, -z)$ give the same observable target-record law, so the absolute orientation is unidentifiable. A known asymmetric source prior would be additional orientation information and a different calibration problem.

The context must supply a preparation of the physical projection P_0 whose label is anchored independently of the unknown reader orientation. Its verification and preparation are resources. If the preparation label is itself inverted by the same unknown C,

the reference cannot remove the ambiguity. Given that permission, prepare $z = +1$ in n_{ref} independent reference trials and measure fresh mixed ports with the same physical C response used for target readout. Estimate C by the sign of their mean, breaking a tie at random. Hoeffding gives

$$\Pr\Big(\widehat{C} \neq C\Big) \leq \delta_{\mathrm{cal}} := \min\{1, e^{-n_{\mathrm{ref}}\alpha^2/2}\}. \tag{9}$$

A change of orientation or response between reference and target runs violates the premise; the bound does not detect arbitrary shared contamination.

Proposition 5.2 (Finite public correctness after calibration)**.** *Let J readers each control m fresh mixed-memory ports of the same target pointer. They measure the memory Z values, take a majority with independent fair tie coins, and multiply their decision sign by $\widehat{C}$. The reference trials are independent of the target ports. With e_m from* (8)*,*

$$A_J := \Pr(\textit{all readers agree}) = (1-e_m)^J + e_m^J, \qquad \Pr(\textit{all agree wrongly}) \leq \delta_{\mathrm{cal}} + e_m^J. \tag{10}$$

Consequently

$$\Pr(\textit{wrong} \mid \textit{all agree}) \leq \min\{1, (\delta_{\mathrm{cal}} + e_m^J)/A_J\}.$$

Proof. Correcting by a common sign does not change agreement. Conditional on the pointer and the true orientation, the fresh ports give independent symmetric majority errors of probability e_m. On the event $\widehat{C} = C$, all reports are wrong with probability e_m^J; on its complement this probability is at most one. Reference–target independence gives the stated bounds. □

For $\alpha = 0.6$, $m = 5$, $J = 2$, the exact private error is $e_5 = 0.05792$ and $A_2 = 0.8908694528$. With $n_{\mathrm{ref}} = 100$ the calibration failure bound is e^{-18}. Even after calibration, the conditional common-error bound is about 0.003766. Increasing the number of readers suppresses their independent mistakes, but a fixed shared calibration error does not disappear through agreement.

The protocol consumes n_{ref} prepared-pointer references and fresh mixed ports, Jm target ports, all corresponding readout bits, access to a common orientation standard, and the communication needed for calibration and agreement. These are prepared resources, not an inference of correctness from consensus. The preparation cost is not identified with the entropy deficit alone. T-DOME III analyzes the remaining common-sign ambiguity [21]; Foundation II distinguishes a shared state or instrument from an independently checked source [7]. Here the reference permission and its finite error are part of one explicit record protocol.

5.3 Many readers: error exponents

Proposition 5.3 (Joint, pooled and majority readout)**.** *Let R disjoint fragments each have overlap $r \in (0, 1)$, and let the prior be $(p, 1-p)$.*

1. *Any measurement on the union of the fragments errs with probability at least $P_e(r^R) = p(1-p)r^{2R} + O(r^{4R})$, and a joint measurement attains this value.*

2. *Let each reader measure $\{|e_0^{F_j}\rangle\langle e_0^{F_j}|,\ \mathbf{1} - |e_0^{F_j}\rangle\langle e_0^{F_j}|\}$ on its own fragment, and decide $\hat{x} = 1$ if at least one reader obtains the second outcome. This pooled local rule errs with probability $(1-p)\, r^{2R}$.*

3. *Let $p = 1/2$, let each reader use its single-fragment Helstrom measurement, and let the decision be the majority of the R reports, with ties broken by a fair coin. The error probability is $r^{R+o(R)}$, and no rule that uses only the R reports has a better exponent.*

Proof. (1) The conditional states of the union are pure with overlap r^R, and the Helstrom bound is attained by the projective measurement onto the positive part of $p|e_0\rangle\langle e_0| - (1-p)|e_1\rangle\langle e_1|$ [15]. (2) Given $x = 0$, every reader obtains the first outcome with certainty. Given $x = 1$, reader j obtains it with probability $|\langle e_0^{F_j}|e_1^{F_j}\rangle|^2 = r^2$, independently by Proposition 5.1, so all R readers obtain it with probability r^{2R}. (3) Each report is wrong with probability $\varepsilon = \varepsilon(r)$ independently. With equal priors and symmetric errors the likelihood ratio of the reports depends only on the number of reports for each value, so the majority rule is optimal among rules that use the reports. Its error is $\Pr(\mathrm{Bin}(R,\varepsilon) > R/2)$ plus half the probability of a tie, which equals $\exp\big[-R\,D(\frac{1}{2}\|\varepsilon) + O(\ln R)\big]$ with $D(\frac{1}{2}\|\varepsilon) = -\ln\big(2\sqrt{\varepsilon(1-\varepsilon)}\big)$ by the standard large-deviation estimate for binomial tails. Finally $4\varepsilon(1-\varepsilon) = 1 - (1-r^2) = r^2$. □

Local readers who only pool their results afterwards thus reach the joint error exponent $2|\ln r|$ per fragment, with a prefactor larger by $1/p$. At $p = 1/2$ and $r = 0.8$, twenty fragments give a joint error of 3.32×10^{-5}, a pooled one-sided error of 6.65×10^{-5}, and a majority error of 1.58×10^{-3}. Readers who are allowed to choose each measurement after learning the earlier outcomes can attain the joint Helstrom error exactly for pure conditional states [17].

The pooled rule requires a matched implementation of the projector onto $|e_0^{F_j}\rangle$. Knowledge of the coupling is one way to supply it; a trusted prepared reference or a previously calibrated fixed reader is another. An incorrect coupling model generally sets up the wrong test. When that relation is unknown, calibration records can change the actual readout permission. T-DOME I computes the value of a readout model in a specified thermodynamic task [6]; the present exponent does not require every successful fixed reader to carry an explicit self-model.

6 Conditions and interfaces

6.1 Preparation and reading permissions

The carrier, pointer basis, collision order, preparation maps and fragment partition enter explicitly. The pure-memory model gives a noncommuting two-state permission lens; the mixed-memory model gives a commuting parallelogram and readable records without complete purity. Both use independent fresh ports. Retained ports must instead be evolved with their actual old correlations, as Section 4.7 demonstrates.

Reader permissions also matter. A one-sided pure-state test requires access to its conditional state vector. The orientation protocol requires an independently anchored pointer preparation and the same stable response in reference and target trials. Neither capability follows from membership of an effect in a generated algebra. Arbitrary shared errors and preparation–reader co-inversions remain outside that calibration certificate.

The global branching state remains a superposition. Born probabilities determine the label distribution used in the readout laws; no unique global outcome is selected by these

calculations. A reader can be any physical device controlling a fragment. The model does not privilege a conscious observer.

Einselection and Paper B's stability analysis address how a given interaction chooses a pointer or protected sector [10, 2]. Paper B's formation constructions add real shared recording ports and Hamiltonian links. Locality and the history of those links must still explain why a particular fragment partition is available. Paper F addresses updating and repeated contact [5]; Q-RAIF C supplies finite acquisition on a carrier retaining quantum history [4]. These are complementary realizations, not permission to replace one model's initial states by another's without a physical bridge.

6.2 Relation to existing approaches

Decoherence and quantum Darwinism. Decoherence theory explains the suppression of interference through entanglement with an environment and identifies pointer observables by their stability [10]. Quantum Darwinism adds that objectivity requires redundant records in many fragments [11], and the exact information content of branching states in spin environments has been analysed in detail [12]. The present paper uses these results and adds the typed distinctions: the permission set of a fragment, the nesting of permission, stability and redundancy in one dynamics, and the difference between Holevo and readable records.

Spectrum broadcast structures. The conditional product form used in Proposition 5.1 is one half of a spectrum broadcast structure, the class of states in which many observers can find the same value independently [13]. The other half, perfect distinguishability of the conditional fragment states, holds here only in the limit $r_F \to 0$. Proposition 5.3 measures how fast readers approach that limit as fragments are added.

QBism and relational quantum mechanics. QBism reads quantum states as an agent's degrees of belief [18], and relational quantum mechanics takes states to be relative to a reference system [19]. Here accessibility is fixed by the interaction structure of a context, and a single objective global state is assumed. The emphasis on relations is shared with relational quantum mechanics; the stable records computed here are candidates for the relata that its relations require.

Algebraic quantum field theory. AQFT treats observable algebras as primary and states as functionals on them [8]. The profile of Definition 2.4 adds typed selection data to that setting. Its entropic statements are finite-dimensional, and no Type III version of the redundancy counts is claimed.

6.3 The actual record interface

Paper A uses these pure fragments as actual readers of its noisy GHZ source [1]. Their visibility $v_i = \sqrt{1 - r_i^2}$ gives the collision-derived law

$$\mathbb{E}(Y_i Y_j) = v_i v_j c_{ij}, \qquad c_{ij} = (1 - 2p_i)(1 - 2p_j).$$

The source sites are correlated; no independence of their answers is assumed. Calibrated finite samples then certify non-isometric shortest-path metrics for two equal-flip-budget noise allocations. This supplies the physical interface a dictionary statement alone lacks.

Paper B's block-constant recorder depends only on sector weights, under its specified commuting noise and recording permissions [2]. Its protected content, fixed algebra and finite channel range differ from this fragment-effect set. The mixed-memory family sharpens the distinction: one unchanged dephasing channel has permissions ranging from trivial to perfect. Paper D and Q-RAIF C add record-controlled physical sources and retained-port acquisition [3, 4]; they grant neither free sharp effects nor fresh inputs in place of spent ports.

The record law supplies two different downstream relations. For Paper F, once-only contact and isolation of spent ports protect earlier records against later operations on the carrier. Redundancy alone supplies no event order: the pointer-controlled gates in Eq. (3) commute, so their full terminal state is unchanged by permuting the contacts. Paper F's order-identification protocol adds actual retained intervention tags and record–swap stations; the tags reveal predecessor relations under its specified one-pass architecture. Neither an order inferred from those relations nor a direction relative to its declared acquisition boundary derives a continuous time parameter [5].

For Foundation II, common content means a common target label and declared record channels, not literal intersection of the readers' separate register algebras. The two fragments can share an accurately readable classical label even when their embedded local algebras intersect only in the scalars. The joint preparation and the response laws must establish this common content; consensus alone cannot fix its orientation or remove a common error [7]. The calibrated protocol above supplies that relation and its error, not a theorem that arbitrary readers agree on a world.

T-DOME I consumes the actual record and its posterior for a task; its action value need not equal the record entropy [6]. Preparation, reading, storage and eventual renewal retain their separate resource conditions. Shared geometry additionally requires source preparation, site labels and a reconstruction rule; the finite orientation certificate here supplies one of those permissions, not all of them.

7 Conclusion

Measurement permissions, stable content and public records are different physical data. In the fresh pure-memory model, the permission lens gives the sharp-test error, while the fixed pointer algebra gives persistence and fragment discrimination gives readable redundancy. Holevo and single-measurement thresholds differ. Pooled local tests attain the joint pure-state error exponent when their readers know the conditional fragment state.

The mixed-memory construction shows why preparation must be accounted for without demanding purity. Its actual readout saturates a purity-deficit budget, its exact permissions form a parallelogram, and its source channel remains complete dephasing even when no record is produced. Reusing the same port reverses its record rather than supplying another independent sample.

Public correctness additionally needs a physical standard. Conditional independence suppresses private common errors, but agreement cannot discover a shared orientation inversion. Prepared-pointer references calibrate that inversion with a finite failure probability; this probability remains in the public error bound and its resources remain in the protocol. Together with Paper A's actual record-to-geometry construction, these results identify a finite operational route from interaction to readable shared structure. They

leave outcome selection, elementary factorization and general spacetime reconstruction as independent explanatory questions.

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Records, Order and Intervention: Conditions for a Temporal Direction

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Abstract

Relations among physical records can identify an event order without a timestamp, but redundancy alone cannot. Pointer-controlled gates on separate fresh ports commute, so every collision permutation gives the same complete endpoint state. We contrast this obstruction with a reversible recording protocol: a carrier visits labelled stations in an unknown order, each station records the arriving classical value and then exchanges it for a fresh random label. Retained copies of those labels and repeated noisy endpoint readings identify the predecessor of every station. For calibrated positive readout bias, a finite-sample correlation certificate recovers the complete directed chain. The local record-before-exchange orientation and one-pass contact architecture are inputs; the global permutation is not.

The direction is relative to a physical preparation boundary. Fresh biased ports and independent label–reference pairs are resources, while inverse evolution starts from the correlated output of the same circuit. A joint purity law bounds the information stored in disjoint fragments, with work and retained side information explicitly counted. Partly mixed ports attain the pointer-controlled bound. Recontact can remove a record and restore source coherence; thermal reset, record transfer and complete history reversal have different targets and prices. Renewing a noisy bank requires removal of its full retained entropy increment, and a specified thermal chiral exhaust adds a finite-rate heat bound. These results separate reconstruction of event order, orientation of a recording process and thermodynamic maintenance. They supply neither a continuous time parameter nor a cosmological initial condition.

1 Introduction

A collection of records says that events left traces. Does it also say which event came first? One can date a record by an external clock, or define influence using interventions at specified times. Neither procedure explains when the relations among the records themselves identify an order. That is the first question of this paper. The second is why a reconstructed chain has a direction relative to a preparation boundary, and what must be supplied to keep making useful records.

The distinction is already sharp in a simple collision model. A pointer controls one gate on each of several isolated memories. The final memories can contain many readable

copies of its value, yet the complete final state is invariant under the order of those gates. No endpoint measurement can recover that order. An intervention can change this conclusion only through the actual interactions it inserts; assigning an influence diagnostic a time argument does not recover an unknown ordering.

We give a contrasting protocol in which endpoint relations do identify the order. A finite carrier visits each labelled station once. At every station a CNOT first writes the arriving classical bit into a biased read port; a SWAP then replaces the carrier by an independent random label whose copy is retained. The public data are the final labels and read ports, indexed by station names, not by their positions in the visit sequence. Their correlation matrix has one nonzero entry per column. It identifies the predecessor chain from the known initial label, including which station was last, without reading the final carrier.

The architecture supplies a directed local operation and a family of possible one-pass histories. It does not supply which of the $n!$ permutations occurred. Conversely, recovering that permutation does not explain the external parameter used to implement the gates, elapsed duration or the origin of the architecture. Reversible gates also admit the inverse history. Its actual input is the forward circuit's output, with its changed correlations, rather than a second independent preparation of fresh ports.

Section 2 proves the endpoint obstruction. Section 3 constructs the relational record and its finite-sample certificate. Section 4 locates its direction and resource boundary. Section 6 then distinguishes this reconstruction from a timed influence test and retains a recontact counterexample. Sections 5–8 supply the record and purity account, and Section 9 states what persistence adds. The appendices retain the acquisition/action, reset and renewal proofs in their original regimes.

2 When the complete endpoint forgets the order

Fix a pointer PVM $\{P_x\}$ on a system S and distinct ports E_i. A collision with port i has $U_i = \sum_x P_x \otimes V_{x,i}$, with each $V_{x,i}$ acting only on E_i. The rotations of Section 5 and the mixed-port CNOTs of Appendix B.1 have this form. No order-dependent drift, intervening operation or retained controller tag is included in the compared composition.

Proposition 2.1 (Endpoint non-identification for commuting record gates)**.** *For one fixed initial joint state of S, all ports and any untouched reference, the complete final state after applying each U_i once is independent of the permutation of the collisions. Thus no endpoint local or joint measurement on these systems, including adaptive measurements, can distinguish the permutations. Redundancy and entropy data cannot do so either.*

Proof. For $i \neq j$, orthogonality of the pointer projections gives $U_iU_j = \sum_x P_x \otimes V_{x,i}V_{x,j} = U_jU_i$. The factors on distinct ports commute. Every ordered product is therefore the same unitary, also after tensoring identity on the reference. Its action on the fixed initial state is identical. Every endpoint instrument consequently has the same outcome law. □

This is stronger than equality of one-port marginals. It holds even for an initially coherent pointer and for a reader controlling all modeled outputs jointly. If the driving controller retains a record of its programmed order, that is an additional order tag, not a counterexample. Intermediate observations or inserted gates are also different data. A high-redundancy record of a fixed pointer value can therefore be completely uninformative about the order in which its copies were made.

3 A chain recorded without timestamps

3.1 The physical preparation and local operation

There are $n \geq 1$ stations, named $1, \ldots, n$, and a qubit carrier S. An unknown permutation $\pi = (\pi_1, \ldots, \pi_n)$ determines a single visit to each station. The station names are identities, not chronological ranks. At station i there is a read port R_i, a replacement label port T_i and a retained label record K_i. A record K_0 accompanies the carrier's initial label.

All bits use the orthogonal computational basis. Prepare independent fair choices $Z_0, \ldots, Z_n$, set $S = K_0 = Z_0$ and $T_i = K_i = Z_i$, and independently prepare $R_i = N_i$ with $\Pr(N_i = 1) = \varepsilon_i < 1/2$. The N_i are independent of one another and of all labels. The correlated pair T_i, K_i is an actual preparation, not knowledge of a secretly chosen value. Only an orthogonal classical encoding is copied. No arbitrary unknown quantum state is cloned.

The known local operation is

$$V_i = \mathrm{SWAP}_{S,T_i}\ \mathrm{CNOT}_{S \to R_i}, \tag{1}$$

with the CNOT acting first. It writes before replacing the carrier. Every exited port remains isolated and retained; there is no erasure or discarded history in the gate sequence. The final readout permits the Z measurements of $K_0, \ldots, K_n$ and $R_1, \ldots, R_n$. Write their outcomes as K_i and Y_j. Neither π nor a timestamp is supplied to the reader, and the final carrier and spent T_i are not measured by this readout. Independent trials repeat the same unknown order and preparation law. The readout pairs all endpoint values from the same trial; unpaired single-port marginals would not supply these relations.

Define $\mathrm{pred}_\pi(\pi_1) = 0$ and $\mathrm{pred}_\pi(\pi_k) = \pi_{k-1}$ for $k > 1$. This is notation for the hidden order, not a readout label.

Theorem 3.1 (Endpoint relations identify the unknown permutation)**.** *For the circuit and preparation above, put $g_j = 1 - 2\varepsilon_j > 0$. The measured endpoint correlation matrix obeys*

$$C_{ij} := \mathbb{E}\,(-1)^{K_i + Y_j} = g_j\, \mathbf{1}\{i = \mathrm{pred}_\pi(j)\}, \qquad i = 0, \ldots, n, \quad j = 1, \ldots, n. \tag{2}$$

Consequently these endpoint relations uniquely identify the directed chain $0 \to \pi_1 \to \cdots \to \pi_n$ among the permitted permutations, without measuring S at the endpoint.

Proof. Immediately before the first visit the carrier is Z_0. At a visit to j, the CNOT sends $R_j \mapsto N_j \oplus S$ and leaves S unchanged. The SWAP then sends $T_j \mapsto S_{\mathrm{old}}$ and $S \mapsto Z_j$. Induction along the actual visits gives

$$Y_j = N_j \oplus Z_{\mathrm{pred}_\pi(j)}, \qquad (T_j)_{\mathrm{out}} = Z_{\mathrm{pred}_\pi(j)}, \qquad S_{\mathrm{out}} = Z_{\pi_n}. \tag{3}$$

The $K_i = Z_i$ never change. For $i = \mathrm{pred}_\pi(j)$ the measured sign is $(-1)^{N_j}$, with expectation g_j. For every other i, the two distinct independent fair labels have zero sign correlation and are independent of N_j, giving zero. This proves Eq. (2). Each column has exactly one positive entry, so it gives one predecessor edge. The assumed single-pass architecture makes these edges one chain starting at the known vertex 0 and visiting every station once. Following that chain recovers the permutation. □

The gates do not merely copy the same pointer repeatedly. The carrier's value is changed at each station, so subsequent records refer to a newly supplied label. The retained K_i reveal which station supplied the value that another station recorded. This is the information missing from the commuting-gate model. The local record-before-exchange direction and the known starting label orient the reconstruction; they do not name the global predecessors in advance.

3.2 A finite-sample certificate

Corollary 3.2 (Calibrated finite-sample order recovery)**.** *Suppose the independent repeated trials have one fixed permutation and a calibrated lower bias* $g_{\min} = \min_j g_j > 0$. *For* N *trials define* $\widehat{C}_{ij} = N^{-1}\sum_{\ell=1}^{N}(-1)^{K_i^{(\ell)}+Y_j^{(\ell)}}$. *Declare an edge* $i \to j$ *when* $\widehat{C}_{ij} > g_{\min}/2$, *and require the declared edges to form one chain from* 0 *through all stations. The probability of failing to recover the correct chain, including rejection, is at most*

$$2n(n+1)\exp\bigl[-Ng_{\min}^2/8\bigr]. \tag{4}$$

For $0 < \zeta < 1$, *a safe integer choice with bound strictly below* ζ *is*

$$N = 1 + \left\lfloor \frac{8}{g_{\min}^2} \log \frac{2n(n+1)}{\zeta} \right\rfloor. \tag{5}$$

Proof. On the strict event $\max_{i,j} |\widehat{C}_{ij} - C_{ij}| < g_{\min}/2$, all false entries fall below threshold and every true entry exceeds it. For one sign variable $A \in [-1, 1]$, the centered log moment-generating function has second derivative equal to a tilted variance, at most one. Its value and first derivative at zero vanish, hence it is at most $\lambda^2/2$. Independence across trials and the Chernoff bound give $\Pr\Bigl(|\widehat{C}_{ij} - C_{ij}| \geq a\Bigr) \leq 2e^{-Na^2/2}$. Take $a = g_{\min}/2$ and a union bound over $n(n+1)$ entries. The added integer in Eq. (5) makes N strictly larger than the real-valued logarithmic bound. □

Equality at the threshold lies outside the sufficient strict event; it is covered by the failure bound, not silently classified as safe. Entries within a trial need not be statistically independent for this union bound. Nor does chain validation certify all preparation and contact assumptions from the same data: those define the tested model.

For $n = 3$, $\varepsilon = (1/10, 1/5, 3/10)$ and the hidden order $(2, 1, 3)$, station-indexed data give

$$C = \begin{pmatrix} 0 & 3/5 & 0 \\ 0 & 0 & 2/5 \\ 4/5 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{6}$$

Rows are $0, 1, 2, 3$ and columns $1, 2, 3$, not chronological positions. The matrix recovers $0 \to 2 \to 1 \to 3$. Here $g_{\min} = 2/5$ and $\zeta = .01$ give $N = 390$ and bound .00983364. Appendix A enumerates all six permutations by the actual reversible gates with rational weights.

3.3 What removes identification

Discarding the K_i makes all Y_j independent fair bits in the stated independent-noise model: they use distinct fair predecessor labels. Their complete joint law, not just

their marginals, is then independent of π. If every $\varepsilon_j = 1/2$, even the (K, Y) joint law is independent of π. With two fair columns, ambiguity can already remain: at $n = 3$, $\varepsilon_1 = \varepsilon_2 = 1/2$ and $0 \le \varepsilon_3 < 1/2$, orders $(1, 3, 2)$ and $(2, 1, 3)$ have the same permitted endpoint law, since both give predecessor 1 for station 3 and the other read ports are independent fair bits. A single missing column need not destroy identification; the remaining chain constraints may force its edge. Thus positive $g_{\min}$ is a sufficient condition for this threshold certificate, not a necessary condition for every reconstruction.

These are readout-relative failures. Equation (3) shows that the retained spent T_j still contain the predecessor labels. Granting their measurement recovers the chain even with completely fair R_j. This differs from Proposition 2.1, whose complete modeled endpoint state is permutation invariant.

Independence of the labels is substantive. If every Z_i is one common fair bit, all labels and record laws are the same for every permutation. Reconnection, repeated visits or hidden inputs can likewise destroy the one-pass predecessor interpretation. Merely correlating the N_j with each other while keeping them independent of all Z_i does *not* destroy Eq. (2): its proof uses only the label independence and each port's bias. The sampling bound also survives within-trial shared noise when trials remain independent. Label-dependent noise or additional unobserved couplings require a new identification argument.

4 Which boundary gives the chain a direction?

The reconstruction has an oriented local operation and a fresh preparation boundary. At entry, R_j is independent of the incoming carrier and its noise is biased toward zero. The labels for different stations are independent; each label nevertheless has its own retained copy. The global order is inferred from the relations produced by these operations, rather than selected by increasing record entropy.

The inverse of every V_j exists: it swaps first and then applies the CNOT. In reverse global order these inverses return the entire actual output, including every spent port and label record, to its input. That reverse input is not a fresh preparation of the declared form. Initially $T_j = K_j$ with certainty. At the forward endpoint, $T_j = K_{\mathrm{pred}_\pi(j)}$ and is independent of K_j, while R_j remains correlated with that spent label when $g_j > 0$. The reverse gates use those actual correlations. Microreversibility therefore coexists with a directed reconstruction relative to the acquisition boundary. No probability suppression of inverse histories follows without a specified ensemble and measure.

For the independent preparation above, there are $3n + 2$ modeled qubits per trial: S, K_0 and the triples T_j, K_j, R_j. No separate register secretly stores chronological ranks. With $h(x) = -x \ln x - (1 - x) \ln(1 - x)$, their complete initial entropy is

$$S_{\mathrm{in}} = (n + 1) \ln 2 + \sum_j h(\varepsilon_j),$$

and the reversible circuit preserves it. Relative to the maximally mixed state of these degenerate registers, its entropy deficit is

$$(3n + 2) \ln 2 - S_{\mathrm{in}} = (n + 1) \ln 2 + \sum_j [\ln 2 - h(\varepsilon_j)]. \tag{7}$$

The first term is the stock of label–copy correlations; each constituent label marginal is fair. The second is the initial read-port purity deficit. This is a state-resource account, not a universal attained work price.

Every independent trial uses a new prepared instance of those registers, or an explicitly paid restoration of the complete needed state. Spent T_j and old records are not erased from that account because the declared readout ignores them. With degenerate endpoint Hamiltonians and pulses off at the endpoints, the ideal gate sequence has zero net endpoint work. Preparing labels and biased ports, implementing controls, reading and retaining results, providing timing, and reusing the device remain physical tasks. Sections 7 and 9 price specific aspects of them. The protocol identifies an unknown event permutation; it does not produce the time parameter or the apparatus that realizes the contacts.

5 Records and how to count them

5.1 The record model

The system S is a qubit with pointer projections P_x, $x \in \{0,1\}$, and initial state $\sqrt{p}\,|0\rangle + \sqrt{1-p}\,|1\rangle$. Memory qubits $E_1, \ldots, E_n$ start in $|0\rangle$, and the k-th collides once with S through $U_k = P_0 \otimes \mathbf{1} + P_1 \otimes R_k$, where R_k is a real rotation with $\langle 0|R_k|0\rangle = c_k \in [0,1]$. After n collisions

$$|\Psi_n\rangle = \sqrt{p}\,|0\rangle|e_0\rangle + \sqrt{1-p}\,|1\rangle|e_1\rangle, \qquad |e_x\rangle = \bigotimes_k |e_{x,k}\rangle, \qquad \langle e_{0,k}|e_{1,k}\rangle = c_k. \tag{8}$$

For a fragment F put $r_F = \prod_{k\in F} c_k$ and

$$H_p(r) = h\left(\frac{1+\sqrt{1-4p(1-p)(1-r^2)}}{2}\right), \qquad h(x) = -x\ln x - (1-x)\ln(1-x).$$

Paper E shows that $\chi(X : E_F) = H_p(r_F)$, that the coherence of S is $\sqrt{p(1-p)}\,r_{[n]}$, and that $S(\rho_S) = S(\rho_{E_{[n]}}) = H_p(r_{[n]})$ [3]. Here X is the value a pointer measurement on S returns. Entropies are in nats.

This pure-port model will be compared with the partly mixed controlled-X model below. The inputs are the pointer basis, the split into system and memories, the fragment partition, the order of collisions with its external parameter, and the preparation of the memories. The collisions can be implemented by couplings $g_k(t)P_1 \otimes \sigma_y$ that vanish at the start and end of each pulse and commute with a pointer Hamiltonian $H_S = \varepsilon P_1$; with degenerate memory Hamiltonians the recording unitary has zero net endpoint work. Preparation of the ports, calibrated pulse control and later bank renewal are separate resources. T-DOME I gives the full thermodynamic account of this protocol [5].

5.2 Redundancy and protocol-relative comparisons

Definition 5.1 (Redundancy index)**.** *For a classical label X, a classical–quantum state ρ_{XE} and a declared family of disjoint fragments $\{E_k\}$, define*

$$\mathcal{R}_\delta^{\chi} = \#\big\{k : \chi(X : E_k) \geq (1-\delta)H(X)\big\}, \qquad \mathcal{R}_\delta^{\mathrm{acc}} = \#\big\{k : I_{\mathrm{acc}}(X : E_k) \geq (1-\delta)H(X)\big\}.$$

The first counts Holevo records and the second records that a permitted single measurement can read; $\mathcal{R}_\delta^{\mathrm{acc}} \le \mathcal{R}_\delta^{\chi}$, and Paper E gives closed-form fragment lengths for both in the collision model. With $p = 1/2$, $c_k = 0.8$ and $\delta = 0.1$, blocks of five memories are Holevo records and blocks of seven are readable, so after n collisions $\mathcal{R}_\delta^{\chi} = \lfloor n/5 \rfloor$ and $\mathcal{R}_\delta^{\mathrm{acc}} = \lfloor n/7 \rfloor$. High redundancy supports public readability of the label; it does not select one global outcome. Here the quoted pure-port counts permit arbitrary fragment POVMs, as in Paper E. A restricted readout menu needs its own accessible-information optimization; ideal algebra closure does not grant those measurements.

Definition 5.2 (Protocol-relative comparisons)**.** *Fix a dynamics, a fragment family and a label X, and let α, β be configurations along the resulting history. Write $\alpha \preceq_{\mathcal{R}} \beta$ if $\mathcal{R}_\delta(\rho_\alpha) \le \mathcal{R}_\delta(\rho_\beta)$, and $\alpha \preceq_{\mathrm{inc}} \beta$ if the algebra of memories that have interacted at α is contained in that at β. Each relation is a preorder on its declared domain.*

In the collision model both comparisons follow the collision order. They need not agree in general: decoherence may increase the number of records while the coherence-sensitive effects of the system, whose expectation values scale with $r_{[n]}$, become unreadable on S itself.

5.3 Temporal inputs

The construction uses temporal orientation at several points: the parameter of the dynamics, the order of collisions, the comparison of records at different parameter values, and a boundary condition of fresh memories at one end of the history. These comparisons describe record growth along that supplied history, not an inverse reconstruction of its unknown permutation. Proposition 2.1 shows that their endpoint values may leave the permutation completely unidentified. The purity account below identifies what the acquisition boundary supplies and what its use costs.

6 Intervention and the agent's time order

The unknown-order reconstruction of Theorem 3.1 uses endpoint statistics and an explicitly restricted family of contact histories. The following diagnostic instead compares interventions at already specified parameter values. Its ordering clause is an input and cannot prove the origin of that order.

Definition 6.1 (Protocol-relative causal-influence diagnostic)**.** *Fix a record-forming protocol with external parameter t, labels X_A, X_B, and a family of interventions on X_A. Write $X_A \rightsquigarrow_{\mathsf{P}} X_B$ when (i) both labels have records in the declared fragments, (ii) the X_A records form no later than the X_B records according to t, and (iii) interventions on X_A, with the rest of the protocol held fixed, change the distribution of X_B or its records.*

The diagnostic is not asserted to be transitive, antisymmetric or independent of the intervention family. Redundancy makes an influence readable across fragments; the intervention structure carries the causal content. The next two propositions compute the diagnostic in the collision model.

Proposition 6.2 (Interventions reach only later records)**.** *In the collision model let an arbitrary channel $\mathcal{T}$ act on S between the j-th and $(j+1)$-th collisions.*

1. *The final reduced state of $E_1, \dots, E_j$ is the same as without $\mathcal{T}$.*

2. *If $\mathcal{T}$ is the pointer flip $\rho \mapsto \sigma_x \rho \sigma_x$, the final state is $\sum_x \sqrt{p_x}\,|1-x\rangle|e_x^{\le j}\rangle|e_{1-x}^{>j}\rangle$: the earlier memories record the value x and the later ones record $1-x$.*

3. *If the Kraus operators of $\mathcal{T}$ are diagonal in the pointer basis, the joint distribution of the pointer value and all records is unchanged.*

Proof. (1) After the j-th collision, $\mathcal{T}$ and the later collisions act on $S \otimes E_{j+1} \otimes \cdots \otimes E_n$, and the partial trace over these systems is invariant under any operation on them. (2) σ_x maps the branch $\sqrt{p_x}|x\rangle|e_x^{\le j}\rangle$ to $\sqrt{p_x}|1-x\rangle|e_x^{\le j}\rangle$, and the later collisions then write the value $1-x$. (3) Diagonal Kraus operators commute with the pointer-controlled collisions and with the pointer measurement, and they leave the populations unchanged. □

For a flip after the second of four collisions with overlaps $(0.3, 0.5, 0.2, 0.4)$ and $p = 1/2$, the reduced states of the first two memories and of the last two are each unchanged, while the joint state of the first and third memories changes by 0.437 in its largest matrix element: the flip is visible only in the relation between earlier and later records. In this fresh-port protocol, interventions on the source can alter later records, while detached earlier marginals remain fixed. The flip example makes the change visible in joint relations even when single-fragment marginals do not change.

Proposition 6.3 (Repeated interaction lets a later intervention rewrite an earlier record)**.** *Let a memory collide with S twice through the same rotation with $c = \cos(\pi/4)$, and let $p = 1/2$. After the first collision the memory holds $H_{1/2}(2^{-1/2}) = 0.4165$ nats about the initial pointer value. If nothing happens between the collisions, the second collision makes the record perfect, $\ln 2$ nats. If S is flipped between the collisions, the second collision leaves the memory with no information about the initial pointer value.*

Proof. Write R for the rotation by $\pi/4$, so $R^2|0\rangle = |1\rangle$. Without the flip, the memory ends in $|0\rangle$ if $x = 0$ and in $R^2|0\rangle = |1\rangle$ if $x = 1$. With the flip, it is in $|0\rangle$ after the first collision if $x = 0$ and is then rotated once, ending in $R|0\rangle$; if $x = 1$ it is in $R|0\rangle$ after the first collision and is not rotated again. Both branches end in $R|0\rangle$. □

For the coherent source $|+\rangle$, the same two-qubit protocol also gives the entropy and population ledger:

$$|\Psi_1\rangle = (|0,0\rangle + |1\rangle R|0\rangle)/\sqrt{2}, \quad |\Psi_{2,\text{noflip}}\rangle = (|00\rangle + |11\rangle)/\sqrt{2}, \quad |\Psi_{2,\text{flip}}\rangle = |+\rangle \otimes R|0\rangle.$$

Source populations stay $1/2, 1/2$, and joint entropy stays zero. Put $a = h[(1+2^{-1/2})/2] = 0.4164955$. At first contact, $S(S) = S(E) = a$ and $I(S:E) = 2a$. The no-flip continuation gives $\ln 2, \ln 2, 2\ln 2$. The flip continuation gives $0, 0, 0$: source coherence and both marginal purities return while the earlier pointer information disappears. For a pointer-diagonal source Hamiltonian, energy stays fixed; source free energy first falls by $k_B T a$ and then recovers it on the flip branch. In fact $U(X_S \otimes I)U = X_S \otimes R$, so the echo decouples every source input. Reading or discarding the port after the first contact would change this coherent protocol. The source trace distance of the inputs $|+\rangle, |-\rangle$ falls to $2^{-1/2}$ at first contact and returns to 1 after the echo: this is distinguishability backflow, not merely an entropy change.

The alignment of the intervention order with the record order thus rests on a specific feature of the dynamics: memories that have recorded do not interact with the source

again. Under repeated interaction, whether an early memory ends as a perfect record or as no record can be decided by an intervention made after it first recorded.

An agent can therefore date an intervention by its effect on records within the stated one-contact architecture. It cannot change a detached port by acting only on the source. Acting directly on that port can relabel, transfer or reset it, with the different balances in Appendix C; reconnecting it to the source can also restore coherence. The alignment comes from the actual interaction menu, fresh stock and external order, not from a universal prohibition on changing the past.

7 The purity of the past

Bath balances below use finite entropies, relative entropies and mean energy changes. An infinite reservoir requires a controlled limit of finite reservoirs preserving those balances, not subtraction of divergent global entropies.

Theorem 7.1 (Purity law for records)**.** *Let a system S, memories $E_1, \dots, E_n$ of dimensions d_k with degenerate Hamiltonians, and a bath B with Hamiltonian H_B start in the product state $\rho_S \otimes \bigotimes_k \sigma_k \otimes \tau_B$, where τ_B is the Gibbs state of H_B at inverse temperature β. Let them evolve under a time-dependent Hamiltonian whose interaction terms vanish at the initial and final times and whose bath and memory terms are then H_B and the given degenerate Hamiltonians, let W be the work done on the whole and ΔE_S the change of the energy of S. Let X be the outcome of a final measurement of a PVM on S with rank-one projections, and let $F_1, \dots, F_R$ be disjoint sets of memories. Then*

$$\sum_{j=1}^{R} \chi(X : E_{F_j}) \le \sum_{k=1}^{n} \big[\ln d_k - S(\sigma_k)\big] + \beta\,(W - \Delta E_S) + \big[H(X) - S(\rho_S)\big]. \tag{9}$$

Without a bath the middle term is absent, and the bound holds for an arbitrary joint unitary on S and the memories, with no condition on their Hamiltonians.

Proof. Let ρ be the final state, and $\rho' = \sum_x (P_x \otimes \mathbf{1})\rho(P_x \otimes \mathbf{1})$ its pinching in the pointer basis. Pinching does not decrease entropy and unitary evolution preserves it, so $S(\rho') \ge S(\rho_S) + \sum_k S(\sigma_k) + S(\tau_B)$. Since the P_x have rank one, $S(\rho') = H(X) + S(E_{[n]}B \mid X)$, where the conditional entropy refers to the classical–quantum state of X, the memories and the bath. Write G for the memories outside $\bigcup_j F_j$. Subadditivity of conditional entropy gives

$$S(E_{[n]}B \mid X) \le \sum_j S(E_{F_j} \mid X) + S(G \mid X) + S(B \mid X) \le \sum_j S(E_{F_j} \mid X) + \ln d_G + S(\rho_B),$$

with ρ_B the final state of the bath. Nonnegativity of $D(\rho_B \| \tau_B)$ gives $S(\rho_B) \le S(\tau_B) + \beta \Delta E_B$. Hence

$$\sum_j S(E_{F_j} \mid X) \ge S(\rho_S) + \sum_k S(\sigma_k) - H(X) - \ln d_G - \beta \Delta E_B.$$

Since $\chi(X : E_{F_j}) = S(E_{F_j}) - S(E_{F_j} \mid X) \le \ln d_{F_j} - S(E_{F_j} \mid X)$, summing over j gives (9) with $\beta \Delta E_B$ in the middle. The memory Hamiltonians are degenerate and the interactions vanish at the endpoints, so the first law gives $\Delta E_B = W - \Delta E_S$. Without a bath the same argument applies with B omitted. □

The bound is attained without a bath. If S starts in $|0\rangle$, a single memory starts maximally mixed, and a controlled-NOT with the memory as control and S as target is applied, then the memories have no purity, $H(X) - S(\rho_S) = \ln 2$, and $\chi(X : E_1) = \ln 2$. In that example the label is created by copying the memory's randomness into a pure system, which is why the system can contribute one record's worth. For a bath-free pointer-controlled unitary, $U = \sum_x P_x \otimes V_x$, the system term is absent: the conditional states of the memories are unitary images of their initial state, and the argument of the proof gives $\sum_j \chi(X : E_{F_j}) \leq \sum_k [\ln d_k - S(\sigma_k)]$. If in addition each V_x is a product over memories, as in the collision model, each fragment is bounded separately by its own purity, $\chi(X : E_F) \leq \sum_{k \in F} [\ln d_k - S(\sigma_k)]$ (Paper E's purity-deficit bound). Without the product form the per-fragment bound can fail: with E_1 maximally mixed, E_2 pure, $p = \frac{1}{2}$ and V_1 the swap of E_1 and E_2, the fragment $\{E_1\}$ holds 0.216 nats about X.

Corollary 7.2 (Purity budget of a record history)**.** *If R disjoint fragments are Holevo records at level δ, then*

$$R(1-\delta)H(X) \leq \beta \sum_k \big[F(\sigma_k) - F(\mathbf{1}/d_k)\big] + \beta\,(W - \Delta E_S) + H(X) - S(\rho_S),$$

where F is the nonequilibrium free energy at temperature T. In particular, memories that start in thermal equilibrium can hold R records only if the work not stored in the system satisfies $\beta(W - \Delta E_S) \geq R(1-\delta)H(X) - [H(X) - S(\rho_S)]$. In a pointer-controlled process without a bath, $R(1-\delta)H(X) \leq \sum_k [\ln d_k - S(\sigma_k)]$.

Proof. Each record contributes at least $(1-\delta)H(X)$ to the left side of (9). For a degenerate Hamiltonian the thermal state of a d-dimensional memory is $\mathbf{1}/d$, and $\beta[F(\sigma) - F(\mathbf{1}/d)] = D(\sigma \| \mathbf{1}/d) = \ln d - S(\sigma)$. The pointer-controlled statement is the summed bound stated after Theorem 7.1. □

The acquisition boundary is thus a quantitative resource, not a requirement that every memory be pure. Stable old records can also serve as standards across time, provided their couplings and orientation connect the old and present instruments [15]. The bound counts simultaneously present information in declared disjoint fragments, not all records that could ever be made, read, erased and remade. With product maximally mixed ports, no bath and pointer-controlled recording, it gives zero; an old correlation can change that boundary, as the conditional bound below shows. The theorem specifies what the acquisition history must supply. It does not explain the primitive origin of that stock.

7.1 Acquisition, action and retained references

Partly mixed ports attain the pointer-controlled purity bound. For a fair pointer and a fresh port with error ε, a computational-basis reading attains $\ln 2 - h(\varepsilon)$ nats. The complete acquisition/action ledger, posterior and proof are in Appendix B.1. For one port, reversible feedback moves its initial purity into the source at zero endpoint gate work; it does not complete a thermal cycle.

The label copies in Section 3 illustrate why marginal purity is not the only stock. The conditional budget of Proposition B.2 includes an old readable reference. Two marginally fair copies of one bit allow a new XOR record to be read jointly even though the new port alone is uninformative. Preparation and preservation of that joint reference are part of the history. A quantum conditional mutual information, by itself, still grants no measurement permission.

8 Records and dissipation

Proposition 8.1 (Records bound the coherence that survives)**.** *In the collision model with $0 < p < 1$, let $0 < \delta < 1$, let q_δ satisfy $h(q_\delta) = (1-\delta)h(p)$ with $q_\delta \le 1/2$, and put*

$$a_\chi = \frac{(1-2q_\delta)^2 - (2p-1)^2}{4p(1-p)} \in (0,1).$$

If R disjoint fragments, of any sizes and any collision overlaps c_k, are Holevo records, then

$$r_{[n]}^2 \le a_\chi^R, \qquad |(\rho_S)_{01}| \le \sqrt{p(1-p)}\, a_\chi^{R/2}.$$

If the pointer Hamiltonian is diagonal in the pointer basis, the system's nonequilibrium free energy falls by

$$F_S(0) - F_S(n) = k_B T\, H_p(r_{[n]}) \ge k_B T\, H_p\big(a_\chi^{R/2}\big).$$

If instead S starts in $pP_0 + (1-p)P_1$, every record quantity is unchanged and $F_S(n) = F_S(0)$.

Proof. A fragment is a Holevo record exactly when $r_F^2 \le a_\chi$ (Paper E's Holevo fragment criterion). The overlaps multiply over disjoint fragments, and the memories outside them contribute factors at most 1, which gives the first two bounds. The collisions preserve the pointer populations, so the energy of S is unchanged and $F_S(0) - F_S(n) = k_B T[S(\rho_S(n)) - S(\rho_S(0))] = k_B T\, H_p(r_{[n]})$; H_p decreases in r. For the diagonal initial state the conditional fragment states are the same, so all $\chi(X : E_F)$ are unchanged, and ρ_S does not change at all. □

Records and dissipation in the system therefore coincide only when the system has coherence to lose. As R grows, $H_p(a_\chi^{R/2}) \to h(p)$ and the system loses all the free energy carried by its coherence. A system that is already diagonal acquires the same records at no cost to itself. In the pure-port model, the marginal purity of written memories is transferred into correlations. The complete unitary preserves joint entropy; the mixed-port ledger makes that transfer explicit.

Proposition 8.2 (Records along a CP-divisible dynamics)**.** *The reduced dynamics of S after n collisions is $\mathcal{D}_n \circ \cdots \circ \mathcal{D}_1$, where $\mathcal{D}_k$ keeps the populations and multiplies the coherence by c_k. Every intermediate map $\mathcal{D}_m \circ \cdots \circ \mathcal{D}_{l+1}$ is completely positive and trace preserving, and for equal overlaps $c \in (0,1)$ applied at intervals Δt the family embeds in the dephasing semigroup generated by $\mathcal{L}(\rho) = \frac{\gamma}{2}(Z\rho Z - \rho)$ with $\gamma = -\ln c/\Delta t$. Along this dynamics $\mathcal{R}_\delta^\chi$ and $\mathcal{R}_\delta^{\mathrm{acc}}$ grow without bound in n.*

Proof. Each $\mathcal{D}_k$ is a dephasing channel, and compositions of channels are channels. The semigroup multiplies the coherence by $e^{-\gamma t}$, which equals c^m at $t = m\Delta t$. The record counts are $\lfloor n/m_\chi \rfloor$ and $\lfloor n/m_{\mathrm{op}} \rfloor$ for equal overlaps. Unbounded record counts refer to a family with growing fresh-port supply, not an unbounded yield from a fixed finite bank. □

Record growth thus does not require memory effects in the reduced dynamics of the system. The memories, once written, have a mixture-of-product marginal correlated through X and are independent conditional on a pointer reading; a Markovian system dynamics says nothing about whether the environment's outputs are correlated. The

converse separation also holds. T-DOME I gives an autonomous two-qubit model in which correlation built between a system and an environment qubit returns to the system at a recurrence time while the environment holds no information at all about the pointer, and it shows that the pure-dephasing spin–boson model with a Lorentz–Drude spectral density has no backflow at any temperature [5]. Backflow and records are logically independent.

Proposition 8.3 (Local and global entropy)**.** *For disjoint fragments $F_1, \dots, F_R$ of the collision model,*

$$\sum_{j=1}^{R} S(\rho_{F_j}) = \sum_{j} \chi(X : E_{F_j}), \qquad S\big(\rho_{\cup_j F_j}\big) \le h(p).$$

If the F_j are Holevo records, the total correlation among them, $\sum_j S(\rho_{F_j}) - S(\rho_{\cup_j F_j})$, is at least $[R(1-\delta)-1]h(p)$.

Proof. The conditional states of every fragment are pure, so $S(\rho_{F_j}) = \chi(X : E_{F_j})$. The union of the fragments has conditional states that are pure as well, so $S(\rho_{\cup F_j}) = H_p(r_{\cup F_j}) \le H_p(0) = h(p)$. Subtracting gives the last claim. □

With $p = 1/2$, $c = 0.8$ and $n = 60$ memories in twelve blocks of five, the local entropy is 7.661 nats while the entropy of all sixty memories is 0.693147 nats, equal to $\ln 2$ to nine digits. A reader with access to single fragments sees entropy grow with every record. The global entropy of the environment saturates at the entropy of the label. Which of the two grows without bound is fixed by the access structure, not by the dynamics.

9 What must be restored to keep recording

Order recovery consumes access to a sequence of prepared trials. It does not grant a renewable device. A protected old record can persist without further entropy removal under compatible noise; independent flips create a different maintenance task.

For a bank of conditionally independent copies of one retained label, with initial error ε, an inaccessible independent flip of probability p changes the error to $q = p + (1-2p)\varepsilon$. Returning the complete label–bank–controller state removes $R[h(q) - h(\varepsilon)]$ nats. Proposition D.1 prices that complete target, rather than the number of distinct labels. Proposition D.2 constructs label-controlled thermal swaps that restore it for positive target error and approach the entropy-removal price. Their gates, gaps, thermal preparations and spent bath ports remain charged; they do not prepare or renew the label boundary for free. The full statements and proofs are in Appendix D.

A sole thermal chiral exhaust of fixed capacity adds the long-window quadratic rate term in Eq. (15). Those exhaust assumptions are separate from the thermal-qubit implementation. They connect actual maintenance to HAFF-D's conditional source and horizon readings, not event order to a gravitational law.

Reset is also distinct from undoing the historical relations. A label flip retains information, a SWAP can move it to another port, and a thermal blank reset has a specified entropy decrease. The inverse of the entire recording circuit restores the complete input when all required systems and gates are accessible. Appendix C retains the different reset permissions, finite-bath obstruction, reference-dependent bound and work account. None of these operations supplies a universal probability measure on reversed histories.

10 Which arrows coincide

Table 1 collects the directions studied above in four specified settings: the collision model with a coherent or a diagonal initial system state, the collision model with one repeated collision, and the autonomous two-qubit model of T-DOME I, in which a system qubit and an environment qubit build and return correlation without any pointer record [5].

Direction	collision, coherent S	collision, diagonal S	one repeated collision	two-qubit recurrence
Record count grows	yes	yes	can fall	no records
Local entropy of fragments grows	yes	yes	can fall	constant E
Marginal memory purity	falls	falls	falls, then returns	no fresh bank
System free energy falls	yes	no	falls, then returns	falls, then returns
Interventions reach only later records	yes	yes	no	no records
Local thermal reset bound $\propto R$	yes	yes	no final record	—
Correlation flows back to S	no	no	yes, with flip	yes

Table 1: Specified protocols, not universal arrows. The repeated-contact column uses the coherent, unmeasured flip continuation of Proposition 6.3. Reset prices refer to blank targets, not arbitrary changes of record content.

With pure fresh ports interacting once, record accumulation, local entropy growth and decreasing marginal port purity align; detached marginals also fix which source interventions can still reach them. Local thermal reset prices that bank relative to a blank target. These relations do not make every record update an entropy-producing operation. The mixed-port ledger shows that acquisition and action redistribute finite stock, while the echo reverses the local entropy and correlation changes. Source free-energy loss depends on initial coherence. Reduced-system backflow, record acquisition and bank renewal remain separate questions.

11 Time in physical theory

In Newtonian mechanics and in the Schrödinger equation time is an external parameter, and the equations have no intrinsic arrow. In canonical quantum gravity the Wheeler–DeWitt equation $\hat{H}|\Psi\rangle = 0$ contains no time parameter [11]. The Page–Wootters construction recovers an effective time from correlations between a clock subsystem and the rest of a stationary universal state [12], but it explains an ordering rather than its asymmetry. The thermodynamic arrow supplies the asymmetry through a coarse-graining and a low-entropy boundary condition [9, 10].

The present dynamical models use an external parameter. In Section 3, a known family of one-pass histories leaves the global permutation unknown to the reader, and endpoint relations recover it. This reconstructs a discrete event order under stated physical

permissions. A continuous clock variable and the asymmetry of its acquisition boundary are additional questions.

12 Relations and interfaces

The purity law gives the record boundary of the past hypothesis a quantitative form, while separating record accumulation from a source's loss of free energy [9, 10]. Mixed-environment redundancy [13] is one instance; the theorem also prices stock drawn from work and a bath. Page–Wootters correlations can supply ordering within a stationary state [12]. A direction of recording additionally needs the acquisition boundary and actual contact architecture specified here.

Complete-access reversal is compatible with retrocausal programmes [16]; restricted thermal reset is a different task. The timed causal-influence diagnostic computes supplied interventions in a physical recording process. It does not turn them into a derived causal partial order. The endpoint theorem instead solves a particular inverse problem: station-labelled intervention records identify an unknown global permutation within a known architecture. The two claims have different data and assumptions.

Paper E supplies the recording instruments [3]. The present posterior, intervention order and reference-dependent budgets are the record-side interfaces to T-DOME I's thermodynamic agent [5]; Paper C implements a separate maintained sensing body [1]. Paper D's heat and horizon bounds apply only under their matched source conditions [2]. The purity law supplies a physical input for Paper G's question of how selection permissions form [4]; it does not derive those permissions, clocks or a geometric algebra.

13 Conclusion

Records identify an order only when their relations depend on that order. Repeated pointer copying into distinct ports does not meet this condition: its commuting gates leave the same complete endpoint state for every permutation. Neither redundancy, entropy nor a more powerful endpoint measurement repairs that obstruction.

A record-before-exchange circuit does meet it. Independent classical labels and their retained copies expose each station's predecessor through a single nonzero correlation entry. Calibrated positive readout bias gives a finite-sample certificate for the unknown global chain, without a timestamp or final-carrier measurement. The known local gate direction and one-pass architecture are indispensable inputs. Fair read ports, missing reference access or correlated labels can remove identification; reading retained spent labels changes that permission boundary.

The direction belongs to a prepared history. Inverse gates act on the correlated actual output and can restore the full input. No cosmological arrow or suppression of inverse histories follows from reversibility and a freshness condition alone. The joint purity law and conditional record budget locate the stock in fresh ports, work or usable reference correlations rather than assuming pure marginals everywhere.

Recontact can remove a record and restore coherence. Sustaining a noisy bank instead requires complete-state renewal, realized here by label-controlled thermal swaps. A thermal exhaust adds its rate price. Order recovery is thereby connected to the resources that make records usable, without deriving its time parameter, contact architecture or primitive acquisition boundary.

A Exact checks of the order protocol

For $n = 3$ and $\varepsilon = (1/10, 1/5, 3/10)$, the check enumerates the 2^7 independent label/noise inputs with rational probabilities. It initializes the full eleven-register basis state, applies each actual CNOT and SWAP in turn, then reads only the endpoint K, Y coordinates in station order. All six permutations have distinct permitted endpoint laws and the correlation formula recovers each chain. The order supplied to the generator is used only to check the reconstruction, not appended to its readout data. The check also enumerates all 2^{11} basis inputs per permutation: the gates are bijective and their reverse sequence restores every input.

For the static obstruction, choose the three controlled rotations

$$(c_1, s_1) = (3/5, 4/5), \qquad (c_2, s_2) = (5/13, 12/13), \qquad (c_3, s_3) = (8/17, 15/17).$$

They give exactly equal full 16×16 rational operators for all six collision orders. This operator check covers coherent states and references as well as diagonal preparations; Proposition 2.1 proves the general statement.

Removing all K makes the six Y laws equal. With every read port fair, all six (K, Y) laws are equal, but the spent-T correlations recover all orders. With only one fair read port the six laws remain distinct in this instance. With two fair columns there are three distinct endpoint laws, including the equal-law pair $(1, 3, 2)$ and $(2, 1, 3)$ stated above. A common label-independent error bit across all read ports retains the predecessor formula. These are exact checks of the same circuit and its boundary changes, not noisy fits or a general chronology theorem.

B Partly mixed acquisition and conditional reference budgets

B.1 Partly mixed ports: acquisition, reading and action

Let a fair classical pointer X be held in S, initially $I/2$. Each of R fresh ports starts independently in $\eta_\alpha = (I + \alpha Z)/2$, with $0 \le \alpha \le 1$. Put $\varepsilon = (1-\alpha)/2$ and $i_\alpha = \ln 2 - h(\varepsilon)$. Apply controlled X from S to each port, using the same physical mixed-port coupling as Paper E. The resulting state is diagonal:

$$P(x, \boldsymbol{y}) = \tfrac{1}{2} \prod_{j=1}^{R} (1-\varepsilon)^{\mathbf{1}_{y_j = x}} \varepsilon^{\mathbf{1}_{y_j \neq x}}. \tag{10}$$

An actual Z reading obtains Y_j. The stated calibration fixes which port value labels which pointer; a common unknown reversal would not be removed by collecting more ports.

Proposition B.1 (A saturated acquisition and action ledger)**.** *Every port has Holevo and accessible information i_α, equal to its initial purity deficit. Write $I_R = I(X : \boldsymbol{Y})$. Then*

$$\begin{aligned} S(SE) &= \ln 2 + Rh(\varepsilon), & S(E \mid X) &= Rh(\varepsilon), \\ S(E_j) &= \ln 2, & S(E) &= Rh(\varepsilon) + I_R, \\ \sum_j S(E_j) - S(E) &= Ri_\alpha - I_R, & H(X \mid \boldsymbol{Y}) &= \ln 2 - I_R. \end{aligned} \tag{11}$$

For $R = 1$, feedback controlled by the actual port, $D = X \oplus Y$, leaves the source in η_α and the port in $I/2$, independently. For degenerate source and port Hamiltonians, the source free energy increases by $k_B T i_\alpha$, at zero net endpoint work for the ideal recording and feedback unitaries.

Proof. Conditional ports have entropy $h(\varepsilon)$, while their fair-label averages are $I/2$. Their states commute, so Z reading attains Holevo information. Thus $\sum_j \chi = R i_\alpha$ saturates the pointer-controlled bound for any number of independent ports, including partly mixed ones. Conditional product structure gives $S(SE)$ and $S(E \mid X)$; the definition of I_R gives $S(E)$. For one port, $Y = X \oplus N$, where N is independent with $\Pr(N = 1) = \varepsilon$. Fair X makes Y independent of N; feedback gives $D = N$. Both gates are basis permutations and preserve the complete joint entropy. The source entropy falls from $\ln 2$ to $h(\varepsilon)$, giving the free-energy increment. □

The posterior is available from the actual record, not just from the algebra it generates. For $k = \sum_j y_j$,

$$\Pr(X = 1 \mid \boldsymbol{y}) = \frac{(1-\varepsilon)^k \varepsilon^{R-k}}{(1-\varepsilon)^k \varepsilon^{R-k} + (1-\varepsilon)^{R-k} \varepsilon^k}, \tag{12}$$

with the deterministic limits interpreted on their support. A reversible gate controlled by $\boldsymbol{y}$ can send X to $X \oplus g(\boldsymbol{y})$. Majority is the minimum-error choice here; ties may be resolved by a fixed label. All original ports remain. Computing and retaining an extra classical copy or a decision register would require its own physical stock, so the ideal implementation uses the ports themselves as controls.

The one-port result locates the resource after action: initial memory purity has moved into the source, not disappeared into an unpriced bath. It supplies free energy for a subsequent task, not work already delivered to a work store. T-DOME I implements such a task and accounts for its reset. Neither the source nor the bank has completed a thermodynamic cycle. Paper C's blank-body transfer is another use of finite stock, with different dynamics; its resource account must be retained when the two mechanisms are composed.

B.2 Old references change the acquisition boundary

Proposition B.2 (Conditional budget with a retained reference)**.** *Let X be classical, K a retained reference, and $E = E_1 \otimes \cdots \otimes E_R$, with $\dim E_j = d_j$. The initial state may correlate all of them. Recording applies $\sum_x |x\rangle\langle x| \otimes I_K \otimes V_x$, with V_x acting on E only. Then*

$$\sum_j I(X : E_j \mid K)_{\text{out}} \le \sum_j \ln d_j - S(E \mid XK)_{\text{in}}. \tag{13}$$

For independent fresh product ports the right side is their summed purity deficit. This bounds final conditional information, not only its increment during recording.

Proof. The controlled unitary preserves $S(E \mid XK)$. Strong subadditivity gives $\sum_j S(E_j \mid XK)_{\text{out}} \ge S(E \mid XK)_{\text{out}}$. Also $S(E_j \mid K) \le S(E_j) \le \ln d_j$. Subtracting these inequalities proves the claim. □

Take independent fair bits X, L, an old reference $K = L$ and a new port $E = L$. Each port is marginally maximally mixed. Controlled X writes $Y = L \oplus X$ into E; E alone carries no information about X, but reading K, E recovers it exactly. Here

$S(E \mid XK)_{\text{in}} = 0$ and $I(X : E \mid K)_{\text{out}} = \ln 2$, attaining Eq. (13). The usable stock is the old correlation and permission to read the reference jointly, not a hidden pure marginal. Creating and retaining the two correlated copies of L belongs to the earlier history's account.

For classical readable K, the conditional information is the Holevo quantity averaged over its records, and in this commuting example it is actually attained. A quantum reference need not admit that interpretation: the entropy inequality remains valid, but its conditional mutual information is not by itself a measurement permission or accessible information. This distinction connects acquisition to T-DOME I's retained side information without identifying unlike tasks.

C Undoing records

Proposition C.1 (Local and joint reset)**.** *Let the pure-port model's memories have degenerate Hamiltonians. Each eraser couples the memories it controls to its own independent Gibbs bath at inverse temperature β, with interactions off at endpoints. A reset with final memory entropy tending to zero has:*

1. *for separate fragment erasers,*

$$\liminf W^{\text{local}} \geq k_B T \sum_j S(\rho_{F_j}) = k_B T \sum_j \chi(X : E_{F_j}) \geq R(1-\delta) k_B T h(p)$$

   when those fragments are Holevo records;

2. *for one eraser controlling all memories but not S, reset work infimum $k_B T S(\rho_{E_{[n]}}) = k_B T H_p(r_{[n]})$, approached with vanishing reset error and increasingly large, engineered thermal reservoirs;*

3. *for an agent controlling S and every memory, the inverse $U_n^{-1} \cdots U_1^{-1}$ restores the initial product state and source coherence at zero net endpoint work in the ideal recording model.*

Proof. For one controlled memory M and its initially independent bath, unitarity and subadditivity give $\Delta S_B \geq S(\rho_M) - S(\rho'_M)$. The Gibbs identity gives $\beta W = \beta Q = \Delta S_B + D(\rho'_B \| \tau_B)$, since the degenerate memory energy is unchanged. Hence $\beta W \geq S(\rho_M) - S(\rho'_M)$. Apply this separately, or to the entire bank. Separate baths cannot exploit correlations between fragments; their marginal inputs fix these entropy decreases. The pure-port bank marginal has rank at most two. A unitary acting only on the degenerate bank maps its two eigenvectors to a logical qubit tensored with blank remaining bits. The logical qubit has eigenvalues $(1-z, z)$, $0 \leq z \leq 1/2$, and entropy $h(z) = H_p(r_{[n]})$. For $z > 0$, thermal swaps through decreasing positive error probabilities to $\epsilon > 0$ approach $k_B T[h(z) - h(\epsilon)]$, as in Proposition D.2. Taking $\epsilon \to 0$ gives the joint reset infimum; for $z = 0$ the bank unitary suffices. The same thermal-swap method applies more generally [14]. The inverse recording unitaries have the same unchanged endpoint energy as the forward gates. □

Exact purification of a mixed memory is not attainable with a finite-dimensional, full-rank Gibbs bath at finite energies and temperature. Its input joint rank is $\text{rank}(\rho_M) d_B > d_B$, whereas a pure final marginal would force the joint state to have rank at most d_B.

Unitary evolution preserves rank. Thus the work infimum is not an exact finite-bath reset guarantee; finite reset error uses the entropy decrease in the proof, with nonnegative bath relative-entropy and correlation terms.

For $p = 1/2, c = 0.8, n = 60$, the corresponding asymptotic reset bounds are $19.505\, k_B T$ for sixty separate ports, $7.661\, k_B T$ for twelve five-port fragments, $5.368\, k_B T$ for eight seven-port readable fragments, and $0.693\, k_B T$ for the joint bank. The local price reflects the access structure, not a larger amount of distinct label content.

Lemma C.2 (Reset with an old reference)**.** *Let M, K have zero endpoint Hamiltonians and the bath be thermal and independent of MK. Allow a joint unitary on MKB that leaves the marginal of K unchanged. Then*

$$\beta W = \beta Q \geq S(M \mid K)_{\text{in}} - S(M').$$

In the pure-reset limit the lower bound is $S(M \mid K)_{\text{in}}$.

Proof. The decrease of the retained joint entropy is $S(MK)_{\text{in}} - S(M'K')_{\text{out}}$. Since $S(M'K') \leq S(M') + S(K') = S(M') + S(K)_{\text{in}}$, it is at least $S(M \mid K)_{\text{in}} - S(M')$. The independent-bath argument above applies to the joint retained state. □

The target here preserves the declared reference marginal. If other reference correlations must also be restored, they belong to the retained target and its ledger. Quantum conditional entropy is not a universal attained work price without an allowed protocol. For two classical copies $M = K = L$ of a fair bit, controlled XOR from K resets M exactly at zero endpoint work and leaves K unchanged. Its conditional entropy is zero, although $S(M) = \ln 2$. Permission to use the reference changes the reset task.

Changing a record is not resetting it. A unitary bit flip reverses its labels at zero endpoint work for a degenerate memory and retains all information. Swapping a record into an independent maximally mixed port makes the old memory uninformative at zero endpoint work, but transfers the record into the other port instead of destroying the joint history. Local thermal reset, such transfer, error correction and inversion of the complete source–bank unitary cannot be assigned one erasure bill. Exact history reversal is allowed when all required gates and systems are accessible. Neither its probability nor a suppression of reversed histories follows without an ensemble and measure [16].

D Renewing records instead of making new ones

A written distinction and a renewable distinction are different resources. The purity law prices acquisition from its declared product boundary. It does not make an old record disappear when fresh blank memories run out. For example, a copied Z label remains unchanged under Z dephasing, although each fair-label record has zero marginal purity. Maintaining it costs no entropy removal in that noise model. Independent bit flips give a different, quantitative renewal problem.

Proposition D.1 (The entropy price of a cyclic record bank)**.** *Let a classical label X be retained unchanged, and let R degenerate binary registers C_i be conditionally independent copies of X with error $\varepsilon \in [0, 1/2)$. Between corrections each undergoes an independent flip of probability $p \in (0, 1/2]$, whose noise port is inaccessible. The controller starts independent of XC and is unchanged by this noise. A correction returns the complete*

joint state of X, the bank and its controller to the pre-noise state. Its bath starts thermal and independent, and its interactions vanish at the endpoints. Put

$$q = p + (1-2p)\varepsilon, \qquad d_{p,\varepsilon} = h(q) - h(\varepsilon) > 0. \tag{14}$$

The entropy removed from the retained state is exactly $Rd_{p,\varepsilon}$ per correction, and its bath heat obeys $\beta Q \geq Rd_{p,\varepsilon}$.

If corrections occur at frequency ν and the sole exhaust has the thermal, zero-charge, extensive-current properties of HAFF-D's chiral exhaust [2], the mean heat current obeys

$$\overline{\dot{Q}} \geq k_B T \nu R d_{p,\varepsilon} + \frac{3\hbar}{\pi c_{\mathrm{ex}}} (\nu R d_{p,\varepsilon})^2. \tag{15}$$

Thus an exhaust-power budget $\overline{\dot{Q}} \leq J$ permits at most

$$R \leq \frac{\sqrt{(k_B T)^2 + 4\alpha J} - k_B T}{2\alpha \nu d_{p,\varepsilon}}, \qquad \alpha = \frac{3\hbar}{\pi c_{\mathrm{ex}}}. \tag{16}$$

Here c_{ex} is dimensionless chiral capacity, and $\nu R d_{p,\varepsilon}$ is the dimensionless entropy removed per unit time, in nats per time. The current bound is a long-window extensive average, not an individual pulse identity. No inference from vanishing budget to vanishing records follows when $d_{p,\varepsilon} = 0$.

Proof. Before noise $S(XC) = H(X) + Rh(\varepsilon)$, and after it $S(XC) = H(X) + Rh(q)$. Cyclicity of the entire retained state prevents a controller or a correlation with the label from absorbing that entropy without charge. For its independent Gibbs bath B, a joint unitary and subadditivity give $\Delta S_B \geq Rd_{p,\varepsilon}$. The Gibbs identity $\beta Q = \Delta S_B + D(\rho'_B \| \tau_B)$ proves the heat bound. The actual net extensive entropy current is at least $\nu R d_{p,\varepsilon}$. Substitution into HAFF-D's increasing finite-rate bound gives Eq. (15); inversion gives Eq. (16). □

Proposition D.2 (Thermal correction with an accessible label)**.** *Under the renewal proposition's conditional-iid error law, assume $0 < \varepsilon < q \leq 1/2$, and allow retained X to control reversible XOR gates on each bank register. Supply an independent Gibbs bath qubit with gap $\Delta_\varepsilon = k_B T \ln[(1-\varepsilon)/\varepsilon]$ for each register. XOR by X, swap with that bath qubit, and XOR by X again restore the complete retained X, C, controller target. Per register the bath heat and mean driven work are*

$$Q = W = k_B T (q - \varepsilon) \ln[(1-\varepsilon)/\varepsilon]. \tag{17}$$

With $q = t_0 > t_1 > \cdots > t_m = \varepsilon$, successive swaps with independent Gibbs qubits of gaps $\Delta_i = k_B T \ln[(1-t_i)/t_i]$ give

$$\beta Q = \beta W = \sum_i (t_{i-1} - t_i) \ln[(1-t_i)/t_i] \longrightarrow h(q) - h(\varepsilon) \tag{18}$$

as the partition mesh tends to zero. All stages have finite dimensions and finite gaps for a fixed positive target error. The finite-step excess is exactly

$$\beta Q - [h(q) - h(\varepsilon)] = \sum_i D(\mathrm{Bern}(t_{i-1}) \| \mathrm{Bern}(t_i))$$

per register; the spent baths end independent of the retained target.

Proof. The first XOR converts $C_i = X \oplus N_i$ into the independent error bit N_i of probability q, without changing X. The thermal swap replaces it by an independent error bit of probability ε; the second XOR restores the intended conditional law. The bath output has probability q, so its energy gain is $\Delta_\varepsilon(q - \varepsilon)$. Controllers remain untouched, and conditional independence is restored. For several swaps, the output of bath qubit i has probability t_{i-1}. Summing its energy gain gives Eq. (18). The right-endpoint Riemann sums tend to $\int_\varepsilon^q \ln[(1-t)/t]\,dt = h(q) - h(\varepsilon)$. The degenerate retained registers have no endpoint energy change, so the first law gives $W = Q$. Each swap's heat equals its error-bit entropy decrease plus $D(\mathrm{Bern}(t_{i-1})\|\mathrm{Bern}(t_i))$. Telescoping gives the stated finite-step excess. All initial error and bath bits are independent after the first XOR, and the swaps merely permute them; hence the spent baths are independent of the final target. □

This is a label-preserving use of the thermal swap construction of Ref. [14]. These swaps are driven gates between degenerate bank bits and nondegenerate bath qubits, not spontaneous energy-conserving collisions. The label's control permission, thermal preparations, clock and switching belong to the implementation. Spent bath qubits retain the exported entropy and are not reset by the construction. It realizes cyclicity of the declared retained bank, not an autonomous battery-complete apparatus or an indefinitely recycled bath. The vanishing-error limit $\varepsilon \to 0$ also requires growing gaps or an appropriate limiting reservoir. This thermal-qubit implementation does not establish the separate sole-chiral-exhaust conditions or sharpness of its finite-rate bound.

For $p = .12$ and $\varepsilon = .05$, one correction removes .2378232484 nat per register. Eight correlated copies of one fair label require 1.9025859874 nats of renewal, despite carrying only one bit of distinct label content. Their independent degradation, not the number of distinct labels, sets this maintenance bill. The cyclic target and the independent noise are substantive inputs. Jointly correlated errors, an accessible noise record, a noncyclic pure controller or a protected bank can change the cost. The heat-current claim additionally needs the actual exhaust assumptions; it is not a Landauer bound on every control pulse.

This gives the record side of a maintained observer. T-DOME III and Brain Anchor let retained history choose which access to repair [17]; Eq. (16) prices one implementation of the retained bank on which such a choice relies. Foundation's exact fixed algebras describe passive protection, not a requirement to export positive entropy indefinitely. Under HAFF-D's further matched horizon-flux conditions, the bank's removal rate can enter the conditional horizon-entropy bound. Those conditions connect a real renewal process to gravity; record existence alone does not.

E Joint-state checks

The mixed-port check constructs the complete diagonal density operator and applies the recording CNOTs, not separately chosen conditional laws. For $\alpha = 0, .2, .8, 1$ and $R = 1, 2, 3, 5$, it verifies Eq. (11), the actual Z-record posterior and one-port feedback. The maximally mixed keyed example is an explicit three-bit permutation. Forty full-density examples with a quantum reference and two initially correlated ports test the conditional bound; the entropy proof establishes it generally.

The coherent echo check applies both collisions and the intermediate flip to the same pure state, verifying $UX_SU = X_S \otimes R$, all marginal and joint entropies, and the recovery of coherence. For renewal, an eight-input XOR–SWAP–XOR permutation verifies the full label–bank target. Thermal chains with $1, 4, 16, 64, 256, 1024$ stages verify the heat

and its relative-entropy excess. At $p = .12, \varepsilon = .05$, one stage costs $0.31799941\, k_B T$ per register; 1024 stages cost $0.23789030\, k_B T$, approaching $0.23782325\, k_B T$. These checks certify the finite-dimensional specified circuits, not the physical chiral exhaust.

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Structural Limits of Unification: Internal Selection, Revisable Permissions and Physical Sources

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Abstract

Can a theory explain the selection rules by which its own physical systems read and act? The operational limit established here is relative to declared permissions, not a universal obstruction to a unified theory. We distinguish an internally executed rule, a history-formed permission and an explanation of the dynamics that makes either possible. An operational equivalence result fixes one boundary: when two dynamics agree on an instrument-preserved observable algebra, no finite adaptive experiment within those permissions can distinguish them. This boundary can move when permissions change; it does not establish a final, unrevisable Cut or prohibit physical self-reference.

The constructive side is a record-driven program with a fixed executor. Controlled couplings select protected algebras, while shared spectra permit coherent choices without central program labels. A qubit loop selects its own protection and a physical field source. Noise limits the duration of history-dependent source contrast. A preceding quantum record can select the two sustained configurations with a maximum contrast equal to its trace distance. Sharp readout selection has a different cost: for the great-circle qubit family, program dimension scales as $\delta^{-1/2}$ under the stated basis-state error criterion, equally for pure and mixed programs. These results support formation of revisable permissions inside a physical architecture. They separate this attainable internal closure from the stronger claim that a model explains every one of its own physical inputs.

1 Introduction

A record can change what a system is able to do next. A damaged body may install a different protective encoding; a reader may move its probe; an internal estimate may lower the authority given to an unreliable response. These changes concern actual operations. To explain them, a theory of selected structure needs a physical account of how a history selects and revises the relations that carry its own activity.

A second question concerns the scope of such an explanation. Does internalizing each selector also explain the laws and physical permissions that implement it? A finite programmer cannot settle this question for every possible theory. Conversely, leaving

an executor in one model's inputs is not evidence that its organization can never be explained. We ask which inputs a construction actually replaces, which distinctions its own experiments can settle, and what a further unification would have to connect. Section 2 makes these three questions separate before constructing the internal selector.

Papers A–F separate parts of this problem. Channels determine which distinctions are read in Paper A; couplings and observation windows determine which survive in Paper B; instruments create records in Paper E; their retention and renewal have resource prices in Paper F [1, 2, 4, 5]. Paper C connects sensing and intervention to a body's viability, while Paper D asks how selected structure and physical sources constrain geometry [25, 3]. Here the question is whether the choice of operation can itself arise from records inside the system.

An operation, a stored description and the device that executes it are different physical objects. We specify an internal program register, a fixed executor and an update instrument. Under these inputs the system selects its couplings, maintains a choice through the records it protects, and changes the matter it emits. The choice need not be a central environmental record. Its origin in a distinguishable history and its effect on a source can be tested separately.

The main construction is a qubit with two record-selected protective configurations. Their history-dependent source weights remain different over a finite noisy window. We then identify the exact information that can pass from a quantum record to this source choice, including a construction that reaches the trace-distance bound. An error bound makes the same finite window robust to imperfect execution. These statements supply a causal interface from history to permission and source; they do not use a reconstructed geometry as a substitute for an actual change of matter.

The cost of selecting operations is separate from the cost of retaining their records. Sharp alternatives require orthogonal program supports. For the one-parameter great-circle qubit family the required register dimension grows as $\delta^{-1/2}$, and mixing cannot improve it under the two basis-state error criterion. Section 7 states this result; Appendix A contains the full proof and numerical bounds. Section 8 places the finite target-register boundary in its proper scope. Section 9 develops the broader formation mechanism and its failure conditions.

Breuer's state self-measurement limit [12] and Wolpert's physical inference limits [13] concern what an embedded system can know. Here we ask what it can choose and execute. The existence of self-reference is not in dispute: self-reproducing automata, recursive programs and fixed-point constructions provide positive examples [14, 21, 22]. What requires explanation is the physical relation between a description, its execution and the history that revises it.

2 Internal closure and explanatory reach

Three claims of completeness require different evidence. *Execution closure* means that the declared operations, records and updates are physical parts of the modeled process. *Formation* means that some usable relation or permission depends on acquired history rather than a preinstalled answer. *Explanatory completeness* would additionally account for the organization supplied to that process. The finite loops below establish the first and particular instances of the second. Their success alone cannot establish the third; nor does their use of specified primitives disprove it.

There is an operational way to locate a model's current boundary. It concerns the entire allowed experiment, not only a snapshot or an internal verbal description.

Proposition 2.1 (An internal experiment can identify only its operational class)**.** *Let $\mathcal{B}$ be a unital observable subalgebra of a finite-dimensional ambient algebra. Two models have initial states ρ_0, ρ_1 with the same restriction to $\mathcal{B}$. For every permitted command a, their Heisenberg evolution channels $\Phi_{0,a}^*, \Phi_{1,a}^*$ preserve $\mathcal{B}$ and agree there. Every outcome map of the permitted instruments also preserves $\mathcal{B}$ and has the same restriction in both models. Use the same adaptive command policy, depending only on the retained outcomes and model-independent private random choices. Then every finite outcome history has the same probability in the two models. With equal prior probabilities, no decision based on those histories identifies the model with success exceeding $1/2$.*

More generally, a property inferred exactly from these experiments must be constant on their operational equivalence classes.

Proof. For a fixed history, pull back the terminal identity through its outcome maps and evolution channels, in reverse composition order. Each intermediate effect remains in $\mathcal{B}$; the two models therefore give the same effect there, and their initial states give it the same expectation. Conditioning on any private random seed and then averaging preserves equality. Adaptive choices do not change the argument because the same realized history selects the same command. Thus the complete finite record laws agree. Any decision function has the same distribution under both hypotheses, giving success $1/2$ at equal priors. If a property differed inside one equivalence class, the same record law could not identify both values exactly. □

A qubit gives a direct example. Compare the identity channel with complete Z dephasing. They agree on $\mathcal{B} = \mathrm{span}\{I, Z\}$, so experiments whose preparation, evolution and outcome maps remain closed on this algebra cannot distinguish them. The statement even holds with the same initial $|+\rangle$ in both models if only those outcome maps are allowed. If an X measurement is admitted after one use, its $+$ probability is 1 versus $1/2$; their record laws now have total variation $1/2$. The boundary was an experimental restriction, not a prohibition imposed on every possible observer. Extra initial correlations or a new transducer must be included in the model before applying the proposition.

This is ordinary operational indistinguishability, used here to separate a closed internal account from a unique account of its underlying dynamics. It provides no theorem that all theories are incomplete: a sufficiently rich instrument family may separate the candidate models. It also provides no theorem that every selection must be written into a central classical record. The coherent program example below denies that additional premise.

A useful unification must replace some independently supplied relation by a joint mechanism with consequences beyond either component alone. HAFF-A and B make stable central sectors depend on a dephasing process, while their spatial interpretation still needs a response dictionary. Paper F treats the distinct identification of event order and its directional boundary. Q-RAIF B and C constrain work and propagation through the same represented generators when their physical alignment is supplied. Paper D's common-source test relates prepared stress, modular data and the geometric response, while modular consistency itself remains a demand on the proposed mechanism. These are explicit, different explanatory tasks. None is discharged by renaming an executor, and none requires deriving every primitive law before an intermediate organization can be explained.

The stronger philosophical proposal is that explanation proceeds through such revisable relations rather than a final internally certified description of everything. The results support revisability and concrete limits on certification. They do not establish that nature has an ontologically necessary last Cut, that an external selector must exist, or that no more unified physical theory is possible. Those conclusions would require premises beyond the operational and programming results.

3 A typed model of internal selection

An element of an observable algebra is an operator, not a description of an algebra, and a formula naming a selection rule is syntax until an encoding maps it into physical degrees of freedom. The following model makes that encoding explicit.

Definition 3.1 (Internal selection model)**.** *A finite internal selection model consists of*

1. *a finite-dimensional system S, on which selected operations act;*
2. *a program register P of dimension d_P and program states $|p\rangle \in P$, $p \in \mathcal{P}$, unit vectors that need not be orthogonal;*
3. *an executor, a fixed physical operation on $P \otimes S$, which for each program state determines an operation on S: a channel $\Phi_p(\rho) = \mathrm{Tr}_P\, \mathcal{G}(|p\rangle\langle p| \otimes \rho)$, a measurement E^p obtained by following $\mathcal{G}$ with a fixed measurement, or a program-controlled coupling to an environment;*
4. *records, the classical outcomes of instruments, and an update rule, a fixed instrument that prepares the next program state from records.*

A selection rule is internalized if the choice of operation is made by program states under a fixed executor and a fixed update rule. A configuration (p, ρ) is self-grounding if ρ is left stable by the operation selected by p and the update rule, applied to the records that ρ produces, returns p.

The space of descriptions is the set of program states, the encoding is the register, the evaluation map is the executor, and the admissible fixed points are the self-grounding configurations. The executor's dispatch map, $p \mapsto \Phi_p$, is physical dynamics; it is not itself a program. Whether it can be made one is the question of Section 8.

4 Selecting one's own accessible algebra

Paper B shows that a system coupled to its environment through commuting operators $A_1, \ldots, A_K$ at positive rates γ_k has, as its stable algebra, the commutant $\{A_k\}'$, and that environmental records can carry only the label of the block of that algebra. We now let the system choose its own couplings. Take an orthonormal program basis and program-controlled couplings

$$A_k = \sum_{p \in \mathcal{P}} |p\rangle\langle p| \otimes A_k^{(p)}, \tag{1}$$

where for each program p the operators $A_1^{(p)}, \ldots, A_K^{(p)}$ on S commute. Let $|p, x\rangle$ be the product of $|p\rangle$ with a joint eigenvector of the $A_k^{(p)}$, with eigenvalues $a_k^{(p)}(x)$.

Proposition 4.1 (Program selection and central records)**.** *For the couplings* (1)*:*

1. *they commute, and the algebra fixed by the generated dephasing semigroup is* $\{A_1, \dots, A_K\}'$*.*

2. *Compressed to the subspace* $|p\rangle \otimes S$ *of a program* p*, this algebra is* $|p\rangle\langle p| \otimes \{A_1^{(p)}, \dots, A_K^{(p)}\}'$*: each program selects the algebra that Paper B assigns to its own couplings.*

3. *The matrix element between* $|p, x\rangle$ *and* $|q, y\rangle$ *is multiplied by* $\exp[-\frac{t}{2}\sum_k \gamma_k(a_k^{(p)}(x) - a_k^{(q)}(y))^2]$*. Coherence between two programs survives exactly between joint eigenvectors with equal eigenvalue tuples. Disjoint joint spectra for* p *and* q *make every operator connecting those two programs decay. The label projector* $|p\rangle\langle p| \otimes \mathbf{1}_S$ *is central in the stable algebra if and only if the joint spectrum of* p *is disjoint from that of every other program. In that case environmental fragments can record its label.*

Proof. The operators are block diagonal in p and commute within each block, so they commute; the basis $|p, x\rangle$ diagonalizes all of them, with eigenvalues $a_k^{(p)}(x)$. Paper B's proposition on commuting couplings then gives the Schur-multiplier form in (3) and the fixed algebra in (1). Its compression to $|p\rangle \otimes S$ consists of the operators there that commute with the $A_k^{(p)}$, which is (2). A program label is central precisely when it is constant on each joint eigenspace, which gives the disjointness condition in (3). The record statement is Paper B's result that fragments carry only block labels, applied to the joint system. □

When the programs have disjoint joint spectra, their labels belong to the centre of the stable algebra and can become redundant environmental records. Shared eigenvalue tuples protect coherence between the corresponding configurations; the spectra need not coincide completely for this exception to occur. With $A = |0\rangle\langle 0| \otimes Z + |1\rangle\langle 1| \otimes X$ on a program qubit and a system qubit, both programs have eigenvalues ± 1, and the superposition $(|0, 0\rangle + |1, +\rangle)/\sqrt{2}$ is an eigenvector of A. Replacing X by $X + 3$ makes the spectra disjoint, and the same superposition decays to an equal mixture of the two selections. Both cases are computed in the accompanying check script.

For more than two programs, disjointness from just one alternative is not enough. With $A = |p\rangle\langle p| \otimes Z + |q\rangle\langle q| \otimes (Z + 3\mathbf{1}) + |r\rangle\langle r| \otimes Z$, the spectra of p and q are disjoint, but $|p, 0\rangle\langle r, 0|$ commutes with A and not with $|p\rangle\langle p| \otimes \mathbf{1}_S$. Thus p is not a central label. The condition in Proposition 4.1 concerns the whole program family.

Internal selection can also change a material source. Use the field normalization of Paper D, in which a coupling through a projector R obeys $U^\dagger j(u)U = j(u) + Rg'(u)/2$. The program controls that projector while the executor, drive and switching schedule stay fixed.

Proposition 4.2 (A program-selected material source)**.** *Let* R_p *be projectors on* S *and put*

$$A = \sum_p |p\rangle\langle p| \otimes R_p, \qquad U = \exp(-i\lambda A \otimes Q), \qquad Q = b + b^\dagger.$$

Take an initial state ρ_{PS} *independent of the field state* σ*. Set* $r = \mathrm{Tr}(\rho_{PS}A)$*. The output field marginal is exactly*

$$\sigma_{\mathrm{out}} = (1 - r)\sigma + r\, e^{-i\lambda Q}\sigma e^{i\lambda Q}.$$

When the displayed current moments exist, its vacuum-subtracted stress is

$$T_{\mathrm{out}}(u) = T_{\mathrm{in}}(u) + r\left[g'(u)\langle j(u)\rangle_\sigma + \frac{g'(u)^2}{4}\right].$$

In a fixed classical program branch p, a non-atomic quadrature spectral law and $\lambda \neq 0$ give the exact multiplicative domain $\{R_p\}'$ of the induced system channel. Thus changing the program can change both the represented algebra and the field source under the same executor.

Proof. Since $A^2 = A$, write $U = (1-A)\otimes 1 + A \otimes e^{-i\lambda Q}$. The off-diagonal terms have zero partial trace over PS, because $\mathrm{Tr}(A\rho_{PS}(1-A)) = 0$. This gives the field mixture even when the program is coherent or correlated with S. Squaring $j + Ag'/2$ gives the stress formula: initial independence supplies the mixed expectation $\langle Aj\rangle = r\langle j\rangle$, and $\langle A^2\rangle = r$. Within branch p, the channel is the two-block dephasing channel of Paper D; a non-atomic spectral law makes its nonzero-gap characteristic value strictly smaller than one, so equality in the Schwarz inequality holds exactly on $\{R_p\}'$. □

For $R_0 = (1+Z)/2$, $R_1 = (1+X)/2$ and $\rho_S = (1+0.8Z)/2$, a program bit prepared in 0 or 1 gives $r = 0.9$ or 0.5, respectively. Their conditional pointer algebras differ, although the two coupling spectra are identical. The field stress therefore need not reveal the program label as a central sector to respond to its choice. With a classical program mixture π, $r = 0.9\pi_0 + 0.5\pi_1$. A record-driven program update can change this weight without an externally assigned emission probability. The global algebra $\{A\}'$ stays fixed; the selected branch algebra on S and the outgoing source vary. This separates internal selection from a change of the executor. Paper D supplies the corresponding JT response for this same wavepacket with matched initial coefficient data; the current statement identifies its physical source, not the origin of the JT action or of a Type III geometric sector. The program, field preparation and switching remain resources of the executor.

5 A self-grounding selection

A selection rule can take as input a record that the selection itself protects. A small model gives two self-grounding fixed points and their history-dependent selection.

A memory qubit M is subject to rotation noise $H = \frac{\omega}{2}Y$. A classical program bit p selects a dephasing coupling of M with rate γ: through Z for $p = 0$ and through X for $p = 1$. Strong dephasing in a basis protects a record in that basis against rotation, by the quantum Zeno mechanism [19, 20]. Every time τ the memory is read in the protected basis, and the update rule sets p equal to the outcome. The rule is internalized in the sense of Definition 3.1, and it is self-applied: the record that sets the selection is a record in the algebra that the selection protects.

After a read the configuration is one of four: $F_0 = (p = 0, |0\rangle)$, $T_1 = (p = 1, |1\rangle)$, $F_1 = (p = 1, |-\rangle)$ and $T_0 = (p = 0, |+\rangle)$. In F_0 and F_1 the protected record agrees with the program it sets; these are the self-grounding configurations. In T_1 and T_0 the record has just switched the program to a basis in which that record is not sharp.

Proposition 5.1 (Self-grounding configurations)**.** *Without rotation noise, F_0 and F_1 are absorbing, and from T_1 and T_0*

$$\Pr(\textit{absorbed in } F_1 \mid T_1) = \tfrac{2}{3}, \qquad \Pr(\textit{absorbed in } F_1 \mid T_0) = \tfrac{1}{3}.$$

From any initial program and memory state the loop reaches F_0 or F_1 with probability one.

Proof. A read in the basis that is already sharp returns the same outcome, so F_0 and F_1 repeat. In T_1 the memory $|1\rangle = (|+\rangle - |-\rangle)/\sqrt{2}$ is read in the X basis, giving F_1 or T_0 with probability $\frac{1}{2}$ each; in T_0 the memory $|+\rangle$ is read in the Z basis, giving F_0 or T_1 with probability $\frac{1}{2}$ each. Writing a and b for the absorption probabilities into F_1 from T_1 and T_0, $a = \frac{1}{2} + \frac{1}{2}b$ and $b = \frac{1}{2}a$, so $a = \frac{2}{3}$, $b = \frac{1}{3}$. Every read from a transient configuration ends in an absorbing one with probability $\frac{1}{2}$, and the first read from an arbitrary initial state produces one of the four configurations. □

Corollary 5.2 (History selects a sustained source)**.** *In the zero-rotation loop, insert before each read a fresh field interaction of Proposition 4.2, using $R_0 = (1 + Z)/2$ and $R_1 = (1 + X)/2$. Each cycle uses a distinct tensor-factor port, in the same state and initially independent of the program, memory and all previous ports. The four-state transition law and its absorption probabilities are unchanged. In F_0 the output field is $e^{-i\lambda Q}\sigma e^{i\lambda Q}$; in F_1 it is σ. Thus the two self-grounding configurations carry different steady field sources under one executor. Starting from T_1 or T_0, the late-cycle mean source weight tends to $1/3$ or $2/3$, respectively.*

Proof. Within either program branch the field coupling is diagonal in the basis of the ensuing read. It changes no outcome probabilities, so the transition matrix of Proposition 5.1 is unchanged. The configurations $F_0 = (0, |0\rangle)$ and $F_1 = (1, |-\rangle)$ are eigenvectors of A with eigenvalues 1 and 0. They remain product states of program and memory, and each pass gives the claimed field output. The source weight therefore converges to the indicator of absorption in F_0. Its expectation is one minus the corresponding probability of absorption in F_1, which gives the stated fractions. □

The read/update history generates the source difference under one executor. Fresh ports and memory operations sustain it. Each cycle supplies a fresh product-state field port, a protected-basis read, the program update and its protecting environment. For a centered field input the coupling's mean switching work is $r_n E_g$, where $E_g = \lambda^2\bar{\omega} = \int g'^2 du/4$ and r_n is the actual source weight. Noncentered inputs add the displayed current cross term. Incoming beam preparation, memory retention and eventual reset have their own charges. The $1/3$ and $2/3$ weights arise from absorption of the two transient histories, not a continuing supply of new source information. Distinct modes of a common bosonic field must be orthogonal and prepared in a product state to realize these ports; mere time translation does not suffice. For Paper D's exact packet, a translation by Δ has overlap modulus $4a^2/(4a^2 + \Delta^2)$. Source accounting for such overlapping pulses requires their joint field state. The full-line packet is a scattering idealization; a record decided at a finite time needs a compact switch scheduled after that decision, as constructed in Paper D. With rotation, the following lifetime uses the stated continuous dephasing generator.

The dephasing plays no role without noise, because the reads are made in the protected basis. With noise it decides whether the self-grounding configurations last.

Proposition 5.3 (Lifetime of a self-grounding selection)**.** *With rotation noise, the probability of leaving F_0 (or F_1) in one cycle is $\varepsilon = \frac{1}{2}(1 - z(\tau))$, where z solves $\ddot{z} + 2\gamma\dot{z} + \omega^2 z = 0$, $z(0) = 1$, $\dot{z}(0) = 0$. For $\gamma > |\omega|$,*

$$z(\tau) = \frac{\lambda_+ e^{\lambda_- \tau} - \lambda_- e^{\lambda_+ \tau}}{\lambda_+ - \lambda_-}, \qquad \lambda_\pm = -\gamma \pm \sqrt{\gamma^2 - \omega^2},$$

so $\varepsilon \approx \omega^2\tau/(4\gamma)$ *when* $\gamma \gg |\omega|$, $\gamma\tau \gg 1$ *and* $\omega^2\tau/\gamma \ll 1$. *Without protection* $\varepsilon = \sin^2(\omega\tau/2)$*; with protection in the wrong basis* ε *approaches* $\frac{1}{2}$ *in the strong-dephasing regime.*

Proof. In F_0 the Bloch vector starts at $z = 1$. The rotation gives $\dot{x} = \omega z$, $\dot{z} = -\omega x$, and dephasing through Z adds $-2\gamma x$ to $\dot{x}$. Eliminating x gives the stated equation; a Z read returns 1 with probability $\frac{1}{2}(1 - z)$. The case F_1 is the same with x and z exchanged, since rotation about y treats both records alike. For $\gamma \gg \omega$, $\lambda_+ \approx -\omega^2/(2\gamma)$ dominates. Without dephasing $z = \cos\omega\tau$. Dephasing in the wrong basis destroys the record at rate 2γ. □

For $\omega = 0.1$ and $\tau = 1$, the escape probability per cycle is 1.42×10^{-3}, 2.37×10^{-4} and 2.49×10^{-5} for $\gamma = 1$, 10 and 100, against 2.50×10^{-3} without protection. Protection in the wrong basis gives 0.433 at $\gamma = 1$ and approaches 0.5 at the larger rates. At $\gamma = 100$ a self-grounding selection lasts on average 4.0×10^4 cycles before the loop passes through a transient configuration and settles again, in F_0 or F_1.

The residence time alone does not determine how long two histories produce different sources. The same four-state process gives that distinction exactly.

Proposition 5.4 (History-dependent source window)**.** *Let* $\gamma > 0$, $\omega \neq 0$ *and* $\tau > 0$. *In each cycle first evolve for time* τ *with the selected rotation–dephasing generator, then apply the fresh field interaction of Corollary 5.2, and then read and update. Put*

$$k = \sqrt{\gamma^2 - \omega^2}, \quad f = e^{-\gamma\tau}\left(\cosh k\tau + \gamma\frac{\sinh k\tau}{k}\right), \quad b = \omega e^{-\gamma\tau}\frac{\sinh k\tau}{k},$$

with analytic continuation when $\gamma < |\omega|$ *and* $\sinh(k\tau)/k = \tau$ *at* $k = 0$. *Set* $\varepsilon = (1 - f)/2$, $q_+ = (1+b)/2$, $q_- = (1-b)/2$. *In the post-read order* (F_0, T_1, F_1, T_0) *the transition matrix and the source-weight vector are*

$$T = \begin{pmatrix} 1-\varepsilon & \varepsilon & 0 & 0 \\ 0 & 0 & q_+ & q_- \\ 0 & 0 & 1-\varepsilon & \varepsilon \\ q_- & q_+ & 0 & 0 \end{pmatrix}, \qquad r = \begin{pmatrix} 1-\varepsilon \\ q_- \\ \varepsilon \\ q_- \end{pmatrix}. \tag{2}$$

For a post-read distribution μ_n*, the next field output has mean stress*

$$T_{\mathrm{in}}(u) + (\mu_n r)\left[g'(u)\langle j(u)\rangle_\sigma + g'(u)^2/4\right].$$

The chain has the unique stationary law

$$\pi = \frac{(q_-, \varepsilon, q_+, \varepsilon)}{1 + 2\varepsilon}, \qquad r_\infty = \pi r = \frac{1}{2} - \frac{b}{2(1 + 2\varepsilon)}.$$

For initial states T_1 *and* T_0*, define*

$$R_1^\circ = \varepsilon + f\frac{q_-^2}{1 - q_+q_-}, \qquad R_0^\circ = \varepsilon + f\frac{q_-}{1 - q_+q_-}.$$

Their actual source weights satisfy, for every integer $n \geq 0$,

$$|\mu_n r - R_i^\circ| \leq n\varepsilon + (q_+q_-)^{\lfloor n/2\rfloor}, \qquad \mu_0 = T_i. \tag{3}$$

Thus a small escape rate preserves different history-selected sources over $1 \ll n \ll \varepsilon^{-1}$*, although both histories ultimately have the same stationary mean.*

Proof. The two Bloch generators on (x, z) are

$$\begin{pmatrix} -2\gamma & \omega \\ -\omega & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & \omega \\ -\omega & -2\gamma \end{pmatrix}.$$

Exponentiating them gives (2). The source weight is the population of the selected projector immediately before the read, not its value in the preceding post-read state. The intervening field coupling preserves that population. It equals the probability that the next program is 0. For nonzero rotation and positive dephasing, the Bloch norm strictly decreases from each of the four pure initial states over any positive interval. Hence $0 < \varepsilon < 1$ and $0 < q_{\pm} < 1$. The graph of T is connected and has self-loops, giving uniqueness and convergence. Direct multiplication gives $\pi T = \pi$ and the stated source weight.

For the window bound, replace the two F rows by absorbing rows, retaining $q_{\pm}$ in the transient rows and retaining the same observable r. In this comparison chain the absorption probabilities into F_0 are $q_-^2/(1 - q_+q_-)$ and $q_-/(1 - q_+q_-)$; the corresponding limiting expectations of r are R_1° and R_0°. Transient survival after n steps is at most $(q_+q_-)^{\lfloor n/2 \rfloor}$. Couple the true and comparison chains until their first escape from an F state. That event has probability at most $n\varepsilon$, and $0 \le r \le 1$, proving (3). □

The history window and the lifetime of one configuration are different. The characteristic polynomial is

$$(z - 1)\left[z^3 + (2\varepsilon - 1)z^2 + (\varepsilon^2 - q_+q_-)z + \varepsilon^2 - \varepsilon + q_+q_-\right].$$

For small ε the slow root is $1-\varepsilon/(1-q_+q_-)+O(\varepsilon^2)$, so the late history contrast decays on a scale of about $(1-q_+q_-)/\varepsilon$ cycles. In the strong-protection regime of Proposition 5.3, this is about $3\gamma/(\omega^2\tau)$, whereas a single configuration has mean residence $1/\varepsilon \approx 4\gamma/(\omega^2\tau)$. At $\gamma = 100$, $\omega = 0.1$, $\tau = 1$, the source weights from T_1 and T_0 are respectively 0.33844 and 0.66090 after 1000 cycles, and 0.49370 and 0.50579 after 10^5 cycles. Both approach 0.4997500.

The loop selects a protection because the record it protects says so, and its history selects a material source through the same program-controlled executor. Finite noise makes that selection metastable rather than permanent. The rotation–dephasing stage, field interaction and read are sequential operations here; simultaneous rotation during a finite field pulse would require a different transition law. The protecting environment, fresh packets, read instrument and update map are supplied. The loop generates the history-dependent choice, not these resources.

6 When a history changes the next source

The four-state loop starts with explicit configurations. An earlier history need not already be a classical choice among them. Let R be a retained physical record, and let $\Lambda : R \to P \otimes M$ be a fixed preparation channel acting after that history has formed. It can include a measurement, an outcome record and a controlled preparation. The next fresh field port remains independent of R, P, M and old emitted ports. This time ordering gives a direct criterion for what the history can cause.

Proposition 6.1 (Record-to-source distinguishability)**.** *For two record states ρ_0, ρ_1 and the projector A in Proposition 4.2, the two source weights satisfy*

$$r_0 - r_1 = \mathrm{Tr}[(\rho_0 - \rho_1)\Lambda^*(A)], \qquad |r_0 - r_1| \le D(\rho_0, \rho_1) := \tfrac{1}{2}\|\rho_0 - \rho_1\|_1.$$

For the zero-rotation loop, the largest contrast over fixed preparation channels Λ equals $D(\rho_0, \rho_1)$. A channel attaining it prepares F_0 or F_1 from an actual binary record measurement. Both subsequent histories remain in their chosen configuration, so the same contrast occurs in every later fresh source port.

Proof. The first equality is the adjoint-channel identity. Complete positivity and unitality of Λ^* give $0 \le \Lambda^*(A) \le 1$. Writing the trace-zero Hermitian operator $\rho_0 - \rho_1$ as its positive part minus its negative part bounds the expectation of any effect by D. To attain the bound, let H be the spectral projector onto its positive part. Measure $\{H, 1 - H\}$ and prepare F_0 on outcome H, F_1 otherwise. The resulting source weight is $\mathrm{Tr}(\rho_i H)$ because F_0, F_1 have A-eigenvalues $1, 0$. Their difference is D. Proposition 5.1 preserves these configurations and Corollary 5.2 preserves their source weights. □

The discrimination identity is a standard trace-distance fact; its role here is to determine exactly how much record information can become a difference in protected configuration and actual source. A nonzero source contrast is possible precisely when the two record states differ, provided this binary measurement and the two preparations are available. A specified executor can fail to use that difference: if $\Lambda^*(A)$ is orthogonal to $\rho_0 - \rho_1$ in trace pairing, its outgoing source is the same. Distinguishable histories need not yield distinguishable sources. Conversely, changing a conditional pointer algebra need not change r; a maximally mixed M gives $r = 1/2$ for both the Z and X selections.

This construction does not preserve an arbitrary coherent input record under measurement. For a classical history stored in orthogonal states the preparation can be controlled without deleting it. An independent old content–reference state Ω_{CE} is preserved jointly, even conditioned on the new outcome, when all preparation and loop operations act on R, P, M and leave C, E untouched. If R is already correlated with CE, the old marginal is preserved by the nonselective channel but conditioning can change its state. This is the distinction required when selection is claimed to retain an earlier body or history.

Ideal execution is not required for a finite history effect. Suppose each complete read/update instrument still outputs the four listed configurations, its actual sampled transition in cycle j is $\widetilde{T}_j$, and its actual source weight is $\widetilde{r}_n$, with

$$\max_x \mathrm{TV}(\widetilde{T}_j(x,\cdot), T(x,\cdot)) \le a_j, \qquad \|\widetilde{r}_n - r\|_\infty \le c_n.$$

These are errors of the stated four-state instrument and observable, not an identification of every physical error with a flip.

Corollary 6.2 (A robust finite source window)**.** *From either transient initial configuration T_i, the actual source weight obeys*

$$|\widetilde{\mu}_n \widetilde{r}_n - R_i^\circ| \le n\varepsilon + (q_+ q_-)^{\lfloor n/2 \rfloor} + \sum_{j=1}^{n} a_j + c_n.$$

Consequently the contrast between the two histories is at least

$$\frac{|f| q_+ q_-}{1 - q_+ q_-} - 2n\varepsilon - 2(q_+ q_-)^{\lfloor n/2 \rfloor} - 2\sum_{j=1}^{n} a_j - 2c_n,$$

whenever the right-hand side is positive.

Proof. At each step, couple the true row and its ideal counterpart with mismatch probability at most a_j. After n steps their distributions differ in total variation by at most $\sum_j a_j$. Expectations of $r \in [0,1]$ differ by at most that amount, and changing the source observable costs at most c_n. Apply Eq. (3). Finally, $|R_0^\circ - R_1^\circ| = |f|q_+q_-/(1-q_+q_-)$, and use the triangle inequality for the two source weights. □

For sufficiently small escape and execution errors, this bound certifies a nonempty interval of different physical consequences. Lower escape rates extend that interval only if execution errors and fresh-port correlations remain controlled. Holding a noisy selection indefinitely would need renewed resources or a stronger protection mechanism. The present bound supplies no free infinite stream.

7 The price of selecting sharp readouts

The fixed executor may be followed by a fixed measurement, with a fixed ancilla. A program selects effects E_z^p on a labelled target basis $\{|b_z^p\rangle\}$. We use the worst-case basis-state error

$$\delta_p = \max_z \left(1 - \langle b_z^p | E_z^p | b_z^p \rangle\right).$$

The program dimension counts the selectable register; a fixed ancillary state does not enlarge it. This criterion concerns the target outcome probabilities, and does not compare complete instruments in diamond norm.

Theorem 7.1 (Cost of sharp selection)**.** *1. If two pure program states select two distinct labelled sharp measurements exactly, they are orthogonal. Exact mixed programs for distinct sharp measurements have orthogonal supports. A register of dimension d_P selects at most d_P distinct labelled sharp measurements exactly.*

2. Let S be a qubit, and suppose that for every Bloch angle θ on a great circle some pure program state selects the sharp measurement along θ with error at most δ, where $0 < \delta < 1$. Then

$$d_P \;\geq\; L(\delta) := \max_{M \geq 1} \frac{M}{1 + \frac{1}{3}\left(2\sqrt{\delta} + \delta\right)^2 (M^2 - 1)}, \tag{4}$$

and $L(\delta)\sqrt{\delta} \to \sqrt{3}/4 \approx 0.433$ as $\delta \to 0$. Conversely, $N = \lceil \pi/(2 \arcsin\sqrt{\delta}) \rceil$ orthogonal program states with a controlled rotation achieve error δ, and $N\sqrt{\delta} \to \pi/2$.

3. Part (2) holds unchanged for mixed program states, with $0 < \delta < 1$. For each target a pure program in the same register has exactly the same two basis-state errors. Consequently $d_P \geq L(\delta)$ and $\liminf_{\delta \to 0} d_P \sqrt{\delta} \geq \sqrt{3}/4$; the same grid gives the upper bound. Mixing cannot reduce the required program dimension under this criterion.

The proof and representative numerical bounds are in Appendix A. The constants $\sqrt{3}/4$ and $\pi/2$ are lower and constructive upper bounds; the result establishes the exponent, not an optimal leading constant. Exact orthogonality is the measurement counterpart of the no-programming theorem for unitaries [15]. The equality of the pure and

mixed costs here follows from the convex joint numerical range of two Hermitian operators [18]. Three independent errors need not admit the same pure replacement. The asymptotic exponent applies to this great-circle family and this criterion; universal processors have other resource bounds [17].

A two-axis reader needs only two orthogonal choices [11]. Tracking a continuum of axes to increasing accuracy needs a growing register. Storing and refreshing the chosen programs has a further purity and free-energy account [4, 5]. Coherently returning a program does not, by itself, assign it a reset heat. Conversely, a large register does not supply the entropy outlet needed to renew noisy records.

8 What the executor leaves specified

Rules, programs, their revision and self-grounding selections can be internal. Within Definition 3.1, a full copy of an executor acting on $P \otimes S$ acts on dimension $d_P d_S$, whereas each selected operation acts on S or part of it, of dimension at most $d_S < d_P d_S$ when $d_P \geq 2$. It cannot be that full executor. A meta-register P' can select among executors on $P \otimes S$ under a fixed meta-executor. At the top of each stipulated finite hierarchy the model still supplies an executor as physical dynamics. This is a feature of the architecture, not a general prohibition on a system containing its own control law.

The top dispatch map is specified dynamics in this architecture. A meta-register can make the current executor selectable, at a resource cost set by the family to be selected. For the one-parameter great-circle qubit family, Theorem 7.1 gives order $\delta^{-1/2}$ register dimension; it assigns no common exponent to every continuous family. Stored classical choices also use the record resources described above. The dimension argument excludes a full executor acting on the larger space from being an operation on the smaller target; it permits storing a description of that executor. In infinite dimension the counting argument fails, since a separable Hilbert space is isomorphic to its tensor product with $\mathbb{C}^2$. Kleene and Lawvere fixed points [21, 22], and the order-theoretic results of Scott and Tarski [23, 24], also permit self-reference. Existence of a self-representing physical executor in that setting remains open.

Every stipulated finite hierarchy specifies its top dispatch map. Self-application is different: the loop of Section 5 applies a rule to its own protected records and reaches fixed points without adding levels. A claim that every operation is selected faces a separate resource test. By Theorem 7.1, an executor selecting every sharp readout exactly would require infinitely many orthogonal program states.

The distinction between structure and knowledge has a concrete meaning here. The dispatch map is physical dynamics; program contents are records. When their coupling spectra are disjoint they can be public environmental records, while overlapping eigenvalue tuples can hide some coherence. Proposition 4.2 adds a further distinction: a program need not become a central label to change the field source through the operation it selects.

9 From selectable rules to formed permissions

The loop forms a choice among supplied couplings. A self-frame needs more: history must establish a relation between a task and the body's own reading or action channels. T-DOME II separates an aligned axis from a signed relation that can actually predict [26].

If opposite couplings produce the same retained axis, the axis can be perfectly aligned while its posterior prediction is zero. Outcome-bearing records supply the missing sign. A useful program must encode whichever distinctions its future task consumes.

Self-readout then becomes a problem of attribution. A changed residual can come from the world, from a damaged body, or from a miscalibrated interface. T-DOME III uses identifiable references and paid maintenance to decide among these causes [9]. Its action is based on retained value and body condition; confidence alone need not determine whether repair is worthwhile. Imprint W supplies a complementary mechanism in which the system uses its own reading history to choose a later inquiry [11]. These are mechanisms for revising programs, with their own record laws and budgets.

Brain Anchor locates formed relations in a body: diagnostic outcomes can alter routes and install a different protected encoding [29]. Such relations can be distributed among components. Calling their aggregate a program register does not imply that a single place stores a full description, or that every relation becomes a central environmental label. Proposition 4.1 already gives a finite counterexample to the latter inference. A mathematical encoding is an interface for analysis; a physical implementation must say where its records, actuators and update couplings reside.

Q-RAIF C gives a further, independent source implementation [27]. Its actual quantum history enters a complete positive Hamiltonian and changes material motion and interacting field source. The fresh-port projector mixture used here is simpler and has different inputs. The two constructions support the same causal question, but their clocks, controllers and source objects cannot be exchanged merely because both have an internal program. Returning a field response to the same body requires the actual sensor and finite-resource operations, as studied in Foundation III [28]. A reconstructed geometry, a field stress and the total apparatus stress remain distinct.

The more ambitious generation problem concerns how these operational relations become available in the first place. A fixed meta-register only relocates a supplied dispatch map. A more informative mechanism is adaptation of the body: records change local couplings, routes or encodings, and the new relations alter which later records and acts are reliable. This can generate a permission from primitive interactions without generating the primitive interactions themselves.

Conjecture 9.1 (Formation of revisable permissions)**.** *Consider a body with locally adjustable couplings and a finite family of reversible encodings of the same logical content. Let actual diagnostic instruments act on relational degrees of freedom outside that logical algebra, and distinguish configurations with different task values. Suppose exploratory operations reach each necessary diagnostic relation, the best configuration has a nonzero value gap on each stationary interval, and paid update and maintenance resources suffice over that interval. There exist local record-driven update dynamics for which task-relevant relations become usable permissions from initially uncalibrated couplings, while previously encoded content remains within a stated error budget. When the task or body changes slowly enough, the same retained relations support detection and revision, rather than only repeating a fixed choice.*

The proposed mechanism is concrete: exploration creates diagnostic records; local retained relations select a route or encoding; an actuator installs it; subsequent errors and maintenance outcomes modify those relations. History-selected protection, signed self-frames and local body learning support different parts of this mechanism. Its quantitative test compares formed relations with supplied ones under the same preparation, diagnostic,

installation and content-loss budgets. Formation must lose its advantage when diagnostics alias task-distinct configurations, exploration cannot reach the relevant coupling, or noise destroys old content before learning and migration finish. A distributed learner that changes only labels while all implemented instruments remain fixed also fails this test.

The conjecture asks for a common finite construction with these inputs and outcomes. It is stronger than the supplied two-choice loop, and is not used in its proof. An infinite stationary guarantee would additionally need a sustainable resource supply and control of rare failures. A finite family chosen by the theorist can demonstrate a formation mechanism; it does not explain the origin of every physical interaction or privilege one universal observable algebra.

10 Conclusion

An internal selector can be physical, self-applied and revisable. In the finite construction, records choose their protection and alter the next field source under one executor. Shared coupling spectra allow coherent choices without a central program label. The transition law separates residence in one configuration, a finite history-dependent source window and eventual loss of contrast. Trace distance fixes how much an earlier record can contribute to source selection, and accumulated execution errors bound the useful window.

These are formation and execution results inside declared dynamics. The great-circle sharp-readout price limits a particular programmable family, not every continuum or physical self-reference. Operational equivalence limits which underlying dynamics the declared experiments can certify; enlarging their permissions can remove that limit. The formation conjecture asks how learned body relations can supply such new permissions while old content and resources survive. It is a stronger target than changing a program label.

The limits therefore attach to identifiable physical relations, not to an untyped claim that a system can never contain its own rule. A common account of formed sectors, event order, control and propagation, and source and geometry would explain more than the present selector. Its missing relations remain assigned to those mechanisms. The source realization here uses specified fresh bosonic ports; it is not Paper D's sustained thermal apparatus or a proof of its joint sharp-exhaust and return conjecture. Neither operational closure nor its extension settles the separate question of first-person experience.

A Proofs and constructions for sharp selection

Here we give the full argument for the readout cost. A readout here is a measurement on S implemented by a fixed executor: a unitary on $P \otimes S \otimes A$, with an ancilla A in a fixed state, followed by a fixed measurement. Purifying the fixed ancilla and dilating the measurement gives projections F_z pulled back to the input space. The embedding $|p\rangle \mapsto |p\rangle|a\rangle$ keeps program overlaps and has span dimension d_P; the ancillary dimension is not a selectable program resource. Write G_z for the induced POVM effects on the original $P \otimes S$, after compression to the fixed ancilla state. A pure program $|p\rangle$ selects effects $E_z^p = (\langle p| \otimes \mathbf{1})G_z(|p\rangle \otimes \mathbf{1})$. Its target is a labelled orthonormal basis $\{|b_z^p\rangle\}$ of S, with selection error $\delta_p = \max_z\big(1 - \langle b_z^p|E_z^p|b_z^p\rangle\big)$. For a mixed program σ, use $E_z^\sigma = \mathrm{Tr}_P[(\sigma \otimes \mathbf{1})G_z]$ and the same error on the target basis states. This is a basis-state error criterion, not a diamond-norm distance between instruments.

Lemma A.1 (Near-orthogonality)**.** *For programs p, q and outcomes $z \neq z'$,*

$$|\langle q|p\rangle|\,|\langle b^q_{z'}|b^p_z\rangle| \leq \sqrt{\delta_p} + \sqrt{\delta_q} + \sqrt{\delta_p\delta_q}.$$

Proof. Put $v = |p\rangle|b^p_z\rangle|a\rangle$ and $w = |q\rangle|b^q_{z'}\rangle|a\rangle$. Then $\langle v|F_z|v\rangle = \langle b^p_z|E^p_z|b^p_z\rangle \geq 1 - \delta_p$, so $\|(\mathbf{1} - F_z)v\| \leq \sqrt{\delta_p}$, and likewise $\|(\mathbf{1} - F_{z'})w\| \leq \sqrt{\delta_q}$. Write $v = F_z v + (\mathbf{1} - F_z)v$ and $w = F_{z'}w + (\mathbf{1} - F_{z'})w$. Since $F_{z'}F_z = 0$, the inner product $\langle w|v\rangle = \langle q|p\rangle\langle b^q_{z'}|b^p_z\rangle$ consists of three cross terms bounded by $\sqrt{\delta_p}$, $\sqrt{\delta_q}$ and $\sqrt{\delta_p\delta_q}$. □

The two-expectation fact used below is the two-Hermitian-operator case of joint numerical-range convexity, equivalent to the Toeplitz–Hausdorff theorem [18]. A rank-reduction proof makes clear why the replacement stays in the original register.

Lemma A.2 (Matching two error rates)**.** *For any two Hermitian operators B_0, B_1 on a finite program register P and any density operator σ on P, there is a unit vector $|p\rangle \in P$ such that*

$$\langle p|B_z|p\rangle = \mathrm{Tr}(\sigma B_z), \qquad z = 0, 1.$$

Proof. If σ has rank $r \geq 2$, the Hermitian operators on its support form a real vector space of dimension $r^2 \geq 4$. The three real linear constraints

$$\mathrm{Tr}\, H = \mathrm{Tr}(HB_0) = \mathrm{Tr}(HB_1) = 0$$

therefore have a nonzero Hermitian solution H supported there. Its zero trace implies that it has both positive and negative eigenvalues. Since σ is positive definite on its support, the interval of t for which $\sigma + tH \geq 0$ contains a neighbourhood of zero and has finite endpoints. At either endpoint the rank strictly decreases, while the trace and both expectations remain unchanged. Repeat on the smaller support until the density operator has rank one. □

Theorem 7.1 is proved below with the same fixed ancilla and basis-state error criterion.

Proof. (1) With $\delta_p = \delta_q = 0$ the lemma gives $\langle q|p\rangle\langle b^q_{z'}|b^p_z\rangle = 0$ for all $z \neq z'$. If $\langle q|p\rangle \neq 0$, each $|b^p_z\rangle$ is orthogonal to every $|b^q_{z'}\rangle$ with $z' \neq z$, hence proportional to $|b^q_z\rangle$, and the two labelled measurements coincide. Each mixed program's errors are averages of nonnegative pure-component errors. Zero average forces every vector in its spectral decomposition to select the same measurement exactly. The pure result makes the supports for distinct measurements orthogonal. Their nonzero dimensions sum to at most d_P.

(2) Choose M targets at angles $\theta_j = 2\pi j/M$ with program states $|p_j\rangle$. For $z = 0$, $z' = 1$ and Bloch angle φ between the targets, $|\langle b^q_1|b^p_0\rangle| = |\sin(\varphi/2)|$, so the lemma gives $|\langle p_j|p_l\rangle| \leq (2\sqrt{\delta} + \delta)/\sin(\pi|j-l|/M)$. The Gram matrix Γ of the $|p_j\rangle$ has $\mathrm{Tr}\,\Gamma = M$ and, by the identity $\sum_{k=1}^{M-1} \csc^2(\pi k/M) = (M^2-1)/3$,

$$\mathrm{Tr}\,\Gamma^2 \leq M + M\,\tfrac{1}{3}(2\sqrt{\delta} + \delta)^2(M^2 - 1).$$

Then $d_P \geq \mathrm{rank}\,\Gamma \geq (\mathrm{Tr}\,\Gamma)^2/\,\mathrm{Tr}\,\Gamma^2$, which is (4). With $c = \frac{1}{3}(2\sqrt{\delta} + \delta)^2$ the maximum over M lies near $M = c^{-1/2}$ and equals $\frac{1}{2}c^{-1/2}(1 + O(\sqrt{c}))$, which gives the limit. For the converse, the executor $\sum_j |j\rangle\langle j| \otimes R(\theta_j)^\dagger$ followed by a measurement in the computational basis selects the sharp measurement along $\theta_j = 2\pi j/N$ exactly; using the nearest θ_j for a target θ misaligns the axes by at most π/N, which costs error $\sin^2(\pi/(2N)) \leq \delta$.

(3) Fix a target basis $\{|b_0\rangle, |b_1\rangle\}$ and define positive operators on the original register P by

$$B_z = \mathbf{1}_P - (\mathbf{1}_P \otimes \langle b_z|)G_z(\mathbf{1}_P \otimes |b_z\rangle), \qquad z = 0, 1.$$

A mixed program σ has target errors $\mathrm{Tr}(\sigma B_z)$. Lemma A.2 supplies a pure program in the original P matching both errors exactly. Choose it for each target angle. The fixed executor then has a pure-program family with the same error bound δ and dimension d_P, so part (2) applies. Conversely, pure programs are included among mixed programs. This equivalence concerns the two target errors, not preparation costs or equality of the complete instruments; arbitrary additional error constraints need not admit a rank-one replacement. □

For k positive error operators $B_1, \ldots, B_k$, a mixed program whose errors are all at most δ always has a pure spectral component with every error at most $k\delta$: some component has the sum of its errors no larger than the mixed sum. Exact matching of all errors need not hold when $k \geq 3$: the three positive operators $(I + \sigma_x)/2$, $(I + \sigma_y)/2$, $(I + \sigma_z)/2$ all have expectation $1/2$ in the maximally mixed qubit, while no pure state matches all three. The two-error equality is therefore stronger than this general factor-k bound.

δ	$L(\delta)$	$N(\delta)$	$L\sqrt{\delta}$	$N\sqrt{\delta}$
10^{-2}	4.15	16	0.415	1.600
10^{-4}	43.1	158	0.431	1.580
10^{-6}	432.8	1571	0.433	1.571

Table 1: Common pure- and mixed-program lower bound (4) and the grid construction for selecting every sharp qubit measurement on a great circle to worst-case basis-state error δ.

D'Ariano and Perinotti give finite exact programming for covariant measurements and polynomial approximate programming for general measurements [16]. This does not remove the sharp-selection constraint.

For N grid programs and a uniformly distributed angle the mean error is $\frac{1}{2} - \frac{N}{2\pi} \sin \frac{\pi}{N} \approx \pi^2/(12N^2)$. Doubling the register dimension therefore divides the leading mean error by four.

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On the Closure of Structure: Origins, Persistence and Renewed Participation

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What would it mean for a world to begin or end as an organized structure? The question is not exhausted by the lifetime of an observer. It concerns the formation of distinctions, the relations through which they become accessible, and the possibility that these relations cease or form again. The Holographic Alaya–Field Framework approaches this cosmic question from within physical processes, without assigning an external observer the work of selecting their world.

A trace can remain after its reader is gone, and spacetime can persist after a region no longer supports active readers. Conversely, a continuing reader need not possess a complete description of itself. These distinctions prevent a local model of maintenance from replacing the question of structural origin and closure.

The world line now separates several steps. Interactions can select persistent sectors whose coherence-decay rates define a coupling-dependent Euclidean relation. A response distance requires a separately identified reversible transfer process; neither that process nor its spatial interpretation follows from dephasing alone [1, 2]. Records can identify an otherwise unknown order of interactions in a specified reversible protocol, but they do not thereby create its underlying evolution parameter [6]. Control can form useful permissions and alter physical sources; common source and modular constraints can then restrict the geometric response under their stated field assumptions [3, 4, 7, 5]. A cosmological account must connect these steps within one dynamics, not extrapolate any one finite construction into a history of the universe.

1 Structural closure and cosmic origin

Three readings of closure should be distinguished. Operational closure means that a declared family of states and operations supports its next step. Explanatory closure means that the structures doing this work are accounted for rather than repeatedly supplied by another apparatus. Cosmic closure concerns the global origin, persistence and possible return of those structures. A finite operating loop establishes the first under its inputs; neither of the other two follows by renaming the loop.

One possible reading of a structural beginning is a regime in which durable subsystem distinctions and useful records have not yet formed. A structural ending would be a regime in which no admissible process can continue the distinctions and revisions at issue. They might be equivalent for a specified restricted set of questions without being

the same physical state, the same boundary condition or the same event. This restricted equivalence keeps the original question of origin and terminus meaningful; it does not assert their identity.

In HAFF G, agreement on an invariant accessible algebra and on every permitted instrument implies agreement of all finite adaptive record laws [7]. That result supplies a precise meaning of equivalence for a fixed experimental class. It does not establish a universal final boundary: enlarge the permissions and the two conditions may become distinguishable. A cosmic identification would therefore need a physically justified class of all relevant interventions, rather than a convenient reader that sees neither difference. Whether such a relational equivalence has an additional ontological meaning remains an interpretive question, not a consequence of the record theorem.

2 A trace is not a continuing reader

Consider a fair classical label X copied perfectly into R two-level memories. Its joint state is

$$\rho_{XC} = \frac{1}{2}\sum_{x=0}^{1} |x\rangle\langle x| \otimes (|x\rangle\langle x|)^{\otimes R}.$$

Every memory alone is maximally mixed; the bank has no marginal purity. Nevertheless each copy determines X. Dephasing in the Z basis leaves this joint record unchanged. The product state with the same maximally mixed marginals carries no such record. The difference is in the correlations, not in the local inventories.

An archive of this kind may outlast the machinery that formed it. A later body with appropriate access can still read it. Without that machinery, however, the archive does not acquire a new observation, test an interpretation or correct its own readout. Passive protection and active participation are different physical achievements. Foundation's distinction between held content and return into another operation makes this difference constructive [9]. The continuing operation must actually be available; a comparison state used to bound an error is not a free preparation of another program.

This is also a philosophical distinction. What persists as a trace need not persist as an agent. The past can remain physically relevant through future interventions even after its original reader is gone. Conversely, continuity of an agent need not mean preserving every past datum or every microscopic component. In the framework's agency account it means maintaining an organization that can carry usable relations into later choices, repair its channels and revise what those relations warrant [3, 8]. This is a proposed criterion of causal participation, not a claim that causal organization alone explains experience.

3 Acquisition and renewal have different limits

HAFF F bounds record acquisition at a declared initial resource boundary [6]. Its assumptions include initially independent, unused memories and a specified thermal work account. The initial purity

$$B = \sum_k [\ln d_k - S(\sigma_k)]$$

is then part of what can pay for new correlations, together with supplied resources and the system's own contribution. The example above explains why one cannot impose this

product boundary anew at each time. Old correlations are resources even when $B = 0$. Running out of fresh purity does not erase them.

Renewal asks another question: what must be done when an existing useful relation degrades? In F's conditional-independent bank, $C_i = X \oplus N_i$, with error probability $\varepsilon < 1/2$, independent flips of probability $0 < p \le 1/2$ give

$$q = p + (1 - 2p)\varepsilon, \qquad d_{p,\varepsilon} = h(q) - h(\varepsilon) > 0,$$

where $h(u) = -u \ln u - (1 - u) \ln(1 - u)$. Returning the complete retained label, bank and controller to their joint target removes $Rd_{p,\varepsilon}$ nats. An independent Gibbs bath at temperature T must receive

$$Q \ge k_B T R d_{p,\varepsilon}.$$

Here the flip record is unavailable, the bath is initially independent of the noisy retained state, and the controller cannot absorb the entropy without being restored. For $p = .12$ and $\varepsilon = .05$, eight copies of one fair label require 1.9025859874 nats of correction. The number of distinct labels is one; the independent degradation of eight carriers still has a price.

A sole thermal chiral exhaust with the additional current assumptions in F and D has a quadratic finite-rate price. That result does not apply to every bath or every control pulse. Exact protection gives $d = 0$ in the bank model and defeats a blanket assertion that records require positive upkeep. Accessible noise records, correlated errors and a controller allowed to consume its own purity alter the boundary as well. The distinction matters before any cosmological extrapolation.

The joint-memory argument extends beyond independent copies. Proposition 1 in the appendix shows that an unavailable partial reset of strength λ destroying a retained relation of at least ι nats requires correction heat at least $k_B T \lambda \iota$, under its independent Gibbs-bath and complete-target conditions. If correction is also cyclic in total retained energy, finite supplied work bounds the number of renewals. This is a conditional resource theorem, not a claim that all bodies suffer that noise or that all cosmic evolution has a positive cost per update.

4 An unfinished self can continue

A demand for complete self-description makes participation seem impossible for the wrong reason. A self-model need only support the inquiries and actions on which continuation depends. T-DOME II gives a concrete example: a finite history forms a two-bit reading frame that beats every history-independent direction at one test read, even when the latter receives its exact regression coefficient for free. Retained capacity and total cost are not matched. History must retain the signed relation between the task and the body's channels; an accurately learned but unsigned axis can leave prediction useless [8]. The useful self-frame therefore consists in how a system can reach its task, not in an inventory of everything it is.

Some distinctions are inaccessible to a body's legitimate references. Foundation II separates that failure of identifiability from disagreement, and separates agreement from correctness [9]. A hidden symmetry can make one parameter unknowable while leaving a valuable intervention completely identifiable. In other cases inquiry resolves the ambiguity, but its preparation and maintenance consume resources. A continuing body must decide which error matters and which probe can repair it before the inquiry destroys the

content it was meant to help. Brain Anchor's learning-versus-survival tradeoff supplies one physical version of this problem [12].

Finite self-knowledge is consequently not equivalent to paralysis. A body may track residual errors, revise its references and retain a margin of viable action without ever obtaining a complete account of itself. The stronger philosophical claim is that an imperfect, historically formed organization can be a continuing subject precisely through this revisability. Its identity lies partly in the maintained relation between history and possible action, not in an immutable perfect image.

A rival account could identify continuation with persistence of the stored pattern. It then misses a relevant difference between two copies of the same archive: one has working probes and can revise its routing after an error, while the other's channels have permanently failed. Their stored descriptions agree; their possible next histories do not. The active criterion explains that difference without demanding exact microscopic identity. Two functioning copies may both inherit a causal history, so this criterion does not by itself select a unique first-person successor. That further identity question, like experience, cannot be decided by the survival of a data pattern alone.

That claim still needs an account of first-person experience. Consciousness III's explanatory gap cannot be crossed by renaming calibration as awareness [10]. A candidate bridge must explain why these particular internal relations matter to experience, distinguish alternatives that make the same reports, and expose a relevant intervention. Neither Proposition 1 nor a finite self-frame proves that a body feels anything. The thermodynamic argument addresses the support for continuation; the experience argument addresses what, if anything, that continuation is like.

5 A cosmological participation-window conjecture

The universe's duration and a body's active history need not have the same limit. Dyson's cooling and scaling scenario explicitly allows indefinitely continued life and communication from finite energy under its assumed laws [13]. Finite resources alone therefore do not establish a finite active window. In asymptotically de Sitter models a detector can encounter cosmological-horizon thermal radiation [14]. Krauss and Starkman examine how horizon restrictions, recoverable energy and memory requirements change the prospect of indefinite activity [15]. These are competing conditional mechanisms, not observational proof of the far future.

The framework suggests a more specific conjecture.

Conjecture 1 (Resource-limited participation)**.** *In a late-time causal domain with finite accessible usable work, a positive environmental temperature floor and unavoidable loss of task-relevant relations, any fixed-capacity body that continues at a fixed nonzero task quality can perform only finitely many successful inquiry–revision–action updates, even while passive traces and spacetime structure persist.*

The proposed mechanism is loss of correlations on which usable permissions depend. A retained history supports diagnosis; diagnosis changes routing or protection; the revised body supports another action and its physical source. Mixing damages the relations required to repeat that loop. Restoration exports entropy or consumes a fresh resource. A horizon may restrict replenishment and a thermal floor may prevent the cost per restoration from tending to zero. Exhaustion then ends this participation without erasing every trace.

The bank and partial-reset propositions establish this mechanism in two specified memory models. G supplies finite history-dependent source differences, while Imprint and Brain connect retained relations to subsequent inquiry and bodily change [7, 11, 12]. The conjecture reaches further: an interacting adaptive body can compress memories, migrate codes and replace its task. To prove it for a proposed body class one must obtain a uniform positive resource cost per successful update after optimizing those operations. Neither a count of bit erasures nor a total marginal purity inventory supplies that result automatically. The temperature floor, accessible work bound and irreducible relation loss must also be established for the actual cosmological domain. They are not consequences of selecting an algebra.

The conjecture is weakened or falsified by a body at the stipulated quality with an indefinitely usable protected relation, by vanishing renewal cost after admissible compression or migration, by a cooling law permitting a summable sequence of positive costs, or by unbounded replenishment. An accessible noise record can break the proposed loss mechanism. Discovery of one such regime would identify which part of the mechanism fails, rather than merely extending a numerical lifetime. A static surviving archive is not a counterexample to this active claim, but a continuing reader with the stated resources would be.

The sustained selection–response conjecture in D has different resource quantifiers [4]: it asks for families of longer runs whose installed energy, size and operating domain may grow, while one body is reused within each run. The participation conjecture here fixes capacity and limits the total accessible work. Neither quantifier refutes the other. Foundation III's finite semiclassical mean-action source–response construction [5] concerns a specified operating window; it does not establish indefinitely renewed thermal operation.

6 Origin, return and what closure would mean

An unused record port and an exhausted one can be locally indistinguishable and have different futures. The former may be independent of the source and backed by a usable gradient. The latter can carry correlations with an old history or lack the resources needed for another operation. Allowed future interventions can distinguish them. Equal local states therefore do not identify origin with terminus.

The stronger origin proposal in G is physical formation of permissions: a body with locally adjustable relations encounters diagnostic tasks, retains their results and uses them to install a useful coupling [7]. Exploration, a distinguishable value difference and enough survival time must be supplied by the primitive dynamics. This moves the problem beyond an external program name, without proving that every supplied executor arises in this way. Enlarging an executor introduces another joint interaction to explain; it does not prove a universal final boundary or forbid a finite system from carrying a useful description of itself.

A cosmological return would require more still. One concrete mechanism candidate is that contraction or another geometric transition creates newly accessible nonequilibrium gradients; interactions then allocate part of that work to fresh record ports and bodies, while old correlations and exported entropy remain accounted for in other degrees of freedom. A repeated active phase would be a renewal of resources and formation, not necessarily recurrence of the global state. The physical transition must have its own dynamics, show that the gradients can be used, and explain why old entropy does not

obstruct the new phase. Local thermal swaps or recurrence of a finite register do not establish any of these cosmological steps. A bounce that restores density but no usable work would fail the mechanism; so would a transition requiring an uncharged deletion of the old history. No such return is forced by the participation-window conjecture.

Conjecture 2 (Formation and return of relational structure)**.** *There exist cosmological dynamics in which effective subsystem distinctions, stable record relations and a spatial response description form in one regime, lose their ability to support continuing participation in another, and become physically available again through a later resource-producing transition. The later regime need not repeat the global state or erase the earlier history.*

This is a joint formation conjecture, not a periodic-universe prediction. Its existential scope is a research target: the failure tests below exclude proposed witnesses, not all possible cosmological dynamics. A testable realization must first specify its dynamical family and admissible transitions, for example a particular bounce or phase-transition model with a closed conservation and entropy account. Its witnesses would have to derive the effective decomposition and response dictionary from the same evolving interactions, identify record order without supplying it as a decoded label, and account for usable gradients, clocks, discarded correlations and entropy across the transition. The geometric law and the transition itself require their own dynamical argument. Existing sector-selection, order-reading and sourced-response results establish components under distinct inputs; they do not yet give that cosmological witness.

The discriminating question is whether renewed activity is accompanied by reformation of the relations that make a world accessible, rather than only by newly charged batteries in a permanently stipulated architecture. A proposed model fails this stronger test if its decomposition, spatial dictionary or record standards are simply reinstalled by hand at each epoch. It also fails if the required transition violates its own conservation or entropy account. By contrast, a model demonstrating those structures and their budgets would turn a metaphor of cosmic return into a physical mechanism. The present record and renewal bounds supply tests for such a model, not a substitute for constructing it.

Closure thus has several meanings that must not be confused. A record can become unreadable to one body; a body can lose the capacity to revise itself; a cosmic domain can cease to support new active readers. None entails the nonexistence of its remaining physical structures. Nor must an imperfect self finish knowing itself before it can live. The framework's most consequential prospect is that history changes what a body can do next, and that this relation can be maintained and revised for a time without being complete.

A history can remain after the means of continuing it have gone. A continuing self need not wait to become a finished account of itself.

A Finite-body renewal bound

The bank is not the only setting in which a retained relation has a renewal cost. The following result treats an entire finite memory as one object, allows correlated and quantum content, and makes the noise assumption explicit. Entropies and mutual informations are in nats.

Proposition 1 (Renewing a useful relation under partial reset)**.** *Let θ_{XMC} be a target state of a retained reference X, a memory M of dimension d, and a controller C. Suppose $I_\theta(M:X) \geq \iota > 0$. During each degradation interval the unavailable noise acts as*

$$\mathcal{D}_\lambda(\theta) = (1-\lambda)\theta + \lambda\, \theta_{XC} \otimes \tau_M, \qquad \tau_M = I_M/d, \quad 0 < \lambda \leq 1.$$

A correction restores the complete target θ_{XMC}, using a bath initially independent of the noisy retained state and Gibbs at temperature T. If the correction is a possibly driven unitary on the retained system and bath, with no other entropy sink and vanishing endpoint interaction energy, then

$$Q \geq k_B T \lambda \iota. \tag{1}$$

If the total retained-system energy, including internal couplings, is equal immediately before and after the correction, and driven work and the counted bath heat are the only energy flows, then $W = Q$. A degenerate memory with additive, unchanged apparatus energies is one sufficient realization. For N renewals with $T_j \geq T_* > 0$, $\lambda_j \geq \lambda_* > 0$, and $I_{\theta_j}(M_j : X_j) \geq \iota_* > 0$, a finite total supplied work $\sum_j W_j \leq E_{\mathrm{av}}$ implies

$$N \leq \frac{E_{\mathrm{av}}}{k_B T_* \lambda_* \iota_*}. \tag{2}$$

For the last statement each interval must satisfy the preceding cyclic correction and energy conditions, even when its target differs.

Proof. Write $Z = XMC$ and $\eta = \mathcal{D}_\lambda(\theta)$. Entropy concavity gives

$$\begin{aligned} S(\eta) - S(\theta) &\geq \lambda[S(\theta_{XC}) + \ln d - S(\theta)] \\ &= \lambda[\ln d - S_\theta(M|XC)] \\ &= \lambda[\ln d - S_\theta(M) + I_\theta(M:XC)] \\ &\geq \lambda I_\theta(M:X) \geq \lambda\iota. \end{aligned}$$

The last line uses $S(M) \leq \ln d$ and monotonicity of mutual information under discarding C. The proof is valid for quantum X and for a controller correlated with the target.

Initially Z and the Gibbs bath B are independent. Entropy conservation under the correction and subadditivity at its end give $\Delta S_B \geq S(\eta) - S(\theta)$. For $\tau_B = e^{-\beta H_B} / \operatorname{Tr} e^{-\beta H_B}$,

$$\beta Q = \Delta S_B + D(\rho'_B \| \tau_B), \qquad \beta = (k_B T)^{-1}.$$

This proves (1). With vanishing endpoint bath coupling, the first law reads $W = \Delta E_{XMC} + Q$. The total-energy and energy-flow assumptions give $\Delta E_{XMC} = 0$ and hence $W = Q$; summing the lower bounds proves (2). □

The reset mixture models a lost alternative, not an accessible branch flag the controller can undo for free. It acts on the whole designated memory, so no protected subsystem within that memory escapes the stated channel. The cost is paid for restoring the joint relation, not for restoring a local marginal. A fair classical copy has a maximally mixed memory marginal and still satisfies the bound. An initially independent maximally mixed memory has $\iota = 0$ and does not.

This extends the renewal argument beyond independent copies. It does not establish that every physical body experiences this noise, that every update needs a fixed amount

of information, or that every target can be restored under a body's allowed access. The bound is necessary where such a correction exists. A consumed controller, an accumulating waste register or an accessible auxiliary noise record must be included among the resources. If the noise environment is retained, its correlations cannot be discarded when defining the target. Calling only the memory cyclic would hide the expenditure. The entropy proof permits an externally scheduled unitary; it is not a proof of autonomous apparatus closure. A physical battery that takes part in the entropy account must be included or separately shown not to supply an additional entropy sink.

Equation (2) bounds successful renewals, not necessarily calendar time. A system can postpone a last update arbitrarily far. A finite upper bound on an uninterrupted active interval additionally requires a positive minimum update rate and a bound on gaps. Hibernation, cooling toward zero temperature, decreasing task quality and improving protection are substantive alternatives, not semantic exceptions.

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Lorentzian Read-out from Local Unitary Propagation: Clifford Relations, Causal Cones and Spectral Mass

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Abstract

A local unitary propagation law can supply the relative sign of a Lorentzian read-out without turning a positive control metric into an indefinite one. For a first-order Hermitian symbol with independent spatial coefficients, all bands share one quadratic shell exactly when those coefficients satisfy the corresponding Euclidean Clifford relations. The resulting form is $\omega^2 - Q(q)$; it describes the full cone when both frequency branches occur. The evolution time is supplied. The internal dimension bounds the number of anticommuting directions, while calibrated coefficient errors give a forward certificate for approximate shells.

A shared shell for one species does not imply one cone for all matter. Fixed-block one-particle mixing cannot merge distinct shells. In a mediator model, sufficiently accumulated interactions align speeds, even when different edges are active on disjoint scale intervals. At a supplied Lorentz-invariant conformal fixed point, nonconserved spin-2 perturbations are irrelevant; additional conserved currents are the marginal exceptions. Neither argument creates the assumed infrared regime.

On a lattice, support, group and rest-and-curvature speeds differ. An actual record protocol certifies the angle of one coined walk and thereby its ideal dictionary parameters, not its noisy long-time transport. The interfaces to physical control and spectral mass require, respectively, calibrated generator alignment and a positive-energy representation. These conditions make the structural connection testable while leaving the formation of the carrier and accessible momentum window as dynamical questions.

Keywords: Lorentzian signature, Clifford algebra, hyperbolic polynomials, quantum cellular automata, quantum walks, Weyl and Dirac equations, causal cone, spectral mass, Q-RAIF

1 Introduction

Programmes that reconstruct geometry from quantum data usually take the Lorentzian signature as given. Holographic entanglement formulas and thermodynamic derivations of the Einstein equation relate entropies to areas and energy fluxes in a spacetime whose causal

structure is already Lorentzian [2, 3]. The Holographic Alaya-Field Framework (HAFF) studies which distinctions a physical reduction can resolve and what geometric data can be reconstructed from them [42, 43]; a spacetime metric needs a further dictionary there as well. This paper asks where the indefinite direction of such a read-out can come from in a model whose dynamics is given explicitly.

Two facts fix the starting point. A positive definite form stays positive under every linear change of variables (Proposition 1), so no control metric, information metric or work form becomes Lorentzian by reparameterization. The companion paper on persistence makes the same distinction physically: its descent metric and field-derived work form are positive. Under an additional all-state work condition and calibrated generator alignment it relates work to the positive spatial shell, not directly to the Lorentzian form [45]. Second, imposing Clifford relations $\{e_\mu, e_\nu\} = 2q_{\mu\nu}\mathbf{1}$ for a chosen Lorentzian q puts the signature into the premise; the universal property of $Cl(V, q)$ then supplies an encoding, rather than a signature-selection mechanism [4, 5].

The model class is local unitary propagation: a particle with an internal Hilbert space $\mathbb{C}^s$ hops on a lattice by a translation-invariant, finite-range unitary step, as in quantum walks and the one-particle sector of quantum cellular automata [9, 10, 11, 12, 13]. Its low-energy generator is a first-order Hermitian matrix symbol. In this setting four objects can be defined separately: a positive norm on generators, which plays the role of a control metric; the causal cones of the dynamics; the algebra of the symbol's coefficients; and the spectral mass. For a first-order Hermitian symbol, unitarity alone makes the characteristic polynomial hyperbolic in the frequency (Theorem 2). For linearly independent spatial coefficients, all bands lie on one quadratic shell if and only if the coefficients satisfy the shell relations of Theorem 4; the characteristic form is then ω^2 minus the Hilbert–Schmidt norm of the spatial symbol and has signature $(1, d)$, and the Lorentzian Clifford generators are $\gamma^0 = \beta$, $\gamma^i = \beta\alpha_i$ (Theorem 4, Corollary 5). The internal dimension bounds the spatial one, $d + [\mu \neq 0] \leq 2\nu_2(s) + 1$. For a qubit carrier three is the largest dimension with an isotropic massless cone, and the only one in which every internal rotation axis is coupled to a direction of motion (Theorem 8, Proposition 10).

On the lattice, group velocities lie in the convex hull of the neighbourhood, so the speed of any massless read-out cone is at most the inradius of that polytope per step (Proposition 12). Exact formulas for the split-step walk and computations for the Weyl automaton then show that the read-out map is not fixed by the dynamics: a band point, a momentum window and the observable that defines the speed all have to be chosen (Section 5).

The mathematical ingredients are standard: Dirac's square root of the Klein–Gordon operator [6], Gårding's hyperbolic polynomials [7], the representation theory of complex Clifford algebras [8], and the derivations of Weyl and Dirac equations from quantum walks and automata [9, 11, 13]. Hyperbolicity as the defining property of a physical dispersion relation is developed in Ref. [14], and an information-theoretic argument linking three spatial dimensions to the qubit is given in Ref. [15]. This paper separates the four objects in one model, identifies which inputs of a Lorentzian read-out follow from propagation, and calculates how the remaining read-out choices enter. The several-species analysis adds a quantitative fixed-block mixing obstruction and applies cut-balanced convergence to interacting speeds that need not couple simultaneously.

The argument proceeds from the symbol and its shell to dimension, lattice windows and

several-species dynamics. A residual bound connects the exact shell criterion to calibrated approximate symbols; a finite angle certificate then connects one such dictionary to actual records. The concluding sections distinguish these contributions from the physical selection of a carrier, an interacting fixed point and a complete source-and-return apparatus.

2 Local unitary propagation and four objects

2.1 The model

Let $\Lambda \subset \mathbb{R}^d$ be a lattice of physical positions and let the one-particle Hilbert space be $\ell^2(\Lambda) \otimes \mathbb{C}^s$. A translation-invariant step with finite neighbourhood $N \subset \Lambda$ acts as

$$(W\psi)(x) = \sum_{h \in N} A_h \, \psi(x - h), \qquad A_h \in M_s(\mathbb{C}), \tag{1}$$

and is unitary exactly when its symbol

$$W(k) = \sum_{h \in N} A_h \, e^{-ik \cdot h} \tag{2}$$

is unitary for every k in the Brillouin zone. One step takes time τ. The eigenvalues of $W(k)$ are $e^{-i\omega_b(k)\tau}$ with quasi-energies $\omega_b(k)$ defined modulo $2\pi/\tau$, and the group velocity of band b is $\nabla_k \omega_b$. With this convention the pure shift $A_h = \mathbf{1}$ for a single h moves every state by $+h$ per step.

Near a point k_* where a gap closes or nearly closes, the step is described by an effective generator. If a parameter θ controls the gap, we say that the walk has *first-order symbol*

$$H(q) = \sum_{i=1}^{d} q_i \alpha_i + \mu\beta, \qquad \alpha_i, \beta \in \mathrm{Herm}(s), \tag{3}$$

at k_* if $(i/\tau) \operatorname{Log} W_\theta(k_* + q) = H(q) + O(|q|^2 + |\mu|^2)$ with μ proportional to θ. This is the scaling limit studied for quantum walks and automata [11, 12, 13]: H generates the continuum dynamics $i\partial_t \psi = H(-i\nabla)\psi$ when lattice spacing and step time go to zero at fixed ratio. For example, the split-step walk of Section 5.2 has $W_\theta(k) = \mathrm{diag}(e^{-ika}, e^{ika}) R(\theta)$ with $R(\theta) = \cos\theta - i \sin\theta \, \sigma_y$, and at $k_* = 0$ its first-order symbol is $H(q) = (a/\tau) q \sigma_z + (\theta/\tau)\sigma_y$.

2.2 Four objects

A control metric. The generators of the internal space form the real vector space $\mathrm{Herm}(s)$, which carries the positive Hilbert–Schmidt norm

$$G_{\mathrm{HS}}(A) = \frac{1}{s} \operatorname{tr}(A^2). \tag{4}$$

Restricted to the span of the coefficients of H, it is the natural positive metric on the controls that the symbol couples to translations. Other positive forms of the same type appear in the series: the descent metric G of a gradient controller and the work form K derived from the stress of a field-coupled actuator [45], and monotone information metrics on states [29].

Causal cones. The step (1) has a strict causal region. After n steps a state supported in S is supported in $S + nN$, the n-fold Minkowski sum, so all influence lies in the cone over the polytope $C_N = \tau^{-1} \operatorname{conv} N$. A first-order symbol has a second cone, its characteristic cone, defined in Section 3. The two need not agree.

A representation. The coefficients α_i, β generate a real algebra inside $M_s(\mathbb{C})$. Whether they satisfy Clifford relations, and with which quadratic form, is a property of the dynamics, not a choice of encoding.

Spectral mass. For a selected logarithm and energy zero, the positive band's value at k_* is the rest energy $E_0 = \hbar|\omega(k_*)|$, and its curvature defines an inertial mass m_* through $\omega(k_* + q) \simeq \omega(k_*) + \hbar q^2/(2m_*)$. A measured phase difference alone does not fix an unknown common energy offset. A Lorentzian read-out identifies $E_0 = m_* c^2$, which defines a speed c; whether this c agrees with either causal cone is a question about the model.

Proposition 1 (Positive forms stay positive)**.** *Let G be a positive definite form on a real vector space V and $T : U \to V$ linear. Then T^*G, given by $(T^*G)(u, u') = G(Tu, Tu')$, is positive semidefinite, and positive definite if T is injective. In particular no positive definite form is congruent to a form with a negative direction.*

Proof. $(T^*G)(u, u) = G(Tu, Tu) \geq 0$, with equality only if $Tu = 0$. The last statement is Sylvester's law of inertia. □

The Lorentzian form of a read-out is therefore not a pullback or a reparameterization of G_{HS}, G or K. In the model it arises as a different kind of object, a form on frequency–momentum space built from the spectrum of the evolution.

3 Where the indefinite direction comes from

For a first-order symbol (3) write $P(\omega, q) = \det(\omega \mathbf{1} - H(q))$ for its characteristic polynomial and $P_0(\omega, q) = \det(\omega \mathbf{1} - \sum_i q_i \alpha_i)$ for the principal part. A homogeneous polynomial p on $\mathbb{R}^n$ is *hyperbolic with respect to e* if $p(e) \neq 0$ and $t \mapsto p(\xi + te)$ has only real roots for every ξ [7].

Theorem 2 (Hyperbolicity in the evolution frequency)**.** *For Hermitian α_i, β, every root in ω of $P(\omega, q)$ is real for every real q, and P_0 is hyperbolic with respect to the frequency direction $e = (1, 0, \ldots, 0)$. The propagation speed of the principal part in a unit direction u is the largest eigenvalue of $\sum_i u_i \alpha_i$.*

Proof. The roots of $P(\cdot, q)$ are the eigenvalues of the Hermitian matrix $H(q)$. For P_0, $P_0(e) = 1$ and the roots of $t \mapsto P_0(\xi_0 + t, \xi')$ are $-\xi_0$ plus the eigenvalues of $\sum_i \xi'_i \alpha_i$, which are real. The characteristic speeds of $i\partial_t \psi = \sum_i \alpha_i (-i\partial_i)\psi$ in direction u are the eigenvalues of $\sum_i u_i \alpha_i$. □

The evolution parameter is supplied by the model. Unitarity makes its conjugate frequency a hyperbolicity direction; it does not choose how many fundamental time parameters

a theory should have. The next lemma determines the signature when the characteristic form is quadratic.

Lemma 3 (Hyperbolic quadratic forms are Lorentzian)**.** *A real quadratic form p on $\mathbb{R}^n$ with $p(e) > 0$ is hyperbolic with respect to e if and only if it has exactly one positive eigenvalue. Then $p = \ell^2 - r$ with ℓ linear and r positive semidefinite, and if p is non-degenerate its signature is $(1, n-1)$.*

Proof. Decompose $\xi = se + \zeta$ with ζ in the p-orthogonal complement of e. Then $p(\xi + te) = p(e)(s+t)^2 + p(\zeta)$, which has real roots in t for every ξ if and only if $p(\zeta) \le 0$ on the complement. That is the statement that p has one positive direction and is negative semidefinite on a complementary hyperplane. □

Theorem 4 (Single shell, Clifford relations and the Lorentzian form)**.** *Let $\alpha_1, \dots, \alpha_d, \beta \in \mathrm{Herm}(s)$ with $\alpha_1, \dots, \alpha_d$ linearly independent, and fix $\mu \in \mathbb{R}$. The following are equivalent.*

(a) There is a quadratic form Q on $\mathbb{R}^d$ such that every eigenvalue ω of $H(q)$ satisfies $\omega^2 = Q(q) + \mu^2$ for all q.

(b) $H(q)^2 = (Q(q) + \mu^2)\mathbf{1}$ for all q.

(c) $\{\alpha_i, \alpha_j\} = 2Q_{ij}\mathbf{1}$, $\mu\{\alpha_i, \beta\} = 0$ and $\mu^2(\beta^2 - \mathbf{1}) = 0$.

If they hold, then:

1. *Q is positive definite and $Q(q) = G_{\mathrm{HS}}\big(\sum_i q_i \alpha_i\big)$;*
2. *the read-out form $\eta(\omega, q) = \omega^2 - Q(q)$ on frequency–momentum space has signature $(1, d)$, and the characteristic set of P_0 lies on its null cone; the sets coincide if $d \ge 2$, if $\mu \neq 0$, or if $d = 1$ and α_1 has both positive and negative eigenvalues;*
3. *if $d + [\mu \neq 0] \ge 2$, then s is even and $P(\omega, q) = (\omega^2 - Q(q) - \mu^2)^{s/2}$;*
4. *every group velocity $v = \nabla_q \omega$ satisfies $Q^{-1}(v, v) = Q(q)/(Q(q) + \mu^2) \le 1$, with equality only for $\mu = 0$.*

Proof. (a)⇒(b): $H(q)$ is Hermitian, hence diagonalizable, and all its eigenvalues square to $Q(q) + \mu^2$. (b)⇒(c): expanding,

$$H(q)^2 = \sum_{i,j} q_i q_j \tfrac{1}{2}\{\alpha_i, \alpha_j\} + \mu \sum_i q_i \{\alpha_i, \beta\} + \mu^2 \beta^2.$$

Comparing the parts of degree two, one and zero in q with $(Q(q) + \mu^2)\mathbf{1}$ gives the three relations. (c)⇒(a): the same expansion gives (b), and (b) gives (a).

For item 1, $Q(q)\mathbf{1} = (\sum_i q_i \alpha_i)^2 \ge 0$; if $Q(q) = 0$ the Hermitian matrix $\sum_i q_i \alpha_i$ squares to zero and so vanishes, and linear independence gives $q = 0$. Taking the normalized trace gives $Q(q) = G_{\mathrm{HS}}(\sum_i q_i \alpha_i)$. For item 2, η is ω^2 minus a positive definite form, and every root of $P_0(\cdot, q)$ belongs to $\{\pm\sqrt{Q(q)}\}$. When $d \ge 2$, the principal symbol at nonzero q anticommutes with an invertible symbol in a Q-orthogonal direction, pairing its two signs.

If $\mu \neq 0$, β provides the pairing. In the remaining one-dimensional case both signs occur exactly when α_1 has both signs. For item 3, choose coordinates with $Q = c^2\mathbf{1}$. The matrices α_i/c together with β (if $\mu \neq 0$) pairwise anticommute and square to $\mathbf{1}$. If at least two are present, each x anticommutes with an invertible g, giving $\operatorname{tr} x = \operatorname{tr}(g^{-1}xg) = -\operatorname{tr} x$. Thus $H(q)$ is traceless, and its eigenvalues $\pm\sqrt{Q+\mu^2}$ have equal multiplicity. For item 4, at a differentiable band point with $\omega \neq 0$, $\nabla_q \omega = Qq/\omega$ and $Q^{-1}(v,v) = Q(q)/\omega^2$. At a massless node the same statement applies to directional band limits. □

Corollary 5 (Lorentzian Clifford generators)**.** *Under Theorem 4 with $\mu \neq 0$, the matrices*

$$\gamma^0 = \beta, \qquad \gamma^i = \beta\alpha_i \tag{5}$$

satisfy $\{\gamma^0,\gamma^0\} = 2$, $\{\gamma^0,\gamma^i\} = 0$ and $\{\gamma^i,\gamma^j\} = -2Q_{ij}$, so they generate a representation of $Cl(1,d)$ on $\mathbb{C}^s$; γ^0 is Hermitian and the γ^i are anti-Hermitian. The eigenvalue equation $H(q)\psi = \omega\psi$ is equivalent to $(\omega\gamma^0 - \sum_i q_i\gamma^i - \mu)\psi = 0$, and $(\omega\gamma^0 - \sum_i q_i\gamma^i)^2 = \eta(\omega,q)\mathbf{1}$. For $\mu = 0$ and $d \geq 1$, the matrices

$$\Gamma^0 = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}, \qquad \Gamma^i = \begin{pmatrix} 0 & \alpha_i \\ -\alpha_i & 0 \end{pmatrix} \tag{6}$$

generate a representation of $Cl(1,d)$ on $\mathbb{C}^s \oplus \mathbb{C}^s$.

Proof. $(\beta\alpha_i)^2 = -\beta^2\alpha_i^2 = -Q_{ii}$ and $\{\beta,\beta\alpha_i\} = \alpha_i - \alpha_i = 0$ use $\{\alpha_i,\beta\} = 0$ and $\beta^2 = 1$; similarly $\{\beta\alpha_i,\beta\alpha_j\} = -\{\alpha_i,\alpha_j\}$. Multiplying $H(q) - \omega$ by β gives the second statement. The block matrices are checked by direct multiplication. □

What has been derived. In the model the indefinite direction of the read-out form is the frequency ω, the spectral variable of the one-parameter group of time steps. The positive part is the Hilbert–Schmidt norm of the spatial symbol, a control-type metric. The Lorentzian form is the difference $\omega^2 - G_{\mathrm{HS}}(\sum_i q_i\alpha_i)$, which by Proposition 1 could not be obtained from any positive form alone. Theorem 2 makes the supplied evolution frequency a hyperbolicity direction for every first-order Hermitian symbol; Lemma 3 says that a quadratic cone with one time direction has signature $(1,d)$; and Theorem 4 says that the cone is quadratic, with all internal states on it, exactly when the propagation coefficients satisfy Euclidean Clifford relations. The Lorentzian Clifford algebra of Corollary 5 is then constructed, not supplied: the time-like generator is the mass term β.

The signature is no longer an input. The Clifford relations are equivalent to an observable property, a single shell, which means no birefringence and no internal state that propagates differently from the others. That property, isotropy of Q (which can be absorbed into the choice of spatial coordinates for one species) and the existence of a band point with a linear symbol remain inputs.

Remark 6 (Two boundary cases). For $d = 1$ and $\mu = 0$, $\alpha_1^2 = Q_{11}\mathbf{1}$ allows unequal multiplicities of $\pm Q_{11}^{1/2}$, and for $s = 1$ the symbol $H(q) = cq$ has a single branch. Its characteristic set is one chiral branch, with one future null direction. This is the chiral field used as an actuator in Ref. [45]: it supplies one null direction, and a full $(1,1)$ cone needs both chiralities. In the other direction, a quadratic cone can occur without a single shell. The Maxwell

field in Riemann–Silberstein form has the symbol $c\,\mathbf{S}\cdot q$ with spin-one matrices; its characteristic polynomial is $\omega(\omega^2 - c^2|q|^2)$, and the flat factor is removed by the Gauss constraint. Theorem 4 concerns symbols for which every internal state lies on the cone.

3.1 A forward certificate for approximate shells

The exact operator identity has a finite-error consequence. It requires bounds on implemented coefficients or their calibration errors, not merely access to effects whose ideal algebra contains those coefficients.

Proposition 7 (Squared-band certificate)**.** *Choose a positive quadratic form Q and Hermitian coefficients α_i, β. Let*

$$R_{ij} = \tfrac{1}{2}\{\alpha_i, \alpha_j\} - Q_{ij} I, \qquad R_{i\beta} = \{\alpha_i, \beta\}, \qquad R_\beta = \beta^2 - I.$$

Suppose their operator norms are bounded by calibrated numbers $\epsilon_{ij}, \epsilon_{i\beta}, \epsilon_\beta$. If the implemented symbol is $\widetilde{H}(q) = H(q) + E(q)$ with Hermitian E and $\|E(q)\| \le e(q)$, put

$$\Delta(q) = \sum_{i,j} |q_i q_j| \epsilon_{ij} + |\mu| \sum_i |q_i| \epsilon_{i\beta} + \mu^2 \epsilon_\beta,$$
$$L(q) = \sum_i |q_i| \|\alpha_i\| + |\mu| \|\beta\|, \qquad B(q) = \Delta(q) + 2L(q)e(q) + e(q)^2.$$

Every eigenvalue $\widetilde{\omega}$ obeys

$$\left|\widetilde{\omega}^2 - Q(q) - \mu^2\right| \le B(q).$$

With $S = Q(q) + \mu^2 \ge 0$, this gives $\left||\widetilde{\omega}| - \sqrt{S}\right| \le \sqrt{B(q)}$. If $S > B(q)$, a sharper radius is $B(q)/(\sqrt{S} + \sqrt{S - B(q)})$. The bounds hold uniformly on any specified window where their input bounds hold.

Proof. Expanding $H^2 - SI$ gives $\sum q_i q_j R_{ij} + \mu \sum q_i R_{i\beta} + \mu^2 R_\beta$, whose norm is at most Δ. The additional square error is $HE + EH + E^2$, of norm at most $2Le + e^2$. Evaluate on an eigenvector of the Hermitian $\widetilde{H}$. Finally, $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x - y|}$ for nonnegative x, y; away from zero use the difference of squares. □

This is a forward dispersion certificate. It does not say that nearly anticommuting coefficients can be repaired to an exact Clifford representation, that the two frequency multiplicities are balanced, or that a finite effect menu implements operator-norm tomography. A calibration valid on a restricted state set is not a norm bound on the entire carrier. The distinction matters: a Hilbert–Schmidt trace identity can hold exactly while different internal states retain different propagation speeds.

3.2 Outputs to physical control and mass read-out

Two physical interfaces use the shell theorem, but require different additional inputs. For control, the spatial map $J(q) = \sum_i q_i \alpha_i$ is an isometry into internal generators with G_{HS} because its square is scalar. The trace identity alone is weaker: the local unitary walk

$W(k) = \mathrm{diag}(e^{-ik}, e^{-2ik})$ has $H(q) = q\,\mathrm{diag}(1,2)$ and $G_{\mathrm{HS}}(H(q)) = 5q^2/2$, yet two distinct speeds. The operator square, not a mean over internal states, supplies one shell.

Q-RAIF B [45] gives that distinction an operational test. For its centered independent field input, a coupling $A(a) = J(Ca)/E_*$ has switching work

$$W_\rho(a) = \kappa_g \,\mathrm{tr}[\rho A(a)^2].$$

If the physical calibration C is invertible, independence of this charge from every unknown internal state is equivalent to a scalar spatial shell. Its value is $\kappa_g Q(Ca)/E_*^2$. The field profile, calibrated map and energy scale are supplied; the encounter requires vanishing or commuting free internal evolution. This is a bridge between two calibrated uses of the same generators. It neither equates the cost with $\omega^2 - Q$ nor identifies the chiral control field with a general multi-dimensional propagation carrier. Q-RAIF C [46] uses Corollary 5's symbol to represent the Lorentzian quadratic space on the closed carrier; discarding a drive gives a different, reduced-channel representation problem.

For mass–energy read-out, the massive continuum shell must also carry the positive-energy Poincare representation specified in Q-RAIF ME [47]. That paper constructs it on the positive band of the Dirac symbol, with the boost Jacobian and spin transport on momentum-space wavefunctions. The symmetric pair of eigenvalues in Theorem 4 alone is not such a representation or a preparation of its positive sector. A finite lattice window is not invariant under arbitrary boosts. The actual coin experiment in Section 5.5 provides a narrower, priced parameter certificate, which remains useful without upgrading it to exact relativistic covariance.

4 Spatial dimension and the internal space

Theorem 8 (Dimension bound)**.** *Suppose Theorem 4 holds on $\mathbb{C}^s$, and let $n = d + [\mu \neq 0]$. Write $s = 2^\nu m$ with m odd, $\nu = \nu_2(s)$. Then $n \leq 2\nu + 1$. Conversely, for every s and every $n \leq 2\nu_2(s) + 1$ there are coefficients satisfying Theorem 4.*

Proof. Choose spatial coordinates in which $Q = c^2\mathbf{1}$; the coefficients transform linearly, so the relations are preserved. Then $e_i = \alpha_i/c$ and, if $\mu \neq 0$, $e_{d+1} = \beta$ are n pairwise anticommuting Hermitian involutions, which define a representation of the complex Clifford algebra $\mathbb{C}l_n$ on $\mathbb{C}^s$. This algebra is isomorphic to $M_{2^{n/2}}(\mathbb{C})$ for even n and to $M_{2^{(n-1)/2}}(\mathbb{C}) \oplus M_{2^{(n-1)/2}}(\mathbb{C})$ for odd n [8]. Every finite-dimensional representation is a direct sum of irreducible ones of dimension $2^{\lfloor n/2 \rfloor}$, so $2^{\lfloor n/2 \rfloor}$ divides s and $\lfloor n/2 \rfloor \leq \nu$. For the converse take an irreducible representation of $\mathbb{C}l_{2\nu+1}$, which has dimension 2^ν, tensor it with $\mathbf{1}_m$, and use any n of its generators. □

Corollary 9 (Qubit carriers)**.** *For $s = 2$: an isotropic single-shell symbol exists for massless propagation in $d \leq 3$ and for massive propagation in $d \leq 2$. Massive single-shell propagation in $d = 3$ requires $4 \mid s$; the minimal case $s = 4$ is the Dirac symbol $\alpha_i = \sigma_x \otimes \sigma_i$, $\beta = \sigma_z \otimes \mathbf{1}$, a pair of Weyl qubits coupled by the mass term.*

For $s = 2$ and $d = 3$ the symbol is $H(q) = c\,\sigma \cdot q$ up to a unitary and an orthogonal change of spatial coordinates, and

$$\det\big(\omega\mathbf{1} - c\,\sigma \cdot q\big) = \omega^2 - c^2|q|^2. \tag{7}$$

This is the Lorentzian form of signature $(1,3)$, obtained from the Pauli relations $\{\sigma_i, \sigma_j\} = 2\delta_{ij}$, which are Euclidean. No operator on $\mathbb{C}^2$ anticommutes with all three Pauli matrices, so $Cl(1,3)$ has no representation on $\mathbb{C}^2$; the Lorentzian form exists anyway, as the characteristic form of the evolution.

Two forms on the same carrier. The identity $\det(x_0 I + x \cdot \sigma) = x_0^2 - |x|^2$ is preserved by determinant-one filtering, not by the positive Hilbert–Schmidt norm. For the massive Dirac carrier, the time-like generator in Corollary 5 is the mass coefficient, not a modular generator.

Proposition 10 (Full coupling of a qubit)**.** *Let $s = 2$, $\mu = 0$, and let the symbol satisfy Theorem 4 with traceless coefficients. The map $q \mapsto \sum_i q_i \alpha_i$ is an injective linear map from $\mathbb{R}^d$ into the three-dimensional space of traceless Hermitian 2×2 matrices. It is onto, so that every internal rotation generator is coupled to some direction of motion, if and only if $d = 3$.*

Proof. Injectivity is the linear independence assumed in Theorem 4; the target has dimension three. □

Proposition 10 is a precise version of a familiar observation: momentum space of a Weyl particle is identified with the Lie algebra of internal rotations of its qubit. It is a characterization under a stated premise, not a derivation of three dimensions. The premise, that no internal axis is left uncoupled, is a physical input. The following conjecture states what would have to be shown to turn it into a selection principle.

Conjecture 11 (Minimal carriers and three dimensions)**.** *In local unitary dynamics whose elementary internal carrier is a qubit and whose low-energy propagation is isotropic and single-shell, propagation that uses the full internal space selects $d = 3$, and massive propagation requires two carriers coupled by a chirality-exchanging term.*

Theorem 8 and Proposition 10 reduce this selection proposal to a physical reason for coupling every internal axis to momentum. Irreducibility alone does not supply it: $H = q_x\sigma_x + q_y\sigma_y$ generates the full matrix algebra, while the omitted σ_z is not a conserved charge. The three-dimensional Weyl automaton [13] and the minimal-directional-system argument of Ref. [15] motivate the proposal, but do not select three dimensions within the present class. A selection principle must distinguish the two-dimensional counterexample by stability, resources or formation, and must explain why a qubit is the relevant elementary carrier.

5 The lattice level: causal polytope, three speeds and nodes

5.1 Group velocities lie in the causal polytope

Proposition 12 (Group velocities and the neighbourhood)**.** *For the step (1), let b be a band and k_0 a point where the eigenvalues of $W(k_0)$ are distinct. Then $\nabla_k \omega_b(k_0) \in \tau^{-1} \operatorname{conv} N$.*

Proof. Take $\tau = 1$. Let X be the position operator and ψ finitely supported. Then $\langle X\rangle_n = \langle W^n\psi, XW^n\psi\rangle$ satisfies $\langle X\rangle_n - \langle X\rangle_0 = \sum_{j<n}\langle W^j\psi, VW^j\psi\rangle$ with $V = W^*[X,W]$. From Eq. (1), $([X,W]\psi)(x) = \sum_h hA_h\psi(x-h)$, so V is bounded and translation invariant with symbol $V(k) = W(k)^* i\nabla_k W(k)$. Write $W(k) = \sum_b e^{-i\omega_b(k)}P_b(k)$ wherever the eigenvalues are distinct, which is almost everywhere once it holds at one point, since the discriminant of the characteristic polynomial is real-analytic in k. For each such k the Cesàro means $n^{-1}\sum_{j<n} W(k)^{-j}V(k)W(k)^j$ converge to $\sum_b P_bVP_b$, and differentiating $W = \sum_b e^{-i\omega_b}P_b$ with $P_b(\nabla P_b)P_b = 0$ gives $P_bVP_b = \nabla\omega_b\, P_b$. By bounded convergence,

$$\frac{\langle X\rangle_n - \langle X\rangle_0}{n} \longrightarrow \bar v(\psi) = \sum_b \int \nabla\omega_b(k)\,\langle\hat\psi(k), P_b(k)\hat\psi(k)\rangle\,\frac{dk}{|B|},$$

where B is the Brillouin zone. On the other hand $W^n\psi$ is supported in $\operatorname{supp}\psi + nN$, so for every unit vector u, $u\cdot\langle X\rangle_n \le \max_{x\in\operatorname{supp}\psi} u\cdot x + n\max_{h\in N} u\cdot h$. Hence $u\cdot\bar v(\psi) \le h_N(u) := \max_{h\in N} u\cdot h$. The map $\hat\psi \mapsto \bar v(\psi)$ is continuous in L^2 because $|\nabla\omega_b| \le \|V(k)\|$ is bounded, and finitely supported states are dense, so the inequality holds for all ψ. Taking $\hat\psi = P_b(k)\varphi$ with φ concentrating at k_0 gives $u\cdot\nabla\omega_b(k_0) \le h_N(u)$ for every u, which is membership in $\operatorname{conv} N$. □

The ballistic scaling used in the proof is the expectation-value form of the weak limit theorem for quantum walks [16].

Corollary 13 (Speed of a massless read-out cone)**.** *If a band has a first-order massless symbol at k_* whose cone is round with speed c, then $c \le \tau^{-1} r_{\text{in}}(\operatorname{conv} N)$, where r_{in} is the inradius.*

Proof. Near k_* the group velocity tends to $c\,\hat q$ along every direction $\hat q$, so $c\hat q \in \tau^{-1}\operatorname{conv} N$ for all $\hat q$. □

5.2 The split-step walk

Consider the coined walk that applies a coin rotation and then a conditional shift,

$$W_\theta(k) = \begin{pmatrix} e^{-ika}\cos\theta & -e^{-ika}\sin\theta \\ e^{ika}\sin\theta & e^{ika}\cos\theta \end{pmatrix}, \qquad 0 < \theta < \frac{\pi}{2}, \tag{8}$$

with neighbourhood $N = \{a, -a\}$ and step time τ.

Proposition 14 (Three speeds of one walk)**.** *For the walk* (8)*:*

1. $\cos(\omega\tau) = \cos\theta\cos(ka)$*;*
2. *the support speed is a/τ, and the largest group velocity is $(a/\tau)\cos\theta$, attained at $ka = \pm\pi/2$;*
3. *near $k = 0$, $\omega(k)^2\tau^2 = \theta^2 + \theta\cot\theta\,(ka)^2 + O((ka)^4)$; the rest energy is $E_0 = \hbar\theta/\tau$, the inertial mass $m_* = \hbar\tau\tan\theta/a^2$, and the Lorentzian read-out speed defined by $E_0 = m_* c_{\text{eff}}^2$ is $c_{\text{eff}} = (a/\tau)(\theta\cot\theta)^{1/2}$;*

4. $\cos\theta < (\theta\cot\theta)^{1/2} < 1$.

Proof. Item 1: $\det W_\theta = 1$ and $\operatorname{tr} W_\theta = 2\cos\theta\cos(ka)$. Item 2: the support statement is Eq. (1) with $N = \{\pm a\}$. Differentiating item 1, $v = (a/\tau)\cos\theta\sin(ka)/\sin(\omega\tau)$, and $v^2 \le (a/\tau)^2\cos^2\theta$ is equivalent to $\cos^2\theta\cos^2(ka) \le \cos^2(ka)$, with equality only at $\cos(ka) = 0$. Item 3: differentiating item 1 twice at $k = 0$, where $\omega\tau = \theta$ and $\omega' = 0$, gives $\tau\sin\theta\,\omega'' = a^2\cos\theta$; then $\frac{1}{2}(\omega^2)'' = \omega\omega''$, $m_* = \hbar/\omega''$, and $E_0/m_* = \omega(0)\omega''(0)$. Item 4 is $\sin\theta\cos\theta < \theta < \tan\theta$. □

The Lorentzian speed read from the rest energy and the inertia of slow particles lies strictly between the largest transport speed and the support speed. No wave packet reaches it, because $v \le (a/\tau)\cos\theta < c_{\text{eff}}$; and the strict causal cone is wider. In the scaling limit $a, \tau \to 0$ with $a/\tau = c$ and $\theta/\tau = \mu$ fixed, all three speeds tend to c and the dispersion tends to $\omega^2 = c^2q^2 + \mu^2$ [11, 12]. At $\theta = 0$ the walk is a pair of exact shifts, with $\omega = \pm k$ and one speed. The deviations at finite mass are $1 - c_{\text{eff}}^2\tau^2/a^2 = \theta^2/3 + O(\theta^4)$ and $1 - \cos^2\theta = \theta^2 + O(\theta^4)$. The consequences for mass read-outs are developed in the companion paper on mass and energy [47].

5.3 The Weyl automaton on the body-centred cubic lattice

The minimal three-dimensional automaton of D'Ariano and Perinotti [13] has $s = 2$, neighbourhood $N = \{\ell(\pm1, \pm1, \pm1)/\sqrt{3}\}$ of eight vectors of length ℓ, and symbol

$$W(k) = e^{-i\kappa_x\sigma_x}\, e^{-i\kappa_y\sigma_y}\, e^{-i\kappa_z\sigma_z}, \qquad \kappa = \frac{\ell k}{\sqrt{3}}, \tag{9}$$

with $\cos(\omega\tau) = c_xc_yc_z - s_xs_ys_z$, where $c_i = \cos\kappa_i$, $s_i = \sin\kappa_i$. Expanding, $c_xc_yc_z = 1 - |\kappa|^2/2 + O(|\kappa|^4)$ and $s_xs_ys_z = \kappa_x\kappa_y\kappa_z + O(|\kappa|^5)$, so

$$\omega\tau = |\kappa| + \frac{\kappa_x\kappa_y\kappa_z}{|\kappa|} + O(|\kappa|^3). \tag{10}$$

At leading order $W(k) \simeq \mathbf{1} - i\sigma\cdot\kappa$, the symbol is $c\,\sigma\cdot k$ with $c = \ell/(\sqrt{3}\tau)$, and Eq. (7) gives a round Lorentzian cone. Isotropy fails at relative order $|\kappa|$ through the parity-odd term in Eq. (10). Table 1 collects analytic properties and numerical unitary wave-packet diagnostics. It is not a table of records obtained from a physical measuring instrument.

5.4 What the read-out map adds

The lattice results show three ways in which the dynamics leaves the Lorentzian read-out underdetermined.

First, at finite mass the speed of the read-out depends on the observable that defines it. In the walk (8), rest energy and inertia give c_{eff}, the fastest wave packets give $(a/\tau)\cos\theta$, and the support front gives a/τ (Proposition 14).

Second, the Lorentzian cone governs low-momentum states only. In the Weyl automaton a state prepared on one site contains all momenta and leaves almost half of its probability

Property	**Value**
Causal polytope $\tau^{-1}\operatorname{conv} N$	cube $\lvert x_i\rvert \le \ell/(\sqrt{3}\tau)$ per unit time (analytic)
Inradius of the cube	$\ell/(\sqrt{3}\tau) = c$: Corollary 13 is saturated (analytic)
Group velocity components	at most c, attained on the cube faces
Largest group speed	$(3/2)^{1/2}c$; the cube corners, at speed $\sqrt{3}c$, are not reached by group velocities
Band-touching points in the Brillouin zone	four: at quasi-energy 0, $k = 0$ (chirality $+1$) and $\kappa = -\frac{\pi}{2}(1,1,1)$ (chirality -1); at quasi-energy π/τ, one of each chirality
One-site initial state	probability outside the cube: zero to machine precision; outside the round cone of radius ct: 0.50 after 16 steps, 0.47 after 24
Gaussian packets near $k = 0$, 24 steps	probability outside radius $ct + R_0$ (R_0 the 99% radius at $t = 0$): 0.084, 0.023, 0.008 for momentum widths $\sigma_\kappa = 0.4$, 0.25, 0.15

Table 1: Weyl automaton (9). Chirality is the sign of the determinant of the linear map $q \mapsto$ coefficient of $-i\sigma$ in $\pm W(k_0 + q)$. The packets carry a fixed spinor and therefore populate both bands.

outside the round cone, while it never leaves the cube; the leakage falls as the momentum width shrinks (Table 1).

Third, the Brillouin zone contains several band-touching points. In the Weyl automaton there are two at quasi-energy zero with opposite chiralities, as the doubling theorem for lattice Hamiltonians would lead one to expect [17], although that theorem does not apply directly to automata. A low-energy observer reads a Weyl cone of one chirality or the other depending on which point the prepared states are near. The dynamics does not choose; the preparation and the instruments do. In one dimension a single chiral cone is possible, since the shift by one site is a legitimate automaton; whether a strictly local isotropic automaton in three dimensions can carry a single node is a question we leave open.

So in this model the read-out map cannot be removed. What can be said precisely is what it consists of: a band point, a momentum window around it, and an observable defining the speed, each with computable consequences. This matches the treatment of access in HAFF, where what a physical reduction resolves depends on the channel and not only on the algebra it generates [42]. Here the Lorentzian geometry read by an observer is the characteristic geometry of the momentum window that its couplings reach.

5.5 Finite records certify one walk's dictionary

The choice of a read-out need not leave all its parameters inaccessible. The companion mass–energy paper realizes the same coin rotation by a fresh Gaussian oscillator, computes its exact open-channel linewidth, and reads its rest phase through controlled mixed-memory instruments [47]. On a supplied finite periodic ring, a normalized zero mode is preserved by a spatially uniform field coupling. Independent X and Z record trials at one and ten steps resolve the phase alias and certify θ. The ring, uniform support, axes, gain, clock, fresh ports

and record archive are priced inputs.

Let that actual protocol give $\theta \in [\theta_-, \theta_+]$ at its stated confidence, with supplied a, τ. Monotonicity of $\tan\theta$, $\cos\theta$ and $C(\theta) = \theta\cot\theta$ gives simultaneous intervals

$$E_0 \in \frac{\hbar}{\tau}[\theta_-, \theta_+], \qquad m_* \in \frac{\hbar\tau}{a^2}[\tan\theta_-, \tan\theta_+],$$
$$c_{\text{eff}}^2 \in \frac{a^2}{\tau^2}[C(\theta_+), C(\theta_-)], \qquad v_{\max} \in \frac{a}{\tau}[\cos\theta_+, \cos\theta_-].$$

At $\theta_+ = \pi/2$ the inertial upper bound is infinite. With the priced angle radius $b = .004296823$ at 95% confidence derived in Ref. [47], true angle .3 and $a = \tau = 1$, every confidence interval on that event lies inside $[.3 - 2b, .3 + 2b]$. The implied intervals are

$$c_{\text{eff}} \in [.9838969, .9856636], \qquad v_{\max} \in [.9527616, .9578408], \qquad c_{\text{support}} = 1.$$

The support speed is supplied, not measured by the phase protocol. The other two are model implications of its angle certificate. The full preparation, pulse, read-memory and reference budget is given with that protocol.

These are parameter inferences for the prescribed ideal walk, informed by an actual open-drive experiment. They are not its noisy long-time transport velocities: nonzero-momentum transport requires the real Bloch channel and inter-momentum coherences. Nor does calibration prove that the dictionary is Lorentzian throughout the band. The companion finite-angle window theorem gives precisely where a Lorentzian mass reading becomes negative and must be rejected. Thus an observer's interface can be calibrated, can distinguish otherwise conflated speeds, and can fail at a testable boundary.

6 Several species: when do the cones become one?

Theorem 4 gives every single-shell species its own form Q. For isotropic species in one frame, a common light cone requires their speeds to agree. In a free model nothing makes them agree: each speed is a parameter of the step. This section asks whether coupling the species can make the speeds equal, so that one cone is an output of the dynamics. The answer depends on the kind of coupling. Off-diagonal mixing cannot produce a single quadratic principal shell while its distinct diagonal shells are held fixed. Interactions can change those diagonal propagation laws: in the mediator model their speeds converge when each coupling has divergent accumulated strength toward low energy.

6.1 Mixing does not merge cones

Write the principal symbol of two species in a direction u as $A_1(u)$ on $\mathbb{C}^{s_1}$ and $A_2(u)$ on $\mathbb{C}^{s_2}$, both single-shell, $A_a(u)^2 = Q_a(u)\mathbf{1}$. A one-particle coupling adds a block off-diagonal term, which may itself depend linearly on q.

Theorem 15 (Mixing and the single shell)**.** *For every $s_1 \times s_2$ matrix V, the symbol $A = \begin{pmatrix} A_1 & V \\ V^* & A_2 \end{pmatrix}$ satisfies*

$$\min_{\lambda\in\mathbb{R}} \big\|A^2 - \lambda\mathbf{1}\big\|_{\text{HS}}^2 \;\geq\; \tfrac{1}{2}\min(s_1, s_2)\,\big(Q_1(u) - Q_2(u)\big)^2. \tag{11}$$

Hence the coupled system has a single shell only if $Q_1 = Q_2$. Moreover the largest eigenvalue of A is at least the largest eigenvalue of either block and the smallest is at most the smallest of either block.

Proof. The diagonal blocks of $A^2 - \lambda$ are $Q_1 - \lambda + VV^*$ and $Q_2 - \lambda + V^*V$, and the Hilbert–Schmidt norm of a block matrix is at least that of its diagonal blocks. If V has rank r, then VV^* and V^*V share the nonzero eigenvalues $\sigma_1^2, \dots, \sigma_r^2$ and have $s_1 - r$ and $s_2 - r$ zero eigenvalues. Using $(x+a)^2 + (x+b)^2 \geq \frac{1}{2}(a-b)^2$ for each shared eigenvalue, and for $k = \min(s_1, s_2) - r$ pairs of zeros, the squared norm is at least $\frac{1}{2}(r+k)(Q_1 - Q_2)^2$. The last statement is Cauchy interlacing for the principal submatrices A_1 and A_2. □

The bound (11) does not depend on the strength of the coupling. A mass-type coupling, independent of q, leaves the principal symbol untouched, but can change low-momentum band slopes. For example,

$$H(q) = \begin{pmatrix} v_1 q & g \\ g & v_2 q \end{pmatrix}, \qquad \omega_\pm(q) = \tfrac{1}{2}(v_1 + v_2)q \pm \sqrt{g^2 + \tfrac{1}{4}(v_1 - v_2)^2 q^2}.$$

For $g \neq 0$ both slopes at $q = 0$ equal $(v_1 + v_2)/2$, although the two distinct principal speeds remain v_1, v_2. Equal slopes at one band point are not a common quadratic characteristic cone. An on-site lattice coin can likewise change the effective logarithm of a step; the fixed-block theorem applies only after its principal blocks have been specified. For a chiral principal velocity matrix with fixed diagonal entries, the Schur–Horn theorem makes the eigenvalues majorize those entries, so off-diagonal mixing cannot reduce their variance. The interacting route below instead changes the propagation coefficients themselves.

6.2 Loops average speeds

Lemma 16 (Loops of one speed)**.** *Let a self-energy be a loop whose internal lines all have Lorentzian propagators of speed c and whose vertices are covariant under the Lorentz group of speed c. Then the coefficient of* $\ln \Lambda$ *in it is Lorentz covariant with speed c, for any fixed regular cutoff shape enclosing the origin and scaled by Λ.*

Proof. The substitution $k \mapsto (k_0, k/c)$ makes the integrand covariant under the standard Lorentz group. For a homogeneous integrand f of degree $-D$, the coefficient of $\ln \Lambda$ is the flux of the divergence-free field $xf(x)$ through a surface enclosing the origin, and it is the same for every such surface. □

The smallest interacting model is a real scalar of speed u coupled to N Dirac fermions of speed v by $g\phi\bar{\psi}\psi$ in $3+1$ dimensions. We integrate out a shell of spatial momenta with the frequency unrestricted; the logarithmic coefficients are scheme independent at this order. By Lemma 16 the fermion loop adds $\frac{Ng^2}{4\pi^2 v^3}(p_0^2 + v^2 p^2)\, dl$ to the inverse scalar propagator $p_0^2 + u^2 p^2$, where $l = \ln(\Lambda/E)$, so the squared scalar speed is replaced by a convex combination of u^2 and v^2. The fermion self-energy contains one fermion and one scalar line. Its frequency integrals are

$$J_0 = \int \frac{t^2\, dt}{(t^2+v^2)(t^2+u^2)^2} = \frac{\pi}{2u(v+u)^2}, \qquad J_s = \int \frac{dt}{(t^2+v^2)(t^2+u^2)^2} = \frac{\pi(v+2u)}{2vu^3(v+u)^2}, \tag{12}$$

and the ratio of the spatial to the temporal coefficient gives the speed toward which v moves.

Proposition 17 (Speed flow in the Yukawa model)**.** *At one loop,*

$$\frac{dv}{dl} = a\Big(\frac{v+2u}{3} - v\Big), \quad a = \frac{g^2}{4\pi^2 u(u+v)^2}, \qquad \frac{du^2}{dl} = \frac{Ng^2}{4\pi^2 v^3}\,(v^2-u^2), \tag{13}$$

and the coupling obeys $dg^2/dl = -(g^4/2\pi^2)\big[1/(vu(v+u)) + N/(2v^3) + 1/(u(v+u)^2)\big]$*. At* $u = v$ *the linearized flow of* $u - v$ *has rate* $(6N+1)g^2/(24\pi^2 v^3)$*, and* $u + 6Nv$ *is conserved to linear order.*

For $N = 1$ these equations coincide with those of Anber and Donoghue [18], who derived them by dimensional regularization. The momentum-shell calculation gives the same logarithmic coefficients. Each speed moves toward a target that lies between the speeds of the lines in its loop: the fermion toward $(v+2u)/3$, the scalar toward v. The scalar moves $6N$ times faster, so the common speed lies close to the initial fermion speed. Since the Yukawa coupling is marginally irrelevant, $g^2 \simeq 8\pi^2 c_*^3/[(2N+3)l]$, where c_* is the common limiting speed and

$$\frac{|u-v|}{v} \;\propto\; \Big[\ln\frac{\Lambda}{E}\Big]^{-(6N+1)/(3(2N+3))}, \tag{14}$$

with exponent $7/15$ for one Dirac fermion. Integrating the full flow to $l = 10^9$ gives log–log slopes -0.4667 for $N = 1$ and -0.7576 for $N = 4$, as predicted.

Theorem 18 (Speeds converge through a common mediator)**.** *Let fermion species of speeds* $v_1, \dots, v_n$ *couple to one scalar of speed* u *with prescribed nonnegative, locally integrable functions* $g_1^2(l), \dots, g_n^2(l)$*, each fermion obeying the speed equations in* (13) *and* u^2 *receiving the sum of the fermion loops. The theorem is conditional on these coupling histories; a multi-species beta function must separately establish their accumulated strengths. Assume positive initial speeds. Let* D *be the difference between the largest and the smallest of* $u, v_1, \dots, v_n$*, and* M_0 *the initial largest speed. Then the largest speed never increases, the smallest never decreases, and*

$$D(l) \le D(0)\exp\Big(-\frac{1}{24\pi^2 M_0^3}\int_0^l \min_a g_a^2\, dl'\Big). \tag{15}$$

If $\int g_a^2\, dl < \infty$*, the speed* v_a *changes in total by at most* $D(0)\int g_a^2\, dl/(24\pi^2 m_0^3)$*, with* m_0 *the initial smallest speed.*

Proof. Each v_a moves toward u at rate $\frac{2}{3}a_a$, and u^2 moves toward a weighted mean of the v_a^2, so u decreases when it is the largest speed and increases when it is the smallest. Consider the upper Dini derivative D^+ of the maximum minus the minimum. If both extremes are fermions $v_a \ge v_b$, then $D^+ \le \frac{2}{3}[a_a(u-v_a) - a_b(u-v_b)] \le -\frac{2}{3}(\min_c a_c)D$; if the maximum is u, then $D^+ \le -\frac{2}{3}a_b(u-v_b)$; the case of u minimal is symmetric. All speeds stay in $[m_0, M_0]$, where $\frac{2}{3}a_a \ge g_a^2/(24\pi^2 M_0^3)$, and Grönwall's inequality gives (15). The last statement follows from $|v_a'| \le \frac{2}{3}a_a D(0)$ and $a_a \le g_a^2/(16\pi^2 m_0^3)$. □

The estimate gives a rate when the weakest simultaneous coupling has divergent integral. Convergence itself needs less.

Corollary 19 (Intermittent interactions)**.** *In the one-loop star model of Theorem 18, suppose the nonnegative functions g_a^2 are locally integrable and initial speeds are positive. Every speed converges. Species connected to the mediator by edges with $\int_0^\infty g_a^2\,dl = \infty$ have the same limiting speed as the mediator. In particular, all speeds converge to one value if each such integral diverges, even when the couplings are never active simultaneously.*

Proof. Write $v_a' = A_a(u - v_a)$ and $u' = \sum_a B_a(v_a - u)$, with

$$A_a = \frac{g_a^2}{6\pi^2 u(u+v_a)^2}, \qquad B_a = \frac{g_a^2(u+v_a)}{8\pi^2 u v_a^3}.$$

For an active edge, $B_a/A_a = \frac{3}{4}(1 + u/v_a)^3$. All speeds stay in $[m_0, M_0]$, so the two directional rates are uniformly comparable, and their integrals diverge exactly when $\int g_a^2$ does. This is a type-symmetric continuous-time consensus system. The cut-balanced convergence theorem [1], applied to the coefficients along the solution, gives convergence and equality of limits within each component of the infinite-integral interaction graph. The theorem allows state-dependent coefficients. It does not provide the same rate as (15) when $\min_a g_a^2$ vanishes. □

The accumulated interaction across renormalization scales decides the outcome. The variable $l = \ln(\Lambda/E)$ is scale, not laboratory time; intermittent edges here are active on different scale intervals. An identically uncoupled species keeps its cone; a finite integrated coupling need not erase an initial difference. Marginally irrelevant couplings remove differences as a power of $\ln(\Lambda/E)$; near an interacting fixed point they fall as a power of E/Λ, with the one-loop exponent $7g_*^2/(24\pi^2 c_*^3)$ for $N = 1$. In $4 - \varepsilon$ dimensions the Yukawa fixed point gives $(6N + 1)\varepsilon/[3(2N + 3)]$ at this order. The weak-coupling model must remain applicable along this flow. A mediator that breaks rotations can instead drive speeds apart, as in nematic criticality of d-wave superconductors [20].

The common speed lies between the smallest and the largest initial speed. Each initial speed of a massless round cone is at most the inradius of conv N per step (Corollary 13), so the emergent cone lies inside the causal polytope.

6.3 Fixed points and the number of cones

Near an interacting fixed point the question has an answer that does not rely on perturbation theory.

Theorem 20 (Speed splittings at a conformal fixed point)**.** *Let the infrared fixed point be a unitary conformal field theory in $D = d + 1 \geq 3$ dimensions, Lorentz invariant for one cone, and consider local perturbations invariant under translations, spatial rotations and CPT. Up to total derivatives the Lorentz-violating perturbations are $\int d^D x\, O_{0\cdots0}$ for symmetric traceless primaries of even spin $\ell \geq 2$. Their couplings satisfy $D - \Delta_O \leq 2 - \ell \leq 0$; equality in the first inequality means that the current is conserved, while a marginal Lorentz-violating coupling requires a conserved spin-2 current. If the stress tensor is the only conserved spin-2 current, all independent spin-2 speed perturbations are irrelevant. When the excited non-conserved spin-2 modes have a smallest gap $\gamma_2 = \min(\Delta_O - D) > 0$, their slowest linearized decay is $(E/\Lambda)^{\gamma_2}$.*

Proof. Descendants integrate to zero. A primary transforms in a finite-dimensional irreducible representation of the Lorentz group, whose complexification is that of $SO(D)$, and because $(SO(D), SO(D-1))$ is a Gelfand pair only the symmetric traceless representations contain a rotation-invariant component, once each. CPT acts on a Hermitian rank-ℓ tensor by $(-1)^\ell$, which removes odd ℓ, and scalars are Lorentz invariant. The unitarity bound $\Delta \geq \ell + D - 2$ for $\ell \geq 1$ holds with equality exactly for conserved currents [21, 22]. The perturbation $\int T_{00}$ rescales time and changes no relative speed. □

The stability argument is established in Refs. [23, 24]; it assumes the Lorentz-invariant fixed point and does not prove that every microscopic dynamics reaches it. Additional conserved spin-2 currents are the marginal exceptions. In three dimensions, for exactly two such currents, a suitable basis generates separate translations and the two families of current correlators factorize [25, Section 8]. This is dynamical separation of the current sectors, not necessarily a tensor product of the full theories: a shared singlet constraint can remain. Nor does counting currents count sectors in general; two free scalar fields already have three conserved spin-2 currents. The result needed here is the local criterion for suppressing relative-speed perturbations, while identifying all independently propagating sectors requires further structure.

In two dimensions several propagation sectors can retain different speeds, as in spin–charge separation. A Sugawara decomposition gives an additional sector only when $T - T_J$ is nonzero. Under its positivity, spectrum and finite-signal-speed assumptions, the scale-to-Lorentz theorem permits either one Lorentz factor or several [26]; it does not forbid a common cone from being selected dynamically. Establishing one cone therefore requires information about the actual sectors and their interactions, rather than the dimensionality alone.

6.4 What the lattice adds

For number-conserving automata whose interaction preserves the supplied neighbourhood, each distinguishable particle stays in $x_j + nN$ after n steps. The centre-of-mass version of Proposition 12 therefore bounds every isolated bound-state band by $\operatorname{conv} N$ per step. Changing the neighbourhood changes that bound. Equalization of low-energy cones does not itself change the support cone.

For a loop dominated by one linear band point, the logarithmic coefficient carries that point's read-out speed by Lemma 16. Finite ultraviolet matching terms need not have the same ratio of temporal and spatial coefficients. A Lorentz-violating regulator can leave corrections of order $g^2/16\pi^2$ [28], subsequently reduced only at the infrared rates above.

Composites need not inherit a constituent's read-out cone. Take distinguishable walkers with $\theta_A = .3$, $\theta_B = .5$, $a = \tau = 1$, first apply both free coined steps, and then multiply the coincident-position sector by $e^{+.4i}$. With total momentum K conjugate to $(x_A + x_B)/2$, the isolated positive band at $K = 0$ has $\omega_0 = .7614121$ and $m_* = 1/\omega_0'' = .9023$, compared with the free threshold .8 and constituent mass sum .8556. Thus $c_{\text{eff}}^2 = \omega_0 \omega_0'' = .8439$, below the constituent values $\theta_A \cot \theta_A = .9698184$ and $\theta_B \cot \theta_B = .9152439$. Periodic relative-coordinate calculations with 91 and 131 sites agree at the quoted precision. They also track a localized band beyond $|K| = .68$; no such endpoint or global maximum speed is

inferred from this local mass calculation. Contact interactions can produce bound molecules in quantum walks [19]. In the example above, the separately computed quasienergy and curvature do not obey the relativistic binding-energy–inertia relation.

A common cone can also be kinematic. The minimal isotropic Weyl automaton is unique on the body-centred cubic lattice [13], so all massless species built from it share $c = \ell/(\sqrt{3}\tau)$; a Dirac automaton built from two of them has first-order speed $c(1-m^2)^{1/2}$ in lattice units, and photons built from pairs of Weyl automata have the dispersion $2|n_{k/2}|$ and the same low-energy cone [27]. For an electron on a lattice of Planck spacing, $m \approx 4 \times 10^{-23}$ and the mass correction is of order 10^{-45}. Loop matching can generate relative-speed corrections of order $g^2/16\pi^2$ unless a symmetry forbids them. Contact binding also changes the composite cone in the example above, but its size is model dependent and carries no general loop suppression factor.

6.5 How much running is needed

With marginal couplings the logarithmic rate (14) is far too slow to produce the observed equality of speeds. Reducing a difference by a factor 10^{-13} requires $\ln(\Lambda/E) \approx 10^{28}$ for one Dirac fermion with $g^2 = 4\pi$ at the cutoff, and about 10^{12} even for a hundred fermions, while $\ln(M_{\mathrm{Pl}}/m_e) = 51.5$. Anber and Donoghue reached the same qualitative conclusion [18].

Conjecture 21 (One cone needs a strongly coupled regime)**.** *In local unitary dynamics with several interacting gapless species, the relative speed difference of two sectors at scale E is $\delta c(E) \simeq \delta c(\Lambda) \exp(-\int_E^\Lambda \gamma \, d\ln\mu)$, where γ is the anomalous dimension of their relative stress tensor and $\delta c(\Lambda)$ is of order $g^2/16\pi^2$ from lattice matching even when the step is common. Equalities of order 10^{-15} then require $\int \gamma \, d\ln\mu \gtrsim 30$ above accessible energies, that is, a strongly coupled regime with γ of order one over many decades, or a symmetry that removes $\delta c(\Lambda)$. The largest residual differences sit between the most weakly coupled sectors.*

Theorems 18 and 20 tie equalization to interaction strength; lattice matching regenerates relative-speed differences, while marginal running removes them only logarithmically. The quantitative estimates above and the power-law emergence in strongly coupled holographic models [24] support the conjecture. Anber and Donoghue's observation that the weakest interactions leave the largest differences also makes gravitational waves against light a sensitive comparison [18].

A viable ultraviolet realization must place the strongly coupled range consistently with the observed weakly coupled particle description and experimental bounds. A composite realization is one possibility, not a necessary consequence of the conjecture. A symmetry route, such as non-relativistic supersymmetry protecting dimension-four terms, can avoid strong coupling but must survive symmetry breaking. Gravity is shared by all species yet is irrelevant at low energy; this flow cannot align its cone there. The relation of the gravitational-wave and matter cones therefore requires an ultraviolet mechanism or a protecting symmetry.

6.6 Consequences

The mediator model makes a common speed a conditional output of divergent accumulated interactions. The conformal argument establishes stability against relative-speed perturbations near a supplied Lorentz-invariant fixed point; it does not establish formation of that fixed point. This separates three inputs to the higher-dimensional Consciousness construction: each species' shell and signature, the alignment of their cones, and the actual infrared regime. Its boosted-ball family requires a common cone. In the two-dimensional HAFF gravity model [44], identifying the matter and Jackiw–Teitelboim cones remains an input; the two-dimensional sector theorem neither supplies nor forbids that identification.

7 Relations to other frameworks and papers

7.1 Algebraic frameworks

Sequential operations are represented here by linear maps under ordinary composition, which is associative by definition. Jordan and octonionic models need additional bracketing and composition rules before tensor products, subsystems or error correction can be transferred [30, 31, 32]; this is a statement of scope, not a no-go theorem. Von Neumann and C^*-algebras are compatible with everything above, since the step (1) and its symbol live in a concrete C^*-algebra; such algebras do not single out a quadratic form until a dynamics is given. Clifford algebras are distinguished in this paper by a dynamical property, the single shell of Theorem 4, rather than by being chosen as an encoding.

7.2 Thermodynamic gravity, modular flow and spectral triples

Jacobson's derivation of the Einstein equation assumes local Lorentzian causal structure, local equilibrium and an entropy–area relation [3], and entanglement derivations of linearized gravity work on a Lorentzian background [33, 34]. The present model supplies local Lorentzian cones for the low-momentum sector of a local unitary dynamics. It supplies neither equilibrium nor the entropy–area relation, and Section 5.4 shows that its cones depend on the momentum window.

Local algebras in relativistic quantum field theory are of Type III [35, 36]. For the vacuum and wedge regions the Bisognano–Wichmann theorem identifies modular flow with the boost subgroup [37, 48]; the identification presupposes the Lorentzian structure it describes. Whether the modular data of states of a lattice automaton reproduce the boosts of its low-energy cone in the scaling limit is open and could be computed in the walk of Section 5.2.

In Connes' noncommutative geometry a Dirac operator with Euclidean Clifford symbol encodes Riemannian geometry [38], and Lorentzian versions are an active problem [39, 40, 41]. Theorem 4 says that using such a Dirac operator as a Hamiltonian, $i\partial_t\psi = D\psi$, produces the Lorentzian characteristic cone of signature $(1, d)$ on time times space. We have not worked out the relation to the Krein-space approaches cited.

7.3 The Q-RAIF papers and HAFF

Section 3.2 specifies the separate outputs to control and spectral mass. The actuator field of Q-RAIF B is the single-chirality case of Remark 6; its outgoing stress is used in HAFF-D [44]. Identifying that matter cone with the gravitational cone is a further physical input, not a consequence of an isometry of internal generators.

8 Discussion

Ingredient of a Lorentzian read-out	**Status in local unitary propagation**
Evolution time	supplied; unitarity makes its frequency a hyperbolicity direction (Theorem 2)
Signature $(1, d)$	derived for any quadratic cone (Lemma 3)
Clifford relations	equivalent to a single shell (Theorem 4); Lorentzian generators constructed (Corollary 5)
Single shell, isotropy, linear band point	inputs; observable and testable in a given model
Spatial dimension d	input; bounded by the internal space (Theorem 8)
Speed c	a property of the dynamics; at finite mass it depends on the defining observable (Proposition 14)
Read-out map	input: band point, momentum window and speed observable (Section 5.4)
Common speed of several species	conditional output of the mediator model; stable against irrelevant relative-speed perturbations near a supplied Lorentz-invariant fixed point, whose formation is not derived (Theorems 18, 20)
Lattice, neighbourhood, step	inputs

Table 2: Inputs of a Lorentzian read-out in local unitary propagation.

Table 2 separates the derived signature from the remaining inputs: lattice data, absence of birefringence and the observer's read-out map. Deriving the first two requires a formation model; no property of the dynamics alone specifies the third.

The problems that follow have the most leverage. Section 6 answers the question of a common cone for several species under stated conditions; what remains there is the anomalous dimension γ_2 in a strongly coupled lattice model, which Conjecture 21 needs. Another is chirality: whether a strictly local, isotropic three-dimensional automaton can have a single node, which would remove one component of the read-out map. The mass-setting interface now has a nonempty field realization and finite certificate, but renewed drives, full noisy spatial transport and covariant support remain physical tasks. A fourth is the modular route: computing the modular flow of a lattice state and comparing it with the boosts of the low-energy cone. Position-dependent coins, which produce effective curved metrics, would connect the model to the gravitational questions of HAFF.

9 Conclusion

Propagation supplies the relative sign in $\omega^2 - G_{\mathrm{HS}}$; a positive control metric alone cannot. The single-shell criterion, dimension bound and calibrated residuals connect this form to a specified carrier. Finite angle records imply separated ideal dictionary-speed intervals, while noisy transport remains a different observable. For several species, the mediator model aligns cones under accumulated-coupling conditions; the conformal argument tests their stability, not the creation of a fixed point. These are the respective interfaces to mass–energy read-out, the common-cone geometry of Consciousness II and the still-supplied matter cone of HAFF-D. The calibrated work bridge constrains how propagation generators can be reused for control, without turning positivity into an indefinite signature. Formation of the carrier, clock and accessible momentum window remains a dynamical question.

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Thermodynamic Stability Constraints on Persistent Open Quantum Subsystems: Control Algebra, Physical Actuation and Energy Cost

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What can make the structure of physical control agree with the structure of propagation? Persistence alone does not select a Clifford algebra. A more specific condition does: for a linear Hermitian control family coupled to one centered, independent field input, switching work is independent of the unknown controlled state exactly when the generators square to a scalar quadratic form. Polarization then gives Clifford relations. If the same generators enter a calibrated propagation symbol, the measured work form is a positive pullback of its spatial shell, not the indefinite spacetime form. This conditional connection separates a physical restriction from a choice of notation.

We realize the coupling with a degenerate carrier and a free chiral field. The same interaction determines the control channel, its error and its added stress. An accurate bit flip requires drive occupation of order $\epsilon^{-1/2}$; squeezed inputs nearly attain the bound, while classical inputs need order ϵ^{-1}. A feedback task with scheduled fresh independent drives joins this actuator to calibration records and has an exact stationary loss and a stability proof. A separate sharp energy bound prices translation-only response readout. For inquiry while quantum content is held, a coherent echo and finite Gibbs bank provide explicit protection, reading and renewal operations. Their clocks, drive preparation and spent outputs remain in the resource account. These finite implementations constrain the control–world interface without deriving the supplied field, propagation time or an indefinitely reusable apparatus.

I. INTRODUCTION

Interaction with a large environment drives a quantum system towards decoherence and thermalization [1, 6]. A subsystem that keeps a non-equilibrium steady state does so through exchanges with its surroundings, which may be regulated by feedback [7–10]. The Q-RAIF question is not only what such regulation costs, but why its control structure should bear any relation to the structure in which the subsystem propagates.

Two distinctions are needed. Positivity of a gradient descent metric does not select a Clifford algebra, and no change of coordinates turns a positive work form into a Lorentzian form. Standard dissipative and coherent controls can stabilize states without a Clifford presentation [15–18]. A connection has to come from an actual interaction and from an operational condition on its use.

Here one field coupling supplies both the operation and its energetic response. For a fixed linear family of Hermitian control generators, a centered independent field input gives a work proportional to the expectation of the generator's square. Requiring that charge to be the same for every unknown carrier state is equivalent to scalar-square, hence Clifford, relations (Proposition 6). This is a condition on a specified calibration experiment, not a consequence of persistence or a claim that all useful controls must satisfy it. A one-state average can conceal its failure. When these generators are also the calibrated generators of propagation, Corollary 7 relates the work form to the spatial propagation shell. Evolution time and the physical alignment of the two uses remain inputs.

The finite actuator also makes the cost of maintaining an interface explicit. A degenerate bit coupled to a free chiral field has a flip channel, outgoing stress and switching work determined by the same unitary. Theorem 8 bounds the drive occupation needed for a given error; squeezed inputs nearly attain it, whereas classical drives do not. Theorem 13 composes this imperfect actuator with bath noise and calibration, and Proposition 14 gives one feasibility condition for records and drive energy. Translation-only response readout has its own sharp mean-energy bound, including retained ancillas and joint decoding (Theorem 16).

An interface can be queried while it carries unknown quantum content. A coherent controlled-parity echo protects that content; a mixed port produces a binary record, and a finite Gibbs bank renews the active oscillator while retaining its spent outputs. Collection loss, control and readout have separate charges. The detailed resonant, packet and finite-band collection constructions are grouped in Appendix A: they support the required access rather than define the main control–propagation question.

The outgoing stress is the matter source used in HAFF-D [29]; the feedback task realizes the supplied flip of the Imprint Chain [36]. These are interfaces between stated models. The chiral field, spatial propagation carrier and external clock are not generated by the persistence calculation. Sections II–III state its algebraic obligations; Sections IV–V establish the physical coupling and the conditional bridge; the energy, persistence and inquiry results then describe what that interface can do at finite resources.

II. PERSISTENCE AS A FEEDBACK PROBLEM

Consider $\mathcal{H}_{\text{tot}} = \mathcal{H}_R \otimes \mathcal{H}_E$ with reduced dynamics

$$\dot{\rho}_R = -i[H_{\text{eff}}, \rho_R] + \mathcal{D}[\rho_R]. \tag{1}$$

The controls must be specified in the composed system dynamics, rather than inferred from a formal stationarity condition on a protocol cost.

Standard quantum control writes $H_{\text{ctrl}} = \sum_k u_k(t) G_k$ with generators of a Lie algebra $\mathfrak{g}$, $[G_i, G_j] = i f_{ijk} G_k$ [14]. The Lie algebra fixes the accessible directions. The size of a correction is quantified by a positive cost, norm or information metric supplied separately. This separation is standard and is not a defect of Lie-algebraic control. Section IV shows where such a cost can come from physically: from the stress of the field that implements the control.

III. WHAT THE OPERATOR ALGEBRA MUST SUPPLY

A. Associativity as a boundary of the model

Remark 1 (Associativity as a control-model boundary)**.** *The master-equation model represents sequential controls by linear maps under ordinary composition, which is associative by definition [1]. This is a declared scope condition, not a theorem excluding every non-associative dynamics. Jordan or octonionic models would need additional bracketing, subsystem and composition rules before a control calculation could be transferred [12, 13]. Q-RAIF Paper A [30] uses the same operator-composition boundary for propagation.*

B. A positive metric for Lyapunov descent

For an auxiliary gradient controller on parameters λ consider the descent rule

$$\dot{\lambda} = -\Gamma\, G^{-1} \nabla_\lambda V, \qquad \Gamma > 0. \tag{2}$$

Proposition 2 (Positive descent metric)**.** *The controller contribution to $\dot{V}$ under Eq. (2) is $-\Gamma(\nabla_\lambda V)^T G^{-1} (\nabla_\lambda V)$. It is nonpositive for every gradient exactly when G^{-1}, equivalently the invertible symmetric matrix G, is positive definite. If G has a negative direction, a gradient along it makes the contribution positive.*

Proof. The first statement is the chain rule. A symmetric quadratic form is nonnegative on all vectors precisely when it is positive semidefinite, and an invertible positive semidefinite matrix is positive definite. □

The descent mechanism therefore selects positive definiteness, not Lorentzian signature. Channel types such as unitary, dephasing and dissipative can be labelled by a discrete index, a direct-sum decomposition or supplied weights; their qualitative difference does not need a negative norm.

Parameter descent is not yet the physical state equation. With drift H_0, dissipator $\mathcal{D}$ and controls $H_{\text{ctrl}}(\lambda) = \sum_i u_i(\lambda) G_i$, the closed loop is

$$\dot{\rho} = -i[H_0 + H_{\text{ctrl}}(\lambda), \rho] + \mathcal{D}[\rho], \qquad \dot{\lambda} = -\Gamma G^{-1} \nabla_\lambda V, \tag{3}$$

and for differentiable $V(\rho, \lambda)$

$$\dot{V} = D_\rho V[-i[H_0 + H_{\text{ctrl}}, \rho] + \mathcal{D}[\rho]] - \Gamma(\nabla_\lambda V)^T G^{-1} (\nabla_\lambda V). \tag{4}$$

A stability proof must control the first term as well. Setting a control amplitude equal to a gradient component gives no general sign.

Dissipative and coherent control can stabilize target states with positive costs and Lie or operator-algebraic controls [15–18]. Von Neumann and C^*-algebras are compatible with this model; neither supplies a preferred quadratic form without a dynamics. These examples also show why persistence alone imposes no indefinite Clifford structure.

C. Clifford presentation and a physical criterion

If one chooses a quadratic space (V, q) and requires control generators to satisfy $v^2 = q(v)\mathbf{1}$, the universal property identifies $Cl(V, q)$ as the minimal algebra for that presentation [11]. This is a definitional fact. It does not show that persistence, channel discrimination or Lyapunov descent imposes those relations. Pauli controls of a qubit satisfy $\{\sigma_i, \sigma_j\} = 2\delta_{ij}\mathbf{1}$, the relations of $Cl(3, 0)$, so the Clifford package is a convenient description of qubit controls. A particular feedback model may favour some representation because of its controls and costs, but that preference has to be shown in its closed-loop dynamics. Proposition 6 below gives a different, explicit criterion: the centered-input switching charge must be independent of the unknown carrier state for every command. That requirement is stronger than positivity or persistence.

a. State stability. Equation (4) is the continuous-time obligation: a parameter descent controls only its own contribution, while drift, dissipator and controlled state evolution remain. For full-rank states and a fixed full-rank target, $\dot{D}(\rho\|\rho_*) = \mathrm{tr}[\dot{\rho}(\log\rho - \log\rho_*)]$. Theorem 13 establishes stability instead for a discrete composed channel with a physically realized, imperfect correction.

IV. A CONTROL FLIP REALIZED BY A FIELD COUPLING

A. The coupling and its channel

The field-coupling and drive bounds in this and the next section use $\hbar = 1$: $\bar{\omega}N$ and $\bar{\omega}\lambda^2$ are energies in these units. In physical units they are $\hbar\bar{\omega}N$ and $\hbar\bar{\omega}\lambda^2$; the field Hamiltonian and stress carry the same overall $\hbar$. Quadratures, occupation and pulse areas are dimensionless. The later delay, entropy and retained-oscillator accounts display $\hbar$ explicitly.

The controlled system is a two-level system with zero free Hamiltonian, Pauli operators X and Z. A bosonic mode with $[b, b^\dagger] = 1$ and quadratures $Q = b + b^\dagger$, $P = i(b^\dagger - b)$, $[Q, P] = 2i$, interacts through

$$U = \exp[-i\lambda XQ], \qquad \lambda > 0. \tag{5}$$

In the eigenbasis $X|x\rangle = x|x\rangle$, $U = \sum_x |x\rangle\langle x| \otimes e^{-i\lambda xQ}$, and $e^{-i\lambda xQ}$ is the displacement $D(-i\lambda x)$. Write $\chi_\sigma(k) = \mathrm{tr}(\sigma e^{-ikQ})$ for the characteristic function of Q in a mode state σ.

Proposition 3 (Channel of the coupling)**.** *Let the mode start in a state σ independent of the bit. The bit channel multiplies the matrix element $\rho_{xx'}$ in the X basis by $\chi_\sigma(\lambda(x - x'))$. For bit states diagonal in Z, its output Z statistics agree with a flip having failure probability*

$$\epsilon = \tfrac{1}{2}\big(1 + \mathrm{Re}\,\chi_\sigma(2\lambda)\big). \tag{6}$$

For a coherent input $|\alpha\rangle$, $\alpha > 0$, the factor is

$$e^{-\lambda^2(1-xx')-2i\alpha\lambda(x-x')}. \tag{7}$$

If $2\lambda\alpha = \pi/2$ and $N = \alpha^2$, the channel on every bit state is

$$\mathcal{F}_N(\rho) = s_N X\rho X + (1 - s_N)\rho, \qquad s_N = \frac{1 + e^{-2\lambda^2}}{2}, \qquad \lambda^2 = \frac{\pi^2}{16N}, \tag{8}$$

so that $\epsilon_N = 1 - s_N = \big(1 - e^{-\pi^2/(8N)}\big)/2$.

Proof. Tracing the mode gives the factor $\mathrm{tr}(\sigma\, e^{i\lambda x'Q}e^{-i\lambda xQ}) = \chi_\sigma(\lambda(x - x'))$. A Z-diagonal state $(I + zZ)/2$ has $\rho_{+-} = z/2$ in the X basis, so the output mean of Z is $z\,\mathrm{Re}\,\chi_\sigma(2\lambda)$, which is $(1-2s)z$ for a flip with success probability s. For the coherent input, $D(\beta)D(\alpha) = e^{(\beta\alpha^* - \beta^*\alpha)/2}D(\alpha + \beta)$ gives $e^{-i\lambda xQ}|\alpha\rangle = e^{-i\alpha\lambda x}|\alpha - i\lambda x\rangle$. The overlap is the coherent-state inner product $\langle\alpha - i\lambda x'|\alpha - i\lambda x\rangle = e^{-\lambda^2(1-xx')-i\alpha\lambda(x-x')}$ times the Weyl phase $e^{-i\alpha\lambda(x-x')}$ from composing the displacements; this is Eq. (7). At $2\lambda\alpha = \pi/2$ the off-diagonal factor is $-e^{-2\lambda^2}$, and $1 - 2s_N = -e^{-2\lambda^2}$ yields Eq. (8) on all inputs. □

The channel $\mathcal{F}_N$ differs from the ideal Ad_X by $\epsilon_N(\mathrm{id} - \mathrm{Ad}_X)$, so its half diamond-norm distance from the ideal flip is ϵ_N, and its average gate infidelity is $2\epsilon_N/3$. These are different error conventions. The coherent family above needs infinite occupation for an exact flip; Section VI shows that this holds for every input state.

B. A local switch on a free field

Equation (5) becomes a local operation once the mode is defined by a switch. Take a right-moving free massless scalar in $1+1$ dimensions with null coordinate u,

$$\phi(u) = \int_0^\infty \frac{d\omega}{\sqrt{4\pi\omega}} \big(a_\omega e^{-i\omega u} + a_\omega^\dagger e^{i\omega u}\big), \qquad [a_\omega, a_{\omega'}^\dagger] = \delta(\omega - \omega'), \tag{9}$$

current $j = \partial_u \phi$ and stress $T_{uu} =: j^2 :$. The current obeys $[j(u), j(v)] = \frac{i}{2}\delta'(u-v)$. For a real smooth compactly supported switch g define

$$Q_g = \int g(u)\, j(u)\, du = \lambda(b + b^\dagger), \qquad \lambda^2 = \int_0^\infty \frac{\omega}{4\pi} |\hat{g}(\omega)|^2 d\omega, \qquad \hat{g}(\omega) = \int g(u) e^{-i\omega u} du, \tag{10}$$

where $b = \int_0^\infty f^*(\omega) a_\omega d\omega$ with $f(\omega) = i\sqrt{\omega/(4\pi)}\, \hat{g}^*(\omega)/\lambda$ is a normalized mode. Its mean frequency is $\bar{\omega} = \int \omega |f|^2 d\omega$, and the current contains it as $j(u) = \psi(u) b + \psi^*(u) b^\dagger + j_\perp(u)$ with $\psi(u) = -i \int_0^\infty \sqrt{\omega/(4\pi)}\, f(\omega) e^{-i\omega u} d\omega$.

A probe at fixed position with interaction Hamiltonian $g(t) X j(t)$ evolves by $U = e^{i\vartheta} \exp(-iXQ_g)$ for a real phase ϑ. The reason is that $[g(t)Xj(t), g(t')Xj(t')]$ is a multiple of the identity, because $X^2 = 1$ and the current commutator is a c-number, so the Magnus series stops at second order and the second-order term is a phase. This uses three conditions: a single probe, a fixed coupling operator X, and a probe free Hamiltonian that is zero or commutes with X. Nonzero field frequencies are harmless. A probe Hamiltonian that does not commute with X makes $X(t)$ time dependent, and several probes keep relative phases; neither case is covered. A point-like limit of the switch needs regularization, so we use smooth profiles. This exact Weyl evolution of a degenerate detector is a known method [21, 22]; what we add is its use as a priced feedback actuator.

C. Stress, work and a derived quadratic cost

Proposition 4 (Stress of one encounter)**.** *For the unitary $U = \exp(-iXQ_g)$,*

$$U^\dagger j(u) U = j(u) + \tfrac{1}{2} X g'(u), \qquad U^\dagger T_{uu}(u) U = T_{uu}(u) + g'(u)\, Xj(u) + \tfrac{1}{4} g'(u)^2. \tag{11}$$

The expected added stress is $g'(u)\langle Xj(u)\rangle + g'(u)^2/4$. The cross term vanishes if the bit and the fresh field input are initially independent and $\mathrm{tr}(\rho X) = 0$, or if their joint state has no coherence in the bit's Z basis. A Z-diagonal bit marginal alone is not sufficient.

Proof. $[Q_g, j(u)] = \int g(v)[j(v), j(u)] dv = -\frac{i}{2} g'(u)$ is a c-number, so $e^{iXQ_g} j(u) e^{-iXQ_g} = j(u) + \frac{1}{2} X g'(u)$ exactly. Squaring, using $X^2 = 1$, and noting that a c-number shift leaves normal ordering unchanged gives the stress identity. Under independence $\langle Xj\rangle = \mathrm{tr}(\rho X)\langle j\rangle = 0$, even when the input beam has $\langle j\rangle \neq 0$. If the joint state is block diagonal in the bit's Z basis, $X \otimes A$ has zero expectation for every field operator A. For the last claim, $(|{\uparrow}\rangle|0\rangle + |{\downarrow}\rangle|1_b\rangle)/\sqrt{2}$ has bit marginal $I/2$ but $\langle Xb\rangle = 1/2$, hence $\langle Xj(u)\rangle = \mathrm{Re}\,\psi(u) \neq 0$; with a relative phase i the integrated cross term equals $-\lambda\bar{\omega}$ and changes the work below. □

Under the stated joint condition the added stress is

$$F_g(u) = \tfrac{1}{4} g'(u)^2 = 4\lambda^2 [\mathrm{Im}\,\psi(u)]^2, \tag{12}$$

the second form following from $2\lambda\,\mathrm{Im}\,\psi = g'/2$. If the interaction is off before and after the encounter and the bit Hamiltonian is zero, the energy delivered by the external switching agent equals the change of field energy,

$$W_{\mathrm{switch}} = \frac{1}{4} \int g'(u)^2 du = \frac{1}{4\pi} \int_0^\infty \omega^2 |\hat{g}(\omega)|^2 d\omega = \lambda^2 \bar{\omega}, \tag{13}$$

by Parseval's identity and $|f|^2 = \omega |\hat{g}|^2/(4\pi\lambda^2)$. This holds for every input state of the field that satisfies the joint condition.

Proposition 5 (Derived quadratic cost)**.** *For $g = \sum_i a_i g_i$ with real smooth compactly supported g_i, $W_{\mathrm{switch}} = \sum_{ij} a_i K_{ij} a_j$ with $K_{ij} = \frac{1}{4}\int g_i' g_j' du$. If the g_i are linearly independent, K is positive definite.*

Proof. $a^T K a = \frac{1}{4}\int (\sum_i a_i g_i')^2 du$ vanishes only if $\sum_i a_i g_i$ is constant; compact support makes the constant zero, and linear independence gives $a = 0$. □

D. Coherent drive

Let the mode b carry a coherent state $|\alpha\rangle$, $\alpha = \sqrt{N}$, with all orthogonal modes in vacuum, independent of a Z-diagonal bit. The output field marginal is then

$$\rho_F^{\text{out}} = \tfrac{1}{2}|\alpha - i\lambda\rangle\langle\alpha - i\lambda| + \tfrac{1}{2}|\alpha + i\lambda\rangle\langle\alpha + i\lambda|, \tag{14}$$

independently of the bit's bias, although bit and field are correlated afterwards. The incoming field energy is $E_{\text{in}} = N\bar{\omega}$ and the outgoing energy is $(N+\lambda^2)\bar{\omega}$. A wave packet is not an eigenmode of the field Hamiltonian, so these are mean values; the field Hamiltonian is not $\bar{\omega}b^\dagger b$. The incoming stress is $4N[\operatorname{Re}\psi(u)]^2$ and the encounter adds F_g of Eq. (12). The gravitational use of these profiles is developed in Ref. [29].

V. CONTROL STRUCTURE AND PROPAGATION

A. State-independent charging selects scalar squares

The fixed Pauli coupling can be generalized without changing the field. Let $A(a) = \sum_i a_i A_i$ be a real linear family of dimensionless Hermitian operators on a finite carrier. During one encounter a is fixed, and the carrier free Hamiltonian vanishes or commutes with $A(a)$. The switch g is fixed, nonzero, smooth and compactly supported. The input field is independent of the carrier and centered, $\langle j(u)\rangle = 0$, with finite energy; the vacuum is one example. Put $\kappa_g = \frac{1}{4}\int g'(u)^2 du > 0$, in the $\hbar = 1$ convention of Section IV.

Proposition 6 (Content-independent switching work)**.** *For the interaction $g(t)A(a)j(t)$ the field shift and switching work are*

$$U^\dagger j(u) U = j(u) + \tfrac{1}{2}A(a)g'(u), \tag{15}$$

$$W_\rho(a) = \kappa_g \operatorname{tr}\big[\rho A(a)^2\big]. \tag{16}$$

The following conditions are equivalent:

1. *$W_\rho(a)$ is independent of the carrier density matrix ρ for every command a;*
2. *$A(a)^2 = Q_{\text{ctrl}}(a)I$ for every a;*
3. *for a real symmetric matrix Q_{ctrl}, $\{A_i, A_j\} = 2(Q_{\text{ctrl}})_{ij}I$.*

The form is positive semidefinite, and positive definite if $a \mapsto A(a)$ is injective. In that case the generators give a representation of its Euclidean Clifford algebra.

Proof. The field commutator is a c-number. The second Magnus term is $i\vartheta A(a)^2$ and all higher terms vanish, since $A(a)^2$ commutes with $A(a)$ and with the field. Thus $U = e^{i\vartheta A(a)^2}e^{-iA(a)Q_g}$. The first factor need not be a global phase, but it commutes with field observables. The Weyl commutator calculation of Proposition 4 gives the stated shift and

$$U^\dagger T_{uu} U - T_{uu} = g'(u)A(a)j(u) + \tfrac{1}{4}g'(u)^2 A(a)^2.$$

Independence and centering remove the cross term for every ρ. The carrier free energy is unchanged and the interaction is off at the endpoints, so integration gives Eq. (16). A Hermitian operator has the same expectation in every density matrix exactly when it is a scalar multiple of I. This proves the first equivalence. Expanding $A(a)^2$ and comparing polynomial coefficients gives the second. Finally $Q_{\text{ctrl}}(a)I = A(a)^2 \geq 0$, and equality implies $A(a) = 0$. □

The all-state condition matters. For $A(a) = a_1 Z + a_2 I$ a maximally mixed carrier has work $\kappa_g(a_1^2 + a_2^2)$, although $A(a)^2 = (a_1^2 + a_2^2)I + 2a_1 a_2 Z$ is not scalar. The two Z eigenstates instead cost $\kappa_g(a_1 \pm a_2)^2$. Thus a positive average charge does not impose Clifford relations. Correlated inputs or a noncentered field introduce the cross term in the proof and need their own account. In particular this centered-input calibration is not a claim of state-independent work for every coherent drive used to rotate the carrier; nor does it guarantee a useful unitary gate.

B. A calibrated bridge to the spatial shell

Write a propagation symbol on the same internal carrier as $J(p) = \sum_i p_i \alpha_i$, with linearly independent Hermitian α_i and $J(p)^2 = Q_{\mathrm{prop}}(p) I$. The scalar-square argument makes Q_{prop} positive definite. Here J has energy units and the spatial carrier is supplied. Let C be a calibrated linear map from commands to momenta and $E_* > 0$ a fixed energy scale. Coupling the actual control through $A(a) = J(Ca)/E_*$ is a physical alignment condition, not a relabelling of two unrelated quadratic forms.

Corollary 7 (Aligned work and propagation)**.** *Under the encounter assumptions of Proposition 6 and this alignment,*

$$W(a) = a^T K_{\mathrm{op}} a, \qquad K_{\mathrm{op}} = \frac{\kappa_g}{E_*^2}\, C^T Q_{\mathrm{prop}} C. \tag{17}$$

Conversely, if C is invertible and this aligned family satisfies the all-state charge condition, it imposes a scalar quadratic shell on J. If in addition a supplied mass generator satisfies $\beta^2 = I$ and $\{\beta, \alpha_i\} = 0$, then

$$[\,\omega\beta - \beta J(p)\,]^2 = [\,\omega^2 - Q_{\mathrm{prop}}(p)\,] I. \tag{18}$$

Proof. Substitution in Eq. (16) proves the work identity. An invertible C ranges over all momenta, so the converse is the scalar-square criterion. For the last identity, $\beta J = -J\beta$ cancels the cross terms and $(\beta J)^2 = -J^2$. □

In Eq. (18), ω denotes energy conjugate to the independently supplied time, with $\hbar = 1$. The relative sign comes from propagation and the mass anticommutator, not from turning a positive work metric into an indefinite one. This is the closed-carrier symbol map of Q-RAIF C [31] with an explicit calibrated cost attached to it.

A nonempty case uses $\alpha_1 = X$, $\alpha_2 = Z$ and $\beta = Y$. In units $E_* = 1$, $Q_{\mathrm{prop}}(p) = p_1^2 + p_2^2$. For $C = \mathrm{diag}(1, 2)$, $K_{\mathrm{op}}/\kappa_g = \mathrm{diag}(1, 4)$ while the propagation shell is unchanged. The unknown calibration cannot be inferred from positivity of the cost. During the encounter the carrier must still obey the commuting-free-Hamiltonian condition. A general massive propagation Hamiltonian does not commute with $A(a)$; pausing or compensating it requires physical control and a separate cost. The example therefore supplies two compatible, calibrated uses of the carrier, not an autonomous spacetime-generating device.

C. Work, descent and spacetime

The descent metric G in Proposition 2 prices changes of controller parameters. Proposition 5 derives a work form K on switch profiles; Eq. (17) instead prices operator amplitudes for one fixed switch. These are different ways to vary a coupling. If controller parameters are the amplitudes being priced and the corresponding work form is positive definite, choosing it as their descent metric makes $-K^{-1}\nabla V$ the steepest direction at fixed quadratic charge, by Cauchy–Schwarz in the K inner product. That is a design choice; the feedback of Section VII has a stability proof without any gradient metric.

The new restriction is not equality of all three forms. It is the testable relation (17), under known coupling, calibration and input conditions. Its failure rejects that aligned model. Without those conditions, persistence supplies no route from a positive cost to a spatial shell or to Lorentzian signature.

VI. THE ENERGY COST OF AN ACCURATE FLIP

How much drive is needed for a flip of given accuracy? The coherent family gives $N = \pi^2/(16\ell)$ with $\ell = -\frac{1}{2}\log(1 - 2\epsilon)$, about $\pi^2/(16\epsilon)$. The following theorem bounds every input state.

Theorem 8 (Energy–accuracy bound)**.** *Let σ be any state of the mode b with finite mean occupation $N = \mathrm{tr}(\sigma b^\dagger b)$, let $\lambda > 0$, and let the coupling* (5) *flip Z-diagonal bit states with failure probability $\epsilon \in (0, \frac{1}{2})$ as in Eq.* (6)*. With $\theta_\epsilon = \arccos(2\epsilon - 1) \in (\pi/2, \pi)$,*

$$N \geq \frac{\theta_\epsilon(1 - 2\epsilon)}{4\sqrt{\epsilon(1 - \epsilon)}} - \frac{1}{2}, \tag{19}$$

$$N + \lambda^2 \geq \frac{\theta_\epsilon}{4\sqrt{\epsilon(1 - \epsilon)}} - \frac{1}{2}. \tag{20}$$

Both right-hand sides equal $\pi/(4\sqrt{\epsilon}) - 1 + O(\sqrt{\epsilon})$. The second inequality is a bound on an occupation-plus-activation budget. It equals the outgoing single-mode energy in units of $\bar{\omega}$ only with the independent zero-cross-term field inputs of Section IV C; correlations can change the actual switching work. An exact flip, $\epsilon = 0$, is impossible at finite occupation.

Proof. Write $C = \cos(2\lambda Q)$ and $U_2 = e^{-2i\lambda Q}$, so that $\langle C\rangle = \operatorname{Re}\chi_\sigma(2\lambda) = 2\epsilon - 1$ and $b^\dagger b = (Q^2 + P^2 - 2)/4$. Finite N makes both second moments finite.

Position quadrature. The function $G(c) = \arccos^2 c$ is convex on $[-1, 1]$: with $c = \cos y$, $y \in (0, \pi)$, one finds $G''(c) = 2(1 - y\cot y)/\sin^2 y \geq 0$, since $y\cot y \leq 1$. For every real q, $(2\lambda q)^2 \geq G(\cos 2\lambda q)$, with equality when $|2\lambda q| \leq \pi$ and strict inequality otherwise because $G \leq \pi^2$. Jensen's inequality for the spectral distribution of Q gives

$$\langle Q^2\rangle \geq \frac{G(\langle C\rangle)}{4\lambda^2} = \frac{\theta_\epsilon^2}{4\lambda^2}. \tag{21}$$

Momentum quadrature. Since $U_2^\dagger P U_2 = P - 4\lambda$, we have $[U_2, P] = 4\lambda U_2$ on the domain of P. Writing $\delta U_2 = U_2 - \langle U_2\rangle$, $\delta P = P - \langle P\rangle$ and applying Cauchy–Schwarz to the Hilbert–Schmidt products $\operatorname{tr}(\sigma\,\delta U_2\,\delta P)$ and $\operatorname{tr}(\sigma\,\delta P\,\delta U_2)$ gives $|\langle [U_2, P]\rangle| \leq 2\,\Delta U_2\,\Delta P$ with $(\Delta U_2)^2 = 1 - |\langle U_2\rangle|^2$. Hence $\Delta P \geq 2\lambda|\chi|/\sqrt{1 - |\chi|^2}$ with $\chi = \chi_\sigma(2\lambda)$. Because $|\chi| \geq |\operatorname{Re}\chi| = 1 - 2\epsilon$ and $x/\sqrt{1 - x^2}$ increases,

$$\langle P^2\rangle \geq (\Delta P)^2 \geq \frac{\lambda^2(1 - 2\epsilon)^2}{\epsilon(1 - \epsilon)}. \tag{22}$$

In particular $|\chi| < 1$ at finite N, which excludes $\epsilon = 0$.

Combination. Adding the two bounds,

$$N \geq \frac{1}{4}\left[\frac{\theta_\epsilon^2}{4\lambda^2} + \frac{\lambda^2(1 - 2\epsilon)^2}{\epsilon(1 - \epsilon)}\right] - \frac{1}{2}, \tag{23}$$

and minimizing over λ by the arithmetic–geometric mean inequality gives Eq. (19). Adding λ^2 changes the coefficient of λ^2 to $\frac{1}{4}[(1 - 2\epsilon)^2/(\epsilon(1 - \epsilon)) + 4] = 1/[4\epsilon(1 - \epsilon)]$, and the same step gives Eq. (20). The expansion uses $\theta_\epsilon = \pi - 2\sqrt{\epsilon} + O(\epsilon^{3/2})$. □

The bound is nearly attained by Gaussian drives.

Proposition 9 (Coherent and Gaussian families)**.** *(a) The coherent family with $2\lambda\alpha = \pi/2$ needs $N = \pi^2/(16\ell)$, $\ell = -\frac{1}{2}\log(1 - 2\epsilon)$. (b) For a displaced pure Gaussian state whose Q has mean 2α and variance $V > 0$, $\chi(2\lambda) = e^{-4i\lambda\alpha}e^{-2\lambda^2 V}$ and $N = \alpha^2 + (V + V^{-1} - 2)/4$. At $2\lambda\alpha = \pi/2$ and failure ϵ,*

$$\min N = \frac{\sqrt{1 + \pi^2/(4\ell)} - 1}{2}, \qquad \min(N + \lambda^2) = \frac{1}{2}\sqrt{\Big(1 + \frac{\pi^2}{4\ell}\Big)(1 + 4\ell)} - \frac{1}{2}, \tag{24}$$

attained at $V = (1 + \pi^2/(4\ell))^{-1/2}$ and $V = [(1 + 4\ell)/(1 + \pi^2/(4\ell))]^{1/2}$ respectively. Both equal $\pi/(4\sqrt{\epsilon}) - \frac{1}{2} + O(\sqrt{\epsilon})$, so their gap to Theorem 8 tends to half a quantum.

Proof. The Gaussian characteristic function gives χ. The failure condition reads $\lambda^2 V = \ell$, so $\alpha^2 = \pi^2/(16\lambda^2) = aV$ with $a = \pi^2/(16\ell)$ and $N = (1 + 4a)V/4 + 1/(4V) - \frac{1}{2}$, minimized at $V = (1 + 4a)^{-1/2}$. Adding $\lambda^2 = \ell/V$ replaces $1/(4V)$ by $(1 + 4\ell)/(4V)$. □

Squeezing is what makes the better scaling possible.

Proposition 10 (Classical drives)**.** *If $\sigma = \int P(\beta)|\beta\rangle\langle\beta|\,d^2\beta$ with a probability density $P \geq 0$, then a flip with failure ϵ needs*

$$N \geq \frac{(\pi - \arccos(1 - 2\epsilon))^2}{16\ell} = \frac{\pi^2}{16\epsilon}\big(1 - 4\sqrt{\epsilon}/\pi + O(\epsilon)\big). \tag{25}$$

Proof. Each coherent component has $\chi_\beta(2\lambda) = e^{-2\lambda^2}e^{-4i\lambda\operatorname{Re}\beta}$, so $\int P\cos(4\lambda\operatorname{Re}\beta) = -(1 - 2\epsilon)e^{2\lambda^2}$. This must be at least -1, hence $\lambda^2 \leq \ell$. The convexity argument of Theorem 8 applied to $y = 4\lambda\operatorname{Re}\beta$ gives $16\lambda^2\int P(\operatorname{Re}\beta)^2 \geq (\pi - \arccos[(1 - 2\epsilon)e^{2\lambda^2}])^2 \geq (\pi - \arccos(1 - 2\epsilon))^2$. Finally $N = \int P|\beta|^2 \geq \int P(\operatorname{Re}\beta)^2$ and $\lambda^2 \leq \ell$. □

ϵ	Classical bound (25)	Coherent, $2\lambda\alpha = \pi/2$	Bound (19)	Grid variational	Gaussian half-turn (24)
0.1	3.496	5.529	1.165	1.847	1.904
0.05	8.589	11.709	2.278	2.921	2.958
0.01	53.53	61.07	6.742	7.314	7.330
0.001	591.7	616.2	23.80	24.32	24.33

TABLE I. Mean drive occupation needed for a flip with failure probability ϵ. Including one activation's switching work at $\epsilon = 0.01$: bound (20) 6.890, grid variational minimum 7.470, Gaussian half-turn 7.487, coherent 61.08. The grid minima change by less than 2×10^{-3} when the spacing is halved; this is not a certified global error bound.

At fixed λ, a constrained minimum can be certified by a supporting ground-state problem. If ψ_μ is the ground state of $b^\dagger b + \mu \cos(2\lambda Q)$ and $\langle C \rangle_{\psi_\mu} = 2\epsilon - 1$, then every σ with the same $\langle C \rangle$ satisfies $\mathrm{tr}(\sigma b^\dagger b) \geq \langle b^\dagger b \rangle_{\psi_\mu}$, because $\mathrm{tr}\, \sigma(b^\dagger b + \mu C)$ is at least the ground energy. Table I lists numerical variational minima on a position grid, further minimized over λ, alongside the analytic bounds and the explicitly attainable Gaussian half-turn family. Halving the grid checks resolution stability, not the continuous global minimum. The computed states are close to displaced squeezed states, while the coherent half-turn family uses substantially more occupation.

Remark 11 (Scope of the occupation)**.** *(i) For a multimode field state, Cauchy–Schwarz applied to $b = \int f^* a_\omega d\omega$ gives $\langle b^\dagger b \rangle \leq \omega_h^{-1} \langle H_F \rangle$ with $\omega_h^{-1} = \int |f|^2 \omega^{-1} d\omega$, so the field energy is at least ω_h times the bound* (19)*; under the zero-cross-term condition of Section IV C, $\omega_h \leq \bar{\omega}$ then makes field energy plus switching work at least ω_h times the bound* (20)*. (ii) The occupation is the drive level present at the encounter, not energy absorbed by the bit. The bit is degenerate; the beam leaves with energy $(N + \lambda^2)\bar{\omega}$ in the coherent case. Whether drive energy is lost depends on whether the outgoing beam is recovered. (iii) Squeezed inputs can have locally negative normal-ordered stress, and their preparation and phase reference are additional resources.*

Recovering the beam is not free of consequences. Two bits placed one after the other on the same chiral packet, with the same switch profile in u, experience $U = \exp[-i\lambda(X_1 + X_2)Q]$.

Proposition 12 (Shared drive correlates errors)**.** *For a shared drive each bit's marginal channel is the single-encounter channel. For two bits starting at $Z = +1$, $\langle Z_1 Z_2 \rangle = \frac{1}{2}[1 + \mathrm{Re}\, \chi(4\lambda)]$. For the coherent family of Proposition 3 this equals $(1 + e^{-8\lambda^2})/2$, which exceeds the product $\langle Z_1 \rangle \langle Z_2 \rangle = e^{-4\lambda^2}$ obtained with independent drives.*

Proof. In the X basis the joint element with indices $(x_1 x_2), (x_1' x_2')$ is multiplied by $\chi(\lambda(x_1 + x_2 - x_1' - x_2'))$; tracing one bit recovers Proposition 3. $Z_1 Z_2$ connects (x_1, x_2) with $(-x_1, -x_2)$, which gives arguments $\pm 4\lambda$ and 0. For the coherent family $\chi(4\lambda) = e^{-8\lambda^2} e^{-2\pi i}$, and $(1 + y^2)/2 > y$ for $0 < y < 1$. □

At $N = 16$ the single failure probability is 0.0371, and the probability that both flips fail is 0.00393, against 0.00138 for independent drives. Drive energy can be shared only at the price of correlated failures. The persistence analysis below uses a fresh drive at each opportunity.

VII. PERSISTENCE WITH A FINITE-ENERGY ACTUATOR

A. The task

We use the feedback task of the Imprint Chain [36] with the corrective flip replaced by the coupling of Section IV. The persistent subsystem is a bit restricted to Z-diagonal states, with value $W = \pm 1$ and target $W = +1$. A bath step flips W with probability $d \in (0, \frac{1}{2})$; put $c = 1 - 2d$. A reading prepares a fresh probe, copies W by a CNOT, rotates the probe by U_a with $a \in \{0, 1\}$, and measures it with effects $E_{y|s} = \frac{1}{2}(I + y\nu\sigma_s)$, $\sigma_0 = Z$, $\sigma_1 = X$, where the instrument label S is hidden and fixed. Then

$$\Pr(Y = y \mid W = w, S = s, a) = \begin{cases} (1 + yw\nu)/2, & a = s, \\ 1/2, & a \neq s. \end{cases} \tag{26}$$

A calibration stage selects a; its mismatch probability $q = \Pr(a \neq S)$ is fixed along a trajectory. In each task step the system reads Y, accepts feedback with an independent probability $h \in (0, 1]$, switches on the coupling if accepted and $Y = -1$, and then the bath acts. The loss is the occupation probability $L_t = \Pr(W_t = -1)$.

Two declarations fix the resource accounting. Each opportunity has its own drive, initially independent of the bit and of earlier records, and the outgoing field never returns to the controller's reading access. The drive is scheduled: the same incoming beam arrives at every opportunity and the command only decides whether g is switched on. On-demand emission of a whole packet would need a router and a timing scheme that we do not model.

B. Closed-loop stability

Theorem 13 (Persistence with a finite-energy actuator)**.** *Let each coupling act on the bit as the flip channel $sX\rho X + (1-s)\rho$, which is the case exactly when $\chi_\sigma(2\lambda)$ is real, for instance for the coherent and Gaussian families at $2\lambda\alpha = \pi/2$. Put $\Lambda = c(1-hs)$ and $B = chs\nu$, and let the task start from W uniform.*

(a) Conditional on $a = S$, the bit evolves by a fixed channel $\mathcal{T}$ on Z-diagonal states, $m \mapsto \Lambda m + B$ for the mean m of W. Conditional on $a \neq S$, $m \mapsto \Lambda m$.

(b) $\mathcal{T}$ has the unique fixed point $\rho_* = (I + m_* Z)/2$, $m_* = B/(1-\Lambda)$. The trace distance to ρ_* contracts by exactly Λ per step, and $D(\rho_t\|\rho_*)$ is nonincreasing and tends to zero.

(c) The stationary loss is

$$L_\infty(q) = \frac{1}{2} - \frac{(1-q)B}{2(1-\Lambda)}. \tag{27}$$

(d) The stationary rate of actual couplings is

$$p(q) = \frac{h}{2}\left[1 - \frac{(1-q)\nu B}{1-\Lambda}\right] = h\nu L_\infty(q) + \frac{h(1-\nu)}{2}, \tag{28}$$

of which a fraction s succeed. Over a window of T steps the expected number of couplings is $\frac{h}{2}\big[T - (1-q)\nu\frac{B}{1-\Lambda}\big(T - \frac{1-\Lambda^T}{1-\Lambda}\big)\big]$.

(e) For two calibrations with $q' < q$, $L_\infty(q) - L_\infty(q') = (q-q')B/[2(1-\Lambda)]$ and $p(q) - p(q') = h\nu[L_\infty(q) - L_\infty(q')]$.

Proof. With $a = S$ a reading has $\mathbb{E}[WY] = \nu$, and the coupling is attempted with probability $h(1-Y)/2$ and flips with probability s. Thus the mean after feedback is $m - 2s\,\mathbb{E}[Wh(1-Y)/2] = (1-hs)m + hs\nu$, and the bath multiplies it by c. With $a \neq S$, Y is independent of W and the mean after feedback is $(1-hs)m$. Fresh probes and fresh drives make each step the same channel on the bit marginal. Trace distance between Z-diagonal states is $|m-m'|/2$, which the affine map multiplies by $\Lambda < 1$. Data processing gives $D(\mathcal{T}\rho\|\rho_*) \leq D(\rho\|\rho_*)$, and $m_t \to m_*$ gives convergence. Starting from $m_0 = 0$, the branch means are $m_*(1-\Lambda^t)$ and 0, weighted by $1-q$ and q; this gives (c). The attempt probability is $h(1-\nu m)/2$ for $a = S$ and $h/2$ otherwise, which gives (d) after averaging and using $(1-q)m_* = 1 - 2L_\infty$. Part (e) follows from (c) and (d). □

Gate failure enters the recursion through hs, while the attempt rate keeps h: a failed gate uses the whole encounter. Stability here does not rely on a gradient rule or a metric G. It follows from the contraction of the composed channel, and the relative entropy to the stationary state is a Lyapunov function of the full closed loop, which is what Section III C 0 a required of a proof. Whether persistence is useful is a separate matter: the stationary polarization is positive only on the calibrated branch and grows with s.

Table II uses $\nu = 0.8$, $c = 0.9$, $h = 0.4$ and coherent drives. The calibrated column uses 32 passive two-read blocks, $q_{32} = 0.0404888$ (Section VII C). With an exact flip the calibrated loss would be 0.199631 [36].

N	ϵ_N	Loss, $q = \frac{1}{2}$	Loss, q_{32}	Rate p, $q = \frac{1}{2}$	Rate p, q_{32}	Added energy per opportunity
1	0.354394	0.360165	0.231654	0.155253	0.114129	0.070401
4	0.132699	0.348517	0.209300	0.151525	0.106976	0.016497
16	0.037104	0.344779	0.202127	0.150329	0.104680	0.004036
64	0.009546	0.343806	0.200259	0.150018	0.104083	0.001003

TABLE II. Stationary loss and coupling rate with a coherent drive of occupation N. The last column is $p\lambda^2$ at q_{32} in units of $\bar{\omega}$; the incoming energy $N\bar{\omega}$ is paid at every opportunity in addition. Reading, probe and calibration costs are not included.

C. Calibration with the same imperfect gate

The Imprint Chain separates calibration blocks by an exact randomization $\mathcal{R}(\rho) = \frac{1}{2}(\rho + X\rho X)$. With an imperfect gate, triggering it with probability $\frac{1}{2}$ leaves a fraction $1 - s$ of the bit's mean. Triggering with probability $1/(2s)$, which is at most one because $s \geq \frac{1}{2}$, and discarding the coin gives

$$\Big(1 - \frac{1}{2s}\Big)\rho + \frac{1}{2s}\big(sX\rho X + (1-s)\rho\big) = \tfrac{1}{2}(\rho + X\rho X) \tag{29}$$

exactly. This presumes that s is known. If the trigger probability is rounded to k binary digits, each randomization has flip probability within $s2^{-k-1}$ of $\frac{1}{2}$. Since $\mathrm{id} - \mathrm{Ad}_X$ has diamond norm 2, the record distributions of a protocol with $n+1$ randomizations differ from the exact ones by at most $(n+1)s2^{-k-1}$ in total variation, or 1.52×10^{-5} for $N = 16$, $n = 32$, $k = 20$.

Passive two-read blocks randomize, read, let the bath act and read again, without a feedback flip. With the compensated randomization they keep the Imprint Chain likelihood: the product of the two readings is positive with probability $p_{\mathrm{cal}} = (1 + c\nu^2)/2$ if $a = S$ and $\frac{1}{2}$ otherwise. After n blocks the optimal mismatch probability is

$$q_n = \frac{1}{2}\sum_{k=0}^{n}\binom{n}{k}\min\big\{p_{\mathrm{cal}}^k(1-p_{\mathrm{cal}})^{n-k}, 2^{-n}\big\}. \tag{30}$$

Active blocks, with a feedback flip after the first reading, change once the flip is imperfect. For $a = S$,

$$\Pr(y_1, y_2) = \tfrac{1}{4}\big[1 + c\nu^2 y_2\big(s + (1-s)y_1\big)\big], \tag{31}$$

and the pair is uniform for $a \neq S$. This law is obtained from the passive pair by an S-independent post-processing that keeps the first reading and relabels with probability s, so gate error creates no calibration information about the sensor. The success probability s itself is readable from the closed loop: in the persistence task the lag-one correlation of readings on gated steps differs from that on ungated steps by a term proportional to s, so the agent's own records determine s, the bath contraction and the squared gain without references, and never the sign [33]. Calibration with n blocks uses $2n$ readings and $n+1$ randomizations with an expected $(n+1)/(2s)$ couplings.

D. Records and drive in one acceptance condition

Proposition 14 (Joint record–drive condition)**.** *If $chs\nu > 0$, the stationary loss satisfies $L_\infty(q) \leq L_*$ exactly when*

$$q \leq 1 - \frac{(1-2L_*)(1-c+chs)}{chs\nu}. \tag{32}$$

If the right-hand side is negative, no amount of calibration achieves L_* with that actuator.

Proof. Rearrange Eq. (27) using $1 - \Lambda = 1 - c + chs$. □

Take $L_* = 0.2$. A coherent drive with $N = 1$ fails even with a known interface, whose loss is 0.220330. For $N = 4$, 16 and 64 the minimal numbers of passive blocks are 57, 36 and 33; since q_n is nonincreasing, checking that $n-1$ blocks fail and n succeed establishes the minimum. With 32 blocks fixed, Eq. (32) requires $s \geq 0.9943781$, that is $\epsilon \leq 0.0056219$. Coherent half-turn drives then need $N \geq 109.10$, classical drives need $N \geq 98.93$ by Proposition 10, and by Theorem 8 every single-mode drive acting as a flip channel needs $N \geq 9.390$ (and $N + \lambda^2 \geq 9.503$). The Gaussian half-turn family reaches 9.957 and the computed grid variational minimum is 9.945. These minima refer to the declared block protocol and feedback rule, not to every possible experiment.

Corollary 15 (Scaling of the persistence cost)**.** *Among drives acting as flip channels, holding the stationary loss within δ of its exact-flip value at a known interface requires a drive occupation per activation that grows as $\delta^{-1/2}$ for optimal drives and as δ^{-1} for classical drives.*

Proof. At $q = 0$, L_∞ is a smooth decreasing function of s with nonzero derivative at $s = 1$, so δ is proportional to ϵ to first order. Theorem 8 and Proposition 9 give occupation of order $\epsilon^{-1/2}$ for optimal drives, and Proposition 10 together with the coherent family gives order ϵ^{-1} for classical drives. □

E. Resource accounting

The resources of the task are kept as a vector. Each task opportunity uses one reading, one probe and CNOT, a gate coin, a bath interval and one incoming drive of occupation N, prepared whether or not it is used. Each actual coupling adds switching work $\lambda^2\bar{\omega}$ at rate p. For $N = 16$ and 32 blocks, calibration uses 64 readings and 33 randomization opportunities carrying incoming energy $528\bar{\omega}$, with an expected 17.14 couplings and switching work $0.6606\bar{\omega}$. Over a task window of 60 steps the expected number of couplings is 6.494. The occupations 61.07 and 7.33 at $\epsilon = 0.01$ are input levels, and 7.49 minimizes input plus one activation's work; none of these is the total energy of an experiment, which would also include reading, preparation, records and the switching apparatus.

F. Finite-energy response readout

An outgoing stress can change a boundary trajectory, which can in turn change signal delays. Consider the particular readout in which these delays are the only branch-dependent signal. Use one calibrated proper clock, nonnegative probe Hamiltonians H_j on separate ports, and a joint probe–ancilla state independent of the unknown branch. The ancilla may be correlated with the probes but is not acted on by the signal. In branch $k = 0, 1$ the probes undergo $U_k = \exp[-i\sum_j H_j\tau_{kj}/\hbar]$. Preparation and final joint measurement can be arbitrary. The charged probe input is $E = \sum_j \operatorname{Tr}(\rho H_j)$, above the declared zero levels.

Theorem 16 (Sharp signed-delay energy bound)**.** *Let* $\Delta\tau = \max_j |\tau_{1j} - \tau_{0j}| > 0$ *and define*

$$\zeta = \max_{x>0} \frac{1-\cos x}{x} = 0.724611\ldots, \qquad z = \frac{\zeta E \Delta\tau}{\hbar}. \tag{33}$$

The trace distance of the two output states, including the retained ancilla, is at most $\sqrt{z(2-z)}$ *for* $0 \le z \le 1$*, and at most one otherwise. For equal priors, error at most* $\epsilon < 1/2$ *therefore requires*

$$E \ge \frac{\hbar}{\zeta\Delta\tau}\left[1 - 2\sqrt{\epsilon(1-\epsilon)}\right]. \tag{34}$$

This lower bound is sharp across the permitted probe families, including oppositely signed delay differences. It need not be sharp for a specified Hamiltonian or a readout with all delay differences of the same sign. If $\Delta\tau = 0$*, no such protocol distinguishes the branches.*

Proof. The scalar maximum occurs at the unique $x_* \in (0, \pi)$ satisfying $\tan(x_*/2) = x_*$; for $x \ge \pi$, $(1-\cos x)/x \le 2/\pi$, whereas the value at $2\pi/3$ gives $\zeta \ge 9/(4\pi) > 2/\pi$. Thus $1 - \cos x \le \zeta|x|$ for every real x. Purify the probe–ancilla input. The overlap of the two resulting pure states is

$$A = \operatorname{Tr}\rho\exp(-iK), \qquad K = \sum_j H_j(\tau_{1j} - \tau_{0j})/\hbar.$$

The Hamiltonians commute strongly, and their positivity gives $|K| \le (\Delta\tau/\hbar)\sum_j H_j$ by joint spectral calculus. Consequently $\operatorname{Re} A \ge 1 - z$. For $z \le 1$, the pure-state trace distance is $\sqrt{1-|A|^2} \le \sqrt{z(2-z)}$; discarding the purifying reference cannot increase it. Equal-prior error is at least $(1-D)/2$ for trace distance D. Solving this inequality gives Eq. (34).

For saturation, take two probes with equal excited energy $e = x_*\hbar/\Delta\tau$, and delay differences $+\Delta\tau, -\Delta\tau$. On the three states $|00\rangle, |e0\rangle, |0e\rangle$ prepare

$$|\psi\rangle = \sqrt{1-p}\,|00\rangle + \sqrt{p/2}\,(|e0\rangle + |0e\rangle), \qquad p = \frac{1 - 2\sqrt{\epsilon(1-\epsilon)}}{1-\cos x_*}.$$

Here $0 \le p < 1$, $E = pe$, and the branch overlap is exactly $1 - p(1-\cos x_*) = 2\sqrt{\epsilon(1-\epsilon)}$. The optimal binary measurement attains error ϵ and Eq. (34). □

Mean-energy quantum speed limits constrain distinguishable evolution, including arbitrary-fidelity targets [34, 35]. Here the relevant resource is the sum of the physical probe energies, not the signed generator's ground-state shift; this accounts for opposite delay differences without assigning a negative probe cost. A readable branch marker, changed output frequency or amplitude, branch-dependent preparation, or a signal interaction other than translation

falls outside the theorem. For example, an unpriced orthogonal marker would distinguish the branches at zero probe energy. Receiver, clock and preparation costs remain additional resources.

In HAFF-D, three echoes determine the coefficients of a supplied JT solution. Its source-compensated inverse permits known stresses in both null directions and propagates timing and source-calibration errors to the next coupled projection [29]. This connects a selection-dependent source, its response and a subsequent operation, conditional on the declared instrument. For its compact-pulse example the largest proper-delay difference is 0.05001338; a translation-only instrument with error 0.1 requires input energy at least $11.0374\hbar$ in that clock's units. The geometric inverse does not construct such a probe or its reflecting boundary. Its supplied bulk source-compensation test is not an energy–resolution demonstration. A joint apparatus must meet the readout bound, the actuator bound and its actual stress and entropy accounts without counting the same outgoing energy twice.

G. Finite-rate entropy export

Drive occupation and exported heat obey different bounds. If the maintained subsystem and its finite reused controller remove a net dimensionless entropy current $\dot{s} > 0$, and all of it leaves through a chiral conformal exhaust of capacity c_{ex} above a thermal input at temperature T, HAFF-D gives [29]

$$\dot{Q} \geq k_B T \dot{s} + \frac{3\hbar}{\pi c_{\mathrm{ex}}} \dot{s}^2.$$

The current must include retained side information and controller consumption; the number of attempted flips alone does not determine it. The bound assumes an extensive long-window entropy-current account, zero chemical potential or the matched conserved-charge account, and one common clock. It is not an exact bound for every short packet. It remains positive at $T = 0$ for a thermal input, where ordinary Landauer heat is zero. For independent parallel exhausts at one temperature, c_{ex} is their total capacity.

A nonthermal drive changes this comparison. If its incoming energy and dimensionless entropy currents are e_{in} and σ_{in}, the same Gibbs argument gives

$$\dot{Q} + e_{\mathrm{in}} - E_{\mathrm{th}}(\sigma_{\mathrm{in}}) \geq k_B T_\sigma \dot{s} + \frac{3\hbar}{\pi c_{\mathrm{ex}}} \dot{s}^2, \qquad E_{\mathrm{th}}(\sigma) = \frac{3\hbar\sigma^2}{\pi c_{\mathrm{ex}}}, \quad T_\sigma = \frac{6\hbar\sigma_{\mathrm{in}}}{\pi c_{\mathrm{ex}} k_B}.$$

The incoming energy above its entropy-matched thermal value can pay for export. In particular, a coherent beam with zero entropy is not a zero-energy reservoir. Its drive occupation and the additional heat must be accounted for separately.

This heat is neither the incoming occupation $N\bar{\omega}$ nor an extra charge to add to an outgoing energy already counted. It is a constraint on that energy when it carries the removed entropy. The scheduled drive calculation above does not establish its exhaust-capacity assumptions. A complete apparatus must price reading, recovery, sensing and entropy transport in the same cycle.

VIII. CALIBRATING AN INTERFACE WHILE IT CARRIES QUANTUM CONTENT

Preparing test states before storage is not the only way to learn an interface. The outgoing field can carry information about the coupling without carrying the logical content that the apparatus is trying to keep. We give the instrument, its loss under incomplete collection, and a coherent implementation.

On the two-qubit code $\mathcal{C} = \mathrm{span}\{|01\rangle, |10\rangle\}$, let Z_L have eigenvalues $+1, -1$. The coupling $Z_1 + \alpha a Z_2$ restricts to mZ_L, $m = 1 - \alpha a$, with unknown $\alpha > 0$ and chosen command a. Write

$$Q_f = \sqrt{k}(b + b^\dagger), \qquad U_m = e^{-imZ_L Q_f}, \qquad k > 0, \quad \Pi = (-1)^{b^\dagger b}.$$

The independently prepared incoming mode is vacuum. A beam splitter collects fraction ξ of this mode, leaving fraction $1 - \xi$ inaccessible.

Proposition 17 (A content-preserving calibration instrument)**.** *Measure the collected mode's parity and apply Z_L for an odd outcome. The corrected instrument, including an arbitrary retained reference, is*

$$\mathcal{I}_\pm(\rho) = p_\pm(a)\mathcal{D}_{\xi,m}(\rho), \qquad p_\pm(a) = \frac{1 \pm e^{-2k\xi m^2}}{2}, \tag{35}$$

where $\mathcal{D}_{\xi,m}$ fixes the populations and multiplies logical coherence by $e^{-2k(1-\xi)m^2}$. At full collection each outcome is a scalar multiple of the identity channel. Fresh-port queries chosen from earlier records can therefore calibrate the interface without replacing its held state.

Proof. The accessible displacement has the expansion $I_L \cos(m\sqrt{\xi}Q_A) - iZ_L \sin(m\sqrt{\xi}Q_A)$. On the vacuum the two field vectors have opposite parity, with squared norms p_+, p_-. The conditional pulse makes both logical factors I_L. The independent inaccessible displacement commutes with this correction and leaves $\mathcal{D}_{\xi,m}$. Each branch is thus Eq. (35), not merely an equality of its unconditional channel. Composition proves the adaptive statement; the same computation tensors with any reference. □

Environment-assisted correction of unitary Kraus branches is a known general mechanism [3, Proposition 2]. Here its records are also the data for learning the same interaction's gain and collection fraction. Repeated ancilla-based parity measurements have been implemented for a cavity mode [5]; collecting the full mode of a travelling chiral field is an additional requirement, not supplied by that experiment.

Partial collection prices the calibration itself. For N queries at each of $a = \pm h$, with fixed α, ξ, the held coherence is multiplied by

$$\exp[-4Nk(1-\xi)(1+\alpha^2 h^2)]. \tag{36}$$

This loss occurs before any subsequent operating benefit. With $k = .2$, $\alpha = 1.1$, $h = .5$ and $N = 4096$, keeping the calibration's dephasing error below .01 needs $1 - \xi \le 4.7335 \times 10^{-6}$. At $\xi = .99$ its error is essentially $1/2$. More records are not automatically better for held content.

The collection fraction in the preceding instrument is realized by the resonant and finite-band protocols of Appendix A. Their finite-packet, waveform and renewal estimates retain the assumptions needed for actual access. The following echo concerns the resulting available mode.

Proposition 18 (Coherent parity echo)**.** *Suppose the relevant mode remains available for joint control. The bounded involution*

$$R = |0_L\rangle\langle 0_L| \otimes I + |1_L\rangle\langle 1_L| \otimes \Pi = I_L \otimes P_{\text{even}} + Z_L \otimes P_{\text{odd}}$$

satisfies, on the quadrature domain,

$$R(Z_L \otimes Q_f)R = I_L \otimes Q_f, \qquad RU_m R = I_L \otimes e^{-imQ_f}. \tag{37}$$

This protects the body's marginal for every mode input and even for an initially correlated body–mode state. For an independent vacuum input, the first R acts trivially; the final R alone disentangles the body and mode.

Proof. $\Pi Q_f \Pi = -Q_f$. In the $+1$ logical block the conjugated generator is Q_f, and in the -1 block it is $-\Pi Q_f \Pi = Q_f$. Functional calculus gives the unitary identity. A unitary on the mode alone cannot change the body's reduced state. On vacuum $\Pi|0\rangle = |0\rangle$. □

One qubit meter implements R coherently. Prepare it in $|0\rangle_C$, apply

$$M = \exp[-i(\pi/2)P_{\text{odd}} \otimes Y_C], \qquad C_Z = I_L \otimes |0\rangle\langle 0|_C + Z_L \otimes |1\rangle\langle 1|_C,$$

and then apply $M^\dagger C_Z M$, with the tensor identities understood. The meter computes parity, controls the body phase and uncomputes parity; it returns exactly to $|0\rangle_C$. Bounded Hamiltonians $g(t)P_{\text{odd}} \otimes Y_C$ and $\gamma(t)(I_L - Z_L)/2 \otimes |1\rangle\langle 1|_C$ suffice, with pulse areas $\pi/2$ and π. Amplitude bounds impose corresponding nonzero durations. No classical syndrome bit has to be erased in this coherent correction. Producing a retained classical calibration record is a separate operation. Pulse area is not a heat bill: the control field, capture, clock and support still need their physical energy account. Half-diamond errors of supplied primitive channels accumulate at most by their sum, by telescoping compositions; such a bounded-channel error alone does not bound an unbounded stress observable.

A retained oscillator offers a direct alternative to the spectral meter. In its interaction picture, with $n = b^\dagger b$ and $P_1 = |1_L\rangle\langle 1_L|$, the dispersive Hamiltonian $H_R(t) = \hbar\chi(t)P_1 \otimes n$ implements R when $\int \chi(t)dt = \pi$. Such number-controlled parity gates are established cavity operations [5, pp. 2–3]. For a degenerate logical body, $\chi \ge 0$ and oscillator energy $\hbar\omega n$, the total Hamiltonian is bounded below. If χ starts and ends at zero and the gate is isolated, $P_1 n$ is conserved, so its net prescribed-switch work is exactly zero:

$$W_R = \hbar \int \dot{\chi}\, \langle P_1 n\rangle dt = 0.$$

The oscillator's stored energy, gate duration and controller are not zero. The integrated expected interaction energy is $\pi\hbar\langle P_1 n\rangle$; $\chi \le \chi_{\max}$ gives duration at least $\pi/\chi_{\max}$. If the pulse area has error δ, its half-trace error, with any reference and an input occupation bound $\langle n\rangle \le \bar{n}$, is at most $\min\{1, \sqrt{2|\delta|\bar{n}}\}$. Indeed the relative unitary is $e^{-i\delta P_1 n}$, and the purified vector error squared is bounded by $2|\delta|\langle P_1 n\rangle$, using $1-\cos x \le |x|$. This is an energy-constrained error, not an unrestricted diamond bound. It gives an explicit retained-oscillator gate account; it does not price trapping a broadband travelling mode or operating the classical switch.

A. Thermal inquiry with a binary record and a finite bank

The echo gives an actual thermal inquiry protocol on the retained oscillator. Let its input be the centered Gibbs state τ_T of $H_b = \hbar\omega n$, with $n_T = (e^{\hbar\omega/(k_B T)} - 1)^{-1}$ and $\nu_T = 2n_T + 1$. This input is independent of the held body, its old reference and earlier records. Both parity gates in $RU_m R$ are used: the first is generally consequential on a thermal input. The result is $\rho_{L\mathrm{old}} \otimes D_m \tau_T D_m^\dagger$, where $D_m = e^{-im\sqrt{k}Q}$. Thus the body and its entire old reference are preserved, while the oscillator has

$$\langle P\rangle = -2m\sqrt{k}, \qquad \mathrm{Var}(P) = \nu_T, \qquad E_b - E_{b,\mathrm{in}} = \hbar\omega k m^2.$$

The displaced Gibbs state has the same entropy as τ_T.

A quadrature mean is not yet a saved record. Prepare a fresh degenerate qubit C in $\eta_v = (I + vZ_C)/2$, $0 < v \le 1$, independently of the oscillator and all earlier outputs. Couple it through

$$V_\kappa = e^{-i\kappa P \otimes Y_C}, \qquad \kappa > 0,$$

then measure X_C and store the outcome $Y = \pm 1$. The preparation, orientation, measurement and storage are declared operations.

Proposition 19 (Thermal binary inquiry)**.** *For this protocol the instrument on the held body and its old reference is*

$$\mathcal{J}_y(\rho_{L\mathrm{old}}) = \frac{1 + y\mu_T(m)}{2}\rho_{L\mathrm{old}}, \qquad \mu_T(m) = -ve^{-2\kappa^2\nu_T}\sin(4\kappa m\sqrt{k}). \tag{38}$$

The read interaction adds oscillator energy $\hbar\omega\kappa^2$ before conditioning on the record. If $m = \pm m_0$ with known $m_0 > 0$, $0 < \kappa \le \pi/(8m_0\sqrt{k})$ gives sign gain $\gamma_T = ve^{-2\kappa^2\nu_T}\sin(4\kappa m_0\sqrt{k}) > 0$. For J fresh trials, the sign estimate from $-\sum_j Y_j$ has error at most $e^{-J\gamma_T^2/2}$, including ties as errors.

Proof. The echo acts only on the independent oscillator, so all subsequent oscillator–port operations factor from $\rho_{L\mathrm{old}}$. Conjugation gives $V_\kappa^\dagger X_C V_\kappa = \cos(2\kappa P)X_C + \sin(2\kappa P)Z_C$. The Gibbs characteristic function is $\langle e^{itP}\rangle = e^{-t^2\nu_T/2}$, and displacement changes it by $e^{-2itm\sqrt{k}}$. Taking the qubit expectation proves Eq. (38). Also $V_\kappa^\dagger Q V_\kappa = Q + 2\kappa Y_C$, while P is fixed. Since the independent port has $\langle Y_C\rangle = 0$ and $Y_C^2 = I$, $n = (Q^2 + P^2 - 2)/4$ increases in mean by κ^2. Each sign-aligned binary variable has mean γ_T; Hoeffding's inequality gives the stated bound. Independence here follows from fresh inputs and the renewal below, rather than from reusing a disturbed mode. □

This record reads a sign that parity cannot resolve. Its orientation reference is an additional permission. The sine is periodic, so one setting does not identify arbitrary continuous m; an unknown initial quadrature mean also confounds it. If the mode initially shares correlations with the body, the echo still preserves the unconditional body state, but measuring the mode may steer it. That case does not give Eq. (38).

The pulse can have bounded amplitude and finite duration. In the laboratory frame choose a smooth real $g(t)$ of compact time support, with $\int g(t)dt = \kappa$, and set

$$H_{\mathrm{read}}(t) = \hbar\omega n + \hbar g(t)[P\cos(\omega t) + Q\sin(\omega t)] \otimes Y_C. \tag{39}$$

Its interaction-picture generator is exactly $\hbar g(t) P \otimes Y_C$. The final laboratory evolution includes a known free rotation; subsequent pulses and read axes use the same clock. Completing the square gives $H_{\mathrm{read}}(t) \ge -\hbar g(t)^2/\omega$. Adding the scalar $\hbar g(t)^2/\omega$ makes it nonnegative and changes only an overall phase; the scalar vanishes at both endpoints. For $|g| \le g_{\max}$ the duration is at least $\kappa/g_{\max}$. The net prescribed-switch work is the endpoint free-Hamiltonian energy increase $\hbar\omega\kappa^2$ in this isolated, degenerate-port model.

The echo's displacement admits the same finite pulse construction, with laboratory force $\hbar m\sqrt{k} f(t)[Q\cos(\omega t) - P\sin(\omega t)]Z_L$ and $\int f(t)dt = 1$. Completing the square bounds its Hamiltonian below by $-\hbar k m^2 f(t)^2/\omega$; a finite command range keeps this bound uniform. The source pulse supplies $\hbar\omega k m^2$ on the Gibbs input. The two parity gates have the finite dispersive realization above. These rotating profiles supply phase and clock access explicitly; their implementation by an autonomous controller is a further physical question.

Proposition 20 (Finite thermal renewal with retained outputs)**.** *Take a bank of J independent same-frequency oscillators in τ_T, independent mixed read ports, and storage for J binary records. After each echo and read, couple b to the next fresh bank oscillator through $\hbar h(t)(b^\dagger c + c^\dagger b)$ with area $\pi/2$. The active oscillator returns exactly to τ_T, independent of all earlier retained outputs. Its entire post-read state and correlations are transferred to the spent bank oscillator, up to known phase rotations. After summing over the new outcome, the entire old body/reference/history state is preserved, including for commands selected from earlier records. If $|h| \le h_{\max} < \omega$, the total contact Hamiltonian is bounded below and its duration is at least $\pi/(2h_{\max})$. A pulse off at both endpoints has zero net prescribed-switch work. For fixed m and κ, the spent oscillator's mean excess energy above its incoming Gibbs value is*

$$\Delta E_{\text{spent}} = \hbar\omega(km^2 + \kappa^2). \tag{40}$$

This is a finite J-trial realization of Eq. (38)*.*

Proof. The contact sends $b \mapsto -ic$, $c \mapsto -ib$. For any joint post-read state ρ_{bA}, where A contains the held body, old reference, port and records, the output is $\tau_{T,b} \otimes \rho'_{cA}$. Gibbs states are invariant under the known phase rotation. This identity transfers correlations instead of tracing them out. The two instantaneous normal frequencies are $\omega \pm h$. Total occupation and $b^\dagger c + c^\dagger b$ are conserved during the contact, which proves the stability and zero switch-work statements as in Proposition 21. The preceding two displacement accounts prove Eq. (40) after averaging over the retained record. For an old classical history r, the new record appends the kernel $p(y \mid m_r)$ to its existing conditional body/reference state. Summing over y leaves each old block, and hence their full joint state, unchanged. Induction proves renewal and this nonselective preservation for adaptive commands. Conditioning only on a new outcome can reweight old history blocks and change the reader's body/reference state; it need not preserve that conditional state. For a fixed unknown sign and fixed settings the new inputs give independent trials. □

For example, $k = .2$, $m_0 = .7$, $v = .6$ and $\kappa = \min\{\pi/(8m_0\sqrt{k}), 1/(2\sqrt{\nu_T})\}$ give sufficient trial counts 178, 247 and 593 for error .05 at $n_T = .2$, .5 and 2. The respective excess oscillator energies per trial, in units of $\hbar\omega$, are .276571, .223 and .148. Higher temperature weakens the record even though this choice reduces the read-pulse energy. A bank consumes one fresh Gibbs oscillator and one mixed read port per trial, retains the spent oscillator and a binary record, and pays both parity gates, the source/read profiles and the resonant contact duration. The initial active Gibbs oscillator is needed in addition to the J fresh bank inputs. Preparation, transport, readout, storage, orientation and the clock/support must be charged in addition.

Neither term in Eq. (40) is an entropy-export certificate. The source displacement preserves entropy; the read operation changes correlations with its port and record. Replenishing the bank or erasing spent outputs requires their joint state, including side information. This construction supplies repeated thermal inquiry for each finite bank. It does not supply the chiral capacity of Section VII, an infinite thermal line or the total covariant stress of a controller.

For a centered independent input the original outgoing source is $m^2 F_f$, $F_f = (f')^2/4$. The coherent echo leaves this coupling-added stress in the field while preserving content. Its implementation requires an available mode and controlled parity; it supplies neither a local capture apparatus for a whole continuous field nor a sustained thermal exhaust. The finite oscillator identity and the moving-field apparatus are different levels of the construction.

IX. RELATIONS TO OTHER WORK

A. Q-RAIF Papers A and C

Paper A [30] looks for the relative sign in translation-invariant local unitary propagation. When all bands share one quadratic shell, the propagation symbol obeys Euclidean Clifford relations. Its form $\eta = \omega^2 - Q(q)$ has one positive frequency direction and d negative spatial directions; the mass term together with the spatial generators yields $Cl(1, d)$. Reversing the overall metric convention reverses these signs, not their relative opposition. By Sylvester's law neither positive G nor K of Section V can be brought to that indefinite form.

Section V relates the spatial shell to measured work under explicit alignment, while keeping their domains and the indefinite spacetime form distinct. Paper C [31] asks what a coupling supplies in place of an assumed representation map. For the coupling XQ_g of this paper the algebra represented exactly is the commutant of X, for every drive state, and the defect on the remaining controls reproduces the energy–accuracy bound of Theorem 8.

The mass–energy paper [32] now uses the same quadrature interaction to realize a quantum-walk coin. With a fresh Gaussian drive each step, its mean angle fixes the rest resonance, while drive fluctuations set its decay rate. At input occupation N the smallest Gaussian rate is $\theta^2/[2\tau N(N+1)]$, attained by a displaced squeezed state. This connects the price of control to the persistence of a spectral readout. The rate is an open-channel linewidth, not a new spectral mass, and shift, preparation and port reset costs are separate.

B. Quantum control theory

Lie-algebraic control characterizes reachability through the algebra generated by drift and control Hamiltonians [14]. Stabilization against perturbations also needs magnitudes and costs. In the realized actuator these come from the physical coupling: the rotation angle from the drive amplitude, the error from the drive's quadrature fluctuations, and the cost from the field stress. A Clifford quadratic form is not needed to supply them.

C. Decoherence-free subspaces

Decoherence-free subspaces [19, 20] are protected passively by symmetry. The feedback task gives the active counterpart. The held-code instrument, coherent echo and finite collection protocols also join passive encoding to priced inquiry and correction: their exact identities protect a supplied code, while collection loss and gate error determine what survives. Brain Anchor [4] uses records of this interface for finite-window inquiry and maintenance decisions. Forming the code and its physical permission from a prior history is a further mechanism, not a consequence of encoding it.

D. The Imprint Chain and HAFF

The Imprint Chain [36] supplies the instrument, the calibration and the task, with an exact supplied flip. Replacing that flip by the field coupling raises the calibrated loss from 0.199631 to 0.202127 at $N = 16$ and makes the drive energy part of the feasibility condition. Each coupling also adds the stress F_g to the outgoing field. The HAFF gravity paper [29] uses this outgoing field as a matter source in a Jackiw–Teitelboim model; there, calibration changes the mean source through the coupling rate p of Eq. (28). Selecting the coupled projection changes its represented algebra and source even at fixed activation schedule. A boundary readout can then update the next projection, with known probe stresses and bounded errors included in the inverse. Section VII F supplies one energy constraint on a specified realization of that return.

E. Quantized control fields

The effect of field quantization on gate accuracy has a substantial literature. For Jaynes–Cummings-type couplings, conservation of excitation number leads to error bounds controlled by photon-number fluctuations, of order $1/\bar{n}$ for coherent fields [24]; coherent-field control limits for such models were analysed in Ref. [25], and simultaneous driving of several atoms by one quantized pulse in Ref. [26]. The coupling studied here is a quadrature coupling of a degenerate bit and conserves no excitation number, which is why squeezing changes the scaling from ϵ^{-1} to $\epsilon^{-1/2}$. Theorem 8 gives the quadrature-specific bound and explicit near-attainment under these assumptions. A covariant stress tensor for a finite detector, including its localization, is constructed in Ref. [23]; it shows why the apparatus stress cannot simply be omitted.

X. DISCUSSION

For gradient-type feedback the algebra must compose associatively and the descent metric must be positive. When control is realized by a field, the structure that limits persistence is the canonical commutation relation of the drive quadratures: $[Q, P] = 2i$ fixes the trade-off between flip accuracy and drive occupation in Theorem 8, and it is the reason squeezed drives do better than classical ones. For the fixed Pauli flip the commutation relation of the driving mode sets the accuracy cost. For a family of control operators, Proposition 6 additionally identifies when their quadratic structure fixes a state-independent centered-input work form. Persistence has a closed-loop proof in this model, with an explicit Lyapunov function, and the drive energy and calibration records enter one feasibility condition.

Several inputs remain supplied. The free chiral field, the degenerate bit, the clock and switch, the drive ports and the instrument family are given. The feedback task also supplies the success probability for compensated randomization and scheduled beams. Its analysis concerns Z-diagonal bit states and flip channels; complex $\chi_\sigma(2\lambda)$ instead rotates Z into Y. The held-content construction concerns arbitrary logical inputs. Its finite-band realization prepares and collects a specified packet, with a finite waveform, clock tolerances and independent vacuum exhausts. It returns a

coherent meter; the carrier, detector, port selection and vacuum supply remain separate inputs. The stress of the switch and its support is not computed; outgoing-field stress is used outside the apparatus.

The thermal bank identifies what renewal has to preserve. A fresh input restores the oscillator only because the previous state, including its record correlations, moves to a spent output. Bank replenishment and memory erasure must therefore be priced on that joint output, as in the information thermodynamic account [9, 10]. The chiral capacity bound constrains a specified entropy flow; the finite oscillator construction gives an implementable inquiry and transfer, rather than that flow's attainability.

Repeated propagation in one field remains a stronger problem. Proposition 12 proves that shared drives can preserve each marginal channel while correlating failures. A persistence calculation on that joint history may therefore have a different energy scaling from the fresh-drive task. The other unresolved connection is formation: which interactions and retained records produce the code, access permissions and controller used here? T-DOME, Imprint and Brain Anchor address that mechanism; the present results constrain the resources of the operations it must supply.

XI. CONCLUSION

The connection between control and propagation becomes precise only after a physical condition is specified. In the centered-field calibration studied here, a linear Hermitian control family has state-independent switching work exactly when it has a scalar quadratic square. A calibrated use of those same generators for propagation then gives a positive pullback relation between work and the spatial shell. The time direction and mass anticommutator enter separately. Persistence alone supplies none of these identifications.

The same field coupling gives a finite actuator whose stress, accuracy and drive energy can be calculated together. Squeezed inputs approach the $\epsilon^{-1/2}$ occupation bound, while classical drives require order ϵ^{-1}. Calibration and physical actuation enter a single feedback feasibility condition. Content-preserving inquiry, finite port renewal and the separate delay-readout bound extend the accounting to both directions of an interface. The remaining formation problem is which physical histories produce the aligned couplings, permissions and controller, rather than merely supplying them. The conditional bridge locates that question at the coupling itself; it does not replace it with the existence of a well-priced device.

Appendix A: Physical collection of the inquiry mode

These constructions support the mode access used in Section VIII. They distinguish a resonant stored mode, a finite packet and a finite-band travelling input; none substitutes a free capture or renewal operation for a physical interaction.

1. A resonant collection port

The collection fraction can itself be realized by a finite-duration interaction. This construction starts with a retained source oscillator b and a receiver oscillator c of the same frequency ω, with c initially in vacuum. It does not assume that a broadband travelling mode has already been trapped by this interaction. In the interaction picture put

$$H_{\mathrm{cap}}(t) = \hbar g(t) C, \quad C = b^\dagger c + c^\dagger b, \quad \theta = \int g(t)\,dt, \quad U_{\mathrm{cap}} = e^{-i\theta C}.$$

Proposition 21 (Finite resonant collection)**.** *For $0 \le \theta \le \pi/2$, the receiver collects $\xi = \sin^2\theta$ of the source mode. Corrected receiver-parity readout realizes Eq. (35) with this ξ. If $|g| \le g_{\max} < \omega$, the total Hamiltonian $\hbar\omega(n_b + n_c) + H_{\mathrm{cap}}$ is bounded below and the pulse needs duration at least $\theta/g_{\max}$. A pulse starting and ending at zero does no net prescribed-switch work in this isolated two-mode model. For the independently prepared vacuum encounter, the mean occupations after collection are ξkm^2 and $(1-\xi)km^2$. Their sum is km^2.*

A pulse-area error δ changes ξ by at most $|\delta|$. For any input with $\langle n_b + n_c\rangle \le \bar{n}$, including a retained reference, its half-trace error is at most $\min\{1, \sqrt{2|\delta|\bar{n}}\}$.

Proof. The Heisenberg transforms are $b \mapsto b\cos\theta - ic\sin\theta$ and $c \mapsto c\cos\theta - ib\sin\theta$. The conditional coherent states produced by U_m therefore split into receiver and remainder amplitudes with squared fractions $\xi, 1-\xi$. The receiver's known phase does not change parity. Applying Proposition 17 proves the instrument and the occupation account, for any logical input.

The normal modes $(b \pm c)/\sqrt{2}$ have instantaneous frequencies $\omega \pm g$, which proves the lower bound. The amplitude bound proves the duration statement. Both C and total occupation commute with the Hamiltonian, so $\int \langle \partial_t H_{\mathrm{cap}} \rangle dt = \hbar \langle C \rangle [g(t_{\mathrm{end}}) - g(t_{\mathrm{start}})] = 0$. Finally $|\sin^2(\theta + \delta) - \sin^2 \theta| \le |\delta|$ and $|C| \le n_b + n_c$. The purified-vector estimate $2\langle 1 - \cos(\delta C) \rangle \le 2|\delta|\bar{n}$ proves the energy-constrained error bound. □

This account includes transferred energy and a gate clock. Receiver parity control, reading and retaining a syndrome, preparing the next empty receiver, and supplying the switching apparatus remain separate resources. Emptying the receiver exports energy; it is not included in the zero net switch work above. Restoring a degraded collection setting also needs a stated physical permission and price. Brain Anchor [4] uses such a permission in a finite model where the same content survives inquiry, operation and maintenance. Resonant port capture, local capture of a continuous field and sustained thermal export are three different implementation questions.

Whole-mode control can be reduced to successive port contacts in a finite packet model. Take n source oscillators of frequency ω and one vacuum receiver c. For real $f_j \ge 0$, $\sum_j f_j^2 = 1$, form the packet mode $b_f = \sum_j f_j b_j$. Sequential source interactions $e^{-im\sqrt{k} f_j Z_L (b_j + b_j^\dagger)}$ produce the same packet displacement as $e^{-im\sqrt{k} Z_L (b_f + b_f^\dagger)}$. After contacting port j, couple only that port to c through the resonant Hamiltonian above, with area $0 \le \theta_j \le \theta_{\max} \le \pi/2$. Released ports are inaccessible. The receiver's packet amplitude, up to its known common phase, obeys

$$y_0 = 0, \qquad y_j = y_{j-1} \cos \theta_j + f_j \sin \theta_j.$$

Proposition 22 (Sequential local-port collection)**.** *The receiver collects fraction $\xi = y_n^2$ of the packet, and its corrected parity instrument is Eq.* (35)*. Among these permitted real resonant contacts, final collection is maximized by*

$$\theta_j = \min\{\theta_{\max}, \arctan(f_j / y_{j-1})\},$$

with the first angle $\pi/2$ before applying the cap when $y_{j-1} = 0 < f_j$. When $y_{j-1} = f_j = 0$, choose $\theta_j = 0$. For a uniform packet and $\theta_{\max} = .5$, the first five contacts are capped and the remaining ones are unconstrained. For $n \ge 5$ the exact optimum is

$$\xi = 1 - \frac{d}{n}, \qquad d = 5 - \cot^2(.25)(1 - \cos^5(.5))^2 = 1.47392688\ldots.$$

Thus 148 uniform ports suffice for collection at least .99, and fewer do not under the same cap. With no cap below $\pi/2$, the angles $\theta_1 = \pi/2$, $\theta_j = \arctan(1/\sqrt{j-1})$ collect a uniform finite packet completely.

Proof. Passive mode mixing sends the packet's conditional coherent displacement to receiver amplitude $-iy_n$ and a remainder of squared norm $1 - y_n^2$. Orthogonal input modes are vacuum. Their overlaps prove the instrument. Each step maximizes $y \cos\theta + f_j \sin\theta$ at the stated angle. The maximum is nondecreasing in $y \ge 0$, because all permitted cosines are nonnegative. Induction therefore makes this stepwise optimum a final optimum. For a uniform packet, the first capped steps give $y_j = n^{-1/2} \cot(.25)(1 - \cos^j(.5))$. After the fifth, the interior optimum has $y_j^2 = y_{j-1}^2 + 1/n$. This proves the formula and threshold. The uncapped recursion gives $y_j = \sqrt{j/n}$. □

Each contact has a Hamiltonian bounded below when $g_{\max} < \omega$, needs duration at least $\theta_j / g_{\max}$ and conserves total occupation. The source prepares mean total occupation km^2, the receiver holds ξkm^2 and the released ports hold $(1-\xi)km^2$. Parity control is needed only on the retained receiver. For contact-area errors δ_j, the half-trace error is at most $\min\{1, \sum_j \sqrt{2|\delta_j|\bar{n}}\}$ when total occupation is bounded in mean by $\bar{n}$; telescope the preceding energy-constrained bound. Source switching, receiver reuse and detector errors require their own account. These are local contacts in a finite collision architecture, not a proof that a specified relativistic wave packet admits that architecture or that its clocks and references can be sustained indefinitely.

2. Positive-frequency continuous collection

A specified continuous field can also supply the collection port. Let $\omega = \Omega + \delta$, $|\delta| \le B < \Omega$, and couple it to a retained oscillator c. With $[a_\delta, a_{\delta'}^\dagger] = \delta(\delta - \delta')$, take a real even form factor g, $J = g^2$ and $G = \|g\|_2$. Put

$$H(h) = \hbar\Omega n_c + \hbar \int_{-B}^{B} (\Omega + \delta) a_\delta^\dagger a_\delta d\delta + \hbar h(t) \int_{-B}^{B} g(\delta)(c^\dagger a_\delta + a_\delta^\dagger c) d\delta, \qquad |h| \le 1. \tag{A1}$$

This is an excitation-preserving effective Hamiltonian with a specified band-smeared coupling. Its frequency form factor gives spatial tails; it is not a strictly point-local relativistic detector. No negative physical frequencies or Markov

extension of the band are used. Matched pulse absorption by a tunable resonator is an established mechanism [2]; the question here is how its finite-band realization enters the held-content instrument and source account.

In the frame rotating at Ω, the one-particle operator is

$$A = \begin{pmatrix} 0 & \langle g| \\ |g\rangle & \delta \end{pmatrix}, \quad S = e^{-iAT}, \quad r(T) = \langle c|S|c\rangle.$$

The capture pulse has $h = 1$ on $[0, T]$ and is otherwise off. Deviations from this prescribed pulse are bounded below.

Proposition 23 (Matched finite-band capture)**.** *If $\Omega > B + G$, every Hamiltonian in Eq.* (A1) *is bounded below by $\hbar(\Omega - B - G)N$. Write $\xi_0 = 1 - |r(T)|^2 > 0$. Among normalized incoming field modes, the mode*

$$f = -P_{\text{field}} S^\dagger |c\rangle / \sqrt{\xi_0}$$

maximizes collection and is collected with fraction ξ_0. It can be prepared from independent vacuum by a finite prescribed body-controlled linear drive with envelope $u(s) = r(T - s)$, $0 \le s \le T$. For even J, r and this envelope are real. The prepared field amplitudes are

$$m\beta w_u(\delta), \qquad w_u(\delta) = -ig(\delta) \int_0^T u(s) e^{i\delta s} ds, \quad w_u = -P_{\text{field}} S^\dagger |c\rangle, \quad \beta = \sqrt{k/\xi_0}. \tag{A2}$$

For an arbitrary prepared envelope and $w_u \neq 0$, replace k by $k_{\text{eff}} = \beta^2 \|w_u\|^2$ and use $\xi = |\langle c|S(0, w_u)\rangle|^2 / \|w_u\|^2$. Corrected receiver-parity readout then realizes Eq. (35) *with these actual parameters.*

Proof. $\|A\| \le B + G$ and second quantization prove the lower bound. The norm of the field part of the receiver row is $\sqrt{\xi_0}$; Cauchy–Schwarz proves the optimum. Its orthogonal complement has zero collection, so the statement is not uniform over unknown packets.

Eliminating the field component of $S^\dagger|c\rangle$ gives Eq. (A2). Reflection $\delta \mapsto -\delta$, field sign reversal and complex conjugation define an antiunitary taking A to $-A$ and leaving its propagator invariant. Hence r is real. One explicit preparation places the band-smeared source at $x = -T$ and the receiver at $x = 0$, with propagation speed one. Drive the source during $t \in [-T, 0]$ with envelope $u(t+T)$ and a prescribed carrier phase Ω. Explicitly its interaction-picture Hamiltonian is

$$H_s^I(t) = \hbar m \beta Z_L u(t + T) \int_{-B}^{B} g(\delta)[a_\delta^\dagger e^{i\delta(t+T)} + a_\delta e^{-i\delta(t+T)}] d\delta.$$

Its displacement is $m\beta w_u$; the receiver is coupled only after this drive. The source Hamiltonian is linear in the band field and proportional to mZ_L. Its time-ordering commutator is a scalar times Z_L^2, so it changes only an overall phase. Completing the square bounds its instantaneous energy below since $\Omega - B > 0$ and $\int J/\omega$ is finite.

Passive field evolution splits the conditional coherent amplitudes into a receiver component of squared fraction ξ and a remainder of fraction $1 - \xi$. Their vacuum overlaps give the claimed instrument for arbitrary logical input and retained reference. Initial independence and parameters fixed within an encounter are required. □

The propagation prescription is within the band-smeared model: a strictly band-limited field operator has no compact spatial support. The construction supplies a drive and a receiver, not their formation from unknown interactions. For real envelopes and even g, the free input and output energies are

$$E_{\text{source}} = \hbar\Omega k_{\text{eff}} m^2, \qquad E_c = \hbar\Omega\xi k_{\text{eff}} m^2, \qquad E_{\text{free,out}} = \hbar\Omega(1 - \xi) k_{\text{eff}} m^2. \tag{A3}$$

The same reflection symmetry makes the mean detuning energy zero before and after collection. Consequently the net prescribed capture-switch work is zero in this family. The complete source drive supplies E_{source}. For other packets the capture work need not vanish: at mean occupation $\bar{n}$, its magnitude is at most $2\hbar B\bar{n}$. Detector, carrier, clock and support costs are separate from these accounts.

Proposition 24 (A finite waveform certificate)**.** *Let $J(\delta) \le J_0$. Approximate $r(T - s)$ by linear interpolation of node values on L uniform intervals, each rounded on a prescribed grid of spacing q. Include a further uniform envelope error η_u. With $\nu = \sqrt{\xi_0}$,*

$$\sigma = \sqrt{2\pi J_0 T} \left[\frac{G^2 (T/L)^2}{8} + \frac{q}{2} + \eta_u \right]$$

bounds the prepared field-vector error relative to Eq. (A2)*. An interval enclosure error for node values is added inside the brackets. Let duration and arrival errors be bounded by* d_T, d_t*, and the capture pulse have* $\int |h_{\mathrm{actual}} - h|dt \le d_h$*. Put* $\varepsilon = (B+G)d_T + Bd_t + Gd_h$*. If* $\sigma < \nu$*, then*

$$1 - \xi \le \left[\frac{|r|\nu + \sigma + \varepsilon(\nu+\sigma)}{\nu - \sigma}\right]^2, \qquad \left(1 - \frac{\sigma}{\nu}\right)^2 \le \frac{k_{\mathrm{eff}}}{k} \le \left(1 + \frac{\sigma}{\nu}\right)^2. \tag{A4}$$

Arrival and duration bounds here concern collection and the logical parity instrument, which are insensitive to the common carrier phase.

Proof. $|r''(s)| = |\langle Ac|e^{-iAs}|Ac\rangle| \le G^2$ gives the interpolation bound. Plancherel over the entire frequency line, followed by restriction to the band, gives $\|w_u - w_{\mathrm{ideal}}\| \le \sigma$. Duhamel bounds the carrier-referenced propagator error by ε. The ideal outgoing field remainder has norm $|r|\nu$. Adding preparation and propagator errors to this norm and dividing by the actual input norm, at least $\nu - \sigma$, proves the first bound. The second follows by the same input-norm bounds. □

For a concrete continuum certificate use

$$J(\delta) = \frac{1}{2\pi}\left[1 - (\delta/B)^2\right]^2, \quad \Omega = 20, \quad B = 5, \quad T = 16.$$

Frequencies are measured in units of $2\pi J_0$ and times in its reciprocal. Then $G^2 = 8/(3\pi)$. The exact even bath moments are $\mu_{2j} = 16J_0B^{2j+1}/[(2j+1)(2j+3)(2j+5)]$. If $M_n = \langle c|A^n|c\rangle$, then

$$M_0 = 1, \qquad M_{2n} = \sum_{j=0}^{n-1} \mu_{2j} M_{2(n-1-j)}, \qquad r(t) = \sum_{n=0}^{\infty} \frac{(-1)^n M_{2n} t^{2n}}{(2n)!}.$$

Decomposing each departure and return to c proves the recurrence. The degree-P Taylor remainder is at most $[(B+G)T]^{P+1}/(P+1)!$ by self-adjoint functional calculus. Directed intervals at 120 decimal digits and $P = 400$ give $r(16) = 0.000061743299929498335 73\ldots$ with remainder below 1.53×10^{-79} and $\xi_0 > 0.99999999618$.

Choose $L = 512$, $q = 1/65536$, $\eta_u = d_T = d_t = d_h = 10^{-5}$. The 513 rounded node values fit the grid $[-1/64, 1]$, whose 66561 labels need 17 bits each. The fixed table uses 8721 logical bits, apart from its decoder, interpolator, carrier and clock. A separately encoded normalization amplitude rounded on a 2^{-24} grid contributes less than 10^{-7} relative error at nominal $k = .2$; it fits within η_u and uses 24 more label bits. Eq. (A4) gives $1 - \xi < 4.42935 \times 10^{-7}$ and $k_{\mathrm{eff}}/k \in [.999030268, 1.000970203]$. For $\alpha = 1.1$, $h = .5$, $N = 4096$ probes at each of $\pm h$ and nominal $k = .2$, the calibration's collection-dependent dephasing error is below .000946. This contrasts with the loss of content at $\xi = .99$ above; it does not include parity-gate or detector errors.

A second numerical representation replaces frequency multiplication by cell midpoints while retaining g itself. The invariant vector in each cell is $g\mathbf{1}_{\mathrm{cell}}/\|g\mathbf{1}_{\mathrm{cell}}\|$, with coupling $\sqrt{\int_{\mathrm{cell}} J}$. Its finite matrix approximates the continuum propagator with operator error at most TB/M for M cells. These cells are a numerical resolution, not fresh physical ports. Dense and sparse matrix calculations agree with the moment enclosure and the prepared waveform. The recurrence and rounding rule specify the waveform without a fitted infinite-precision function.

After an encounter, swapping c with an independent same-frequency vacuum oscillator returns c exactly to vacuum, even when its old state is correlated with a reference. The swap exports E_c and has the finite duration and occupation account of Proposition 21. It supplies a reusable receiver for a finite bank of encounters, not a free thermal sink. Repeating the query law needs independent field preparations as well as independent vacuum exhausts. Independent field sectors selected one at a time give a finite-bank realization whose Hamiltonian remains bounded below. 8192 queries consume 8192 such preparations and exhausts, while sharing the waveform program. In this calibration example the nominal source energy is $E_{\mathrm{probes}}/[\hbar(2\pi J_0)] = 4096(.2)(20)[(1 - .55)^2 + (1 + .55)^2] = 42680.32$. It is predominantly collected and later exported, not eliminated by the small content-damage bound. Serial preparation and capture take $8192(2T) = 262144$ time units before parity control and receiver reset. Time windows in one band-limited waveguide are not automatically independent; that alternative requires a return/reset bound. Thermal exhaust resets c to a thermal state and changes the readout law. For example, for the exactly matched mode, a thermal receiver of mean occupation n_T gives even-parity probability

$$\frac{1}{2}\left[1 + \frac{e^{-2\xi k m^2/[1+2(1-\xi)n_T]}}{1 + 2(1-\xi)n_T}\right]$$

after mixing with the vacuum source packet, rather than the vacuum formula. For an unmatched packet, the thermal occupation term instead uses $|\langle c|S|c\rangle|^2 n_T$, which need not equal $(1-\xi)n_T$.

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Representing Control in the World: Retained History, Physical Sources and Renewal in Q-RAIF

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Abstract

What does a physical coupling actually represent? For a system coupled to a discarded port, the reduced map is completely positive but need not be multiplicative. We determine its exact algebra for finite-spectrum dephasing: a non-atomic port leaves precisely the coupling operator's commutant, which can be noncommutative. Retaining an oscillator permits exact closed-loop gates; retaining translational degrees of freedom gives a faithful representation in the closed propagation carrier. A single quadratic shell supplies the spatial Clifford relations, while evolution time and the mass anticommutator enter separately in the Lorentzian symbol.

A centered-input charging condition connects these structures without identifying their quadratic forms. Under an actual calibrated alignment of control and propagation generators, the work form is a positive pullback of the spatial shell. The same encounter can satisfy that all-state charge condition while its reduced exact algebra remains only a commutant. Neither persistence nor a positive cost alone selects a spacetime representation.

The physical realization retains the records and resources that the algebraic statement omits. A positive Gaussian material Hamiltonian supports acquisition, record-controlled force, sensing and another action, with bounds on physical energy moments and smooth interacting sources. A paid Gibbs bank renews a specified receiver, and one finite cursor approximates a sequence while retaining the complete old history. Full-field ring renewal, neutral compression and thermal sorting, local massive-body gates and a spatial work clock provide further finite implementations in their stated layouts. A common action includes clock and support reaction. Matching these quantum layouts to one covariant supported apparatus, sustained thermal renewal and jointly sharp exhaust remains a separate construction.

Keywords: Q-RAIF, physical representation, multiplicative domain, propagation symbol, quantum records, material response, thermal renewal

Technical appendices and their original numbering are retained in Q-RAIF C technical supplement (`QRAIF_C_supplement.pdf`). The certificate guide at the end of this paper identifies their claims.

1 Introduction

A control algebra specifies how operations compose. A physical interface must also say which operations its coupling transports exactly, what happens to the degrees of freedom carrying its records, and what resources allow another operation. An assumed homomorphism answers the first algebraic question by definition; it does not construct the interface.

Q-RAIF A identifies a Lorentzian read-out under propagation-symbol conditions, Q-RAIF B derives actuation and work from a field coupling, and the mass–energy paper relates spectral mass to finite read-outs [25, 26, 27]. Here we put the representation question between these structures and their physical implementation. The distinction is whether the relevant degrees of freedom are discarded or retained, not whether the environment has a large Hilbert space.

Exact transport. Sections 2–4 begin with the algebra supplied by a coupling. Its reduced Heisenberg map has an exact algebra, the multiplicative domain [29, 30, 31]. Theorem 4 computes it for a dephasing interaction. With a non-atomic discarded port it is the commutant of the coupling observable, including its full degeneracy blocks. The field-coupled flip then relates its remaining defect to drive energy and escaped information. These are architecture-specific statements, not a universal conservation-law prohibition on control [34, 35, 36].

Propagation and charging. When translations remain in the closed carrier, the internal algebra embeds faithfully. Section 5 distinguishes that exact embedding from the quadratic-space structure of a first-order propagation symbol. The latter requires a single shell, and its Lorentzian extension uses the supplied time and a mass anticommutator. Section 6 brings in Q-RAIF B's conditional work bridge: one centered independent field input charges every unknown carrier state equally precisely when each control generator combination has a scalar square. An actual calibration between these generators and the propagation symbol then relates the positive work form to the spatial shell. It does not turn that form into an indefinite spacetime metric. We show why the same calibrated encounter still need not represent the full control algebra after the port is discarded.

Retention and physical return. Section 7 gives the two decisive controls on the discarded-port argument: a genuinely fresh lost port leaves an irreversible defect, whereas a retained oscillator closes exact Weyl loops. The continuum loop and its finite adaptive error budget are developed in Appendix B. The retained case requires coherent routing, quadratures and a clock; an abstract embedding supplies none of those permissions. These are two realizations, not an exhaustive classification of physical interfaces.

The material construction in Sections 8–10 then follows one complete Gaussian Hamiltonian through acquisition, force, sensing, receiver replacement and another action. Records remain operators in their old joint history. Physical energy moments and smooth-source forms control unbounded field readings, rather than inferring stress convergence from trace distance alone. Receiver replacement uses a prepared Gibbs bank and transfers the spent correlations into its ledger.

The longer certificates are grouped by the physical interface they establish. Appendix A constructs the charged regulator, a finite mode with its remainder and the same-Hamiltonian source bounds. Appendix C treats the complete ring body, neutral gates, packet loading and thermal sorting. Appendix D treats the spatial work field and the separate massive-body gate. The remaining appendices retain the battery, outlets, thermal preparations, contact recoil, actual diagnostic return and current-stress compiler. Their distinct carriers are not identified by a common vocabulary. Section 11 gives a common action with clock/support reaction and states the matching still required.

Three resource limits matter. Bounded-energy receiver renewal takes longer and uses more clock states; fixed-duration finite rounds use growing clock energy; a spatial work clock uses horizon-dependent inertia and a supplied material frame. The ring converter's complete-output thermal deficit can approach zero in a prescribed finite layout. These results do not yet form one indefinitely renewed, fixed-capacity, covariant apparatus.

Unless stated otherwise, $\hbar = c = k_B = 1$. Energies expressed as frequencies acquire a factor $\hbar$ when units are restored, times and pulse areas use the corresponding frequency units, and entropy is in nats. The combinations $b + b^\dagger$, $i(b^\dagger - b)$ and occupation numbers are dimensionless; material smearings use the normalizations specified with their Hamiltonians below. All joint-state assertions include the stated old references; a preserved nonselective marginal need not remain unchanged after conditioning on a new record.

2 The algebraic statement

Let (V, q) be a finite-dimensional real vector space with a non-degenerate quadratic form and $Cl(V, q)$ the real Clifford algebra generated by V with $v^2 = q(v)\mathbf{1}$. When generators are to act as operators, those with $q(e) > 0$ are taken Hermitian and those with $q(e) < 0$ anti-Hermitian; this fixes an involution with $e^* = \operatorname{sign}(q(e))\, e$ on an orthogonal basis.

Definition 1 (Representation input)**.** A *representation* of a control algebra $\mathcal{O}_{\text{int}} \cong Cl(V, q)$ in an environmental C^*-algebra $\mathcal{A}_{\text{ext}}$ is a unital $*$-homomorphism $\phi : \mathcal{O}_{\text{int}} \to \mathcal{A}_{\text{ext}}$.

Lemma 2 (Semisimplicity)**.** *$Cl(V, q)$ is semisimple. If $\dim V$ is even it is simple, and every nonzero algebra homomorphism out of it is injective.*

Proof. By the classification of real Clifford algebras [40, 41], $Cl(V, q)$ is a matrix algebra over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$, or a direct sum of two such algebras, according to signature and dimension modulo eight. For even $\dim V$ it is a single matrix algebra. The kernel of a homomorphism is a two-sided ideal, and a simple algebra has none besides 0 and itself. □

Proposition 3 (Image closure)**.** *Let ϕ be as in Definition 1. Then $\operatorname{Im}\phi$ is a $*$-subalgebra of $\mathcal{A}_{\text{ext}}$ in which the Clifford relations hold. If $Cl(V, q)$ is simple, in particular when $\dim V$ is even, ϕ is injective. If it is the sum of two simple algebras, its image is isomorphic to the full algebra or to one of the two summands.*

Proof. Images of $*$-homomorphisms are $*$-subalgebras and relations descend. Simplicity gives injectivity. The only proper ideals of a sum of two simple algebras are zero and either summand; a unital representation has a proper kernel. □

Non-degeneracy is needed: for $q(x,y)=x^2$ on $\mathbb{R}^2$, the map $e_1 \mapsto \mathrm{diag}(1,-1)$, $e_2 \mapsto 0$ is a nonzero unital homomorphism into $M_2(\mathbb{R})$ that is not injective. An algebra homomorphism also does not compare signatures. That requires a quadratic-space morphism $j : (V,q) \to (W,Q)$ with $Q(jv) = q(v)$, and, where relevant, preservation of the involution and grading by the induced map of Clifford algebras. Proposition 3 provides neither the map ϕ nor such a j. Sections 3–5 ask where they can come from.

3 Couplings give channels

Let a finite-dimensional system S interact with an environment $\mathcal{E}$ through a unitary U on $\mathcal{H}_S \otimes \mathcal{H}_\mathcal{E}$, with the environment in a normal state σ. The induced Heisenberg map on system observables is

$$\Phi^*(A) = \mathrm{tr}_\mathcal{E}\big[(\mathbf{1}\otimes\sigma)\,U^*(A\otimes\mathbf{1})U\big]. \tag{1}$$

It is unital and completely positive, and by Stinespring's theorem every such map arises in this way [29]. It is in general not multiplicative. Its *multiplicative domain*

$$\mathcal{M}_\Phi = \{A :\ \Phi^*(A^*A) = \Phi^*(A)^*\Phi^*(A),\ \Phi^*(AA^*) = \Phi^*(A)\Phi^*(A)^*\} \tag{2}$$

is a C^*-subalgebra on which Φ^* is a $*$-homomorphism, and $\Phi^*(AB) = \Phi^*(A)\Phi^*(B)$ whenever $A \in \mathcal{M}_\Phi$ [30, 31]. It is the largest part of the system's algebra that the coupling represents exactly. For other elements we measure the failure by the *defect*

$$\delta(A) = \Phi^*(A^*A) - \Phi^*(A)^*\Phi^*(A) \ \geq 0. \tag{3}$$

Theorem 4 (Exactly represented algebra of a dephasing coupling)**.** *Let $X = \sum_x x P_x$ have finite spectrum, let E be self-adjoint on $\mathcal{H}_\mathcal{E}$, and $U = \exp(-iX\otimes E)$. Write $\chi(t) = \mathrm{tr}(\sigma e^{-itE})$ and $A_{xx'} = P_x A P_{x'}$. Then*

$$\Phi^*(A) = \sum_{x,x'} \chi(x'-x)\, A_{xx'}, \tag{4}$$

the relation $x \sim x'$ defined by $|\chi(x'-x)| = 1$ is an equivalence relation on the spectrum of X, and

$$\mathcal{M}_\Phi = \{A :\ A_{xx'} = 0 \text{ whenever } |\chi(x'-x)| < 1\}. \tag{5}$$

On $\mathcal{M}_\Phi$, Φ^* is an inner automorphism $A \mapsto W^*AW$ with $W = \sum_x u_x P_x$ for phases u_x. If $|\chi(t)| < 1$ for every nonzero difference t of eigenvalues of X, then $\mathcal{M}_\Phi = \{X\}'$ and Φ^* is the identity on it.

Proof. $U = \sum_x P_x \otimes e^{-ixE}$, so $U^*(A\otimes\mathbf{1})U = \sum_{x,x'} A_{xx'} \otimes e^{ixE}e^{-ix'E}$, and taking the expectation in σ gives Eq. (4).

If $|\,\mathrm{tr}(\sigma e^{-itE})| = 1$, write $\sigma = \sum_i p_i |\psi_i\rangle\langle\psi_i|$; each $|\langle\psi_i|e^{-itE}|\psi_i\rangle| \leq 1$, so equality forces every ψ_i to be an eigenvector of e^{-itE} with one common eigenvalue $\chi(t)$. Hence e^{-itE} acts as the scalar $\chi(t)$ on the support of σ. If this holds for t_1 and t_2, it holds for $t_1 + t_2$ with $\chi(t_1+t_2) = \chi(t_1)\chi(t_2)$. Together with $\chi(0) = 1$ and $\chi(-t) = \overline{\chi(t)}$ this shows that $\sim$ is an equivalence relation, and that within a class with base point x_0 the phases $u_x = \chi(x - x_0)$ satisfy $\chi(x'-x) = \overline{u_x}u_{x'}$.

The diagonal blocks of the defect are

$$P_y\big[\Phi^*(A^*A) - \Phi^*(A)^*\Phi^*(A)\big]P_y = \sum_x \big(1 - |\chi(y-x)|^2\big)A^*_{xy}A_{xy}, \tag{6}$$

a sum of positive operators with nonnegative coefficients. It vanishes for all y only if $A_{xy} = 0$ whenever $|\chi(y-x)| < 1$; the condition with AA^* is the same with the roles of rows and columns exchanged. Conversely, if A has this block pattern, Eq. (4) gives $\Phi^*(A) = \sum \overline{u_x}u_{x'}A_{xx'} = W^*AW$, and these A form a $*$-algebra on which Φ^* is multiplicative. If no nonzero difference has $|\chi| = 1$, the classes are single eigenvalues, the algebra is $\bigoplus_x P_xB(\mathcal{H}_S)P_x = \{X\}'$, and W is scalar on each block. □

Corollary 5 (Continuous spectrum forbids exact coupled control)**.** *If the spectral distribution of E in σ has no atoms, then $|\chi(t)| < 1$ for every $t \neq 0$, so $\mathcal{M}_\Phi = \{X\}'$. In particular this holds for every normal state when E has purely continuous spectrum. Every exactly represented element A commutes with X, and hence with the interaction on its natural domain. No nonzero rescaling of the coupling changes this conclusion.*

Proof. The equality $|\int e^{-ite}\,d\mu(e)| = 1$ requires the phase e^{-ite} to be constant μ-almost everywhere. For $t \neq 0$ its level sets are translates of the countable lattice $(2\pi/t)\mathbb{Z}$, and a non-atomic probability measure gives each such set measure zero. Absolute continuity is sufficient but unnecessary. □

A field quadrature, or the smeared current Q_g of Paper B, has purely absolutely continuous spectrum, so Corollary 5 applies to every normal drive state, whatever its energy. For other couplings, $|\chi(t)| = 1$ requires support in an eigenspace of e^{-itE}, not necessarily an eigenstate of E. Periodic spectra can give exact revivals: with $X = \pi Z$, $E = \mathrm{diag}(0,1)$ and $\sigma = |+\rangle\langle+|$, the interaction unitary is $I \otimes \mathrm{diag}(1,-1)$ and the system channel is identity, although σ is not an E eigenstate. A Fock state in a number-type coupling gives another finite-energy possibility. The quadrature obstruction concerns the specified single coupling; extending it to composite local controls requires a separate argument.

4 The field-coupled flip

Paper B couples a degenerate bit to one mode of a free chiral field through $U = \exp(-i\lambda XQ)$, $\lambda > 0$, with $Q = b + b^\dagger$, $P = i(b^\dagger - b)$ and $[Q,P] = 2i$ [26]. Here $E = \lambda Q$ and the eigenvalues of X are ± 1. Write $\chi = \mathrm{tr}(\sigma e^{-2i\lambda Q})$ and $\sigma_\pm = e^{\mp i\lambda Q}\sigma e^{\pm i\lambda Q}$ for the field states conditioned on $X = \pm 1$. A flip of Z-diagonal states fails with probability $\epsilon = (1 + \mathrm{Re}\,\chi)/2$.

Theorem 6 (Representation defect of the field-coupled flip)**.** *Let the drive mode be in a normal state σ with mean occupation $N = \mathrm{tr}(\sigma b^\dagger b) < \infty$.*

1. *In the X eigenbasis, Φ^* multiplies the two off-diagonal entries by χ and $\overline{\chi}$, and $\mathcal{M}_\Phi = \{X\}'$, the algebra generated by X.*

2. *For every A with vanishing diagonal in the X basis, in particular Z and Y, $\delta(A) = (1 - |\chi|^2)\,\mathrm{diag}(|A_{-+}|^2, |A_{+-}|^2)$; thus $\delta(Z) = \delta(Y) = (1 - |\chi|^2)\mathbf{1}$.*

3. $1-|\chi|^2 \;\geq\; \dfrac{\arccos^2(\operatorname{Re}\chi)\,|\chi|^2}{(2N+1)^2}$.

4. $1-|\chi|^2 \geq D(\sigma_+,\sigma_-)^2$*, with equality when* σ *is pure, where* D *is the trace distance.*

Proof. Item 1 is Theorem 4 with eigenvalue differences ± 2 and Corollary 5. Item 2 follows from Eq. (4) by multiplying out 2×2 matrices.

Item 3 uses Paper B's two quadrature bounds, valid for a normal (density-operator) drive state of finite occupation and $[Q,P]=2i$ [26]: with $G(c)=\arccos^2 c$,

$$\langle Q^2\rangle \geq \frac{G(\operatorname{Re}\chi)}{4\lambda^2}, \qquad (\Delta P)^2 \geq \frac{4\lambda^2|\chi|^2}{1-|\chi|^2}.$$

The first is spectral Jensen; the second is the unitary uncertainty bound. Since $4N+2=\langle Q^2+P^2\rangle$,

$$\frac{4\lambda^2|\chi|^2}{1-|\chi|^2} \leq \langle P^2\rangle = 4N+2-\langle Q^2\rangle \leq 4N+2-\frac{G(\operatorname{Re}\chi)}{4\lambda^2}.$$

With $u=4\lambda^2$, $a=G(\operatorname{Re}\chi)$ and $A=4N+2$, this reads $1-|\chi|^2 \geq u^2|\chi|^2/(Au-a)$ for some $u>a/A$. The right side is minimized over u at $u=2a/A$, where it equals $4a|\chi|^2/A^2$.

Item 4. If $\sigma=|\psi\rangle\langle\psi|$, the conditioned states are pure with overlap $\langle e^{i\lambda Q}\psi, e^{-i\lambda Q}\psi\rangle=\chi$, and $D^2=1-|\chi|^2$. For mixed σ, applying $e^{\mp i\lambda Q}\otimes\mathbf{1}$ to a purification of σ gives purifications of $\sigma_\pm$ with overlap χ, so Uhlmann's theorem gives fidelity $F(\sigma_+,\sigma_-)\geq|\chi|$, and $D\leq(1-F^2)^{1/2}$ [32, 33]. □

Corollary 7 (The energy–accuracy bound as a representation statement)**.** *For a flip with failure probability* $\epsilon\in(0,\frac{1}{2})$ *and* $\theta_\epsilon=\arccos(2\epsilon-1)$*,*

$$N \;\geq\; \frac{\theta_\epsilon(1-2\epsilon)}{4\sqrt{\epsilon(1-\epsilon)}}-\frac{1}{2}. \tag{7}$$

Proof. $\operatorname{Re}\chi=2\epsilon-1$ gives $|\chi|^2\geq(1-2\epsilon)^2$, hence $1-|\chi|^2\leq 4\epsilon(1-\epsilon)$. Combining with item 3 of Theorem 6 gives $4\epsilon(1-\epsilon)\geq\theta_\epsilon^2(1-2\epsilon)^2/(2N+1)^2$. □

This is exactly the occupation bound of Paper B. Read in the present terms, an accurate flip requires the coupling to represent Z almost multiplicatively, and the defect $1-|\chi|^2$ cannot vanish at finite occupation. Item 4 identifies the defect with information: for pure drives it is the squared distinguishability of the two field states that record the value of X. The field is then a fragment in the sense of the HAFF paper on measurement, with conditional overlap $|\chi|$; that paper determines which pointer effects such a fragment can implement [28], and the same overlap fixes the representation defect here. The exactly represented algebra $\{X\}'$ is the pointer algebra of the decoherence literature [37]: the control interface transports exactly the part of the system that the environment records.

For the phase-fixed half-turn squeezed family, Paper B gives occupations 2.958, 7.330 and 24.329 at $\epsilon=.05,.01,.001$, against bound values 2.28, 6.74 and 23.80. Its separately optimized finite quadrature-grid values are variational minima, not certified global optima; the continuous supporting ground-state argument has its own assumptions. Mixtures of coherent states require occupation of order $\pi^2/(16\epsilon)$ [26].

4.1 Same clock as mode matching

The requirement that control and environment share a clock, so that their operational windows overlap, has a precise form in the field-coupled flip. The switch g defines a normalized mode function f and the mode operator $b_f = \int \overline{f(\omega)}\, a_\omega \, d\omega$ with $Q_g = \lambda(b_f + b_f^\dagger)$ [26]. A drive is prepared in some normalized mode f', by a source with its own clock.

Proposition 8 (Mode overlap)**.** *Let the drive occupy mode f' with mean occupation N', all modes orthogonal to f' being in the vacuum, and let $c = \int \overline{f(\omega)} f'(\omega)\, d\omega$. Then the bit channel equals the channel of Theorem 6 for the reduced state of mode f, which is the output of a pure-loss channel of transmissivity $|c|^2$ and has mean occupation $|c|^2 N'$. A flip with failure ϵ therefore needs*

$$N' \;\geq\; \frac{1}{|c|^2}\left[\frac{\theta_\epsilon(1-2\epsilon)}{4\sqrt{\epsilon(1-\epsilon)}} - \frac{1}{2}\right]. \tag{8}$$

If $f'(\omega) = f(\omega)e^{i\omega\Delta}$, a copy of f delayed by Δ, then $c = \int |f(\omega)|^2 e^{i\omega\Delta} d\omega$. For a positive-frequency field this integral is over $\omega > 0$. Extending a narrowband Gaussian spectral density of width Ω to the full real line gives $|c| \simeq e^{-\Omega^2\Delta^2/2}$ and an occupation penalty $e^{\Omega^2\Delta^2}$ in the range where this approximation controls the overlap. The exact positive-frequency integral, not this Gaussian approximation, determines the cost at long delays.

Proof. Write $f = \bar{c} f' + r$ with $\int \overline{f'}\, r\, d\omega = 0$ and $\|r\|^2 = 1 - |c|^2$. Since $g \mapsto b_g$ is antilinear, $b_f = c\, b_{f'} + b_r$, where $[b_r, b_r^\dagger] = 1 - |c|^2$ and b_r commutes with $b_{f'}$ and $b_{f'}^\dagger$. The mode $r/\|r\|$ is orthogonal to f' and in the vacuum, independent of mode f', so the state of mode f is that of a beam splitter of transmissivity $|c|^2$ with vacuum in the other port, and its occupation is $|c|^2 N'$. Q_g involves only mode f, so the bit channel depends only on this reduced state, and Corollary 7 applies to it. Translation gives the exact Fourier overlap. The Gaussian specialization follows only after the stated extension of the frequency support. □

The prepared drive has energy $N'\bar{\omega}'$, while only the fraction $|c|^2$ of its occupation reaches the selected mode. This is a charged preparation resource, not automatically dissipated heat. The source/switch clock offset therefore enters Paper B's feedback acceptance as a mode-matching loss.

5 A representation generated by propagation

The situation is different when the "environment" is part of the same closed dynamics. In the local propagation model of Paper A, a particle with internal space $\mathbb{C}^s$ moves on a lattice Λ by a unitary step $W = \sum_{h \in N} T_h \otimes A_h$, where T_h translates by h [25]. The internal algebra is represented by

$$\phi : M_s(\mathbb{C}) \to B\big(\ell^2(\Lambda) \otimes \mathbb{C}^s\big), \qquad \phi(A) = \mathbf{1} \otimes A, \tag{9}$$

a faithful unital $*$-homomorphism, and $[\phi(A), W] = \sum_h T_h \otimes [A, A_h]$ is nonzero unless A commutes with every A_h. The requirement of Definition 1 together with nontrivial coupling is met exactly, because nothing is traced out: the translational degrees of freedom and the internal ones evolve under one unitary. The contrast with Section 4 is the content of

Corollary 5: discarding a non-atomic port restricts exact multiplicativity to the coupling commutant.

The propagation also supplies the quadratic-space morphism that Section 2 found missing. Near a band point the step has a first-order symbol $H(q) = \sum_i q_i\alpha_i + \mu\beta$ with Hermitian coefficients [25].

Proposition 9 (The propagation symbol as a quadratic-space morphism)**.** *Let $j(q) = \sum_i q_i\alpha_i$ with $\alpha_1, \dots, \alpha_d$ linearly independent. Then $j(q)^2 = Q(q)\mathbf{1}$ for a quadratic form Q if and only if, for $\mu = 0$, every eigenvalue of $H(q)$ lies on one shell $\omega^2 = Q(q)$. In that case Q is positive definite, j is an isometry from $(\mathbb{R}^d, Q)$ into* $\mathrm{Herm}(s)$ *with the normalized Hilbert–Schmidt form, and j extends to a unital $*$-homomorphism $Cl(\mathbb{R}^d, Q) \to M_s(\mathbb{C})$. If moreover $\mu \neq 0$, $\beta^2 = \mathbf{1}$ and $\{\alpha_i, \beta\} = 0$, then*

$$J(\omega, q) = \omega\beta - \sum_i q_i\beta\alpha_i \tag{10}$$

satisfies $J(\omega, q)^2 = (\omega^2 - Q(q))\mathbf{1}$ and extends to a unital homomorphism $Cl(\mathbb{R}^{1+d}, \eta) \to M_s(\mathbb{C})$ with $\eta(\omega, q) = \omega^2 - Q(q)$, which preserves the involution in which β is Hermitian and the $\beta\alpha_i$ are anti-Hermitian.

Proof. The first statement is the single-shell theorem of Paper A [25] with $\mu = 0$; isometry follows from $\frac{1}{s}\operatorname{tr} j(q)^2 = Q(q)$, and the extension from the universal property of Clifford algebras. For J, expand J^2 using $\beta^2 = \mathbf{1}$ and $\beta\alpha_i\beta = -\alpha_i$: the cross terms $\omega q_i\{\beta, \beta\alpha_i\}$ vanish and $\sum q_i q_j \frac{1}{2}\{\beta\alpha_i, \beta\alpha_j\} = -Q(q)$. Hermiticity of β and anti-Hermiticity of $\beta\alpha_i = -\alpha_i\beta$ follow from $(\beta\alpha_i)^* = \alpha_i\beta$. □

J is the map $j : (V, q) \to (W, Q)$ of Section 2 with $(V, q) = (\mathbb{R}^{1+d}, \eta)$, a Lorentzian quadratic space, and W the real span of β and the $\beta\alpha_i$ in $M_s(\mathbb{C})$, with the form defined by $Y^2 = Q_W(Y)\mathbf{1}$. The spatial scalar-square condition is equivalent to the specified single-shell property, so no internal band has a different squared dispersion. The Lorentzian extension additionally uses the time parameter and the mass anticommutator. A massless one-dimensional carrier can have only one chiral branch; a full two-sided cone requires both frequency signs, as Q-RAIF A explains [25].

The canonical inclusion ϕ remains exact on the lattice. It is the linear quadratic-space description of its propagation symbol that generally holds only at leading order. For $s = 2$ and $W(k) \in SU(2)$, the effective generator $H_{\mathrm{eff}}(k) = (i/\tau)\operatorname{Log} W(k)$ is traceless, so $H_{\mathrm{eff}}(k)^2 = \omega(k)^2\mathbf{1}$ holds at every k; what fails is linearity and isotropy of $k \mapsto H_{\mathrm{eff}}(k)$. In the Weyl automaton of Paper A, $H_{\mathrm{eff}}(k) = c\,\sigma \cdot k$ up to corrections of relative order $|k|\ell$, which include a parity-odd anisotropic term. In the split-step walk the quadratic identity $\omega^2 = c_{\mathrm{eff}}^2 q^2 + \mu^2$ fails at order $(qa)^4$, with coefficient $-\theta^2/45 + O(\theta^4)$ in lattice units [27]. These orders are the precise sense in which a lattice propagation gives an approximate representation.

6 Calibrated work and exact access

The propagation symbol and an actuator can use the same internal generators without being the same physical system. Q-RAIF B supplies a specific way to test their alignment [26].

Let $A(a) = \sum_i a_i A_i$ be a fixed real linear family of dimensionless Hermitian operators. During an encounter the command a is held fixed, the carrier free Hamiltonian is zero or commutes with $A(a)$, and a nonzero real smooth compact switch g couples it to one chiral current as $g(t)A(a)j(t)$. The field input is independent, centered and of finite energy. With $\kappa_g = \frac{1}{4}\int g'(u)^2 du > 0$, the same encounter has

$$W_\rho(a) = \kappa_g \operatorname{tr}[\rho A(a)^2]. \tag{11}$$

This is field energy delivered by switching, not a total-device heat charge. A nonzero commuting carrier Hamiltonian has no energy change during this encounter; cancelling a non-commuting drift is an additional operation.

The charge is independent of every carrier state ρ for every command precisely when $A(a)^2 = Q_{\mathrm{ctrl}}(a)I$. Polarization gives $\{A_i, A_j\} = 2(Q_{\mathrm{ctrl}})_{ij} I$. For an injective family Q_{ctrl} is positive definite. The implication is from an all-state charging requirement, not from persistence, reachability or positivity of an average cost. The field shift is $j(u) \mapsto j(u) + A(a)g'(u)/2$. The second Magnus term can be a system operation $e^{i\vartheta A(a)^2}$ rather than a global phase; it commutes with the field and does not change this work calculation [26].

To relate it to propagation, write the spatial symbol as $j_{\mathrm{prop}}(p) = \sum_i p_i \alpha_i$, with energy units and $j_{\mathrm{prop}}(p)^2 = Q_{\mathrm{prop}}(p)I$. Let C be the measured map from commands to momenta and $E_* > 0$ an energy scale. The actual coupling must satisfy

$$A(a) = j_{\mathrm{prop}}(Ca)/E_*.$$

The equality specifies a physical alignment. It cannot be inferred by renaming two positive forms.

Corollary 10 (Aligned charging and the reduced exact algebra)**.** *Under these encounter and alignment assumptions,*

$$W(a) = a^T K_{\mathrm{op}} a, \qquad K_{\mathrm{op}} = \frac{\kappa_g}{E_*^2} C^T Q_{\mathrm{prop}} C. \tag{12}$$

For a normal field input and a discarded field port, the encounter's multiplicative domain is nevertheless

$$\mathcal{M}_a = \{j_{\mathrm{prop}}(Ca)\}'. \tag{13}$$

If C is invertible, the common exact algebra for all commands is $\bigcap_a \mathcal{M}_a = \{\alpha_1, \ldots, \alpha_d\}'$. For an irreducible complex matrix representation this common algebra is scalar.

Proof. The scalar-square identity inserted in Eq. (11) gives Eq. (12), the aligned-work result of Q-RAIF B. The nonzero switched current is a field quadrature with non-atomic spectral distribution in every normal input. Corollary 5 applies with $X = A(a)$ and $E = Q_g$. The commuting Magnus factor is scalar for this aligned family, so it does not change the domain. Commuting free carrier evolution preserves the same block algebra. Commuting with every linear combination is equivalent to commuting with each generator when C is invertible. The last assertion is the matrix commutant criterion for irreducibility. □

The corollary combines the charging result of Q-RAIF B with the multiplicative-domain calculation; it is not a further selection principle for Clifford algebras. For one command the domain retains full degenerate eigenspace blocks and can be noncommutative. Over all commands it can become much smaller. Exact closed-carrier representation, state-independent work and exact reduced control are therefore different properties.

Two failures keep the calibration operational. For $A(a) = a_1 Z + a_2 I$, the maximally mixed state costs $\kappa_g(a_1^2 + a_2^2)$, but the two Z eigenstates cost $\kappa_g(a_1 \pm a_2)^2$; one average-state test does not establish the all-state condition. For $j_{\mathrm{prop}}(p) = p_1 X + p_2 Z$ and $C = \mathrm{diag}(1, 2)$ in units $E_* = 1$, the unchanged Euclidean propagation shell gives $K_{\mathrm{op}}/\kappa_g = \mathrm{diag}(1, 4)$. Without C, work does not identify the propagation form.

The time direction and mass anticommutator remain the additional conditions of Proposition 9. In $\omega^2 - Q_{\mathrm{prop}}(p)$, ω is conjugate to the supplied evolution time. An independent, noncentered drive or a correlated input has a stress cross term $g'(u)\langle A(a)j(u)\rangle$; a coherent gate cannot inherit the centered calibration price for all states. Nor does the calibration construct a shared propagation medium, pause its noncommuting drift for free, or maintain its controls indefinitely.

7 Discarded ports and coherent return

A discarded record port and a retained environment permit different controls. Write $\chi(t) = \mathrm{Tr}(\sigma_E e^{-itE})$ for the port's spectral characteristic function. Section 3 derives the single-coupling channel and its multiplicative domain; the following bound survives arbitrary later controls that cannot reach the lost port.

Proposition 11 (A fresh port leaves an irreversible defect)**.** *Suppose a finite-dimensional system first undergoes ideal unitary controls and then one coupling $e^{-iX\otimes E}$ to an initially independent port, with X having at least two distinct eigenvalues and the environment's spectral distribution non-atomic. Let F collect every auxiliary port accessible later; just before this coupling the joint state is $\rho_S \otimes \sigma_E \otimes \tau_F$ for every system input. If E is thereafter inaccessible, no subsequent system channel can make the complete channel unitary on the full system algebra. More precisely, let Λ include all the preceding and subsequent controls, and let $\mathcal{V}$ be any unitary channel. For every pair of distinct eigenvalues x, x',*

$$\frac{1}{2}\|\Lambda - \mathcal{V}\|_\diamond \;\geq\; \frac{1 - |\chi(x' - x)|}{2}. \tag{14}$$

Proof. Choose eigenvectors $|x\rangle, |x'\rangle$ and the orthogonal states $|\pm\rangle = (|x\rangle \pm |x'\rangle)/\sqrt{2}$. Their trace distance is one before the coupling and $|\chi(x' - x)| < 1$ afterwards. All subsequent operations induce a system channel and cannot increase this distance. Pull the two inputs back through the preceding unitaries and call them $\rho_\pm$. If $e_\pm = D(\Lambda\rho_\pm, \mathcal{V}\rho_\pm)$, the triangle inequality gives $1 \leq e_+ + |\chi(x' - x)| + e_-$. Thus $\max e_\pm \geq (1 - |\chi|)/2$, which also bounds half the diamond norm from below. □

The independence concerns the entire later-accessible auxiliary system jointly, including any purification of the discarded port. A later register correlated with that port can reveal which phase was applied and allow its reversal; the system's reduced channel alone then

does not describe the available recovery operation. For a qubit and no later noise, the bound is attained: remove the phase of χ, and the remaining dephasing is the mixture $(1+|\chi|)\mathcal{I}/2 + (1-|\chi|)\mathcal{Z}/2$. Its half diamond distance from $\mathcal{I}$ is $(1-|\chi|)/2$.

Proposition 12 (An exact interface from a retained oscillator)**.** *Let $Q = b+b^\dagger$, $P = i(b^\dagger - b)$ satisfy the Weyl relations, so $[Q,P] = 2i$, and let X be Hermitian with finite spectrum. Then*

$$e^{-i\lambda XQ}e^{-i\mu XP}e^{i\lambda XQ}e^{i\mu XP} = e^{-2i\lambda\mu X^2} \otimes I. \tag{15}$$

Every initial oscillator state is returned exactly. The induced system map has multiplicative domain $M_s(\mathbb{C})$, including operators that do not commute with the coupling observable X.

Proof. In an X eigenblock, the two pulse generators have scalar commutator

$$[-i\lambda xQ, -i\mu xP] = -2i\lambda\mu x^2.$$

The Weyl group commutator is therefore that scalar phase. Summing the spectral blocks gives (15). □

For a qubit choose $X = (I+\sigma_x)/2$. With $\lambda\mu = \pi/2$, the loop is $e^{-i\pi X} = -\sigma_x$: an exact corrective flip, even with a vacuum or thermal oscillator input. The intermediate system–oscillator correlations are returned, not discarded. Conditional displacement loops of this kind are established geometric gates [24]; the point here is their implication for the interface. Continuous spectrum alone cannot prohibit exact control of a retained port.

Corollary 13 (One retained oscillator gives universal qubit control)**.** *Suppose the four pulses of Proposition 12 can be applied through either $P_z = (I+\sigma_z)/2$ or $P_x = (I+\sigma_x)/2$, with arbitrary real pulse areas. Every qubit unitary, up to global phase, is a sequence of at most three closed loops, and the oscillator returns after each loop.*

Proof. A loop with $2\lambda\mu = \alpha$ implements $e^{-i\alpha P_z} = e^{-i\alpha/2}e^{-i\alpha\sigma_z/2}$, and similarly for P_x. The z–x–z Euler decomposition of a qubit unitary therefore gives three loops. For example the three angles $\pi/2, \pi/2, \pi/2$ give a Hadamard up to global phase. Each identity holds on the full system–oscillator tensor product, so composition returns even an oscillator initially correlated with another system. □

For a larger system the same construction supplies the one-parameter groups e^{-itP_j}. Full unitary control follows when the real Lie algebra generated by iP_j is $\mathfrak{u}(s)$; generation of the associative matrix algebra alone is not the required controllability condition. The pulse permissions, including the choice between noncommuting projectors, remain physical inputs.

The loop is not costless. For a centred input and a mode of frequency ω, the largest conditional occupation increase along the four corners is $(\lambda^2+\mu^2)x^2$. Minimizing it for the qubit flip gives $\lambda = \mu = \sqrt{\pi/2}$ and peak mean stored energy $\pi\omega\langle X\rangle \le \pi\omega$. The mode's final energy is unchanged; an ideal reversible switch can recover the intermediate work. This is a stored-energy statement, not a total-device dissipation estimate. It assumes access to both quadratures, their controlled pulse order, and coherent retention until closure. A travelling field that has left the apparatus need not supply these permissions. An implementation with smooth local spacetime couplings must establish them separately.

Thus a record carried away by an inaccessible port constrains future control, while a coherently returned record can disappear without thermal erasure. Retaining an interface can replace a preparation-accuracy requirement by a closure requirement. The oscillator, its routing, clock and switching controls are part of that retained interface; they are not supplied by an abstract algebra homomorphism.

8 What must be retained by a material interface

The representation results answer an algebraic question about a specified interaction. A material interface must additionally let an acquired record change a later operation on the same body. The source then belongs to that joint dynamics; it is not assigned from the program's name.

For the next construction, fix a finite Dirichlet interval and retain all modes of the positive Gaussian operator K built in Appendix A.1. Its energy form contains the complete carrier square $\|\phi' + a\Theta\|^2$, the material gradient and a strictly positive material mass matrix. The interval gives a spectral gap. The field Hamiltonian is $H_F = d\Gamma(\sqrt{K})$ with one fixed vacuum subtraction. Projecting onto the bound material mode does not remove the orthogonal modes: their reaction is explicitly bounded in Proposition 23. The charged regulator in that appendix is a separate finite-resolution construction, not a proof that this static Gaussian realization equals the nonlinear charged continuum.

The following section keeps this complete K, the finite controller, its records and all old references. Completing the controlled field squares gives a positive Hamiltonian and a physical rms bound. The operator retarded equation makes acquisition affect the force, the actual field affects another sensor, and its new record can control the next force. The smooth interacting-source estimate retains both cross terms and connected fluctuations.

Receiver renewal has a narrower but attainable target. A prepared matched-gap Gibbs cell can replace one force-idle receiver, moving its old correlations into the spent bank. The material and the bank's purifier remain present. Section 10 gives this certificate and compiles a specified finite sequence on one cursor. Renewing the entire interacting equilibrium while preserving arbitrary held quantum content is a different task, with a product-extension obstruction.

The longer implementations establish additional permissions: full-ring reset and spectral sorting (Appendix C), a spatial work field and a separate massive-body gate (Appendix D), a guarded battery (Appendix E), local export (Appendix F), thermal preparation (Appendix G), and contact reaction/recoil (Appendix H). Their lattice, ring, copropagating and bidirectional carriers have different supports and Hamiltonians. Their estimates must be matched before they describe one apparatus.

9 Acquired quantum records on the complete material

The force calculation in Appendix A.2 conserves a classical record during its drive. An acquired record need not be classical in the full old history. Keep the complete finite-interval K and $H_F = d\Gamma(\omega)$, $\omega = \sqrt{K}$, of Section A.3. Let the finite controller include probes,

record cells and program cursor. Its nonnegative Hamiltonian H_D need not commute with the force amplitudes. The following construction therefore includes acquisition and a later record-controlled response within one autonomous dynamics.

9.1 A positive common Hamiltonian and a uniform physical moment

Let B_a be finitely many commuting bounded self-adjoint controller operators and h_a real smooth compact material smearings. Define

$$F = \sum_a B_a h_a, \qquad C_{\min} = \frac{1}{2}\sum_{ab} B_a B_b \langle h_a, K^{-1} h_b\rangle,$$
$$H_{\rm tot} = H_D + H_F - \sum_a B_a Q(h_a) + C_{\min}. \tag{16}$$

The full canonical material fields obey $[Q(h), P(k)] = i\langle h, k\rangle$, with $Q(h) = [a(\omega^{-1/2}h) + a^*(\omega^{-1/2}h)]/\sqrt{2}$. This is the canonical normalization of the material Hamiltonian, distinct from the earlier dimensionless $[b + b^\dagger, i(b^\dagger - b)] = 2i$ convention. The amplitudes commute at equal Schrödinger time, not necessarily at different Heisenberg times. In particular a cursor-controlled record force and a probe force can commute while acquisition edges in H_D change those same records.

Proposition 14 (Positive quantum interface and physical rms bound)**.** *The Hamiltonian* (16) *is self-adjoint and nonnegative. Its domain is equivalent to* $\operatorname{Dom} H_F$*. For a normalized initial vector* ψ *in that domain, including all old references, put*

$$L_F = \sqrt{2}\sum_a \|B_a\|\|\omega^{-1}h_a\|, \qquad c_F = \frac{1}{\sqrt{2}}\sum_a \|B_a\|\|\omega^{-1/2}h_a\|,$$
$$A_F = \|H_{\rm tot}\psi\| + \|H_D\| + \|C_{\min}\| + c_F, \qquad M_F = \left[\frac{L_F + \sqrt{L_F^2 + 4A_F}}{2}\right]^2.$$

Then $\|H_F e^{-itH_{\rm tot}}\psi\| \le M_F$ *for every* t*. If the initial cursor lies in a force-idle region and* $\|H_D\|$*, coupling norms and initial physical rms stay fixed, this certificate does not grow with the finite program length.*

Proof. Resolve the common spectrum of the B_a into projections P_b and write $f_b = \sum_a b_a h_a$, $z_b = -\omega^{-3/2} f_b/\sqrt{2}$. The finite interval has a gap; hence the finite controlled Weyl unitary $D = \sum_b P_b \otimes W(z_b)$ exists and $D^* H_F D = H_F - \sum B_a Q(h_a) + C_{\min}$. It follows that $H_{\rm tot} = D^* H_F D + H_D \ge 0$. The linear field bound

$$\|Q(h)\zeta\| \le \sqrt{2}\|\omega^{-1}h\|\|H_F^{1/2}\zeta\| + \|\omega^{-1/2}h\|\|\zeta\|/\sqrt{2}$$

also makes the force infinitesimally H_F-bounded, giving the graph domain statement by the closed operator realization. For $\psi_t = e^{-itH_{\rm tot}}\psi$ and $x = \|H_F\psi_t\|$, conservation of $\|H_{\rm tot}\psi_t\|$ gives $x \le A_F + L_F\sqrt{x}$, since $\langle H_F\rangle_t \le x$. Solving this quadratic inequality proves the bound. Spectral-core approximation extends the calculation to the stated domain. In a force-idle initial region, $\|H_{\rm tot}\psi\| \le \|H_F\psi\| + \|H_D\|$. Every step tensors with the actual reference identity. □

The displacement D proves positivity; it is not a free preparation of a dressed vacuum. This finite-interval argument is not an infinite-line static-dressing theorem. A positive nearest-neighbor Feynman cursor with bounded edge gates can have $\|H_D\| \le 4J + \|H_{\mathrm{data}}\|$ independently of its site count [4]. This removes a program-length factor from the physical moment bound, not from cursor space, execution time or preparation cost. For the uniform-J cursor, Foundation III's lemma "Wholly idle input and a finite completion window" in the appendix "A Finite Arrival Bound at Fixed Hopping Energy" bounds arrival for a packet supported wholly in the input idle region of the pure gate-dressed hopping comparison [50]. Cell gaps and forces require the subsequent Duhamel control; this is neither the spin-clock nor the spatial work-clock estimate below. Linear finite-system/field Hamiltonians and their domain analysis are established tools [5]; the issue here is their common retained-record interface.

The field obeys the operator retarded equation

$$Q(t) = \cos(\omega t)Q(0) + \omega^{-1}\sin(\omega t)P(0) + \int_0^t \omega^{-1}\sin[\omega(t-s)]F(s)\,ds. \tag{17}$$

It follows by solving $\dot{Q} = P$, $\dot{P} = -KQ + F(t)$ as weak smeared identities on the energy core. The $B_a(s)$ in $F(s)$ are the actual joint Heisenberg operators. Their expectations are not an externally supplied force history. With material-only force, the exterior carrier formula of Section A.2 holds with the complete operator $\Delta\Theta$:

$$\Delta\phi_R(u) = -\tfrac{1}{2}\int a(y)\Delta\Theta(y, u+y)\,dy, \qquad \Delta\phi_L(v) = \tfrac{1}{2}\int a(y)\Delta\Theta(y, v-y)\,dy.$$

An additional carrier-smeared probe contributes its own retarded source. Quadratic stress uses correlations of these operators, not only the square of their mean.

9.2 Actual acquisition, sensing and the next action

For a probe P and a blank record R, the unitary $C_Y = P_-^Y \otimes I_R + P_+^Y \otimes X_R$ acquires the probe's Y reading. Retain P, R and their old references. During a subsequent block that conserves R, the force of Section A.2 acts as $V = \sum_{r=0,1} |r\rangle\langle r| \otimes e^{ir^2\Phi_h} W(rz_h)$. For actual initial material/reference blocks $\sigma_{rr'}$, the output is

$$\sum_{rr'} |r\rangle\langle r'| \otimes e^{i(r^2 - r'^2)\Phi_h} W(rz_h)\sigma_{rr'}W(r'z_h)^*. \tag{18}$$

This is not a sampled classical force. For a pure complete input with both record branches present, dephasing R changes the full retained state by half trace distance $\sqrt{p(1-p)}$, where $p = \Pr(R=1)$. Indeed the deleted off-diagonal block connects two orthogonal vectors of norms $\sqrt{p}, \sqrt{1-p}$ and has two eigenvalues $\pm\sqrt{p(1-p)}$.

Now let a fresh sensor S start in $|+x\rangle$, couple by $e^{igZ_S Q(b)}$ to the material after its actual propagation, and acquire its Y value into a cell E. The latter starts with thermal occupation ν, independently prepared with its purifier retained. For a centered Gaussian

material initially independent of the first probe, let μ_r be the actual conditional mean of $Q(b)$ and V_b its conditional variance after the force. Then

$$\Pr(E = 1 \mid R = r) = \frac{1}{2} - \frac{1 - 2\nu}{2} e^{-2g^2 V_b} \sin(2g\mu_r). \tag{19}$$

The sensor's Y mean is $-\operatorname{Im}\langle e^{2igQ(b)}\rangle$; the Gaussian characteristic function and the mixed cell's visibility $1-2\nu$ give the formula. The acquisition acts on the same disturbed material, and the actually acquired n_E can control the next force. Its energy and source are part of that dynamics. For a correlated or non-Gaussian input use its actual characteristic function, not this Gaussian specialization.

Before sensing, a conserved source record and centered product input give $\langle \Delta j\rangle = p j_h$ but $\Delta\langle : j^2 : \rangle = p j_h^2$, where j_h is the complete driven carrier current. The connected contribution is $p(1-p)j_h^2$, not an extra energy charge. Similarly the full added material/carrier energy is pE_h, of which $p^2 E_h$ is the energy of the mean displacement. Record dephasing leaves this immediate field marginal unchanged while changing the old joint history. Equal mean source therefore does not establish equal later control.

The formulas describe the sequential block. Under the autonomous Hamiltonian (16), acquisition, force and sensing instead occupy cursor regions that a dispersive cursor can revisit. Its deadline statistics require the actual joint evolution, without postselection or assignment of the sequential probabilities.

A one-mode check retains the old probe reference and the second cell's purifier: $p = .2766071$, $\Pr(E = 1 \mid R = 1) = .4267590$, dephasing distance .4473205, and first-acquisition probe/reference change below 4.1×10^{-16}. Cutoffs $6, 10, 14$ reduce Gaussian-law error from 1.04×10^{-7} to 1.11×10^{-16}. Field energy .0418368 splits into mean .0115724 and connected .0302645; sensing adds $.04805 = g^2/2$ at $g = .31$. This checks the sequence, not the full spatial K theorem.

A separate 31-site positive cursor diagnostic executes actual first acquisition, record force, sensing, second acquisition, receiver/bank SWAP and a later force controlled by the spent cell's actual record. With both cell purifiers retained, its unpostselected deadline $t = 40$ has field energy .0300724, physical rms at most .186211 against the uniform bound 3.01330, and energy residual below 1.9×10^{-13}. But only .857524 of its cursor probability is after the SWAP region; its receiver occupation is .157712, exceeding the prepared bank value .13 by .027712. Unfinished and returning cursor branches retain pre-SWAP data and can revisit acquisition and force regions. The program therefore fails an exact deadline-renewal claim. The force-idle window condition and its completion error must be certified separately, as done below; the sequential probability formula is not assigned to this autonomous output.

9.3 The local field source during interaction

The free-output qualification in Corollary 26 is unnecessary for a specified smooth test under the same H_{tot}. Let $B(x)$ be a real symmetric smooth matrix, supported strictly inside the Dirichlet interval, with $\|B\|_\infty = G$; let $w \in C_c^\infty$ be real and supported in $[0, T]$. Use the same free-K vacuum subtraction and energy factors $\mathcal{Y} = (P, L_s Q)$, $L_s^* L_s = K$. Define

$$O_{\mathrm{int}} = \int w(t) \int [\mathcal{Y}(t,x)^T B(x) \mathcal{Y}(t,x)/2 - \text{free vacuum}_K]\, dx\, dt.$$

This is the material/carrier field source, not the controller's stress. By (17),

$$\mathcal{Y}(t) = \mathcal{Y}_0(t) + \Delta\mathcal{Y}(t), \qquad \Delta\mathcal{Y}(t) = \sum_a \int_0^t M_{t-s} h_a B_a(s)\, ds,$$

where $M_\tau h = (\cos(\omega\tau)h, L_s\omega^{-1}\sin(\omega\tau)h)$ and $\|M_\tau h\| = \|h\|$. As a vector of joint-state operators, $\|\Delta\mathcal{Y}(t)\psi\| \le W(t)\|\psi\|$, $W(t) = t\sum_a \|B_a\|\|h_a\|$.

Proposition 15 (An interacting-window field-source bound)**.** *Assume smooth coefficients of K and the smooth compact h_a above. For each a, define finite constants*

$$D_{P,a} = \sup_{0\le\tau\le T} \|\omega^2 [B M_\tau h_a]_P\|,$$
$$D_{S,a} = \sup_{0\le\tau\le T} \|\omega^2 L_s^* [B M_\tau h_a]_S\|.$$

With $\omega_i \ge c_- i$ as in Section A.3, set

$$C_0(t) = \frac{t}{\sqrt{2}} \sum_a \|B_a\|[D_{P,a} c_-^{-3/2}\zeta(3/2) + D_{S,a} c_-^{-5/2}\zeta(5/2)],$$
$$C_1(t) = \frac{t}{\sqrt{2}} \sum_a \|B_a\|[D_{P,a} c_-^{-2}\zeta(2) + D_{S,a} c_-^{-3}\zeta(3)].$$

For any $\Lambda > 0$, O_{int} is a Hermitian form with

$$\|(H_F + \Lambda)^{-1/2} O_{\mathrm{int}} (H_F + \Lambda)^{-1/2}\| \le C_{\mathrm{int}}, \tag{20}$$

where

$$C_{\mathrm{int}} = C_O + \int |w(t)| \left[\frac{C_1(t)}{\sqrt{\Lambda}} + \frac{C_0(t) + GW(t)^2/2}{\Lambda}\right] dt.$$

Here C_O is the free source constant with time test w. All references and the finite controller are retained. For two initial joint vectors of rms at most M_F and vector distance e, their actual interacting-window source difference is at most

$$2C_{\mathrm{int}}\sqrt{\Lambda e^2 + 2M_F e}\sqrt{\Lambda + M_F}. \tag{21}$$

Proof. The free part is Proposition 25. The response-square term is bounded by $GW(t)^2/2$ without assuming commuting response components. For the symmetrized cross term, let $v_i(t,x)$ be the free annihilation coefficient of mode i and $C_i(t) = \int v_i(t,x)^T B(x)\Delta\mathcal{Y}(t,x)\, dx$. Projection of the two response components onto e_i and insertion of ω^2 give

$$\|C_i(t)\| \le \frac{t}{\sqrt{2}} \sum_a \|B_a\|[D_{P,a}\omega_i^{-3/2} + D_{S,a}\omega_i^{-5/2}].$$

Interior multiplication by smooth B removes boundary compatibility conditions. Iterating the elliptic operator $K = -I\partial_x^2 +$ smooth lower-order terms controls the necessary Sobolev orders of propagated compact h_a, proving finiteness of the displayed graph norms uniformly over the compact time interval. Thus $\sum_i \|C_i(t)\| \le C_0(t)$ and $\sum_i \|C_i(t)\|/\sqrt{\omega_i} \le C_1(t)$.

For $E = \langle H_F\rangle_\psi$, $\|a_i\psi\| \le \sqrt{E/\omega_i}$ and $\|a_i^*\psi\| \le \sqrt{E/\omega_i} + 1$. Bounding $\langle a_i C_i\rangle$ by $\|a_i^*\psi\|\|C_i\psi\|$, and $\langle a_i^* C_i^*\rangle$ by $\|a_i\psi\|\|C_i^*\psi\|$, bounds the cross expectation by $2C_1(t)\sqrt{E}+C_0(t)$. No commutation between C_i and a_i has been used. The C_0 term is the creation contribution; treating the record as a c-number would incorrectly remove it. Apply $2\sqrt{E} \le (E+\Lambda)/\sqrt{\Lambda}$ and add the free and response bounds. Absolute convergence of the cross terms, the bounded response vector and the established free form give spectral-core convergence. Polarization proves (20). The weighted-form difference estimate of Corollary 26, with $\|H_F(\psi-\varphi)\| \le 2M_F$, proves (21). □

The constants do not depend on cursor site count if its coupling norms stay fixed. They do depend on the box, smooth test, profiles and time window. Rough spatial multipliers, boundary distributions and unsmeared stress are not covered. This result compares retained inputs under one common $H_{\rm tot}$; it does not compare arbitrary different apparatus Hamiltonians or supply the controller's missing stress.

9.4 A local contact energy and the interacting thermal reference

The inverse-K counterterm in (16) is useful for proof but is not a local physical density. For smooth compact material profiles $w_a(x)$ choose $h_a = (0, M(x)w_a(x))$ and replace it by

$$C_{\rm loc} = \frac{1}{2}\sum_{ab} B_a B_b \int w_a^T M w_b \, dx.$$

The actual mass-potential density becomes

$$V_{\rm loc}(x) = \tfrac{1}{2}[u - \sum_a B_a w_a]^T M [u - \sum_a B_a w_a]. \tag{22}$$

All gradient and current squares of the complete material remain. For each joint spectral branch $w = \sum_a b_a w_a$, the inverse-form variational formula gives

$$\langle (0,Mw), K^{-1}(0,Mw)\rangle = \sup_X \{2\langle w, Mu\rangle - q[X]\} \le \langle w, Mw\rangle,$$

because $q[X] \ge \langle u, Mu\rangle$. Consequently $C_{\rm loc} - C_{\rm min} \ge 0$ is a bounded controller operator. The actual local-contact Hamiltonian is $D^* H_F D + H'_D$ with $H'_D = H_D + C_{\rm loc} - C_{\rm min} \ge 0$. The positivity, rms and source proofs therefore apply to this physical contact, using its actual $C_{\rm loc}$ and the adjusted controller norm. No local stress is assigned to a nonlocal inverse operator.

Proposition 16 (Interacting equilibrium and retained record capacity)**.** *For either contact, put $d = \dim \mathcal{H}_D$, $d_0 = \|H'_D\|$, and $Z_F(\beta) = \operatorname{Tr} e^{-\beta H_F}$. Then the actual interacting Gibbs state exists and*

$$\begin{aligned} d e^{-\beta d_0} Z_F(\beta) \le Z_{\rm tot}(\beta) \le d Z_F(\beta), \\ \langle H_{\rm tot}^2\rangle_\beta \le \frac{16 e^{\beta d_0}}{e^2\beta^2} \frac{Z_F(\beta/2)}{Z_F(\beta)}. \end{aligned} \tag{23}$$

If independent uniform input bits $X_1, \dots, X_N$ are retained in the accessible controller and material with field mean energy at most M, and each is later readable with error $\epsilon_j \le 1/2$, then

$$\sum_j [\log 2 - h_2(\epsilon_j)] \le \log d + \inf_{\beta>0} [\beta M + \log Z_F(\beta)]. \tag{24}$$

Any decoder-accessible outside reference is an additional resource.

Proof. The eigenvalues of $D^* H_F D + H'_D$ lie between those of $D^* H_F D$ and those eigenvalues plus d_0, by min–max. Summing scalar exponentials proves the partition bounds; operator monotonicity of the exponential is neither available nor needed. For $x \ge 0$, $x^2 e^{-\beta x/2} \le 16/(e^2\beta^2)$. The partition ratio then proves the moment bound; its controller multiplicity cancels. Subadditivity and the field Gibbs variational inequality give $S(\rho) \le \log d + \beta M + \log Z_F(\beta)$. Binary Fano, conditional entropy subadditivity for the independent classical inputs, and data processing give $\sum_j [\log 2 - h_2(\epsilon_j)] \le I(X^N : \text{apparatus}) \le S(\rho)$. Take the infimum. The frequency bound also yields $\log Z_F(\beta) \le q/(1-q)^2$, $q = e^{-\beta c_-}$, so the capacity is finite for fixed resources. □

The same-H Gibbs purification is a graph-domain input, not the product Gaussian in (19). Stationary equilibrium supplies neither directed cursor nor blank records; $\beta^{-1} D(\rho_{\text{prep}} \| \rho_\beta)$ is a finite running state's free-energy excess, not its preparation protocol. At fixed energy, cursor dimension increases record capacity only logarithmically. Any outside history bank or entropy channel remains a resource.

10 Finite renewal and continuing return

10.1 Replacing a port is not renewing the interacting material

There are three distinct thermal operations below: replacement of a specified finite port; local thermal restriction on a free-carrier window; and preparation of the whole interacting Gibbs state. Only the first two have the stated finite operating protocols.

Proposition 17 (Universal held-content preservation forces a product extension)**.** *Let a channel $\mathcal{E}$ append a finite-dimensional port B to a held d-dimensional system A, whose marginal is the identity channel on every input, including all reference extensions. Then $\mathcal{E}(\rho) = \rho \otimes \tau$ for one fixed port state τ. Thus such a universal nondisturbing preparation cannot reproduce a correlated interacting Gibbs state. For $H = J Z_A Z_B$, $x = \beta J$, its Gibbs state is $\gamma = (I - \tanh(x) Z_A Z_B)/4$. At held input $I/2$ every such extension has half trace distance from γ at least $|\tanh x|/2$, attained by $\tau = I/2$.*

If instead only the classical Z_A label is conserved while conditional port Gibbs states $\tau_\pm = (I \mp \tanh(x) Z_B)/2$ are installed, the largest retained coherence factor is $\operatorname{sech} x$*. After the best held-system phase correction, an input $|+\rangle$ therefore suffers half trace distance at least $(1 - \operatorname{sech} x)/2$.*

Proof. Dilate the channel and purify its fixed ancilla. A pure held input must have a pure unchanged held marginal; hence the isometry sends $|\psi\rangle$ to $|\psi\rangle|e_\psi\rangle$. Applying this to basis

states and their superpositions forces one common environment vector. Tracing the environment gives the product extension. Equivalently the identity channel has rank-one Choi state. For the Gibbs example the observable $Z_A Z_B$ has norm one, expectation zero in $(I/2)\otimes\tau$ and $-\tanh x$ in γ, giving the distance bound; direct diagonal subtraction proves attainment. In a Z-nondemolition conditional preparation the two environment/port vectors purify τ_+, τ_-. Their overlap is at most their root fidelity, $\sum_z \sqrt{p_+(z)p_-(z)} = \operatorname{sech} x$, with equal Schmidt bases attaining it. That overlap multiplies the held off-diagonal matrix element. The qubit trace-distance formula proves the last assertion. □

This is a universal identity-channel condition, not a prohibition on preparing one specified state or maintaining a selected classical content. It supplies no universal constant for arbitrary approximate identity channels. An entire-body SWAP can prepare a fresh correlated copy, but it moves the old body's quantum content and correlations to the spent copy; it is not that identity channel.

For an attainable finite update, specify instead one receiver cell E with $H_E = \delta n_E$ and a fresh bank cell B with the same gap, Gibbs state η_β, and retained purifier. All held material, earlier records and old references form $\mathcal{O}$; the previous state of E may be correlated with all of them. The true S_{EB} commutes with $H_E + H_B$. The positive pulse $g(t)(I - S_{EB})$, $g \geq 0$, $\int g = \pi/2$, implements the true SWAP at finite duration, at least $\pi/(2g_{\max})$. At its endpoint E is η_β independently of the old $\mathcal{O}E$ state. That complete old state now belongs to $\mathcal{O}B$, with its correlations retained. The new cell remains correlated with its own Gibbs purifier. The $\mathcal{O}$ marginal is unchanged nonselectively; old records stored in E have moved to the spent slot rather than being erased. For an isolated prescribed pulse its switch work is zero because the bare cell energy is conserved and the pulse vanishes at the endpoints. This is not zero total controller or preparation cost.

Such a SWAP is also a bounded edge gate in H_D of the positive common material Hamiltonian. The cell must be force-idle while it is replaced. If n_E is still coupled to the material, $[S_{EB}, n_E Q(h)] \neq 0$; the matched gap alone gives no exact reset for that interacting contact. A finite-window estimate can retain this imperfection explicitly.

Proposition 18 (Finite-window port renewal with a live material)**.** *Compare the actual autonomous interface with an ideal renewal evolution having the same H_D, matched cell gaps and SWAP edge gate, but no material force in its renewal window $[0, T_R]$. Suppose their difference on the reference trajectory is $-\sum_a \Delta B_a Q(h_a) + \Delta C$, where $\Delta B_a = P_{\text{leak}} \Delta B_a$, $\Delta C = P_{\text{leak}} \Delta C$ and these controller operators commute with H_F. Here P_{leak} is a controller orthogonal projection; the differences are self-adjoint, so their displayed one-sided support is also two-sided support. Write $b_a = \|\Delta B_a\|$, $c = \|\Delta C\|$, $p(s) = \|P_{\text{leak}} \psi_s^i\|^2$, and assume $\|H_F \psi_s^i\| \leq M$. Define*

$$L_\Delta = \sqrt{2} \sum_a b_a \|\omega^{-1} h_a\|, \qquad c_\Delta = c + \sum_a b_a \|\omega^{-1/2} h_a\| / \sqrt{2}.$$

Including the bank and every old reference, the vector error is at most

$$\epsilon_R \leq \int_0^{T_R} [L_\Delta \sqrt{M}\, p(s)^{1/4} + c_\Delta \sqrt{p(s)}]\, ds. \tag{25}$$

Any specified incomplete ideal-SWAP or clock error is added separately. The same bound controls half trace distance of the renewed cell and held-reference marginal from their ideal outputs.

Proof. Duhamel bounds the vector difference by the integrated norm of the Hamiltonian difference on ψ_s^i. Apply the linear-field estimate to $P_{\text{leak}}\psi_s^i$. Its norm is $\sqrt{p}$ and

$$\|H_F^{1/2} P_{\text{leak}}\psi_s^i\|^2 \le \|P_{\text{leak}}\psi_s^i\|\|H_F P_{\text{leak}}\psi_s^i\| \le M\sqrt{p}.$$

The field and counterterm contributions give (25). Pure-vector distance bounds half trace distance, which contracts under the stated restrictions. □

Corollary 19 (A constructive finite receiver-renewal family)**.** *Fix $A > 0$, $0 < p < 1/2$, a matched receiver/bank pair and a material smearing h. For even m, use the positive spin-history clock with $J = A/m$, a true receiver/bank SWAP on the first edge and identities thereafter. Put $\chi_m = \mathbf{1}_{\{j \ge m/2\}}$ and*

$$H_m = H_{\text{hist}} + H_E + H_B + H_F - \chi_m n_E Q(h) + \tfrac{1}{2}\chi_m n_E \langle h, K^{-1}h\rangle.$$

Its initial cursor is $j = 0$; the material/old receiver state and all old references may be correlated, with initial physical rms $M < \infty$. The independent bank is Gibbs with its own purifier retained. This is a positive autonomous Hamiltonian. At $t_R = (m/A)\arcsin\sqrt{p}$ its joint half trace distance from a completed receiver renewal, with an explicitly specified comparison clock, is at most

$$2(1-p)^{m/2} + \frac{m \arcsin\sqrt{p}}{A}[L_\Delta \sqrt{M}\, e^{-mD/4} + c_\Delta e^{-mD/2}], \quad D = D_{\text{Ber}}(1/2\|p) > 0. \tag{26}$$

It tends to zero while the clock mean energy A, its norm bound $2A$ and the common-H physical rms certificate remain bounded independently of m. Clock space and duration grow; the bank is not cycled for free.

Proof. The field/contact terms are the positive controlled-displacement Hamiltonian of Proposition 14; the history clock and matched cell energies are nonnegative. For the force-free reference, each prefix is either the true SWAP or identity and commutes with $H_E + H_B + H_F$. Prefix conjugation therefore gives the same binomial cursor law for every correlated input. For $s \le t_R$ its binomial parameter is at most p; Chernoff gives $\Pr(j \ge m/2) \le e^{-mD}$. The ideal physical rms is the unchanged M. Proposition 18 proves the second term. The no-SWAP probability is $(1-p)^m$. As comparison use the same untrimmed force-free cursor with the completed SWAP and freely evolved material/reference state. Only its $j = 0$ branch differs from the ideal joint state; the retained-purification vector error is at most $2\sqrt{(1-p)^m}$. Half trace distance is no larger. No clock outcome is postselected. Triangle inequality proves (26). Finally $\|H_{\text{hist}}\| \le 2A$ and its initial mean is A. The cell norms and force/counterterm norms are fixed, so the initial total rms and the physical bound of Proposition 14 are uniform. □

This condition has a finite nonempty realization. Take the positive spin-history clock with $m = 64$, $J = .02$, the matched SWAP on its first edge, and a material force $-.2\,\mathbf{1}_{\{j\ge 32\}} n_E Q(h)$

with $\omega = 1$ and unit one-mode smearing. At $T_R = \arcsin\sqrt{.1}/J = 16.08753$ the force-free reference cursor has Bin$(64, .1)$ law; throughout this window its source tail is at most 7.04652×10^{-16}. Its uncompleted first-edge probability is $.9^{64} = .00117902$. For vacuum material input $M = 0$, (25) is below 6.9×10^{-8}. Completing the ideal SWAP on the same untrimmed cursor costs at most $2\sqrt{.9^{64}} = .068674$ in vector norm. The clock's mean energy is 1.28. Direct sparse evolution of the full force-on Hamiltonian at Fock cutoffs $6, 10, 14$, retaining a correlated held-body/receiver input, its old reference and the fresh bank purifier, gives vector difference 6.54×10^{-10} from the same force-free reference. The receiver's half trace distance from its bank state is .00043624; the held-body/reference marginal changes by less than 2.5×10^{-13} and energy conservation residual is below 4.1×10^{-13}. These are finite-regulator diagnostics of the nonempty renewal window, not a continuum proof. This is a paid receiver-renewal block with given correlated input, not an assertion that a previous program produces its cursor for free.

Actual leakage and completion require paid force-idle padding. The receiver certificate preserves held material and old content, but neither prepares its interacting Gibbs state nor implements an entropy outlet. A J-cell bank retains J spent slots, outputs and purifiers; each isolated replacement transfers excess energy $\delta(\langle n_E\rangle - \nu)$ of either sign. Preparation, clock, storage and removal remain charged. Q-RAIF B's oscillator-bank formulas [26] concern its different material. Repeated noisy readings of one command also do not justify $J \log 2$ in the independent-bit capacity bound (24).

10.2 Several rounds on one running material

The next reading can be installed on the same cursor that performs replacement. Fix a finite sequence of permitted data gates U_b, including receiver–bank swaps, and a nonnegative data energy H_0 with $[U_b, H_0] = 0$. Degenerate logic and matched-cell exchanges meet this condition. Keep the body, all banks, their purifiers, previous records and reversible workspace in the input. Between gates use the linear material forces and completing terms of Eq. (16).

Choose rational $0 < a_1 < \cdots < a_q < 1$ and integers m for which $k_b = ma_b$ are integral. Put U_b on edge k_b of the spin clock; other edges are identities. Each cursor interval has its own fixed force amplitudes, commuting within that interval. Orthogonal cursor supports make the full amplitudes commute. With J fixed, the resulting $H_m \geq 0$ is time independent and has common domain $\operatorname{Dom} H_F$. Its comparison schedule is

$$p(t) = \sin^2(Jt), \qquad t_b = J^{-1} \arcsin\sqrt{a_b}, \qquad T = \pi/(2J).$$

It applies the specified gates at t_b while the same complete material evolves throughout. Only the comparison uses these switching times; the actual apparatus has its gates and forces fixed in space along its cursor.

Proposition 20 (A finite sequence with its history retained)**.** *Let the smooth force profiles satisfy $h_\ell, \omega^{1/2} h_\ell \in L^2$, and fix an input purification with $\|H_F\psi\| \leq M_0$, including arbitrary old references. The actual H_m approximates the scheduled sequence jointly with its complete, untrimmed cursor, at every fixed checkpoint away from the t_b. At T all data gates have been included. The joint vector error tends to zero as $m \to \infty$, uniformly on this input class.*

Both actual and comparison material rms energies are bounded by a constant independent of m.

More explicitly, set

$$L_P = \sqrt{2}\sum_\ell \|B_\ell\|\|h_\ell\|, \qquad c_P = \frac{1}{\sqrt{2}}\sum_\ell \|B_\ell\|\|\omega^{1/2}h_\ell\|.$$

For any energy $E_* > 0$ a common rms bound is

$$M_T = \left[\sqrt{M_0 + E_*} + \frac{T}{2}\left(L_P + \frac{c_P}{\sqrt{E_*}}\right)\right]^2 - E_*. \tag{27}$$

In the prefix gauge, let H_b be the material/data Hamiltonian in interval b, and choose K_b so that $\|(H_b - H_{b-1})\zeta\| \le K_b$ for normalized vectors with $\|H_F\zeta\| \le M_T$. These constants are finite by the linear field bound. Then the gauge-state error satisfies

$$e_m(t) \le \sum_{b=1}^{q} K_b \int_0^t e^{-m[p(s)-a_b]^2}\, ds. \tag{28}$$

At a physical checkpoint add $2\sum_b e^{-m[p(t)-a_b]^2}$; this additional term is zero at T. For $0 < \alpha_b < \min(a_b, 1-a_b)$ and $v_b = 2J\sqrt{(a_b-\alpha_b)(1-a_b-\alpha_b)}$,

$$e_m(T) \le \sum_b K_b \left[\frac{\sqrt{\pi}}{v_b\sqrt{m}} + Te^{-m\alpha_b^2}\right]. \tag{29}$$

Proof. The prefix conjugation used for the history clock transforms its hopping term to $mJ + 2JJ_x$ and each force to the corresponding prefix-conjugated data amplitude. The transformed material Hamiltonian is piecewise constant in j. Compare it with the product of the untrimmed spin-clock state and the data/material sequence H_b. The clock distribution is $\mathrm{Bin}(m, p(t))$. Writing both piecewise Hamiltonians as sums of their jumps gives

$$H_m^{\mathrm{gauge}} - H_{\mathrm{cmp}}(t) = \sum_b (\mathbf{1}_{\{j\ge k_b\}} - \mathbf{1}_{\{t\ge t_b\}})(H_b - H_{b-1}).$$

The norm of each clock factor on the comparison state is the square root of its wrong-side probability. Hoeffding's bound is $e^{-m[p(t)-a_b]^2}$. Duhamel's formula therefore proves (28). On $|p - a_b| \le \alpha_b$ one has $p' \ge v_b$; integrate the Gaussian there and bound its complement by $Te^{-m\alpha_b^2}$ to obtain (29). Undoing the gauge at a checkpoint costs at most twice the square root of the probability of an incorrect prefix. At T the reference cursor is exactly $|m\rangle$, giving no prefix error.

For the energy estimate, all controller terms commute with H_F, even though their norms grow with m. On the energy core, $[H_F, Q(h)] = -iP(h)$ and

$$\left\|\sum_\ell B_\ell P(h_\ell)\zeta\right\| \le L_P\|H_F^{1/2}\zeta\| + c_P\|\zeta\|.$$

For $x(t) = \|H_F\psi_t\|$ this gives $D^+x \le L_P\sqrt{x} + c_P$. Hence $D^+\sqrt{x + E_*} \le (L_P + c_P/\sqrt{E_*})/2$, which proves (27). The data gates commute with H_F and cause no jump in this bound. Finite-mode energy-core regularization gives the same estimate before removing the cutoff; the common-domain propagator limit and closed-operator lower semicontinuity extend it to $\operatorname{Dom} H_F$. The linear field bound then supplies the K_b used above. Every estimate acts on the full purification and tensors with the old reference identity. □

For prescribed finite N, compile acquisition, decision, record force and bank swap on this path. Proposition 58 supplies one decision circuit from actual meter records and retained posterior weights; its repair flag can drive the force. Registers and thermal pairs are prepared initially, and spent correlations remain. At checkpoints, (21) and (27) certify a later reading and source test under the same subsequent H_m.

As a finite diagnostic, two probe readings of one oscillator, an intervening receiver–bank swap and a later force controlled by both records give endpoint joint vector errors .0701594, .0355814, .0192136, .0107206, .00393182 at $m = 16, 32, 64, 128, 512$, with $J = 1$. An old probe reference and the spent bank are retained. At $p = 7/16$, after the swap and before the second probe interval, receiver excitation is .0321012 for $m = 128$ and .00089552 for $m = 512$; endpoint accuracy alone would miss this timing error. The $m = 128$ final coordinate is .0393364 against the scheduled .0393304; removing the record-controlled forces gives -2.89×10^{-6} against zero. Fock cutoffs 8 and 10 change the endpoint vector error by 1.1×10^{-11}. These are regulator checks of the construction.

The clock uses $m + 1$ states, mean energy mJ, variance mJ^2, norm $2mJ$ and maximal edge coefficient at most $J(m+1)/2$. Fixed duration thus costs growing energy; fixed-energy renewal instead costs growing duration. Neither is indefinite fixed-capacity operation, and this material-source bound excludes controller covariant stress.

Appendix K gives a further use of this program: local linear-current pulses implement stress flows in the neutral $c = 1$ carrier, uniformly on bounded mean-energy inputs. The construction supplies a physical permission used in finite modular-geometry tasks, with explicit control and coherence conditions.

The comparison clock need not have the actual clock's energy. For one oscillator take $H_F = \omega a^\dagger a$, $Q = (a + a^\dagger)/\sqrt{2}$, $V = -bQ + C$ and $C = b^2/(2\omega)$. Use identity clock edges and turn on V for cursor sites $j \ge m/2$, with even m. The positive Hamiltonian is $H_m = H_{\mathrm{clock,m}} + H_F + \mathbf{1}_{j\ge m/2}V$. Starting in $|0\rangle_{\mathrm{clock}}|0\rangle_F$, its conserved total energy is mJ. The scheduled endpoint is $|m\rangle \otimes \zeta$ up to phase, where $\zeta = e^{-i(T/2)(H_F+V)}|0\rangle$ and $\langle H_F + V\rangle_\zeta = C$. Consequently

$$\langle H_m\rangle_{\mathrm{scheduled}} - \langle H_m\rangle_{\mathrm{actual}} = C \quad \text{for every even } m. \tag{30}$$

The joint-state error still tends to zero. The finite work is paid by the actual clock and missed by its unperturbed product comparison.

This defect can be repaired for the total-energy observable. Rotate the terminal spin toward negative S_x by an angle θ. Its clock energy is $mJ - mJ\sin\theta$, and its vector distance from $|m\rangle$ is at most $|\theta|\sqrt{m}/2$. If $\epsilon_m(\theta)$ is its probability below $m/2$, the product state's total-energy excess is $-mJ\sin\theta + C - \epsilon_m(\theta)\langle V\rangle_\zeta$. It equals C at zero and is negative at $\theta = \arcsin[2C/(mJ)]$ for sufficiently large m: the last probability is at most $e^{-3m/8}$ once $\theta \le \pi/6$. Continuity therefore supplies an exactly energy-matched comparison state with an additional $O(m^{-1/2})$ vector error. This is a physical comparison vector, not an extra gate

supplied by the original dynamics. Matching one energy expectation does not match an extended stress distribution. HAFF-D uses the mass-work identity of its common action to keep this debit in a finite source comparison [23].

10.3 Local thermal windows and a continuing diagnostic return

Port replacement is distinct from using a later region of a correlated field. Appendix I.1 executes a local thermal preparation with a paid finite autonomous clock. Appendix I.2 proves its record-selected same-field conditional error from vacuum mutual information and controls the source with second moments. The complete outside purification stays present; a spatially separate window is not declared globally fresh. These free-CAR certificates require a new thermal matching proof before use on the nonlinear interacting material.

Appendix J.1 gives a separate finite-horizon return in which the actual meter word, not an inaccessible fault label, selects repair of the held body and a thermal exchange. Appendix J.2 supplies an actual terminal record and replacement of its specified probe. The full input bank, posterior arithmetic, reversible garbage, clock energy and retained output history belong to that horizon. The actual mixed-port quantum-material update above does not inherit their different deadline or mean-support guarantees.

11 A common covariant ledger: controller, clock and support

The local potential (22) has potential stress $-g_{\mu\nu}V_{\mathrm{loc}}$ if its variables are Lorentz scalars. That term alone omits the mechanism which moves its profile and changes its amplitude. A common local action makes this omission explicit and offers a candidate completion at the classical action and regulated canonical levels.

Take dynamical scalars $X = (\phi_1, \phi_2, Z, \theta, y, \chi, r_1, \dots, r_d)$ on a $(1+1)$-dimensional background of signature $(-,+)$. Here Z is a support texture, χ a clock coordinate, and r_j memory/control coordinates. Specify the positive target coframe

$$\begin{aligned} e_{\phi_a} &= d\phi_a + \lambda_a \theta f(Z) dZ, & e_Z &= m_Z dZ, \\ e_\theta &= \sqrt{I}\, d\theta, & e_y &= \sqrt{J}\, dy, \\ e_\chi &= \sqrt{I_\chi}\, d\chi, & e_{r_j} &= \sqrt{I_j}\, dr_j. \end{aligned}$$

with all diagonal coefficients strictly positive, and put $G = \sum e \otimes e$. Let $b_a(\chi, r)$ be smooth bounded functions, $w_a(Z)$ smooth profiles compact in the texture value, and $M(Z)$ a smooth strictly positive material matrix. With $u = (\sqrt{I}\theta, \sqrt{J}y)$ and $v = u - \sum_a b_a(\chi, r) w_a(Z)$, choose

$$S = -\int \sqrt{-g}\, [\tfrac{1}{2} G_{AB}(X) g^{\mu\nu} \partial_\mu X^A \partial_\nu X^B + V(X)]\, d^2x, \tag{31}$$

$$V = V_Z(Z) + V_{\mathrm{mem}}(r) + V_{\mathrm{clock}}(\chi) + \tfrac{1}{2} v^T M(Z) v \geq 0. \tag{32}$$

The last square replaces the material mass potential rather than adding a second copy of it. Z, χ, r are actual fields in this candidate, not prescribed positions, waveforms or cursor

amplitudes. A prepared finite-energy texture can localize the contact; localization is not produced by writing $w_a(Z)$.

Proposition 21 (Common-action stress and clock reaction)**.** *For smooth finite-energy classical data of* (31)*, metric variation gives*

$$T_{\mu\nu} = G_{AB}\partial_\mu X^A \partial_\nu X^B - g_{\mu\nu}[\tfrac{1}{2} G_{AB}\partial X^A \cdot \partial X^B + V]. \tag{33}$$

The Euler–Lagrange equations are

$$\nabla_\mu (G_{AB}\partial^\mu X^B) - \tfrac{1}{2}(\partial_A G_{BC})\partial X^B \cdot \partial X^C - \partial_A V = 0. \tag{34}$$

They imply $\nabla^\mu T_{\mu\nu} = 0$ for the complete matter action. In specified Minkowski coordinates its canonical energy is

$$H = \int [\tfrac{1}{2}\Pi^T G^{-1}\Pi + \tfrac{1}{2} X'^T G X' + V]\, dx \geq 0.$$

At finite Fourier resolution, the following specified canonical ordering has a nonnegative Friedrichs realization on the supplied charge sectors and finite real coordinate space.

Proof. Vary the inverse metric with all scalar variables held fixed and use $T_{\mu\nu} = -2(\delta S/\delta g^{\mu\nu})/\sqrt{-g}$. Integration by parts in a field variation gives (34); contraction with $\partial_\nu X^A$ gives the stress divergence. In Minkowski coordinates $\Pi_A = G_{AB}\dot{X}^B$, so the Legendre transform has the displayed positive squares. The coframe is nonsingular and triangular. For a spatial Fourier regulator, let q denote its finite real coordinates and compact scalar zero modes. The reduced Lagrangian has positive kinetic matrix $G_K(q)$; put $A_K = G_K^{-1}$ and let $W_K(q) \geq 0$ be its integrated gradient and potential energy. Use Lebesgue measure on the real coordinates and the scalar zero-mode line bundles of Proposition 22, with their charge/winding direct sum. On smooth compactly supported sections choose the divergence-form kinetic ordering

$$\mathfrak{q}_K[\psi] = \tfrac{1}{2}\int (D_i\psi)^*(A_K)^{ij}(D_j\psi)\, dq + \int W_K|\psi|^2\, dq.$$

The D_i are coordinate derivatives with the stated zero-mode boundary conditions. Smooth locally elliptic coefficients make this densely defined nonnegative form closable; its Friedrichs realization is the specified operator. If the interaction acts only on retained low modes, the complete free carrier tail is added positively, as in the charged regulator. This choice of canonical measure and ordering is not a claim of regulator-independent covariant quantization. □

In particular,

$$I_\chi \Box\chi = \partial_\chi V = V'_{\text{clock}} - \sum_a (\partial_\chi b_a) w_a^T M v.$$

The amplitude change reacts on the clock. The memory equations have the corresponding $\partial_{r_j} b_a$ force, and the support equation includes derivatives of w_a, M and the target coframe. Dropping these terms returns an externally driven model. For a free massless clock initially

having $\partial_t \chi = \Omega$ over a proper-length interval ℓ, its initial kinetic energy is at least $I_\chi \Omega^2 \ell/2$. Smooth spatial tapers and contact reaction add costs and change that free trajectory. A locally timelike clock gradient, texture, memory preparation and boundary conditions are specified initial data; they are not guaranteed throughout an arbitrary run.

The action includes clock/support reaction; canonical positivity is only in the chosen coordinates and finite regulator. Curved-space quantum positivity, renormalized continuum stress and causal propagation after cutoff removal require further proofs. Matching (16) requires quantum acquisition and transported gates, the prepared texture/charge sectors, and a common finite-cursor/source estimate. The Gaussian K needs static supplied Z and controlled freezing or small perturbations of r, χ; dynamical cross terms and backreaction otherwise change H. That approximation needs an error bound, as does matching HAFF D's dilaton coordinates and full supported quantum state.

12 Discussion

The exact representation problem and the material return problem now have different tests. A discarded-port channel is checked by its multiplicative domain; a closed propagation symbol by its shell and coefficient relations. An aligned actuator can share the latter generators and charge their spatial form while still losing exact reduced control. The common-action and finite-return constructions ask whether the records, permissions and sources needed for these operations can be implemented together.

Acquisition is a physical change of the old history, not a supplied source label. The actual record drives the same complete material and produces another reading. Its connected fluctuations contribute to the source; dephasing can preserve an immediate field marginal while changing later joint control. Physical moments and smooth-source forms certify that return without replacing an unbounded stress by a trace-distance estimate.

Renewal depends on what is held. Universal identity-channel preservation cannot append a correlated interacting equilibrium. A specified receiver can nevertheless be replaced by a paid Gibbs bank, with spent correlations retained; the full ring can be reset and its mixed target reinstalled through four thermal outlets. Neutral compression, packet loading and spectral sorting supply a complete-output converter, not merely thermal mean energy. Isospectral axes need not retain a heat-source contrast after optimal sorting, and accurate direct reset alone leaves its purifier redundancy. Persistent sharp-exhaust gain remains a joint implementation question.

The clock distinctions in the introduction fix the resource claims: bounded-energy slow receiver renewal, fixed-duration energy-growing finite rounds, and a spatial work field with horizon-dependent inertia. The full joint-state estimate for the latter retains every old reference and charges finite clock work in the source comparison. Static frame/endpoints and dynamical support are still different problems. The neutral-current stress compiler gives another finite physical control permission, with coherence and precision budgets; it is not noise-protected control over every charged sector.

Foundation III supplies a complementary finite semiclassical mean-action scalar/JT return at common initial constraints: a learned frame protects old content, its record changes the material source, and a later X-basis reading of the old probe carries the geometric re-

Interface question	Established construction	Required input or remaining bridge
Exact operations	Multiplicative domain; propagation symbol; retained loops	Spectral law, access and quadratic-space conditions
Calibrated charging	State-independent work with actual generator alignment	Centered independent input; commuting drift; a measured command–momentum map
Actual quantum return	Acquisition, full material response and second record under one positive H	Prepared ports, allowed gates, physical transport and finite cursor
Local source	Same-H smooth interacting field bound; local positive contact potential	Smooth interior tests and moments; controller/support stress is separate
Finite maintenance	Receiver-bank SWAP; correlated-window and diagnostic protocols	Paid bank, storage and clock; not full interacting-material Gibbs renewal
Repeated body renewal	Full ring contraction, mixed-target preparation, neutral code, packet loading and finite-carrier sorting	Four thermal outlets, paid buffer, prescribed gates and energy-constrained capacity
Massive spatial body	Guarded local full-angle gate and weighted physical stress tests	Paid mass profile and state; actual outlet renewal and moving support remain
Field–clock evolution	Retained-state finite schedule with matrix Ward and delayed boundary	Static walls/frame; smooth input norms and horizon-dependent inertia
Geometric flow tasks	Local linear-current synthesis of neutral $c = 1$ stress gates	Circle recurrence, coherent control and growing precision budgets
Covariant apparatus	Common local action with dynamic clock and support reaction	Quantum gate matching, charged continuum and renormalized total stress

Table 1: The main interfaces and their distinct burdens. Auxiliary apparatus proofs are collected in the appendices.

sponse into the spent cell [50]. Its first-order signal arises after acquisition; the cursor and old purifications remain. Matching that run to the complete quantum material here requires a physical state/source bridge. HAFF D's rest-frame mean-stress proposition identifies one excess-radiation sector under fixed covariance and linear contact assumptions, not the entire acquired-record apparatus. Neither construction treats a spent port as a fresh bath.

A world model must retain the carrier and conserved total source; an interface must specify actual records and executable access; a continuing observer must maintain the content and resources those records serve. The common-action candidate connects these responsibilities at the classical and regulated canonical levels. Its quantum acquisition, charged continuum, moving supports and total renormalized stress still need matching. Neither exact multiplicativity nor finite repair derives experience or public geometric agreement. Conservation-law and multimode bounds [36, 38, 39] do not settle all these architectures at once. Split/modular inclusions and virtual subsystems [43, 44, 42] organize retained access; formation and maintenance of its permissions remain questions for T-DOME and Imprint.

13 Conclusion

A coupling represents an algebra exactly only under specific conditions. After a non-atomic dephasing port is discarded, its multiplicative domain is the coupling commutant. Retention can change that answer: closed oscillator loops return their port exactly, while closed propagation retains a faithful copy of the whole internal algebra. The single-shell condition supplies the spatial quadratic structure, and the supplied time and mass anticommutator supply its Lorentzian extension.

An actual alignment of propagation and actuator generators gives a conditional work–shell relation. It does not identify a positive cost with an indefinite metric, and it does not remove the reduced operation's multiplicative defect. This separates a testable physical compatibility condition from an assumed representation map.

The finite material construction makes the records, source and next action part of one retained quantum history. Receiver renewal and smooth-source accuracy have explicit clock, preparation and energy conditions. The further ring, massive-body and spatial-clock certificates extend what can be implemented, but their distinct layouts still need a common supported quantum realization. Foundation III's finite semiclassical return and HAFF-D's sustained renewal and joint sharp-exhaust questions retain their separate burdens. Exact representation is an interface condition; it does not by itself generate the world, a permanent apparatus or a subject.

Certificate guide: Q-RAIF C technical supplement

The representation bridge, physical mechanism and core results remain in this paper. The complete technical appendices are in `QRAIF_C_supplement.pdf`, *Retained Material, Work Clocks and Renewal Certificates: Technical Supplement to Q-RAIF C*. This is supplementary documentation, not a separate paper. Appendix letters and all existing result and equation numbers are retained. References below link to the supplement, which links back to the main argument.

Main claim or question	Technical certificate	Scope of support
Complete charged material	Appendix A; Propositions 22, 25	Regulated carrier and smooth source tests with weighted tails.
Finite thermal-field return	Appendix B; Propositions 27, 28	The same field and old correlations are retained.
Ring-body renewal and conversion	Appendix C; Propositions 29, 30, 32	Paid outlets and buffers; finite-carrier assumptions.
Spatial work clock and massive gate	Appendix D; Propositions 35, 37	Finite joint-state control under the supplied walls and frame.
Conserved control and local outlet	Appendices E, F; Propositions 38, 41	Battery, timing, output window and outlet costs are retained.
Preparation, contact and recoil	Appendices G, H; Propositions 45, 54	The selected source is not its mean covariance; reaction is physical.
Actual acquired-record return	Appendices I, J; Propositions 57, 58	Actual programs, delayed readings and charged finite windows.
Neutral stress-flow control	Appendix K; Proposition 60	Local current synthesis with stated recurrence and precision resources.

These certificates concern distinct finite layouts. They do not establish one supported quantum apparatus with sustained renewal and jointly sharp informative exhaust; the remaining matching conditions are stated in the main discussion.

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Memory as a Usable Resource: Correlations, Records and Complete Endpoint Accounting

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Abstract

A stored correlation has value only through an operation that can use it. We ask which correlations can fund a free-energy gain, who can read them and what reading, feedback and memory reset cost. The survival functional $\mathcal{S} = \Delta F - W$ obeys an exact balance retaining mutual information, environmental displacement from equilibrium, control energy and interaction energy. Three models distinguish their roles: Lorentz–Drude dephasing has no backflow at any temperature but a nonzero interaction energy; an autonomous two-qubit recurrence saturates the correlation budget without a pointer record; fresh-memory collisions create records available to several readers. A classical memory used for feedback is worth at most $k_B T I(X{:}M)$, with matched quasistatic feedback approaching the value. Every feedback protocol assumes a posterior and loses at least $k_B T$ times its expected divergence from the true posterior. A wrong readout model can therefore make a record worse than no memory. For a tail-controlled estimate from n references per sign, calibration has leading loss $k_B T(1+g^2/(1-b^2))/(4n)$ and pays through reuse. Within a reset menu with no added purity, positive read–harvest–reset work requires accessible correlated side information. Two partially mixed record ports realize this possibility: one drives matched feedback and the other reduces the reset cost through a reversible parity operation. These results establish the memory–self–introspection route's first functional link: relevant memory permits a gain, a model of one's readout realizes its full value, and calibration builds that model at a calculable cost. A separate finite-history classification bound proves that, for each fixed observation window, a longer finite episode can improve the attainable decision risk.

Keywords: open quantum systems, information thermodynamics, system–environment correlations, quantum records, feedback control, self-calibration, interaction energy, non-Markovian dynamics

1 Introduction

1.1 What memory can pay for

A system coupled to a thermal bath can lose the free energy needed for further action. An autonomous detailed-balance semigroup makes that loss precise; a structured environment can instead return correlations at a later time. Persistent structures, from molecular machines to organisms, maintain themselves by importing low-entropy energy and exporting entropy [2]. Within that general picture a narrower question can be answered exactly: when can correlations that a system has already built with its environment fund a later interval, and what does it take for an agent to turn a stored record into usable work?

Two errors are easy to make here. One is to treat any memory effect in the reduced dynamics as a resource. A memory kernel represents retained history; it does not guarantee that the history can be used, and the sign of the free-energy balance is fixed by the resources and the endpoint terms, not by Markovianity as such. The other is to treat information as free once it is stored. A record has value only for an agent that can read it, act on it with a correct model of its own reading, and dispose of the memory afterwards; each step has a price. The endpoint balance and the feedback task below specify which terms must be paid.

1.2 Position within the series

This paper is the first of the T-DOME (Thermodynamic Dynamics of Observer-Memory Entanglement) series, the third line of a programme whose other two lines are HAFF and Q-RAIF.

Line	**Question**		**Role here**
HAFF [25, 26]	How do stable, public records form?	Ocean	Supplies the collision model of records
Q-RAIF [27]	How does a subsystem read and act on its world?	Fish	Reading and control interfaces; costs of persistence
T-DOME I (this work)	What can stored correlations fund, and for whom?	Seed	Ledger, correlation types, value of memory and readout model
T-DOME II [28]	How can a bounded agent organize what it keeps?	Ego	Frame and feature selection under a budget
T-DOME III [29]	How does an agent keep its model of itself calibrated?	Loop	Detection, tracking and body-gated repair

The three T-DOME papers form a conditional sequence of models along what the programme calls the memory–self–introspection route: an agent keeps records of its world, models its own access to that world, and checks that model against references. The present paper supplies the first link and the thermodynamic currency for the other two. It shows

in one explicit task that a memory is informationally necessary for a work gain, that every protocol acts on an implied posterior and obtaining the full value requires that posterior to match the agent's readout, and that calibrating the model from references is an achievable mechanism whose cost and benefit can be computed. Paper II shows that under a reading budget a self-frame formed from records is informationally necessary for the task, prices it and identifies the blind spot it creates; Paper III develops drift detection, priced tracking and history-selected repair of body-gated access.

1.3 Results

- An exact accounting identity for the survival functional with every endpoint term retained (Theorem 8), a budget bound and a charge–discharge interval closure (Corollary 9, Proposition 10), and a restricted autonomous ceiling for a detailed-balance semigroup (Theorem 6).

- Three exact models with different roles (Section 5): no backflow at any temperature and a nonzero interaction-energy term in the Lorentz–Drude pure-dephasing model (Propositions 11 and 12); exact saturation of the correlation budget without any record in an autonomous two-qubit model (Proposition 13); the complete ledger of the collision model of records (Proposition 14).

- The work value of a record held in a classical memory and its information-theoretic ceiling (Theorem 16); the price of acting on a wrong posterior, which every protocol implicitly assumes (Theorem 18); the local metric of that price and the loss after calibration from references (Proposition 19), with the resulting amortization rule (Corollary 21) and a drift law for self-calibration, proved in Paper III (Proposition 22).

- A read–harvest–reset bound for a specified reset menu (Proposition 23), and a two-mixed-port realization with a constructive conditional reset (Proposition 24). The task restores the private memory and the retained record's marginal, not the complete record source.

Sections 8–10 distinguish temporal influence from kernel memory, price specified archives and compare retained history with fixed observation windows.

2 Thermodynamic setting

2.1 Open systems and memory kernels

Consider $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E$ with total Hamiltonian

$$H(t) = H_S(t) \otimes \mathbb{1}_E + \mathbb{1}_S \otimes H_E + H_{\mathrm{int}}(t), \qquad H_S(t) = H_R + H_{\mathrm{ctrl}}(t), \tag{1}$$

where H_R is the fixed bare system Hamiltonian, H_{ctrl} the control and H_{int} the coupling. Under the Born–Markov and secular approximations the reduced dynamics obeys a GKSL equation [3, 4]. For a fixed environmental reference σ_E, set $\mathcal{P}A = \mathrm{tr}_E(A) \otimes \sigma_E$, $\mathcal{Q} = \mathrm{id} - \mathcal{P}$

and $\mathcal{L}(t)A = -i[H(t), A]$. Eliminating the $\mathcal{Q}$ component gives the exact Nakajima–Zwanzig equation

$$\dot{\rho}_S(t) = -i[H_{\text{eff}}(t), \rho_S(t)] + \int_0^t ds\, \mathcal{K}(t,s)\, \rho_S(s) + \mathcal{I}_0(t), \tag{2}$$

where $H_{\text{eff}}(t) = H_S(t) + \text{tr}_E[H_{\text{int}}(t)(\mathbb{1}_S \otimes \sigma_E)]$ and

$$\mathcal{K}(t,s)A = \text{tr}_E[\mathcal{L}(t)\mathcal{G}_Q(t,s)\mathcal{Q}\mathcal{L}(s)(A \otimes \sigma_E)]\,, \qquad \mathcal{I}_0(t) = \text{tr}_E[\mathcal{L}(t)\mathcal{G}_Q(t,0)\mathcal{Q}\rho_{SE}(0)]\,.$$

Here $\mathcal{G}_Q$ propagates the $\mathcal{Q}$ subspace with generator $\mathcal{Q}\mathcal{L}(t)\mathcal{Q}$ [5, 6]. The inhomogeneous term vanishes for $\rho_{SE}(0) = \rho_S(0) \otimes \sigma_E$; correlated inputs allowed below generally retain it. Non-Markovian dynamics can show information backflow, a temporary increase of the trace distance between two evolving states [14]. Neither the kernel nor backflow is used as a resource below; the accounting identity applies directly to the joint unitary dynamics.

2.2 Conventions

Definition 1 (Thermodynamic setup)**.** *Let $\beta = (k_B T)^{-1}$. The reference state of the system is $\rho_{\text{eq}} = e^{-\beta H_R}/Z_R$, and its nonequilibrium free energy is*

$$F(\rho) = \text{tr}(\rho H_R) - \beta^{-1} S(\rho), \qquad D(\rho\|\rho_{\text{eq}}) = \beta\big(F(\rho) - F_{\text{eq}}\big) \geq 0. \tag{3}$$

The environmental reference is the fixed Gibbs state $\rho_E^{\text{th}} = e^{-\beta H_E}/Z_E$. The inclusive work over $[0,\tau]$ is $W = \int_0^\tau \text{tr}\big(\rho(t)\,\partial_t H(t)\big)dt$, so that $W = \Delta\langle H\rangle$ exactly.

Definition 2 (Standing assumptions)**.** *Unless stated otherwise:*

(A1) *$\mathcal{H}_S \otimes \mathcal{H}_E$ is finite dimensional and the total evolution is unitary, generated by* (1)*. Infinite baths are treated as limits of finite mode sets in which the quantities below have finite limits.*

(A2) *Every energy change is booked: $W = \Delta\langle H_R\rangle + \Delta\langle H_{\text{ctrl}}\rangle + \Delta\langle H_E\rangle + \Delta\langle H_{\text{int}}\rangle$. Dropping $\Delta\langle H_{\text{int}}\rangle$ is a declared approximation, never a consequence of autonomy.*

(A3) *H_E is time independent; the actual initial state of the environment need not be ρ_E^{th}.*

(A4) *The initial state $\rho_{SE}(0)$ may be correlated.*

(A5) *Relative entropies are finite (full rank, or restriction to supports).*

2.3 The survival functional

Definition 3 (Survival functional)**.** *For an interval $[a,b]$,*

$$\mathcal{S}[a,b] := F(\rho_S(b)) - F(\rho_S(a)) - W[a,b], \qquad \beta\,\mathcal{S} = -\Sigma, \quad \Sigma := \beta(W - \Delta F). \tag{4}$$

The name records the intended use: $\mathcal{S} > 0$ means the system gained more free energy than the external protocol invested, a thermodynamic profit; $\mathcal{S} < 0$ means it paid more than it gained. Mathematically $\mathcal{S}$ is a generalized entropy-balance functional. For an uncorrelated thermal initial environment and zero changes in the endpoint control and interaction energies, $\Sigma \geq 0$ and $\mathcal{S} \leq 0$. When those endpoint terms change, their signs enter the balance of Theorem 8. For correlated initial conditions Σ can be negative [15], and the next sections say what pays for it.

In measurement-based feedback, the generalized second law $\Sigma \geq -I_{\mathrm{fb}}$ allows $\mathcal{S} > 0$ at the expense of the information I_{fb} gained by measurement [7, 8, 9]. Pre-existing correlations enter the balance in an analogous role. A memory kernel is not itself a measurement record, and the two protocols are kept distinct; Section 6 returns to feedback with an explicit record.

3 A restricted autonomous baseline

Definition 4 (Autonomous Davies baseline)**.** *The autonomous baseline has $H_{\mathrm{ctrl}} = 0$, fixed H_R, and a time-independent GKSL generator $\mathcal{L}$ obtained as the weak-coupling limit of coupling to one thermal bath, satisfying quantum detailed balance with respect to the faithful Gibbs state ρ_{eq} [11, 1]. No measurement, feedback or initial system–bath resource is included.*

Lemma 5 (Spohn [11])**.** *For a GKSL semigroup $\Lambda_t = e^{\mathcal{L}t}$ with invariant state ρ_{eq}, $t \mapsto D(\Lambda_t\rho\|\rho_{\mathrm{eq}})$ is nonincreasing.*

Proof. Relative entropy is contractive under CPTP maps and $\Lambda_t\rho_{\mathrm{eq}} = \rho_{\mathrm{eq}}$. □

Theorem 6 (Restricted autonomous ceiling)**.** *Under Definition 4, $\mathcal{S}[0,\tau] = F(\rho(\tau)) - F(\rho(0)) \leq 0$.*

Proof. $W = 0$, and by (3) and Lemma 5, $\beta\Delta F = \Delta D(\rho\|\rho_{\mathrm{eq}}) \leq 0$. □

The theorem is a free-energy monotonicity statement for one autonomous detailed-balance class. It is not a ceiling for driven or feedback-controlled Markovian agents, which can persist under continuous driving; time-dependent protocols need an instantaneous detailed-balance hypothesis and a different theorem. The reduced semigroup omits explicit inter-epoch correlations, which is a modelling loss, not a proof that such correlations are unusable.

4 Initial-correlation accounting

4.1 The identity

The quantum mutual information $I(S{:}E) = S(\rho_S) + S(\rho_E) - S(\rho_{SE}) = D(\rho_{SE}\|\rho_S \otimes \rho_E)$ measures all correlations between system and environment. For any full-rank σ_E,

$$D(\rho_{SE}\|\rho_S \otimes \sigma_E) = I(S{:}E) + D(\rho_E\|\sigma_E), \tag{5}$$

an algebraic identity that holds for correlated ρ_{SE} [16].

Lemma 7 (Information–thermodynamic identity)**.** *For unitary evolution from any initial state,*

$$\Delta I(S{:}E) + \Delta D(\rho_E \| \rho_E^{\mathrm{th}}) = \Delta S_S + \beta\, \Delta\langle H_E \rangle. \tag{6}$$

Proof. By (5) with $\sigma_E = \rho_E^{\mathrm{th}}$ and $\ln \rho_E^{\mathrm{th}} = -\beta H_E - \ln Z_E$, the quantity $I + D(\rho_E \| \rho_E^{\mathrm{th}})$ equals $-S(\rho_{SE}) + S(\rho_S) + \beta\langle H_E \rangle + \ln Z_E$. Unitarity keeps $S(\rho_{SE})$ fixed, and taking differences gives (6). □

Theorem 8 (Survival balance)**.** *Under* (A1)–(A5)*, for any interval and any initial state,*

$$\boxed{\beta\, \mathcal{S} = -\Delta I(S{:}E) - \Delta D(\rho_E \| \rho_E^{\mathrm{th}}) - \beta\, \Delta\langle H_{\mathrm{ctrl}} \rangle - \beta\, \Delta\langle H_{\mathrm{int}} \rangle.} \tag{7}$$

Proof. By (A2) and $\Delta F = \Delta\langle H_R \rangle - \beta^{-1} \Delta S_S$, $\Sigma = \beta(W - \Delta\langle H_R \rangle) + \Delta S_S = (\Delta S_S + \beta \Delta\langle H_E \rangle) + \beta \Delta\langle H_{\mathrm{ctrl}} \rangle + \beta \Delta\langle H_{\mathrm{int}} \rangle$. Lemma 7 replaces the bracket by $\Delta I + \Delta D$, and $\beta\mathcal{S} = -\Sigma$. □

The identity is exact. It is the Esposito–Lindenberg–Van den Broeck decomposition [15] written with arbitrary initial correlations and with both endpoint terms kept. It is dynamics-agnostic: Markovianity does not enter, and a memory kernel matters only if a model shows that it preserves correlations across an interval boundary and later allows their consumption. Autonomy removes the control term but not the interaction term; Section 5.1 gives an autonomous model in which the interaction energy changes at every time.

4.2 Regimes and the budget

Corollary 9 (Regimes and budget)**.** *Suppose* $\Delta\langle H_{\mathrm{ctrl}} \rangle = \Delta\langle H_{\mathrm{int}} \rangle = 0$ *on the interval, as for a switched protocol whose control and interaction Hamiltonians are zero at both endpoints.*

1. *If the interval starts from a product state with a thermal environment,* $\beta\mathcal{S} = -I(S{:}E; b) - D(\rho_E(b) \| \rho_E^{\mathrm{th}}) \le 0$.
2. *If correlations are consumed,* $\Delta I < 0$*, and* $|\Delta I| > \Delta D(\rho_E \| \rho_E^{\mathrm{th}})$*, then* $\mathcal{S} > 0$.
3. *In every case* $\beta\mathcal{S}[a, b] \le I(S{:}E; a) + D(\rho_E(a) \| \rho_E^{\mathrm{th}})$.

Proof. Insert the hypotheses into (7) and use $I \ge 0$, $D \ge 0$ at the end of the interval. □

When the endpoint terms do not vanish they are added to the budget with their signs. The budget in (iii) is the profit ceiling: correlations present at the start of the interval plus the environment's displacement from equilibrium.

Proposition 10 (Charge–discharge interval closure)**.** *Under the hypotheses of Corollary 9 at the times* 0, t^* *and* τ*, and with* $I(S{:}E; 0) = 0$ *and* $\rho_E(0) = \rho_E^{\mathrm{th}}$*,*

$$\begin{aligned} \beta\mathcal{S}[0, t^*] &= -I(t^*) - D(t^*) \le 0, \\ \beta\mathcal{S}[t^*, \tau] &= -[I(\tau) - I(t^*)] - [D(\tau) - D(t^*)], \\ \beta\mathcal{S}[0, \tau] &= -I(\tau) - D(\tau) \le 0, \end{aligned}$$

with $D(t) = D(\rho_E(t) \| \rho_E^{\mathrm{th}})$.

Proof. ΔF and W are additive over concatenated intervals, and Theorem 8 applies to each interval. □

A positive second interval is funded by the correlations built in the first, and their concatenation cannot have positive $\mathcal{S}$. No restoration of the complete joint state is assumed in this accounting statement. This charge–discharge structure is realized exactly by the two-qubit model of Section 5.2. Under the Born approximation every step restarts from a product state, so the resource is removed by construction; a reduced semigroup law by itself does not say that every microscopic dilation is uncorrelated. BLP backflow and correlation consumption can coincide in particular models, but neither implies the other in general [10], so backflow is not a witness for the sign of $\mathcal{S}$ without the full balance.

5 Which correlations can be used: three exact models

5.1 Pure dephasing with a Lorentz–Drude bath: no backflow, nonzero interaction energy

Let $\hbar = 1$ and

$$H = \frac{\omega_0}{2}\sigma_z + \sum_k \omega_k b_k^\dagger b_k + \sigma_z \sum_k (g_k b_k + g_k^* b_k^\dagger), \qquad J(\omega) = \sum_k |g_k|^2 \delta(\omega - \omega_k) = \frac{2\lambda\gamma\omega}{\omega^2 + \gamma^2}, \tag{8}$$

with $\lambda, \gamma > 0$. With this normalization $\int_0^\infty J(\omega)\omega^{-1} d\omega = \pi\lambda$. The initial state is any system state times the thermal state of the bare bath at inverse temperature $\beta \in (0, \infty]$. Removing the free rotation, the coherence is multiplied by a factor of modulus

$$|p(t)| = e^{-G_\beta(t)}, \qquad G_\beta(t) = 4\int_0^\infty \frac{J(\omega)}{\omega^2}(1 - \cos\omega t)\coth\frac{\beta\omega}{2}\, d\omega, \tag{9}$$

with $\coth = 1$ at $\beta = \infty$. This is the standard exact solution: the two eigenvalues of σ_z displace each bath mode in opposite directions, the displacement difference is $2g_k(1-e^{-i\omega_k t})/\omega_k$, and the thermal characteristic function gives the factor $\exp[-|\delta\alpha_k|^2 \coth(\beta\omega_k/2)/2]$ per mode [1, 17]. The integral converges for every finite t, so $|p(t)| > 0$.

Proposition 11 (No backflow at any temperature)**.** *For every $\beta \in (0, \infty]$ and $\lambda, \gamma > 0$, $G'_\beta(t) > 0$ for all $t > 0$. Hence $|p(t)|$ decreases strictly, the trace distance of any two system states does not increase, the reduced dynamics is CP-divisible, and for $\beta < \infty$, $\mathcal{S}[a, b] < 0$ on every interval $0 \le a < b$ when the system starts with nonzero coherence.*

Proof. Differentiating under the integral sign, which is allowed because $|\sin\omega t| \le \min(1, \omega T)$ for $t \le T$ gives an integrable majorant, yields

$$G'_\beta(t) = 8\lambda\gamma \int_0^\infty f(\omega)\sin(\omega t)\, d\omega, \qquad f(\omega) = \frac{\coth(\beta\omega/2)}{\omega^2 + \gamma^2}.$$

The function f is positive and strictly decreasing on $(0, \infty)$ and tends to 0; near $\omega = 0$ it behaves as $2/(\beta\gamma^2\omega)$, and $f(\omega)\sin\omega t$ stays bounded there. Substituting $u = \omega t$ and splitting

$(0,\infty)$ into half-periods,

$$\int_0^\infty f(\omega)\sin(\omega t)\,d\omega = \frac{1}{t}\sum_{j=0}^{\infty}(-1)^j a_j, \qquad a_j = \int_0^\pi f\Big(\frac{j\pi+x}{t}\Big)\sin x\,dx.$$

The a_j are positive, strictly decreasing and tend to 0, so the alternating sum is at least $a_0 - a_1 > 0$. Between times $s < t$ the coherence factor is multiplied by $p(t)/p(s)$ with modulus less than 1, which is a CPTP dephasing map, and qubit trace distances shrink under it. Populations are conserved and $W = 0$, so $\beta\mathcal{S}[a,b] = -[S(\rho_S(b)) - S(\rho_S(a))]$, which is negative because the entropy of a qubit with fixed populations increases as its coherence decreases. □

For $\lambda = 1$ and $\gamma = 0.5$ at zero temperature, $|p| = 0.313$, 0.0360 and 2.2×10^{-4} at $t = 0.5$, 1 and 2; at $\beta = 2$ the values are 0.138, 1.43×10^{-3} and 1.0×10^{-9}. Changing γ/λ or the temperature does not produce a backflow interval in this model.[1]

Proposition 12 (Interaction energy)**.** *For any initial system state and a bath that starts in its bare thermal state with vanishing first moments,*

$$\langle H_I\rangle_t = -2\int_0^\infty \frac{J(\omega)}{\omega}(1-\cos\omega t)\,d\omega = -2\pi\lambda\big(1-e^{-\gamma t}\big), \qquad \Delta E_B[0,t] = 2\pi\lambda\big(1-e^{-\gamma t}\big),$$

independently of the temperature, while $\Delta E_S = 0$ *and* $W = 0$*.*

Proof. The Heisenberg equations give $b_k(t) = e^{-i\omega_k t}b_k(0) + (g_k^*/\omega_k)\sigma_z(e^{-i\omega_k t} - 1)$. With $\sigma_z^2 = \mathbb{1}$ and $\langle b_k(0)\rangle = 0$, $\langle\sigma_z(g_k b_k(t) + g_k^* b_k^\dagger(t))\rangle = -2|g_k|^2(1-\cos\omega_k t)/\omega_k$. Summing with the spectral density and using $\int_0^\infty \cos(\omega t)/(\omega^2+\gamma^2)\,d\omega = \pi e^{-\gamma t}/(2\gamma)$ gives the formula. The total energy is conserved and σ_z commutes with H, so $\Delta E_B = -\Delta\langle H_I\rangle$. □

The balance on $[a,b]$ is therefore $\beta\mathcal{S} = -\Delta I - \Delta D_B - \beta\Delta\langle H_I\rangle = -\Delta S_S$, with $\beta\Delta\langle H_I\rangle = -2\pi\beta\lambda(e^{-\gamma a} - e^{-\gamma b}) \neq 0$. The evolution is autonomous and the interaction-energy term still cannot be dropped. At zero temperature the dynamics and energies are well defined, but the relative-entropy terms of the finite-temperature ledger cannot be used with β^{-1} treated as a finite energy unit.

5.2 An autonomous two-qubit recurrence: correlation without a record

Let a system qubit and an environment qubit evolve under the time-independent, mutually commuting Hamiltonian

$$H = \frac{\omega_0}{2}\sigma_z + \frac{\Omega}{2}\tau_z + g\,\sigma_z\tau_z, \qquad \rho_{SE}(0) = \frac{\mathbb{1} + r\sigma_x}{2}\otimes\tau_E, \quad \tau_E = \frac{e^{-\beta\Omega\tau_z/2}}{Z_E}, \tag{10}$$

with $g, \Omega, \beta > 0$ and $0 < r < 1$. Put $m = \tanh(\beta\Omega/2)$, $t_c = \pi/(4g)$ and $t_r = \pi/(2g)$.

[1] The oscillatory decoherence function, survival-gain table and figure in the July 2026 example were not solutions of its stated pure-dephasing model. Propositions 11 and 12 replace them and restore the interaction-energy term.

Proposition 13 (Exact saturation of the correlation budget)**.** *The coherence of the system is* $\frac{r}{2}e^{-i\omega_0 t}\big(\cos 2gt + im\sin 2gt\big)$*, of modulus* $\frac{r}{2}\sqrt{\cos^2 2gt + m^2\sin^2 2gt}$*. The environment marginal stays* τ_E*, and* $\Delta E_S = \Delta E_E = \Delta\langle H_I\rangle = W = 0$*. On the discharging interval*

$$\beta\mathcal{S}[t_c, t_r] = h\Big(\frac{1+rm}{2}\Big) - h\Big(\frac{1+r}{2}\Big) = I(S{:}E; t_c) > 0,$$

where h *is the binary entropy in nats, so the budget of Corollary 9(iii) is attained with equality. The charging interval* $[0, t_c]$ *has the opposite value and* $\mathcal{S}[0, t_r] = 0$*. The two conditional environment states for the two pointer values coincide, so the environment holds no information about the pointer:* $\chi(\text{pointer}{:}E) = 0$ *at all times.*

Proof. The Hamiltonian is diagonal in the product basis. The off-diagonal element of S acquires the phase $e^{-i\omega_0 t}e^{-2igst}$ when the environment is in the τ_z eigenstate $s = \pm 1$, which has weight $(1 \mp m)/2$; averaging gives the coherence. The diagonal blocks are unchanged, so the energies and the environment marginal are constant and the global entropy is constant; hence $I(S{:}E; t) = S(\rho_S(t)) - S(\rho_S(0))$. At t_r the coupling factor of the evolution operator is $e^{-i(\pi/2)\sigma_z\tau_z} = -i\sigma_z \otimes \tau_z$, so the evolution operator is a product of local unitaries and the correlation vanishes. With the energies constant, $\beta\mathcal{S} = -\Delta S_S$, which gives the displayed values. The environment state conditional on either pointer value is τ_E, because τ_E commutes with τ_z. □

For $(\beta, \Omega, g, r) = (1.2, 1.1, 0.3, 0.7)$, $m = 0.578363$, $t_c = 2.617994$, $t_r = 5.235988$ and $\beta\mathcal{S}[t_c, t_r] = 0.1860846$. The model realizes the charge–discharge cycle of Proposition 10 with an autonomous constant coupling, and it is exact. Its environment remembers in the sense that the correlation returns at a fixed time; it does not remember in the sense of holding a readable record. The reduced dynamics is not CP-divisible, since the coherence revives.

5.3 Records: the collision model

The collision model of the companion papers supplies correlations of a third kind [25, 26]. A pointer qubit with $H_R = \varepsilon P_1$ starts in $\sqrt{p}\,|0\rangle + \sqrt{1-p}\,|1\rangle$; memories $E_1, \ldots, E_n$ with degenerate Hamiltonians start in $|0\rangle$, and the k-th collides once through $P_0 \otimes \mathbb{1} + P_1 \otimes R_k$ with $\langle 0|R_k|0\rangle = c_k$. The collision can be generated by $g_k(t)P_1 \otimes \sigma_y$, which commutes with H_R and vanishes at both ends of the pulse, so $W = 0$ and $\Delta\langle H_{\text{int}}\rangle = 0$. With $r = \prod_k c_k$ and $s_n = H_p(r)$, where $H_p(r) = h\big((1 + \sqrt{1 - 4p(1-p)(1-r^2)})/2\big)$, a fragment F holds $\chi(X{:}E_F) = H_p(r_F)$ nats about the pointer label X [25].

Proposition 14 (Ledger of the collision model)**.** *With the environmental reference* $\mathbb{1}/2^n$*:*

1. *from the coherent initial state,* $\Delta I(S{:}E) = 2s_n$*,* $\Delta D = -s_n$ *and* $\beta\mathcal{S} = -s_n$*;*
2. *from the diagonal initial state* $pP_0 + (1-p)P_1$*, every record quantity is the same,* $\Delta I = s_n$*,* $\Delta D = -s_n$ *and* $\beta\mathcal{S} = 0$*;*
3. *the exact inverse protocol, which needs access to* S *and all memories, gives* $\Delta I = -2s_n$*,* $\Delta D = +s_n$ *and* $\beta\mathcal{S} = +s_n$*, and the full forward–backward cycle has* $\mathcal{S} = 0$*.*

Proof. The global state is pure in case (1), so $I(S{:}E) = 2S(\rho_E) = 2s_n$, and $D(\rho_E \| \mathbb{1}/2^n) = n \ln 2 - s_n$. Populations of S are conserved and $W = 0$, so $\beta\mathcal{S} = -\Delta S_S = -s_n$, in agreement with (7). In case (2) the global entropy is $h(p)$ throughout, $S(\rho_S) = h(p)$ is unchanged and $S(\rho_E) = s_n$, so $I = s_n$. Case (3) reverses case (1). □

For $p = 1/2$, $c_k = 0.8$ and $n = 60$, $s_{60} = 0.693147$ nats, equal to $\ln 2$ to nine digits. Records form along a CP-divisible reduced dynamics, and the diagonal case shows that the formation of records does not by itself cost the system free energy; the memories' initial purity deficit supplies the record budget [26].

5.4 Who can use which correlation

	Lorentz–Drude dephasing	Two-qubit recurrence	Collision records
Correlation built	yes	yes	yes
Returns to S by itself	never	at t_r, fixed by g	no
Pointer information in E	yes, spread over a continuum of modes	none ($\chi = 0$)	yes, in many separate copies
Who can use it	an agent with coherent control of the bath	the autonomous dynamics, at one time	an agent with fragment readout and feedback permissions
Endpoint terms	$\Delta\langle H_I \rangle \neq 0$	zero	zero

Table 1: Three forms of system–environment correlation.

In the accounting identity, consuming one nat of mutual information contributes $+k_B T$ to $\mathcal{S}$, with the other terms retained. The models show that the identity does not say who can consume it. Correlation created by a structured bath can be dissipated without ever returning, although the bath holds information about the pointer spread over its modes. Correlation held by a single environmental degree of freedom with a commensurate spectrum can return at a fixed time, without anyone reading anything. Correlation copied into records can be read by many agents at their own times, and the rest of this paper studies what such an agent must do to turn a record into work.

6 The value of a record, and of a model of one's own readout

6.1 The task

The pointer S of Section 5.3 has a degenerate Hamiltonian during the task ($\varepsilon = 0$) and has been decohered by memories the agent does not read, so that its state is diagonal: $\rho_S = pP_0 + (1-p)P_1$, correlated with the memories through X. The agent reads a fragment

F by a measurement N and stores the outcome in a classical memory M. It then runs a feedback protocol on S and a bath B at inverse temperature β, chosen according to the stored value m, in which the system Hamiltonian returns to zero and the coupling vanishes at both ends. The work extracted is $W_{\text{ext}} = -W$. The task is to extract as much work as possible on average.

Lemma 15 (Extraction from a qubit)**.** *For a qubit with degenerate Hamiltonian in a state ρ, every such protocol extracts on average at most $k_B T\, D(\rho\|\mathbb{1}/2) = k_B T[\ln 2 - S(\rho)]$. If the diagonal of ρ is a distribution P and Q is a distribution with full support, the protocol that quenches the levels to $E_x = -k_B T \ln Q(x)$, lets the system thermalize and returns the levels quasistatically to degeneracy extracts*

$$k_B T\big[\ln 2 - H(P) - D(P\|Q)\big].$$

Proof. For the bound: the joint state of system and bath evolves unitarily from $\rho\otimes\tau_B$; since the system Hamiltonian is zero at both ends and the coupling vanishes there, $W = \Delta E_B$. Subadditivity gives $S(\rho') + S(\rho'_B) \geq S(\rho) + S(\tau_B)$, and $D(\rho'_B\|\tau_B) \geq 0$ gives $\beta\Delta E_B \geq S(\rho'_B) - S(\tau_B) \geq S(\rho) - S(\rho') \geq S(\rho) - \ln 2$. For the protocol: the quench costs $\sum_x P(x)E_x = k_B T[H(P) + D(P\|Q)]$; thermalization at fixed levels costs no work; the quasistatic return costs the free-energy difference $-k_B T \ln 2 - (-k_B T \ln \sum_x Q(x)) = -k_B T \ln 2$. The work done on the system is $k_B T[H(P) + D(P\|Q) - \ln 2]$. Off-diagonal elements do not affect it, since the levels are diagonal and the bath relaxes the system to the Gibbs state of the quenched levels. □

Quasistatic values throughout this section are ideal limiting values. A full-support posterior needs finite quenched gaps, but reversible thermalization and return are still limits of physical protocols. A posterior on the boundary additionally needs a divergent level gap; it is treated by a full-support approximation.

6.2 Memory is necessary, and its value is the accessible information

Theorem 16 (Value of a stored record)**.** *In the task above, let $W_0 = k_B T[\ln 2 - h(p)]$ be the largest average work obtainable without the record. For every measurement N, every processing of its outcome into M and every feedback protocol,*

$$\mathbb{E}[W_{\text{ext}}] - W_0 \leq k_B T\, I(X{:}M) \leq k_B T\, I_{\text{acc}}(X{:}E_F) \leq k_B T\, \chi(X{:}E_F) = k_B T\, H_p(r_F).$$

The first inequality is approached by the quasistatic protocol of Lemma 15 with $Q = P(\cdot \mid m)$ when every posterior $P(\cdot \mid m)$ has full support; otherwise it is the supremum over such protocols. For $p = 1/2$ the Helstrom measurement gives $I(X{:}M) = I_{\text{acc}}(X{:}E_F) = \ln 2 - h\big((1 - \sqrt{1 - r_F^2})/2\big)$.

Proof. Given $M = m$ the state of S is $\sum_x P(x \mid m)P_x$, because S is diagonal and correlated with F only through X. Lemma 15 bounds the extracted work by $k_B T[\ln 2 - H(X \mid M = m)]$, with equality in the matched quasistatic limit. Averaging over m and subtracting W_0 gives $k_B T[h(p) - H(X \mid M)] = k_B T\, I(X{:}M)$. Since M is obtained from a measurement on F

followed by processing, $I(X{:}M) \le I_{\mathrm{acc}}(X{:}E_F)$, and Holevo's bound gives $I_{\mathrm{acc}} \le \chi$ [18]. The value of χ and the equal-prior accessible-information formula are the pure-fragment results of Paper E [25]. For mixed record ports the first inequality follows from the same extraction argument; their actual information replaces $H_p(r_F)$. □

The theorem gives the memory link of the route in the strongest form available for this task. A work gain ΔW is possible only if the agent holds a memory with $I(X{:}M) \ge \beta\Delta W$, and only if it reads a fragment with $H_p(r_F) \ge \beta\Delta W$; the fragment lengths of Paper E then say how many environmental memories it must control. Conversely, the gain $k_B T\, I(X{:}M)$ is the matched quasistatic limit. With $p = 1/2$ and $c_k = 0.8$, one memory gives at most $0.1927\, k_B T$ to an agent with a classical memory although its Holevo information is 0.3251; five memories give $0.5668\, k_B T$ against 0.6385.

The gap between I_{acc} and χ has a thermodynamic reading. An agent that does not measure the fragment but keeps it and acts on S and F jointly can, in the asymptotic limit, use F as quantum side information, for which the relevant quantity is the conditional entropy $S(X \mid F) = H(X) - \chi(X{:}E_F)$ [19]; the Holevo value is then the ceiling. The accessible information is the value of the same record to an agent that first writes it into a classical memory.

6.3 What a compressed history must preserve

A record need not retain every observation that produced it. The exact price of deleting distinctions depends on the future task, not on the length of the archive.

Corollary 17 (Task loss under history compression)**.** *In the diagonal-pointer feedback task, let H be a finite classical history obtained through the permitted record channels, and let M be a compression with $X \to H \to M$ a Markov chain. Use the same source, thermal bath and feedback menu. The supremum gross work gain over the blind harvest lost by replacing H with M is*

$$W_H^* - W_M^* = k_B T\, I(X : H \mid M).$$

It vanishes exactly when $P(X \mid H) = P(X \mid M)$ almost surely under the joint history–compression law.

Proof. Theorem 16 gives the two suprema as $k_B T I(X : H)$ and $k_B T I(X : M)$ above the same blind value. The chain rule and $I(X : M \mid H) = 0$ give their difference. Conditional mutual information is zero precisely when the conditional laws agree on positive-probability pairs. Boundary posteriors use the same full-support limiting protocols as that theorem. □

This is the information-processing identity applied to a physical feedback task, not a new universal law of memory. Acquisition, compression, retention and reset costs have not been subtracted from these gross work values. A compression that saves storage may lose work; one that preserves the current posterior need not support the next posterior update or a different task.

Paper II asks the further question of how retained history selects the channel through which the next input will be read [28]. Its foreground is an output of learning about the reader's own coupling, rather than an archive of all past inputs. The information required

for that prediction task is not automatically the pointer information priced here. One must specify the joint target, record law and permitted action before converting a frame's capability into work. This distinction lets task-specific compression explain a useful organization without making raw-archive overload a general necessity theorem for a self.

6.4 A model of one's own readout is necessary

Theorem 18 (Every protocol assumes a posterior)**.** *For every feedback protocol* Π *on* S *and a bath, of the kind allowed in the task, there is a distribution* Q_Π *with full support such that for every diagonal state with populations* P

$$\mathbb{E}_\Pi[W_{\mathrm{ext}} \mid P] \le k_B T\big[\ln 2 - H(P) - D(P\|Q_\Pi)\big],$$

with equality for all P *when* Π *is the protocol of Lemma 15 designed for* Q_Π*. Consequently, if the agent runs* Π_m *after reading* m*,*

$$\mathbb{E}[W_{\mathrm{ext}}] - W_0 \le k_B T\Big[I(X{:}M) - \sum_m P(m)\, D\big(P(\cdot \mid m) \,\|\, Q_{\Pi_m}\big)\Big],$$

with equality for the matched two-step protocols. The record has negative value whenever the expected divergence exceeds $I(X{:}M)$*, and the value* $k_B T\, I(X{:}M)$ *is attained only if* $Q_{\Pi_m} = P(\cdot \mid m)$ *for every* m *with* $P(m) > 0$*.*

Proof. For a fixed protocol the extracted work is an affine function of the initial system state, $E(\rho) = -\operatorname{tr}\big[(U^\dagger H_{\mathrm{end}} U - H_{\mathrm{start}})(\rho \otimes \tau_B)\big]$, and by Lemma 15 it lies below the convex function $f(\rho) = k_B T\, D(\rho\|\mathbb{1}/2)$. On diagonal states $f(P) = k_B T[\ln 2 - H(P)]$, whose gradient $k_B T \ln P$ (up to a constant) takes every value as P ranges over distributions with full support. Choose Q_Π so that the tangent $T(P) = f(Q_\Pi) + \langle \nabla f(Q_\Pi), P - Q_\Pi\rangle = k_B T[\ln 2 - H(P) - D(P\|Q_\Pi)]$ has the same slope as E. Then $E - T$ is constant, and at $P = Q_\Pi$ it equals $E(Q_\Pi) - f(Q_\Pi) \le 0$. So $E \le T$ everywhere. The quasistatic two-step protocol for Q approaches T by Lemma 15. Averaging over m and subtracting W_0 gives the second display, and the divergence vanishes only when the distributions coincide. □

The extra dissipation $k_B T\, D(P\|Q)$ of a protocol tuned to Q and run on P is a special case of the mismatch cost of Kolchinsky and Wolpert, for a process whose output does not depend on its input [20]; the tangent argument adds that every protocol, tuned or not, has such a Q. The distribution Q_Π is what the protocol assumes about the system, whether or not anything in the agent represents it explicitly. An agent that obtains $P(\cdot \mid m)$ from the prior and its readout channel $P(m \mid x)$, that is, from how its own apparatus responds to the two pointer values, acts on a correct assumption only if its model of that channel is correct where the posterior depends on it.

What the agent needs is therefore not a full description of its apparatus but the part on which the posterior depends: given the prior, the likelihood ratio of its readout. Getting that part wrong has a price that no choice of protocol avoids. With one memory of overlap 0.8 and the Helstrom readout, the true error is 0.2 and the record is worth $0.1927\, k_B T$. An agent that believes its error is 0.01 loses $0.4287\, k_B T$ and ends $0.2359\, k_B T$ below an agent with no memory; one that believes 0.1 keeps $0.1483\, k_B T$; one that believes 0.4 keeps $0.1012\, k_B T$. Overconfidence in one's own readout is more expensive than underconfidence of the same size because the quench pushes the level of the excluded value up by $-k_B T \ln Q$.

6.5 Calibration from references

Let the readout be the binary channel of T-DOME III, $P(y=+1 \mid x) = (1+gx+b)/2$ with pointer values written $x = \pm 1$, gain g and bias b, $|g|+|b|<1$, and prior $1/2$ [29]. The agent estimates $\theta = (g, b)$ from n reference reads of each known pointer value.

Proposition 19 (Metric and calibrated loss)**.** *Let $P_\theta(m \mid x)$ be a smooth family and $L(\hat\theta) = \sum_m P_\theta(m) D(P_\theta(\cdot \mid m) \| P_{\hat\theta}(\cdot \mid m))$ the expected divergence of Theorem 18. Then*

$$L(\hat\theta) = \tfrac{1}{2}(\hat\theta - \theta)^\top G\,(\hat\theta - \theta) + O(|\hat\theta - \theta|^3), \qquad G = J_{XM} - J_M,$$

where J_{XM} is the Fisher information of the joint distribution of (x, m) with x drawn from the prior and J_M that of the marginal of m. If $\hat\theta$ is estimated from n reference reads per pointer value with second-moment matrix $\mathbb{E}(\hat\theta - \theta)(\hat\theta - \theta)^\top = (2nJ_{XM})^{-1} + o(1/n)$, third absolute moments $O(n^{-3/2})$, and $\mathbb{E}\big[L(\hat\theta)\mathbf{1}\{|\hat\theta - \theta| > \epsilon\}\big] = o(1/n)$ for some $\epsilon > 0$, then

$$\mathbb{E}\, L = \frac{1}{4n}\Big[\dim\theta - \operatorname{tr}\big(J_M J_{XM}^{-1}\big)\Big] + o(1/n).$$

For the gain–bias channel this is $\mathbb{E}\, L = \dfrac{1}{4n}\Big(1 + \dfrac{g^2}{1-b^2}\Big) + o(1/n)$. The tail condition is needed: the plug-in frequency estimate meets the other conditions but returns a boundary value with positive probability, at which the model posterior excludes an outcome the true posterior allows, so its expected loss is infinite for every n; add-one-half estimates satisfy it.

Proof. For each m, $D(P_\theta(\cdot \mid m) \| P_{\hat\theta}(\cdot \mid m)) = \tfrac{1}{2}\delta^\top J_{X|m}\delta + O(|\delta|^3)$ with $\delta = \hat\theta - \theta$ and $J_{X|m}$ the Fisher information of the posterior family. The chain rule for Fisher information, $J_{XM} = J_M + \sum_m P(m) J_{X|m}$, gives G. On a fixed interior neighborhood, Taylor's remainder has expectation $O(n^{-3/2})$. Outside it, the assumed loss tail is $o(n^{-1})$, and the third-moment bound makes the quadratic tail $O(n^{-3/2})$. Thus $\mathbb{E}\, L = \frac{1}{2}\operatorname{tr}(G\Sigma) + o(n^{-1})$, with Σ the error second-moment matrix including bias. With $\Sigma = (2nJ_{XM})^{-1}$, $\frac{1}{2}\operatorname{tr}(G\Sigma) = \frac{1}{4n}\operatorname{tr}\big((J_{XM} - J_M)J_{XM}^{-1}\big)$. For the gain–bias channel with prior $1/2$, the efficient reference estimator of T-DOME III has covariance $\frac{1}{2n}\begin{pmatrix} 1-g^2-b^2 & -2gb \\ -2gb & 1-g^2-b^2 \end{pmatrix} = (2nJ_{XM})^{-1}$, the marginal of m is Bernoulli with parameter $(1+b)/2$, so $J_M = \operatorname{diag}(0, (1-b^2)^{-1})$, and $\operatorname{tr}\big(J_M J_{XM}^{-1}\big) = (1-g^2-b^2)/(1-b^2)$. □

Lemma 20 (Tail control for add-one-half calibration)**.** *For a fixed interior gain–bias channel, let independent $K_\pm \sim \mathrm{Bin}(n, p_\pm)$, $p_\pm = (1+b\pm g)/2$. Set $\hat p_\pm = (K_\pm + 1/2)/(n+1)$, $\hat g = \hat p_+ - \hat p_-$ and $\hat b = \hat p_+ + \hat p_- - 1$. These estimates meet all moment and tail hypotheses of Proposition 19.*

Proof. The estimates are a linear transform of two independent frequencies. Their bias is $O(n^{-1})$ and their second-moment matrix is $(2nJ_{XM})^{-1} + O(n^{-2})$; the fourth centred binomial moment is $O(n^2)$, so their third absolute error moment is $O(n^{-3/2})$. Every estimated likelihood is at least $1/[2(n+1)]$. For the fair prior every model posterior probability is therefore at least $1/[4(n+1)]$, giving $0 \le L \le \log[4(n+1)]$. For any fixed sufficiently small neighborhood of the interior true parameter, Hoeffding's inequality bounds its complement by Ce^{-cn}, with $c > 0$. Its contribution to expected loss is at most $C\log[4(n+1)]e^{-cn} = o(n^{-1})$. □

The matrix G is the nonnegative posterior Fisher form on the agent's model of itself, a metric on its identifiable directions, and Proposition 19 gives it a thermodynamic meaning: a calibration error δ costs $\frac{1}{2}k_BT\,\delta^\top G\delta$ of work per use. The subtraction of J_M has a simple content. Unlabelled readings already carry information about the bias, which enters the marginal of m, and the posterior is less sensitive to that direction. For $g = 0.55$ and $b = 0$ the leading term is $1.3025/(4n)$; exact enumeration over the reference outcomes, with estimates $(k + \frac{1}{2})/(n + 1)$ for the two response probabilities, gives $\mathbb{E}\,L = 0.02064$, 0.005113 and 0.001273 for $n = 16$, 64 and 256, ratios 1.014, 1.005 and 1.001 to the leading term. The same channel carries a record worth $0.1600\,k_BT$, and an agent that assumes $g = 0.8$ without calibrating loses $0.0666\,k_BT$ per use.

Corollary 21 (When calibration pays)**.** *Suppose each reference read costs w_{ref} and the calibrated model is used U times before it is renewed. To leading order the total cost $2nw_{\mathrm{ref}} + Uk_BT\kappa/(4n)$, with $\kappa = 1 + g^2/(1 - b^2)$, is minimized at $n^* = \sqrt{U\kappa k_BT/(8w_{\mathrm{ref}})}$, and the overhead per use is $\sqrt{2\kappa k_BTw_{\mathrm{ref}}/U}$. If the known pointer states must be prepared from the thermal state of the degenerate pointer, $w_{\mathrm{ref}} \geq k_BT\ln 2$ by Landauer's bound. Calibration is worthwhile when this overhead is smaller than the loss of the uncalibrated model.*

Proof. Minimize $2nw + Uk_BT\kappa/(4n)$ over $n > 0$. Preparing a pure qubit state from the maximally mixed state with a bath at temperature T costs at least $k_BT\ln 2$ [12, 21]. □

With $g = 0.55$, $b = 0$ and $w_{\mathrm{ref}} = k_BT\ln 2$, the overhead per use is $0.134\,k_BT$ for $U = 100$, $0.0425\,k_BT$ for $U = 1000$ and $0.00425\,k_BT$ for $U = 10^5$, against a record value of $0.1600\,k_BT$. A model of one's own readout is worth building only if it is used many times before it goes stale. How long it stays fresh depends on how fast the apparatus drifts, which is the subject of T-DOME III.

Proposition 22 (Drift law for self-calibration; proved in Paper III)**.** *Let the scalar parameter follow a Gaussian random walk with increment variance ν^2 per use. Suppose each reference has conditional Fisher information at most J, unlabelled observations provide no information about that parameter, and the loss is globally $\frac{1}{2}k_BTG(\hat{\theta}-\theta)^2$ with constant $G > 0$. References cost w_{ref} each. Every predictable calibration policy has long-run average loss plus reference cost at least $c\,w_{\mathrm{ref}}^{1/3}(k_BTG\nu)^{2/3}J^{-1/3} - \frac{1}{2}k_BTG\nu^2$, where $c = 3\cdot 4^{-2/3}$. With a Gaussian prior and Gaussian observations of controllable real precision at linear cost, a persistent Kalman estimate at rate $\rho^* = \big(k_BTG\nu/(4w_{\mathrm{ref}}\sqrt{J})\big)^{2/3}$ attains $3\rho^*w_{\mathrm{ref}}$ to leading order [29]. For whole reading counts, the continuous-precision optimum is a relaxation bound.*

The rate follows from the steady state of a Kalman filter: with ρ references per use the posterior variance is $\nu/\sqrt{\rho J}$ to leading order, and optimizing $w_{\mathrm{ref}}\rho + \frac{1}{2}k_BTG\nu/\sqrt{\rho J}$ gives the exponent. Paper III proves the lower bound by telescoping the posterior Cramér–Rao recursion, including predictable choices made from earlier records. It also proves the exact predictive-relative-entropy optimum of Consciousness III under a Gaussian prior and controllable precision. Free information changes the optimum: in this static readout experiment, unlabelled readings inform the directions where $J_M \neq 0$; in an acting body, the entire correlated history may inform additional parameters. Such a history needs its joint likelihood. For the gain–bias channel, the work loss is a local divergence, so the quadratic comparison

requires Proposition 19's expansion and tail control. A bounded physical gain also needs its actual transition law; clipping a Gaussian walk changes the filtering problem. Smoother drift and vector parameters remain open.

7 Reading, harvesting and resetting

After a harvest the private memory must be retained, reset or replaced. A pure replacement of dimension d_M carries free energy $k_B T \ln d_M$ relative to its degenerate thermal state. Moving the old record to a spent register postpones reset and consumes storage; it does not restore a finite memory bank. We first price reset with a declared side-information menu.

Proposition 23 (Read–harvest–reset task bound)**.** *Let the agent read F into M, extract work from S with feedback on M, and reset M to a standard state by a protocol that couples M and possibly a second fragment G to a bath, leaves the marginal of G unchanged, and vanishes at the endpoints, all registers having degenerate Hamiltonians. The reset bath is thermal and initially independent of MG. No additional purity-bearing workspace or correlated side information is available to the reset. Exact blanking denotes the vanishing-error limit when a finite full-rank thermal bath cannot realize it. Then the net average work satisfies*

$$\beta\big(\mathbb{E}[W_{\mathrm{ext}}] - W_0 - W_{\mathrm{reset}}\big) \le I(X{:}M) - S(M \mid E_G) = I(X{:}M) - H(M) + \chi(M{:}E_G).$$

Without G, the right side is $-H(M \mid X) \le 0$, so this task has no positive net gain over the blind harvest. If the records are perfect ($r_F = r_G = 0$), the bound equals $h(p)$ and is approached by reading with a controlled-NOT, extracting with the matched protocol and resetting M by a controlled-NOT from G; with exactly perfect records it is reached only as a limit.

Proof. Theorem 16 bounds the extraction gain by $k_B T\, I(X{:}M)$. For the reset, the joint state of M, G and the bath evolves unitarily from a state in which the bath is thermal and uncorrelated with MG; with degenerate Hamiltonians and vanishing coupling at the endpoints, $W_{\mathrm{reset}} = \Delta E_B$. As in Lemma 15, $\beta \Delta E_B \ge S(\rho_{MG}) - S(\rho'_{MG})$, and with M pure at the end and the marginal of G unchanged, $S(\rho'_{MG}) = S(\rho_G)$, so $\beta W_{\mathrm{reset}} \ge S(M \mid E_G)$. For classical M, $S(M \mid E_G) = H(M) - \chi(M{:}E_G)$. Without G, $I(X{:}M) - H(M) = -H(M \mid X)$. With perfect records, M is a copy of X, $\chi(M{:}E_G) = H(M) = h(p)$, and the controlled-NOT from G maps M to its standard state at no work. □

For pure conditional fragment states with $p = 1/2$, $c_k = 0.8$ and Helstrom readings of five-memory fragments, $I(X{:}M) = 0.5668$, $H(M) = \ln 2$ and $\chi(M{:}E_G) = 0.5231$, so the net task gain is at most $0.3968\, k_B T$ with a second five-memory copy and at most $-0.1263\, k_B T$ without one; with seven-memory fragments the bounds are 0.5511 and −0.0611. Quantum conditional-entropy erasure has asymptotic realizations with the corresponding collective control and side-information permissions [19]. This does not give an arbitrary finite local eraser those permissions. The following classical record pair supplies a direct realization within the present task. If pure blanks are supplied instead, their entropy capacity can lower reset work and can itself fund work extraction; their consumed free energy must enter the input ledger. The harvest's information ceiling alone does not cap harvest minus a work-producing reset.

7.1 Two mixed ports: feedback and an actual conditional reset

Use the fair diagonal pointer and the controlled-X recording protocol of Paper F [26]. The two independent initial ports are $\eta_\alpha = (\mathbb{1} + \alpha Z)/2$, with $0 < \alpha < 1$ and $\epsilon = (1-\alpha)/2$. Their endpoint Hamiltonians are zero. The recording gates act as

$$(x, n_1, n_2) \longmapsto (x, m = x \oplus n_1, g = x \oplus n_2), \qquad X \sim \mathrm{Bern}(1/2), \quad N_1, N_2 \overset{\mathrm{iid}}{\sim} \mathrm{Bern}(\epsilon).$$

Actual Z readout uses the diagonal port M as the feedback control; no extra free copy of its value is introduced. The other physical port G is retained and is accessible to the reset. Preparation, gates and any controller that reads their values remain part of the apparatus.

Proposition 24 (Mixed-record task realization)**.** *In this protocol, put $t = 2\epsilon(1-\epsilon)$. Matched feedback on M has work gain over the blind harvest $k_B T[\ln 2 - h(\epsilon)]$. With G available as a reversible reset control, the infimum reset work is $k_B T h(t)$, and the supremum net task gain is*

$$k_B T\big[\ln 2 - h(\epsilon) - h(t)\big]. \tag{11}$$

Without G, the infimum reset cost is $k_B T \ln 2$ and the net gain is $-k_B T h(\epsilon)$. The supremum in (11) is approached as a quasistatic, vanishing-reset-error limit and is positive for sufficiently small ϵ. It does not restore the prepared record source or the thermal bath supply.

Proof. The actual joint record law is $P(m, g \mid x) = P_\epsilon(m \mid x) P_\epsilon(g \mid x)$. Thus M is fair, $H(X \mid M) = h(\epsilon)$ and $I(X{:}M) = \ln 2 - h(\epsilon)$. Lemma 15 with the two full-support posteriors gives the stated harvest. The matched branch protocol ends with a thermal source independent of the records, and leaves the classical joint law of MG unchanged.

The reversible controlled-NOT from G to M changes its physical content to $Z = M \oplus G = N_1 \oplus N_2$. The pair (Z, G) has law $\mathrm{Bern}(t) \otimes \mathrm{Bern}(1/2)$: for each fixed pair of noises the fair X makes G uniform. Hence $S(M \mid G) = h(t)$ before the reset, and after this gate the noisy logical bit is independent of the retained control.

Choose $t = q_0 > q_1 > \cdots > q_\ell = \zeta > 0$. At stage i, swap the degenerate logical memory with a fresh independent Gibbs qubit whose gap is $\Delta_i = k_B T \ln[(1 - q_i)/q_i]$. This is a driven swap, with endpoint interaction off. It resets the logical memory to error ζ, leaves G unchanged and independent of it, and has

$$\beta W_{\mathrm{reset}} = \beta Q_{\mathrm{bath}} = \sum_{i=1}^{\ell} (q_{i-1} - q_i) \ln \frac{1 - q_i}{q_i} \longrightarrow h(t) - h(\zeta).$$

The limit is the thermal-swap construction of Paper F and Reeb–Wolf [26, 21]. Letting $\zeta \downarrow 0$ reaches the entropy lower bound $h(t)$. The post-parity register is reset to the standard zero value, so the parity gate is not undone. Without G the same construction starts from the fair M and approaches cost $\ln 2$. Combining these reset costs with the matched harvest proves the claims. □

At $\epsilon = 0.05$ ($\alpha = 0.9$), the dimensionless harvest is 0.494632, conditional reset costs 0.313956 and net task gain is 0.180676; without the second record the net value is −0.198515. Both ports are partially mixed. Their total initial purity deficit is $2[\ln 2 - h(\epsilon)]$, paid before

acquisition; their one-port record values saturate that budget in Paper F. The recording gates themselves have zero net endpoint work. Harvesting is a separate source–bath operation, not work extracted by the recording gate.

The retained G need not be a perfect copy of X: its correlation with the private memory is the useful reset resource. In the present menu it can fund a positive task balance while its marginal is unchanged. Its correlation with the harvested source, the two prepared ports, the driven gaps, clock and fresh Gibbs qubits have not all returned to their input states. A subsequent use must supply a new compatible source–record state and account for that preparation. More generally, an upkeep cost w can fit within the task only if $k_B T[I(X{:}M) - S(M \mid E_G)] \geq w$, before charging any omitted record supply or apparatus costs. This connects readable records to an action budget without identifying it with a sustainable complete-device cycle.

8 Operational temporal relations

The accounting identity does not derive a temporal order. HAFF-F now separates a genuine endpoint inverse problem from influence at supplied times: commuting pointer-record gates have an order-independent complete final state, whereas retained labels in its record–swap circuit identify an unknown visit permutation. Local gate orientation, initial references and the one-pass architecture remain inputs [26]. The two influence relations below instead use an already specified time parameter; neither should be confused with that reconstruction.

Definition 25 (Kernel memory and intervention influence)**.** *For a reduced model with Nakajima–Zwanzig kernel $\mathcal{K}$ and tolerance $\epsilon > 0$, write $s \rightsquigarrow_{\mathcal{K}} t$ if $\|\mathcal{K}(t,s)\| > \epsilon$ for some $s < t$; this is kernel memory. For a record-forming protocol, write $s \rightsquigarrow_{\mathrm{rec}} t$ if an intervention on the system at time s changes the records written at time t, as in the diagnostic of Paper F [26].*

The total linear response of a state at t to a perturbation at s is not a memory diagnostic: a Markov semigroup propagates a perturbation $\delta\rho$ as $e^{(t-s)\mathcal{L}}\delta\rho \neq 0$. The two relations of Definition 25 come apart in both directions. In the collision model the reduced dynamics is a product of dephasing channels with a time-local generator, so there is no kernel memory beyond a single step, while an intervention after the j-th collision changes every later record and none of the earlier ones (the fresh-fragment intervention result of Paper F). In the two-qubit model the coherence revives, so the reduced dynamics is not a semigroup and its Nakajima–Zwanzig kernel is not local in time, while the environment never holds a record. A correspondence between a memory-influence relation and nested record algebras therefore holds in neither direction in general; in fresh-memory collision models the record relation and the intervention order coincide, and kernel memory is absent.

9 The cost of keeping records

Landauer's principle prices logically irreversible resets, not passive storage, sensing, correlation maintenance or reversible computation. If a memory implementation irreversibly

erases N_{erase} unbiased bits into a bath at temperature T, its reset work obeys $W_{\text{reset}} \geq k_B T \ln 2\, N_{\text{erase}}$; reversible storage and measurement can postpone this cost until an irreversible operation is performed [12, 13]. No lower bound proportional to stored mutual information follows without an implementation model, and Proposition 23 shows how strongly the reset cost depends on what the eraser can access.

To describe how much of a history is worth keeping, let $\{X_t\}$ be a stationary process describing the environment as seen by the agent. Its entropy rate is $h_\mu = \lim_{k\to\infty} h_k$ with $h_k = H(X_k \mid X_1, \ldots, X_{k-1})$ and $h_1 = H(X_1)$, and its excess entropy, or predictive information, is

$$\mathbf{E} = I(\overleftarrow{X}; \overrightarrow{X}) = \sum_{k=1}^{\infty} \big(h_k - h_\mu\big) \tag{12}$$

whenever the series converges [22, 24, 23]. For a symmetric two-state Markov chain with flip probability $0 < \varepsilon < 1/2$, $h_1 = \ln 2$ and $h_k = h(\varepsilon)$ for $k \geq 2$, so $\mathbf{E} = \ln 2 - h(\varepsilon)$. Finite excess entropy does not prove that a finite physical memory captures every useful feature, and a positive entropy rate does not bound the predictive information from below by h_μ times a memory depth.

Remark 26 (Raw-archive overload)**.** *Consider a specified implementation: an uncompressed archive receives records at r_{raw} bits per unit time, has capacity M bits and evaluates the entire archive once per control cycle. Without deletion it fills after M/r_{raw}. If it irreversibly resets discarded records at rate r_{erase}, Landauer gives $\dot{W}_{\text{reset}} \geq k_B T \ln 2\, r_{\text{erase}}$. For a long-memory or non-stationary source with divergent $\mathbf{E}$, the predictively relevant history may also grow; that is an assumption about the source. The archive can then exceed a finite storage, throughput or reset budget. This is an engineering overload scenario for one implementation.*

An agent using such an archive can respond by enlarging capacity, sampling less, uncomputing reversibly, keeping sufficient statistics, compressing lossily or accepting stale data. Each choice of what to keep needs a criterion that ranks stored correlations by their value. Section 6 supplies one such value in work units for a single record; Paper II studies what a bounded agent reads, where a preferred frame arises from the agent's history rather than from the task ensemble. Neither compression, nor Clifford coordinates, nor symmetry breaking, nor a "self" is shown here to be thermodynamically necessary in general.

10 A classical filtering illustration

The following hidden Markov model compares finite observation windows with a Bayes filter in an aliased environment. Its reward is classification accuracy, not the survival functional, so it illustrates the task-level value of retained history rather than testing a theorem.

Proposition 27 (A finite-history classification separation)**.** *Let a fair binary hidden state flip with probability ε per step. Conditional on the hidden path, observations are independent, with $\Pr(O = 1 \mid X = 1) = 1/2 + \delta$ and $\Pr(O = 1 \mid X = 0) = 1/2 - \delta$, where $0 < \delta < 1/2$. A decision using only the last k observations, the known model and independent private randomness has classification risk at least*

$$R_k \geq \tfrac{1}{2}(1 - 2\delta)^k.$$

For odd n, the majority of n observations ending at the decision time has risk

$$R_{\mathrm{maj,n}} \le (n-1)\varepsilon + \exp\bigl(-2n\delta^2\bigr).$$

For every fixed k, some finite n and positive ε make the second bound smaller than the first.

Proof. Every observation pattern of length k, conditional on either terminal hidden value, has probability at least $(1/2-\delta)^k$: each emission factor has that lower bound, and averaging over hidden paths preserves it. For the fair prior the optimal risk is $\frac{1}{2}\sum_o \min\{P(o \mid X = 0), P(o \mid X = 1)\}$. Summing the lower bound over 2^k patterns gives the first inequality; randomization cannot improve the optimal rule.

With probability at least $1-(n-1)\varepsilon$ there is no hidden flip in the n-observation episode. On that event the observations are independent samples with mean $1/2+\delta$ or $1/2-\delta$, so Hoeffding's inequality [31] bounds the majority error by $\exp(-2n\delta^2)$. The complement is charged its full probability. Choose odd n so that the exponential is below half the first bound, then choose $\varepsilon > 0$ small enough to make the remaining term smaller than that half. □

For $\delta = 0.05$, $k = 8$, $n = 1001$ and $\varepsilon = 10^{-4}$, these bounds are $R_{\mathrm{maj,n}} \le 0.106704 < 0.215234 \le R_8$. The count can be stored in $\lceil \log_2(n+1) \rceil = 10$ bits, in addition to the episode clock, decision and calculation resources; keeping every raw reading would require n bits instead. The comparison excludes additional correlated side information for the window rule. Acquisition, counter updates and any reset of its workspace still require an implementation budget; the classification bounds do not price them. It establishes a finite task separation, not a fixed physical memory that beats all window lengths, and not a bound on thermodynamic profit.

The hidden state $s_t \in \{0,1\}$ keeps its value with probability $1-\varepsilon$, $\varepsilon \in [10^{-3}, 10^{-0.7}]$, so the correlation length is $\ell \sim 1/\varepsilon$. Observations are aliased: $\Pr(o_t = A \mid s_t = 0) = 0.5+\delta$ and $\Pr(o_t = A \mid s_t = 1) = 0.5-\delta$ with $\delta = 0.05$, so a single observation carries 0.0072 bits about the state. The reward is 1 when the action equals s_t. All agents know the model and compute exact posteriors; Markov-k agents ($k \in \{1,2,4,8\}$) use only the last k observations from a uniform prior, and the memory agent runs the full Bayes filter. The supplied script runs $T = 10^5$ total steps, discards the first 2000, and uses 10 seeds at 16 logarithmically spaced values of ε. The supplementary 16-row summary reports this sweep.

The reward of the Bayes filter increases with the correlation length, while each Markov-k agent saturates at a level ordered by k, all below the filter for $\ell \gtrsim 20$ (Figure 1). The Markov-1 and Markov-2 curves nearly coincide at small ℓ because short windows add little in this aliasing regime. The sampled long-correlation regime shows an advantage over the four specified windows. For the given model and observation history, the full posterior Bayes decision minimizes classification risk. The example does not establish tight saturation constants or superiority across different reading and resource menus. Theorem 16 supplies the conversion to work for a different task, in which the retained statistic is used for feedback on a system whose state it predicts. The supplementary source record gives the script, historical sweep and legacy-data distinction; these classification data do not computationally verify a survival-functional theorem.

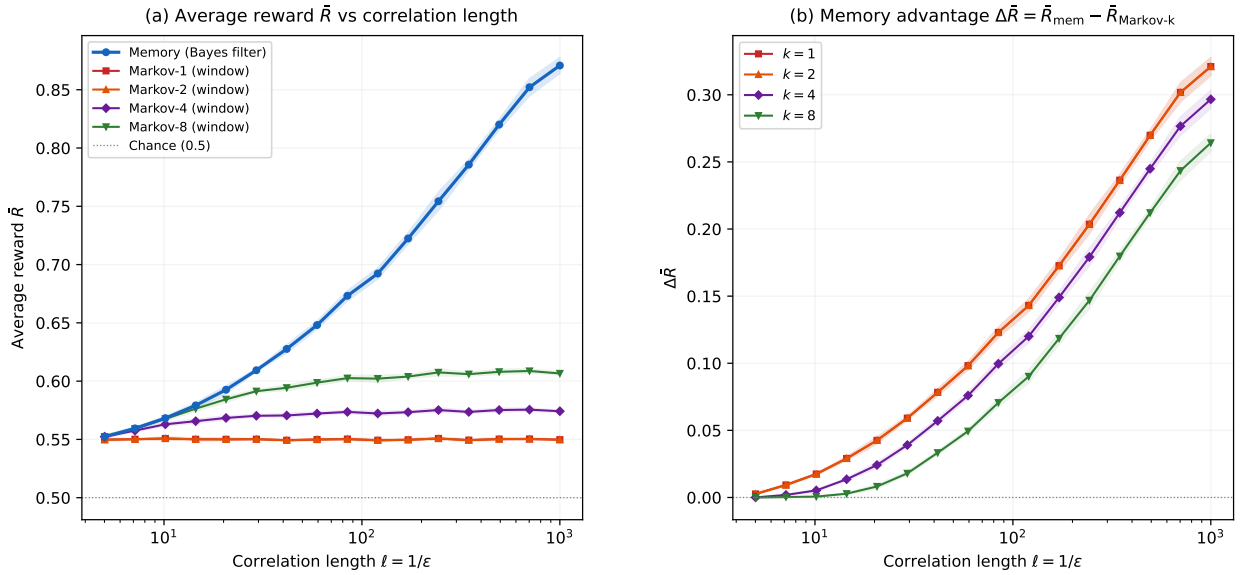


Figure 1: **Finite-window saturation and Bayes-filter advantage.** Historical sweep: $T = 10^5$ total steps, 2000 discarded, 10 seeds. Shading uses 1.96 times the across-seed standard deviation (with NumPy's population convention) divided by $\sqrt{10}$; these are normal-approximation error bands, not rigorous finite-sample confidence intervals. **(a)** Average reward $\bar{R}$ against correlation length $\ell = 1/\varepsilon$. The Bayes filter (blue, bold) rises monotonically; Markov-k agents saturate at k-dependent levels. **(b)** Gap $\Delta\bar{R} = \bar{R}_{\text{mem}} - \bar{R}_{\text{Markov-}k}$, which increases with ℓ and is larger for smaller k.

11 Discussion

11.1 The memory–self–introspection route

In the task of Section 6, a work gain requires a memory carrying the relevant correlation. A protocol matched to its posterior realizes the value $k_B T\, I(X{:}M)$; every error in that posterior has a computable price. Reference calibration builds the required readout model, and reuse pays its preparation cost through the square-root amortization rule. Paper III proves the stated scalar Gaussian drift law. These are the concrete links in the series' memory–self–introspection route. Proposition 23 adds the reset condition for its allowed menu, and Proposition 24 realizes positive task value with two partially mixed records. A useful second record reduces the reset price; a full apparatus cycle still has to renew the source, recording stock and bath supply. The conditions belong to this task and its allowed protocols. Driven Markovian systems can persist without memory or a self-model.

The results concern function. They show what an agent must hold and do to cash a record, and they do not show that holding a model of one's own readout is, or produces, experience. The relation between such functional self-models and first-person organization is treated in the Consciousness series with its own arguments.

The task fixes the record source, pointer basis and feedback device. Paper II addresses how an agent selects what it reads; Paper III addresses detection, the price of tracking, and repair selected by the history of its body-gated access. The three correlation models

here remain distinct: the Lorentz–Drude bath carries pointer information without designated fragments, the two-qubit environment carries none, and collision records have no autonomous recurrence. Paper III combines the collision records, a bounded changing instrument, a field-coupled actuator, harvest and reset on one ledger [29]. It exhibits a positive operating margin after body upkeep under supplied food, record purity, drive preparation, phase reference and timing. The calculation does not make those inputs products of the loop. Paper III also constructs a continuing finite joint filter: a bounded changing instrument and controlled body generate the history that revises their model, with a uniform quantization-risk bound and adaptive reference timing. The updating mechanism is thus explicit under its stated transition law and resource permissions. Its processor, reference supply and drive preparation still need one sustainable apparatus ledger.

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Reference Frames Selected by History: The Capability and Cost of a Bounded Agent's Self-Frame

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Abstract

When a task's coupling to its channels is unknown, a bounded reader can use retained history to choose its next evidence. We call the retained relation and the foreground it selects a self-frame. In a two-channel Gaussian task, a finite training history forms a signed-coordinate frame retained in two bits. With one test read, its certified risk is below every history-independent foreground's risk, even when the latter is given its exact regression coefficient. Replaying a frame from an independent task destroys this advantage; freezing the learned frame in its original static task preserves it. For general dimension we derive the optimal posterior reading subspace, the information price of alignment, and the additional signed information needed to predict. In-frame evidence cannot locate omitted directions and can confound noise with misalignment. Exact drift curves and a calibration policy quantify the value and cost of off-frame probes. For eight channels and three reads, that policy lowers static risk from 0.86 to 0.26. A quantum reference supplies a separate realization: its bounded-task advantage is controlled by asymmetry, and a qubit phase token has an exact discrimination value under dephasing. These results give the memory–self–introspection route a task-specific mechanism: retained history chooses future access, and that access limits the evidence available for revising the frame.

1 Introduction

1.1 The question

A system that persists by using its records cannot use all of them at once. It reads some channels and not others, keeps some correlations and lets others go. Whatever rule makes that choice embodies a view of where the task lies in the system's own coordinates. If the system's coupling to the world is not known in advance, that view has to be learned, and once learned it decides which evidence the system will see next. This paper asks four questions

about such a view. When is it informationally necessary? How can it be formed from the agent's own records? What capability does it buy? What does it cost, including what it prevents the agent from seeing?

The series calls this organized view the ego. The word is used here for a definite object: a retained model of how a task-relevant direction sits in the agent's own channels, together with the foreground of reads that the model selects. We call the pair a self-frame. It is a model of the agent's own coupling, because the direction it describes is the image of a world direction under that coupling.

1.2 Position within the series

This paper is the second of the T-DOME (Thermodynamic Dynamics of Observer-Memory Entanglement) series, the third line of a programme whose other lines are HAFF and Q-RAIF.

HAFF-E supplies the actual record protocol [5]; Q-RAIF supplies physical interfaces and their prices [6]. Here we ask how the retained history organizes the reader itself.

Paper I bounds the extra work available from a record by $k_B T\, I(X{:}M)$ in a matched feedback task, explains when that bound is attained, and includes preparation, updates and old correlations in the complete account [3]. It leaves open how an agent comes to select what it keeps. Paper III studies drift detection, the price of tracking a changing instrument, and history-selected repair of body-gated access [4]. The present paper sits between them. Its self-frame is formed from records, it decides which records are gathered next, and its blind spot is the reason a calibration loop is needed at all.

The main results start with the posterior capability (Theorem 2) and the information needed to select and use a foreground (Theorem 4 and Proposition 5). Proposition 9 then constructs a frame from finite observations with bounded working registers and a two-bit retained code. It compares a history-selected reading direction with a preselected direction, granting the latter its exact regression coefficient. The comparison isolates the value of changing the next reading; it does not match retained memory or total implementation cost, or distinguish a self-frame from general learning. The subsequent blind-spot and drift results explain when that selected access needs revision. Section 9 gives an explicit revision policy with exact stationary risk. Section 6 treats quantum references separately; Section 10 connects history-selected access to learned protection without identifying the two implementations.

1.3 Related work

Bounded rationality [11], rate–distortion theory [12, 13], rational inattention [14] and the information bottleneck [15] treat the choice of what to keep as an optimization under an information constraint. Budgeted learning studies learners that may observe only a few attributes of each example; Cesa-Bianchi, Shalev-Shwartz and Shamir show, among other results, that a budget on the attributes available at prediction time can make arbitrary accuracy unattainable even when a perfect linear predictor exists [16] . The present setting is close to theirs, with a different emphasis: the unknown is the agent's own coupling, the foreground is a function of the agent's memory, and the questions are the information that the

foreground must carry and the evidence it excludes. Reference frames as physical resources are reviewed by Bartlett, Rudolph and Spekkens [17]; Skotiniotis and Gour give the accessible information of frame-alignment protocols an operational role for the G-asymmetry [18] . Theorem 4 is the classical counterpart for a direction on a sphere: an application of the Gibbs variational principle, which we do not claim as new mathematics. Its use here is to price the self-frame of a bounded agent and to connect that price to the agent's capability and blind spot.

2 The budgeted reader

2.1 Channels, task and coupling

At each step t the agent's D channels register a fresh input $x_t \sim \mathcal{N}(0, I_D)$, independent of everything before. After the agent responds, the consequence

$$y_t = \rho\, u_t \cdot x_t + \xi_t, \qquad \xi_t \sim \mathcal{N}(0, s^2), \tag{1}$$

is revealed, with $\rho > 0$ and $|u_t| = 1$. Unless stated otherwise the agent knows ρ and s; Theorem 13(iii) drops this. The agent is scored by the squared error of its anticipation $\hat{y}_t$ of y_t. More generally the consequence may be a vector $y_t = A_t x_t + \xi_t \in \mathbb{R}^p$ with noise covariance of trace ν; we write $W = A^\top A$.

The direction u_t is expressed in the agent's channel coordinates. If e is the task-relevant direction in world coordinates and $R_t \in O(D)$ is the agent's coupling, which maps its channels to the world, then $u_t = R_t^\top e$. A model of u_t is therefore a model of the agent's own coupling as far as this task is concerned. The records cannot separate a change of R_t from a change of e; only their relation enters. Distinguishing the two requires a reference with a known world direction, the role played by reference reads in Paper III, or a known push through an actuator that shares the agent's sensing coupling; Paper III shows that pushes through an independent coupling do not help [4].

Symmetric ensemble. The initial coupling is Haar distributed, so u_0 is uniform on the sphere S^{D-1}, and more generally the law of A_0 is invariant under $A_0 \mapsto A_0 O$ for $O \in O(D)$. Nothing in the task specification prefers any channel.

2.2 The reading budget

At each step the agent may make at most b linear reads $w_1 \cdot x_t, \ldots, w_b \cdot x_t$ with orthonormal w_j chosen from what it retains. The span of the reads used for the response is its *foreground* P_t, an orthogonal projection of rank $k \le b$. Everything the agent retains from past steps is its memory M_t; it may contain past reads, past outcomes and any summaries of them. The foreground and the response are functions of M_t and of the current reads only.

A full probe reads all D channels. With $b < D$ it cannot be made at every step. The resource protocol charges reads per step, start-up probes and retained numbers or bits; it does not assign a thermodynamic price. Section 9.3 relates it to the work currency of Paper I.

Definition 1 (Self-frame)**.** A self-frame is a retained state from which the agent computes, at each step, an estimate of how the task direction sits in its own channels and the foreground that this estimate selects. Its alignment at step t is $c_t = \mathbb{E}|P_t u_t|^2$ for rank-one tasks.

2.3 Comparators

Except for Proposition 9, the comparisons in this paper charge the same b reads per step and the same start-up probes. That proposition fixes one test read while granting the history-independent direction its exact regression coefficient for free; it does not match retained capacity or total cost.

- *No memory*: $\hat{y} = 0$.
- *No self-frame*: a foreground fixed independently of the history (for definiteness the first b channels), with in-frame coefficients learned from outcomes.
- *Fixed self-frame*: a foreground formed once from the start-up probes, with in-frame coefficients learned afterwards.
- *Frozen self-model*: frame and coefficients formed once and never revised.
- *Calibratable self-frame*: $b - 1$ coordinate probes plus one orthogonal residual read, which together supply the response along the current frame and revise it.

3 What a foreground is worth

Theorem 2 (Capability of a budgeted reader)**.** *Let M be any memory independent of the current input, let P be a rank-k foreground that is a function of M, and let $\hat{y}$ be any function of (Px, M). For the vector task,*

$$\mathbb{E}|y - \hat{y}|^2 \;\geq\; \nu + \mathbb{E}\,\mathrm{tr}\,W - \mathbb{E}\,\mathrm{tr}\big(P\,\widehat{W}_M P\big) \;\geq\; \nu + \mathbb{E}\,\mathrm{tr}\,W - \mathbb{E}\sum_{i\leq k}\lambda_i(\widehat{W}_M), \tag{2}$$

where $\widehat{W}_M = \mathbb{E}[A \mid M]^\top \mathbb{E}[A \mid M]$ and $\lambda_1 \geq \lambda_2 \geq \cdots$ are its eigenvalues. The first bound is attained by $\hat{y} = \mathbb{E}[A \mid M]Px$; the second by taking P to be a top-k eigenspace of $\widehat{W}_M$. For the rank-one task (1)*, with $m = \mathbb{E}[u \mid M]$,*

$$\mathbb{E}(y - \hat{y})^2 \;\geq\; s^2 + \rho^2\big(1 - \mathbb{E}|Pm|^2\big) \;\geq\; s^2 + \rho^2\big(1 - \mathbb{E}|Pu|^2\big). \tag{3}$$

Proof. Write $x = Px + (I - P)x$. Given M, the input x is independent of (A, M), and for fixed P the two Gaussian components are independent of each other. Hence $\mathbb{E}[(A(I-P)x) \mid Px, M] = \mathbb{E}[A \mid M]\,\mathbb{E}[(I-P)x] = 0$ and $\mathbb{E}[APx \mid Px, M] = \mathbb{E}[A \mid M]Px$. The Bayes response is therefore $\mathbb{E}[A \mid M]Px$, and its risk is $\mathbb{E}|y|^2 - \mathbb{E}|\mathbb{E}[A \mid M]Px|^2 = \nu + \mathbb{E}\,\mathrm{tr}\,W - \mathbb{E}\,\mathrm{tr}(P\widehat{W}_M P)$, using $\mathbb{E}[xx^\top] = I$. No other response has smaller risk. Ky Fan's maximum principle gives $\mathrm{tr}(P\widehat{W}P) \leq \sum_{i\leq k}\lambda_i(\widehat{W})$ for every rank-k projection, with equality on a top-k eigenspace. For the rank-one task $\widehat{W} = \rho^2 mm^\top$, which gives the first inequality in (3). For the second, $\mathbb{E}|Pu|^2 = \mathbb{E}\,\mathbb{E}[|Pu|^2 \mid M] \geq \mathbb{E}|P\,\mathbb{E}[u \mid M]|^2$ by Jensen's inequality, since P is fixed given M. □

Corollary 3 (Three limits)**.** *(a)* No history. *If M is independent of the coupling, then $\mathbb{E}[A \mid M] = 0$ in the symmetric ensemble and every response has risk at least $\nu + \mathbb{E}\,\mathrm{tr}\,W$, the risk of $\hat{y} = 0$.*

(b) Known task. *If M determines A, the best rank-k risk is $\nu + \sum_{i>k} \lambda_i(W)$, attained by the top-$k$ eigenspace of W. Each added dimension strictly helps while $\lambda_{k+1} > 0$, and the optimal subspace is unique when $\lambda_k > \lambda_{k+1}$. If $W = \lambda I$, every rank-k subspace is optimal.*

(c) Frame-free level. *If the foreground is independent of the coupling, then $\mathbb{E}|Pu|^2 = k/D$ and every policy has risk at least $s^2 + \rho^2(1 - k/D)$ in the rank-one task.*

Proof. (a) Invariance gives $\mathbb{E}[A] = \mathbb{E}[A]O$ for all $O \in O(D)$, hence $\mathbb{E}[A] = 0$ for $D \geq 2$; use (2). (b) Here $\widehat{W} = W$. (c) For P independent of the uniform u, $\mathbb{E}|Pu|^2 = \mathrm{tr}\,P/D$. □

Theorem 2 has the same form as Paper I's theorem that every feedback protocol assumes a posterior [3]: whatever a budgeted policy represents, its performance is bounded by the alignment of the foreground it actually uses. The agent need not store an explicit direction. It does need some retained state from which its foreground is computed, and no response computed from that state can do better than the alignment of its foreground allows.

Part (b) is the known-task case of rank selection, a spectral variational fact. It locates the source of any preferred subspace. When W is anisotropic with a known eigenbasis, the selection is fixed by the task, an explicit asymmetry. When the ensemble is symmetric, as here, no subspace is preferred before records arrive. A preferred foreground can then come only from the agent's history, which is the subject of the next section.

4 A frame is a record of the coupling

For uniform u and a fixed rank-k projection P, the overlap $t = |Pu|^2$ has the Beta law $\beta_{k,D} = \mathrm{Beta}(k/2, (D-k)/2)$ with mean k/D and moment generating function $\mathbb{E}e^{\kappa t} = {}_1F_1(k/2; D/2; \kappa)$. Define

$$\Lambda^*_{D,k}(c) = \sup_{\kappa \geq 0} \Big[\kappa c - \ln {}_1F_1\big(\tfrac{k}{2}; \tfrac{D}{2}; \kappa\big) \Big] \tag{4}$$

for $0 \leq c < 1$. It is the upper half of the Legendre transform of the log moment generating function of $\beta_{k,D}$: convex, zero for $c \leq k/D$ and increasing above.

Theorem 4 (The information a self-frame must carry)**.** *In the symmetric rank-one ensemble, let P be a rank-k foreground that is a function of the agent's memory. Then*

$$I(U; P) \;\geq\; \Lambda^*_{D,k}\big(\mathbb{E}|PU|^2\big). \tag{5}$$

For $k/D \leq c < 1$ the bound is attained: if P is Haar distributed and, given P, U has density proportional to $e^{\kappa|PU|^2}$ with respect to the uniform measure, with $\kappa \geq 0$ chosen so that $\mathbb{E}|PU|^2 = c$, then U is uniform and $I(U; P) = \Lambda^*_{D,k}(c)$. (For $\kappa < 0$ the same construction

gives the lower branch of the Legendre transform, which (5) *does not use.) Consequently, a policy whose risk is at most* $r < s^2 + \rho^2(1 - k/D)$ *has a foreground with*

$$I(U;P) \;\geq\; \Lambda^*_{D,k}\Big(1 - \frac{r - s^2}{\rho^2}\Big) > 0. \tag{6}$$

Proof. Since U is uniform, $I(U;P) = \mathbb{E}_P\, D(\mathcal{L}(U \mid P)\|\mathrm{Unif})$. Fix a value of P and map $u \mapsto |Pu|^2$. Under the uniform law the image is $\beta_{k,D}$; let Q_P be the image of $\mathcal{L}(U \mid P)$. The data-processing inequality gives $D(\mathcal{L}(U \mid P)\|\mathrm{Unif}) \geq D(Q_P\|\beta_{k,D})$. The Gibbs variational inequality $D(Q\|\beta) \geq \kappa\mathbb{E}_Q t - \ln \mathbb{E}_\beta e^{\kappa t}$ for every κ gives $D(Q_P\|\beta_{k,D}) \geq \Lambda^*_{D,k}(\mathbb{E}_{Q_P} t)$. Averaging over P and using convexity of Λ^* (Jensen) proves (5). In the attaining construction, rotation invariance of the uniform measure and of the Haar law of P makes the marginal of U uniform. Given P, the law of U is the exponential tilt of the uniform law by $\kappa|Pu|^2$; its image is the corresponding tilt of $\beta_{k,D}$, both inequalities are equalities, and the relative entropy is $\kappa c - \ln {}_1F_1(k/2; D/2; \kappa)$ with κ chosen so that the tilted mean is c. For (6), Theorem 2 gives $\mathbb{E}|PU|^2 \geq 1 - (r - s^2)/\rho^2$, and Λ^* is increasing on $[k/D, 1)$. □

Proposition 5 (Information needed to use an aligned frame)**.** *Let* U *be uniform on* S^{D-1}*,* $D \geq 2$*, and let* $m_M = \mathbb{E}[U \mid M]$*. Set*

$$\Psi_D(a) = \sup_{t \geq 0}\{ta - \log \mathbb{E}e^{tU_1}\}, \qquad 0 \leq a \leq 1,$$

with $\Psi_D(1) = +\infty$*; the inequalities use extended relative entropy. For any memory and any allowed rank-k foreground,*

$$I(U;M) \geq \mathbb{E}\Psi_D(|m_M|) \geq \frac{D}{2}\mathbb{E}|m_M|^2 \geq \frac{D}{2}\mathbb{E}|Pm_M|^2. \tag{7}$$

Consequently a predictor with risk at most $r < s^2 + \rho^2$ *needs* $I(U;M) \geq \frac{D}{2}[1 - (r - s^2)/\rho^2]$ *nats. This bound concerns the signed conditional mean needed for prediction; it is not asserted to be sharp.*

Proof. For each posterior, the relative-entropy variational inequality applied to $tv \cdot U$, with $v = m_M/|m_M|$, gives $D(\mathcal{L}(U \mid M)\|\mathrm{Unif}) \geq \Psi_D(|m_M|)$. At $m_M = 0$ take $t = 0$. The uniform sphere has odd moments zero and

$$\mathbb{E}U_1^{2j} = \frac{(2j-1)!!}{D(D+2)\cdots(D+2j-2)} \leq \frac{(2j-1)!!}{D^j}.$$

Termwise comparison of the moment-generating series gives $\log \mathbb{E}e^{tU_1} \leq t^2/(2D)$. Choosing $t = D|m_M|$ gives the second inequality in Eq. (7); averaging gives the mutual information. Projection contracts the norm. Finally Theorem 2 implies $\mathbb{E}|Pm_M|^2 \geq 1 - (r - s^2)/\rho^2$ for any such predictor. □

The attaining law in Theorem 4 is invariant under $U \mapsto -U$ conditional on P. Its posterior mean is therefore zero: a memory consisting only of that P has no predictive advantage, despite its high alignment. At the limit, knowing the unoriented line $UU^\top$ gives alignment one and still leaves the sign undetermined. Outcome-bearing records, not merely

D	$c = 0.5$	$c = 0.9$	$c = 0.99$	ratio to $\frac{D}{2}\log_2\frac{1}{1-c}$ at $c = 0.9$
2	0	1.18	2.92	0.354
3	0.21	2.80	6.19	0.561
8	2.46	11.07	22.76	0.833
16	6.45	24.35	49.33	0.916
64	30.49	104.07	208.78	0.979
256	126.50	422.97	846.58	0.995

Table 1: Minimum frame information $\Lambda^*_{D,1}(c)$ in bits for a one-dimensional foreground (Theorem 4). For $D = 2$, $c = 1/2$ is the frame-free alignment.

an axis label, supply the signed relation required by the response. The alignment table below is an information bound for choosing a subspace, not an achievable risk–information curve. For a finite memory its entropy bounds both the alignment price and the signed price. The step-capacity argument below likewise applies to either necessary information bound.

Corollary 6 (Price of forming a frame)**.** *(a) A memory from which a foreground of alignment c is computed holds at least $\Lambda^*_{D,k}(c)/\ln 2$ bits.*

(b) Each step, whatever reads are made, conveys at most $C_1 = \frac{1}{2}\ln(1 + \rho^2/s^2)$ nats about a static coupling. Forming a foreground of alignment c from scratch therefore takes at least $\Lambda^*_{D,k}(c)/C_1$ outcome-bearing steps.

Proof. (a) $I(U;P) \le I(U;M) \le H(M)$. (b) Let H_{t-1} be the history. The reads at step t are functions of x_t and H_{t-1}, so $I(U;\text{reads}_t, y_t \mid H_{t-1}) \le I(U;y_t \mid x_t, H_{t-1})$. Given (x_t, H_{t-1}) the outcome has variance at most $\rho^2 x_t^\top \Sigma x_t + s^2$, with Σ the posterior covariance of u, and given U as well its entropy is that of $\mathcal{N}(\cdot, s^2)$. The Gaussian maximum-entropy bound and Jensen's inequality give at most $\frac{1}{2}\ln(1 + \rho^2 \operatorname{tr}\Sigma/s^2) \le C_1$, since $\mathbb{E}x_t^\top \Sigma x_t = \operatorname{tr}\Sigma \le 1$. The chain rule sums the steps, and $I(U;P) \le I(U;H_n)$. □

Table 1 lists $\Lambda^*_{D,1}$. For $D = 2$ the transform is $\sup_\kappa[\kappa(2c-1) - \ln I_0(\kappa)]$, the von Mises form on the doubled angle; the two expressions agree to 10^{-15} in the check. A Laplace evaluation of the moment generating function for large D gives $\Lambda^*_{D,1}(c) = \frac{D}{2}\ln\frac{1}{1-c}\,(1 + o(1))$ at fixed c, and the last column shows the approach. A self-frame thus costs about half a natural logarithm of the inverse misalignment per channel, the same per-direction precision cost that appears in finite self-models generally [7].

Remark 7 (Symmetry broken by the record)**.** If the agent's update rule is equivariant, so that rotating all channel coordinates rotates its records, memory and foreground alike, then the marginal law of the foreground over histories is Haar: the ensemble prefers no subspace. Conditional on one history the foreground is concentrated, and $I(U;P)$ measures how far. This is the precise sense in which a reference frame is selected spontaneously here. The symmetry of the task ensemble is intact; each history breaks it, and the frame is the trace that the history leaves. The reading budget makes a selection valuable (Corollary 3(c)); the records decide which frame is selected.

Remark 8 (Group-theoretic and quantum settings)**.** Nothing above uses the full orthogonal group beyond its transitivity on directions. If the channels carry a representation of a smaller group acting transitively on the sphere, for example $SO(3)$ acting on the pure quaternions of $Cl(0,2) \cong \mathbb{H}$, the overlap law and Theorem 4 are unchanged. In quantum settings an agent without a frame is restricted to group-averaged operations, and alignment resources are quantified by asymmetry measures [17, 18]. Section 6 gives a quantum task bound and an exact finite reference. The spherical information price is not identified with quantum asymmetry.

5 A finite history selects a two-bit reading frame

The information prices above do not by themselves construct a foreground from finite observations. We now do so in the same static Gaussian task, with $D = 2$ and U uniform on the unit circle. Training consists of n independent full probes: both coordinates of x_ℓ and the revealed y_ℓ are read. The comparison below restricts the alternative's reading direction to be independent of this history; it does not restrict the alternative's regression coefficient to ignore history. During the separate test phase exactly one linear read is allowed per fresh input; U remains fixed, and no further coefficient training is required by the constructed policy.

The supplied reading menu is $\mathcal{C} = \{e_1, -e_1, e_2, -e_2\}$. A retained code $M = w \in \mathcal{C}$ specifies both which coordinate to read and which signed prediction to make:

$$\widehat{y} = \rho\mu\, w \cdot x, \qquad \mu = \mathbb{E}\max(|U_1|, |U_2|) = \frac{2\sqrt{2}}{\pi}. \tag{8}$$

There are four codes, so only two learned bits enter the test phase. The codebook, read hardware and constants are supplied. One bit chooses the foreground axis and the other its prediction sign; an unsigned coordinate alone would not specify this response.

Proposition 9 (A finite history-selected frame beats every history-independent foreground)**.** *Let $\rho > 0$, $s \geq 0$, and use the training and test task above. For $L, q > 0$, preprocess each observed sample by*

$$z_\ell = y_\ell x_\ell/\rho, \qquad v_\ell = \mathrm{proj}_{\{|v|\leq L\}}\, z_\ell, \qquad \widetilde{z}_{\ell,i} = q\ \mathrm{round}(v_{\ell,i}/q),$$

with fixed nearest-integer tie rules independent of U. Set $g = n^{-1}\sum_{\ell=1}^{n} \widetilde{z}_\ell$ and choose $w \in \arg\max_{a\in\mathcal{C}} a \cdot g$, again with a fixed tie rule. With

$$M_2 = 4 + \frac{2s^2}{\rho^2}, \qquad e_{n,L,q} = \sqrt{\frac{M_2}{n}} + \frac{M_2}{L} + \frac{q}{\sqrt{2}}, \tag{9}$$

the prediction risk, averaged over U, training and the independent test input, satisfies

$$R_{\mathrm{formed}} \leq s^2 + \rho^2[1 - \mu^2 + 4\mu e_{n,L,q}]. \tag{10}$$

Every rank-one foreground chosen independently of the training history and U, even with its exact U-dependent regression coefficient supplied, has risk at least $s^2 + \rho^2/2$. Thus the finite formed frame is strictly better whenever $4\mu e_{n,L,q} < \mu^2 - 1/2$. For $n = 4096$, $\rho = s = 1$, $L = 256$, $q = 1/1024$, the upper bound is $1.414155 < 1.5$.

Proof. Condition on U. Gaussian second and fourth moments give $\mathbb{E}[z_\ell \mid U] = U$ and

$$\mathbb{E}[|z_\ell|^2 \mid U] = \mathbb{E}[(U \cdot x_\ell)^2 |x_\ell|^2] + (s^2/\rho^2)\mathbb{E}|x_\ell|^2 = 4 + 2s^2/\rho^2 = M_2.$$

The radial projection has mean bias bounded by

$$|\mathbb{E}[v_\ell \mid U] - U| \le \mathbb{E}[|z_\ell| \mathbf{1}_{\{|z_\ell| > L\}} \mid U] \le M_2/L.$$

Independence of the clipped samples and $|v_\ell| \le |z_\ell|$ give

$$\mathbb{E}\left[\left|\frac{1}{n}\sum_\ell v_\ell - \mathbb{E}[v_\ell \mid U]\right| \,\middle|\, U\right] \le \sqrt{M_2/n}.$$

Coordinate rounding changes each vector by at most $q/\sqrt{2}$. Consequently $\mathbb{E}[|g - U| \mid U] \le e_{n,L,q}$. The preprocessing uses only actual (x_ℓ, y_ℓ), not U.

Put $h(U) = \max(|U_1|, |U_2|)$ and let $w_*(U)$ maximize $a \cdot U$ over the same four codes. Since w maximizes $a \cdot g$,

$$U \cdot w \ge g \cdot w - |g - U| \ge g \cdot w_* - |g - U| \ge h(U) - 2|g - U|.$$

The circular mean of h is $(4/\pi)\int_0^{\pi/4} \cos\theta \, d\theta = 2\sqrt{2}/\pi = \mu$. The current test input x is independent of U and the retained code. Its conditional risk is $s^2 + \rho^2|U - \mu w|^2$. Averaging gives $R_{\text{formed}} = s^2 + \rho^2[1 + \mu^2 - 2\mu\mathbb{E}(U \cdot w)]$, which proves Eq. (10).

For a history-independent unit vector v, even an oracle coefficient cannot remove the Gaussian component orthogonal to v. Its risk is $s^2 + \rho^2[1 - \mathbb{E}[(U \cdot v)^2]] = s^2 + \rho^2/2$, also for a randomized v independent of U. This relaxation only helps the comparator: it need not pay memory or estimation error for that coefficient. Substituting the declared finite parameters gives the strict inequality. □

The ideal code $w_*(U)$ is not supplied to the learner. It describes the limiting comparison: by symmetry $\mathbb{E}[U \mid w_*] = \mu w_*$, and its risk is $s^2 + \rho^2(1 - \mu^2)$, or 1.189430531 at $\rho = s = 1$. The finite learner instead estimates from the actual probes and has the certified error above. No optimality among all two-bit encoders is claimed. The symmetric four-code hardware is also not a continuously rotation-equivariant policy; the example does not establish spontaneous breaking of $SO(2)$.

5.1 The retained code and the working registers

The selection need not store a real-valued g. Accumulate the two integers $A_i = \sum_\ell \text{round}(v_{\ell,i}/q)$; the signed largest coordinate of A gives the same w. For $L/q = 2^{18}$, nearest rounding of a coordinate in $[-L, L]$ has integer magnitude at most 2^{18}, including endpoint and rounding ties. Every partial sum through $n = 2^{12}$ therefore has magnitude at most 2^{30}. Two signed 32-bit accumulators suffice, and a 13-bit counter stores every value from 0 through 4096. These are 77 working bits; a separately allocated two-bit output register brings those declared registers to 79 bits during handoff. Only the two-bit code is needed for the specified static test.

This is not a two-bit complete apparatus. The analog input readings and revealed outcomes, multiplication, radial clipping, rounding hardware, fixed codebook and constants, control and temporary processing have their own implementation costs. The constructed policy uses $2n$ coordinate reads and n revealed outcomes for training, followed by one read per test input. The comparison grants the fixed-direction reader an exact coefficient without charging for its estimation or retention; the theorem is therefore not a comparison at equal total resource cost. A history-independent foreground may use its training records to estimate coefficients; the oracle comparison already grants it a stronger coefficient resource. Reusing or clearing the training registers also needs an account and is not made free by retaining a small final code. A finite implementation of the fixed coefficient must keep its error below the strict risk margin, as in Section 9.3.

The theorem identifies a use of memory: retained information about the task–channel relation changes which measurement is made next. Calling this code a self-frame specifies that functional role, not a new statistical class distinct from learned control parameters. The independent-donor intervention below can retain the same two-bit format while removing its relation to the current task. It tests the value of that relation, but does not compare the code with every equally small history-dependent representation or establish a distinctively subjective organization.

5.2 What a history intervention tests

Replay a code from an independently trained trial with independent coupling, so that the installed w is independent of the current U. It is now a history-independent foreground and cannot beat $s^2 + \rho^2/2$, even if its correct current regression coefficient is supplied. If the original response coefficient $\rho\mu$ is kept instead, its risk is

$$R_{\text{replay}} = s^2 + \rho^2(1 + \mu^2), \tag{11}$$

since $\mathbb{E}(U \cdot w) = 0$. At $\rho = s = 1$ this is 2.810569469, not 1.5. The oracle baseline and an actual stale signed response are different interventions.

Freezing the code learned for the original static U does *not* remove its value: the construction already freezes it through the test phase. What matters is the learned relation to this task, not updating for its own sake. Off-frame reads become necessary to revise omitted directional information when the coupling changes (Section 7); the static theorem does not certify a continuing drift filter.

The two-bit code is a decision summary for this test loss and reading menu, not a sufficient statistic for arbitrary future posterior updates. The same distinction applies to the conditional mean in Theorem 2: it determines a current Bayes response, but Bayesian updating need not close on that mean alone. The formed code changes which physical reading the agent uses next from an already supplied menu. It neither creates that menu nor identifies whether $U = R^\top e$ changed through the body coupling R or the world direction e.

A deterministic angular sum and one small finite-training diagnostic are described in Appendix B. The finite-n guarantee is the moment and truncation proof, not the diagnostic's sample mean.

6 A quantum reference changes the available reading

The classical foreground is a policy selected by memory. A retained quantum reference changes what a symmetry-restricted policy can measure. This is an operational bridge to a fixed-point algebra, rather than an identification of that algebra with a classical subspace.

Proposition 10 (Bounded task value of a retained reference)**.** *Let a compact group G act on a finite reference system, with twirling $\mathcal{G}$. In a specified task the reference starts in τ, independent of the hidden task label; target states and permissions are unchanged when the reference is replaced by $\mathcal{G}(\tau)$. The same class of physical protocols is allowed in both experiments. If losses lie in $[0, L]$, their optimal Bayes risks satisfy*

$$|R^*(\tau) - R^*(\mathcal{G}(\tau))| \leq \frac{L}{2}\|\tau - \mathcal{G}(\tau)\|_1 \leq L\sqrt{\frac{\mathcal{A}_G(\tau)}{2}}, \tag{12}$$

where $\mathcal{A}_G(\tau) = D(\tau\|\mathcal{G}(\tau))$ uses natural logarithms. For jointly G-invariant measurement permissions, an invariant reference induces only invariant effects on the target.

Proof. For each hidden label and each allowed protocol, the input trace distance is $\|\tau - \mathcal{G}(\tau)\|_1/2$. Contractivity bounds the total variation of its outcome laws by the same number. A loss in $[0, L]$ changes by at most L times that variation. Averaging and taking the infimum over the unchanged protocol class gives the first bound. Quantum Pinsker gives the second. The twirled-state relative entropy equals the entropy increase under twirling [2, Proposition 2 and Theorem 3]; the task bound is its combination with distinguishability. If a joint effect commutes with $U_S(g) \otimes U_R(g)$ and τ is invariant, partial expectation over R gives an effect commuting with $U_S(g)$. □

Proposition 11 (One finite phase reference and its passive lifetime)**.** *Two equiprobable target states are $\rho_\pm = (I \pm X)/2$. The retained reference $\tau_r = (I + rX)/2$, $0 \leq r \leq 1$, is prepared relative to their source phase. Allowed joint effects commute with $Z_S + Z_R$. The exact optimal success probability is*

$$P^*(r) = \frac{1}{2} + \frac{r}{4}. \tag{13}$$

With independent phase-dephasing of the unused reference, $r(t) = r_0 e^{-\kappa t}$. A required advantage $\Delta > 0$ above chance can persist without renewal only until $t \leq \kappa^{-1} \ln[r_0/(4\Delta)]$, for $4\Delta \leq r_0$.

Proof. The optimum invariant measurement is the Helstrom measurement of the two jointly twirled input states. Their difference is

$$\mathcal{G}_{SR}(\rho_+ \otimes \tau_r - \rho_- \otimes \tau_r) = \frac{r}{4}(X \otimes X + Y \otimes Y).$$

Its nonzero eigenvalues are $r/2$ and $-r/2$, in the common-charge space spanned by $|01\rangle, |10\rangle$. The trace norm is r, giving Eq. (13); its eigenprojections are permitted. Dephasing multiplies the reference's transverse Bloch vector by $e^{-\kappa t}$, and substitution proves the lifetime. □

Corollary 12 (One noisy carrier as a record and a frame)**.** *Use $X_S + X_R$ as the conserved charge and $(I \pm Z)/2$ as the equiprobable target states. A retained reference with Bloch vector $(0, y, z)$ gives optimal invariant-measurement success*

$$P^*_{\text{frame}} = \frac{1}{2} + \frac{\sqrt{y^2 + z^2}}{4}.$$

Take the reference to be the initially positive record in HAFF-B's transverse-qubit model [1], with known elapsed time and known generator. If $E^*_{\text{record}}(t)$ is the optimal error for reading its two possible initial signs directly, then

$$P^*_{\text{frame}}(t) = \frac{3}{4} - \frac{1}{2} E^*_{\text{record}}(t). \tag{14}$$

Thus the same physical carrier loses its record distinguishability and its one-use reference advantage on the same slow decay scale, even though its exact fixed algebra is scalar.

Proof. A rotation about the charge axis brings the transverse reference vector to one axis without changing its length. It commutes with the symmetry and maps allowed effects to allowed effects. The calculation of Proposition 11, with the Pauli axes relabelled, therefore gives the first equation. The two noisy record states have opposite Bloch vectors of length $\sqrt{y^2 + z^2}$, so their Helstrom error is $(1 - \sqrt{y^2 + z^2})/2$, proving Eq. (14). □

The baseline reference $r = 0$ permits only diagonal target effects and cannot distinguish ρ_+ from ρ_-. Retaining $r > 0$ makes a phase-sensitive reading possible without a non-invariant joint measurement. It does not create an absolute phase: the entire input may be jointly twirled with no change of the reading probabilities. This is the established relational-reference mechanism [17], here used to connect task capability and passive noise. A fresh symmetric history cannot create $r > 0$ by covariant operations alone; each conditional output of an invariant-outcome covariant instrument is still invariant. The source-relative preparation is therefore a priced input, not a frame created from an uninformative record.

The lifetime concerns an unused token followed by one reading. Repeated measurement can disturb it further and requires a reuse account. If a protocol must keep its advantage in this drifting task, it must obtain new phase information, protect that information or refresh the reference; renaming a stale token a self-model does not change its capability. This is a finite quantum counterpart of the self-frame's calibration problem, not a sufficient condition for experience or a theorem that all tasks need asymmetry. Charge-label tasks, for example, can be solved with invariant effects.

7 The frame shapes its own evidence

Once a foreground is chosen, the agent's next records are read through it. The following theorem says what those records can and cannot reveal.

Theorem 13 (In-frame identifiability and its blind spot)**.** *Fix a rank-k foreground P and consider in-frame records (Px_t, y_t) from a static coupling.*

(i) Their joint law depends on (ρ, s, u) only through $b = \rho P u \in \operatorname{ran} P$ and $V = \rho^2 + s^2$.

(ii) With ρ and s known, the likelihood is constant on each fibre $\{u : Pu = w\}$, a sphere of dimension $D-k-1$ in the background. The posterior law of the background direction $(I-P)u/|(I-P)u|$ is not updated by any number of in-frame records.

(iii) With ρ and s unknown, every triple (ρ', s', u') with $\rho' Pu' = b$, $\rho'^2 + s'^2 = V$ and $|b| \le \rho' \le \sqrt{V}$ produces the same law. The risk reduction available by reorienting the frame, $\rho'^2 - |b|^2$, ranges over $[0, V-|b|^2]$ within this family: the unexplained variance may be entirely world noise or entirely the agent's own misalignment.

(iv) No rule using only in-frame records and the frame achieves risk below $V - |b|^2 = s^2 + \rho^2(1-|Pu|^2)$, and in-frame regression attains it asymptotically.

Proof. (Px, y) is Gaussian with mean zero, $\mathrm{Cov}(Px) = P$, $\mathrm{Cov}(Px, y) = \rho Pu$ and $\mathrm{Var}\, y = \rho^2 + s^2$, which proves (i). Parts (ii) and (iii) follow, since the likelihood of any sample is a function of (b, V); with ρ, s known, b fixes Pu and nothing else, and the conditional law of the background direction given Pu is therefore unchanged by the data. For (iv), given u the part $\rho(I-P)u \cdot x + \xi_t$ of y_t has variance $V - |b|^2$ and is independent of Px_t and of all earlier records, so no rule based on them removes it; regression of y on Px estimates b consistently and attains $V - |b|^2$. □

Theorem 13 states the cost side of a self-frame. A frame that serves the agent well also decides which evidence it will receive, and that evidence cannot show where the missing part of the task has gone. Part (iii) adds a second limitation: from its own in-frame records the agent cannot know whether anything is missing. A large residual is compatible with a well-aligned frame in a noisy world and with a misaligned frame in a quiet one. Separating the two requires either a known noise level or reads outside the frame. This is the agent-level form of the distinction between a change of the world and a change of the agent's own coupling.

Corollary 14 (The two-channel case: an even residual)**.** *Let $D = 2$, $k = 1$, foreground v and $u = \cos\theta\, v + \sin\theta\, v_\perp$. The in-frame law depends on θ only through $\cos\theta$, so θ and $-\theta$ are indistinguishable, and the in-frame residual variance $s^2 + \rho^2 \sin^2\theta$ is even in θ. The Fisher information of one in-frame record about θ is*

$$F(\theta) = \frac{\rho^2 \sin^2\theta \left(\rho^2 + s^2 + \rho^2 \cos^2\theta\right)}{\left(s^2 + \rho^2 \sin^2\theta\right)^2} = \frac{\rho^2(2\rho^2 + s^2)}{s^4}\,\theta^2 + O(\theta^4). \tag{15}$$

One off-frame read along $v_\perp$ gives $\mathrm{Cov}(v_\perp \cdot x, y) = \rho \sin\theta$, which is odd in θ and resolves the sign.

Proof. The covariance of $(v \cdot x, y)$ is $\Sigma = \begin{pmatrix} 1 & a \\ a & V \end{pmatrix}$ with $a = \rho\cos\theta$. For a Gaussian family with parameter-dependent covariance, $F = \frac{1}{2}\mathrm{tr}(\Sigma^{-1}\Sigma'\Sigma^{-1}\Sigma')$; with $a' = -\rho\sin\theta$ and $\det\Sigma = V - a^2$ this equals $a'^2(a^2+V)/(V-a^2)^2$, which is (15). □

Paper III takes this even residual law as its detection model, proves that it cannot identify the sign of the mismatch, and shows that a regular even record law with a nonzero quadratic

leading perturbation has Fisher information quadratic and relative entropy quartic in the mismatch near calibration, which fixes the time needed to detect drift [4]. Corollary 14 derives the structure from a budgeted self-frame: the residual is even because the frame reads only one of two channels, and the sign is the unobservable fibre of Theorem 13. A single off-frame read is the minimal probe that breaks the symmetry. The derivation fixes the structure (evenness, a two-point fibre, quadratic information), not the Gaussian mean model used in Paper III's computations.

8 Drift: fixed and frozen frames

Let the coupling drift. In continuous time we take u_t to be Brownian motion on S^{D-1} with generator $\sigma\Delta_{S^{D-1}}$. A time-homogeneous discrete Markov step equivariant under $O(D)$, including reflections, has the corresponding scalar structure: $\mathbb{E}[u_{t+1} \mid u_t] = \lambda_1 u_t$, and the degree-two harmonics have multiplier λ_2. Equivariance under $SO(D)$ suffices for $D \geq 3$, but not for a directed drift on the circle.

Theorem 15 (Fixed and frozen frames under drift)**.** *For spherical Brownian motion with generator $\sigma\Delta$,*

$$\mathbb{E}[u_t \cdot v] = e^{-(D-1)\sigma t}\, u_0 \cdot v, \qquad \mathbb{E}[(u_t \cdot v)^2] - \tfrac{1}{D} = e^{-2D\sigma t}\big((u_0 \cdot v)^2 - \tfrac{1}{D}\big). \tag{16}$$

(a) A frozen self-model with response $\hat{y} = g{\cdot}x$, $g = \rho u_0$ fixed, has risk $s^2{+}2\rho^2\big(1{-}e^{-(D-1)\sigma t}\big)$. It is worse than having no memory after $t^* = \ln 2/((D-1)\sigma)$, and worse than the frame-free level once $2(1 - e^{-(D-1)\sigma t}) > 1 - b/D$. If g is estimated from n_0 full probes, the risk is $s^2 + 2\rho^2 + \epsilon_0 - 2\rho^2 e^{-(D-1)\sigma t}$ with $\epsilon_0 = (D(\rho^2 + s^2) + \rho^2)/n_0$.

(b) A fixed foreground spanned by u_0 and $b-1$ orthonormal directions orthogonal to it, with exact in-frame coefficients, has risk $s^2 + \rho^2(1 - b/D)\big(1 - e^{-2D\sigma t}\big)$. It never rises above the frame-free level and approaches it at rate $2D\sigma$.

For an $O(D)$-equivariant discrete step the same formulas hold with $e^{-(D-1)\sigma}$ and $e^{-2D\sigma}$ replaced by λ_1 and λ_2 per step. The discrete risks need not be monotone when these multipliers are negative.

Proof. The functions $u \mapsto u \cdot v$ and $u \mapsto (u \cdot v)^2 - 1/D$ are spherical harmonics of degrees one and two, with Laplace–Beltrami eigenvalues $-(D-1)$ and $-2D$; their expectations decay accordingly. For an $O(D)$-equivariant step the stabilizer of u forces $\mathbb{E}[u' \mid u] = \lambda_1 u$ and $\mathbb{E}[u'u'^{\top} \mid u] = aI + \lambda_2 uu^{\top}$, with $Da + \lambda_2 = 1$. Iteration gives the same first- and second-moment formulas with the corresponding powers of λ_1 and λ_2. (a) The risk is $s^2 + \mathbb{E}|\rho u_t - g|^2 = s^2 + \rho^2 + |g|^2 - 2\rho\, g \cdot \mathbb{E}u_t$. With g the mean of $y_i x_i$ over n_0 probes, $\mathbb{E}[g \mid u_0] = \rho u_0$ and $\mathbb{E}|g - \rho u_0|^2 = \epsilon_0$ by the Isserlis identity $\mathbb{E}[(u \cdot x)^2 x_i^2] = 1 + 2u_i^2$. (b) The risk is $s^2 + \rho^2(1 - \mathbb{E}|Pu_t|^2)$, and (16) gives $\mathbb{E}(u_t \cdot u_0)^2 = \frac{1}{D} + (1 - \frac{1}{D})e^{-2D\sigma t}$ and $\mathbb{E}(u_t \cdot w)^2 = \frac{1}{D}(1 - e^{-2D\sigma t})$ for $w \perp u_0$; summing over the b directions gives the stated curve. □

The two Brownian curves separate two failure modes that a single visibility angle cannot tell apart. A self-model that keeps both its frame and its coefficients becomes confidently

wrong: its predictions remain as large as before while their direction decorrelates, and after t^* it does worse than predicting nothing. A frame that keeps learning its in-frame coefficients cannot become worse than the frame-free reader, because in-frame regression shrinks the coefficient on a stale direction, but it loses its whole advantage. Neither can recover the advantage without information from outside the frame, by Theorem 13. Paper III compares these curves with the time needed to detect drift and finds that a residual-variance test detects slow drift long before t^* [4].

Proposition 16 (Directed drift of a planar frame)**.** *Let $D = 2$ and $u_t = (\cos\varphi_t, \sin\varphi_t)$, with $\varphi_{t+1} = \varphi_t + \Delta_t$ modulo 2π, where the increments are independent with one common law. Write $\mu_j = \mathbb{E}e^{ij\Delta_t}$. For a known initial direction, a frozen predictor $g = \rho u_0$ has risk*

$$R_{\text{frozen}}(t) = s^2 + 2\rho^2[1 - \operatorname{Re}(\mu_1^t)].$$

A fixed one-dimensional foreground along u_0, using its exact current regression coefficient, has risk

$$R_{\text{axis}}(t) = s^2 + \frac{\rho^2}{2}[1 - \operatorname{Re}(\mu_2^t)].$$

These laws are $SO(2)$-equivariant. Reflection symmetry makes both multipliers real and recovers the scalar discrete formulas.

Proof. Independence gives $\mathbb{E}e^{ij(\varphi_t - \varphi_0)} = \mu_j^t$. The two risks are respectively $s^2 + 2\rho^2\mathbb{E}[1 - \cos(\varphi_t - \varphi_0)]$ and $s^2 + \rho^2\mathbb{E}\sin^2(\varphi_t - \varphi_0)$. Taking real parts proves both identities. For deterministic increments $\Delta_t = \phi$, they involve $\cos(t\phi)$ and $\cos(2t\phi)$, not $(\cos\phi)^t$ and $(\cos 2\phi)^t$. □

A directed rotation can repeatedly return a stale frame to alignment. The Brownian crossing time is therefore a property of that diffusion, not a universal lifetime inferred from rotational symmetry.

9 A calibratable frame and its price

9.1 The policy and its exact stationary risk

The calibratable agent stores a vector $g_t \in \mathbb{R}^D$, its estimate of $\theta_t = \rho u_t$. Before the input arrives it chooses a set S_t of $b - 1$ channel indices uniformly without replacement, independently of the world. It reads those coordinate directions and, as its last read, the normalized component of g_t orthogonal to their span. If this component vanishes it fills the slot with any unused orthogonal direction. These are exactly b orthonormal reads. From them it computes $g_t \cdot x_t$ and responds $\hat{y}_t = \gamma g_t \cdot x_t$, with $\gamma \in [0, 1]$. No repeated or additional read is required: coordinate values and the orthogonal residual reconstruct the noiseless linear functional algebraically. Each coordinate is probed with probability $p = (b-1)/D$. For each probed channel i it updates $g_{t+1,i} = g_{t,i} + \alpha(y_t x_{t,i} - g_{t,i})$; unprobed coordinates are kept. The frame-free comparator stores coefficients β_t on the fixed channels $S = \{1, \ldots, b\}$, reads them every step, updates them by the same rule with $p = 1$, and responds $\gamma\,\beta_t \cdot x_{t,S}$.

Proposition 17 (Exact stationary risk)**.** *Assume $1 < b < D$, $0 < \alpha < 2$, u_0 uniform, a drift step with $\mathbb{E}[u_{t+1} \mid u_t] = \lambda_1 u_t$ and a uniform invariant marginal, with factor $\lambda_1 = 1 - \kappa \in [0, 1]$, fresh inputs and noise, and probe choices independent of the world (correlations among the probe indicators of different channels do not matter). Let m be the number of tracked coordinates ($m = D$ for the calibratable frame, $m = b$ for the frame-free reader), p the probe probability ($p = 1$ for the frame-free reader), $T = \rho^2 m/D$ and $a = p\alpha(2 - \alpha)$. The tracking error e_t of the tracked coordinates has stationary moments*

$$\begin{aligned} c_\infty &= \mathbb{E}[e \cdot \theta] = -\frac{\kappa T}{1 - \lambda_1(1 - p\alpha)}, \\ A_\infty &= \mathbb{E}|e|^2 = \frac{p\alpha^2 m\big(\rho^2 + s^2 + \rho^2/D\big) + 2\kappa T + 2\kappa(1 - p\alpha)c_\infty}{a}, \end{aligned} \tag{17}$$

and the stationary risk with weight γ is

$$R(\gamma) = s^2 + \rho^2 - 2\gamma\,(T + c_\infty) + \gamma^2\,(T + 2c_\infty + A_\infty). \tag{18}$$

In particular $R(1) = s^2 + (\rho^2 - T) + A_\infty$, and the reliability weight $\gamma^* = (T + c_\infty)/(T + 2c_\infty + A_\infty)$ gives $R(\gamma^*) = s^2 + \rho^2 - (T + c_\infty)^2/(T + 2c_\infty + A_\infty) \le s^2 + \rho^2$; the weight lies in $[0, 1]$ under the stated nonnegative first-harmonic assumption.

Proof. Coordinate i is probed with indicator χ_i, $\mathbb{E}\chi_i = p$. The probe value yx_i has conditional mean θ_i given the past and the current coupling, and conditional variance $\rho^2 + s^2 + \rho^2 u_i^2$ by the Isserlis identity; its average over the uniform marginal of u_t, which holds at every t, is $\rho^2 + s^2 + \rho^2/D$. After the update the error is $e_i' = (1 - \alpha\chi_i)e_i + \alpha\chi_i n_i$ with n_i conditionally centred, so $\mathbb{E}|e'|^2 = (1 - a)\mathbb{E}|e|^2 + p\alpha^2 m(\rho^2 + s^2 + \rho^2/D)$ and $\mathbb{E}[e' \cdot \theta] = (1 - p\alpha)\mathbb{E}[e \cdot \theta]$. The drift step then replaces θ by $\theta + \Delta$ and e' by $e' - \Delta$, where Δ is independent of e' given u_t, $\mathbb{E}[\Delta \mid u_t] = -\kappa\theta$ on the tracked coordinates, $\mathbb{E}|\Delta|^2 = 2\kappa T$ and $\mathbb{E}[\Delta \cdot \theta] = -\kappa T$. Hence

$$\mathbb{E}|e_{t+1}|^2 = \mathbb{E}|e'|^2 + 2\kappa(1 - p\alpha)\mathbb{E}[e_t \cdot \theta_t] + 2\kappa T, \qquad \mathbb{E}[e_{t+1} \cdot \theta_{t+1}] = \lambda_1(1 - p\alpha)\mathbb{E}[e_t \cdot \theta_t] - \kappa T.$$

Both recursions are affine with contraction factors $1 - a$ and $\lambda_1(1 - p\alpha)$ of modulus less than one, so the moments converge to the fixed point (17) from any start. The response $\gamma\, g \cdot x$ has risk $s^2 + \mathbb{E}|\theta - \gamma g|^2$ with g supported on the tracked coordinates; expanding with $g = \theta_m + e$ gives (18). The minimizing weight follows by completing the square, and $R(\gamma^*) \le R(0) = s^2 + \rho^2$. □

Only the one-step correlation $\lambda_1 = \mathbb{E}[u_{t+1} \cdot u_t]$ of the drift enters. For spherical Brownian motion sampled at unit time, $\kappa = 1 - e^{-(D-1)\sigma}$. The weight γ^* is a property of the agent's own tracking error. An agent that knows its drift factor and noise level can compute it from (17); an agent that knows only its residuals cannot, because by Theorem 13(iii) its in-frame residual mixes its tracking error with world noise. The reliability weight is thus a small self-assessment, and its effect is measurable.

9.2 Capability and benefit threshold

Table 2 compares the five policies of Section 2 under one protocol. The exact stationary risk agrees with the simulation to within the Monte Carlo error at every drift level, including

κ	Frozen	No frame	Fixed	Calib.	Calib. (th.)	Calib. γ^*	No frame γ^*	Alignment
0	0.414	0.863	0.340	0.2608	0.2610	0.2607	0.863	0.991
1.40×10^{-5}	0.812	0.872	0.559	0.2856	0.2848	0.2852	0.872	0.967
8.75×10^{-5}	1.934	0.897	0.856	0.3367	0.3361	0.3350	0.897	0.921
3.50×10^{-4}	2.448	0.915	0.920	0.4188	0.4190	0.4116	0.914	0.849
1.40×10^{-3}	2.426	0.938	0.941	0.5743	0.5742	0.5477	0.935	0.718
5.54×10^{-3}	2.464	0.994	0.995	0.8411	0.8396	0.7568	0.985	0.506
1.23×10^{-2}	2.449	1.049	1.049	1.0383	1.0402	0.8969	1.030	0.353
2.15×10^{-2}	2.430	1.095	1.096	1.1740	1.1736	0.9944	1.064	0.238
4.61×10^{-2}	2.423	1.169	1.169	1.2516	1.2518	1.1043	1.116	0.128

Table 2: Time-averaged risk in the closed loop ($D = 8$, $b = 3$, $\rho = 1$, $s = 0.5$, start-up $n_0 = 64$ full probes, 24 000 steps, first 8 000 discarded, 200 trajectories with common random numbers). No memory gives 1.25. "Calib. (th.)" is $R(1)$ from Proposition 17; learning rates are set from the formulas, not tuned on the simulation. Alignment is $\mathbb{E}(u \cdot g/|g|)^2$ of the calibratable frame. Standard errors are at most 0.016 for the frame-free and fixed columns (dominated by the sample of couplings), at most 0.05 for the frozen column, and at most 0.0015 for the calibratable columns; the paired difference between the calibratable and frame-free readers has standard error at most 0.016.

levels where the calibratable frame loses. With a static coupling the calibratable frame reaches alignment 0.991 and risk 0.261, against 0.863 for the frame-free reader and 0.340 for the frame fixed after start-up. The frozen self-model is worse than no memory from $\kappa = 8.75 \times 10^{-5}$ on. The fixed frame becomes indistinguishable from the frame-free reader at moderate drift, as Theorem 15(b) predicts.

Without the reliability weight the calibratable frame loses to the frame-free reader between $\kappa = 1.23 \times 10^{-2}$ and 2.15×10^{-2}: it keeps responding at full strength along a direction it can no longer track. With γ^* for both readers it remains better through $\kappa = 4.61 \times 10^{-2}$. Table 3 gives the crossing drift κ_c from the exact formulas for other dimensions and budgets. Weighting by self-assessed reliability raises κ_c by factors from 2.1 ($D = 4$, $b = 3$) to 13.8 ($D = 32$, $b = 3$). Without weighting, $\kappa_c D^2/(b-1)$ stays between 0.14 and 0.94 over the computed cases: the sampled scaling is $\kappa_c \propto (b-1)/D^2$, or D^{-2} at fixed b, rather than $[(b-1)/D]^2$.

Corollary 18 (Static limit)**.** *With $\kappa = 0$ the stationary risk of the calibratable frame is $s^2 + \alpha D(\rho^2 + s^2 + \rho^2/D)/(2-\alpha)$ at weight 1, which tends to s^2 as $\alpha \to 0$, while the frame-free reader stays at or above $s^2 + \rho^2(1 - b/D)$. Under a reading budget $2 \le b < D$, a calibratable self-frame is therefore strictly better than every frame-free policy for all sufficiently slow drift.*

Proof. Set $\kappa = 0$ in (17). The frame-free bound is Corollary 3(c). Continuity of (17) in κ at fixed α extends the strict inequality to small κ. □

9.3 What the agent pays

The exact curves use noiseless linear reads and real-valued coefficients. Orthonormalizing the probe span is a computation on retained g and chosen indices before the input arrives. It

D	b	κ_c, weight 1	κ_c, weight γ^*
4	2	1.47×10^{-2}	3.82×10^{-2}
4	3	1.73×10^{-2}	3.68×10^{-2}
8	2	7.66×10^{-3}	3.49×10^{-2}
8	3	1.29×10^{-2}	5.78×10^{-2}
8	5	1.85×10^{-2}	7.19×10^{-2}
16	2	2.85×10^{-3}	2.23×10^{-2}
16	3	5.07×10^{-3}	4.00×10^{-2}
16	5	8.45×10^{-3}	6.44×10^{-2}
32	2	9.16×10^{-4}	1.26×10^{-2}
32	3	1.68×10^{-3}	2.31×10^{-2}
32	5	2.93×10^{-3}	3.93×10^{-2}

Table 3: Drift $\kappa_c = 1 - \lambda_1$ at which the calibratable frame and the frame-free reader have equal stationary risk, each with learning rate optimized from Proposition 17 ($\rho = 1$, $s = 0.5$). Below κ_c the calibratable frame is better.

needs arithmetic and working storage, not another observation. These resources are counted separately from the b reads. If the implemented response coefficient differs by a vector with mean squared norm at most ϵ^2, the squared-risk increase is bounded by $2\epsilon\sqrt{R - s^2} + \epsilon^2$, by Cauchy–Schwarz. Thus a strict benefit margin survives sufficiently precise finite arithmetic; precision is an additional paid input. Independent noisy hardware reads require their own covariance and calibration account.

All readers above use b reads per step and the same $n_0 D$ start-up reads. They differ in retained state. The frame-free reader keeps b coefficients. The fixed frame keeps b coefficients and its b directions. The calibratable frame keeps the D numbers of g, and Theorem 4 says that some such growth is unavoidable: alignment 0.991 in eight channels requires at least $\Lambda^*_{8,1}(0.991)$ nats in its foreground, 23.3 bits. In time the calibratable frame spends $b - 1$ of its b reads off its own frame. Paper I prices these resources when they are implemented thermodynamically. In Paper I's matched task, a record adds at most $k_B T$ times its task mutual information. Isothermal erasure of independent unbiased degenerate bits without usable side information costs at least $k_B T \ln 2$ per bit; general updates instead require the conditional-entropy and free-energy ledger (a known bit need not cost that amount). Reference preparation requires the free-energy difference for its specified Hamiltonian and initial state [3]. A pure degenerate one-bit reference has price $k_B T \ln 2$; an arbitrary reference does not have that universal price. The present paper keeps reads, probes and stored numbers as separate entries and does not convert them into a single price, because the conversion depends on the implementation.

10 Connections within the theory

10.1 A frame can change the body's protective relations

Foundation III gives a finite body implementation [10]. A diagnostic record of an unknown noise sign selects and later decodes an odd or even protective code on two body bits holding an unknown logical qubit and its inaccessible old reference. The controller sees the record, never the sign. In that protocol the learned code beats both fixed codes when $p > .51430317693$; at $p = .9$ its ideal loss is .0527213975, with early damage included. The full gate program, paid ports and source/return construction belong to Foundation III. This is a separate implementation, not an identification of Gaussian subspace alignment with quantum protection. The available encoders and diagnostic coupling remain inputs; Paper III studies history-selected maintenance when body access becomes unreliable. The positive subject-formation criterion proposed here is that learned relations organize the same system's future access, protection and revision, rather than merely naming a preferred basis. Paper III's body-gated repair gives this criterion an acting interpretation: the acquired model selects which future access is worth restoring. This is an interpretive connection between the models, not a theorem of general subject formation.

Paper I. Paper I establishes the task value of correlations and the conditions under which their preparation and maintenance pay [3]. Its feedback protocols imply a posterior; a record is not valuable independently of its task and implementation. Theorem 2 is the reading-side counterpart: every budgeted policy implies a frame, and its capability is the alignment of that frame. Theorem 4 prices the frame in information, as Paper I prices the record in work.

Paper III. Corollary 14 derives the even residual model that Paper III assumes, including the sign ambiguity and the quadratic Fisher signal. Paper III's reference reads play the role of the known world direction in Section 2: they separate a change of the agent's coupling from a change of the world, which in-frame records cannot do (Theorem 13(iii)). Its memory-under-drift proposition and Proposition 17 have the same structure, one for a scalar instrument parameter and one for a frame in D channels; the second shows that the cost of tracking a frame grows with the number of channels the frame excludes.

Consciousness III. Theorem 13 is an instance of a self-shaped unobservable fibre: what the agent cannot learn about itself is set by its own current frame and moves when the frame moves. Consciousness III develops the general theory of such fibres for finite self-models, including a floor for self-knowledge under drift [7].

Imprint and Brain Anchor. The reader of *The World as Imprint* learns which of two instrument axes it has and retains the corresponding information [8]. The self-frame here is its continuous, budgeted generalization, with the added feature that the learned model controls which records arrive. Brain Anchor's learned relation weights are another case of a frame formed from records and selecting later use [9].

11 Discussion

11.1 What is established

Within the budgeted reader, risk below $s^2 + \rho^2(1 - b/D)$ requires a history-dependent foreground (Corollary 3 and Theorem 4). Corollary 6 gives necessary information, storage and outcome-bearing sample costs; it is not itself a finite learning construction. Proposition 9 supplies one: with 4096 full probes, 77 declared working bits and a separate two-bit output, the two-channel reader selects a foreground whose one-read risk is at most 1.414155, below the fixed-foreground oracle level 1.5. The apparatus and its processing remain supplied and separately chargeable. The code's value is its relation to the current coupling: independent replay loses that relation, whereas static freezing preserves it. With $b = D$, selecting a foreground gains nothing, although predicting still requires the task coefficients.

Theorem 13 describes the omitted directions that in-frame records cannot locate. Off-frame probes can revise the frame; Proposition 17 computes the risk and cost of one such policy under drift. The finite static construction and the drifting calibration policy solve different stages of this problem: the former establishes formation with finite registers, while the latter quantifies continued revision without claiming the same finite-register implementation.

For the memory–self–introspection route, this gives the self link a definite content. Task-relevant memory becomes a self-frame here when it selects reads through a learned model of the system's own coupling. The blind spot then gives a reason for acquiring new relations outside that frame. A system may implement this calibration function without representing a human-like self. Each statement concerns this task class. It is not claimed that every persistent system needs a self-frame; a system that can read everything it needs, or whose coupling is known, does not.

The self-frame is a functional object. Nothing here shows that holding one is, or produces, experience. The Consciousness series treats the relation between such functional self-models and first-person organization with its own arguments.

11.2 Drift boundaries and two conjectures

Proposition 19 (A drift factor does not determine the tracking problem)**.** *Let $U_t = S_t V$ with V initially uniform on the sphere and fixed, while $S_t \in \{-1, 1\}$ flips independently with probability $\kappa/2$ each step. This rotation-invariant Markov law has the same $\lambda_1 = 1 - \kappa$ as a spherical diffusion with that first harmonic. Given a retained exact axis V, a legal reader of that axis has excess risk at most*

$$R - s^2 \le 4\rho^2 \left(\frac{w\kappa}{2} + e^{-wC_1} \right), \qquad C_1 = \frac{1}{2} \ln(1 + \rho^2/s^2), \tag{19}$$

using only the last w outcome-bearing reads. In particular it has $O(\kappa \log(1/\kappa))$ excess risk at fixed signal-to-noise ratio.

Proof. Read $z_j = V \cdot x_j$ and estimate the current sign by the sign of $\sum_{j=t-w}^{t-1} z_j y_j$. Predict $\rho \widehat{S}_t V \cdot x_t$. A flip in the w transitions joining these outcomes to the current step has probability at most $w\kappa/2$. On the no-flip event, relabel the common sign positive. For one sample,

$y = \rho z + s\eta$, with z, η independent standard normals. At $a = \rho/s^2$,

$$\mathbb{E}e^{-azy} = \mathbb{E}e^{-a\rho z^2 + a^2 s^2 z^2/2} = (1 + \rho^2/s^2)^{-1/2}.$$

Chernoff's inequality bounds a wrong sign by e^{-wC_1}. A sign error costs $4\rho^2$ in squared prediction loss; the current Gaussian input is independent of the sign estimate. Taking $w = \lceil \log(1/\kappa)/C_1 \rceil$ proves the order. □

An exact axis is a supplied resource in this statement. A paid initialization with sufficiently small axis error adds a corresponding prediction error; finite precision is not free. The axis can also be learned without access to S_t: for $i \neq j$, $\mathbb{E}[y_t^2 x_{t,i} x_{t,j}] = 2\rho^2 V_i V_j$, while $\mathbb{E}[y_t^2 x_{t,i}^2] = \rho^2 + s^2 + 2\rho^2 V_i^2$. Cycling through coordinate pairs uses two legal reads per training step. These expectations are unchanged by sign flips and identify $VV^\top$ as training grows. Training time, stored moments and the sign window are resources. This law preserves an axis rather than diffusing $D - 1$ independent tangent directions. Thus λ_1 suffices for the moment formula of Proposition 17, but cannot justify a universal diffusion tracking lower bound.

Conjecture 20 (Rate optimality of probe calibration)**.** *For the rank-one task with $2 \leq b < D$ noiseless linear reads per step, let U_t be spherical Brownian motion sampled at unit intervals, with generator $\sigma\Delta$ and $\kappa = 1 - e^{-(D-1)\sigma}$. Policies may retain their paid observed history and randomize, but receive no external coupling truth or additional drift reference. Their long-run average risk is at least $s^2 + c_0\rho^2 \min\{1, D\sqrt{\kappa(\rho^2 + s^2)/(\rho^2(b-1))}\}$ for a constant $c_0 > 0$ independent of D, b, κ, in the local diffusion regime. Initialization is charged and does not supply future increments.*

The mechanism is that each off-frame read informs one background direction, and $D - 1$ directions drift independently. For a single coordinate observed with probability p per step under a random walk with increment variance q, the posterior Cramér–Rao recursion gives a stationary error of order $\sqrt{q/(pF)}$, with F the Fisher information per observation; summing over $D-1$ coordinates with $p = (b-1)/D$, $q \simeq 2\rho^2\kappa/D$ and $F \simeq (\rho^2 + s^2)^{-1}$ gives the stated scaling, which Proposition 17 attains with optimized α for small κ. The strongest objection is that the coordinates are not independent: an adaptive policy can probe the directions in which its uncertainty is largest, and the response read itself carries information about the magnitude of the misalignment (Corollary 14). These could improve the constant; the conjecture would fail if they changed the exponent of κ or the power of D.

Conjecture 21 (Sharp quantum counterpart of the frame price)**.** *For a specified quantum direction-estimation task and a compact group acting transitively on its directions, the minimum retained reference resource needed for a given capability admits a sharp task-dependent variational bound whose classical direction limit is Theorem 4.*

Proposition 10 supplies a general bounded-risk inequality, and Proposition 11 supplies an exact quantum task. What remains is sharpness and the classical limit, not the existence of any risk bound. Equal asymmetry need not give equal capability in different tasks, so the task family cannot be omitted. The noise-selected algebra of Consciousness I is a symmetry-fixed algebra; the present reference enlarges effective measurement permissions through a state, not by claiming that every fixed algebra becomes a larger abstract algebra.

11.3 Open problems

The task class is linear and Gaussian with a rank-one or fixed-rank target. Several competing frames and structured non-Gaussian inputs remain open. Paper III joins a binary self-frame to a world-changing loop: acting cannot identify parameters moved by its record-law symmetry, while explicit experiments in its stationary nondegenerate model identify persistence, squared gain and flip reliability [4]. Symmetry invariance is therefore a necessary condition, not a general identification theorem. A continuous frame with matrix-valued loop gains, and a thermodynamic implementation of the reading and retention budgets, are next steps.

A Simulation protocol for Table 2

Parameters: $D = 8$, $b = 3$, $\rho = 1$, $s = 0.5$. The coupling starts uniform and moves by the rotation-invariant step $u \mapsto (u + \eta g_\perp)/|u + \eta g_\perp|$ with $g_\perp$ the projection of a standard Gaussian vector orthogonal to u, for $\eta \in \{0, 0.002, 0.005, 0.01, 0.02, 0.04, 0.06, 0.08, 0.12\}$; $\lambda_1 = \mathbb{E}(1 + \eta^2 \chi^2_{D-1})^{-1/2}$ is computed by quadrature. Each agent receives the same $n_0 = 64$ start-up probes, from which g_0 is the mean of $y_i x_i$. The frozen agent responds $g_0 \cdot x$. The frame-free agent reads channels 1–3; the fixed agent reads $g_0/|g_0|$ and two fixed random orthonormal directions orthogonal to it; both learn their three coefficients by the update of Section 9 with $p = 1$. The calibratable agent chooses two coordinate probes uniformly without replacement and reads those axes plus the normalized component of g orthogonal to them. It reconstructs $g \cdot x$ from these three orthonormal reads and updates only the probed coordinates. The recorded noiseless simulation used the algebraically equivalent nonorthogonal functionals; this equivalence need not hold for hardware with independent read noise. Learning rates and weights are set by minimizing the risks of Proposition 17 over $\alpha \in [10^{-4}, 0.999]$ (floor 0.002). Risks are evaluated exactly given the state, $s^2 + |\rho u - c|^2$ with c the agent's effective coefficient vector, and averaged over steps 8 000–23 999 and over 200 trajectories that share the world's random numbers across agents. The code is `td2_loop.py`; the thresholds of Table 3 are computed by `td2_threshold.py`, and the analytic checks of Theorems 2–15 by `td2_checks.py`.

B Checks of the finite formation construction

The accompanying `check_formation.py` evaluates the constants in Proposition 9 and implements its actual sample preprocessing. At the declared parameters it gives $M_2 = 6$, $e_{n,L,q} = 0.06240131120$ and risk upper bound 1.41415420535, a margin greater than 0.08584 below the fixed-foreground oracle risk. Integer bounds are checked exactly: each quantized coordinate is at most 262144 in magnitude and each partial sum at most 1073741824. An equally spaced midpoint angular sum with 65536 points gives $\mu = 0.900316316502$, within $3.5\,10^{-10}$ of $2\sqrt{2}/\pi$.

For a diagnostic, 128 independently sampled couplings each receive 4096 Gaussian training samples, with seed 1729. The encoder is passed only the observed x_1, x_2, y; it clips, rounds and accumulates integers before selecting its code. Nearest rounding uses a fixed

positive-infinity tie rule. The test input and noise are then integrated analytically: the conditional loss is $1 + |U - \mu w|^2$. The diagnostic mean is 1.1831812333 with between-history standard error 0.0134057633; it is not a confidence certificate for the risk bound or evidence of improving on the ideal population value 1.1894305309. The script separately checks the exact independent replay risk 2.8105694691, the replay foreground's oracle level 1.5, and invariance under freezing a code in the original static task. These checks do not implement a thermodynamic apparatus. The analytic proof, not this small sample, supplies the guarantee.

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Keeping a Self-Model Calibrated: Drift Detection, the Price of Keeping Up, and Reliability in a Closed Loop

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Abstract

A self-frame learned from history can choose the next reading; when its coupling drifts, the same history must also support diagnosis and decisions about future access. We study this continued revision and its price. Smooth even record laws delay linear-drift detection to order $\varepsilon^{-4/5}$, compared with $\varepsilon^{-2/3}$ for first-order probes. For scalar Gaussian transitions, no free background information and quadratic loss, the all-policy long-run cost lies between $C - \frac{1}{2}k_B T G \nu^2$ and C, with $C \propto \nu^{2/3}$; Gaussian priors and controllable continuous precision attain the slow-drift leading term. An acting body supplies some relational standards, while jointly relabelled orientation remains unidentifiable. When that body's working state gates reading and processing, the retained model is held through disability and a latched repair changes the kernel between acquired records. Within declared permissions, body history strictly raises harvest. At positive gain and sign drift, legally acquired histories with equal sign confidence can require opposite paid repairs. Two posterior moments determine terminal access value, but continuing enquiry can require a third moment and the joint belief. Thus one acquired model can authorize a task, price further evidence and select maintenance without reducing those decisions to one confidence score. Exact adaptive reference timing and finite-register filters supply further realizations under stated budgets. The resource accounts distinguish supplied preparations, records and drive from retained model bits and the remaining processor costs.

Keywords: self-calibration, self-models, Fisher information, change detection, posterior Cramér–Rao bound, drift tracking, information thermodynamics, feedback control

1 Introduction

1.1 The question

A frame formed from records is not thereby correct forever. Nor is continual updating valuable regardless of what the task changes. The relevant question is how the acquired

task–interface relation continues to organize useful access. Paper I prices task information and the loss from acting on a mismatched posterior [1]. Paper II constructs a two-bit reading frame from a finite unknown-coupling history and compares it with a history-independent direction at one test read [2]. Its retained code selects a subsequent channel; it does not supply a complete continuing model.

Here records check a represented model against references or against the pointer value selected by the body's working state. The model also determines whether an available task is worth executing, what a further reading could reveal and whether access should be restored after disability. These are different decisions from the same history. A high confidence in orientation can accompany low maintenance value; even equal confidence can require opposite repairs.

We ask how quickly a formed frame's error becomes detectable, what revision costs, and how history learns the value of maintaining the physical access on which later revision depends.

1.2 Position within the series

This paper completes the T-DOME (Thermodynamic Dynamics of Observer-Memory Entanglement) series, the third line of a programme whose other lines are HAFF and Q-RAIF.

The three T-DOME papers follow a route that the programme calls memory–self–introspection. An agent keeps records of its world (Paper I), organizes its access to them through a model of itself (Paper II), and checks that model against references while it acts (this paper). Several interfaces match exactly without identifying every model as one device. The gain–bias channel whose calibrated loss Paper I computes is the instrument of Section F; the even residual law that Paper II derives from a budgeted frame is the detection model of Section 3; the conditional drift law cited in Paper I is proved in Section 4. Each link is established in a stated model.

1.3 Main results

The main line runs from a formed frame to diagnosis, priced revision and history-selected maintenance. Section 3 distinguishes noticing a mismatch from identifying its direction. Section 4 gives an all-policy Gaussian comparison for the cost of keeping up. Section 6 then supplies the body, records, actuator and complete declared ledger, including the limits of what its own actions identify.

Section 7 is the main acting result. The working body determines whether reading and processing can occur. Its actual acquired history raises task value and can change the optimal paid repair at equal sign confidence. Propositions 29–30 separate the summary sufficient for a last task from the belief needed for further enquiry. General prior estimation, natural-gradient comparisons, the long three-parameter reference analysis and a static repair witness are retained in Appendices A–D. The numerical banks, physical budgets and finite-filter certificates remain in the subsequent appendices.

1.4 Relation to internal selection

Paper G of HAFF gives a typed model of internal selection: a program register holds a choice, a fixed executor turns program states into couplings or readouts, records are outcomes of instruments, and a fixed rule rewrites the program from the records [8]. A finite system can select its own readouts and revise that selection. Within this target/program architecture, a selected target operation is not a complete implementation of the larger executor acting on both registers; a stored description or an additional selector is not excluded. The calibration loop uses this architecture. The self-model is the program, decoding and command rules are the executor, and calibration is the revision rule. History changes the program and the operations it selects. The executor and the labelled standard specify how those changes are used.

1.5 Related work

The Fisher information metric was introduced by Rao [13] and characterized by Čencov [14]; Amari developed the natural gradient and information geometry [15, 16]. The Bayesian Cramér–Rao (van Trees) inequality [17] and its recursive form for filtering, the posterior Cramér–Rao bound of Tichavský, Muravchik and Nehorai [18], are the estimation tools of the Gaussian comparison in Section 4 and Appendix A; the Kalman filter [19] attains the filtering bound in the linear Gaussian case. The trade-off between noise and lag in adaptive estimation, and the persistent-excitation condition, are standard in adaptive control [21]. Thermodynamic length and minimum-dissipation protocols are due to Crooks and Sivak [22, 23], and Fisher information enters entropy production through thermodynamic uncertainty relations and information geometry [24, 25]. Second-order cybernetics, the control of the observing system itself, goes back to von Foerster, building on Ashby's cybernetics [26, 27]. The data-processing bound for hypothesis tests used in Section 3 is standard [28]. We do not claim that the telescoping inequality behind Theorem 9 is new to filtering theory; its use here is to settle the drift law of Paper I for all policies and to give it a thermodynamic reading.

2 Setting

2.1 Inputs from Papers I and II

Paper I supplies the record's matched work value, the posterior mismatch loss and its calibrated expansion [1]. Its gain–bias example has leading loss $k_B T(1 + g^2/(1 - b^2))/(4n)$ after n add-half reference reads per sign. Paper II supplies a retained foreground, an in-frame blind spot and, for its two-channel reader, the even information law [2]

$$F(\theta) = \frac{\rho^2 \sin^2\theta\,(\rho^2 + s^2 + \rho^2\cos^2\theta)}{(s^2 + \rho^2\sin^2\theta)^2}. \tag{1}$$

An off-frame read resolves its sign fibre. The Gaussian drift law cited by Paper I is proved here; its assumptions do not identify either reader with the bounded acting body below.

Paper II also supplies a genuine finite formation comparison. In its static two-channel Gaussian task, 4096 full probes select a signed-coordinate code retained in two bits. At unit signal and noise, its one-read test risk is at most 1.414155, below the history-independent foreground's oracle level 1.5. Both use one test read; the oracle coefficient is free to the comparator, so retained memory and total cost are not matched. The result demonstrates the value of using history to select a reading, not a separation from every general learning rule. Its four-code menu, instrumentation and processing are given. The 77 declared working bits and a separate two-bit output are not the complete physical apparatus. The code's task-relatedness matters: independent-coupling replay loses its value, while freezing it in its original static task preserves it.

That result establishes formation, not the filter used in this paper. The bounded joint instrument below has its own transition law, labelled standards and reference supply. Its posterior is revised from its actual acquired history, rather than receiving the instrument's true state. Foundation III separately demonstrates history-selected protection of already-held content and physical return [11]; its encoding and resource protocol are not this body's maintenance controller.

A current-task compression need not be a recursive model. In Paper I's diagonal-pointer task the gross work lost by compressing history H to M is $k_B T I(X : H \mid M)$. Preserving that posterior removes the current loss, not necessarily the information needed for the next update. The distinction becomes a physical decision here: the sufficient summary for a last available invocation can fail to price a later enquiry.

2.2 Fisher information and the Cramér–Rao bounds

Definition 1 (Fisher information)**.** Let $\{p(x \mid \theta) : \theta \in \Theta \subset \mathbb{R}^d\}$ be a family of densities for which differentiation and integration may be interchanged. Its Fisher information matrix is

$$g_{ij}(\theta) := \mathbb{E}_\theta\big[\partial_i \log p(x \mid \theta)\, \partial_j \log p(x \mid \theta)\big] = -\mathbb{E}_\theta\big[\partial_i \partial_j \log p(x \mid \theta)\big]. \tag{2}$$

Where it is positive definite it is a Riemannian metric on Θ.

By Čencov's theorem, on finite probability simplices the Fisher metric is, up to a constant factor, the unique Riemannian metric invariant under statistically sufficient embeddings [14]. This makes it the canonical local measure of distinguishability; it does not make a Fisher-based detector physically mandatory, and it does not choose the agent's likelihood or objective. The metric is the Hessian of relative entropy,

$$D\big(p_\theta \,\|\, p_{\theta+d\theta}\big) = \tfrac{1}{2}\, g_{ij}(\theta)\, d\theta^i\, d\theta^j + O(|d\theta|^3), \tag{3}$$

and it bounds estimation: for an unbiased estimator from n independent observations, $\mathrm{Cov}(\hat{\theta}) \succeq g(\theta)^{-1}/n$ [13, 28]. For a correlated stream use the Fisher information of its joint likelihood. A decorrelation time alone does not justify replacing the sample count by T/τ_E.

2.3 Standing assumptions

Definition 2 (Standing assumptions)**.** Sections 3–5 assume the following. Section F states its own inputs.

(C1) **Self-model.** The agent's model of itself is a parameter ϑ in a finite-dimensional manifold Θ: a frame of Paper II, the gain and bias of an instrument, or a scalar mismatch coordinate. The family of records $p(e \mid \vartheta)$ that the agent uses to assess it is specified.

(C2) **Drift.** The parameter that best describes the instrument, $\vartheta^*(t)$, moves. Each result states its drift law: linear, spherical Brownian, a Gaussian random walk, or bounded speed.

(C3) **Regularity.** Fisher informations are finite and differentiation under the integral sign is allowed to the order used. Uniform lower bounds on information are assumed separately where they are needed.

3 Detecting drift from one's own records

A self-model goes stale when the instrument it describes changes. The agent's first question is whether it can notice this from the records it already gathers. Paper II shows that records read through a frame depend on the frame's misalignment only through an even function of it. This section works out what evenness implies.

3.1 Even models

Proposition 3 (Sign non-identifiability)**.** *If a record family satisfies $p(e \mid \theta) = p(e \mid -\theta)$ for all θ, then the joint law of any number of conditionally independent records is invariant under $\theta \mapsto -\theta$, and no statistic of the records distinguishes θ from $-\theta$.*

Proof. The joint likelihood is a product of factors that are each invariant under the sign change. □

Paper II derives this structure instead of assuming it: a frame that reads one of two channels sees an in-frame law that depends on the angle θ to the coupling only through $\cos\theta$, and the sign is the unobservable fibre of its blind spot [2]. Evenness has a second consequence, which governs how fast drift can be noticed at all.

Proposition 4 (Information and relative entropy of an even model)**.** *Let $p(e \mid \theta)$ be even in θ and smooth enough near $\theta = 0$ that the expansions below may be integrated term by term (this holds for the Gaussian families of Corollary 5). Write $\ell(\theta) = \log p(e \mid \theta)$ and*

$$\kappa := \mathbb{E}_0\big[\ell''(0)^2\big].$$

Assume $\kappa > 0$ for the sharp detection exponents below. Then the score vanishes at $\theta = 0$, so the Fisher information $\mathcal{I}_1(0)$ of one record is zero, and

$$\mathcal{I}_1(\theta) = \kappa\,\theta^2 + O(\theta^4), \qquad D\big(p_\theta \,\|\, p_0\big) = \frac{\kappa}{8}\,\theta^4 + O(\theta^6). \tag{4}$$

Proof. For each e the function ℓ is even, so $\ell'(0) = \ell'''(0) = 0$ and $\ell'(\theta) = \theta\,\ell''(0) + O(\theta^3)$. Since $p_\theta - p_0 = O(\theta^2)$, $\mathcal{I}_1(\theta) = \mathbb{E}_\theta[\ell'(\theta)^2] = \theta^2\mathbb{E}_0[\ell''(0)^2] + O(\theta^4)$. For the relative entropy write $p_\theta/p_0 = 1+\eta$ with $\eta = \frac{1}{2}\theta^2 H + O(\theta^4)$ and $H = p_0''/p_0 = \ell''(0) + \ell'(0)^2 = \ell''(0)$. Normalization gives $\mathbb{E}_0\eta = 0$, and $(1+\eta)\ln(1+\eta) = \eta + \frac{1}{2}\eta^2 + O(\eta^3)$, so $D(p_\theta\|p_0) = \frac{1}{2}\mathbb{E}_0\eta^2 + O(\theta^6) = \frac{1}{8}\theta^4\mathbb{E}_0 H^2 + O(\theta^6)$. □

Corollary 5 (Three record families)**.** *(a)* Gaussian mean model, $e \mid \theta \sim \mathcal{N}(a\theta^2, \sigma_e^2)$*:* $\mathcal{I}_1(\theta) = 4a^2\theta^2/\sigma_e^2$ *and* $D(p_\theta\|p_0) = a^2\theta^4/(2\sigma_e^2)$ *exactly, so* $\kappa = 4a^2/\sigma_e^2$*.*

(b) Paper II's in-frame record $(v\cdot x, y)$ *with* $D = 2$*:* $\mathcal{I}_1 = F(\theta)$ *of* (1)*, and*

$$D(p_\theta\|p_0) = \frac{\rho^2(1-\cos\theta)}{s^2} - \frac{1}{2}\ln\Big(1 + \frac{\rho^2\sin^2\theta}{s^2}\Big), \qquad \kappa = \frac{\rho^2(2\rho^2+s^2)}{s^4}. \tag{5}$$

(c) An off-frame probe record $(v_\perp\cdot x, y)$*:* $D(p_\theta\|p_0) = -\frac{1}{2}\ln\big(1 - \rho^2\sin^2\theta/(\rho^2+s^2)\big)$*, and its Fisher information*

$$F_\perp(\theta) = \frac{\rho^2\cos^2\theta\,(\rho^2\sin^2\theta + \rho^2 + s^2)}{(\rho^2 + s^2 - \rho^2\sin^2\theta)^2} \tag{6}$$

is positive at calibration, $F_\perp(0) = \rho^2/(\rho^2+s^2)$*.*

Proof. (a) is the normal location family with mean $a\theta^2$. For (b) and (c) the record is a zero-mean Gaussian pair with covariance $\Sigma_\theta = \left(\begin{smallmatrix} 1 & c(\theta) \\ c(\theta) & V \end{smallmatrix}\right)$, $V = \rho^2 + s^2$, where $c = \rho\cos\theta$ in (b) and $c = \rho\sin\theta$ in (c). Then $D(p_\theta\|p_0) = \frac{1}{2}[\mathrm{tr}\big(\Sigma_0^{-1}\Sigma_\theta\big) - 2 - \ln(\det\Sigma_\theta/\det\Sigma_0)]$ gives the two relative entropies, and the Fisher information of a Gaussian family with parameter-dependent covariance, $\frac{1}{2}\,\mathrm{tr}(\Sigma^{-1}\Sigma'\Sigma^{-1}\Sigma') = c'^2(c^2+V)/(V-c^2)^2$, gives (1) and (6). Expanding (5) to fourth order gives $\kappa/8 = \rho^2(2\rho^2+s^2)/(8s^4)$, in agreement with $F(\theta)/\theta^2 \to \rho^2(2\rho^2+s^2)/s^4$. □

Model (a) is a convenient assumption; model (b) is what a budgeted frame actually produces. Proposition 4 shows that the quadratic onset of information does not depend on the choice: evenness gives the expansion; a nonzero κ gives its quadratic onset. The constant κ is the only feature of the model that enters the leading order. For records at rate r under linear drift $\theta(t) = \theta_0 + \varepsilon t$, the accumulated local information $J_T := \sum_{t_m \le T}\mathcal{I}_1(\theta(t_m)) \simeq r\int_0^T \mathcal{I}_1(\theta_0 + \varepsilon t)\,dt$ is

$$J_T = \frac{\kappa r\varepsilon^2 T^3}{3} + O(\varepsilon^4 T^5) \quad (\theta_0 = 0), \qquad J_T = \frac{4a^2 r}{3\sigma_e^2\varepsilon}\big[(\theta_0+\varepsilon T)^3 - \theta_0^3\big] \quad \text{in model (a).} \tag{7}$$

It enters the estimation bound of Section A. Detection is governed by a different quantity, the accumulated relative entropy.

3.2 How long detection takes

Fix a horizon T and let records arrive at rate r, conditionally independent given the mismatch path $\theta_m = \theta(m/r)$. A test is a function of the records up to time T with the values "calibrated" and "drifted". It has errors (α, β) if it says "drifted" with probability at most α when $\theta_m = 0$ for all m, and "calibrated" with probability at most β under the drifted law; for a random drift path the drifted law is the mixture over paths. Write $d(p\|q) = p\ln\frac{p}{q} + (1-p)\ln\frac{1-p}{1-q}$.

Proposition 6 (Detection time)**.** *Every test with errors* (α, β)*,* $\alpha + \beta < 1$*, satisfies*

$$\sum_{m \le rT} \mathbb{E}\, D\big(p_{\theta_m} \,\|\, p_0\big) \;\ge\; d(1-\beta \,\|\, \alpha), \tag{8}$$

with the expectation over the drift path if it is random. Consequently, as the drift rate tends to zero:

(a) for an even model with $D(p_\theta \| p_0) \le \kappa\theta^4/8 + C\theta^6$ *for all* θ *and linear drift* $\theta(t) = \varepsilon t$*,* $T \ge \big(40\, d/(\kappa r \varepsilon^4)\big)^{1/5}(1 + o(1))$*;*

(b) if probe positions are fixed in advance with counting function $N_\perp(t) = prt + o(t)$ *for* $p > 0$*, their record law has* $D(p_\theta \| p_0) \le \frac{1}{2}F_1\theta^2 + C\theta^4$ *with* $F_1 > 0$ *its Fisher information at calibration, as off-frame probes do, and the others are as in (a),* $T \ge \big(6\, d/(pF_1 r\varepsilon^2)\big)^{1/3}(1 + o(1))$*;*

(c) for Paper II's reader with $D = 2$ *under spherical Brownian drift with generator* $\sigma\Delta$*,* $T \ge \big(2\, d/(\kappa r \sigma^2)\big)^{1/3}(1 + o(1))$*,*

where $d = d(1-\beta\|\alpha)$*.*

Proof. Let P_0 and P_1 be the laws of the records up to T under the two hypotheses. The test's decision is a function of the records, so by the data-processing inequality $D(P_1\|P_0)$ is at least the relative entropy between the two laws of the decision, which are Bernoulli with parameters $p_1 \ge 1-\beta$ and $p_0 \le \alpha$; since $d(p_1\|p_0)$ increases in p_1 and decreases in p_0 for $p_1 > p_0$, it is at least $d(1-\beta\|\alpha)$ [28]. For a fixed path the chain rule gives $D(P_1\|P_0) = \sum_m D(p_{\theta_m}\|p_0)$; for a random path P_1 is a mixture and joint convexity gives $D(P_1\|P_0) \le \mathbb{E}\sum_m D(p_{\theta_m}\|p_0)$. This proves (8). For (a), summing the hypothesis over $m \le rT$ gives $\kappa r\varepsilon^4 T^5/40\,(1 + O(\varepsilon^2T^2))$, and the solution $T \propto \varepsilon^{-4/5}$ has $\varepsilon T \to 0$. For (b), summation by parts gives $\sum_{t_m \le T,\, m \text{ probe}} t_m^2 = prT^3/3 + o(T^3)$. The probe records therefore give $pF_1 r\varepsilon^2 T^3/6\,(1 + o(1) + O(\varepsilon^2T^2))$ and the in-frame records only $O(\varepsilon^4T^5)$, which vanishes at $T \propto \varepsilon^{-2/3}$. For (c), the generator $\sigma\Delta$ on the circle makes the unwrapped angle a Brownian motion with variance $2\sigma t$, and the relative entropy (5) is periodic and bounded, so $D(p_\theta\|p_0) \le \kappa\theta^4/8 + C\theta^6$ for all real θ. With $\mathbb{E}\theta_t^4 = 12\sigma^2t^2$ and $\mathbb{E}\theta_t^6 = 120\sigma^3t^3$ the sum is $\kappa r\sigma^2T^3/2\,(1 + O(\sigma T))$, and $\sigma T \to 0$ at $T \propto \sigma^{-2/3}$. The families of Corollary 5 satisfy the hypotheses of (a) and (b): in (a) with $C = 0$, in (b) and (c) because the relative entropies are bounded and smooth. □

The bound holds for any test, whatever statistic it uses. For a known drift it is also of the right order: the log-likelihood ratio has mean equal to the accumulated relative entropy and, for small drift, variance about twice that mean, so the likelihood-ratio test succeeds at a constant multiple of the bound. In the example of Section E it reaches equal errors of 0.05 at about 1.3 times the bound. Evenness therefore costs time. A frame that sees only its own channel needs a time of order $\varepsilon^{-4/5}$ to notice a drift of rate ε; a stream that includes probes outside the frame needs a time of order $\varepsilon^{-2/3}$, which is shorter for slow enough drift at any fixed fraction p of probes.

Corollary 7 (Detection before failure)**.** *For Paper II's reader with $D = 2$ under spherical Brownian drift, a frozen self-model becomes worse than no memory at $t^* = \ln 2/\sigma$. Let the agent compare the mean S_T of the squared in-frame residuals $e_m = y_m - \rho\, v \cdot x_m$ over a window of length T with the threshold that gives type I error α under calibration. For every $\beta > 0$ there is a constant K, depending on α, β, ρ, s and r, such that at $T = K\sigma^{-2/3}$ the type II error is at most β for all small enough σ. Together with Proposition 6(c), the detection time is of order $\sigma^{-2/3}$, and its ratio to t^* tends to zero: slow drift can be detected from in-frame records long before the frozen model fails.*

Proof. Paper II's frozen-model curve with $D = 2$ gives t^*. Given the path, $e_m \sim \mathcal{N}(0, s^2 + 2\rho^2(1 - \cos\theta_m))$ independently, and under calibration nS_T/s^2 is χ^2_n with $n = rT$, so the threshold is $s^2(1 + z_\alpha\sqrt{2/n}\,(1 + o(1)))$. Write $\theta_{uT} = \sqrt{2\sigma T}\,B_u$ with B a standard Brownian motion. Then the conditional mean of $S_T - s^2$ is $2\sigma T\rho^2 Z_n(1 + O(\sigma T))$ with $Z_n \to Z = \int_0^1 B_u^2\,du$, and its conditional fluctuation is $s^2\sqrt{2/n}\,(1 + o(1))$ times a standard normal variable. At $T = K\sigma^{-2/3}$ the ratio of the shift to the null scale is $\sqrt{2}\,\rho^2\sqrt{r}\,K^{3/2}Z/s^2$, independent of σ, so the type II error tends to $P\big(N + \sqrt{2}\rho^2\sqrt{r}K^{3/2}Z/s^2 \le z_\alpha\big)$ with N standard normal and independent of Z. Since $Z > 0$ almost surely, this tends to zero as $K \to \infty$. □

For a general predetermined probe schedule, let χ_m indicate a probe at $t_m = m/r$. Under the linear drift of Proposition 6(b), the leading term of the probe entropy upper bound is

$$\frac{F_1\varepsilon^2}{2}\sum_{m\le rT}\chi_m t_m^2,$$

with remainder $O(\varepsilon^4 rT^5)$ under its global entropy bounds. The prefix-density assumption makes this sum $prT^3/3 + o(T^3)$. If only the horizon count $\sum\chi_m = prT + o(T)$ is specified, placing all probes first gives $p^3rT^3/3$, while placing them last gives $[1 - (1-p)^3]rT^3/3$, to leading order. These are the sharp extremes because the weights t_m^2 increase. A probe fraction alone therefore does not determine the detection constant. The placement is a genuine use of the access budget, even though all three schedules have the same $\varepsilon^{-2/3}$ scaling of the necessary detection time at fixed $p > 0$.

Detection is not correction. An even record can signal that the frame has moved but not in which direction (Proposition 3). One off-frame read or one reference read supplies the sign and a first-order signal at the same time. This is the division of labour in Paper II's calibratable frame and in Section F: most reads serve the task through the frame, and a small budget of reads outside it keeps the frame correct.

4 The price of keeping up

4.1 The drift law

Let a scalar self-parameter, for instance the gain of the agent's instrument, drift as a Gaussian random walk $\theta_k = \theta_{k-1} + \omega_k$, with independent $\omega_k \sim \mathcal{N}(0, \nu^2)$ per use and θ_0 drawn from a prior with finite Fisher information Q_0 ($Q_0 = 1/P_0$ for a Gaussian prior of variance P_0). In use k the agent makes $\rho_k \ge 0$ reference reads, each costing w and carrying Fisher information at

most J about θ_k under the pointwise or Bayesian conditional-information budget of Lemma 8, together with any unlabelled reads, which carry expected information at most J_0 per use at no extra cost. It then forms an estimate $\hat{\theta}_k$ from all its records so far and incurs a loss. The number ρ_k may be a real reading intensity, and it may depend on the records of earlier uses. Define $\bar{\rho}_k = \mathbb{E}\rho_k$, $\psi(Q) = (Q^{-1} + \nu^2)^{-1}$ and

$$Q_k = \psi(Q_{k-1}) + J_0 + J\bar{\rho}_k, \qquad P_k = Q_k^{-1} \qquad (k \geq 1). \tag{9}$$

Lemma 8 (Predictable design and stopped reading scores)**.** *Fix a use k with θ_k unchanged during its reads. Let $\mathcal{H}_{j-1}$ contain the preceding complete acquired history, selected designs and private coins, including earlier reads of this use, and set $\mathcal{G}_{j-1} = \mathcal{H}_{j-1} \vee \sigma(\theta_0, \ldots, \theta_k)$. Assume a parameter-independent, nonanticipating design rule and regular dominated likelihoods. The next reading score s_j has conditional mean zero given $\mathcal{G}_{j-1}$, and put $I_j = \mathbb{E}[s_j^2 \mid \mathcal{G}_{j-1}]$. Either the pointwise bound $I_j \leq J$, or the weaker Bayesian bound $\mathbb{E}[I_j \mid \mathcal{H}_{j-1}] \leq J$ under the actual joint prior and experiment, suffices. For a stopping count ρ_k with $\mathbb{E}\rho_k < \infty$ and $\{\rho_k \geq j\} \in \mathcal{H}_{j-1}$, the Bayesian expected reading information is at most $J\mathbb{E}\rho_k$. A separately declared free record channel contributes its joint conditional-information budget J_0.*

Proof. The joint likelihood factors as $p(\theta_0)\prod_k p(\theta_k \mid \theta_{k-1})\prod_j p(z_{kj} \mid \theta_k, d_{kj})$ times design probabilities. At the realized record the selected designs are fixed under differentiation, so design contributes no parameter score. The scores are martingale differences in the enlarged filtration $\mathcal{G}_j$, not necessarily in $\mathcal{H}_j$. Writing $A_j = \mathbf{1}\{\rho_k \geq j\}$, orthogonality and the tower property give

$$\mathbb{E}S^2_{\rho_k \wedge m} = \sum_{j \leq m} \mathbb{E}[A_j I_j] = \sum_{j \leq m} \mathbb{E}[A_j \mathbb{E}(I_j \mid \mathcal{H}_{j-1})] \leq J\mathbb{E}(\rho_k \wedge m).$$

The same argument bounds the squared L^2 tail by $J\mathbb{E}(\rho_k - m)_+ \to 0$. The transition information blocks of the posterior Cramér–Rao recursion are unchanged by this factorization. The conditional-average alternative proves a Bayesian information bound, not a separate bound at every θ_k. Parameter-dependent design or informative status records need their own scores; an unconditional per-read average alone does not justify stopping on the record. □

Theorem 9 (Drift law)**.** *Let $\ell : (0, \infty) \to [0, \infty)$ be a continuous function such that the expected loss in use k is at least $\ell(P_k)$ for every estimator. Then every reading schedule and every estimator satisfy*

$$\liminf_{K \to \infty} \frac{1}{K} \sum_{k=1}^{K} \big(w\bar{\rho}_k + \mathbb{E}[\mathrm{loss}_k]\big) \;\geq\; \inf_{0 < P \leq P_{\mathrm{free}}} h(P), \qquad h(P) = \frac{w}{J}\Big(\frac{\nu^2}{P(P+\nu^2)} - J_0\Big) + \ell(P), \tag{10}$$

where $P_{\mathrm{free}} = 1/Q_{\mathrm{free}}$ and Q_{free} is the positive fixed point of $Q = \psi(Q) + J_0$ ($P_{\mathrm{free}} = \infty$ if $J_0 = 0$). For the quadratic loss $\mathbb{E}(\hat{\theta}_k - \theta_k)^2$ times a constant c the hypothesis holds with $\ell(P) = cP$. If the reads are Gaussian, the loss of the posterior-optimal decoder is exactly $\ell(P_k)$, and arbitrary nonnegative observation precision is an allowed control, and the initial prior is Gaussian, a constant precision attaining a minimizing P^* attains the bound. When reads must be counted in whole units, the same expression is a lower bound from a continuous-precision relaxation; equality requires an admissible schedule.

Proof. Information bound. For a random walk with increment variance ν^2 the posterior Cramér–Rao recursion of Tichavský, Muravchik and Nehorai [18] has $D_{11} = -D_{12} = -D_{21} = \nu^{-2}$ and $D_{22} = \nu^{-2} + I_k$, where I_k is the expected Fisher information of the records of use k about θ_k, and it reads $\mathcal{J}_k = D_{22} - D_{21}(\mathcal{J}_{k-1} + D_{11})^{-1}D_{12} = \psi(\mathcal{J}_{k-1}) + I_k$, with $\mathbb{E}(\hat{\theta}_k - \theta_k)^2 \geq \mathcal{J}_k^{-1}$. Lemma 8 supplies $I_k \leq J_0 + J\bar{\rho}_k$ for predictable designs, including finite-mean within-use stopping counts. Whole-use Gaussian precision has its declared conditional information, without a discrete stopping argument. Since ψ is increasing, induction gives $\mathcal{J}_k \leq Q_k$, so the mean squared error is at least P_k, and the quadratic loss is at least cP_k.

Telescoping. By (9), $J\bar{\rho}_k = Q_k - \psi(Q_{k-1}) - J_0$. Summing,

$$\sum_{k=1}^{K}\big(Q_k - \psi(Q_{k-1})\big) = \sum_{k=1}^{K-1}\big(Q_k - \psi(Q_k)\big) + Q_K - \psi(Q_0) \;\geq\; \sum_{k=1}^{K-1}\big(Q_k - \psi(Q_k)\big) - \psi(Q_0),$$

and $Q - \psi(Q) = \nu^2/(P(P+\nu^2))$ for $P = 1/Q$. Hence

$$\sum_{k=1}^{K}\big(w\bar{\rho}_k + \mathbb{E}[\mathrm{loss}_k]\big) \;\geq\; \sum_{k=1}^{K-1} h(P_k) + \ell(P_K) - \frac{w}{J}\big(\psi(Q_0) + J_0\big) \;\geq\; \sum_{k=1}^{K-1} h(P_k) - \frac{w}{J}\big(\psi(Q_0) + J_0\big).$$

Each term of the sum is paired with a single P_k: no convexity is needed.

Range. Since $\bar{\rho}_k \geq 0$, $Q_k \geq f(Q_{k-1})$ with $f(Q) = \psi(Q) + J_0$ increasing, so $Q_k \geq f^k(0)$, and $f^k(0)$ increases to Q_{free}. For every $\eta > 0$, $P_k \leq (1+\eta)P_{\mathrm{free}}$ for all large k. Dividing by K, letting $K \to \infty$ and then $\eta \to 0$, which is allowed because h is continuous, gives (10).

Attainment. With a Gaussian prior and Gaussian measurements, the allowed constant precision makes Q_k converge to $1/P^*$. A posterior-optimal decoder with loss exactly $\ell(P_k)$ then has average cost converging to $h(P^*)$. □

Corollary 10 (Paper I's drift law)**.** *Take Paper I's quadratic loss $\frac{1}{2}k_BTG(\hat{\theta} - \theta)^2$ per use and $J_0 = 0$, and let*

$$C = 3 \cdot 4^{-2/3}\, w^{1/3}\, (k_BTG\nu)^{2/3}\, J^{-1/3}. \tag{11}$$

Every schedule and estimator has long-run average reference cost plus loss at least $C - \frac{1}{2}k_BTG\nu^2$ per use. With a Gaussian prior and reads of controllable real precision, the best stationary policy achieves at most C, so the least achievable average L^* in that class satisfies $C - \frac{1}{2}k_BTG\nu^2 \leq L^* \leq C$. The optimal reference rate is $\rho^* = \big(k_BTG\nu/(4w\sqrt{J})\big)^{2/3}(1 + o(1))$ as $\nu \to 0$, and $C = 3w\rho^*$ to leading order. The optimal estimator is a persistent memory: the steady-state Kalman filter updates its estimate with the constant weight $\alpha^* = P^*J\rho^* = \nu\sqrt{\rho^* J}\,(1 + o(1))$, which is the memory of Proposition 43 with that weight.

Proof. With $c = \frac{1}{2}k_BTG$ and $Q = P + \nu^2$, the inequalities $\nu^2/Q^2 \leq \nu^2/(P(P+\nu^2)) \leq \nu^2/P^2$ give $h(P) \geq (w/J)\nu^2/Q^2 + cQ - c\nu^2$ and $h(P) \leq (w/J)\nu^2/P^2 + cP$. Both $\min_{Q>0}[(w/J)\nu^2/Q^2 + cQ]$ and $\min_{P>0}[(w/J)\nu^2/P^2 + cP]$ equal $3 \cdot 2^{-2/3}c^{2/3}(w\nu^2/J)^{1/3} = C$. Theorem 9 gives the lower bound and, with Gaussian reads, the upper bound. At the stationary point $\nu^2/(P^*(P^*+\nu^2)) = J\rho^*$ and $P^* = \nu/\sqrt{\rho^* J}\,(1+o(1))$; minimizing $w\rho + c\nu/\sqrt{\rho J}$ gives ρ^* and the value $3w\rho^*$. The stationary Kalman gain is the posterior variance times the precision of the use's reads, $P^*J\rho^*$. □

Paper I conjectured this law with an unspecified universal constant, the rate ρ^* and the value $3w\rho^*$ [1]. Corollary 10 proves it, with the constant $3 \cdot 4^{-2/3} \approx 1.191$, for every schedule including time-varying and adaptive ones in its scalar Gaussian-transition model, for any read family whose Fisher information is bounded by J, and for the quadratic loss defined by Paper I's metric G. For Paper I's exact divergence loss the statement holds to leading order in the calibration error; a global version holds with G replaced by any $G' > 0$ such that the loss is at least $\frac{1}{2}k_B T G'(\hat{\theta}-\theta)^2$ on the explored range, under a uniform quadratic separation of posteriors on that range. Injectivity alone does not guarantee a positive quadratic minorant.

Corollary 11 (The relative-entropy floor of Consciousness III for all policies)**.** *In the Gaussian self-model of Consciousness III [3], let the initial prior be Gaussian and the self-state $\mathcal{N}(\theta_t, v)$ drift with increment variance q, with readings $\theta_t + \mathcal{N}(0, r)$ costing w each. Every predictable observation-precision schedule and decoder has long-run average relative-entropy loss plus reading cost at least*

$$\min_{P>0}\Big[\frac{wrq}{P(P+q)} + \tfrac{1}{2}\ln\Big(1+\frac{P}{v}\Big)\Big],$$

For a Gaussian initial prior and controllable real observation precision at linear cost, the constant-rate Kalman schedule attains it. Integer reading counts give the corresponding relaxation bound. The stationary constant-rate optimum computed in Consciousness III is therefore optimal among all policies.

Proof. Let $\mathcal{H}_k$ contain the full readings, chosen precisions, and private randomness through use k. A precision chosen from $\mathcal{H}_{k-1}$ does not contribute an extra parameter-dependent likelihood factor. Gaussian conjugacy therefore gives, path by path,

$$P_k^{-1} = (P_{k-1} + q)^{-1} + \rho_k/r, \qquad \theta_k \mid \mathcal{H}_k \sim \mathcal{N}(m_k, P_k).$$

The posterior predictive law is $\mathcal{N}(m_k, v + P_k)$, with conditional optimal relative-entropy loss $\ell(P_k) = \frac{1}{2}\ln(1 + P_k/v)$. Put $Q_k = P_k^{-1}$ and $\psi(Q) = (Q^{-1} + q)^{-1}$. For every realized observation schedule,

$$\sum_{k=1}^{K}[w\rho_k + \ell(P_k)] = \sum_{k=1}^{K-1}\left[\frac{wrq}{P_k(P_k+q)} + \ell(P_k)\right] + \ell(P_K) + wr[Q_K - \psi(Q_0)].$$

The terminal terms are at least $-wr\psi(Q_0)$. Take expectations and divide by K. A constant allowed precision at the minimizer attains the resulting bound. □

4.2 What the floor means

Remark 12 (Local thermal comparison)**.** At a fixed interior gain g, the Bernoulli reference information and Paper I's local loss metric are $J = G = 1/(1-g^2)$. Substitution into the exact Gaussian comparison yields

$$3 \cdot 4^{-2/3}(\ln 2)^{1/3}\frac{k_B T \nu^{2/3}}{(1-g^2)^{1/3}} - \frac{k_B T \nu^2}{2(1-g^2)}. \tag{12}$$

This expression gives the local quadratic comparison at thermally prepared reference cost $k_B T \ln 2$. A Gaussian random walk leaves the physical gain interval almost surely, so this substitution alone does not prove an infinite-time lower bound for the Bernoulli instrument. A bounded-drift theorem or a uniform finite-window limit is needed for that conclusion. The Gaussian theorem itself retains its all-policy strength. An alternative coordinate is $p = \sin^2 \phi$, whose Bernoulli Fisher information is 4 where $0 < p < 1$. This alone cannot supply a global quadratic KL minorant on the circle: ϕ and $\pi - \phi$ give identical laws while their circle distance can be nonzero. A bounded-instrument extension must use an identifiable quotient or chart, establish its loss separation and treat its actual transition law; the Gaussian information recursion is not coordinate-invariant under that change.

Paper I's amortization rule says that the overhead of a static calibration falls as $U^{-1/2}$ with the number U of uses, and so can be made as small as desired. The exact Gaussian comparison removes that freedom: with no free information its price per use has order $k_B T \nu^{2/3}$. For $g = 0.55$ the local Gaussian comparison value is $0.012\, k_B T$ per use at $\nu = 10^{-3}$, where the calibration error at the optimum is about 0.1 in g and the leading-order statement is accurate to a few per cent; the quadratic-loss value at $\nu = 10^{-2}$ is $0.055\, k_B T$, but there the error is about 0.23 and the divergence loss is about 7% below its quadratic approximation (Section E). A record read through the same channel is worth $0.160\, k_B T$ per use in Paper I's example. For a bounded physical gain, a uniform information bound and a quadratic loss minorant must be paired with the actual transition law; clipping the Gaussian walk changes its information recursion. With the quadratic loss the two would be equal near $\nu = 0.05$, where calibration errors are no longer small, so that crossing is only indicative. A physical persistence calculation must combine the actual record reset and the cost of its admissible calibration policy; Section K.1 supplies such a bounded example.

Corollary 13 (Optimal rigidity under drift)**.** *In the optimal stationary policy of Corollary 10, the information that the self-model carries into each use from earlier uses is* $\mathcal{I}^*_{\text{self}} = 1/(P^* + \nu^2)$*, the information gathered in the use is* $J\rho^*$*, and their ratio is* $(1 - \alpha^*)/\alpha^* = (\nu\sqrt{\rho^* J})^{-1}(1 + o(1))$.

Proof. At stationarity $Q^* = \psi(Q^*) + J\rho^*$ with $\psi(Q^*) = 1/(P^* + \nu^2)$, and the Kalman weight is $\alpha^* = J\rho^*/Q^*$. □

A more rigid self-model would ignore real drift; a softer one would spend references on noise. The trade-off that Section A could only assert exists has, under drift, an exact optimum set by the drift speed and the price of a reference.

Remark 14 (What the drift law does not cover)**.** (i) *Free information.* If unlabelled reads inform the parameter ($J_0 > 0$), the floor is lower, and when $\nu/\sqrt{J_0}$ is small enough the agent can track without references at an error of about $\nu/\sqrt{J_0}$; the $\nu^{2/3}$ law then no longer describes the optimum. In Paper I's setting with prior 1/2, unlabelled reads inform the bias and not the gain. In a world that the agent changes, an unlabelled read has mean $gx + b$ with the world state x unknown, so world reads alone do not separate the instrument from the world, which is Paper II's blind spot in another form. Section F uses references for this reason; Section 6 shows that a closed loop supplies free information about some invariant self-parameters in its stationary nondegenerate model, while the symmetry obstructs absolute sign identification.

(ii) *Linear drift.* Under the linear drift of Proposition 43, a memory with a constant weight has mean squared error $\nu^2(1-\alpha)^2/\alpha^2 + \alpha\sigma^2/(2-\alpha)$; optimizing the weight and the number of reads gives a cost proportional to $\nu^{2/5}$, worse than $\nu^{2/3}$. A memory that also estimates the drift rate drives the error to zero under exactly linear drift, so linear drift has no positive floor. (iii) *Vector parameters.* For a self-model with several drifting parameters the same telescoping, applied to traces, gives a lower bound by relaxation; whether it is attained when one reference informs several parameters at once is open.

Conjecture 15 (Smoother drift)**.** *Let the self-parameter be an m-fold integrated random walk, so that its m-th differences are independent $\mathcal{N}(0,\nu^2)$ variables, and let the other inputs be as in Theorem 9 with quadratic loss. Then every policy pays at least $c_m\, w^{(2m-1)/(4m-1)}\, \nu^{2/(4m-1)}$ per use, where $c_m > 0$ depends only on m, J and the loss constant, and the stationary Kalman filter with a constant reference rate attains this order.*

The stationary part is a steady-state computation: for an m-fold integrator observed with noise variance R the stationary position error scales as $\nu^{1/m} R^{(2m-1)/(2m)}$, and optimizing $R = 1/(\rho J)$ gives the exponent $2/(4m-1)$, which is $2/3$, $2/7$ and $2/11$ for $m = 1, 2, 3$; a numerical solution of the Riccati equation reproduces these slopes. The open part is the lower bound for time-varying and adaptive schedules. The telescoping of Theorem 9 does not extend directly, because the information recursion is matrix-valued and the reads inform only the first component. The conjecture would fail if a schedule that alternates bursts of reads with silence could exploit the smoothness to beat the stationary exponent.

5 The calibration loop

A formed frame remains useful only while its relation to the task supports the next decision. Residuals can reveal drift without locating its cause; a reference can supply the missing distinction. The local natural-gradient comparison in Appendix B makes this concrete: in-frame information degenerates at calibration, while an off-frame probe gives a nondegenerate metric. Its exact update, assumptions and proof are retained there. Appendix A separately prices the information in a prior self-model. Neither computation makes a correct likelihood or persistent excitation an automatic product of having memory.

The body loop below adds a different question. Restoring access is valuable only through what an acquired model expects that access to permit. A record can therefore change the next task, the value of further evidence and the repair decision without making any one scalar confidence sufficient for all three.

5.1 The four links of the loop

The series uses the following model architecture. It is a template, not a necessity theorem.

Definition 16 (Four links)**.** (I) **Records.** A world leaves readable, stable records, which an agent can store and use (HAFF; Paper I gives their value).

(II) **Reading and control interface.** Instruments read the world and actuators change it, each with a cost and a legal form (Q-RAIF; Foundation III's legal feedback channels [11]). The programme's running example of an internal algebra, a Clifford representation, can serve as the channel space; nothing here depends on it.

(III) **Self-frame.** A retained model of how the task sits in the agent's own channels, formed from records and selecting later reads (Paper II).

(IV) **Calibration.** References, memory and reliability weighting that keep (III) correct while the instrument drifts, and that decide how much authority over the world each reading carries (this paper).

Each link has an exact result in a stated model, listed in Section 9. Section F joins (II)–(IV) in one finite model in which the agent's actions change what it reads next. Section 7 makes the body's working state a condition of access and identifies a task domain where body history is necessary for a higher harvest. The template is not shown to be necessary for persistence: a system that can read everything it needs, or whose coupling never changes, needs neither (III) nor (IV).

A separate-register implementation can add reset, reference preparation and slow-driving actuation costs without double counting. Appendix J states that conditional bound and its full proof. These implementation charges are not a universal persistence budget.

6 A joint loop and its reference standard

The memory–self–introspection route can be joined around a body that the observer reads and changes. Records consume memory purity; a model of the instrument selects feedback and harvest levels; calibration revises that model. We first give the ledger and its identification boundary, then a finite-block operating regime and a continuing joint filter that uses its body history to revise both its instrument model and its reference decisions. A further construction lets the body determine whether those readings and updates can occur; latched recovery, when physically available, preserves later access. A labelled operation supplies orientation relative to its dispatch map, the calibration instance of HAFF Paper G's protocol-relative dispatch boundary [8]. Record purity, field preparation and timing are supplied resources in these constructions.

6.1 The model

Definition 17 (Joint model)**.** One cycle is one use; entropies are in nats and energies in units of k_BT.

(i) *World.* A persistent *body* bit $B_t = \pm 1$, flipped by a bath with probability d per cycle ($c = 1 - 2d$), and a *food* pointer $X_t = \pm 1$, fresh each cycle with $P(X_t = +1) = \frac{1}{2}$. Both have degenerate Hamiltonians and are diagonal in the pointer basis.

(ii) *Records.* Each cycle fresh pure memories collide with each carrier through pointer-controlled couplings [5, 6]: the food leaves two disjoint fragments F and G of overlap

r, the body a fragment F_B and a copy G_B. Record formation costs a diagonal carrier nothing and consumes memory purity (Paper I, ledger of the collision model).

(iii) *Instrument.* Reading a fragment through the setting that matches its record gives $P(m \mid x) = (1+\sigma g\, mx)/2$ with $0 \le g \le \sqrt{1-r^2}$, the Helstrom gain. The *magnitude* g follows a specified bounded transition law; the unbounded Gaussian model of Section 4 is an exact comparison and requires a separate bridge to this physical instrument; the *sign* $\sigma = \pm 1$ may invert with probability η per use; the *setting* is one of D, and a read with the wrong setting returns a fair coin. The same instrument reads food and body. (This is the gain–bias channel of Section F.2 with $b = 0$, and the setting is a discrete restriction of Paper II's direction family.) For uniform settings, the D choices are coordinate axes $\{e_1, \ldots, e_D\} \subset S^{D-1}$; exact identification removes $\ln D$ nats of setting uncertainty. This is not Paper II's continuous spherical alignment price, and the present model supplies these discrete settings rather than deriving their formation.

(iv) *Actions.* The flip of Q-RAIF Paper B, $U = e^{-i\lambda XQ}$, which with drive occupation N acts as $s\, X\rho X + (1-s)\rho$, $s = 1 - \epsilon_N$ [9], and costs $\theta_\omega(N + a\lambda^2)$ per opportunity, $\theta_\omega = \hbar\bar{\omega}/k_B T$, $a = 1$ if switched on. On the food: flip if $M = -1$, then harvest with fixed levels $E_\pm = -\ln \hat{q}(\pm)$ set from the agent's estimate, by the level protocol of Paper I. On the body: with a gate probability p_{g} known to the agent, flip if the reading is -1.

(v) *Memory and calibration.* The record register M is reset against the copy G, as in the cycle bound of Paper I, or against blank qubits. References are fragments of pointers prepared in a known state and read with the same instrument. A *self-label* is the recorded quench work of the harvest, which equals the energy of the level that the flipped food occupies and so records the food value relative to the agent's own level assignment.

With an ideal actuator, no drift, the matched setting and no body, the food cycle is Paper I's read–harvest–reset cycle; the body alone with a known instrument is the persistence task of Q-RAIF Paper B; the instrument alone is the drifting channel of Section 4. After the flip the food has $P(X' = +1) = q = (1+s\sigma g)/2$, and the harvest returns $\ln 2 - H_2(q) - d(q\|\hat{q})$, with $H_2(x) = -x \ln x - (1-x)\ln(1-x)$, by Paper I's lemma on extraction from a qubit, so $N \to \infty$ recovers the value $I(X{:}M)$ of Paper I.

6.2 The ledger

Every read leaves an outcome that the agent must eventually erase.

Proposition 18 (Reading tax)**.** *Let a classical register R with finite alphabet be updated once per cycle by a deterministic map $(R_t, Y_t) \mapsto R_{t+1}$, after which the outcome register is returned to a standard state, all registers having degenerate Hamiltonians and the update coupling them to a thermal bath with couplings that vanish at the endpoints and no other system correlated with (R_t, Y_t). Then $\beta \sum_{t \le K} W_t \ge \sum_{t \le K} H(Y_t \mid R_t) - \ln |R|$. Under these isolation assumptions, every reference read costs at least $\mathbb{E}\, H_2((1+g)/2)$ per read in the long run, whatever its preparation costs.*

Proof. As in the cycle bound of Paper I, unitarity and $D(\rho'_B\|\tau_B) \geq 0$ give $\beta W_t \geq H(R_t, Y_t) - H(R_{t+1}) = H(Y_t \mid R_t) + H(R_t) - H(R_{t+1})$; the last two terms telescope to at least $-\ln|R|$. Condition also on the sign, known input and register history. Fresh sensing noise gives $H(Y_t \mid R_t) \geq \mathbb{E}H_2((1+g_t)/2)$. A retained correlated outcome copy or newly consumed pure workspace must instead enter the reset ledger. □

The tax distinguishes a free reference input from a free cyclic processor. Under Proposition 18's isolation assumptions, an outcome stream still has a reset cost even when reference preparation is free. In a valid scalar Gaussian comparison this can supply the positive reading price w and preserve the $\nu^{2/3}$ order. For the local Bernoulli comparison, adding preparation and separate outcome erasure changes w from $\ln 2$ to $\ln 2 + H_2((1+g)/2)$; the leading comparison rises by a factor 1.21 at $g = 0.55$. The bounded physical transition and joint likelihood must still be established.

Proposition 19 (Purity available to a reset)**.** *Suppose the reset may couple M, a copy G whose marginal is retained, and n_B initially pure, independent blank qubits to a thermal bath. All registers have degenerate endpoint Hamiltonians. Then*

$$\beta W_{\text{reset}} \geq S(M \mid E_G) - n_B \ln 2.$$

Consequently, for a uniform food pointer,

$$\beta \,\text{net} \leq \chi(M{:}E_G) - H(M \mid X) + \Pi_B \leq \Pi_G + \Pi_B - H(M \mid X), \qquad \Pi_B = n_B \ln 2,$$

provided the record model gives $\chi(X{:}E_G) \leq \Pi_G$ and $M - X - E_G$ is the stated conditional product model. If the allowed reset is additionally required to consume nonnegative work, the first upper bound can also be capped by $I(X{:}M)$.

Proof. The entropy balance for reset gives the first inequality. Subtract it from the harvest bound $\beta W_{\text{harvest}} \leq I(X{:}M)$, and use $S(M \mid E_G) = H(M) - \chi(M{:}E_G)$ and $I(X{:}M) - H(M) = -H(M \mid X)$. Data processing gives $\chi(M{:}E_G) \leq \chi(X{:}E_G) \leq \Pi_G$. Pure blanks can themselves fund work during reset; only the added nonnegative-reset-work condition permits the cap by the harvest alone. □

In the uniform-pointer task, the second copy and pure blanks are resources for reset. Preparing all consumed purity from the agent's own thermal resources removes that contribution to profit. The five-memory copy of Paper I supplies $\chi(M{:}E_G) = 0.5231$ from $5\ln 2 = 3.466$ nats of initial purity, while a blank qubit supplies $\ln 2$ of entropy capacity. The copy also serves as a readable record.

Theorem 20 (Ledger of the joint model)**.** *Use the food feedback of Definition 17, reset against the declared copies with no additional pure blanks, and account separately for every remaining processor reset and reference preparation. Let $\mathcal{C}_t$ denote the classical information retained and available to a reset, whose marginal the reset preserves. Condition each register's state on that information. With separate baths and no shared erasure accounting, the long-run average net free energy per cycle satisfies*

$$\beta\,\overline{\text{net}} \leq \mathbb{E}\big[\ln 2 - H_2(q_t)\big] - \mathbb{E}\big[S(M_t \mid E_G, \mathcal{C}_t)\big] - \mathbb{E}\big[\ell_t + w\rho_t\big] - \bar{\tau} - \theta_\omega \mathbb{E}\big[N_t + a_t\lambda_t^2\big] - U_{\text{body}}, \quad (13)$$

where $\ell_t = d(q_t\|\hat{q}_t)$ is the harvest mismatch, $w \geq \ln 2$ the reference preparation cost, $\bar{\tau}$ the cost of the remaining outcome and model-register resets, and $U_{\text{body}} \geq \mathbb{E}[S(Y_t \mid E_{G_B}, \mathcal{C}_t)] + \theta_\omega \mathbb{E}[N_B + a\lambda^2]$ the body upkeep. Without retained side information the conditional entropy reduces to $H(M_t) - \chi(M_t{:}E_G)$. Proposition 18 applies to an isolated processor among the remaining resets. To apply Theorem 9, the complete likelihood, transition law and loss must meet its hypotheses. For constant known s and fixed sign, the local loop-gain coordinate $\vartheta = s\sigma g$ has reference information $1/(s^2(1-g^2))$; a marginal self-label has information $1/(1-\vartheta^2)$. Retaining the food reading changes the label likelihood and requires its full conditional information.

Proof. Apply Paper I's extraction lemma and conditional reset bound to each state conditioned on $\mathcal{C}_t$, then average. Book the other processor resets, reference preparation and Q-RAIF B's drive and switching work once each. The displayed Bernoulli informations follow by differentiating their likelihoods. Local quadratic harvest loss alone does not supply the global loss premise of Theorem 9. Under the separate-register assumptions the actual costs add to the displayed bound. □

Beyond leading order the calibration loss has a closed form. An agent with a belief of variance P about its gain sets its levels to $\mathbb{E}q$, and its expected harvest is $\ln 2 - H_2(\mathbb{E}[q \mid \text{records}])$: the agent earns what its belief allows, and the loss relative to a known gain is the Jensen gap $H_2(\mathbb{E}q) - \mathbb{E}H_2(q)$, which tends to $Ps^2/(2(1-s^2g^2))$ as $P \to 0$ and saturates for broad beliefs. The harvest is increasing in s, for a fixed gain belief. Its Jensen-gap mismatch penalty can also increase with s, so total harvest and mismatch penalty have different monotonicities.

6.3 What acting can reveal

Let Z_t be the hidden world state, θ the hidden parameters (instrument, actuators, world dynamics, record formation, initial law μ_θ), and let the agent's record be $(Y_1, A_1, \ldots, Y_t, A_t)$, with readings $Y_t \sim K^{\mathrm{s}}_\theta(\cdot \mid Z_t)$, actions A_t chosen by an arbitrary adaptive policy with private randomness, and transitions $Z_{t+1} \sim K^{\mathrm{w}}_\theta(\cdot \mid Z_t, A_t)$.

Definition 21 (Loop symmetry)**.** A group $\mathcal{G}$ acting on world states and parameters is a symmetry of the loop if $K^{\mathrm{s}}_{g\theta}(y \mid gz) = K^{\mathrm{s}}_\theta(y \mid z)$, $K^{\mathrm{w}}_{g\theta}(g\,\cdot \mid gz, a) = K^{\mathrm{w}}_\theta(\cdot \mid z, a)$ and $\mu_{g\theta} = g_*\mu_\theta$ for all $g \in \mathcal{G}$.

Theorem 22 (Acting cannot see the symmetry of its loop)**.** *If $\mathcal{G}$ is a loop symmetry, then for every policy the law of the agent's record is the same under $g\theta$ as under θ, and $\mathbb{E}_{g\theta}L(Z_t) = \mathbb{E}_\theta L(gZ_t)$ for every loss L. A parameter function that $\mathcal{G}$ moves has no consistent estimator from the agent's records, however it acts. If $\gamma \in \mathcal{G}$ exchanges a binary loss, $L(\gamma z) = 1 - L(z)$, then $\mathbb{E}_\theta L(Z_t) + \mathbb{E}_{\gamma\theta} L(Z_t) = 1$ for every policy and every t.*

Proof. Write the joint density of hidden variables and records as $\mu_\theta(z_0)\prod_t K^{\mathrm{s}}_\theta(y_t \mid z_t)\,\pi_t(a_t \mid \cdot)\,K^{\mathrm{w}}_\theta(z_{t+1} \mid z_t, a_t)$ and substitute $z \mapsto gz$ in every hidden variable. The policy factors do not involve z, and the others are invariant by Definition 21. Integrating out the hidden variables gives equal record laws, and the same substitution gives the loss identity. □

In the joint model, $\gamma : (B, X, \sigma) \mapsto (-B, -X, -\sigma)$ with everything else fixed is a loop symmetry for any agent that only reads and flips: the instrument depends on σx, the flip commutes with $x \mapsto -x$, the bath and the collisions are symmetric, and the gate coin is independent. The body loss $\mathbf{1}\{B = -1\}$ is exchanged. Every such agent therefore satisfies $P_\theta(B_t = -1) + P_{\gamma\theta}(B_t = -1) = 1$: if it holds its body at loss L it holds it at $1 - L$ in the world related to it by γ, with identical records. Exact enumeration over twenty random adaptive policies of five steps gives record laws that agree to 1.2×10^{-16} in total variation; one reference read per step separates them by at least 0.30. This is the observation of the Imprint Chain that its calibration statistic depends only on ν^2 [7], extended to every policy and to the task loss. For a linear loop, $x_{t+1} = ax_t + \kappa u_t + f + w_t$, $y_t = gx_t + b + e_t$, the affine maps $x \mapsto \lambda x + \mu$ with $(g, b, \kappa, f) \mapsto (g/\lambda,\ b - g\mu/\lambda,\ \lambda\kappa,\ \lambda f + \mu(1 - a))$ form a loop symmetry when the initial and process-noise laws are transformed as well; acting can identify the loop gain $g\kappa$ and never g or κ separately (Kalman likelihoods under random affine maps and dithered adaptive policies agree to 6×10^{-14}).

Proposition 23 (What the closed loop identifies)**.** *In the stationary body loop with constant gain and sign, $0 < g \le 1$, $d \in (0, 1) \setminus \{1/2\}$, and the matched setting, gate coin A_t of known probability $p_{\mathrm{g}} \in (0, 1)$ and flip success s,*

$$\mathbb{E}[Y_{t+1}Y_t \mid A_t = 0] = cg^2, \qquad \mathbb{E}[Y_{t+2}Y_t \mid A_t = A_{t+1} = 0] = c^2g^2,$$
$$\mathbb{E}[Y_{t+1}Y_t \mid A_t = 1] = cg^2(1 - s) + cs\,\mathbb{E}Y_t.$$

Writing the three moments as K_0, K_2, K_1, and $\mu = \mathbb{E}Y_t$, one has

$$c = K_2/K_0, \qquad g^2 = K_0^2/K_2, \qquad s = (K_1 - K_0)/(c\mu - K_0).$$

Stationarity gives $\mu = cp_{\mathrm{g}}sg^2/(1 - c + cp_{\mathrm{g}}s)$, so the denominator is nonzero. Thus c, g^2 and s are determined by the agent's own records, without references, and σ is not. With a wrong setting the first and third moments vanish.

Proof. $B_{t+1} = \xi_t B_t \Phi_t$ with $\mathbb{E}\xi_t = c$ independent and $\Phi_t = -1$ if the flip is switched on and succeeds. For $A_t = 0$, $\mathbb{E}[Y_{t+1}Y_t] = \sigma gc\,\mathbb{E}[B_tY_t] = cg^2$. For $A_t = 1$, $\mathbb{E}[B_t\Phi_tY_t] = \sigma g + 2s\,\mathbb{E}[B_t\mathbf{1}\{Y_t = -1\}]$ and $\mathbb{E}[B_t\mathbf{1}\{Y_t = -1\}] = (\mathbb{E}B_t - \sigma g)/2$; multiply by σgc and use $\sigma g\,\mathbb{E}B_t = \mathbb{E}Y_t$. The lag-two identity iterates the first. □

A frame error is thus visible in the agent's own loop as a loss of loop correlation; locating the right setting still requires trying the others, which is Paper II's blind spot in binary form. What no record reveals is fixed by the following statement. Call an operation *symmetry-breaking* if its action singles out a world state in a way that the symmetry does not respect: a reference prepared in the state $+1$, a reset to $+1$, harvest levels with the higher level on -1, a bath relaxing toward a known state. Each carries a label ζ saying which state it singles out, and that label belongs to the operation's dispatch map in the sense of Paper G.

Proposition 24 (No internal standard)**.** *Let $\mathcal{G}$ act also on the labels of all symmetry-breaking operations by relabelling. If the labelled sensing, transition and initial kernels are jointly covariant, $\mathcal{G}$ remains a loop symmetry. No policy identifies an absolute jointly relabelled orientation. A system can form and retain relational standards.*

Proof. Apply the assumed joint covariance in Theorem 22. Set-theoretic closure of a kernel family alone does not imply covariance. □

The model locates an operational reference standard in a labelled kernel: an external reference, an actuator reset or level assignment, a known environmental asymmetry, or a previously interpreted record. Jointly relabelling every such kernel preserves the record law. A system can nevertheless construct relational standards and carry them between operations. Paper G distinguishes a supplied dispatch map from a universal boundary of explanation [8]; here the protocol-relative calibration boundary is the labelled operation whose orientation the loop uses. The symmetry theorem does not derive the physical origin of that operation.

6.4 Identification and reference timing

Acting supplies some relational information and leaves other distinctions unresolved. In the stationary nondegenerate body model, Proposition 23 identifies persistence, squared gain and actuator success from its record correlations. It does not identify absolute sign or supply a joint information rate for a drifting instrument. Appendix C retains the three-parameter comparison and all its proofs. Fresh-record read-and-flip policies with independent sign and bath inversions approach chance (Theorem 37). Earlier stable records or labelled standards change that experiment.

Reference timing is itself a decision from history. Predetermined gaps and record-adaptive requests have different bounds (Proposition 38). At equal inversion and bath rates 1/10, ideal reads and flips, and reference price 1/10, the exact all-adaptive optimum is $2089311/43698622 = 0.0478118280\ldots$. It requests a reference after an apparent innovation or five quiet transitions (Proposition 40). The full Bellman certificate and noisy comparison remain in that appendix. This finite-rate result is not a universal drift exponent or an optimal repair policy for the access-gated body.

6.5 Operating and continuing regimes

A bounded two-gain block model has a supplied-resource operating margin of $0.02396\,k_BT$ per food use with a squeezed drive, body loss 0.08725 and explicit purity input. Its complete ledger and comparison are in Appendix K.1. A separate continuing eight-state filter carries instrument and body history through every use. Under the same reference cap, adaptive timing gives body loss .10319, versus .11726 for periodic timing. Its transition/likelihood, two-seed check, 350-bit quantization certificate and resource boundary are retained in Appendix K.2. The block margin cannot be imported into that continuing protocol without its own apparatus ledger.

7 When the body controls access

The preceding loop receives a reading and can process it at every use, even when its body is at the loss state. A body that supports those operations changes the problem. Keep the bounded instrument, record couplings, field drive and body bath, but let $B = +1$ enable the main instrument, reference ports, model logic and food protocol. At $B = -1$ these ports are

disconnected. The model register is held, not updated. A disabled reader cannot acquire a reference or ask a running estimator to repair it.

Definition 25 (Access-gated joint loop)**.** At each working use, read the body with likelihood $(1 + y\sigma g_G)/2$, update the model, perform the food task and write a recovery command $\ell \in \{0, 1\}$ to a latch. The working body is not actively flipped; the bath disables it with probability d. During disability the latch and model register remain unchanged. A separate standby port, if present, executes that stored command through the same field drive; it obtains no new instrument reading. The recovery probability per disabled slot is

$$r_\ell^{\mathrm{rec}} = d + \ell s(1 - 2d), \qquad 0 < d < \tfrac{1}{2}.$$

The instrument continues its gain and sign transitions in all wall-clock slots. Disabled-slot fragments and enable flags leave through output streams that the reader cannot later acquire; there is no bank of delayed readings on resumption. The reader obtains no elapsed-time counter on recovery. A resumed use is another working invocation, with its prior obtained from the event kernel below. Harvest work enters the physical ledger but is not read back as an additional self-label unless a protocol explicitly permits that read. The latch is written after all records acquired in its working invocation.

An independent enable flag copies the pointer value of B and controls the standby drive while the main processor is held. In the diagonal body model the repair leaves $+1$ fixed and sends -1 to $+1$ with probability s. A CPTP extension has Kraus operators

$$D_0 = P_+, \qquad D_1 = \sqrt{1 - s}\, P_-, \qquad D_2 = \sqrt{s}\, |+\rangle\langle -|, \qquad \sum_j D_j^\dagger D_j = I.$$

It dephases the body blocks and is nonunital. The enable flag, its preparation or reset, and the body's free-energy change enter the ledger. One explicit resource model supplies a fresh pure flag per wall-clock slot and retains the used flag in a charged output stream, consuming $\ln 2$ of flag purity per slot. Cyclic reuse requires a conditional reset and protection protocol. Table 1 collects these resource declarations. With no standby, the symmetric bath still recovers the body at rate d.

Proposition 26 (The kernel between working uses)**.** *Let* K *be the four-state gain/sign factor of* T_a *and fix a stored command with recovery rate* $r_{\mathrm{rec}} > 0$*. If* L *is the number of wall-clock slots between successive working invocations, then*

$$P(L = 1) = 1 - d, \qquad P(L = k) = d r_{\mathrm{rec}} (1 - r_{\mathrm{rec}})^{k-2} \quad (k \geq 2), \tag{14}$$

$$\mathbb{E}L = 1 + d/r_{\mathrm{rec}}, \qquad \pi_{\mathrm{w}} = \frac{r_{\mathrm{rec}}}{r_{\mathrm{rec}} + d}, \tag{15}$$

$$\mathsf{K}_\ell = (1 - d)\mathsf{K} + d r_{\mathrm{rec}} \mathsf{K}^2 [I - (1 - r_{\mathrm{rec}})\mathsf{K}]^{-1}. \tag{16}$$

π_{w} *is the long-run working fraction. A right eigenmode of* K *of eigenvalue* λ *has event eigenvalue*

$$f_{r_{\mathrm{rec}}}(\lambda) = (1 - d)\lambda + \frac{d r_{\mathrm{rec}} \lambda^2}{1 - (1 - r_{\mathrm{rec}})\lambda}.$$

The event kernel remains valid conditionally when the latch is chosen from the preceding working record.

Proof. The body either remains working for the next slot or enters a geometric recovery wait. This gives (14) and its mean. The stationary flow balance is $\pi_{\mathrm{w}} d = (1 - \pi_{\mathrm{w}}) r_{\mathrm{rec}}$; renewal averaging gives the same fraction. Instrument noise is independent of the body and recovery noise conditional on the selected latch. Sum K^L with the displayed gap probabilities. Since $\|(1 - r_{\mathrm{rec}})\mathsf{K}\|_\infty = 1 - r_{\mathrm{rec}} < 1$, the sum is the positive resolvent in (16). Applying it to an eigenvector gives $f_{r_{\mathrm{rec}}}$. No recursion is executed in disabled slots: the finite controller uses the matrix for its last stored command on the next invocation. □

The working body provides a relative calibration standard. With static sign and gain, its working readings are independent samples of $(1 + y\sigma g)/2$; their mean identifies σg for $g > 0$. With a fixed latch their Fisher information rate is $\pi_{\mathrm{w}}/(1 - g^2)$ in the signed-gain coordinate. Under a history-dependent latch, information is counted on actual acquired uses, as in Consciousness III [3]. The transformation $(B, X, \sigma) \mapsto (-B, -X, -\sigma)$ exchanges enabled and disabled blocks and hence fails to preserve this access kernel. The orientation comes from the declared body–permission coupling. Jointly relabelling that coupling, every operation and the task preserves the absolute-orientation obstruction.

The readout must be informative: $g = 0$ or wrong-setting reads give no instrument information. Continued access is a separate condition. In a hardware variant with an absorbing disabled state and conditional failure hazard at least $h > 0$ on every working use, every policy has

$$P(N \geq n) \leq (1 - h)^{n-1}, \qquad \mathbb{E}N \leq h^{-1},$$

where N is its total number of working readings. Successive conditioning proves the bound, so only finitely many readings occur almost surely. If two static parameters have positive overlapping likelihoods on finite stopping records, their stopped-record laws have positive overlap and their equal-prior Bayes error is strictly positive. Waiting longer cannot identify them. The absorbing variant removes passive recovery as well as standby.

Proposition 27 (Calibration through unobserved disability)**.** *Take known $0 < g < 1$, sign inversion probability $0 < \eta < 1/2$, one body reading on resumption and no additional current directional observation. Conditional on a stored recovery command, every sign estimator has average error at least*

$$\min\{e_{\mathrm{rec}}, (1 - g)/2\}, \qquad e_{\mathrm{rec}} = \frac{1 - f_{r_{\mathrm{rec}}}(1 - 2\eta)}{2} > 0.$$

Increasing r_{rec} lowers this oracle floor and increases the working fraction. For small η, $e_{\mathrm{rec}} = \eta/\pi_{\mathrm{w}} + O(\eta^2)$.

Proof. Grant the reader the true sign at its preceding working invocation. Across an unobserved gap L, its sign bias is multiplied by $(1 - 2\eta)^L$. Averaging gives the prior error e_{rec}. The current body reading is a binary symmetric channel of error $e = (1 - g)/2$. For prior error $q \leq 1/2$, its two-output Bayes error is

$$\min\{(1 - q)(1 - e), qe\} + \min\{(1 - q)e, q(1 - e)\} = \min\{q, e\}.$$

An actual history is less informative than the oracle history. For $0 < \lambda < 1$, differentiation gives

$$\frac{\partial f_{r_{\mathrm{rec}}}(\lambda)}{\partial r_{\mathrm{rec}}} = \frac{d\lambda^2(1 - \lambda)}{[1 - (1 - r_{\mathrm{rec}})\lambda]^2} > 0.$$

The working fraction is also increasing. A paid elapsed-time counter or another current reference changes the available record and therefore changes this bound. □

Filtering at actual working invocations uses the positive four-state event kernels of Proposition 26. The certificate in Appendix L needs 150 ideal model bits and bounds asymptotic total-variation error by 1.247×10^{-7} for the stated switching rates. Its precision is fixed; identifying a continuous parameter to arbitrarily growing precision requires growing memory.

Proposition 28 (A domain where history is necessary for the food task)**.** *Fix the recovery command, a known static $0 < g < 1$, an unknown static sign with equal prior, one body reading per working invocation and no further directional sample before the food action. Within the joint food protocol, choose harvest levels from the body model before the fresh food reading M, then choose whether to flip from M. Without retained history, the maximal harvest is*

$$J_1 = \ln 2 - H_2((1 + sg^2)/2).$$

Let $a_g = (1+g)/2$, $b_g = (1-g)/2$, and retain the latest three body readings. For their number k of positive results set

$$p_k = \tfrac{1}{2}\binom{3}{k}(a_g^k b_g^{3-k} + b_g^k a_g^{3-k}), \quad m_k = \frac{a_g^k b_g^{3-k} - b_g^k a_g^{3-k}}{a_g^k b_g^{3-k} + b_g^k a_g^{3-k}}.$$

The achievable harvest

$$J_3 = \sum_{k=0}^{3} p_k[\ln 2 - H_2((1 + sg|m_k|)/2)]$$

is strictly larger than J_1. Thus a target harvest in $(J_1, J_3]$ requires history within these permissions. An already learned sign encoded in a program, latch or weight counts as retained history.

Proof. The fair food reading has no marginal sign information. A body-only record of posterior sign mean m allows a food polarization of at most $sg|m|$: for arbitrary flip probabilities depending on M, its magnitude is $sg|m|$ times the difference of two probabilities, whose absolute value is at most one. Choosing the appropriate orientation attains that endpoint, and matched levels attain the stated harvest. Without history $m = gY$. Three-sample Bayes gives p_k, m_k. The function $F(m) = \ln 2 - H_2((1 + sgm)/2)$ is strictly convex, with $F''(m) = (sg)^2/[1 - (sgm)^2] > 0$. Posterior means form a martingale, and two extra informative noisy samples make its conditional value nonconstant. Strict Jensen therefore gives $\mathbb{E}F(m_3) > \mathbb{E}F(m_1)$. A three-bit sliding record acquires no additional samples and remains held during disability. □

For $g = 3/5$ and $s = 49/50$, exact finite body–food–drive path enumeration gives $J_1 = 0.0635939$ and $J_3 = 0.1202672$. If levels may instead depend on the current food reading, the posterior-matched protocol family of Paper I replaces sg by g in both formulas and retains the strict history gap.

Performance is averaged over wall-clock slots. The fixed-latch harvest gain is $\pi_{\mathrm{w}}(J_3 - J_1)$ before the extra memory cost. The drive of Section K.2 changes the working fraction from

1/2 to about 0.949083 when standby is enabled. Body readings at known $B = +1$ contain fresh instrument noise uncorrelated with the body copy. Table 1 therefore charges their accessible reset information separately, together with holding and processing. The finite-block operating margin uses a different body protocol.

Resource	Declared supply or charge	Availability and accounting
Food and body fragments	$4n \ln 2$ purity nats per wall slot if continuously allocated	Acquired and reset only on working uses; disabled outputs are not banked
Standby enable flag	$\ln 2$ supplied purity nats per wall slot in the fresh-flag implementation	Used outputs leave the reader; cyclic reuse needs a reset/protection ledger
Two drive packets	$2\theta_\omega N$ per wall slot	Unused prepared packets remain charged
Drive switching	$\theta_\omega \lambda^2[\ell(1-\pi_{\mathrm{w}}) + \pi_{\mathrm{w}}/2]$	Recovery on disabled slots; food on working slots
Body outcomes and model	Conditional erasure, update and reliable-hold costs	No processing during disability; the body copy does not erase fresh sensing noise

Table 1: Resource declarations for access-gated comparisons. Purity is an input, not a proved total energy cost. Preparation from thermal resources and any remaining apparatus work must also be charged.

For a fully known instrument and fixed active pipeline, let v denote net value per working use after its active costs, and put $\kappa = \theta_\omega \lambda^2$. If all base supplies, including fresh flags, are allocated equally in wall time, the two fixed latches have residual values

$$U_\ell = \pi_{\mathrm{w},\ell} v - (1 - \pi_{\mathrm{w},\ell})\kappa\ell - C_{\mathrm{base}}, \qquad U_1 - U_0 = \frac{(r_1^{\mathrm{rec}} - d)v - 2d\kappa}{2(r_1^{\mathrm{rec}} + d)}.$$

This identifies the priced gain from making future calls possible. Flag costs that depend on the latch must be added, not hidden in a common base. For an uncertain instrument, the future task value and the evidence lost while disabled must also enter the control problem. The known-parameter price threshold leads to a history-dependent repair comparison.

An exact static witness in Appendix D already gives equal sign confidence and opposite paid repairs from legally acquired histories. The main comparison now uses the close gains and positive drift of the continuing instrument, including the stated body-reading and food-reset charges. The object to retain is the task-relevant joint belief, not confidence alone.

Proposition 29 (Two moments determine terminal access value)**.** *Immediately before a final working read let $X = \sigma g_G$ have posterior law b, and put $m = \mathbb{E}_b X$ and $v = \mathbb{E}_b X^2$. Use the body reading $P(Y = y \mid X) = (1 + yX)/2$, charge its conditional entropy, and permit either the posterior-matched food protocol or abstention. For overlap r, flip success s and switching*

charge $\kappa/2$, the expected working-use value under these separate conditional-reset prices is

$$A(m,v) = -H_2((1+|m|)/2) + \sum_{y=\pm 1} \frac{1+ym}{2}\, M(m_y), \qquad m_y = \frac{m+yv}{1+ym}, \tag{17}$$
$$M(z) = \max\{0, S_r(0) - S_r(|z|) - H_2((1+s|z|)/2) - \kappa/2\}.$$

Here S_r is defined in Section K.1; a term of zero observation probability is taken by continuity. A post-record decision with two wall slots remaining has

$$Q_1(b) - Q_0(b) = d[(r_1 - d)A(m_2, v_2) - \kappa(2 - r_1)], \qquad r_1 = d + s(1-2d), \tag{18}$$

where m_2, v_2 are the moments of $b\mathsf{K}^2$.

Proof. Bayes' rule gives $P(Y=y) = (1+ym)/2$ and $\mathbb{E}[X \mid Y=y] = (m+yv)/(1+ym)$. The food harvest is $\ln 2 - H_2((1+s|m_y|)/2)$, its copy-assisted reset costs $\ln 2 - S_r(0) + S_r(|m_y|)$, and switching costs $\kappa/2$. Their difference, optimized against abstention, is $M(m_y)$. The body-outcome charge is $H_2((1+|m|)/2)$. These terms give A. For the two-slot difference repeat the branch calculation in (26); the instrument undergoes two hidden transitions before the final reading. □

Two moments suffice for this terminal task, not for the continuing joint filter. The prices are conditional-entropy limits for specified separate resets. Base packets, flags, clock, reliable holding and processor implementation have the common supplied charges of Table 1; the expression is not a total cyclic apparatus profit. Additional action-dependent costs must be included in its value before deciding to repair.

One formed model, different decisions. The body model carried between invocations is a joint belief over (G, σ), formed by the permitted record history. It has three uses within this protocol. The food controller acts only when $M(m_y) > 0$; otherwise it withholds permission to spend the switching and reset resources. The repair controller compares (18) with zero. A controller that can use a later invocation for further evidence must also forecast what today's record will make learnable. These are not three independently supplied assessments of the observer. They use different functions of the same acquired belief. A finite horizon and read quota, or a specified continuing budget, determine which functions are worth retaining and computing.

Proposition 30 (Terminal compression need not close enquiry)**.** *Use the same body likelihood as Proposition 29 and write $t = \mathbb{E}_b X^3$. Conditional on the present working read,*

$$P(Y=y) = \frac{1+ym}{2}, \qquad m_y = \frac{m+yv}{1+ym}, \qquad v_y = \frac{v+yt}{1+ym}. \tag{19}$$

For a static instrument and two available working invocations, consider the body-only retained-history protocol: use the terminal food/abstention rule on each invocation, retain the body record, and do not use harvest-work labels or the food record as an extra calibration record. Its two-invocation value, before common supplied charges, is

$$A_2(b) = -H_2((1+|m|)/2) + \sum_{y=\pm 1} \frac{1+ym}{2}[M(m_y) + A(m_y, v_y)]. \tag{20}$$

Thus (m, v) is sufficient for the last invocation but need not determine the value of retaining a record for the next. With the four distinct supported values $\{-g_+, -g_-, g_-, g_+\}$, moments of orders zero through three determine the entire belief; fewer need not. Under positive drift, propagate that belief with the legal event or wall-slot kernel before applying the next likelihood.

Proof. Multiply the prior by $(1 + yX)/(1 + ym)$ and integrate X and X^2. This gives (19). The first invocation pays its outcome entropy and chooses $M(m_y)$; its retained outcome determines the prior of the second, whose value is $A(m_y, v_y)$. Conditioning and averaging prove (20). For the four-state support, moment recovery is an invertible Vandermonde system. To see the failure of two-moment closure without relying on a dimension count, put $a = g_-$, $c = g_+$ and perturb any strictly positive belief in the stated order by $e(a/c, -1, 1, -a/c)$. Normalization, m and v are unchanged, whereas t changes by $2ea(a^2 - c^2) \neq 0$. For example, with $a = 0.90\sqrt{1 - 0.8^{14}}$, $c = 0.98\sqrt{1 - 0.8^{14}}$, $s = 0.979992$ and $\kappa = 0.00225388$, start from $b = (.02, .08, .10, .80)$ and take $e = .02$. Both beliefs have $(m, v) \simeq (.765001391, .892279800)$, but $\sum_y P(y)A(m_y, v_y)$ is respectively .181752936 and .180874313. The continuous entropy expressions in (17) give this strict difference. These positive-belief examples establish failure of universal compression; they are not claimed as the two histories reached from the uniform prior in Proposition 31. □

The invocation quota in this proposition is real: one body reading per invocation, no extra calibrated input and no read during a blackout. Record retention has holding, processing and eventual reset costs; the gain in (20) pays for it only if it exceeds those prices. An optional additional enquiry must be compared with foregoing that invocation or using it for the current task, including its drive, switching and record charges. The formulas are conditional task values, not a free probe or a claim of recurrent optimality. In a finite-horizon control problem the legal belief, remaining quota and available clock information suffice for the usual Bellman recursion. If the clock is absent, the disabled duration must instead be marginalized with (16); it cannot be recovered from unobserved slots. The fixed-support filter and its precision certificate implement this common model; they do not supply processor work or a cyclic reset for free.

Proposition 31 (Paid-repair decisions under positive drift)**.** *Set $r = 0.8^7$, $h = \sqrt{1 - r^2}$, $g_- = 0.90h$, $g_+ = 0.98h$, $\delta = 3 \times 10^{-4}$, $d = 1/20$, $s = 0.979992$ and $\kappa = 0.00225388$. Start from the uniform four-state prior. Take a priced clock, horizon $H = 12$, and a working decision at slot ten. Use the terminal protocol of Proposition 29, with no harvest-work read-back or further processing surcharge. At $\eta = 2 \times 10^{-4}$, a lower-confidence history optimally repairs while a higher-confidence history declines. There is also a strictly positive η_* in*

$$.00019999976122194292 < \eta_* < .00019999976122194293$$

at which two reachable histories have exactly equal sign confidence and opposite optimal repairs. The histories and rounded values are shown in Table 2.

Proof. Hold latch zero through the preceding slots. Each displayed gate path then has positive probability, independently of the instrument. All reading likelihoods and instrument transitions are positive. Starting from the uniform prior, propagate with K between slots

Inversion rate	Slot history	Sign confidence	$Q_1 - Q_0$
0.0002	$- \times - - - - + + ++$	.969070314	$+.002191561$
0.0002	$++ - + - - - - - --$	.999973536	$-.000178194$
η_*	$- + + + \times \times \times + - +$	.999514604	$+.000256112$
η_*	$+ - - \times - + - - + -$	.999514604	$-.000845366$

Table 2: Working body readings are + or −; × is a disabled slot. All decisions use the same two-slot horizon, prices and action family. In the last pair confidence is exactly equal at a root in the stated interval, not merely equal after rounding.

and multiply by $(1 + y\sigma g_G)/2$ on working readings. Thus these are reachable beliefs, not assigned posteriors.

The confidence difference of the last two histories is a continuous function of η. Outward interval evaluation gives opposite signs at the two stated rational endpoints, proving existence of an exact root. Over that whole interval the two differences in (18) lie respectively in $(.00025611, .00025612)$ and $(-.00084537, -.00084536)$. At $\eta = .0002$ the first pair lies in $(.00219156, .00219157)$ and $(-.00017820, -.00017819)$. The entropy, square-root and Bayes operations in (17) enclose these values directly; every posterior denominator is strictly positive. □

The enclosures evaluate the displayed Bayes and entropy equations directly, rather than using sampled trajectories to decide the four histories.

The first pair needs no tuned parameter and rules out an increasing confidence-threshold repair rule at the continuing instrument's own rates. The second rules out sign confidence as a sufficient decision statistic in a nearby positive-rate model. It does not assert a unique root. The restoring reader assigns more weight to the higher gain; the declining reader expects less value from a working invocation. Confidence in orientation and the value of accessible tasks are different properties of the same acquired history.

The finite horizon is material. A continuing observer must also price evidence gained on later invocations, lost evidence during disability, and the registers that carry its model. Recurrent optimal maintenance without an elapsed-time counter remains a stronger target. The two moments above explain a terminal decision without adding history as an independent reward.

8 Consequences for the other links

Paper II. Let the agent add a known push p_t, independent of the input, through an actuator with gain k, so that $y_t = \rho\, u \cdot x_t + k\rho\, u_A \cdot p_t + \xi_t$, with one in-frame read per step. If sensing and acting share one coupling ($u_A = u$), the regressions of y on the push and on the in-frame read, with the residual variance, determine u including its background, ρ, k and s; the blind spot of Paper II disappears with a single read per step when $u \cdot v \neq 0$, $k\rho \neq 0$ and sign conventions for k, ρ are fixed, and the remaining symmetry between actuator gain and input scale is fixed by the known input law. If the actuator coupling is independent of the sensing coupling, pushes determine $k\rho u_A$ and nothing about the background of u. In a simulation

with $D = 6$ the shared coupling recovers ρ and k within 2% and the direction of u within 0.016; with an independent coupling the direction error is 1.3. A shared sensing/action frame supplies this constructive route. At $u \perp v$ the scale parameters remain confounded; varying the read direction repairs the degeneracy.

HAFF Paper C. The viability bound of an agent depends on it only through the product of its sensing and control influences [4], which is invariant under the sign symmetry, while viability itself is not: an inverted agent has the same influences and fails with probability $\frac{1}{2}(1 + (1 - 2\nu)(1 - 2\mu)(1 - 2q))$. Positive influences can be certified from the agent's own records; viability cannot be certified without a standard.

Several observers. Applying Theorem 22 with the sign symmetry acting on two agents at once, their relative sign is invariant and detectable from agreement, and a common inversion is not. The common error that agreement can never expose, which Foundation II measures [10], is the direction shared by the observers' standards.

9 Discussion

9.1 The memory–self–introspection route

The route now has three distinct constructive roles. Paper I identifies which correlations a specified task consumes. Paper II forms a finite retained task–channel relation that selects later reads. This paper asks how that relation remains revisable when the instrument and the body change. Even records can detect drift before a frozen frame fails without identifying the correction's sign. References and relational standards provide additional distinctions; they are physical permissions with prices, not an automatic consequence of noticing error.

The acting results add maintenance value to calibration confidence. A history-dependent body model determines whether a task should use its switching and reset budget, what another acquired outcome can change, and whether a disabled body is worth making callable again. Proposition 31 gives opposite paid repairs from equal sign confidence at positive drift. Proposition 30 shows why terminal compression need not close further enquiry. These are uses of one acquired relation, not separately preinstalled answers for each decision.

The mechanisms have different supplied structures. The Gaussian formation task gives a finite menu; the bounded acting model supplies its transition family, labelled standards, body gate and resource streams. The finite filters certify approximation of that model, not its truth. A shared body–world inference problem would need an explicit joint likelihood before these separate results became one larger device. Freezing a correct static frame remains a useful control, not a generic failure of subject formation.

9.2 Established results and stronger targets

The detection bound applies to every test. Its known-path check reaches errors at about 1.3 times the necessary horizon, and the residual variance test has the sharp Brownian order. The scalar quadratic drift law bounds every admissible schedule and estimator; Gaussian priors/readings and controllable real precision attain its slow-drift leading term. Whole-read schedules retain the relaxation boundary. The joint model supplies exact identifiability

obstructions, a finite-rate adaptive-reference optimum and a complete ledger for its specified resources. Its body-gated extension supplies an acquired-history harvest gap, an event kernel through unobserved disability and positive-drift repair counterexamples. The bank's one-step task-risk formula in Appendix F remains exact for every joint report law within its linear action family.

Several stronger claims still need work. The scalar Gaussian drift law does not settle vector or smoother drift (Conjecture 15). For non-Gaussian increments its information recursion uses increment Fisher information, not variance alone. Paper II's probe-calibration conjecture concerns spherical Brownian or local isotropic diffusion, not every rotation-invariant law; independent scalar recursions cannot simply be assigned to a frame's coupled directions. No online rule here learns the true-error quantities in Proposition 45; the gain proxy is not proved task-optimal, and no organism is required to compute Fisher information.

The repair witness is finite-horizon, not a recurrent optimality theorem. Continued maintenance without an elapsed-time counter must account for later evidence, evidence lost in disability and finite processing. Holding, computation, reference preparation and heat transport still need one attainable cyclic ledger. The continuing filter's operational advantage cannot inherit the finite-block margin from a different body protocol. BRAIN's several-port maintenance task [12] has its own repair clock and cannot inherit the single-body event kernel without a mapping.

9.3 Function and reflective organization

The self-model here is not merely an external observer's estimate. Its retained distinctions select the same system's subsequent task, enquiry and access repair. This gives the memory–self–introspection route a functional organization whose formation and continued value can be tested. It does not make every persistent system need a self-frame: a known static interface or unrestricted access can remove the corresponding need.

Consciousness III proposes that reflective continuity depends on maintained dispositional knowledge and causal return [3]. The present task supplies a mechanism such a proposal must respect: matching content or confidence can miss cause-specific action, whereas matching all permitted interventional laws leaves no further discriminator in that class. The bridge requires independent evidence about reflection. Working and disabled states here are physical permissions, not models of sleep, death or phenomenal presence. The contribution is a learned relation that guides its own continued access, not a proof that a calibrated controller experiences.

A Estimating the mismatch with a prior self-model

For an agent that retains a model of its instrument, earlier calibration records supply a prior π on the drift parameter $\delta\vartheta$. Its Fisher information measures how firmly that estimate is held. Paper II's foreground uses such a retained model; Paper I's implicit protocol distribution alone does not require one.

Definition 32 (Rigidity of a self-model)**.** The rigidity of a prior self-model π is its Fisher information

$$\mathcal{I}_{\text{self}} := \int \big(\partial_{\delta\vartheta} \log \pi(\delta\vartheta)\big)^2 \, \pi(\delta\vartheta)\, d(\delta\vartheta). \tag{21}$$

Theorem 33 (Van Trees bound with a prior self-model)**.** *Let $\delta\vartheta$ lie in a compact interval $[a, b]$, let the prior π be C^1 with finite rigidity and vanish at a and b, and let the record consist of n_{eff} conditionally independent samples (or a joint experiment with that exact information), each with Fisher information $\mathcal{I}_1(\delta\vartheta)$ under a regular likelihood. Then every estimator $\delta\hat{\vartheta}$ based on the record satisfies*

$$\mathbb{E}\big[(\delta\hat{\vartheta} - \delta\vartheta)^2\big] \;\geq\; \frac{1}{n_{\text{eff}}\, \mathbb{E}_\pi[\mathcal{I}_1(\delta\vartheta)] + \mathcal{I}_{\text{self}}}. \tag{22}$$

Proof. The van Trees inequality [17] bounds the Bayesian mean squared error of any estimator by the inverse of the prior-averaged Fisher information of the record plus the Fisher information of the prior; the boundary conditions remove the boundary term of its integration by parts. Conditional independence makes the record's information $n_{\text{eff}}\mathcal{I}_1$. □

The prior helps when it is well placed and hinders when it is not. In the conjugate Gaussian location model, with constant per-sample precision $\bar{\mathcal{I}}_1$, the posterior mean is the precision-weighted average

$$\hat{\vartheta} = \frac{n_{\text{eff}}\bar{\mathcal{I}}_1\, \hat{\vartheta}_{\text{MLE}} + \mathcal{I}_{\text{self}}\, \vartheta_0}{n_{\text{eff}}\bar{\mathcal{I}}_1 + \mathcal{I}_{\text{self}}} \tag{23}$$

of the maximum-likelihood estimate and the prior centre ϑ_0. A rigid self-model centred on the wrong value pulls every estimate toward it and resists recalibration; a soft one follows the noise. Without a model of the drift one can only say that an optimal rigidity exists when the loss is lower semicontinuous on a compact set of rigidities. Under a random-walk drift the optimum is exact (Corollary 13).

B Natural-gradient calibration

This local comparison uses assumptions (C1)–(C3) of Section 2.3.

B.1 The natural-gradient update

A loop that keeps the self-model current can move ϑ along the steepest descent of a frame loss in the geometry of its own records,

$$\dot{\vartheta} = -\eta\, g(\vartheta)^{-1} \nabla_\vartheta L(\vartheta, t), \tag{24}$$

where $\eta > 0$ is the adaptation rate, g is the Fisher metric of the records the loop uses, and $L(\vartheta, t)$ is the risk of acting at time t with self-model ϑ. For Paper II's rank-one frame v, $L(v, t) = s^2 + \rho^2\big(1 - (v \cdot u_t)^2\big)$, which is smallest for the frame aligned with u_t. The update is invariant under reparametrization of Θ; this matters because no coordinate system on a space of frames or instrument models is preferred. It is a definite algorithm. An

implementation must supply the likelihood, evaluate or approximate its score and realize the update law; accumulating an unlabelled second moment of residuals is not in general a Fisher computation.

The metric must be nondegenerate where the loop operates, and records read through the frame fail exactly there.

Proposition 34 (The in-frame metric degenerates at calibration)**.** *For Paper II's two-channel reader with the frame at angle θ to the coupling, the metric of in-frame records is $F(\theta)$ of (1), and the natural gradient of $L = s^2 + \rho^2 \sin^2\theta$ in this metric is*

$$\frac{\partial_\theta L}{F(\theta)} = \frac{\rho^2 \sin 2\theta}{F(\theta)} = \frac{2\rho^2}{\kappa\,\theta}\big(1 + O(\theta^2)\big),$$

which diverges at $\theta = 0$. If a fraction $p \in (0, 1]$ of the records are off-frame probes, the metric $(1-p)F(\theta) + pF_\perp(\theta)$ is at least $p\rho^2 \cos^2\theta/(\rho^2 + s^2)$, and for $|\theta| < \pi/2$ its absolute natural gradient is bounded by $2(\rho^2 + s^2)|\tan\theta|/p$.

Proof. The first statement follows from $F(\theta) = \kappa\theta^2 + O(\theta^4)$ (Corollary 5). In (6) the factor $\rho^2 \sin^2\theta + \rho^2 + s^2$ is at least the denominator's base $\rho^2 + s^2 - \rho^2 \sin^2\theta$, so $F_\perp(\theta) \geq \rho^2 \cos^2\theta/(\rho^2 + s^2 - \rho^2 \sin^2\theta) \geq \rho^2 \cos^2\theta/(\rho^2 + s^2)$; divide $\rho^2 \sin 2\theta$ by p times this. □

A loop that uses only in-frame records therefore takes ever larger natural-gradient steps as it approaches calibration, and by Proposition 3 its records do not even fix the sign of the step. This is the geometric form of Paper II's blind spot, and the remedy is the same: a fixed fraction of reads outside the frame makes the metric positive at calibration, with value $pF_\perp(0) = p\rho^2/(\rho^2 + s^2)$.

The local strong-convexity tracking theorem and proof are retained in Appendix I.1. Its assumptions require a nondegenerate metric and a smooth geodesic neighborhood; they do not repair the in-frame degeneracy. Persistent excitation is a separate input.

Definition 35 (Persistent excitation)**.** The records are persistently exciting with constant $\alpha_{\mathrm{PE}} > 0$ if for some window T

$$\frac{1}{T}\int_t^{t+T} g(\vartheta(s))\,ds \;\succeq\; \alpha_{\mathrm{PE}}\, I \qquad \text{for all } t. \tag{25}$$

Pointwise regularity and a positive entropy rate of the environment do not by themselves give a uniform lower bound of this kind, and no equality between α_{PE} and m_L is assumed. Proposition 34 is a concrete failure of persistent excitation: in-frame records near calibration have $\alpha_{\mathrm{PE}} \to 0$, and probes restore it.

C Standards and adaptive reference timing

C.1 Three self-parameters, three laws

The magnitude and the setting are invariant under the sign symmetry; the sign is not.

Proposition 36 (Free loop information changes the exponent)**.** *In Theorem 9 let unlabelled reads carry $J_0 > 0$ per use at no extra cost, and let the loss be cP. Put $P_f = \frac{1}{2}\big(-\nu^2 + \sqrt{\nu^4 + 4\nu^2/J_0}\big)$. If $c \le (w/J)J_0(2P_f + \nu^2)/(P_f(P_f + \nu^2))$, which holds for $\nu \le \nu_\times = 2wJ_0^{3/2}/(Jc)\,(1+o(1))$, every policy pays at least $cP_f = c\nu/\sqrt{J_0}\,(1+O(\nu\sqrt{J_0}))$ per use, and with Gaussian reads this is attained with no references.*

Proof. $\phi(P) = \nu^2/(P(P+\nu^2))$ is convex and decreasing, so h of Theorem 9 is convex. P_f solves $\phi(P) = J_0$, and $|\phi'(P_f)| = J_0(2P_f + \nu^2)/(P_f(P_f + \nu^2))$, so under the condition $h' \le 0$ on $(0, P_f]$ and the infimum is $h(P_f) = cP_f$. As $\nu \to 0$, $P_f = \nu/\sqrt{J_0} - \nu^2/2 + O(\nu^3)$. The Kalman filter on the free reads alone has stationary variance P_f. □

The body moments establish static identifiability under Proposition 23's nondegeneracy conditions. A joint information-rate bound for the correlated record stream is a further step: the information of one overlapping product is not that rate. Proposition 36 gives an exact Gaussian realization of the linear-in-ν price when a genuine free observation channel is supplied. For self-labels the marginal Bernoulli information is $1/(1-\vartheta^2)$; retaining the earlier food read changes the conditional likelihood and its information.

Theorem 37 (Sign inversions)**.** *Suppose that sign inversions and body-bath flips are independent Bernoulli events with probabilities $\eta, d \in (0,1)$ per use. The agent knows the initial sign, has no symmetry-breaking operation, and reads only records formed in the current cycle. Put*

$$b_* = \max\{|1-2\eta|, |1-2d|\}.$$

For every read-and-flip policy,

$$\big|P(B_t = -1) - \tfrac{1}{2}\big| \le \tfrac{1}{2} b_*^t.$$

In particular the loss converges to $1/2$. If $\eta \le \min\{d, 1-d\}$, its lower bound is $\frac{1}{2}[1-(1-2\eta)^t]$.

Proof. Let a_j be the exclusive-or of the bath flip and the instrument inversion. Condition on the apparent-body trajectory, the a_j, and all independent reading and actuation randomness. This is more information than the agent receives. Conditional inversion probabilities are

$$q_1 = \frac{\eta(1-d)}{\eta(1-d) + d(1-\eta)}, \qquad q_0 = \frac{\eta d}{\eta d + (1-\eta)(1-d)}.$$

Conditional on that information, the absolute sign bias is $\prod_{j=1}^t |1 - 2q_{a_j}|$. The absolute conditional body bias has the same value, since the apparent body is then fixed. Hence

$$|\mathbb{E}B_t| \le \left(\sum_{a=0}^{1} P(a)|1-2q_a|\right)^t = \big(|1-\eta-d| + |\eta-d|\big)^t = b_*^t.$$

The a_j are independent with a law unaffected by the policy. This proves both sides of the bound. The special case follows from $b_* = 1 - 2\eta$ under the stated condition. □

The limit $\frac{1}{2}$ is the loss of an agent that does nothing. An inversion is the image under γ of a world kick that the loop corrected, so the records never show it. The food fares worse: after an inversion the flip pushes it the wrong way, the harvest runs at a loss, and an agent that never reads its energy store cannot tell. Reading the store is a self-label, whose standard is the agent's own level assignment. The Bayes-optimal policy, computed by backward induction over all adaptive policies for eight cycles, respects the bound in every case with $\eta \le d$ (for $g = 0.9$, $d = 0.1$, $s = 0.95$, $\eta = 0.05$: 0.307 against 0.285 at $t = 8$). Consciousness III shows that an observer can be confidently and undetectably wrong on a two-point fibre of its protocol [3]; Theorem 37 adds that acting does not change the protocol in the relevant sense, and that inversions make the error grow to chance. The hypothesis of fresh records matters. If a stored record of an earlier body state, made by the environment's fixed couplings, is read again later, the two reads compare σ at the two times, so the world's stable records detect changes of the sign though not the sign. Records of Paper F's kind [6] thus carry an agent's calibration across time, and the standard sits in the past.

Proposition 38 (Reference cost under exogenous and adaptive timing)**.** *Let a reference cost $w > 0$ and reveal the current sign exactly, and let body loss cost $V > 0$ per use. For $0 \le r < 1$, define*

$$F_r(G) = \frac{G}{2} - \frac{1 - r^G}{2(1 - r)}, \qquad C_r = \inf_{G \in \mathbb{N}} \frac{w + V F_r(G)}{G}.$$

Reference times fixed in advance, or randomized independently of the record, give long-run average cost at least C_{b_*}, where b_* is defined in Theorem 37. For reference times chosen from the records, the bound C_{r_*} holds with

$$r_* = \max_{a=0,1} |1 - 2q_a|.$$

If $\eta \to 0$ with $d \in (0, 1)$ and w, V fixed, these benchmarks satisfy

$$C_{b_*} = \sqrt{2wV\eta}\,(1 + o(1)), \qquad C_{r_*} = \sqrt{2wV\eta\, \min\{d/(1-d), (1-d)/d\}}\,(1 + o(1)).$$

When a finite gap attains the first infimum, an ideal periodic read-and-control policy attains it; otherwise the no-reference limit gives its value $V/2$. It need not bound record-adaptive schedules.

Proof. With an exogenous gap, the mean loss at age j after a reference is at least $(1 - b_*^j)/2$, by the proof of Theorem 37. For an adaptive gap one cannot use this unconditional mean after conditioning on the absence of another reference. Instead, the same proof gives the conditional bound $(1 - r_*^j)/2$ for every apparent-event history. The event that a policy has not yet requested another reference is determined by such a history and its private randomness, so this bound also holds on each adaptive gap. Summing over complete gaps and the terminal incomplete gap gives at least $KC_r - O(w)$ over K uses. Equivalently, the piecewise linear interpolation of F_r is convex, and an exogenous mixture of gap lengths cannot beat the best single length for the corresponding benchmark. Writing $r = 1 - 2a$ gives $F_r(G) = aG(G-1)/2 + O(a^2G^3)$; minimization has $G \sim \sqrt{2w/(Va)}$ and cost $\sqrt{2wVa}(1 + o(1))$. For small η, $(1 - b_*)/2 = \eta$ and $(1 - r_*)/2 = \eta \min\{d/(1-d), (1-d)/d\}(1 + O(\eta))$. Ideal reads reveal the apparent state, and ideal flips can hold it aligned with the optimal sign estimate, attaining the exogenous benchmark. □

Remark 39 (Record-adaptive references improve the benchmark)**.** Take $\eta = d = 1/10$, $w = 1/10$ and $V = 1$. Read the apparent body exactly. After accounting for the preceding commanded flip, request a reference precisely when the apparent body has changed sign. Then control the body using the last reference's sign estimate. Such events have probability $p = 2\eta(1-\eta) = 9/50$. Between them the conditional inversion probability is $q_0 = 1/82$. The stationary posterior sign bias is

$$z = p + (1-p)(1-2q_0)z = \frac{9}{10}.$$

The loss is $1/20$, and references cost $9/500$, giving total cost $17/250 = 0.068$. The exogenous-gap benchmark is $1/10$. An exact finite-state computation with sensing gain and flip success both 0.999 gives cost 0.0687253, so the separation persists away from the ideal limit. The bound for adaptive timing is $C_{r_*} = 0.0429971$ here.

Proposition 40 (Exact adaptive value at equal inversion and bath rates)**.** *In Proposition 38, take $\eta = d = 1/10$, $w = 1/10$ and $V = 1$. The minimum long-run average cost, with ideal apparent-body reads and ideal flips, over all adaptive reference policies is*

$$\rho_* = \frac{2089311}{43698622} = 0.0478118280\ldots.$$

Request a reference after an apparent innovation or after five consecutive quiet transitions since the last reference. This policy attains the value, which also bounds policies with less informative sensing or control.

Proof. An apparent innovation has probability $p = 9/50$ and conditional sign bias zero; a quiet transition has probability $z = 41/50$ and multiplies the bias by $u = 40/41$. The oracle therefore has states $k = 1, 2, \ldots$ for k quiet transitions after a reference and a zero-bias state. At state k, continued control has loss $l_k = (1-u^k)/2$; a reference resets the bias and costs w. For threshold $K = 5$, a renewal cycle has

$$D_K = \sum_{j=0}^{K-1} z^j, \qquad C_K = w + \sum_{j=1}^{K-1} z^j l_j, \qquad \rho_* = C_K/D_K.$$

To verify global optimality, set $v_k = w$ for $k \geq 5$ and set $v_k = l_k - \rho_* + pw + zv_{k+1}$ backwards for $k = 4, 3, 2, 1$. Exact rational arithmetic gives

$$-\rho_* + pw + zv_1 = 0, \quad v_k < w \ (1 \leq k < 5), \quad l_5 > \rho_*.$$

Thus $v_k = \min\{w, l_k - \rho_* + pw + zv_{k+1}\}$ for every k; monotonicity of l_k proves the stopping inequality for all $k \geq 5$. The zero-bias state also stops, since $1/2 > \rho_*$. These bounded Bellman inequalities telescope over every adaptive, randomized or nonstationary policy, leaving a bounded terminal term. Dividing by the horizon proves the lower bound. Ideal readings reveal the apparent innovations, and ideal flips attain their conditional Bayes body loss, so the threshold policy is an admissible implementation of the oracle optimum. Less informative records are a garbling of this oracle. A less reliable flip can be simulated by an ideal actuator using a private coin, with its storage and reset charged to the policy. Neither restriction improves the oracle value. □

Self-parameter	Sign symmetry	Without any standard	With references or labels
Setting (switches)	invariant	detected by the loop (Proposition 23); found by search	not needed
Magnitude (random walk ν)	invariant	static body moments identify gain; a valid Gaussian free channel gives price $c\nu/\sqrt{J_0}$ (Proposition 36)	$\nu^{2/3}$ (Theorem 9); marginal self-labels avoid pointer preparation; full retained history determines their information
Sign (inversion rate η)	variant	loss $\to \frac{1}{2}$ for fresh-record read-and-flip policies (Theorem 37)	positive square-root lower bound for adaptive timing; exact finite-rate optimum in Proposition 40

Table 3: What acting buys for each self-parameter of the joint model.

In the exogenous benchmark with $V = 1$, the small-η calculation is within 4% of $\sqrt{2w\eta}$ over the tested $\eta \le 10^{-4}$ range. A periodic noisy-reference policy gives excess cost within a factor 2–3 of $\sqrt{2wV(1-2L_0)\eta}$, where $L_0 = 0.052$ is its no-inversion loss. Its fitted exponent is 0.41–0.43, consistent with the growing reference-block size. Proposition 40 settles the stated finite-rate ideal example. Proposition 38 supplies the general adaptive lower bound; its constant need not be sharp.

D A static witness for history-dependent repair

Proposition 41 (Equal sign confidence, opposite paid repairs)**.** *There is a static, clocked version of Definition 25 in which two legally acquired histories at the same working slot have equal sign confidence and opposite optimal repair decisions. Take $d = 1/20$, $s = 49/50$, $H = 12$, and a uniform prior on the four static states $(\sigma, g) \in \{\pm 1\} \times \{3/20, g_+\}$. At slot $t = 10$, compare the history with positive readings at slots 1 and 10 and disability at slots $2, \dots, 9$, with ten working readings of which eight are positive. Choose g_+ so that their sign confidences agree, with*

$$0.959375749239844 < g_+ < 0.959375749239846.$$

Both confidences are $0.876966618\dots$, whereas their posterior means of the unsigned gain are $0.678161237\dots$ and $0.249069192\dots$.

Each working invocation acquires one body reading and permits the food protocol of Proposition 28, or abstention from that food task. This witness does not acquire or read back the harvest-work self-label of Definition 17: its histories contain body readings, gate outcomes and the priced clock. The harvest work still enters the physical ledger. Food switching costs

$\kappa/2$ per executed task and latched recovery costs κ per disabled slot. Every working invocation also pays a declared mandatory processing charge c_{p}. A paid clock signals the slot and horizon; base record, flag, drive and clock supplies have the same wall-time charges for both latch choices. In this witness, body-reading erasure and food reset are supplied operations, excluded from $V(b)$ below rather than assigned zero universal cost. Proposition 31 establishes the corresponding opposite-repair phenomenon with the terminal reading and reset charges included. At $\kappa = 2/25$, $c_{\mathrm{p}} = 0$, the optimal latch differences satisfy

$$Q_1(b) - Q_0(b) \in (0.00559146, 0.00559147), \qquad Q_1(b') - Q_0(b') = -0.004272.$$

At $\kappa = 9/4000$, $c_{\mathrm{p}} = 3/25$, their differences satisfy

$$\begin{aligned} Q_1(b) - Q_0(b) &\in (0.00587980, 0.00587982), \\ Q_1(b') - Q_0(b') &\in (-0.00432522, -0.00432520). \end{aligned}$$

Proof. For a gain g put $a_g = (1+g)/2$ and $b_g = (1-g)/2$. The posterior after n readings, k positive, is proportional to $a_g^k b_g^{n-k}$ for sign $+1$ and $b_g^k a_g^{n-k}$ for sign -1. Let $c_{n,k}(x)$ be its positive-sign probability with gains $3/20, x$. The rational function $c_{2,2}(x) - c_{10,8}(x)$ has negative and positive values at the displayed rational endpoints. Its denominator is positive, so continuity gives an exact root. Both histories can be generated by the same legal policy that holds latch zero until slot ten. Their gate paths have probabilities $d^2(1-d)^7$ and $(1-d)^9$, independent of the static instrument; every required reading has positive likelihood. Gate and clock records therefore leave the count-based posteriors unchanged.

Write $m(b) = \mathbb{E}_b[\sigma g]$ and

$$F(x) = \ln 2 - H_2((1+x)/2), \qquad V(b) = \max\{0, F(s|m(b)|) - \kappa/2\} - c_{\mathrm{p}}, \qquad A(b) = \mathbb{E}_b V(b^Y),$$

where b^Y is the posterior after one further working reading. Only two future slots remain. If slot eleven works, its optimal continuation is common to the two current latches. It writes latch zero: a recovery switched on at slot twelve cannot create a working opportunity within this horizon. If slot eleven is disabled, the stored latch pays $\kappa\ell$ there; recovery makes slot twelve working with probability r_ℓ, otherwise the same charge is paid once more. Consequently

$$Q_1(b) - Q_0(b) = d\big[(r_1 - d)A(b) - \kappa(2 - r_1)\big], \qquad r_1 = d + s(1-2d). \tag{26}$$

This is the full Bellman difference for the stated action family.

The numerical intervals follow from rational enclosures, including the gain-root bracket. For $0 \le x < 1$ use

$$F(x) = \sum_{j=1}^{24} \frac{x^{2j}}{2j(2j-1)} + R_{24}(x), \qquad 0 \le R_{24}(x) \le \frac{x^{50}}{50 \cdot 49(1-x^2)}.$$

Rational interval evaluation gives, for the first price pair, $A(b) \in (0.2236, 0.2237)$ and $A(b') = 0$. For the second it gives $A(b) \in (0.1360, 0.1361)$ and $A(b') \in (-0.0954, -0.0953)$; retaining more digits gives the stated intervals. Substitution in (26) proves the strict decisions. □

Linear drift ε	10^{-2}	3×10^{-3}	10^{-3}
In-frame records, exact (asymptotic)	54 (49)	137 (129)	324 (312)
Probe records, exact (asymptotic)	58 (58)	130 (130)	271 (271)
Brownian drift σ	10^{-2}	10^{-3}	10^{-4}
In-frame records, exact (asymptotic)	17 (11)	68 (53)	282 (245)
Residual-variance test reaches errors 0.05	84	356	1565
Frozen self-model fails at $t^* = \ln 2/\sigma$	69	693	6931

Table 4: Detection times in steps for Paper II's reader ($\rho = 1$, $s = 0.5$, one record per step, errors 0.05). The rows labelled "exact (asymptotic)" are necessary times: "exact" sums the exact relative entropies, with the unwrapped angle normal of variance $2\sigma t$ under Brownian drift, and "asymptotic" is the formula of Proposition 6. The test row is the smallest horizon at which the residual-variance test of Corollary 7, with the exact χ^2 threshold, reaches both errors 0.05 (Monte Carlo).

The histories distinguish how much a future callable task is worth even when their confidence about its orientation is identical. The processing charge 3/25 is a stipulated implementation price, not a universal erasure lower bound. The first witness needs no such charge. Proposition 31 in the main text gives a further comparison at positive drift with the stated reading and reset charges.

E Worked example

E.1 Detecting drift in Paper II's two-channel reader

Take Paper II's reader with $D = 2$, $\rho = 1$, $s = 0.5$ and one record per step ($r = 1$). Then $\kappa = 36$ for in-frame records and $F_\perp(0) = 0.8$ for probes. For errors $\alpha = \beta = 0.05$, $d(0.95\|0.05) = 2.650$. Table 4 gives the smallest horizon at which the exact relative entropies of Corollary 5, summed over the records, reach d. By Proposition 6 no test with these errors can succeed earlier.

For known linear drift, the likelihood-ratio test reaches errors 0.05 at about 1.3 times the necessary horizon (4000 repetitions per point). The path-free residual-variance test needs roughly five times the Brownian lower bound; it precedes frozen-model failure at $\sigma = 10^{-3}, 10^{-4}$, but not at 10^{-2}. Its $T\sigma^{2/3}$ approaches the predicted constant 3.28. Near calibration the in-frame natural gradient is $0.0556/\theta$, whereas a probe fraction p supplies metric $0.8p$ at calibration.

E.2 The local Gaussian comparison for a drifting gain

Let the gain of the gain–bias channel drift with $g \approx 0.55$, $b = 0$, so $J = G = 1.4337$, and let references cost $w = k_B T \ln 2$. Table 5 gives the local Gaussian comparison of Remark 12, its leading term C, the optimal reference rate and the optimal memory weight.

ν per use	Floor (k_BT)	C (k_BT)	ρ^* (references per use)	α^*
10^{-2}	0.05511	0.05515	0.0265	1.95×10^{-3}
3×10^{-3}	0.02471	0.02471	0.0119	3.9×10^{-4}
10^{-3}	0.01188	0.01188	0.0057	9.1×10^{-5}
10^{-4}	0.002560	0.002560	0.0012	4.2×10^{-6}

Table 5: The local Gaussian comparison for a drifting gain ($g = 0.55$, $b = 0$, $w = k_BT \ln 2$). "Floor" is the Gaussian quadratic comparison $\min_P h(P)$ of Theorem 9 for the quadratic loss; C is the leading term of Corollary 10. For Paper I's divergence loss the entries are leading-order values, accurate for $\nu \lesssim 10^{-3}$. A record read through the same channel is worth $0.160\, k_BT$ (Paper I).

At $w = J = c = 1$ and $\nu = 10^{-2}$, average-cost value iteration on a variance grid agrees with the Gaussian floor to relative error 1.3×10^{-5}; 4000 Kalman trajectories agree within Monte Carlo error. Integer binomial reads provide a distinct comparison: at $w = 10^{-4}k_BT$, $\nu = 3 \times 10^{-3}$, the best tested grid policy costs 1.10 times the relaxed floor. For Corollary 11, $q = 10^{-4}$, $r = v = 1$, $w = 10^{-3}$ gives optimum 0.0054878. The source map identifies the full scans and fixed-seed outputs.

F A drifting instrument in a closed loop

Two finite bank models couple the estimate $(\hat{g}, \hat{b})$ to an action that changes the next world reading. Their instruments, baths and results remain separate. They specify their own assumptions and do not use (C1)–(C3).

F.1 The world and the observation map

The bank has M binary variables $z_j \in \{\pm 1\}$ with mean $x = M^{-1} \sum_j z_j$. In each epoch every variable is updated independently: it keeps its value with probability a, is replaced by a variable biased toward the agent's command u with probability c, and is replaced by a variable biased toward the environmental input d with probability r, where $a + c + r = 1$ and the biased variable is $+1$ with probability $(1 + v)/2$ for the bias $v = u$ or $v = d$. For $|u|, |d| \le 1$ this is a convex combination of the identity and two resets, hence a legal channel (Foundation III, section on legal finite feedback [11]). The probability of $+1$ is therefore $(1 + az_j + cu + rd)/2$, and

$$x_{k+1} = a\, x_k + c\, u_k + r\, d_k + \zeta_k, \qquad \mathbb{E}[\zeta_k \mid \text{past}] = 0, \quad \mathbb{E}[\zeta_k^2 \mid \text{past}] \le M^{-1}. \tag{27}$$

The count of $+1$ variables is updated exactly from two binomial draws. The simulations hold that count for efficiency and never give it to the agent. We use $(a, c, r) = (0.65, 0.25, 0.10)$.

An instrument with gain g and bias b, $|g| + |b| \le 1$, reads a uniformly random variable and returns $y = \pm 1$ with probability $(1 + gz_j + b)/2$. The index is not returned. A reference read presents an input known to be $+1$ or -1 and returns $+1$ with probability $(1 \pm g + b)/2$.

Gain and bias are fixed within an epoch and drift between epochs. From n_s world reads with mean $\bar{y}$ and the current estimates the agent forms

$$\hat{x} = Q\left(\frac{\bar{y} - \hat{b}}{\max(\hat{g},\, 0.30)}\right), \tag{28}$$

where Q clips to $[-1, 1]$ and rounds to a grid of 1025 points, and it acts with $u = -h\hat{x}$, $h = 0.8$. The total number of reads per agent and epoch is 512. World loss is $x^2/4$, averaged over trajectories and over epochs after a burn-in of 200.

F.2 What reference reads measure

Proposition 42 (Reference reads)**.** *Let $p \sim \mathrm{Bin}(n, \pi_+)$ and $m \sim \mathrm{Bin}(n, \pi_-)$ be independent, with $\pi_\pm = (1 \pm g + b)/2$ and $|g| + |b| < 1$. Then $(\hat{g}, \hat{b}) = \big((p - m)/n,\ (p + m)/n - 1\big)$ is unbiased with covariance*

$$\Sigma_{\mathrm{ref}} = \frac{1}{2n}\begin{pmatrix} 1 - g^2 - b^2 & -2gb \\ -2gb & 1 - g^2 - b^2 \end{pmatrix},$$

which is the inverse of the Fisher information matrix of the $2n$ reads about (g, b). The estimator is therefore efficient.

Proof. $\mathrm{Var}\, p = n(1-(g+b)^2)/4$ and $\mathrm{Var}\, m = n(1-(b-g)^2)/4$, which gives the covariance by linearity. The pair $(p/n, m/n)$ is the maximum-likelihood estimator of the mean parameters (π_+, π_-) of a product of binomials and attains the Cramér–Rao bound $\mathrm{diag}(\pi_\pm(1 - \pi_\pm))/n$. The map to (g, b) is linear and invertible, and efficiency is preserved under it. □

The frequency estimator above is unbiased and efficient for mean-square error, but its boundary values make the plug-in expected divergence infinite at every finite sample size. To use Paper I's calibrated-work expansion, replace each estimated response probability k/n by $(k + 1/2)/(n + 1)$, or use another regularization satisfying its tail condition. The leading covariance is unchanged, and the $1/n$ work loss is then an asymptotic statement for that regularized estimator.

For that regularized estimator, Paper I computes the leading calibrated loss: acting on the estimated readout model costs $k_B T(1 + g^2/(1 - b^2))/(4n)$ per use to leading order, $0.00254\, k_B T$ for $n = 128$ at $g = 0.55$, $b = 0$. A reference read costs a world read, so n_r is a budget. The estimates are also kept in memory. A persistent memory with weight α on the new reference estimate, $\hat{\theta}_k = Q[(1 - \alpha)\hat{\theta}_{k-1} + \alpha\tilde{\theta}_k]$ for $\theta \in \{g, b\}$, trades noise against lag.

Proposition 43 (Memory under drift)**.** *Let $0 < \alpha \le 1$, $\beta = 1 - \alpha$ and $e_k = \hat{\theta}_k - \theta_k$. Suppose the reference estimate has the form $\tilde{\theta}_k = \theta_k + \xi_k$, where θ_k is determined by the information available before the reference reads of epoch k (a deterministic drift qualifies, and the true instrument state may enter without giving the observer access to it) and ξ_k has conditional mean zero given that information and variance at most σ^2, that $\theta_k \in [-1, 1]$ and $|\theta_k - \theta_{k-1}| \le \nu$, and that Q clips to $[-1, 1]$ and rounds to a grid of spacing Δ. Then*

$$\|e_k\|_{L^2} \le \sqrt{\big(\beta(\|e_{k-1}\|_{L^2} + \nu)\big)^2 + \alpha^2\sigma^2} + \frac{\Delta}{2}, \qquad \|e_k\|_{L^2} \le X_\sharp + \beta^k\big(\|e_0\|_{L^2} - X_\sharp\big)_+,$$

with $X_\sharp = \dfrac{\Delta/2 + \beta^2\nu + \sqrt{\beta^2(\Delta/2+\nu)^2 + (1-\beta^2)\alpha^2\sigma^2}}{1-\beta^2}$. *If* $0 < \alpha < 1$, *the noise variance is exactly* σ^2, *and the drift is the idealised linear drift* $\theta_k = \theta_0 + \nu k$ *without rounding or clipping, the steady-state error has mean* $-\nu(1-\alpha)/\alpha$ *and variance* $\alpha\sigma^2/(2-\alpha)$. *Hence the persistent memory has smaller mean-square error than a memoryless estimate (weight* $\alpha = 1$, *error variance* σ^2*) exactly when*

$$\nu^2 < \frac{2\alpha^2\sigma^2}{(1-\alpha)(2-\alpha)}.$$

Proof. Put $y_k = \beta\hat{\theta}_{k-1} + \alpha\tilde{\theta}_k$, so that $y_k - \theta_k = \beta e_{k-1} - \beta(\theta_k - \theta_{k-1}) + \alpha\xi_k$. The noise ξ_k is orthogonal in L^2 to the first two terms, which are determined before it, so $\|y_k - \theta_k\|^2 \le \beta^2(\|e_{k-1}\| + \nu)^2 + \alpha^2\sigma^2$. Clipping to an interval that contains θ_k does not increase the distance to θ_k, and rounding adds at most $\Delta/2$; this is the first inequality. Write $\varphi(X)$ for its right-hand side, which is increasing with $\varphi'(X) = \beta^2(X+\nu)/\sqrt{\beta^2(X+\nu)^2 + \alpha^2\sigma^2} \le \beta$. The larger root $X_\sharp$ of $(X - \Delta/2)^2 = \beta^2(X+\nu)^2 + \alpha^2\sigma^2$ is the displayed expression, and it satisfies $X_\sharp \ge \Delta/2$, so $\varphi(X_\sharp) = X_\sharp$ and $\varphi(X_\sharp + t) \le X_\sharp + \beta t$ for $t \ge 0$. The second inequality follows by induction. For linear drift without rounding or clipping the recursion is linear, $e_k = \beta e_{k-1} - \beta\nu + \alpha\xi_k$, which gives the stated mean and variance. The mean-square comparison is $\alpha\sigma^2/(2-\alpha) + \nu^2(1-\alpha)^2/\alpha^2 < \sigma^2$. □

For Model A's $\alpha = 0.25$ and $n_r = 128$ reads per sign, the unrounded linear-drift comparison gives $\nu^* \approx 0.0161$ per epoch. It places the crossover at the scale of the observed four-to-eight-fold drift bracket; it omits the loop, gain floor and periodic drift. The Gaussian-transition optimum of Theorem 9 instead uses the stationary Kalman gain and real precision. A constant-weight estimator under linear drift has the different $\nu^{2/5}$ price of Remark 14.

The world is read through the same estimates. The next statement bounds the error of the decoded state in terms of the errors of the estimates.

Proposition 44 (Error of the decoded state)**.** *Let* $g \ge g_{\min} > 0$ *and let* $\bar{y}$ *be the mean of* n_s *readings of the bank, which are independent given the state* x. *Let* $\hat{x} = Q_x\big[\mathrm{clip}\big((\bar{y} - \hat{b})/\max(\hat{g}, g_{\min})\big)\big]$, *where* Q_x *rounds to a grid of spacing* Δ_x *that contains* ± 1. *Then*

$$\|\hat{x} - x\|_{L^2} \le \min\left(2,\ \frac{n_s^{-1/2} + \|\hat{g} - g\|_{L^2} + \|\hat{b} - b\|_{L^2}}{g_{\min}} + \frac{\Delta_x}{2}\right).$$

Proof. Write $\bar{y} = gx + b + \nu$ with $\mathbb{E}[\nu \mid x] = 0$ and $\mathbb{E}[\nu^2 \mid x] \le n_s^{-1}$, and $\hat{g}' = \max(\hat{g}, g_{\min}) \ge g_{\min}$. Then $(\bar{y} - \hat{b})/\hat{g}' - x = [(g - \hat{g}')x + (b - \hat{b}) + \nu]/\hat{g}'$. Since $g \ge g_{\min}$ one has $|g - \hat{g}'| \le |g - \hat{g}|$, and $|x| \le 1$, so the numerator has L^2 norm at most $\|g - \hat{g}\| + \|b - \hat{b}\| + n_s^{-1/2}$, and the division by $\hat{g}' \ge g_{\min}$ gives the bound before clipping. Clipping to $[-1, 1] \ni x$ does not increase the distance to x, and rounding adds at most $\Delta_x/2$. Both $\hat{x}$ and x lie in $[-1, 1]$, so the error is at most 2. □

Model A has $\nu_g \le 0.005357$, $\nu_b \le 0.006585$ and $g \ge 0.30$. With $\alpha = 0.25$, $n_r = 128$, $\sigma^2 = 1/(2n_r)$ and grid spacing $2/1024$, Proposition 43 gives limiting L^2 errors at most 0.0351 and 0.0376. Proposition 44, with $n_s = 256$, then bounds decoded-state error by 0.4518, compared with observed epoch-averaged error 0.135. These are expectation bounds for the stated drift, not simultaneous guarantees or bounds for the zero-gain fault.

F.3 One observer: the loop closes on the world

Model A. A single agent, a drifting instrument and a bath $d_k = 0.60 + 0.25\sin(2\pi k/177)$. Normal drift is

$$g(k) = 0.55 + 0.17\sin(2\pi k/420) + 0.03\sin(2\pi k/67),$$
$$b(k) = 0.14\sin(2\pi k/300) + 0.025\sin(2\pi k/43),$$

and a drift factor s replaces k by sk. The static instrument has $g = 0.55$, $b = 0$ and the same bath. Three policies share the interface and the budget of 512 reads. *Frozen*: 512 world reads, calibrated once from 2048 reads per sign at the start and never again. *Memoryless*: 256 world reads and 128 reads per sign each epoch, using only that epoch's references ($\alpha = 1$). *Persistent*: the same reads, $\alpha = 0.25$. Each policy pays the 4096 start-up reads. The agent does not know the drift functions.

Table 6 retains the static, normal-drift and fast-drift comparisons. A separate implementation and larger normal-drift run agreed within 1.7 combined standard errors; their full scans are identified in the source map.

Instrument	Frozen	Memoryless	Persistent
static instrument	3.426±0.036	3.589±0.009	3.520±0.008
drift ×1	5.299±0.043	3.652±0.010	3.591±0.010
drift ×4	4.642±0.040	3.654±0.010	3.622±0.009
drift ×8	4.543±0.042	3.643±0.008	3.802±0.008

Table 6: World loss $x^2/4$ ($\times 10^{-3}$, ± one standard error over 160 independent trajectories, 1200 epochs, burn-in 200).

At normal drift, persistent calibration lowers loss by 1.7% relative to memoryless calibration and 32.2% relative to the frozen policy. Freezing is best for the static instrument. The persistent-minus-memoryless difference changes from -0.032 ± 0.013 at four-fold drift to $+0.159 \pm 0.012$ at eight-fold drift ($\times 10^{-3}$). Memory therefore helps only within a drift range at this weight and read allocation.

F.4 Two observers: reliability and authority

Model B. Two agents read the same bank through different instruments (the instruments and the public event of Section 5 of Foundation II [10], here with $d_k = 0.6\,c_k$), each with the calibration of Section F.2 and the same budget of 512 reads. Each agent forms $\hat{x}_i$ by (28) and issues $u_i = -h\hat{x}_i$. The world receives the joint command $u = wu_1 + (1-w)u_2$, so $|u| \le h$ whenever $w \in [0,1]$. The agents may also move their bias estimates toward the partner's report, either with equal weight or weighted by $\mathrm{clip}(\hat{g}_j/0.4, 0, 1)^2$, which requires exchanging $\hat{g}$ as well as $\hat{x}$ (22 bits per sender and epoch instead of 11, so 44 for the pair instead of 22). The fixed rule is $w = \frac{1}{2}$. The convex rule is $w = s_1/(s_1 + s_2)$ with $s_i = \mathrm{clip}(\hat{g}_i/0.4, 0, 1)^2$, and $w = \frac{1}{2}$ if both vanish. In the convex rule the more reliable agent receives more of the authority, and the total never exceeds the cap h.

One-step task risk. Let the reports be $\hat{\boldsymbol{x}} = x\mathbf{1} + \epsilon$ with errors ϵ of any joint law, and let the weights $w \geq 0$, $\mathbf{1}^\top w = 1$, be determined by the information $\mathcal{F}$ available when they are chosen, which may include the reference reads of the epoch. Write $q = ch$, $z = (a-q)x + rd$, $R = \mathbb{E}[\epsilon\epsilon^\top \mid \mathcal{F}]$ (an uncentred second moment) and $b = \mathbb{E}[z\epsilon \mid \mathcal{F}]$. Weights are compared with the conditional laws of the reports and of the world held fixed.

Proposition 45 (Task risk of a weight vector)**.** *Suppose $|u| \leq 1$ and $|d| \leq 1$, so that all branch probabilities of the bank step lie in $[0,1]$, and the action is the linear rule $u = -h\, w^\top \hat{\boldsymbol{x}}$. Then*

$$\mathbb{E}\big[\tfrac{1}{4}x'^2 \,\big|\, \mathcal{F}\big] = K_{\mathcal{F}} + \frac{1 - M^{-1}}{4}\left(q^2\, w^\top R w - 2q\, b^\top w\right),$$

where $K_{\mathcal{F}} = \frac{1}{4}\big[(1-M^{-1})\mathbb{E}[z^2 \mid \mathcal{F}] + M^{-1}\mathbb{E}[1 - a^2 + a^2x^2 \mid \mathcal{F}]\big]$ does not depend on w.

Proof. Given x, u and d the variables move independently, so $\mathbb{E}[x' \mid x,u,d] = ax + v$ and $\mathrm{Var}(x' \mid x,u,d) = M^{-1}(1 - a^2 - v^2 - 2axv)$ with $v = cu + rd$. Hence $\mathbb{E}[x'^2 \mid x,u,d] = (1-M^{-1})(ax+v)^2 + M^{-1}(1-a^2+a^2x^2)$. With $u = -h(x + w^\top\epsilon)$ one has $ax + v = z - q\, w^\top\epsilon$. Taking the conditional expectation over ϵ gives the formula. □

The proposition needs no assumption that the errors are unbiased, centred, or independent between the two agents. Quantisation and the gain floor can sit inside ϵ. If the action is saturated or non-linear the formula no longer applies, and for $q = 0$ the weights do not change the one-step risk. When the errors are unbiased given x, d and $\mathcal{F}$, $b = 0$ and R is the error covariance. The risk is a convex quadratic in w. For two agents with $w = (t, 1-t)$, $q > 0$ and $D = R_{11} + R_{22} - 2R_{12} > 0$, its minimiser over $t \in [0,1]$ is

$$t_* = \mathrm{clip}\left(\frac{R_{22} - R_{12} + (b_1 - b_2)/q}{D},\, 0,\, 1\right). \tag{29}$$

If $R = \mathrm{diag}(v_1, v_2) + \tau^2 \mathbf{1}\mathbf{1}^\top$ and $b = 0$, the best combined error is $\tau^2 + v_1v_2/(v_1+v_2)$: weighting removes the private part of the error and cannot touch the common part.

Proposition 46 (Estimation-optimal and task-optimal weights differ)**.** *Take $M = 20$, $x = 0.2$ (twelve variables at $+1$), $(a,c,r) = (0.65, 0.25, 0.10)$, $h = 0.8$, $d = -0.7$. Let ϵ_1 be 0 or 0.2 and ϵ_2 be -0.1 or 0.1, each with probability $\frac{1}{2}$ and independent. The weight that minimises the joint estimation error is $t = \frac{1}{3}$; the weight that minimises the next-step world loss is $t = \frac{2}{3}$.*

Proof. The estimation error is $t^2\, 0.02 + (1-t)^2\, 0.01$, minimised at $t = \frac{1}{3}$. Here $q = 0.2$, $z = 0.02$, $R = \mathrm{diag}(0.02, 0.01)$ and $b = (0.002, 0)$, so (29) gives $t_* = \frac{2}{3}$. In every branch $|u| \leq 0.32$, so the step is legal. The values in Table 7 come from enumerating all four error branches and both binomial draws exactly, and they agree with Proposition 45. □

Agent 1's mean error 0.1 offsets $z = 0.02$, making more authority useful despite worse estimation. Task-optimal allocation needs the effects of action as well as sensor reliability. The observable gain proxy below is not the covariance R; no online learner for the true-error quantities R, b is supplied.

The linear term $-2q\, b^\top w$ also marks where Paper I's metric applies. Paper I prices a calibration error by $\frac{1}{2}k_B T\, \delta^\top G \delta$ when the downstream protocol is matched to the posterior

Allocation rule	t	Joint estimation error	Next-step world loss
Smallest joint estimation error	0.3333	0.006667	0.007525
Inverse centred variance	0.5000	0.007500	0.007501
Smallest next-step world loss	0.6667	0.010000	0.007493

Table 7: The example of Proposition 46, by exact enumeration.

the agent assumes, and then a calibrated self-model is a stationary point of the loss. With a fixed linear action rule it need not be one, and a biased report can lower the world loss, as in Proposition 46.

Four cells. We cross the two interventions that can be applied separately: the update rule (equal or weighted alignment) and the authority rule (fixed or convex). One agent's instrument is damaged for 100 epochs at a random start: a bias fault raises its bias by a nominal 0.30 before the legal clip, a gain fault sets its gain to 0.05 or 0 ("zero gain": the current reading carries no information about x). Table 8 gives the increase in world loss during the window relative to a matched run without the fault. The commands in all runs are $u_i = -h\hat{x}_i$ with $\hat{x}_i$ on the 1025-point grid, so $|u_i| \le h$ exactly; the largest $|u|$ applied over 105 runs was 0.8000 and no epoch exceeded the cap. Every cell is charged the same message budget, 22 bits per sender and epoch ($\hat{x}$ and $\hat{g}$) and 44 for the pair, although the unweighted, fixed cell does not use $\hat{g}$.

Policy	Bias fault	Gain 0.05	Gain 0
Unweighted, fixed $w = \frac{1}{2}$	$+0.01 \pm 0.01$	$+2.05 \pm 0.02$	$+2.53 \pm 0.02$
Unweighted, convex authority	$+0.03 \pm 0.01$	$+1.11 \pm 0.02$	$+1.30 \pm 0.02$
Weighted, fixed $w = \frac{1}{2}$	$+0.01 \pm 0.01$	$+1.32 \pm 0.02$	$+1.58 \pm 0.02$
Weighted, convex authority	$+0.01 \pm 0.01$	$+0.34 \pm 0.02$	$+0.31 \pm 0.02$
References only, no coupling	$+0.00 \pm 0.01$	$+1.46 \pm 0.02$	$+1.80 \pm 0.02$

Table 8: Increase of world loss in the fault window ($\times 10^{-3}$, mean of three seed blocks of 800 trajectories $\pm$ combined standard error; the different policies use different random draws, so the comparison between rows is not a paired test).

Weighted alignment limits a failed report's influence on the other estimate; convex authority limits its influence on the world while preserving the command cap. Together they reduce the zero-gain fault increase by 87.8% relative to equal alignment and fixed authority. The interaction, -0.042 ± 0.035 ($\times 10^{-3}$), is unresolved. Fault-free losses range from 3.156 to 3.167 ($\times 10^{-3}$), without a claim of equality.

F.5 What the two models give

Both models are finite banks with a declared instrument family, a declared bath and a declared cost vector (reads, start-up references, bits of memory and message). Model A shows a self-model built from reference reads improving the agent's effect on its world, with a memory whose useful length is limited by the drift. Model B shows that reliability can be

used in two separate places, in what an agent believes about its partner and in how much of the world it is allowed to move, and that the second place accounts for the larger part of the reduction when a sensor fails. The two models use different instruments and baths and are not one experiment.

G Simulation protocols for Section F

Model A. Use the parameters and policies of Section F.3 and start with $n_+ = \lfloor 0.6M \rfloor$, $M = 4096$. Each epoch reads the pre-action bank and references, updates the estimates, computes (28), then applies (27). The agent receives only read counts and its stored estimates. It retains 22 estimate bits and an 11-bit command index, pays 4096 startup references, 256 references per adaptive epoch and 4096 bank-step applications. Counters, index generation and arithmetic workspace are additional resources; no heat or work is inferred from these counts. Run lengths and errors are specified in Table 6.

Model B. The same bank and step, with the public event c_k (switching with probability 0.01 per epoch) and $d_k = 0.6\, c_k$. The instruments are

$$\begin{aligned} g_i(k) &= 0.55 + 0.17 \sin(2\pi k/420 + \phi_i) + 0.03 \sin(2\pi k/67 + \phi_i), \\ b_i(k) &= 0.10 \sin(2\pi k/250) + 0.03 \sin(2\pi k/53) \\ &\quad + 0.10 \sin(2\pi k/300 + \psi_i) + 0.025 \sin(2\pi k/43 + \psi_i), \end{aligned}$$

with $\phi = (0, 1.7)$ and $\psi = (0.4, 2.9)$, clipped to $|b| \le 0.98 - |g|$.

Each agent makes 512 reads per epoch, of which $2n_r$ with $n_r = 16$ are reference reads, and starts from 2048 reference reads per sign. Order within an epoch: both agents read the world and references on the same pre-action state; each updates its own calibration and forms $\hat{x}_i$; one synchronous exchange; the commands $u_i = -0.8\, \hat{x}_i$ and the joint command are formed from the values before the exchange; the world takes one joint step; the alignment update for the next epoch uses the exchanged values. Equal alignment: $\hat{b}_i \leftarrow Q(\hat{b}_i + 0.25\, e_i)$ with $e_i = \bar{y}_i - (\hat{g}_i \hat{x}_j + \hat{b}_i)$. Weighted alignment multiplies the step by $\mathrm{clip}(\hat{g}_j/0.4, 0, 1)^2$. The fault window lasts 100 epochs from a start drawn uniformly from $[800, 2400)$ for each trajectory; faulted and no-fault runs share it. Runs of 3000 epochs, burn-in 200, 800 trajectories per run, three seed blocks. The world and each agent draw from separate random streams addressed by seed, epoch and event, so a fault in one instrument does not shift the draws of the world or of the other agent; because the agents act on the world, faulted and no-fault worlds differ afterwards.

Reproducibility. The supplemental source map lists checks by manuscript table, proposition or equation label, with named scripts, archived outputs and hashes, including the two Model A implementations and Model B. Checks cover detection, drift prices, gauge symmetry, sign inversion, body moments, finite-block margins, continuing filtering and repair. Their individual assumptions and seeds are retained with the scripts; one experiment's output is not a certificate for a different model.

H Thermodynamic implementation inputs

H.1 Thermodynamic length

Let $\lambda(t)$, $t \in [0,\tau]$, be a protocol in control-parameter space. For conjugate forces $F_i = -\partial_i H$, define the friction tensor by

$$\zeta_{ij}(\lambda) = \beta \int_0^\infty \langle \delta F_j(0)\delta F_i(t)\rangle_\lambda \, dt.$$

The thermodynamic length is $\mathcal{L} = \int_0^\tau \sqrt{\zeta_{ij}\dot\lambda^i\dot\lambda^j}\, dt$ [22]. Within this slow-driving linear-response approximation, $W_{\rm ex} \geq \mathcal{L}^2/\tau$. Constant thermodynamic speed saturates the bound along a fixed path; minimizing also over paths gives a shortest geodesic. Where the normalized correlation integrals τ_{ij} are defined, $\zeta_{ij} = k_B T\, \tau_{ij} g_{ij}$ entry by entry. A common scalar relaxation time gives $\zeta = k_B T \tau_{\rm relax} g$ only when all the relevant entries have that same time [23].

I Local tracking of a moving optimum

I.1 Tracking a moving optimum

Let $\vartheta^*(t)$ be the minimizer of $L(\cdot,t)$, differentiable in t, and let $\|\dot\vartheta^*\|_g$ be its speed in the metric g. As a monitoring function take the squared geodesic distance

$$V(t) := d_g\big(\vartheta(t),\vartheta^*(t)\big)^2. \tag{30}$$

Theorem 47 (Loop tracking bound)**.** *Suppose the trajectory of* (24) *stays in a region where $d_g(\cdot,\vartheta^*(t))^2$ is smooth and $L(\cdot,t)$ is geodesically m_L-strongly convex for some $m_L > 0$. Then*

$$\frac{dV}{dt} \;\leq\; -2\eta m_L\, V + 2\sqrt{V}\,\|\dot\vartheta^*\|_g. \tag{31}$$

Consequently: (a) V decreases whenever $\sqrt{V} > \|\dot\vartheta^*\|_g/(\eta m_L)$; (b) if $\|\dot\vartheta^*\|_g \leq \Lambda$, then

$$\limsup_{t\to\infty} V(t) \;\leq\; \frac{\Lambda^2}{(\eta m_L)^2}; \tag{32}$$

(c) if ϑ^* is constant, $V(t) \leq V(0)\, e^{-2\eta m_L t}$.

Proof. The natural gradient $g^{-1}\nabla L$ is the Riemannian gradient $\operatorname{grad} L$. At ϑ the gradient of $d_g(\cdot,\vartheta^*)^2$ is $-2\exp_\vartheta^{-1}\vartheta^*$. Strong convexity along the geodesic from ϑ to ϑ^* gives $L(\vartheta^*) \geq L(\vartheta) + \langle \operatorname{grad} L(\vartheta), \exp_\vartheta^{-1}\vartheta^*\rangle + \frac{1}{2}m_L V$, and since ϑ^* is the minimizer, $L(\vartheta) \geq L(\vartheta^*) + \frac{1}{2}m_L V$. Adding the two, $\langle \operatorname{grad} L(\vartheta), -\exp_\vartheta^{-1}\vartheta^*\rangle \geq m_L V$. The contribution of the update to dV/dt is $\langle -2\exp_\vartheta^{-1}\vartheta^*, -\eta \operatorname{grad} L\rangle \leq -2\eta m_L V$. The contribution of the moving optimum is $\langle -2\exp_{\vartheta^*}^{-1}\vartheta, \dot\vartheta^*\rangle \leq 2\sqrt{V}\|\dot\vartheta^*\|_g$ by Cauchy–Schwarz. This gives (31) and (a). For (b), $W = \sqrt{V}$ obeys $\dot W \leq -\eta m_L W + \Lambda$ wherever $V > 0$, so $\limsup W \leq \Lambda/(\eta m_L)$. For (c) set $\dot\vartheta^* = 0$. □

For the two-channel frame the loss $s^2 + \rho^2 \sin^2\theta$ has $L'(0) = 0$ and $L''(0) = 2\rho^2$, so its Riemannian Hessian at calibration is $2\rho^2/g(0)$. When probes make $g(0) > 0$ (Proposition 34), continuity gives strong convexity in a neighbourhood of calibration with m_L close to $2\rho^2/g(0)$, and the theorem applies there. Without probes $g(0) = 0$ and it does not apply.

Under drift the loop cannot converge to zero mismatch; it holds the mismatch within the neighbourhood (32), which shrinks with ηm_L and grows with the drift speed. Raising η is not free: a physical implementation has to relate the adaptation rate to its energy budget (Section J), and in a stochastic loop a faster update also follows the noise, which is the trade-off Section 4 prices exactly. A second constant governs how well the metric itself is conditioned.

J Thermodynamic cost

J.1 Cost components

A physical calibration loop senses, computes, actuates and prepares references. What each costs depends on how it is realized.

Sensing and computing. Measurement, copying and logic can in principle be reversible, so reading a stream carries no positive cost by itself. If sensor and work registers irreversibly erase Shannon information at rates R_s and R_c bits per unit time, Landauer's bound gives $\dot{W}_{\text{sense}} \geq k_B T \ln 2\, R_s$ and $\dot{W}_{\text{compute}} \geq k_B T \ln 2\, R_c$.

References. A reference read requires an input in a known state. Preparing a known pointer state from the thermal state of a degenerate pointer costs at least $k_B T \ln 2$ per reference (Paper I). At R_{ref} references per unit time this is $k_B T \ln 2\, R_{\text{ref}}$, unless references are recycled.

Actuation. Moving the self-model from ϑ to a new estimate $\hat{\vartheta}^*$ is a finite-time transformation of whatever physical parameter stores it. In linear response, a protocol of duration τ_{recalib} and thermodynamic length $\mathcal{L}(\vartheta, \hat{\vartheta}^*)$ dissipates at least $\mathcal{L}^2/\tau_{\text{recalib}}$, a rate of at least $\mathcal{L}^2/\tau_{\text{recalib}}^2$ if repeated every τ_{recalib} [23].

J.2 A conditional implementation bound

Proposition 48 (Conditional implementation cost)**.** *Suppose that (i) sensing, computing, reference preparation and actuation use separate physical degrees of freedom with no shared erasure accounting; (ii) sensor and work registers erase Shannon information at rates R_s and R_c, and R_{ref} references per unit time are prepared from thermal states; and (iii) actuation follows a linear-response protocol of duration τ_{recalib} and thermodynamic length $\mathcal{L}$, repeated every τ_{recalib}. Then*

$$\dot{W}_{\text{loop}} \;\geq\; k_B T \ln 2 \left(R_s + R_c + R_{\text{ref}}\right) + \frac{\mathcal{L}^2}{\tau_{\text{recalib}}^2}. \tag{33}$$

Proof. Landauer's bound applies to the two declared erasure streams and to each reference preparation; the Sivak–Crooks bound divided by the protocol duration gives the actuation

rate. Separability allows the terms to be added without double counting. Dropping separability, or using reversible sensing and computation without resets in the window, removes the corresponding terms. □

The proposition and the drift law count different things. Remark 12 counts references and the loss of acting on an imperfect self-model, in the exact Gaussian comparison. Proposition 48 adds the costs of the registers and actuators of one implementation. Together they bound what a physically realized calibration loop pays per use. Papers I and II supply no universal maintenance or processing cost that could be added to these without a concrete implementation, so no general persistence budget is claimed.

K Finite-block margins and continuing filtering

K.1 A resource-conditioned operating margin

A lower bound on drive occupation cannot by itself supply an achievable gate. We instead choose the displaced squeezed Gaussian family of Q-RAIF B and use its exact characteristic function for both error and energy. If its mean occupation is N, set

$$V_Q = \frac{1}{2N+1}, \quad \lambda^2 = \frac{\pi^2(2N+1)}{16N(N+1)}, \quad \alpha^2 = \frac{\pi^2}{16\lambda^2}, \quad s = \frac{1+\exp(-2\lambda^2 V_Q)}{2}.$$

Then $\alpha^2 + (V_Q + V_Q^{-1} - 2)/4 = N$. For the coherent comparison use $V_Q = 1$, $\lambda^2 = \pi^2/(16N)$ and the same formula for s. Each opportunity receives a fresh independent scheduled packet, costing $\theta_\omega N$ whether or not it is switched on. Actual switching adds $\theta_\omega \lambda^2$. Incoming field energy and switching work thus refer to the same state.

The gain is physical and bounded. In blocks of $L = 4096$ food uses, draw independently one of $g_- = 0.90h$ and $g_+ = 0.98h$, with equal prior probabilities, where $r = 0.8^n$ and $h = \sqrt{1-r^2}$. Keep that gain constant through the block, with matched setting and known positive sign. At the start take $m = 32$ references of the input $+1$. The count K has likelihood

$$p_g(k) = \binom{m}{k}[(1+g)/2]^k[(1-g)/2]^{m-k}; \qquad \bar{g}_k = \frac{g_- p_{g_-}(k) + g_+ p_{g_+}(k)}{p_{g_-}(k) + p_{g_+}(k)}.$$

The reference pointer preparation costs $m \ln 2$. We conservatively erase its outcomes separately for another $m \ln 2$, and erase the $(m+1)$-state count register for $\ln(m+1)$ per block. This finite-memory filter incurs no Gaussian approximation or truncation.

Write

$$S_r(x) = H_2\left(\frac{1+\sqrt{r^2+(1-r^2)x^2}}{2}\right).$$

The posterior-optimal food levels give income $\ln 2 - H_2((1+s\bar{g}_k)/2)$. The food reset against the untouched quantum copy costs $\ln 2 - S_r(0) + S_r(\bar{g}_k)$; the switching probability is exactly $1/2$. Its net, averaged over the exact law of K, is

$$A = \mathbb{E}_K[S_r(0) - S_r(\bar{g}_K) - H_2((1+s\bar{g}_K)/2)] - \theta_\omega(N + \lambda^2/2) - \frac{2m\ln 2 + \ln(m+1)}{L}.$$

n	θ_ω	Drive	N	A: margin	B: margin	Body loss	B: purity
3	0.01	coherent	6.220	-0.1607	-0.5798	0.1402	8.334
3	0.01	squeezed	4.220	-0.0910	-0.4899	0.1377	8.334
5	0.01	coherent	7.343	0.0533	-0.2522	0.1037	13.890
5	0.01	squeezed	4.758	0.1393	-0.1402	0.1013	13.890
7	0.01	coherent	7.865	0.1552	-0.0985	0.0896	19.446
7	0.01	squeezed	5.019	0.2492	0.0240	0.0873	19.446
5	0.05	coherent	2.522	-0.1104	-0.4709	0.1097	13.890
5	0.05	squeezed	2.454	0.0036	-0.3540	0.1032	13.890
7	0.05	coherent	2.708	-0.0196	-0.3317	0.0955	19.446
7	0.05	squeezed	2.587	0.1069	-0.1999	0.0891	19.446
10	0.10	coherent	1.587	-0.0810	-0.3884	0.0945	27.780
10	0.10	squeezed	1.919	0.0410	-0.3007	0.0840	27.780

Table 9: Achievable conditional-entropy limit in units of k_BT. Margins are harvest minus the listed reset, reference and drive costs after food and record resources have been supplied. Each row optimizes food-only drive within its family, then uses that same state for food and body. The B: purity column is external record-memory purity per food use, $(4n+mn/L)\ln 2$, in addition to the listed operating costs; A uses $(2n + mn/L)\ln 2$. Body loss is averaged after each step over the full block. Calibration costs 0.011684 per food use.

The reset is the reversible conditional-entropy limit for the stated classical–quantum state, retaining the copy marginal. Its attainability uses quasistatic processing across independent replicas of the complete block protocol, conditioned on the retained count. An implementation with reset overhead below a positive margin has the same sign of net work.

Economy B adds one body read and drive opportunity per food use. The body starts each block uniform and is thermalized by a work-free swap with a fresh equilibrium bit at the block boundary. With bath flip probability $d = 0.05$ and feedback on every negative read,

$$x_{g,t+1} = 0.9[(1 - s)x_{g,t} + sg], \qquad x_{g,0} = 0.$$

Given $K = k$, define $x_k = \mathbb{E}[x_{g,t} \mid k]$, $y_k = \mathbb{E}[gx_{g,t} \mid k]$, and $p_\pm = (1 \pm y_k)/2$. The conditional body-copy biases are $b_\pm = (x_k \pm \bar{g}_k)/(1 \pm y_k)$. Body erasure therefore costs

$$H_2(p_+) - S_r(x_k) + \sum_\pm p_\pm S_r(b_\pm),$$

and its drive costs $\theta_\omega[N + p_-\lambda^2]$. Subtract their exact block average from A to get B. This policy retains only K, ignores body readings as calibration data, and requires no unproved body-stream information rate.

At $n = 7$ and $\theta_\omega = 0.01$, the squeezed construction leaves $0.02396\,k_BT$ per food use after body upkeep, with body loss 0.08725; the coherent comparison leaves $-0.09847\,k_BT$. A sufficiently small irreversible overhead preserves the positive squeezed margin. This is a maintained body supplied by a record and drive stream, with a gain that changes between blocks. It is not an infinite Gaussian-gain process.

Record formation consumes $2n \ln 2$ of fresh memory purity per food use in A and $4n \ln 2$ in B, plus $mn \ln 2/L$ for reference fragments. Those memories are supplied by the environment and their purity is explicitly an input, not work produced by the cycle. Replenishing them from the agent's own thermal resources debits that input and eliminates the apparent profit. The drive additionally presupposes its preparation, phase reference and scheduled port; uncomputed apparatus overhead can be tolerated only within the displayed positive margin. The example identifies a nonempty region of positive operating margin under these resource declarations. It does not derive the resources or the reference standard from the loop.

K.2 A continuing joint filter and adaptive references

The block calculation redraws a prior and resets the body at known times. A continuing observer instead carries its model through the changes that its own history reveals. Keep the $n = 7$ fragments and the same single-pass squeezed drive, $N = 5.01898$, $s = 0.979992$ and $\lambda^2 = 0.225388$. Let the gain switch between $g_- = 0.90h$ and $g_+ = 0.98h$ with probability $\delta = 3 \times 10^{-4}$ per use, and the sign invert with probability $\eta = 2 \times 10^{-4}$. The body bath has $d = 0.05$. These are bounded transitions valid for arbitrarily long operation. The body, instrument and stored model continue through every reference-budget boundary.

The hidden state $Z_t = (G_t, \sigma_t, B_t)$ has eight values. Each use gives one body reading with likelihood $(1 + y\sigma g B)/2$. A requested reference gives one known-input reading with likelihood $(1 + z\sigma g)/2$. Its request can depend on the body reading and earlier records. The observer then selects a body flip by the sign of its posterior body mean, and food levels by its posterior signed gain. It receives two fresh scheduled drive packets per use, one for each task, and a maximum of $m = 32$ reference ports per $L = 4096$ uses. The ports, their timing and their input labels are supplied equally to the compared policies; an unused port creates no additional income.

For body command $a \in \{0, 1\}$, let T_a be the eight-state transition matrix. Its gain and sign factors are the symmetric switching kernels of rates δ and η. The body flips with probability d for $a = 0$, and with probability $d + s(1 - 2d)$ for $a = 1$. All entries of T_a are positive. Given the selected actions and realized records, the filter predicts with T_a and multiplies by the two likelihoods, omitting the second when no reference is requested. The request rule contributes no extra parameter-dependent likelihood factor. The fair food reading alone supplies no instrument information and is used only for its task.

Positive transition and observation kernels give a uniform finite-register certificate for this filter (Proposition 49, Appendix L). It controls quantization error even when actions and reference requests are chosen from the rounded posterior.

This filter forms its next prior from the preceding posterior and the actual transition law. It does not request a newly assigned gain prior or reprepare the body. For the stated parameters, $a_0 = 3 \times 10^{-9}$, $D = 39.14572$ and $\tau = 0.999999993681$. A log grid with $\Delta = 2^{-44}$ uses 350 model bits; its asymptotic total-variation bound is 2.25×10^{-6} and its one-step excess-risk bound is 4.50×10^{-6}. These conservative uniform bounds concern quantization of a correct filter. They do not certify a wrong transition family or the arithmetic implementation. The 350-bit state is an abstract finite controller; its workspace, update throughput and erasure protocol remain to be implemented.

For a resource-matched comparison, the periodic policy uses one reference every 128 steps. The adaptive policy requests a reference whenever the posterior probability of the less likely sign exceeds 0.04, subject to the same 32-port cap. Both update the joint model from every body reading and use the same body-control rule. Starting with the uniform eight-state prior, 128 independent trajectories of 65536 steps, with the first 8192 discarded, give

	Body loss	References/use	Food mismatch
Periodic	0.11726 ± 0.00083	0.0078125	0.06726 ± 0.00048
Adaptive	0.10319 ± 0.00066	0.005704 ± 0.000020	0.05226 ± 0.00014.

Errors are one standard error across independent trajectories. Food mismatch is the posterior Jensen gap $H_2((1+s|\mathbb{E}[\sigma g \mid \mathcal{H}_t]|)/2) - \mathbb{E}[H_2((1+sg)/2) \mid \mathcal{H}_t]$ in nats. The adaptive gain mean has mean-square error 0.000353, compared with the stationary prior variance 0.001529. A separate seed and shorter horizon give body losses 0.11730 and 0.10344. Requesting too early is costly under the cap: threshold 0.01 exhausts references prematurely and gives body loss 0.19567 in that shorter run. Adaptive timing needs an allocation rule as well as a confidence statistic.

Every policy is allocated the same food and record stream, drive state and 32-port reference budget. The external record purity is at most $(4n + mn/L)\ln 2 = 19.44603$ nats per food use, and reference preparation plus conservative separate erasure is charged at the same supplied-cap rate $2m \ln 2/L$. The posterior is updated from the body's own records and the policy acts on that body, closing the inference–control loop. This establishes continuing model revision and an operational benefit at finite memory. It does not close the apparatus resource loop: conditional-copy reset, model-register processing, field preparation, phase reference and the clock still need one attainable cyclic implementation. The block margin of Table 9 cannot be imported into this continuing protocol without that ledger.

L Finite-filter certificates

L.1 Continuing eight-state filter

Proposition 49 (Finite continuing filter)**.** *Let $a_0 = \min_{a,i,j}(T_a)_{ij} > 0$, and let every observation likelihood lie in $[l_-, l_+]$ with $l_- > 0$. Suppose at most two observations precede each filter update. Set*

$$\epsilon_0 = a_0(l_-/l_+)^2, \quad A = -\ln \epsilon_0, \quad D = \max_{a,i,j,k,l} \ln \frac{(T_a)_{ik}(T_a)_{jl}}{(T_a)_{il}(T_a)_{jk}}, \qquad \tau = \tanh(D/4) < 1.$$

After the first prediction, every exact update has probability at least ϵ_0 in each state. Round each of the seven log ratios $\ln(p_i/p_8)$ to a grid of spacing Δ, then normalize their exponentials. This uses at most

$$7 \left\lceil \log_2 \left(\frac{2A}{\Delta} + 3 \right) \right\rceil$$

bits for the model. For the exact posterior p_t^* conditioned on the same realized history and the rounded filter $\tilde{p}_t$,

$$d_H(\tilde{p}_t, p_t^*) \le \tau^t d_H(\tilde{p}_0, p_0^*) + \frac{\Delta(1-\tau^t)}{1-\tau}, \qquad \|\tilde{p}_t - p_t^*\|_{\mathrm{TV}} \le \tanh\big(d_H(\tilde{p}_t, p_t^*)/4\big).$$

Here $d_H(p,q) = \max_i \ln(p_i/q_i) - \min_i \ln(p_i/q_i)$. A one-step action minimizing a loss in $[0,1]$ under $\tilde{p}_t$ has excess conditional risk at most $2\|\tilde{p}_t - p_t^*\|_{\mathrm{TV}}$. With a finite reference counter and clock, the eight-state plant and rounded controller form a finite Markov chain and possess an invariant distribution under the supplied streams.

Proof. Prediction gives each probability at least a_0. The product of at most two positive likelihoods reduces that bound by at most $(l_-/l_+)^2$. Hence the seven log ratios lie in $[-A, A]$, and rounding increases their range by at most $\Delta/2$ at each end. In the Hilbert metric, positive diagonal likelihood multiplication and normalization are isometries, and multiplication by T_a contracts by at most τ, the Birkhoff coefficient. This is the standard projective stability mechanism for finite hidden Markov filters [20]. Rounding changes the oscillation of the log ratios by at most Δ. The triangle inequality gives $e_t \le \tau e_{t-1} + \Delta$, which iterates to the displayed bound even when actions and reference requests depend on earlier rounded states. To obtain the total-variation constant, let $m \le p_i/q_i \le M$, with $M/m = e^e$. Normalization and the chord bound for $(x-1)_+$ give

$$\|p - q\|_{\mathrm{TV}} \le \frac{(M-1)(1-m)}{M-m} \le \tanh(e/4),$$

where maximizing over m at fixed M/m gives $m = e^{-e/2}$; distance zero is immediate. Comparing a minimizing action under each distribution costs at most twice that distance for a bounded loss. The model grid, budget counter, clock and eight-state plant are finite, so their joint transition has an invariant probability. This last statement need not give a unique invariant distribution or an optimal reference policy, and it concerns the declared plant and controller rather than the whole resource-supplying apparatus. □

L.2 Filter on working invocations

The hard access gate has zero likelihoods on excluded body states, so Proposition 49's eight-state certificate cannot simply be inherited. Filtering at actual working invocations avoids this issue. Each of the two event matrices is positive. For the present switching rates,

$$\min_{\ell,i,j} (\mathsf{K}_\ell)_{ij} \ge a = (1-d)\delta\eta = \frac{57}{10^9}, \qquad g_{\max} < 0.98.$$

One body likelihood multiplication gives a posterior lower bound $a/99$, hence a log-ratio range less than 22. Event cross-ratios are at most a^{-2}, so their common projective contraction coefficient is at most $(1-a)/(1+a)$. Round the three log ratios on a grid $\Delta = 2^{-44}$. The proof of Proposition 49, now applied once per working invocation, gives 150 ideal model bits and

$$\limsup \|\tilde{p}_n - p_n^*\|_{\mathrm{TV}} \le \frac{\Delta(1+a)}{8a} < 1.247 \times 10^{-7}.$$

For a one-step loss in $[0,1]$, regret is at most twice the total-variation distance and hence less than 2.494×10^{-7}. The latch, startup state, arithmetic, workspace and protection are separate from these model bits. In wall time the plant, held finite register and latch form a finite Markov chain; they have an invariant probability under the supplied streams. The claim neither processes new data during disability nor certifies a cyclic processor.

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